MUHS final-year BPTh (IV BPTh), 2012 syllabus

Cardio-Vascular and Respiratory Physiotherapy

Complete exam notes: Chapter 6 conditions, technique cards, the other syllabus chapters, and exam skills.



Built from	20 MUHS question papers, Summer 2016 to Summer 2026 (code 51413)
Contents	23 Chapter 6 topics and 34 cards and chapter pages, with past questions, answer skeletons and self-tests
Content checked	19 October 2026
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/

Educational notes for exam preparation. Not medical advice. Written in original words from current textbooks and dated guidelines; each page lists its sources.

How to use these notes

1. **Read a topic once**, slowly, including the "Why?" boxes.
2. **Close it and recall.** Write the keywords and draw the diagrams on paper from memory, then compare.
3. **Test yourself** with the questions at the end of each topic. Cover the answer before you look.
4. **Come back** on day 1, day 7 and day 21. Testing yourself again beats re-reading.

> **What the real paper looks like (20 papers, 2016–2026)**
>
> 80 marks in 3 hours. **No MCQs**, even though the syllabus printed a scheme with them.
>
> - Short answers: any 5 of 6 at **3 marks**, and any 5 of 6 at **7 marks**.
> - Long answers: two questions, each any 1 of 2 at **15 marks**. Every option is a patient case, and almost all ask for ICF, a functional diagnosis or impairments.
>
> Since 2023, ventilated ICU patients appear as long cases and an ICU short question is in every paper.

How this book is arranged

- **Part A**: the 23 Chapter 6 conditions in three tiers, by how often they are asked.
- **Part B**: technique cards (ICU, airway clearance, breathing, oxygen and aerosols, therapy skills, exercise, investigations and scales).
- **Part C**: cardiac and pulmonary rehabilitation, the revision chapter (anatomy, physiology, drugs) and basic life support.
- **Part D**: answer writing (ICF, goals, documentation) and the practical exam.

Each topic opens with its syllabus heading, priority, the sessions it was asked in (LAQ = 15-mark long answer, 7m and 3m = short answers; S = Summer, W = Winter, for example S24 = Summer 2024), and the guidelines it was checked against. Page references such as (p. 123) point to related pages. Abbreviations are listed at the end.

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Syllabus map

Where each chapter of the MUHS CVR Physiotherapy syllabus is covered in this book.

Chapter	Syllabus heading	Pages in this book
1	Review of applied anatomy, physiology and pharmacology	Respiratory anatomy, physiology and mechanics of the thorax (p. 343) Cardiac anatomy and physiology for physiotherapists (p. 348) Cardiac and respiratory drugs: what physiotherapists need to know (p. 352)
2	Investigations and exercise testing	Investigations: chest X-ray, PFT, ECG and bedside tests (p. 317) ABG interpretation and why patients desaturate (p. 326) Abnormal breathing patterns and chest signs (p. 330) Exercise testing: contraindications, stop criteria, Bruce, cycle, walk and step tests (p. 308)
3	Exercise physiology	Exercise physiology and prescription: METs, energy systems, intensity, training (p. 313) Effects of bed rest and early mobilisation (p. 241)
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10	Functional scales	Scales: Borg, mMRC, NYHA, CCS and quality-of-life questionnaires (p. 322)

Chapter	Syllabus heading	Pages in this book
11	Basic life support	Basic life support (CPR) for physiotherapists (p. 356)
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Part A

Chapter 6: Physiotherapy management

Conditions, in the order they score in the real papers.

Tier 1: full notes, asked almost every year

IHD, myocardial infarction, angioplasty and CABG, with cardiac rehab 7
Burns: acute care, positioning, splinting and inhalation injury 31
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Ischaemic heart disease, myocardial infarction and CABG

> The most-asked long answer in this paper. Every case asks the same routine: pull the ICF out of the story, set goals that fit the patient's work and home, then plan cardiac rehab with doses.

Syllabus	Ch 6(a)(ii) I.H.D., myocardial infarction; surgery from 6(a) and Ch 7 cardiac rehabilitation
Priority	**Tier 1.** In 18 of 20 papers (2016–2026). 16 long-answer cases, more than any other topic.
Asked as	LAQ S16 · S17 · W17 · S18 · W18 · S19 · S20 ×2 · W20 · S21 · W22 · W19 · S23 · S24 · S25 · S26 · 7m W16 · W17 · S16 · S21 · S23 · W24 · W25 · 3m S17 · W17 · S20 · W20 · S24 · W24 · W25 ×2
Usual split	ICF 5 + cardiac rehab 10, or 3 + 5 + 7, or 5 + 7 + 3 (ICF, rehab, lifestyle)
Checked against	ESC 2023 and ACC/AHA 2025 acute coronary syndrome guidelines; ACSM and AACVPR rehab criteria; AHA 2023 resistance-training statement; Cochrane reviews. Full list under Sources. Content checked 18 October 2026.

1 Definition

IHD, also called coronary artery disease (CAD), is a condition in which the heart muscle gets less blood and oxygen than it needs. The cause is usually atherosclerosis of the coronary arteries.

Angina pectoris is chest pain or discomfort caused by *temporary* ischaemia of the heart muscle. The muscle is not permanently damaged.

Myocardial infarction (MI) is death of heart muscle caused by prolonged ischaemia.

> **Exam point: the modern definition of MI**
>
> The Fourth Universal Definition of MI (2018) needs two things:
>
> 1. a rise and/or fall of cardiac troponin, with at least one value above the 99th percentile of normal, and
> 2. evidence of ischaemia: typical symptoms, new ECG changes, new Q waves, a new wall-motion defect on imaging, or a clot seen on angiography.

ACS is the umbrella term for sudden ischaemia. It covers unstable angina, NSTEMI and STEMI.

PCI, also called angioplasty or PTCA, opens the blocked artery from inside with a balloon, and usually leaves a stent behind.

CABG is open-heart surgery. The surgeon joins a graft vessel around the blocked part of the coronary artery so blood can bypass the block.

2 Classification and grading

The coronary spectrum

Type	Pain	ECG and troponin
Stable angina	On exertion; settles within minutes with rest or nitrate	ECG often normal at rest; troponin normal
Unstable angina	New, worsening, or at rest	May show ST depression; troponin **normal**
NSTEMI	Prolonged, at rest	ST depression or T inversion; troponin **raised**
STEMI	Severe, over 20 minutes, not relieved by rest	**ST elevation**, later Q waves; troponin raised

MI can also be described by **site** (anterior, inferior, lateral, posterior) and by **depth**. A full-thickness (transmural) infarct usually shows ST elevation and later Q waves. A subendocardial infarct affects the inner layer only.

Grading scales examiners ask for

Grade	NYHA (heart failure symptoms)	CCS (angina)
I	No limitation. Ordinary activity causes no symptoms.	Angina only with strenuous, rapid or prolonged exertion
II	Slight limitation. Ordinary activity causes fatigue, palpitations or breathlessness.	Slight limitation: walking fast, climbing stairs quickly, after meals, in cold or stress
III	Marked limitation. Comfortable at rest; less than ordinary activity causes symptoms.	Marked limitation: walking one or two blocks on the level, or one flight of stairs
IV	Symptoms at rest. Any activity increases discomfort.	Unable to do any activity without discomfort; angina may occur at rest

Killip class grades heart failure in the first days after MI: I no failure; II crackles in the lower lungs or a third heart sound; III frank pulmonary oedema; IV cardiogenic shock.

Universal Definition types of MI: type 1 is plaque rupture with clot; type 2 is a supply–demand mismatch (for example severe anaemia or fast arrhythmia); type 3 is sudden cardiac death; types 4 and 5 are related to PCI and CABG.

3 Causes and risk factors

Main cause: atherosclerosis. Fatty plaque builds up in the artery wall and narrows the lumen. Less common causes include coronary spasm, an embolus, a tear in the artery (dissection), and extra demand from severe anaemia or a very fast heart rate.

Risk factors (asked as 3 marks in S17 and W17)

Non-modifiable	Modifiable
• Age: men over 45, women over 55 • Male sex (women catch up after menopause) • Family history of early heart disease: father or brother before 55, mother or sister before 65 • South Asian origin: heart disease starts earlier and is more severe	• Smoking and all tobacco: cigarettes, bidi, gutkha • High blood pressure • Diabetes and insulin resistance • Abnormal lipids: high LDL, low HDL, high triglycerides • Obesity, especially a large waist • Physical inactivity • Unhealthy diet: excess salt, fried food, vanaspati and other trans fats • Excess alcohol • Stress and depression

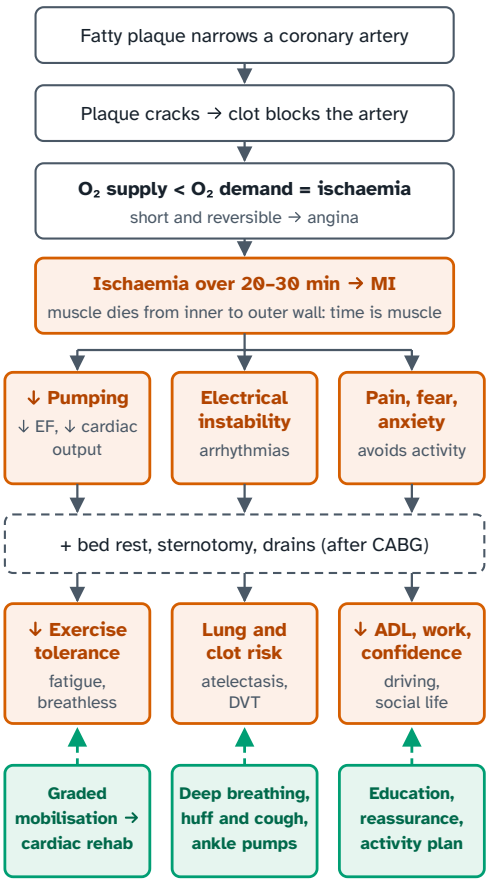
> **Exam point**
>
> For "3 modifiable and 3 non-modifiable", write the three strongest in each column: age, male sex, family history and smoking, hypertension, diabetes. Abnormal lipids is an equally good fourth.

4 Pathophysiology to impairments

Examiners never ask for pathophysiology alone. They ask for "clinical reasoning for the impairments". This chain is that reasoning.

The heart muscle's oxygen demand rises with **heart rate**, **blood pressure** and the force of contraction. Supply falls when a coronary artery narrows. Ischaemia starts when demand is greater than supply.

From plaque to the problems physio treats



Green dashed arrows show where physiotherapy acts

In the exam, draw a quick sketch in about 2 minutes: the four-step chain down the middle, then three problems, three impairments and three physio actions.

Text version (write this as the answer)

1. Atherosclerotic plaque narrows the coronary artery.
2. The plaque ruptures and a clot (thrombus) forms.
3. Oxygen supply falls below demand, which is ischaemia. If it is brief, it causes angina.
4. If ischaemia lasts more than 20–30 minutes, heart muscle dies (MI), from the inner wall outwards over a few hours.
5. Results: reduced pumping (lower ejection fraction and cardiac output), electrical instability (arrhythmias), and pain, fear and anxiety.
6. Bed rest, and after CABG the sternotomy and drains, add deconditioning, shallow breathing and clot risk.
7. Impairments: low exercise tolerance with fatigue and breathlessness; risk of atelectasis and DVT; reduced ADL, work and confidence.
8. Physiotherapy: graded mobilisation leading to cardiac rehab; breathing exercises, huff and cough, ankle pumps; education and reassurance.

Why physio cares about healing time

Dead muscle is slowly replaced by scar. The scar takes about **6–8 weeks** to become firm. Early activity is safe and helpful, but heavy straining is avoided until the scar matures.

? Why does heart rate × blood pressure matter?

> Heart rate multiplied by systolic BP is the **rate–pressure product**. It is a bedside estimate of how much oxygen the heart muscle needs. Angina tends to start at the same rate–pressure product in the same patient. Keep exercise below that level, and progress slowly so the heart can adapt.

Extra problems after CABG

- **Pain** from the sternotomy and graft sites makes breathing shallow and cough weak.
- **Anaesthesia, ventilation and lying flat** reduce lung volumes. This leads to basal atelectasis.
- **Guarding** of the chest stiffens the shoulders and thoracic spine and causes a rounded posture.
- **Graft sites** in the leg or forearm can swell, hurt or get infected.

Recall check. Close your eyes and say three keywords from this section. Then look back and compare.

5 Clinical features

Angina

- Central chest tightness, heaviness or pressure. Patients often describe it as a weight rather than a sharp pain.
- Spreads to the left arm, jaw, neck or back.
- Triggered by exertion, cold, heavy meals or emotion.
- Stable angina settles within a few minutes of rest or a nitrate under the tongue.

Myocardial infarction

- Severe chest pain lasting **more than 20 minutes**, not relieved by rest or nitrate.
- Sweating, nausea and vomiting, breathlessness, weakness, and a sense of impending doom.
- Signs: fast or slow pulse, low blood pressure, extra heart sounds, and crackles at the lung bases if the left ventricle is failing.
- Inferior MI often causes a **slow pulse** or heart block, because the right coronary artery usually supplies the SA and AV nodes.

> **Silent or atypical MI**
>
> People with diabetes, older people and women may have little chest pain. They present with breathlessness, tiredness, upper-abdominal pain or confusion. The S20 case of a woman with diabetes and an EF of 25% is a typical example.

After CABG

Expect a patient with sternotomy and graft-site pain, chest drains, shallow breathing, reduced breath sounds at the lung bases, a weak cough, tiredness, a swollen leg or forearm at the graft site, and stiff shoulders. The S19 case ("unable to raise his hands above head") is this last problem.

6 Investigations

Write only what changes physio decisions. That is what examiners look for in a case answer.

ECG (3 marks in S24: criteria for acute and evolved MI)

- **STEMI**: new ST elevation in two neighbouring (contiguous) leads. The threshold is at least **1 mm** in most leads. In V2–V3 it is **2 mm** for men aged 40 or over, **2.5 mm** for men under 40, and **1.5 mm** for women.
- **Sequence in STEMI**: tall (hyperacute) T waves → ST elevation → pathological Q waves → T-wave inversion.

- **NSTEMI and unstable angina**: ST depression or T-wave inversion.
- **Evolved (established or old) MI**: **pathological Q waves** in two neighbouring leads, with T-wave inversion and the ST segment back to baseline. A Q wave is pathological when it is at least `0.03 s` wide and `1 mm` deep, or any Q wave wider than `0.02 s` in V2–V3 (Fourth Universal Definition). Many textbooks use the older rule: wider than `0.04 s` (one small square) or deeper than a quarter of the following R wave.

Which leads, which wall, which artery (also a practical spot)

Leads with changes	Wall	Usual artery
II, III, aVF	Inferior	Right coronary (sometimes circumflex)
V1–V2	Septal	Left anterior descending (LAD)
V3–V4	Anterior	LAD
I, aVL, V5–V6	Lateral	Circumflex or a diagonal branch of the LAD
ST depression V1–V3 with tall R waves	Posterior	Right coronary or circumflex

Other investigations and what they mean for physio

Test	What it shows	Physio meaning
Troponin I or T	Most specific marker of heart-muscle damage. Rises within a few hours, peaks around a day, stays raised for up to about two weeks.	A rising troponin means the event is still evolving. Start mobilising once values are stable or falling.
CK-MB	Rises within hours, returns to normal in 2–3 days	A fresh rise suggests a new infarct
Myoglobin; LDH (older markers)	Myoglobin rises earliest but is not specific to the heart; LDH rises late and stays up for about a week. Rarely used now.	Know them for the viva
BNP or NT-proBNP	Released by a stretched, failing ventricle	High or rising values mean heart failure: lower intensity, slower progression, closer monitoring
Echocardiogram	Ejection fraction (EF), wall-motion defects, valve problems, clot in the ventricle	EF below 40% puts the patient in the high-risk group: start lower, progress slower, monitor more
Coronary angiography	Site and number of blocks: single, double or triple vessel disease	Decides PCI or CABG. Triple vessel or left main disease usually means CABG.
Exercise test	Low-level test before discharge; symptom-limited test a few weeks later	Gives the safe heart-rate range and ischaemic threshold for Phase II
6-minute walk test	Distance walked in 6 minutes	Simple field test of functional capacity and an outcome measure
Blood tests	Lipids, blood sugar and HbA1c, haemoglobin, electrolytes	Low haemoglobin or abnormal potassium raises arrhythmia and fatigue risk. Check before exercise.

Cardiac markers and their importance in cardiac rehab (7 marks in W25)

- **Diagnosis:** a rise and fall of troponin, with symptoms or ECG changes, confirms MI and separates NSTEMI from unstable angina.
- **Size of the infarct and risk:** higher peaks mean more muscle lost, usually a lower EF, and a higher risk group for exercise.
- **When to start:** Phase I mobilisation begins once markers are stable or falling and there is no new chest pain.
- **Safety during rehab:** new chest pain during progression is checked with ECG and troponin; a fresh rise means a new infarct, so exercise stops and the plan is reset.

- **After CABG:** some troponin rise is expected from the surgery itself; the surgeon decides whether it is more than expected.
- **Heart failure:** BNP helps judge severity and response to treatment, and so how hard to train.

7 Medical and surgical management

Emergency treatment of STEMI

- **Reperfusion is the priority.** Primary PCI should be done within **90 min** of first medical contact, or within **120 min** if the patient must be transferred to another hospital.
- If PCI cannot be done within 120 minutes, give **thrombolysis** (clot-dissolving drugs), ideally within **30 min**. The W22 case was "thrombolysed".
- Aspirin plus a second antiplatelet drug, and an anticoagulant.
- Give **oxygen only if SpO_2 is below 90%**.
- Nitrates for pain. Use an opioid only for severe pain.

Classic answer vs current guideline: MONA

Classic (older books)	Current (ESC 2023; ACC/AHA 2025)	Write in the exam
Every MI gets MONA: Morphine, Oxygen, Nitrates, Aspirin.	Oxygen only if SpO_2 is below 90%. Opioids only for severe pain. The key steps are antiplatelets and **urgent reperfusion**.	"Aspirin and early reperfusion by primary PCI or thrombolysis; oxygen only if SpO_2 < 90%."

Long-term drugs and what they mean for exercise

Drug group	Effect that matters	What physio does
Beta-blockers (metoprolol, bisoprolol)	Lower resting and exercise heart rate; heart rate rises less with effort; tiredness	Do not rely on age-predicted heart rate. **Use RPE and the talk test.**
Nitrates (GTN spray, isosorbide)	Widen vessels; headache; drop in BP on standing	Stand up slowly. Patient keeps the GTN spray within reach during exercise.
Antiplatelets, anticoagulants, thrombolytics	Bleeding and bruising	Check puncture sites before moving the patient. Use gentle manual techniques.
ACE inhibitors, ARBs, diuretics	Low BP, dizziness, dehydration, cramps, frequent urination	Check BP when sitting and standing. Plan sessions around the timing of diuretic doses.
Statins	Occasional muscle pain	Ask about new muscle aches and report them

PCI (angioplasty with stent)

- A catheter enters through the **radial artery** at the wrist or the **femoral artery** in the groin. A balloon opens the block and a stent keeps it open.
- After femoral access, the leg is kept straight for a few hours according to hospital protocol. Check the groin for bleeding or swelling before walking.
- After radial access, walking can start early. The wrist is protected from heavy loading for a few days.
- Driving: the UK DVLA allows car driving **1 week** after successful PCI for an acute coronary syndrome, if EF is at least 40% and no further urgent procedure is planned; otherwise 4 weeks. India has no specific rule, so this is a useful reference. The S20 bank manager who must drive one hour fits here.

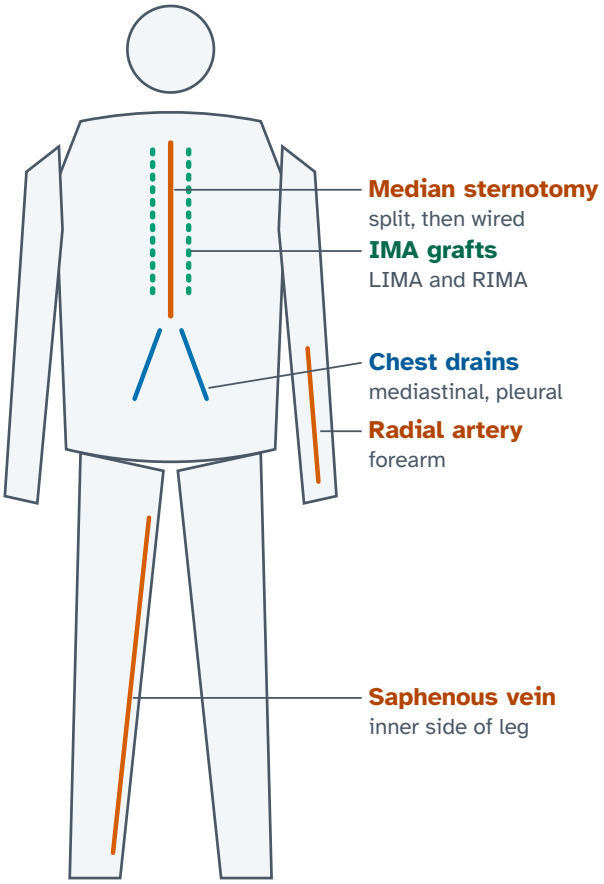
CABG

Indications (3 marks in W25):

- Significant **left main** coronary artery disease.

- **Triple vessel disease**, especially with diabetes or a weak left ventricle (low EF).
- Two-vessel disease involving the **proximal LAD**.
- Angina not controlled by medicines when PCI is unsuitable (long, diffuse or complex blocks).
- Failed PCI, or a complication during PCI.
- At the time of other heart surgery (for example valve replacement), or for mechanical complications of MI (ventricular septal rupture, papillary muscle rupture, ventricular aneurysm).

CABG: incision, drains and graft sites



Vermillion lines are wounds. Green dots are the internal mammary arteries, which stay attached at the top and are redirected to the heart. Blue lines are drains.

- **Approach:** median sternotomy. The sternum is split vertically and later closed with steel wires. Unlike a thoracotomy, no major muscle belly is cut, so the main post-op problems are sternal pain and protecting the healing bone.
- **Grafts (3 marks in W24):** left internal mammary artery (**LIMA**) to the LAD is the standard because it stays open longest. Other grafts: right internal mammary (**RIMA**), radial artery, and great saphenous vein.
- **On-pump or off-pump:** on-pump surgery uses a heart–lung machine while the heart is stopped. Off-pump surgery is done on the beating heart.
- **Lines and tubes after surgery:** endotracheal tube on a ventilator for a few hours, mediastinal and pleural drains, temporary pacing wires, arterial and central lines, urinary catheter.

Typical uncomplicated course after CABG (varies by hospital)

When	What usually happens
Day of surgery	ICU, ventilated, extubated within hours once awake and stable
Day 1	Breathing exercises, sitting out of bed, drains often removed on day 1–2
Days 2–4	Ward; walking in the room, then the corridor; pacing wires removed

When	What usually happens
Before discharge	Corridor walking and one flight of stairs; discharge often around day 5–7

8 Complications and prevention

Complication	What to look for	Physio role
Arrhythmias (commonest in the first 48 hours after MI)	Irregular or very fast or slow pulse, palpitations, dizziness	Check pulse and monitor before, during and after activity. Stop if a new arrhythmia appears.
Heart failure, cardiogenic shock	Breathlessness lying flat, crackles, ankle swelling, low BP, cold clammy skin	Sit the patient upright. Keep activity low. Report.
Re-infarction, recurrent angina	Chest pain at rest or with low effort	Stop. Rest. GTN as prescribed. Report.
Mechanical rupture (papillary muscle, septum, wall)	Sudden collapse or new loud murmur in the first week	Medical emergency. Call for help.
Pericarditis (early) or Dressler's syndrome (weeks later)	Sharp pain that worsens lying flat and eases sitting forward	Recognise it and report. Not a reason to panic, but needs a doctor's review.
Clot in the ventricle, embolism, stroke	Sudden weakness, speech change, cold painful limb	Emergency. Call for help.
DVT and pulmonary embolism (immobility)	Calf swelling and pain; sudden breathlessness	Prevent: early walking, ankle pumps, stockings when prescribed
Lung complications after CABG: atelectasis, effusion, pneumonia	Reduced breath sounds at the bases, fever, low SpO_2	Prevent: early sitting and walking, deep breathing, supported huff and cough
Atrial fibrillation after CABG (common, often days 2–4)	Irregular pulse, palpitations, tiredness	Feel the pulse before each session. Hold exercise until the medical team has reviewed.
Sternal wound infection or instability	Clicking or movement of the sternum, increasing pain, redness, discharge	Teach sternal care. Report at once. Higher risk with diabetes, obesity and when both IMAs are used.
Graft-site problems in the leg or arm	Swelling, pain, wound breakdown	Elevate the leg when sitting, use ankle pumps and walking, and teach wound care
Shoulder and thoracic stiffness, poor posture	Cannot raise arms overhead (S19 case), rounded shoulders	Prevent with early pain-free shoulder and trunk movement and posture advice
Anxiety and depression	Fear of activity, low mood, poor sleep	Explain each step, give graded success, refer when needed

9 PT assessment

Subjective

- **Present complaints:** chest pain (site, character, spread, triggers, relief), breathlessness, palpitations, fatigue, wound pain.
- **Event and treatment:** date of MI, thrombolysis, PCI or CABG with the post-operative day, grafts used, complications in ICU.
- **Risk factors and past history:** diabetes, hypertension, lipids, previous heart events.
- **Medications**, especially beta-blockers and nitrates.
- **Social history (this fills the ICF):** job and its physical demands, travel and driving, earning role in the family, home floor and lift, Indian or Western toilet, family support, smoking, tobacco, alcohol, diet.
- **Patient's own goals and fears.**

Objective

- **Vital signs at rest and with activity:** heart rate and rhythm, BP lying and standing, respiratory rate, SpO_2, temperature.
- **Observation:** monitors, lines, drains, oxygen, breathing pattern, posture, colour, swelling.
- **Chest:** expansion, auscultation (listen especially at the bases), cough effectiveness.
- **Wounds:** sternotomy and graft sites.
- **Musculoskeletal:** shoulder and thoracic range, posture, muscle strength, calf for DVT signs.
- **Function:** bed mobility, sit-to-stand, walking distance with vital signs, stairs.

Outcome measures to name

Measure	What it tells you
Borg RPE scale (6–20) or CR10	How hard exercise feels. The main intensity guide with beta-blockers.
6-minute walk test	Functional exercise capacity; compare before and after rehab
Incremental shuttle walk test	Symptom-limited walking capacity
NYHA and CCS grade	Symptom severity in daily life
Duke Activity Status Index	Self-reported functional capacity
HADS	Anxiety and depression screen
MacNew or HeartQoL questionnaire	Heart-specific quality of life

Risk stratification for exercise (asked as 7 marks in S21)

Before Phase II, sort the patient into low, moderate or high risk of a cardiac event during exercise. The AACVPR scheme is the standard. **One high-risk feature is enough to make the patient high risk.**

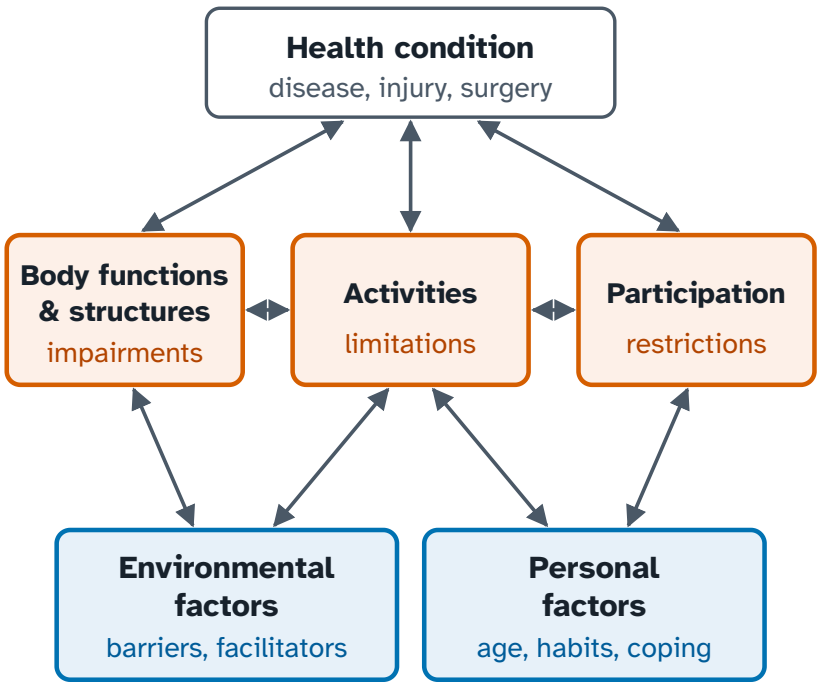
Risk	Features	What it means for rehab
Low	• EF **50% or more** • Functional capacity **7 METs or more** • No angina or significant symptoms on exercise test • No complex arrhythmias; normal HR and BP response • Uncomplicated MI or revascularisation; no heart failure; no depression	Standard supervised programme; can progress faster
Moderate	• EF **40–49%** • Angina or symptoms only at high effort (7 METs or more) • Mild to moderate silent ischaemia (ST depression under 2 mm) • Functional capacity below 5 METs	Closer supervision; ECG monitoring for the first sessions
High	• EF **below 40%** • Angina or symptoms at low effort (under 5 METs) or in recovery • ST depression 2 mm or more; BP fails to rise or falls with effort • Complex arrhythmias; survived cardiac arrest • Complicated MI or surgery; heart failure; signs of ongoing ischaemia; clinical depression	Continuous ECG monitoring; start at low intensity; progress slowly

To assess risk of developing coronary disease in the first place, count the risk factors in section 3 and use a risk chart such as the WHO risk charts for South-East Asia.

10 PT diagnosis by ICF

Nearly every long-answer option in 2016–2026 asks for ICF, a functional diagnosis or impairments. Master this section first.

The ICF model (draw this in every case answer)



Six boxes, all arrows two-way. In the exam, draw this first, then fill each box from the case.

Worked example from a real paper (W17, 5 + 3 + 7 marks)

> "50 years old male businessman had angina since 1 month. Angiography showed major blocks in LAD and other 2 arteries. He was operated for CABG 2 days back. Grafts taken from LIMA, RIMA and radial artery. He needs to drive to work everyday. Write impairments, activity and participation limitation and short term and long term management."

Spot the ICF clues before you write

businessman and drive to work every day are participation and environment. CABG 2 days back means Phase I and an acute post-op chest. LIMA, RIMA means both internal mammary arteries were used, which raises the risk of sternal wound problems. Radial artery means a forearm wound too.

ICF component	From this case	Clinical reasoning
Health condition	Triple vessel coronary artery disease; CABG day 2 with LIMA, RIMA and radial artery grafts	Sets the phase: Phase I, acute post-operative
Body structure	Median sternotomy (wired sternum); both internal mammary arteries harvested; forearm wound	Healing bone and wounds need protection

ICF component	From this case	Clinical reasoning
Body function impairments	• Incisional pain • Reduced lung expansion and shallow breathing; risk of basal atelectasis • Ineffective cough; risk of retained secretions • Reduced exercise tolerance and fatigue • Risk of arrhythmia (AF) • Guarded, reduced shoulder and thoracic movement • Anxiety about the heart	• Pain and fear of the wound lead to guarded breathing • Anaesthesia and lying flat reduce lung volumes • Pre-op angina, surgery and bed rest cause deconditioning
Activity limitations	Bed mobility and sit-to-stand without pushing on the arms; walking distance; stairs; upper-body dressing and bathing; lifting	Pain, sternal protection and low exercise tolerance
Participation restrictions	Cannot drive to work; cannot run his business; reduced social and family roles	Driving is usually avoided for about 4 weeks after CABG (DVLA), so he cannot reach work alone
Environmental factors	**Facilitators:** family support, hospital rehab services, medicines. **Barriers:** daily long drive, traffic, business stress. Ask about home stairs and toilet type.	Barriers become goals; facilitators are used in the plan
Personal factors	50-year-old man; businessman; motivated to return to work; ask about smoking, diet and stress	Motivation helps adherence; habits are targets for lifestyle change

One-line functional diagnosis

A 50-year-old businessman, day 2 after CABG (LIMA, RIMA, radial grafts) for triple vessel disease, with incisional pain, reduced lung expansion, weak cough, low exercise tolerance and guarded shoulder movement. These limit bed mobility, walking, stairs and upper-body self-care, and restrict driving to work and running his business. Family support is a facilitator; the daily drive is a barrier.

ICF code numbers are not needed. No paper has asked for them.

Recall check. On paper, draw the six ICF boxes from memory. Then fill them for the W22 case in the exam corner.

11 Problem list and goals

Goals must be **functional, measurable and time-bound**, and tied to the patient's own life. Use the examiner's words: short-term goals and long-term goals.

Goals for the W17 businessman

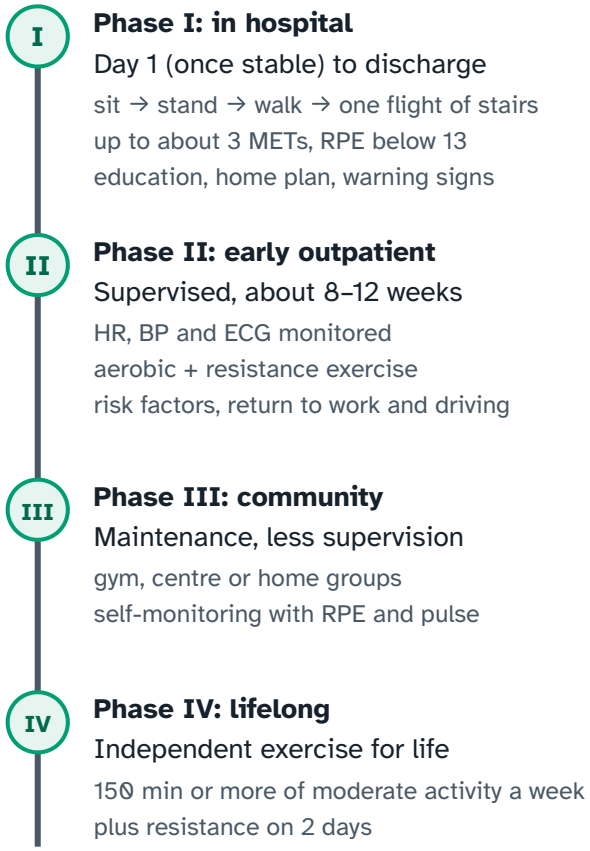
Problem	Short-term (by discharge, about 1 week)	Long-term (6–12 weeks)
Pain and guarded breathing	Pain controlled so he can take deep breaths and cough with the wound supported	Pain-free daily activity
Lung expansion and cough	Clear chest, no atelectasis; effective supported huff and cough	Normal breathing pattern and chest mobility
Mobility and exercise tolerance	Independent sternal-safe bed mobility; walks in the corridor and climbs one flight of stairs within Phase I limits	Walks 30 minutes continuously at RPE 12–13 on most days; better 6-minute walk distance
Shoulder and posture	Pain-free active shoulder movement with arms working together	Full shoulder range and upright posture

Problem	Short-term (by discharge, about 1 week)	Long-term (6–12 weeks)
Participation	He and his family know the home programme, warning signs and sternal care	Returns to driving after medical clearance (about 4 weeks), then to business work; resumes social life
Risk factors	Understands his own risk factors	Risk factors controlled; independent lifelong exercise (Phase III–IV)

12 PT management

Two layouts, because papers ask both ways: by phase ("plan his cardiac rehabilitation") and by impairment ("physiotherapy for each impairment", S19).

Phases of cardiac rehabilitation



Some books merge Phases III and IV into one maintenance phase. Write four phases for MUHS, because the syllabus lists "Phases I, II, III and IV".

Phases with their aims (S26)

Phase	When	Aims
I: inpatient	From the first stable day to discharge	Prevent bed-rest effects and lung complications; safe self-care and walking up to about 3 METs; education on risk factors, warning signs and GTN; home plan; referral to Phase II
II: early outpatient (convalescence)	From discharge, about 8–12 weeks (includes the first 6 weeks at home)	Supervised, monitored exercise to raise functional capacity; risk-factor control; psychological support; return to work, driving and sexual activity
III: community	After Phase II, for several months	Keep and improve fitness with less supervision; self-monitoring; continued risk-factor control

Phase	When	Aims
IV: lifelong	For life	Independent regular exercise and a healthy lifestyle to prevent another event

Before surgery (when CABG is planned)

- Explain the operation, the ICU, the drains and the plan for the first days. This reduces anxiety.
- Teach deep breathing, holding a full breath in for a few seconds, and a supported huff and cough with a pillow hugged over the chest.
- Teach sternal-safe bed mobility and ankle pumps.
- Record baseline walking distance and vital signs.
- In high-risk patients, inspiratory muscle training before surgery reduced lung complications after CABG in a randomised trial (Hulzebos, JAMA 2006).

Phase I: in hospital

Start when stable. After MI this means no new chest pain, no new ECG changes or dangerous arrhythmias, troponin stable or falling, and no uncontrolled heart failure. This is usually within the first day. After CABG, start on day 1.

Memory hook: Phase I limits (ACSM)

13 / 20 / 30 / 120

- **13**: keep RPE below 13 on the 6–20 Borg scale
- **20**: heart rate no more than 20 above resting after MI
- **30**: heart rate no more than 30 above resting after surgery
- **120**: and below 120 beats per minute after MI

Checked against the ACSM Phase I prescription table as reproduced in two independent secondary sources. The primary ACSM text was not available.

Phase I dose (ACSM)

Part	Dose
Intensity	RPE **< 13** ; HR **≤ rest + 20** and **≤ 120** after MI; **≤ rest + 30** after surgery; to tolerance if no symptoms
Duration	Bouts of **3–5 min** with **1–2 min** rest (rest shorter than exercise), about **20 min** in total
Frequency	**3–4 times a day** on days 1–3, then **twice a day**
Progression	Build up to **10–15 min** continuous first, then raise intensity
Activities	Up to about **3 METs** : self-care, sitting, standing, walking

Phase I progression (a typical uncomplicated course; follow the hospital's own protocol)

Stage	After MI	Add after CABG
ICU or CCU (about 1–2 METs)	Breathing exercises, ankle pumps, active limb movements, sitting up, feeding and washing the face, bedside commode	Deep breaths with a 3-second hold, 5–10 every hour while awake; supported huff and cough; sitting out of bed on day 1; sternal-safe log-roll; pain-free shoulder movement with both arms together
Ward (about 2–3 METs)	Standing, walking in the room and then the corridor, washing in the bathroom, sitting out for meals	Corridor walks 3–4 times a day; posture correction; leg elevation and walking for the graft-site leg
Before discharge (about 3 METs)	10–15 minutes of continuous walking; one flight of stairs under supervision; taking own pulse and using the RPE scale	Same, plus sternal care, wound care and a home walking plan

Monitor before, during and after every session: heart rate and rhythm, BP, SpO_2, RPE and symptoms. Teach the patient the warning signs and how to use GTN before discharge. Refer to Phase II.

Factors that decide progression of exercise (S24)

Move to the next step only when the patient tolerated the last step:

- **Heart rate:** rise within the limits (rest + 20 after MI, rest + 30 after surgery); a resting HR above about 120 is a reason to wait.
- **Blood pressure:** systolic rises a little with effort; a fall of more than 10 mmHg, or diastolic 110 or more, stops progression.
- **Rhythm and ECG:** no new arrhythmia (atrial fibrillation is common after CABG) and no ischaemic changes.
- **Symptoms:** no angina, dizziness or marked breathlessness; RPE below 13; SpO_2 stays stable.
- **Surgery-related:** incision pain controlled (S24: VAS 4 at rest, so treat after analgesia), stable sternum, drains and pacing wires, haemoglobin, fluid balance, chest X-ray (effusion or collapse).
- **Recovery:** HR and BP back near resting within a few minutes; not unduly tired the next morning.
- **Patient factors:** age, previous fitness, risk group (EF), diabetes and other diseases, confidence and anxiety.

Classic answer vs current evidence: bed rest after MI

Classic (older books)	Current (Cochrane; ACSM)	Write in the exam
Strict bed rest for a week or more, then a slow, many-step mobilisation programme over about two weeks.	Mobilise as soon as the patient is stable, often within the first day. Short bed rest was no less safe than long bed rest.	"Early mobilisation within 12–24 hours once clinically stable, progressing from 1–2 METs to about 3 METs by discharge."

Classic answer vs current evidence: incentive spirometry after CABG

Classic	Current (AARC 2011)	Write in the exam
Every post-CABG patient gets an incentive spirometer.	Incentive spirometry is **not recommended alone or routinely** after CABG. Deep breathing, supported cough and early walking are the core.	"Deep breathing exercises and early mobilisation; incentive spirometer may be used as an adjunct, not alone."

Phase II: early outpatient, supervised

Starts soon after discharge and usually runs for **about 8–12 weeks**. It has four parts: **exercise, education, risk-factor control and psychological support**. Begin with risk stratification (section 9 (p. 15)) and an exercise test or 6-minute walk test.

Each session: warm-up **5–10 min**, conditioning **20–60 min**, cool-down **5–10 min**. A longer warm-up and cool-down lower the risk of ischaemia and arrhythmia at the start and end.

Phase II prescription using FITT-VP

Part	Aerobic	Resistance
Frequency	**3–5 days** a week (at least 3)	**2–3** non-consecutive days a week
Intensity	**40–80% HRR**, starting at the low end; or RPE **12–16**, starting at 11–13. Stay at least **10 beats** below any heart rate that brought on angina or ST change.	Light: **10–15 reps** at RPE **11–13**; a starting load around **40–60% of 1RM** or less
Time	**20–60 min**; start with short bouts and build up	**8–10 exercises** for major muscle groups, **1–3 sets**

Part	Aerobic	Resistance
Type	Walking, treadmill, cycle ergometer; arm ergometer once the sternum allows	Bands, light dumbbells, machines, body weight. After CABG, start with the legs.
Volume	Build towards **150 min** or more a week	–
Progression	Increase duration first, then intensity	"2 for 2": add load when 2 extra reps are done in 2 sessions in a row (AHA 2023)

> **Worked target heart rate (Karvonen)**
>
> Age 50, resting HR 70. Maximum HR = 220 − 50 = 170. Heart-rate reserve (HRR) = 170 − 70 = 100.
>
> 40% intensity = 70 + 40 = **110**. 60% intensity = 70 + 60 = **130**. Target range: **110–130 beats per minute**.
>
> If the patient takes a beta-blocker, this formula overestimates the target. Use RPE 11–13 and the talk test instead.

Phase II after PCI for a young, active patient (7 marks in W24)

- **No sternal precautions:** PCI has no chest wound, so arm work and resistance training can start early. Protect the puncture site: after radial access avoid heavy gripping and wrist loading (racquet and bat sports) for about a week; after femoral access avoid heavy lifting and strenuous leg work for about a week.
- **Exercise test first:** a symptom-limited test shows the safe heart-rate range and any remaining ischaemia; it sets the intensity.
- **Aerobic:** 3–5 days a week, starting at moderate intensity (40–60% HRR, RPE 12–13) and progressing towards 60–80% HRR; add interval training once continuous exercise is comfortable.
- **Resistance:** 2–3 days a week, all major muscle groups, progressing the load gradually.
- **Sport-specific stage:** skill work at low intensity first, then match-like efforts with warm-up and cool-down; avoid sudden all-out bursts in the early months.
- **Return to sport:** recreational sport once he trains at the needed intensity without symptoms, ischaemia or arrhythmia; competitive sport only after a cardiologist's assessment (ESC 2020 sports cardiology guideline).
- **Other points:** antiplatelet drugs cause easy bruising and bleeding, so avoid collision injuries; never stop these drugs without the cardiologist; risk factors still need control even in a fit person.
- **Breathe out on effort.** Never hold the breath while lifting (Valsalva), because it causes sudden swings in BP.
- **Talk test:** the patient should be able to talk in short sentences while exercising.

Classic answer vs current evidence: sternal precautions

Classic	Current ("Keep your move in the tube")	Write in the exam
• Strict precautions for about 6–8 weeks • No lifting heavy objects • No pushing or pulling with the arms, no pushing up from a chair • No reaching both arms back or overhead against resistance	Keep the upper arms close to the body, as if inside an imaginary tube around the trunk. Move within pain limits, with both arms working together. No fixed weight limit. A randomised trial and before-and-after studies found this as safe as strict precautions, with earlier return of function.	"Sternal precautions for 6–8 weeks: avoid heavy lifting, pushing and pulling, and one-sided arm loading; keep arm movements close to the body and pain-free." This covers both versions.

Lifting heavy objects is usually avoided for about **3 months** after CABG (NHS advice). Light, pain-free upper-limb exercise with the arms kept close to the body can begin earlier.

Phases III and IV: maintenance for life

- Exercise in the community, at home or in a gym with less supervision.
- Aim for at least **150 min** a week of moderate activity plus resistance exercise on **2 days**.
- Self-monitoring with RPE and pulse. Regular review of risk factors.

> **Yoga (syllabus: "applied yoga in cardio-respiratory conditions")**
>
> The Indian Yoga-CaRe trial (3,959 patients after MI; JACC 2020) found that yoga-based cardiac rehab improved self-rated health and return to pre-infarct activities. It did not significantly reduce major cardiac events. It is a good option when conventional rehab is not available or acceptable. Avoid breath-holding and straining postures in the early weeks.

By impairment (for "physiotherapy for each impairment")

Impairment	Technique and dose	Why
Incisional pain	Hug a pillow over the chest for cough and movement; positioning; analgesia before sessions. TENS beside the wound if allowed (not with pacing devices).	Less pain allows deeper breaths and earlier walking
Reduced lung expansion, atelectasis risk	Deep breathing with a 3-second hold, thoracic expansion exercises, 5–10 breaths every hour while awake; upright sitting; early walking	Large breaths reopen collapsed alveoli at the bases
Weak cough, secretions	Active cycle of breathing technique (ACBT); supported huff and cough	A huff moves secretions with less pain and less pressure on the sternum than a full cough
Low exercise tolerance	Graded walking within the Phase I limits; then Phase II aerobic training	Gradual overload trains the heart and muscles without triggering ischaemia
Shoulder and thoracic stiffness (S19)	Active shoulder flexion and abduction with both arms together, thoracic rotation and extension, scapular setting; 10 repetitions, 3–4 times a day; posture correction	Restores the range lost to guarding without one-sided stress on the sternum
Muscle weakness, deconditioning	Active exercises in Phase I, then light resistance training in Phase II	Muscle strength makes daily tasks use a smaller share of capacity
DVT risk, leg graft swelling	Ankle pumps 10 every hour; early walking; elevate the leg when sitting; stockings if prescribed	The calf muscle pump moves venous blood and reduces swelling
Anxiety, fear of activity	Explain each step; relaxation; graded activity with success at each stage; involve family	Confidence improves when the patient sees safe progress
Risk factors (smoking, obesity, diabetes)	Lifestyle counselling (section 14); aerobic training for weight and blood-sugar control	Controls the disease process itself

Recall check. Without looking, write the Phase I limits hook and the FITT-VP for Phase II aerobic training.

13 Precautions, contraindications and stop criteria

Do not start or continue exercise if (ACSM)

- Unstable angina
- Resting systolic BP above **180** or diastolic above **110 mmHg** (decide case by case)
- BP falls more than **20 mmHg** on standing, with symptoms
- Significant aortic stenosis
- Uncontrolled arrhythmias, or uncontrolled sinus tachycardia above **120**
- Uncompensated heart failure
- Third-degree heart block without a pacemaker
- Active pericarditis or myocarditis
- Recent embolism; acute thrombophlebitis or DVT
- Fever or acute illness
- Uncontrolled diabetes
- Severe orthopaedic problems that prevent exercise

Stop the session if (ACSM, inpatient)

- Diastolic BP reaches **110 mmHg** or more
- Systolic BP **falls** more than **10 mmHg** as the workload increases
- Significant atrial or ventricular arrhythmias
- Second- or third-degree heart block
- Angina, marked breathlessness, or ECG changes of ischaemia

Also stop for dizziness, confusion, pallor, cold sweat, or any sign of sternal movement or wound bleeding.

- **Angina during exercise:** stop, sit the patient down, give GTN as prescribed. If pain is not better 5 minutes after the first dose, call for emergency help.
- Avoid exercise for 1–2 hours after a heavy meal, and in very hot or cold conditions.
- Avoid heavy isometric holds and breath-holding early on.
- With beta-blockers, use RPE rather than heart rate.
- After femoral PCI, check the groin before walking.

The ACSM thresholds above were checked against two independent reproductions (a WHO-hosted rehabilitation guideline and an insurer's clinical policy). The primary ACSM book was not available.

14 Education, lifestyle and home programme

Asked in S16 (7 marks), W20 (3 marks), S23 (7 marks, education for lifestyle disorders), and as the last part of S17, S19 and W19.

Lifestyle modification with reasons

Change	Advice	Reason
Stop all tobacco	Cigarettes, bidi, gutkha and second-hand smoke. Offer cessation support.	Tobacco damages the artery lining, makes blood clot easily and narrows grafts
Heart-healthy diet	More vegetables, fruit, whole grains, pulses; salt under **5 g a day** (WHO); less fried food, vanaspati, ghee and sugar	Lowers BP, LDL and weight
Regular exercise	At least **150 min** a week of moderate activity, after the rehab programme	Improves fitness, BP, sugar and lipids, and mood

Change	Advice	Reason
Healthy weight	Indian targets: BMI below **23**; waist below **90 cm** (men), **80 cm** (women)	Abdominal fat drives insulin resistance and high BP
Control BP	Target below **130/80**; take medicines regularly	Lower pressure reduces the heart's workload
Control diabetes and lipids	Regular medicines and tests (HbA1c, lipid profile)	Slows the progression of plaque
Limit alcohol	Avoid or keep to a minimum	Raises BP and triglycerides, and can trigger arrhythmias
Manage stress	Relaxation, yoga, regular sleep; seek help for low mood	Stress and depression worsen outcomes and adherence

Grades of obesity (asked in S21)

Category	WHO BMI (kg/m²)	Asian Indian BMI (kg/m²)
Normal	18.5–24.9	18.5–22.9
Overweight	25–29.9	23–24.9
Obese	Class I 30–34.9; Class II 35–39.9; Class III 40 or more	25 or more

A newer staged definition of obesity for Indians was proposed in 2025. It adds waist size and health effects to BMI. Use the cut-offs above in the exam.

Return to activities

Activity	Usual advice
Driving a car	About **4 weeks** after CABG; **1 week** after successful PCI for ACS if EF is at least 40%, otherwise 4 weeks (UK DVLA). Commercial drivers wait longer.
Sexual activity	Reasonable when the patient can do **3–5 METs** without symptoms. After CABG, usually after **6–8 weeks**, avoiding positions that strain the sternum (AHA 2012).
Lifting	No heavy lifting for about **3 months** after CABG
Work	Depends on the job. Most normal activities return by about 6 weeks after CABG; full recovery takes 2–3 months. Desk work returns sooner than heavy manual work.

Home programme at discharge

- Walk on level ground twice a day, starting with the distance managed in hospital. Add a few minutes each week, keeping RPE at 11–13.
- Deep breathing exercises several times a day for the first weeks after surgery.
- Pain-free shoulder and trunk exercises, with posture correction.
- Check your pulse and use the RPE scale. Carry GTN.
- Avoid straining and heavy lifting. Rest between tasks (energy conservation).
- Report chest pain, unusual breathlessness, palpitations, fainting, wound redness or discharge, or sternal clicking.

Do's and don'ts after discharge (S25)

Do	Don't
Walk every day, building up slowly at RPE 11–13	Don't exercise to exhaustion or hold your breath while lifting
Take all medicines regularly; carry GTN	Don't stop medicines (especially antiplatelets) on your own

Do	Don't
Check your pulse and use the RPE scale	Don't exercise within 1–2 hours of a heavy meal, or in very hot or cold weather
Eat a low-salt, low-fat diet; keep a healthy weight	Don't smoke, chew tobacco or sit in smoky rooms
Join the Phase II programme and keep follow-up visits	Don't lift heavy loads, push cars or do heavy work until cleared
Rest between tasks; sleep well; ask for help with stress or low mood	Don't drive until cleared (about 1–4 weeks, depending on treatment)
Sit down and take GTN for chest pain; call for emergency help if it is not better 5 minutes after the first dose	Don't ignore chest pain, breathlessness, palpitations or fainting

15 Prognosis and discharge criteria

Prognosis depends on the size of the infarct (EF), arrhythmias, risk-factor control and taking part in rehab. Exercise-based cardiac rehab reduces heart attacks by about 28% and hospital admissions by about 35%, probably reduces deaths, and improves quality of life; home-based and digitally supported programmes work about as well as centre-based ones (Cochrane review update, 2026: 107 trials, 26,886 people).

Discharge from Phase I: stable vital signs at rest and with activity; independent in self-care and walking at about 3 METs; managed one flight of stairs without symptoms; knows the home programme, warning signs and wound care; referred to Phase II.

Discharge from Phase II: goals met; exercises safely and independently with self-monitoring; risk factors addressed; ready for Phase III–IV.

16 Exam corner

Past questions (MUHS, 2016–2026)

Wording cleaned up from scanned papers.

S16 Q3(a), 5 + 10 marks

55-year-old male farmer, chronic alcoholic, myocardial ischaemia 2 days back. (i) Functional diagnosis. (ii) Physiotherapy management in detail.

S17 Q4(a), 5 + 7 + 3 marks

60-year-old male bank manager, CABG for triple vessel disease 3 days back; chronic smoker, hypertensive, BMI 30. Discuss ICF, Phase 1 and Phase 2 cardiac rehabilitation, and lifestyle modifications.

W17 Q4(a), 5 + 3 + 7 marks

The worked example in section 10.

S18 Q4(a), 15 marks

60-year-old male shopkeeper with anginal pain, breathlessness and anxiety, in ICU for 3 days; chronic alcoholic, obese, hypertensive. Discuss ICF and cardiac rehabilitation.

W18 Q4(a), 5 + 10 marks

60-year-old shopkeeper, CABG 1 week back, with hypertension and diabetes. Functional diagnosis and cardiac rehabilitation in detail.

S19 Q4(a), 5 + 7 + 3 marks

55-year-old male bank manager, CABG for triple vessel disease 3 weeks back; discharged on day 3 but did not follow physio advice; chronic smoker not ready to quit; trying to change his diet; BMI 30, low HDL; wound healed but cannot raise his hands above his head. (i) Impairments with clinical reasoning. (ii) Physiotherapy for each impairment. (iii) Lifestyle modification with reasons.

S20 Q4(a), 3 + 5 + 7 marks

65-year-old retired female nurse with stable angina and ventricular dysfunction, EF 25%, diabetic, refused surgery, on conservative treatment. Impairments, activity and participation limitation, short- and long-term management.

S20 Q4(b), 3 + 5 + 7 marks

50-year-old male bank manager, angioplasty 2 days back after an inferior wall MI a week back; wants to return to work early; drives 1 hour to work. Same question as above.

W20 Q4(a), 5 + 10 marks

50-year-old male executive, chronic smoker, hypertensive; sudden chest pain spreading to the left arm and upper back; ST elevation with pathological Q waves; raised CK-MB and troponin; now stable. Discuss ICF and plan cardiac rehabilitation in detail.

S21 Q4(b), 5 + 7 + 3 marks

43-year-old businessman, CABG 20 days back, type 2 diabetes for 5 years, BMI 32. Plan Phase II cardiac rehabilitation with ICF; give grades of obesity.

W22 Q4(b), 5 + 5 + 5 marks

50-year-old in ICU with breathlessness and severe chest pain; ST elevation and T inversion in II, III, aVF; thrombolysed. Discuss ICF and plan short- and long-term physiotherapy.

W19 Q3(a), 5 + 5 + 5 marks

55-year-old senior executive, CABG for triple vessel disease 2 days ago; stopped smoking 6 months ago; hypertensive; married with 2 children, wants to resume his job. Functional diagnosis; Phase I cardiac rehabilitation; importance of lifestyle modification for him.

S23 Q3(b), 15 marks

50-year-old schoolteacher admitted with chest pain and hypertension, treated conservatively. Functional diagnosis and physiotherapy management.

S24 Q3(a), 15 marks

55-year-old executive, on-pump CABG 3 days ago; incision pain VAS 4/10 at rest; resting HR 96/min, BP 140/90 mmHg; cannot sit up or sleep; dyslipidaemia and hypertension for 15 years. Functional diagnosis; Phase I cardiac rehabilitation; factors you would consider for progressing exercise.

S25 Q3(a), 5 + 5 + 5 marks

60-year-old man with severe chest discomfort spreading to the left shoulder, jaw and upper back; deep Q waves in II, III and aVF; echo shows inferior wall hypokinesia, EF 40%. ICF; Phase I physiotherapy; do's and don'ts after discharge.

S26 Q4(a), 15 marks (four parts)

55-year-old office manager with an inferior wall MI, haemodynamically stable; chronic smoker, hypertensive, diabetic for 2 years, poor compliance. (i) Phases of cardiac rehabilitation with their aims; (ii) physiotherapy in hospital until discharge; (iii) convalescence from discharge to 6 weeks; (iv) rehabilitation after 3 months.

7-mark questions

Phase I cardiac rehabilitation following MI (W16, W17). Importance of lifestyle modification in preventing IHD (S16). Risk assessment and stratification for coronary artery disease (S21). Phase 2 cardiac rehabilitation after PTCA to the LAD in a 45-year-old executive who wants to play sport (W24). Patient education for lifestyle disorders (S23). Cardiac markers and their importance in cardiac rehabilitation (W25).

3-mark questions

Risk factors for coronary artery disease (S17). Three modifiable and three non-modifiable risk factors (W17). Phase I of cardiac rehabilitation (S20). Lifestyle modifications after CABG (W20). ECG criteria for acute and evolved MI (S24). Grafts used in CABG (W24). Karvonen's formula in cardiac rehabilitation (W25). Indications of CABG (W25).

Technique cards for this topic: exercise testing (p. 308), exercise physiology (p. 313), scales (p. 322), bed rest (p. 241), cardiac rehabilitation (p. 334), ECG and investigations (p. 317), cardiac drugs (p. 352).

Keywords to underline in your answer

ICF impairment activity limitation participation restriction facilitator barrier Phase I–IV early mobilisation METs RPE HR rest + 20 FITT-VP Karvonen risk stratification sternal precautions supported cough stop criteria secondary prevention

Answer skeleton: any CABG or MI long answer

1. Restate the case in two lines: age, job, diagnosis or surgery with day, other conditions, key social detail.
2. **[ICF, 5]** Draw the ICF model, fill every box from the case, add a one-line functional diagnosis and 2–3 reasons.
3. Problem list, then short- and long-term goals linked to his work and home.
4. **[Management, 7–10]** Phase I with the "13 / 20 / 30 / 120" limits and progression; Phase II FITT-VP with a worked target heart rate; Phase III–IV in two lines.
5. Stop criteria and precautions in a short list (sternal precautions if CABG).
6. **[Add-on, 3]** Whatever was asked: lifestyle with reasons, grades of obesity, outcome measures.

Draw these

- The ICF model (section 10): in every long answer
- The phases timeline (section 12): in any cardiac rehab answer
- The plaque-to-impairment chain (section 4): for "impairments with clinical reasoning"

Viva questions

1. Why do you use RPE rather than target heart rate in a patient on a beta-blocker?
2. What is the rate–pressure product and why does it matter?
3. Why is the LIMA preferred for grafting the LAD?
4. Which leads show an inferior MI, and what complication do you watch for during mobilisation?
5. When will you stop a Phase I session?
6. What are sternal precautions? What does "keep your move in the tube" mean?

Linked practical spots

- ECG with ST elevation in II, III and aVF: inferior STEMI, right coronary artery.
- ECG with pathological Q waves: an old or evolving infarct.
- Incentive spirometer: name it, its use, and its limits after CABG.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **List three modifiable and three non-modifiable cardiac risk factors.**

Answer

Non-modifiable: age (men over 45, women over 55), male sex, family history of early heart disease.

Modifiable: smoking or tobacco, hypertension, diabetes. Abnormal lipids, obesity and inactivity are also correct.

3 m **Write Phase I cardiac rehabilitation in brief.**

Answer

1. In-hospital phase from the first stable day to discharge. Aims: prevent bed-rest effects, educate, and prepare for home.
2. Early mobilisation from 1–2 METs to about 3 METs: breathing exercises, limb exercises, sitting, standing, walking, one flight of stairs before discharge.
3. Limits: RPE below 13; HR within rest + 20 (MI) or rest + 30 (surgery). Monitor HR, BP, SpO_2 and symptoms. Teach warning signs and the home programme.

7 m **Phase I cardiac rehabilitation following MI.**

Answer

Use this structure. Each line expands to a short paragraph.

1. Definition and aims of Phase I.
2. Criteria to start: no new chest pain, no new ECG changes or dangerous arrhythmias, troponin stable or falling, no uncontrolled heart failure.
3. Dose table: intensity (RPE < 13, HR ≤ rest + 20 and ≤ 120), bouts of 3–5 min, 3–4 times a day, then twice; progress duration then intensity.
4. Progression by stage: CCU (1–2 METs), ward (2–3 METs), before discharge (about 3 METs, stairs).
5. Monitoring and stop criteria.
6. Education: risk factors, GTN use, warning signs, home walking.
7. Discharge criteria and referral to Phase II.

Draw the phases timeline.

LAQ **W22: inferior STEMI, thrombolysed, in ICU. Fill the ICF boxes.**

Answer

- **Health condition:** acute inferior wall STEMI (II, III, aVF), after thrombolysis.
- **Impairments:** chest pain; reduced cardiac reserve and exercise tolerance; risk of bradycardia and heart block (right coronary territory); breathlessness; bleeding risk after thrombolysis; anxiety; deconditioning from ICU stay.
- **Activity limitations:** bed mobility, self-care, walking, stairs.
- **Participation restrictions:** work and family roles, social activities (the case gives no job, so write "to be asked").
- **Environmental factors:** ICU setting with monitoring as a facilitator; ask about home, stairs and family support.
- **Personal factors:** 50 years old; ask about smoking, diet, stress.

Then STG: safe Phase I progression to corridor walking and stairs. LTG: Phase II training and risk-factor control.

MCQ **Day 3 after CABG, resting HR 82. While walking in the corridor HR reaches 118 and RPE is 14. What is the best next step?**

A. Continue; this is a normal response after surgery

B. Stop, rest, then resume at a slower pace

C. Increase speed to reach a training effect

D. Switch to the arm ergometer

Answer
B. HR is 36 above resting, beyond the "rest + 30" limit after surgery, and RPE is above 13. Reduce the intensity. Nothing suggests an emergency, so stopping completely and calling for help is not needed.

MCQ **A patient on bisoprolol starts Phase II. How do you best set the exercise intensity?**

A. 85% of 220 minus age

B. RPE 11–13 and the talk test

C. Resting HR plus 40

D. A maximal test after stopping the drug

Answer
B. A beta-blocker blunts the heart-rate response, so formulas based on age overestimate the target. Never stop cardiac drugs to test.

MCQ **ECG shows ST elevation in II, III and aVF. Which problem do you watch for most during early mobilisation?**

A. Bradycardia or heart block

B. Pneumothorax

C. Sternal instability

D. Compartment syndrome

Answer
A. Inferior MI usually involves the right coronary artery, which supplies the SA and AV nodes in most people.

MCQ **"Keep your move in the tube" after sternotomy means:**

A. No arm movement for 6 weeks

B. Upper arms kept close to the body, pain-free, both arms working together

C. Lifting up to 20 kg from the first week

D. Pushing up from a chair with both arms

Answer
B. It replaces fixed weight limits with pain-guided movement close to the trunk.

Last recall check. Write three keywords for each: ICF, Phase I, Phase II, sternal care. Then check them against this page.

Sources

Notes written in original words. Numbers were checked against the sources below.

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Burns: acute care, positioning, splinting and inhalation injury

> Burns came up in 19 of the 20 papers from 2016 to 2026 (all except Summer 2023). The long answers give a burn site and percentage, then ask for ICF, positioning and splinting, and a phase-wise plan. Recent cases add a child, a full-thickness facial burn with breathing difficulty and a pregnant woman.

Syllabus	Ch 6(g) Burns: head, face, neck and thorax, inhalation burns; acute care management
Priority	**Tier 1.** In 19 of 20 papers (2016–2026). 12 long-answer cases, second only to IHD and CABG.
Asked as	LAQ S16 · W16 · W17 · W20 · S21 · S22 · W22 · W19 · W21 · S24 · W24 · S26 · 7m S17 · W17 · S18 · W18 · S20 · W22 · S24 · S25 · W25 · 3m W16 · W18 · S19 ×2 · S20 · S24 · S25
Usual split	5 + 5 + 5 (ICF, short-term, long-term), or 6 + 6 + 3 and 5 + 7 + 3 with an add-on such as rule of nines or wound healing
Checked against	ISBI Practice Guidelines for Burn Care (2016, 2018); American Burn Association fluid resuscitation guidance (2023, summarised 2025); Cochrane review of pressure garments (2024); burn-unit mobilisation protocols. Full list under Sources. Content checked 19 October 2026.

1 Definition

A **burn** is an injury to the skin and deeper tissues caused by heat, electricity, chemicals, radiation or friction.

An **inhalation injury** is damage to the airway and lungs from breathing in hot gases, smoke or toxic products of burning.

The burn wound is described by three things: its **cause**, its **depth**, and its **extent** as a percentage of total body surface area (TBSA).

2 Classification and grading

By cause

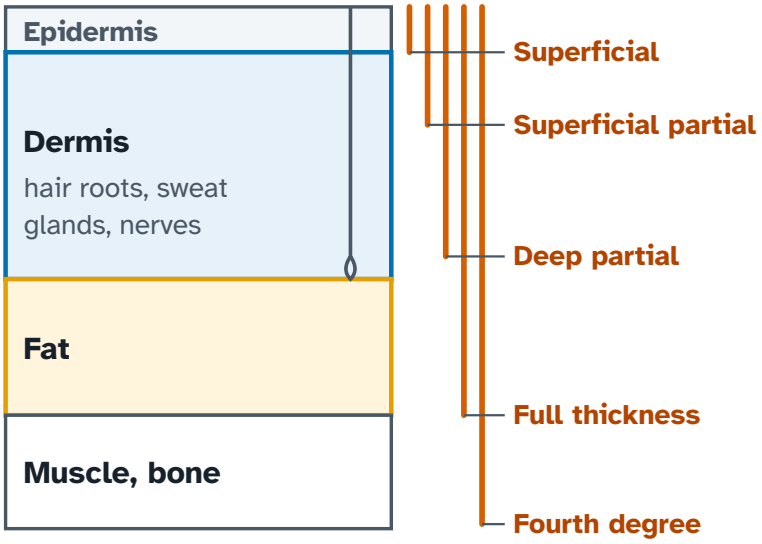
- **Thermal:** flame (stove, kerosene, firecrackers), scald (hot liquids, commonest in children), contact (hot objects).
- **Electrical:** low or high voltage, and lightning. Small skin wounds can hide deep muscle damage.
- **Chemical:** acids and alkalis. Alkalis penetrate deeper.
- **Radiation** and **friction**.
- **Inhalation injury**, which often comes with face and neck burns.

By depth (asked as 3 marks in W16 and S20; 7 marks in W17)

Depth	Look and feel	Healing
Superficial (first degree): epidermis only	Red, dry, painful, no blisters. Blanches on pressure. Example: sunburn.	About **3–7 days**, no scar
Superficial partial thickness (second degree): epidermis + upper dermis	Blisters, pink and moist, **very painful**, blanches and refills quickly	Within about **2–3 weeks**; little scarring
Deep partial thickness (second degree): into deep dermis	Blotchy red and white, drier, **less painful** (pressure felt more than pain), slow or no blanching	More than **3 weeks**; high risk of hypertrophic scar; often grafted

Depth	Look and feel	Healing
Full thickness (third degree): whole dermis	White, brown or black, leathery (eschar), **painless** because nerve endings are destroyed, no blanching	Cannot heal except from the edges; needs **skin grafting**
Fourth degree	Through fat into muscle, tendon or bone	Needs flaps, sometimes amputation

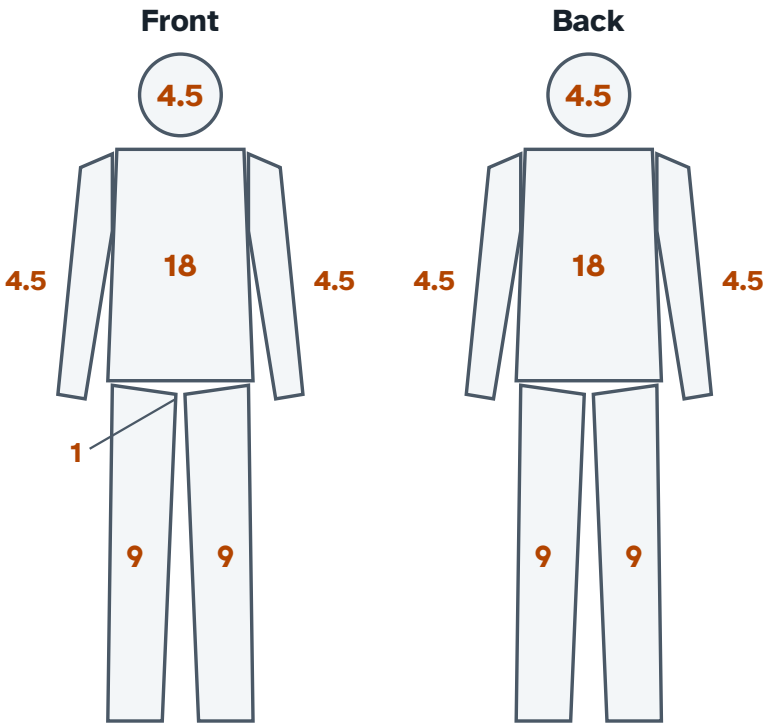
Burn depth through the skin



Each red bracket shows how deep that burn goes. Deeper burns destroy more nerve endings, so they hurt less.

By extent: Wallace's rule of nines (3 marks in W18; 3 marks inside S21)

Rule of nines, adult (front and back)



Front 50.5 + back 49.5 = 100

Whole regions: head and neck 9, each arm 9, front of trunk 18, back of trunk 18, each leg 18, perineum 1.

Text version (write this as the answer)

1. Head and neck: **9%** (front 4.5, back 4.5).
2. Each upper limb: **9%** (front 4.5, back 4.5), so both arms 18%.
3. Front of trunk (chest and abdomen): **18%**. Back of trunk: **18%**.
4. Each lower limb: **18%** (front 9, back 9), so both legs 36%.
5. Perineum and genitals: **1%**.

- **Palm method:** the patient's own palm with fingers is about **1%** TBSA. Use it for small or scattered burns.
- **Children:** the head is relatively bigger and the legs smaller, so use the **Lund and Browder chart**, which adjusts each region for age.
- Count only partial- and full-thickness areas. Superficial (first-degree) burns are not counted.

Exam point: working out a case

S21: face, upper chest and both arms. If the whole of both arms (9 + 9), the front of the head (4.5) and the upper half of the front trunk (9) are burnt, the total is about **31.5%**. Show the working; examiners give marks for the method.

By severity

A burn needs a **specialist burn unit** when any of these apply (classic American Burn Association referral criteria):

- Partial-thickness burns over **10%** TBSA
- Any full-thickness burn
- Burns of the face, hands, feet, genitals, perineum or over major joints
- Electrical, chemical or inhalation injury

- Burns in a patient with other serious illness or injury (for example the W17 electrician with rib fractures)

3 Causes and risk factors

- **Kitchen burns:** kerosene or gas stoves, cooking on the floor, loose synthetic clothing such as a dupatta or saree catching fire. Women are at high risk (S21 housemaid, S22 stove refuelling).
- **Scalds** in children from hot water, tea and milk.
- **Firecrackers** around festivals (W16 boy).
- **Electrical injury** in electricians and linemen (W17, W20).
- **Chemical burns** at work, and acid attacks.
- Higher risk with epilepsy, alcohol use, old age and young childhood, crowded housing and poor fire safety.

4 Pathophysiology to impairments

Papers ask "discuss the impairments with clinical reasoning" (W22). This chain is that reasoning.

Local wound: Jackson's three zones

- **Zone of coagulation** (centre): tissue is dead and cannot recover.
- **Zone of stasis** (around it): blood flow is reduced. It can recover with good resuscitation, or die if the patient is under-resuscitated or infected.
- **Zone of hyperaemia** (outer): increased blood flow. Usually recovers fully.

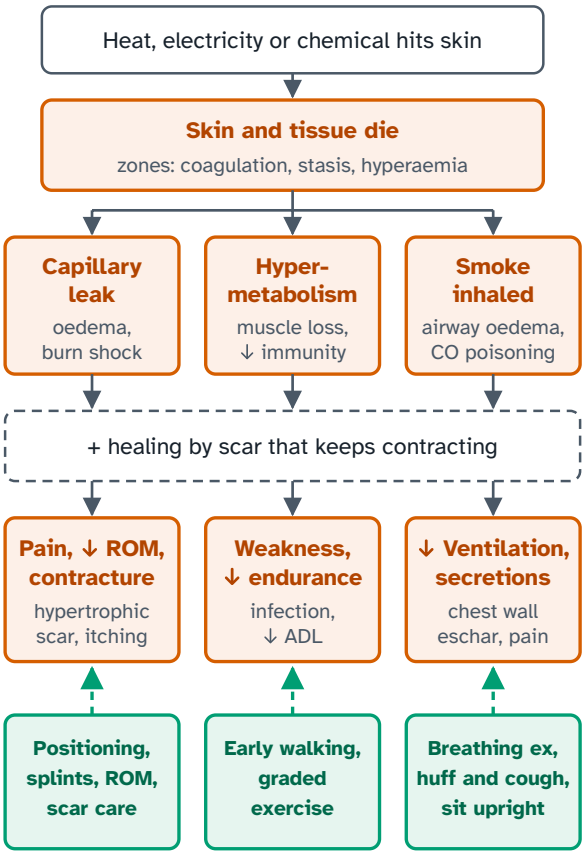
Whole-body effects

- **Capillary leak:** fluid and protein leak out of vessels, so the tissues swell and blood volume falls. In large burns (over about 20% TBSA) this causes **burn shock** in the first 24–48 hours.
- **Hypermetabolism:** the body burns energy and breaks down muscle for months, which causes weight loss and weakness.
- **Lowered immunity** and an open wound lead to infection and sepsis.
- **Inhalation injury:** heat swells the upper airway; smoke chemicals damage the lower airways and alveoli; carbon monoxide and cyanide poison oxygen delivery.

Healing causes the long-term problems

Deep burns heal with scar. Cells in the scar called myofibroblasts pull the wound edges together. Scar keeps contracting for months, so it pulls joints into the position of comfort, which is usually flexion and adduction. This is why **the position of comfort is the position of contracture**.

From burn to the problems physio treats



Green dashed arrows show where physiotherapy acts

In the exam, draw a quick sketch: the injury, three body responses, scar, three impairments, three physio actions.

Text version (write this as the answer)

1. Heat, electricity or chemicals kill skin and deeper tissue.
2. Capillary leak causes oedema and, in large burns, burn shock.
3. Hypermetabolism breaks down muscle; immunity falls, so infection risk rises.
4. Inhaled smoke causes airway oedema, lung damage and carbon monoxide poisoning.
5. Deep burns heal by scar, and the scar keeps contracting for months.
6. Impairments: pain, reduced range and contracture with hypertrophic scar; weakness, low endurance and reduced ADL; reduced ventilation with retained secretions.
7. Physiotherapy: positioning, splinting, range of motion and scar care; early mobilisation and graded exercise; breathing exercises, airway clearance and upright positioning.

Stages of wound healing (3 marks in S19; 3 marks inside W22)

Stage	When	What happens
1. Haemostasis	Minutes to hours	Vessels constrict, platelets form a clot, which releases growth factors
2. Inflammation	About days 1–5	Neutrophils and macrophages clean the wound; redness, heat, swelling, pain
3. Proliferation	About day 4 to week 3	Granulation tissue, new vessels, collagen laid down, epithelium covers the wound, wound contracts
4. Remodelling (maturation)	Week 3 to 1–2 years	Collagen is reorganised and strengthened; scar flattens and fades. Scar is never as strong as normal skin.

Factors affecting wound healing (7 marks in S25)

Type	Factors that slow healing
Local (in the wound)	Infection; poor blood supply (arterial disease, pressure, tight dressings); oedema; dead tissue or foreign bodies; repeated shear, friction or pressure; a wound that is too dry or too wet; a large or deep wound; previous radiotherapy
General (in the patient)	Older age; poor nutrition (protein, vitamin C, zinc); diabetes with high sugar; smoking; anaemia and low oxygen; obesity; drugs (steroids, chemotherapy); kidney or liver failure; reduced immunity; stress and poor sleep

The modalities that help healing (S24, S25) are on the diabetes and wound page (p. 153).

Recall check. Write the three zones of Jackson, the four stages of wound healing and five factors that slow healing from memory.

5 Clinical features

- Burn wound appearance by depth (see section 2).
- Pain: severe in superficial partial thickness; less in deep burns.
- Swelling peaks around 24–48 hours, especially in the hands, face and neck.
- Circumferential deep burns of the chest can restrict breathing; on a limb, they can cut off circulation.
- Fever, fast heart rate and fast breathing from hypermetabolism.

Suspect inhalation injury when

- The burn happened in a closed space (kitchen, room, vehicle).
- There are burns of the face and neck, singed nasal hair or eyebrows, or soot in the mouth or nose.
- Sputum is black (carbonaceous).
- The voice is hoarse, or there is stridor, wheeze or breathlessness.
- The patient is confused or drowsy (carbon monoxide).

Airway swelling can worsen over the first day, so these patients are watched closely and often intubated early.

Electrical burns

- Entry and exit wounds can look small while muscles along the current's path are badly damaged.
- Risks: heart arrhythmias, compartment syndrome, kidney damage from muscle breakdown (dark urine from myoglobin), nerve damage, and fractures from falls or strong muscle contraction.

6 Investigations

Test	What it shows	Physio meaning
TBSA and depth chart	Extent and depth of the burn	Shows which joints are at risk of contracture
ABG with carboxyhaemoglobin	Oxygenation, CO level, acidosis	Pulse oximeter SpO_2 can read **falsely normal** in CO poisoning. Use the blood gas.
Chest X-ray	Often normal at first; later collapse, pneumonia or ARDS	Guides airway clearance and positioning
Bronchoscopy	Soot and airway damage in inhalation injury	Confirms the need for intensive chest physiotherapy
ECG, CK, urine myoglobin	Arrhythmia and muscle damage after electrical injury	Mobilise only when the heart is stable

Test	What it shows	Physio meaning
Haemoglobin, electrolytes, albumin, wound swabs	Blood loss, fluid balance, nutrition, infection	Low haemoglobin or infection limits exercise tolerance

7 Medical and surgical management

First aid

- Stop the burning: remove the person from the source, take off burnt clothing and jewellery.
- Cool the burn under **cool running water for 20 minutes**. Do not use ice, toothpaste or butter.
- Cover with clean cling film or a clean cloth; keep the patient warm.

Hospital care

- **Airway and breathing** first; early intubation if inhalation injury is suspected; high-flow oxygen for CO poisoning.
- **Fluid resuscitation** for large burns (about 20% TBSA or more in adults), guided by urine output.
- Pain relief, tetanus cover, high-protein and high-calorie feeding.
- **Wound care:** cleaning, dressings (for example silver-based dressings), and daily review.
- **Escharotomy:** cutting through tight full-thickness eschar on the chest or a limb to free breathing or circulation. **Fasciotomy** if a compartment is tight.
- **Early excision and grafting:** dead tissue is removed and the wound is covered, usually with a **split-thickness skin graft** taken from a donor site such as the thigh. Grafts may be meshed to cover a larger area. Full-thickness grafts and flaps are used for the face, hands and joints.
- **Later reconstruction:** contracture release, Z-plasty and further grafts.

Classic answer vs current guideline: fluid formula

Classic (Parkland formula)	Current (American Burn Association)	Write in the exam
4 mL × body weight (kg) × %TBSA of Ringer's lactate in 24 hours. Half in the first 8 hours from the time of the burn, half over the next 16 hours.	Start lower, at **2 mL/kg/%TBSA**, and adjust the rate to keep urine output around **0.5 mL/kg/hour** in adults. Giving too much fluid causes swelling, compartment syndrome and lung problems.	"Parkland formula 4 mL/kg/%TBSA, half in 8 hours; current guidance starts at 2–4 mL/kg/%TBSA and titrates to urine output 0.5 mL/kg/h."

8 Complications and prevention

Complication	What to look for	Physio role
Contractures	Loss of range across burnt joints, especially neck, axilla, elbow, hand, knee, ankle	Prevent from day 1: anti-deformity positioning, splints, regular stretching and ROM
Hypertrophic scar and keloid	Raised, red, itchy, tight scar within the wound (hypertrophic) or spreading beyond it (keloid)	Pressure garments, silicone, massage, stretching
Chest problems: pneumonia, collapse, ARDS	Rising breathing rate, falling oxygen, fever, crackles	Upright positioning, deep breathing, airway clearance, early mobilisation
Infection and sepsis	Fever, wound changes, confusion, low BP	Strict hand hygiene and gowns; report changes
Heterotopic ossification (bone in soft tissue, commonest at the elbow)	Painful, warm joint with sudden loss of range	Active, gentle movement; avoid forceful passive stretching of that joint

Complication	What to look for	Physio role
Compartment syndrome	Tight, very painful limb, pain on passive stretch, numbness	Elevate limbs; report urgently
Exposed tendon injury, especially the finger extensor hood (boutonnière deformity)	Tendon visible in the wound over the finger joints	Protect: splint the finger straight; avoid combined finger flexion
Weakness, deconditioning, pressure sores, DVT	Long bed rest, weight loss	Early walking, strengthening, turning schedule, ankle pumps
Nerve damage and neuropathy	Numbness, weakness (common after electrical burns)	Protect numb areas; strengthen; splint if needed
Psychological problems	Anxiety, low mood, nightmares, body-image distress	Encourage, set achievable goals, refer for counselling
Marjolin's ulcer (late)	A non-healing ulcer in an old burn scar, years later	Teach skin checks; refer any non-healing scar ulcer

9 PT assessment

Subjective

- How, when and where the burn happened: closed space? electrical? first aid given?
- Pain (rest, movement, dressings), itching, breathlessness, cough and sputum.
- Surgery done or planned: grafts and donor sites, escharotomy.
- **Social history (fills the ICF):** occupation and hand dominance, earning role, children at home, house type, toilet, family support.
- Patient's goals and fears, including appearance.

Objective

- **Vital signs:** HR, BP, RR, temperature, SpO_2 (remember the CO caution).
- **Chest:** breathing pattern, chest expansion (eschar may restrict it), auscultation, cough and sputum.
- **Burn chart:** sites, depth, TBSA, areas crossing joints, grafts and donor sites.
- **Oedema** of the hands, face and limbs.
- **ROM** with a goniometer (active first; passive only where allowed), muscle strength, sensation.
- **Function:** bed mobility, transfers, walking, self-care, hand function.
- **Scar** (rehabilitation phase): height, flexibility, colour and blood supply.

Outcome measures to name

Measure	What it tells you
Vancouver Scar Scale	Scar pigmentation, vascularity, pliability and height
POSAS (Patient and Observer Scar Assessment Scale)	Scar quality from both the patient's and the therapist's view, including itch and pain
Goniometry	Range at each burnt joint
Burn Specific Health Scale – Brief	Burn-specific quality of life
Pain (VAS or NRS), grip strength, 6-minute walk test, Barthel or FIM	Pain, hand strength, endurance and independence

10 PT diagnosis by ICF

Draw the six-box ICF model first (it is drawn on the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (S21, 6 + 6 + 3 marks)

> "35 year old housemaid suffered accidental stove burns 3 days back affecting her upper chest, face & bilateral upper limbs. Discuss her ICF & PT management. Write the Wallace's Rule of Nine."

Spot the ICF clues before you write

housemaid means heavy two-handed work: participation. 3 days back means the emergent-to-acute phase, with swelling and a fluid shift. face plus a stove fire in a kitchen raises the question of inhalation injury. bilateral upper limbs means both hands may be involved, so self-care is affected.

ICF component	From this case	Clinical reasoning
Health condition	Thermal (flame) burns of the face, upper chest and both upper limbs, day 3; about 30% TBSA if all of the face, upper chest and both arms are burnt; depth to be charted	Day 3: oedema is at its peak; grafting may follow
Body structure	Skin loss over face, anterior chest, axillae, elbows, wrists and hands	Burns crossing flexor surfaces will contract
Body function impairments	• Pain • Oedema of the hands and face • Reduced neck, shoulder, elbow, wrist and hand range; risk of contracture • Reduced chest expansion; risk of inhalation injury • Risk of microstomia (small mouth opening) from facial burns • Reduced strength and endurance • Anxiety, worry about appearance	• Capillary leak causes the oedema • Pain and the position of comfort (neck flexion, arms adducted, elbows bent) start contractures • Chest burns and pain limit deep breaths
Activity limitations	Feeding, washing, dressing, grooming and toileting (both hands); reaching overhead; lifting; walking long distances	Both hands affected, so she cannot compensate with one hand
Participation restrictions	Cannot work as a housemaid (sweeping, washing clothes and vessels, cooking); family roles; social life	Housework needs repeated gripping and reaching, which are lost
Environmental factors	**Barriers:** loss of daily wage, cooking on a stove at floor level, likely Indian toilet. **Facilitators:** burn unit care, family support (to be asked).	Barriers become education goals (kitchen safety, return-to-work plan)
Personal factors	35-year-old woman; manual worker; possible distress about facial scarring	Young, so good potential; psychological support affects adherence

One-line functional diagnosis

A 35-year-old housemaid, day 3 after flame burns of the face, upper chest and both upper limbs, with pain, oedema and risk of neck, axillary, elbow and hand contractures and reduced chest expansion. These limit self-care, reaching and gripping, and restrict her work and family roles. Burn-unit care is a facilitator; loss of daily wages is a barrier.

Recall check. Fill the six ICF boxes for the W22 case in the exam corner (40% burns of both thighs and legs).

11 Problem list and goals

Goals for the S21 housemaid

Problem	Short-term (2–3 weeks)	Long-term (3–6 months and beyond)
Breathing	Clear chest, normal SpO_2 on air, effective cough	Normal chest expansion

Problem	Short-term (2–3 weeks)	Long-term (3–6 months and beyond)
Oedema and pain	Hand oedema reduced; pain controlled for exercise	Pain-free daily activity
Range of motion	Maintain full neck extension, shoulder abduction, elbow extension and hand range; no contracture	Full functional range; mouth opening normal
Scar	Wounds closed; grafts taken	Flat, pliable scar using pressure garment and silicone
Function	Independent feeding, washing face, toileting	Full self-care; return to housework and paid work
Prevention and coping	Understands positioning and splint schedule	Practises kitchen fire safety; coping with appearance

12 PT management

Write it phase by phase. Papers ask for short-term and long-term management, which map onto these phases.

Phase 1: emergent or resuscitative (first 72 hours)

- **Respiratory care:** head of bed raised (sit up at least 30°), deep breathing, thoracic expansion exercises, ACBT, assisted cough, suction if needed. If ventilated: positioning, secretion clearance, manual techniques where skin allows.
- **Oedema control:** elevate burnt limbs above the heart (hands on pillows or slings), with active finger and ankle exercises.
- **Positioning and splinting** in anti-deformity positions from day 1 (table below).
- **Range of motion:** active exercises of every burnt joint, about every hour while awake. Use passive or active-assisted movement if the patient cannot move. Give analgesia before sessions.
- **Mobility:** sitting out and walking as soon as the patient is stable.

Phase 2: acute (until the wounds are closed)

- Continue ROM, positioning and splinting. Time exercise with dressing changes, when the wound is visible and analgesia is given.
- **After skin grafting:** immobilise the grafted area, usually for about **3–5 days** (the surgeon decides). Keep moving all other joints. Then restart gentle active ROM over the graft. Avoid friction and shear on new grafts.
- **Donor sites** (often the thigh) hurt but can usually be moved early.
- **Lower-limb burns or grafts:** wrap the leg in a compression bandage before standing, because dependent legs swell and grafts can fail.
- Strengthening, walking, and self-care training (eating, washing) with built-up handles if grip is weak.

Phase 3: rehabilitation (after wound closure, while scar matures over 1–2 years)

- **Stretching:** slow, sustained stretches across scar until it blanches, held 20–30 seconds or longer, several times a day.
- **Scar management:** pressure garments, silicone gel sheets, scar massage once the skin is strong enough, moisturising.
- **Splints:** night splints to hold gains; dynamic splints or serial casting for established contractures.
- **Strength and endurance:** progressive resistance, aerobic training, hand therapy, fine motor practice.
- **Return to work:** work-specific tasks, graded loading of the hands, sun protection outdoors.
- Psychological support and peer support; cosmetic camouflage advice.

Anti-deformity positioning (7 marks in S17 and W18; part of S16)

Memory hook

Position of comfort = position of contracture

Burns pull joints into flexion and adduction. So position the opposite way: **stretch the burnt surface**.

Burnt area	Tends to contract into	Position or splint in
Anterior neck	Flexion	Neck **extension**: no pillow, small roll under the shoulders; neck conformer collar
Axilla and shoulder	Adduction, internal rotation	Shoulder **abduction about 90°** with slight forward flexion (about 15°) to protect the brachial plexus; axillary ("aeroplane") splint
Front of elbow	Flexion, pronation	Elbow **extension**, forearm neutral or supinated
Back of hand	Claw: MCP extension, IP flexion, thumb adduction	**Safe position** ("intrinsic plus"): wrist extended about 20–30°, MCP joints flexed about 70–90°, finger joints straight, thumb abducted
Palm	Finger and thumb flexion	Fingers and thumb **extended** (palm stretched out)
Chest and trunk	Forward flexion, rounded shoulders	Lie flat, trunk extended, shoulders back
Hip and groin	Flexion, adduction	Hip **extension**, slight abduction (about 10–20°), neutral rotation; prone lying when allowed
Back of knee	Flexion	Knee **extension**; no pillow under the knee
Ankle	Plantarflexion (foot drop)	Ankle **neutral (90°)**; ankle–foot splint
Face and mouth	Small mouth (microstomia), eyelid pulling	Mouth-stretching exercises, mouth splint; no pillow pressing on burnt ears

Directions are standard in all burn rehabilitation texts. Angles are typical values; they vary slightly between units.

Splints (S19: three splints with their purpose)

Splint	Purpose
Resting hand splint in the safe position	Prevents claw hand after dorsal hand burns; protects the extensor tendons
Neck conformer (soft collar)	Keeps the neck extended; prevents neck flexion contracture
Axillary (aeroplane) splint	Holds the shoulder abducted; prevents axillary contracture
Elbow and knee extension splints	Hold extension across flexor-surface burns
Ankle–foot splint	Holds the ankle at 90°; prevents foot drop and heel pressure sores
Mouth splint	Stretches the mouth; prevents microstomia
Dynamic splints and serial casts (later)	Gradually stretch established contractures

Role of splinting (S18): splints hold the anti-deformity position when the patient cannot, protect exposed tendons and new grafts, and keep the range gained by exercise. In the acute phase they are worn at rest and at night and removed for exercise. Check them daily for pressure and fit, because swelling changes.

Inhalation burns (7 marks in W22)

- Upright positioning to reduce airway swelling and help breathing.
- Humidified oxygen; high-flow oxygen for carbon monoxide.
- Deep breathing, ACBT, huffing, assisted cough, suction. Mobilise early.
- If ventilated: regular repositioning, airway clearance, careful bagging with suction.

- Watch for worsening: stridor, rising breathing rate, falling oxygen. Call the team early.

Lower-limb burns (7 marks in W25; long cases W22 and S24)

Depth	What to expect	Physiotherapy
Superficial and superficial partial-thickness	Heal on their own in about 1–3 weeks; painful; little scarring	Elevate the leg; ankle pumps and active hip, knee and ankle movement every hour; walk early, with a compression bandage if the leg swells when down; moisturise once healed
Deep partial and full-thickness	Need grafting; risk of hip and knee flexion and ankle plantarflexion contractures, oedema, DVT and heel pressure sores	Hip extension with slight abduction, knee straight, ankle at 90°; knee extension splint and ankle–foot splint; elevation; heels off the bed. After grafting, rest the grafted area as the surgeon advises, then start standing and walking in a compression bandage, with a walking aid if needed. Later: stretches of hip flexors, hamstrings and calf; strengthening of quadriceps, glutes and calf; pressure garments; stairs and work tasks

Deep burns all the way round a limb can tighten like a band and cut off the blood supply. Check toe colour, warmth and pulses, and report quickly (the surgeon may release the eschar).

Burns in pregnancy (S24: full-thickness burns of the right leg, 6 months pregnant)

There are two patients: the mother and the baby. The baby does well only if the mother's oxygen, blood pressure and nutrition stay good, so involve the obstetric team and report any warning sign at once.

- **Position:** from about 20 weeks, do not leave her flat on her back, because the uterus presses on the inferior vena cava and lowers her blood pressure. Use **left side-lying** or a wedge under the right hip (left tilt). With a right-leg burn, left side-lying also keeps the burnt leg on top, supported on pillows with the knee straight and the ankle at 90°.
- **DVT:** pregnancy, burns and bed rest all raise the risk. Ankle pumps and active exercise of the good leg every hour, early walking, and anticoagulants as prescribed.
- **Chest:** the uterus pushes the diaphragm up, so lung volumes fall. Deep breathing and thoracic expansion in side-lying or sitting.
- **Burnt leg:** the same lower-limb plan as above, but no electrotherapy or heat over the abdomen, pelvis or lower back, and no shortwave diathermy.
- **Exercise:** moderate effort, avoid breath-holding and straining. Stop and report bleeding, abdominal pain, contractions, leaking fluid or reduced baby movements.
- **Emotional support:** she is crying and afraid for her baby. Explain each step, give treatment after analgesia, and involve her family.

By impairment (for "physiotherapy for each impairment", W22)

Impairment	Technique and dose	Why
Pain	Exercise after analgesia; explain each step; relaxation and distraction	Pain is the main reason patients stop moving
Oedema	Elevate limbs above the heart; active pumping exercises 10 repetitions every hour; compression wraps before standing	Muscle pump and gravity move fluid back
Contracture risk, reduced ROM	Anti-deformity positioning, splints; active ROM of every burnt joint hourly; sustained stretches in rehab	Holds scar in its lengthened position while collagen is laid down
Reduced chest expansion, secretions	Deep breathing, thoracic expansion, ACBT, assisted cough, upright sitting	Prevents collapse and pneumonia
Weakness and low endurance	Early sitting and walking; progressive resistance; aerobic exercise in rehab	Counters hypermetabolic muscle loss and bed rest

Impairment	Technique and dose	Why
Hypertrophic scar	Pressure garment about 23 hours a day; silicone sheets; massage; moisturiser	Pressure and hydration flatten and soften scar
Reduced function	ADL training, adapted handles, work-specific practice	Restores independence and return to work

Classic answer vs current evidence: pressure garments

Classic	Current (Cochrane 2024)	Write in the exam
Every deep burn gets a custom pressure garment, worn about 23 hours a day at about 20–25 mmHg, for 12–18 months until the scar matures.	Still standard practice, but the evidence that garments improve scars is **very uncertain**. Comfort and wearing them regularly matter. Silicone and massage are used alongside.	"Pressure garment therapy about 23 hours a day for 12–18 months, with silicone gel and scar massage."

Electrotherapy for wound healing (S16, S22) is covered on the diabetes and wound-healing page.

Recall check. Without looking, write the anti-deformity position for the neck, axilla, back of the hand and ankle.

13 Precautions, contraindications and stop criteria

Take care

- **Fresh grafts:** no movement of the grafted area until the surgeon allows; no shearing or friction.
- **Exposed tendons:** no aggressive or combined stretching; splint the finger straight over an exposed extensor hood.
- **Heterotopic ossification:** no forceful passive movement of a hot, painful, stiffening joint.
- **Haemodynamic instability** in the first days: keep exercise gentle and monitored.
- **CO poisoning:** pulse oximetry can be falsely normal.
- **Electrical burns:** check ECG monitoring and cardiac stability before mobilising.
- **Infection control:** wash hands, wear gloves and a gown, clean equipment between patients.
- **Associated injuries:** fractures, rib fractures and head injury (W17 case) change the plan.

Stop and report if

- Breathing becomes harder, stridor appears, or oxygen falls
- Blood pressure falls, or the heart rate becomes fast or irregular
- A graft bleeds, lifts or shears
- A limb becomes tight, numb or very painful

14 Education, lifestyle and home programme

Do's and don'ts for the patient

- Do your stretches and exercises several times every day, even after discharge. Scar keeps tightening for 1–2 years.
- Wear your splints and pressure garments as advised. Wash garments daily; get them refitted when they loosen.
- Moisturise healed skin and massage the scar several times a day.
- Protect healed skin from the sun with clothing and sunscreen; new skin burns and darkens easily.

- Do not scratch itchy scars. Use moisturiser, cool compresses and medicines the doctor gives.
- Protect numb areas from heat and injury.
- Report any blister, wound or ulcer in a scar that does not heal.

Preventing burns at home

- Cook on a raised platform, not on the floor. Never refuel a lit stove.
- Wear close-fitting cotton clothes while cooking; tuck in the dupatta or saree end.
- Keep children away from the kitchen, hot liquids and firecrackers.
- Know first aid: stop, drop and roll; cool running water for 20 minutes.

15 Prognosis and discharge criteria

Prognosis is worse with larger TBSA, deeper burns, older age and inhalation injury. The revised Baux score estimates risk of death as **age + %TBSA + 17 if there is an inhalation injury**.

Discharge from the burn unit: wounds healed or manageable at home; independent or supported self-care and walking; family taught positioning, splints, exercises and skin care; follow-up arranged for scar and contracture review.

16 Exam corner

Past questions (MUHS, 2016–2026)

S16 Q4(a), 5 + 5 + 5 marks

30-year-old man, 60% burns of neck, chest and both arms, in ICU. Functional diagnosis; physiotherapy; role of positioning and splinting.

W16 Q4(a), 5 + 5 + 5 marks

13-year-old boy, firecracker explosion: inhalation burns and 30% partial-thickness burns of the neck, chest and hand. Functional diagnosis, short- and long-term management.

W17 Q3(a), 5 + 5 + 5 marks

30-year-old electrician, electrical burns 2 days ago and a fall from height; 3 right ribs fractured; unconscious and ventilated. Impairments, activity and participation, short-term management.

W20 Q4(b), 5 + 5 + 5 marks

22-year-old electrician, 30% mixed burns of the right upper limb and front of the chest from wiring; in the burn unit, stable. ICF, short- and long-term management.

S21 Q4(a), 6 + 6 + 3 marks

The worked example in section 10.

S22 Q3(b), 15 marks

35-year-old man, inhalation burns while refuelling a stove; deep partial-thickness burns of the right arm, front and back of the trunk, side of the neck, right face and thigh. Functional diagnosis, goals and management in detail.

W22 Q3(b), 5 + 7 + 3 marks

23-year-old mother of a young child, 40% burns of both thighs and legs. Impairments with clinical reasoning; physiotherapy for each impairment; stages of wound healing.

W19 Q4(a), 5 + 5 + 5 marks

12-year-old boy, 40% electrical burns of the face, neck, front of the chest and right arm from a live wire; medically stable in the burns ward. Functional diagnosis; short- and long-term physiotherapy.

W21 Q4(b), 15 marks

25-year-old fashion designer, full-thickness burns of the face, neck and chest 2 days ago; difficulty breathing; swollen face. Likely complications; functional diagnosis; management.

S24 Q4(b), 15 marks

35-year-old clerk, full-thickness burns of the right leg from thigh to foot 3 days ago; 6 months of amenorrhoea (she is pregnant); crying and not moving the limb. Functional diagnosis and management. See "Burns in pregnancy" in section 12.

W24 Q4(b), 15 marks

12-year-old boy, superficial partial-thickness burns of both arms and the front of the trunk 2 days ago; pain VAS 6/10 at rest; cannot turn or move his limbs. Functional diagnosis and management.

S26 Q4(b), 15 marks (three parts)

25-year-old woman, hot-water burns of the left arm, chest and neck, treated with closed dressings; complains of pain. (i) Extent of the burns; (ii) impairments using ICF; (iii) short- and long-term physiotherapy.

7-mark questions

Deformities in burns and anti-deformity positioning (S17). Classification of burns and common deformities (W17). Role of splinting (S18). Positioning to prevent deformities (W18). Management after skin grafting (S20). Inhalation burns and management (W22). Modalities for wound healing (S24). Factors affecting wound healing, with the modalities used (S25). Physiotherapy for superficial and deep burns of the lower limb (W25).

3-mark questions

Deep partial-thickness burns (W16). Rule of nine (W18). Three splints with their purpose (S19). Stages of wound healing (S19). Superficial burns (S20). Define inhalation burns (S24). Three complications of inhalation burns (S25).

Technique cards for this topic: suctioning (p. 231), oxygen therapy (p. 290), ICU physiotherapy (p. 223), positioning (p. 236), electrotherapy (p. 298).

Keywords to underline

TBSA rule of nines depth inhalation injury oedema anti-deformity positioning position of comfort is position of contracture splinting graft immobilisation hypertrophic scar pressure garment emergent, acute, rehabilitation phase

Answer skeleton: any burns long answer

1. Restate the case: age, job, cause, sites, %TBSA (work it out), depth, day, inhalation or electrical injury.
2. **[ICF, 5–6]** ICF diagram filled from the case; one-line functional diagnosis; reasons for each impairment.
3. **[Short-term, 5]** Emergent and acute phase: respiratory care, oedema control, anti-deformity positioning table, splints, hourly ROM, graft precautions, early mobilisation.
4. **[Long-term, 5]** Rehabilitation phase: stretching, scar management, strengthening, hand therapy, ADL and return to work, psychological support, prevention.
5. **[Add-on, 3]** Rule of nines, wound-healing stages or splints, whichever is asked.

Draw these

- The rule of nines figure (section 2)

- The burn-depth skin diagram (section 2)
- The ICF model, filled from the case (IHD page, section 10)

Viva questions

1. Why is a full-thickness burn painless?
2. What are the signs of inhalation injury?
3. Why can SpO_2 be misleading after a smoke inhalation?
4. What is the safe position of the hand, and why?
5. How long do you immobilise a fresh skin graft?
6. What is the Parkland formula, and what does current guidance change?

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Write the rule of nine.**

Answer

Head and neck 9%; each upper limb 9%; front of trunk 18%; back of trunk 18%; each lower limb 18%; perineum 1%. Total 100%. Draw the front and back figure.

3 m **Characteristics of deep partial-thickness burns.**

Answer

- Reaches the deep dermis.
- Blotchy red and white, drier, blisters may be absent; slow or no blanching.
- Less painful than superficial partial thickness; pressure is felt.
- Heals in more than 3 weeks with a high risk of hypertrophic scar; often needs grafting.

7 m **Positioning strategies to prevent deformities in burns.**

Answer

Principle: the position of comfort is the position of contracture, so stretch the burnt surface. Then write the anti-deformity table: neck extension, shoulder abduction about 90°, elbow extension, safe hand position, hip extension with slight abduction, knee extension, ankle at 90°, mouth stretching. Add splints, the wearing schedule (rest and night, off for exercise) and daily checks.

LAQ **W22: 23-year-old mother, 40% burns of both thighs and legs. List the impairments with clinical reasoning.**

Answer

- Pain: exposed nerve endings in partial-thickness burns.
- Oedema of the legs: capillary leak; worse when the legs hang down.
- Risk of hip flexion, knee flexion and ankle plantarflexion contractures: position of comfort and scar contraction.
- Reduced walking and standing tolerance: pain, oedema, weakness; grafts on the legs need compression before standing.
- Risk of DVT and pressure sores: bed rest.
- Weakness and fatigue: hypermetabolism.
- Participation: cannot care for her young child or do housework; anxiety about appearance.

MCQ **A patient with dorsal hand burns should be splinted with:**

A. Wrist flexed, fingers flexed

B. Wrist extended, MCP joints flexed, finger joints straight, thumb abducted

C. Wrist neutral, all finger joints straight

D. Hand in a fist around a roll

Answer
B. This "safe position" keeps the collateral ligaments of the MCP joints stretched and protects the extensor tendons over the PIP joints.

MCQ **After a kitchen fire, a patient's SpO_2 reads 99% but he is confused and has soot in his nose. What is the best next step?**

A. Reassure; oxygenation is normal

B. Arterial blood gas with carboxyhaemoglobin

C. Start a 6-minute walk test

D. Begin chest percussion

Answer
B. Carbon monoxide makes the pulse oximeter read falsely high. A blood gas measures carboxyhaemoglobin directly.

MCQ **A split-thickness skin graft was placed over the elbow yesterday. What should physiotherapy do for that elbow?**

A. Full passive stretching twice daily

B. Keep it immobilised as the surgeon advises, and exercise other joints

C. Resisted elbow flexion

D. Massage the graft

Answer
B. A new graft needs a few days without movement or shear to "take". Movement restarts after the surgeon clears it, usually around day 3–5.

Last recall check. Write the burn depth table, the rule of nines and four anti-deformity positions. Then check them against this page.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

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4. American Burn Association guidance on fluid resuscitation, as summarised in *American Family Physician* 2025.
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6. Early mobility after lower-limb split-thickness skin grafts: burn-unit protocols reporting immobilisation for about 3–5 days (conference abstract, *J Burn Care Res* 2025).
7. Dewey WS, Richard RL, Parry IS. Positioning, splinting, and contracture management. *Phys Med Rehabil Clin N Am* 2011.
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9. Structure and classical content: Frownfelter & Dean, *Cardiovascular and Pulmonary Physical Therapy*, 6th ed. (2022); Main & Denehy, *Cardiorespiratory Physiotherapy*, 5th ed. (2016); Cash's Textbook of General Medical and Surgical Conditions for Physiotherapists.

Mastectomy, lymphoedema and cancer surgery

> Breast cancer surgery was asked in 16 of 20 papers (2016–2026), almost always about a working woman: a farm labourer, a teacher, a tiffin cook. Answer with the ICF, the week-by-week shoulder programme, lymphoedema prevention and lifelong arm care, and give the reason for each step.

Syllabus	Ch 6(f) General abdominal and oncological surgeries: pre- and post-operative care, complications and management
Priority	**Tier 1.** In 16 of 20 papers (2016–2026). 6 long-answer cases; a 7-mark question in S25, W25 and S26.
Asked as	LAQ W16 · S17 · S19 · W20 · W21 · W24 · 7m S16 · S18 · S20 · S21 · W19 · S25 · W25 · S26 · 3m S21 · W22 · S23
Usual split	5 + 5 + 5 (functional diagnosis or ICF, short-term, long-term), 5 + 7 + 3 (ICF, long-term, prosthesis), or 15 (complications, rehab, lifelong care)
Checked against	Cochrane review of exercise after breast cancer surgery (2010); International Society of Lymphology consensus (2020); ACSM international roundtable on exercise for cancer survivors (2019). Full list under Sources. Content checked 19 October 2026.

1 Definition

Mastectomy is surgical removal of the breast, usually for breast cancer. It is often combined with removal or sampling of the lymph nodes in the armpit (axilla).

Lymphoedema is chronic swelling caused by protein-rich fluid collecting in the tissues because the lymphatic system cannot drain it. After breast cancer treatment it affects the arm, hand, breast or chest wall on the operated side.

2 Classification and grading

Types of breast surgery

Operation	What is removed	Physio significance
Radical mastectomy (Halsted)	Breast, pectoralis major and minor, all axillary nodes	Rarely done now. Big loss of shoulder strength, chest wall deformity, high lymphoedema risk.
Modified radical mastectomy (MRM)	Breast and axillary nodes; pectoralis major kept (pectoralis minor may be removed or retracted)	Commonest in the papers (W16, W20, S20). Shoulder stiffness, axillary scar, lymphoedema risk.
Simple (total) mastectomy	Breast only	Less shoulder and lymph problems
Breast-conserving surgery (lumpectomy, wide local excision)	The lump with a margin; followed by radiotherapy	Radiotherapy can cause fibrosis and stiffness
Sentinel node biopsy vs axillary clearance	One or a few first nodes, vs many axillary nodes	Full axillary clearance carries a much higher lymphoedema risk
Reconstruction	Implant, or own tissue: latissimus dorsi flap or abdominal flap	Extra precautions at the donor site (back or abdomen)

Lymphoedema stages (International Society of Lymphology)

Stage	Features
0 (latent)	Lymph transport impaired but no visible swelling; may feel heavy. Can last months or years.
I	Early swelling with protein-rich fluid; may pit; goes down with elevation

Stage	Features
II	Elevation alone rarely reduces it; pitting present early, then fibrosis makes the tissue firm
III	Lymphostatic elephantiasis: no pitting, thickened skin, warty overgrowth, skin folds

Lymphoedema is commonly diagnosed when the affected arm is **2 cm or more** larger in circumference at any point, or about **10% or more** larger in volume, than the other arm.

3 Causes and risk factors

Breast cancer risk factors

- Female sex, increasing age, family history, BRCA gene mutations.
- Early first period, late menopause, no children or a first child after 30, not breastfeeding.
- Obesity after menopause, alcohol, inactivity, hormone therapy, previous chest radiation.

Lymphoedema risk factors

- Axillary node clearance (more nodes removed, higher risk) and radiotherapy to the axilla.
- Infection or seroma in the arm or wound; obesity; reduced arm use; cellulitis.
- Heavy, sudden overuse of the arm without building up gradually.

4 Pathophysiology to impairments

1. Surgery cuts skin, fascia, muscle attachments and lymph vessels, and removes axillary nodes.
2. **Pain and fear** make the patient hold the arm against the chest, so the shoulder stiffens (capsule and scar tighten), especially in flexion, abduction and external rotation.
3. **Scar and radiotherapy fibrosis** tie down the chest wall and axilla.
4. **Lymph drainage is reduced.** When the lymph load is more than the remaining vessels can carry, protein-rich fluid collects: lymphoedema. The protein attracts water and later causes fibrosis and infection risk.
5. **Nerve injury** during axillary surgery: long thoracic nerve (serratus anterior, so winging of the scapula), thoracodorsal nerve (latissimus dorsi), intercostobrachial nerve (numb inner upper arm).
6. **Axillary web syndrome:** tight, painful cords from the armpit down the inner arm in the first weeks, from damaged lymph vessels; they limit abduction.
7. **Posture:** the shoulder on the operated side rises and rolls forward; weight imbalance after removing a large breast can strain the neck and spine.
8. **Chemotherapy and radiotherapy** add fatigue, low blood counts, nerve damage in the hands and feet, and skin and lung changes.

Recall check. Name the three nerves at risk in axillary surgery and the deficit each causes.

5 Clinical features after surgery

- Pain and tightness across the chest wall and axilla; drains in the first week.
- Limited shoulder elevation, abduction and external rotation (S17: "shoulder joint restriction").
- Swelling of the arm or hand on the operated side (S17: "mild oedema"; W20: "swelling in right upper limb").
- Numbness of the inner upper arm; scapular winging if the long thoracic nerve was injured.
- Cord-like bands in the axilla (axillary web syndrome).
- Fatigue, nausea and hair loss during chemotherapy; skin redness during radiotherapy.
- Anxiety, low mood, altered body image.

6 Investigations

Test	What it shows	Physio meaning
Mammography, ultrasound, biopsy	Diagnosis, tumour type and hormone receptor status	Decides the treatment plan (surgery, chemotherapy, radiotherapy, hormones)
Staging scans	Spread to bones, lungs, liver	Bone metastases: avoid heavy loading of affected bones
Blood counts during chemotherapy	Low white cells, haemoglobin or platelets	Infection, fatigue or bleeding risk: reduce or hold exercise
Arm circumference and volume	Early lymphoedema	Baseline before surgery, then regular checks
Lymphoscintigraphy, bioimpedance	Lymph flow; early fluid change	Detects lymphoedema before it is visible

7 Medical and surgical management

- **Surgery:** mastectomy or breast-conserving surgery, sentinel node biopsy or axillary clearance, with or without reconstruction.
- **Chemotherapy:** before (to shrink the tumour) or after surgery (W20 case).
- **Radiotherapy** to the chest wall, breast or axilla.
- **Hormone therapy** (tamoxifen, aromatase inhibitors) for hormone-positive cancers; aromatase inhibitors cause joint pain and bone loss.
- **Targeted therapy** such as trastuzumab, which can affect the heart.

8 Complications and prevention (7 marks in S20)

Complication	When	Prevention and physio role
Pain	Immediate	Analgesia before exercise; supported positioning; relaxation
Chest complications (atelectasis, infection)	Immediate	Deep breathing, supported cough, early walking
Seroma, haematoma, wound infection, flap necrosis	First 1–2 weeks	Limit shoulder elevation to about 90° while drains are in; report swelling or redness
DVT	Early	Early walking, ankle pumps
Frozen or stiff shoulder	Weeks	Graded ROM from day 1; full range after drains removed; daily stretching
Axillary web syndrome (cording)	Weeks 1–8	Gentle sustained stretching, soft-tissue work, continue ROM
Lymphoedema	Months to years, lifelong risk	Measure regularly, skin care, graded exercise, healthy weight, early treatment
Cellulitis	Any time	Protect skin from cuts and burns; urgent antibiotics if the arm becomes red and hot
Nerve injury, winging, numbness	From surgery	Serratus anterior and scapular strengthening; protect numb skin
Postural changes, neck and back pain	Weeks to months	Posture correction, scapular exercises, breast prosthesis
Radiation fibrosis, brachial plexopathy	Months to years	Keep stretching during and after radiotherapy
Cancer-related fatigue, low mood, body-image distress	Any time	Aerobic and resistance exercise, support groups, counselling

9 PT assessment

- **Subjective:** type of surgery and side, number of nodes removed, drains, reconstruction; chemotherapy or radiotherapy plans; pain, heaviness, numbness, tightness; hand dominance; occupation and household load (cooking, washing, lifting); children; patient's goals.
- **Observation:** wound, drains, posture (shoulder height, protraction), scapular winging, swelling, skin colour.
- **Shoulder range:** goniometry of flexion, abduction, external rotation, both sides.
- **Strength:** shoulder and scapular muscles, grip.
- **Arm measurements:** circumference at fixed points (for example every 4 cm from the wrist), compared with the other arm; volume calculation; pitting test; skin and tissue texture.
- **Sensation** of the inner arm; axillary cords on palpation.
- **Outcome measures:** DASH (arm function), shoulder ROM, arm volume, FACT-B (quality of life), fatigue scale, 6MWT.

Clues in the story (W21): heaviness of the arm and sleeves or rings that feel tight are early signs of lymphoedema, so measure both arms the same day. Redness, heat and fever in the arm suggest cellulitis, which needs urgent medical treatment before any massage or bandaging. Neck pain after surgery usually comes from guarding: a raised, protracted shoulder and a stiff neck.

10 PT diagnosis by ICF

Draw the six-box ICF model (see the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (W20, 5 + 5 + 5 marks)

> "A 40 year old female, right handed, teacher by occupation has undergone right modified mastectomy three weeks back and is receiving chemotherapy. She has swelling in right upper limb and pain in right shoulder. She is married, has two children and wishes to resume her job. Discuss ICF with clinical reasoning. Plan her short term and long term physiotherapy management."

Spot the ICF clues before you write

right handed and right mastectomy means her dominant arm is affected. three weeks means drains are out and full ROM can start. receiving chemotherapy means fatigue and infection risk. swelling in right upper limb is early lymphoedema. teacher means writing on a board with the arm raised.

ICF component	From this case	Clinical reasoning
Health condition	Carcinoma of the right breast; right MRM 3 weeks ago; on chemotherapy	Wound healed; now the stiffness and lymphoedema phase
Body function and structure impairments	• Right shoulder pain • Reduced shoulder flexion, abduction, external rotation • Right arm swelling (early lymphoedema, probably stage I) • Scar and axillary tightness, possible cording • Reduced strength and endurance; chemotherapy fatigue • Anxiety, altered body image	• Guarding after surgery stiffens the capsule • Axillary nodes removed, so lymph drainage is reduced • Chemotherapy causes fatigue and deconditioning
Activity limitations	Reaching overhead, writing on a blackboard, combing hair, dressing (fastening behind), lifting and carrying, cooking	Dominant arm is painful, stiff and swollen
Participation restrictions	Cannot teach yet; reduced care of her two children; social withdrawal	Teaching needs prolonged arm elevation and stamina

ICF component	From this case	Clinical reasoning
Environmental factors	**Facilitators:** husband and family, hospital care, a job she values. **Barriers:** chalk-board work, heavy household chores, school bag and shopping loads.	Plan graded return to work and help at home
Personal factors	40-year-old woman, motivated to return to work, worried about cancer and appearance	Motivation helps adherence; reassurance needed

One-line functional diagnosis

A right-handed teacher, 3 weeks after right MRM and on chemotherapy, with right shoulder pain and stiffness, early lymphoedema of the right arm and fatigue, limiting reaching, writing on the board, self-care and lifting, and restricting her teaching and family roles. Family support and her motivation are facilitators; heavy household tasks are a barrier.

Recall check. Fill the ICF for the S19 case: Sunita, right radical mastectomy 3 days ago, right-handed, cooks tiffin boxes for ten clients twice a day.

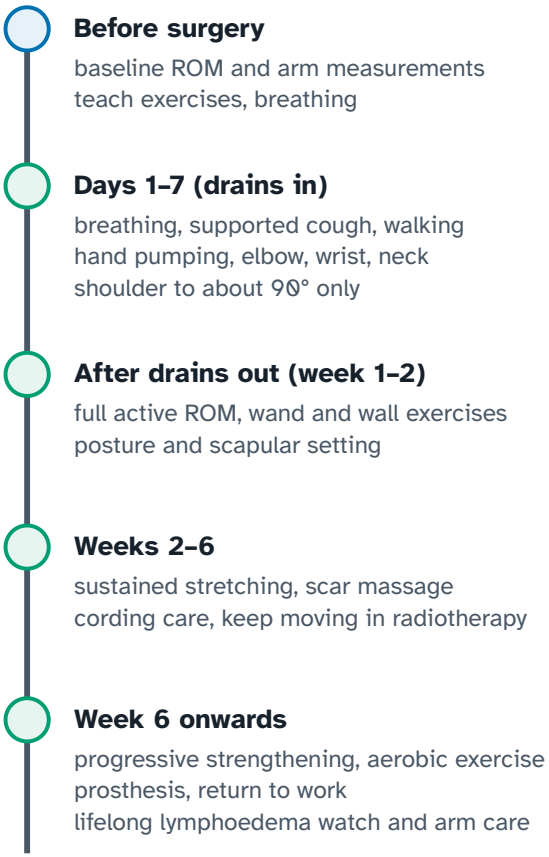
11 Problem list and goals

Goals for the W20 teacher

Problem	Short-term (4 weeks)	Long-term (3–6 months)
Shoulder pain and stiffness	Pain reduced; shoulder flexion and abduction to about 150°	Full, pain-free range; writes on the board comfortably
Arm swelling	Arm circumference reduced; knows skin care and self-massage	Volume difference controlled with a compression sleeve and exercise
Strength and fatigue	Walks 20 minutes most days during chemotherapy	Aerobic and resistance programme 2–3 times a week; energy for full workdays
Posture	Shoulders level; scapular setting learned	Good posture with a fitted prosthesis
Participation	Independent in self-care with the right arm	Back to teaching and household roles

12 PT management

Rehabilitation timeline after mastectomy



The step from "about 90°" to "full range" comes when the drains are out, usually around a week. Follow the surgeon's instructions.

Before surgery

- Measure shoulder range and both arms (baseline for lymphoedema).
- Teach the first-week exercises, breathing exercises and posture; explain drains and what to expect.

Days 1–7 (drains in)

- Deep breathing, supported cough, early walking.
- Hand pumping (open and close the fist), wrist and elbow movements, neck and shoulder shrugs; 10 repetitions several times a day.
- Shoulder flexion and abduction kept to about **90°** while drains are in.
- Rest the arm on a pillow when lying, with the hand slightly higher than the elbow.

? Why limit the shoulder to about 90° in the first week?

> A Cochrane review found that early exercise improves shoulder range, but starting full-range exercise straight after surgery increases wound drainage and seroma. Delaying full elevation by about a week reduces drainage, and shoulder function still recovers within about a month.

After drains are removed

- Full active and active-assisted range: wand (stick) exercises, wall climbing with the fingers, pulley, hands behind the head, hands behind the back. Hold stretches 20–30 seconds, 3–5 repetitions, 2–3 times a day.
- Scapular setting, retraction and posture correction; serratus anterior work if winging.

- Scar massage once the wound is fully healed; stretches for axillary cords.
- Continue stretching through radiotherapy to prevent fibrosis.

From about 6 weeks: strength, fitness and return to work

- **Progressive resistance training** of the arms, shoulders and legs: start light under supervision and progress slowly.
- **Aerobic exercise** such as brisk walking or cycling.
- Work-specific practice: board writing, cooking, lifting children, carrying.

Exercise during chemotherapy and for cancer-related fatigue

ACSM roundtable 2019 doses for cancer-related fatigue

Part	Aerobic	Resistance
Frequency	**3 times a week**	**2 times a week**
Intensity	Moderate: about 65% of maximum heart rate, RPE about 12	About **60% of 1RM**, 2 sets of 12–15 repetitions
Time	**30 min**	Major muscle groups
Duration	At least **12 weeks**. Reduce the dose on bad days rather than stopping completely.	

Lymphoedema: complete decongestive therapy (S21, 7 marks)

Phase	Components
1. Intensive (reduction), daily for about 2–4 weeks	• **Manual lymph drainage (MLD):** light, slow, skin-stretching strokes; first clear the neck, the other axilla and the trunk, then move fluid from the upper arm towards working lymph nodes, then the forearm and hand • **Multilayer short-stretch compression bandaging**, worn day and night • **Exercises with the bandage on**: hand pumping, elbow and shoulder movement, deep breathing (muscle pump) • **Skin care**: keep clean and moisturised; treat cuts and fungal infection early
2. Maintenance, lifelong	• **Compression sleeve (and glove if the hand swells)** worn during the day, measured and replaced regularly • Simple self lymph drainage at home • Regular exercise, including supervised progressive resistance training • Skin care and weight control

Bandages used for lymphoedema (3 marks in S23)

- **Short-stretch (low-stretch) bandages:** the main lymphoedema bandage. They stretch only a little, so they give **low pressure at rest** (safe to wear at night) and **high pressure when muscles contract** against them, which pumps fluid out of the arm.
- **Multilayer lymphoedema bandaging:** tubular stockinette next to the skin, narrow gauze bandages on each finger, padding (soft synthetic wool or foam) to even out the arm's shape and protect bony points, then several layers of short-stretch bandage with more pressure at the hand than the upper arm.
- **Adjustable compression wraps** (hook-and-loop, Velcro-type): the patient can put them on and tighten them herself; useful for self-management and maintenance.
- **Long-stretch (high-stretch, elastic) bandages** are not used for lymphoedema: they keep high pressure even at rest and can act like a tourniquet.

Also: elevate the arm when resting; deep diaphragmatic breathing (helps central lymph flow); pneumatic compression only as an add-on. A 33-year-old receptionist (S21) needs workstation advice too: arm supported on the desk, regular movement breaks, sleeve during work.

Classic answer vs current evidence: lifting and the "at-risk" arm

Classic	Current (ACSM 2019; supervised trials)	Write in the exam
Avoid lifting anything heavy with the arm for life; no weights; no BP cuffs, injections or blood tests on that side.	**Slowly progressive, supervised resistance training is safe** for women with or at risk of lymphoedema and does not make it worse. "Start low, progress slow." The evidence for avoiding BP cuffs and needles is weak, but many centres still advise using the other arm when possible.	"Avoid sudden heavy or unaccustomed loading; build up arm strength gradually with supervised resistance exercise; protect the skin; use the other arm for injections where possible."

Breast prosthesis (3 marks in S17)

- **Temporary soft prosthesis** (light, fibre-filled) worn in a soft bra from soon after surgery, because it does not press on the scar.
- **Permanent silicone prosthesis** fitted once the wound has healed and swelling has settled (often around 6–8 weeks, or after radiotherapy skin has recovered). It matches the weight of the other breast. Worn in a mastectomy bra with a pocket; partial and adhesive types exist.
- **Why it matters:** restores body symmetry and balance, prevents the shoulder rising and the spine tilting, helps clothes fit, improves confidence, and protects the chest wall.

By impairment

Impairment	Technique and dose	Why
Shoulder stiffness	Wand, wall-climb and pulley exercises; stretches held 20–30 s, 3–5 reps, 2–3 times a day	Lengthens capsule and scar
Pain	Positioning, analgesia timing, gentle movement, TENS away from the tumour site	Allows exercise
Lymphoedema	Decongestive therapy; sleeve; hand pumping and elevation; skin care	Moves fluid through remaining lymph routes
Weakness, winging	Scapular and serratus exercises; progressive resistance from about 6 weeks	Restores shoulder control
Posture	Mirror work, scapular retraction, prosthesis	Corrects elevated, protracted shoulder
Fatigue	Aerobic 30 min × 3 a week; resistance 2 a week	Exercise reduces cancer-related fatigue

Rationale of physiotherapy after MRM (7 marks in S25 and W25)

What we do	Why
Pre-operative baseline and teaching	Gives a baseline for shoulder range and arm size, so early lymphoedema or stiffness is spotted; a patient who knows the exercises starts them sooner
Breathing exercises and early walking	Prevent chest complications after anaesthesia and pain-limited breathing; prevent DVT
Hand pumping, elbow and wrist movement from day 1	The muscle pump moves lymph and blood, limiting swelling; keeps the distal joints free
Shoulder only to about 90° while the drains are in	Vigorous early shoulder movement increases wound drainage and seroma; limiting it for about a week reduces this without losing range in the long term

What we do	Why
Full active range and stretching after the drains are out	Prevents frozen shoulder and adhesion of the scar and pectoral tissues; treats axillary cording
Scar massage after healing	Keeps the scar mobile, so it does not tether the skin and limit the shoulder
Posture and scapular exercises	Patients guard with a raised, protracted shoulder; breast loss alters balance; long thoracic nerve injury causes winging
Lymphoedema education, skin care and arm measurement	Node removal and radiotherapy reduce lymph drainage for life; infections and sudden overload can trigger lymphoedema; early treatment works best
Progressive resistance and aerobic exercise from about 6 weeks	Restores strength for work (lifting, cooking); reduces cancer-related fatigue; safe for lymphoedema when progressed slowly
Prosthesis and return-to-work plan	Restores symmetry, posture and confidence; work and income matter in every MUHS case

Recall check. Write the timeline: what changes when the drains come out, the two phases of decongestive therapy, and the three types of lymphoedema bandage.

13 Precautions (3 marks in S21)

Precautions after mastectomy

- Shoulder elevation limited to about 90° while drains are in; no vigorous stretching over a fresh wound.
- Watch for seroma, wound infection and flap problems; report swelling, redness or discharge.
- Protect the arm on the operated side: avoid cuts, burns, insect bites and tight sleeves or bangles.
- No heat (hot packs, saunas) or vigorous massage on a lymphoedematous arm.
- Build up lifting gradually; avoid sudden heavy loads.
- Reconstruction: protect the flap donor site; follow the surgeon's limits on arm and trunk movement.
- During chemotherapy: reduce or hold exercise on days with fever, very low blood counts, severe fatigue or vomiting; avoid crowded gyms when white cells are low.
- Bone metastases: avoid heavy loading and twisting of affected bones.
- No electrotherapy over active tumour sites.

Stop and refer urgently if

- The arm becomes red, hot and painful with fever (cellulitis)
- Sudden calf or arm swelling with pain (DVT)
- New bone pain, or breathlessness

14 Education and lifelong arm care

S19 asked "Why and how does she need to take care of her hand life long?"

Why: the lymph nodes removed from the armpit do not grow back. The arm always has less capacity to drain fluid, so any extra load (infection, burn, injury, heavy unaccustomed work) can start lymphoedema, even years later.

How:

- Keep the skin clean and moisturised; dry well between the fingers.
- Wear gloves for cooking, gardening and washing vessels. Use oven gloves and long-handled spoons (important for the tiffin cook).

- Treat every cut, scratch or insect bite at once with antiseptic; see a doctor if it becomes red or hot.
- Avoid tight bangles, rings, watches and tight blouse sleeves on that arm.
- Build up heavy tasks slowly; take breaks; carry loads with both hands or a trolley.
- Stay active and keep a healthy weight.
- Measure or check the arm regularly; report heaviness, tightness or swelling early.
- Wear the compression sleeve if prescribed, especially for long travel or heavy work.

15 Prognosis and discharge criteria

Prognosis for shoulder function is good when exercise starts early. Lymphoedema is a lifelong risk, and once established it can be controlled but rarely cured. Cancer prognosis depends on stage and tumour type.

Discharge from physio: full functional shoulder range; independent home programme; knows arm care and lymphoedema warning signs; compression garment fitted if needed; back to work and household roles.

16 Exam corner

Past questions (MUHS, 2016–2026)

W16 Q3(b), 5 + 5 + 5 marks

45-year-old female farm labourer, right MRM 2 days ago; one of the earning members; wants to return to farm work. Functional diagnosis, short- and long-term management.

S17 Q4(b), 5 + 7 + 3 marks

45-year-old female teacher, married with 2 children, radical mastectomy 15 days ago; mild oedema of the operated arm and shoulder restriction. ICF; long-term management; prosthesis.

S19 Q4(b), 15 marks

40-year-old Sunita, right radical mastectomy 3 days ago; right-handed; cooks tiffin boxes for ten clients morning and evening. Immediate and late complications and prevention; rehab with reasoning; why and how to care for the arm lifelong.

W20 Q3(b), 5 + 5 + 5 marks

The worked example in section 10.

W21 Q3(b), 15 marks

40-year-old woman, right MRM 2 days ago; neck pain and heaviness of the right arm, VAS 6/10; her sleeves look tight. Subjective and objective evaluation parameters; functional diagnosis; management.

W24 Q3(b), 15 marks

55-year-old cook, right-hand dominant, right MRM 2 days ago. Functional diagnosis and physiotherapy management.

7-mark questions

Physiotherapy after radical mastectomy (S16, S18). Complications after MRM (S20). Physiotherapy for a 33-year-old receptionist with lymphoedema (S21). Physiotherapy after MRM (W19, S26). Rationale of physiotherapy after MRM (S25, W25).

3-mark questions

Precautions after radical mastectomy (S21). Three goals of physiotherapy in radical mastectomy (W22). Three types of bandages used for lymphoedema (S23).

Technique cards for this topic: breathing exercises (p. 275), exercise physiology (p. 313), swelling and posture (p. 298).

Keywords to underline

MRM axillary clearance drains 90° until drains out full ROM axillary web syndrome lymphoedema complete decongestive therapy MLD compression progressive resistance training prosthesis lifelong arm care

Answer skeleton: mastectomy long answer

1. Restate: age, side and hand dominance, type of surgery and day, chemo or radiotherapy, job and home load.
2. **[ICF, 5]** ICF diagram; one-line diagnosis; reasons (guarding, nodes removed, chemotherapy).
3. **[Short-term, 5]** By day: breathing, hand pumping, shoulder to 90° with drains, posture; then full ROM after drains out.
4. **[Long-term, 5–7]** Stretching, scar care, strengthening, aerobic exercise, lymphoedema prevention and decongestive therapy, prosthesis, return to work.
5. **[Add-on, 3]** Prosthesis, precautions or lifelong arm care, as asked.

Draw these

- The rehabilitation timeline (section 12)
- An arm outline with measuring points every 4 cm (for lymphoedema)

Viva questions

1. What is the difference between radical and modified radical mastectomy?
2. Why is shoulder elevation limited in the first week?
3. Which nerve injury causes winging after mastectomy?
4. What are the stages of lymphoedema?
5. What are the components of complete decongestive therapy?
6. Is weight training safe after breast cancer surgery?

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Three goals of physiotherapy in radical mastectomy.**

Answer

1. Prevent chest complications and restore full, pain-free shoulder range.
2. Prevent and manage lymphoedema.
3. Restore strength, posture and function for return to work and daily roles.

7 m **Complications after modified radical mastectomy.**

Answer

Immediate: pain, chest complications, seroma, haematoma, infection, flap necrosis, DVT. Late: shoulder stiffness, axillary web syndrome, lymphoedema, cellulitis, nerve injury (winging, numbness), postural changes, radiation fibrosis, fatigue, psychological distress. Give prevention for each.

LAQ **S19: why must Sunita care for her arm lifelong, and how?**

Answer

Why: axillary nodes are gone, so drainage capacity is permanently reduced; any extra load can trigger lymphoedema. How: gloves and oven mitts while cooking, immediate care of cuts and burns, no tight bangles or sleeves, gradual build-up of heavy cooking duties with breaks, moisturise, exercise, healthy weight, report swelling early.

MCQ **Day 3 after MRM, drains in place. Which exercise is most appropriate?**

A. Full overhead pulley stretches

B. Hand pumping and shoulder flexion to about 90°

C. Heavy resistance shoulder press

D. No movement until drains are removed

Answer

B. Gentle early movement prevents stiffness; full elevation waits until the drains are out to reduce seroma.

MCQ **The first phase of complete decongestive therapy includes:**

A. Hot packs and massage

B. MLD, multilayer short-stretch bandaging, exercise and skin care

C. Only a compression sleeve

D. Complete rest of the arm

Answer

B. The intensive phase reduces volume; a compression sleeve is used in the maintenance phase.

MCQ **Tight cords from the axilla down the inner arm 3 weeks after surgery are most likely:**

A. DVT

B. Axillary web syndrome

C. Stage III lymphoedema

D. Long thoracic nerve palsy

Answer

B. Cording comes from damaged lymph vessels and responds to gentle stretching.

Last recall check. Write the timeline, lymphoedema stages, decongestive therapy and lifelong arm care. Then check them against this page.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

1. MUHS BPTh syllabus (2012), Cardio-Vascular & Respiratory Physiotherapy. muhs.ac.in
2. MUHS IV BPTh question papers, code 51413, Summer 2016 to Summer 2026: muhsonline.com (unofficial archive, 2016–2022) and the Dhole Patil College of Physiotherapy past-paper page (2019–2026)
3. McNeely ML, et al. Exercise interventions for upper-limb dysfunction due to breast cancer treatment. Cochrane Review 2010.
4. International Society of Lymphology. The diagnosis and treatment of peripheral lymphedema: 2020 consensus document. *Lymphology* 2020.
5. Campbell KL, et al. Exercise Guidelines for Cancer Survivors: Consensus Statement from International Multidisciplinary Roundtable. *Med Sci Sports Exerc* 2019;51:2375–90. PMC
6. Schmitz KH, et al. Weight lifting in women with breast-cancer-related lymphedema (PAL trial). *N Engl J Med* 2009.

7. Lymphoedema Framework. Best Practice for the Management of Lymphoedema. International consensus. London: MEP Ltd, 2006 (compression bandaging).

8. Structure and classical content: Cash's Textbook of General Medical and Surgical Conditions for Physiotherapists; Main & Denehy (2016); Hillegass (2022).

Peripheral arterial disease, Buerger's disease and other arterial disorders

> Arterial disease was in 18 of 20 papers (2016–2026), mostly as 3-mark tests (ABI, special tests) and 7-mark questions; ABI alone was asked in S23, S24, W25 and S26. The long answers describe a smoker whose calf pain stops him walking; the answer is ICF plus a dosed walking programme.

Syllabus	Ch 6(a)(viii) Other arterial disorders; arterial ulcers from 6(h)
Priority	**Tier 1.** In 18 of 20 papers (2016–2026). 3 long-answer cases (none since 2022) and many short questions.
Asked as	LAQ S16 · W17 · W22 · 7m S17 · W17 · S19 · S22 · W24 · S25 · 3m W16 · S17 · S18 · W18 · S20 · W20 · S21 · S23 · S24 · W25 · S26
Usual split	5 + 5 + 5, 7 + 8 (functional diagnosis + management), or 5 + 3 + 7 (impairments, complications, management)
Checked against	2024 ACC/AHA multisociety guideline on lower-extremity PAD; AHA and SVS exercise recommendations; 2023 ESC-led consensus on exercise in PAD; Shionoya and Olin criteria for Buerger's disease. Full list under Sources. Content checked 19 October 2026.

1 Definition

Peripheral arterial disease (PAD) is narrowing or blockage of the arteries supplying the limbs, usually the legs, so the muscles and skin get too little blood. The cause is almost always atherosclerosis.

Intermittent claudication is cramping pain in a muscle group, usually the calf, that comes on with walking a fairly fixed distance and goes away within a few minutes of rest. It is the main symptom of PAD.

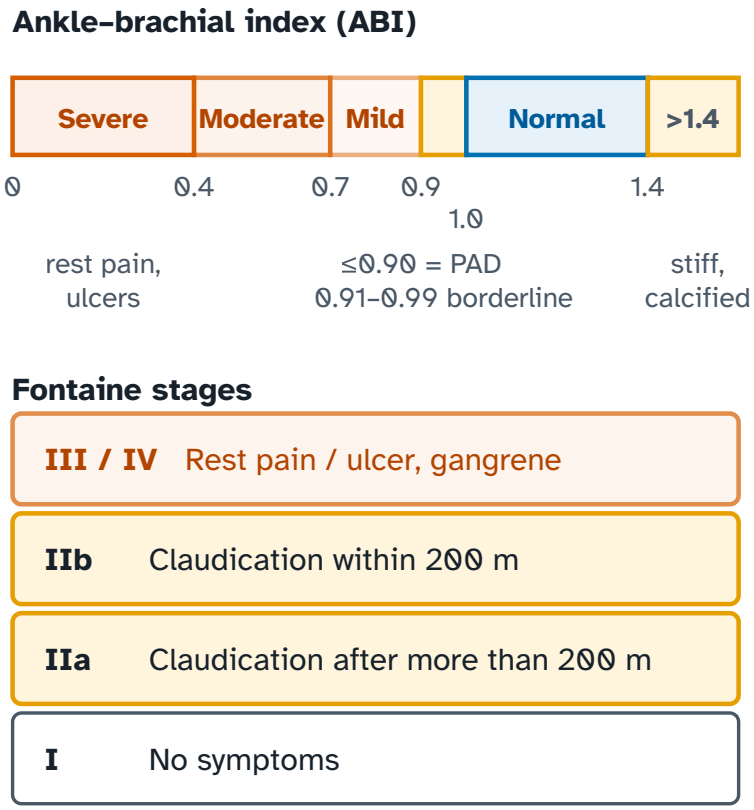
Buerger's disease (thromboangiitis obliterans) is an inflammatory, non-atherosclerotic disease of the small and medium arteries and veins of the hands and feet. It affects young smokers.

2 Classification and grading

Clinical presentation

- **Asymptomatic** PAD: found only on testing.
- **Chronic symptomatic** PAD: intermittent claudication.
- **Chronic limb-threatening ischaemia (CLTI):** rest pain, non-healing ulcer or gangrene.
- **Acute limb ischaemia:** sudden loss of blood supply from an embolus or thrombosis. This is a surgical emergency.

ABI number line and Fontaine stages



Severity climbs upwards. Stages III and IV are chronic limb-threatening ischaemia.

ABI interpretation (3 marks in S17, W18, S23, S24, W25 and S26)

ABI	Meaning
Above 1.40	Non-compressible, calcified arteries (common in diabetes and kidney disease). Result unreliable; measure the toe–brachial index (0.70 or less = PAD).
1.00–1.40	Normal
0.91–0.99	Borderline
0.90 or less	**PAD.** Textbooks commonly grade severity as 0.70–0.90 mild, 0.40–0.69 moderate, and below 0.40 severe (rest pain and ulcers likely).

Fontaine and Rutherford

Fontaine	Symptoms	Rutherford category
I	Asymptomatic	0
IIa	Mild claudication, walks more than about 200 m	1
IIb	Moderate to severe claudication, less than about 200 m	2–3
III	Ischaemic rest pain	4
IV	Ulceration or gangrene	5–6

Other arterial disorders in the syllabus

- **Buerger's disease** (sections 3–5 below).
- **Raynaud's phenomenon:** fingers turn white, then blue, then red on exposure to cold or stress, because small arteries spasm. Primary (no cause) or secondary (for example to scleroderma).
- **Acute limb ischaemia:** embolus (often from atrial fibrillation) or thrombosis on an existing plaque.

- **Aneurysm:** a balloon-like widening of an artery, most often the abdominal aorta or popliteal artery. It can rupture or throw clots.

3 Causes and risk factors

Atherosclerotic PAD

- **Smoking** (cigarettes and bidi): the strongest risk factor.
- **Diabetes:** disease is more distal (below the knee) and often combined with neuropathy.
- Age over 50–65, hypertension, abnormal lipids, chronic kidney disease, family history.

Buerger's disease: diagnostic criteria

Shionoya (1998)	Olin (2000)
• Smoking history • Onset before age 50 • Blocks in the arteries below the knee • Arm involvement or migrating superficial phlebitis • No atherosclerotic risk factors other than smoking	• Age under 45 • Current or recent tobacco use • Distal limb ischaemia (claudication, rest pain, ulcers or gangrene) confirmed by tests • Autoimmune disease, clotting disorders and diabetes ruled out • No source of emboli; typical angiogram

Exam point

The S16 and W17 case (a 25-year-old male chronic smoker with claudication and cold feet with sweating) is Buerger's disease. Say so in the first line of your answer.

4 Pathophysiology to impairments

For "impairments with clinical reasoning" (W22), explain why walking hurts and why exercise helps.

1. Atherosclerotic plaque narrows the artery. At rest, blood flow through the narrowing is still enough.
2. Walking raises the calf muscles' oxygen demand, but the narrowed artery cannot raise its supply.
3. The muscle becomes ischaemic, produces lactic acid and other metabolites, and pain starts: **claudication**. Resting lowers the demand, so the pain goes.
4. The patient walks less, so muscles weaken and fitness falls. The distance he can walk shrinks further.
5. When the narrowing becomes severe, even resting flow is not enough: **rest pain**, then skin breakdown, ulcers and gangrene.

Where the block is decides where the pain is: aorto-iliac disease gives buttock and thigh pain; femoro-popliteal disease (the commonest site, and the S19 popliteal case) gives calf pain; tibial disease gives foot pain.

? Why does walking exercise help if it causes pain?

> Repeated walking to moderate pain improves the muscle's ability to use oxygen, opens up collateral vessels around the block, improves the lining of the arteries, and makes walking more efficient. The block itself does not go away, but the patient walks much further before pain starts.

? Why is rest pain worse at night and better with the leg hanging down?

> Lying flat removes the help of gravity, and the heart pumps a little less during sleep. Hanging the leg down uses gravity to push more blood into the foot.

Recall check. Explain claudication in three sentences: what causes the pain, why rest relieves it, and why exercise helps.

5 Clinical features

Symptoms

- Calf, thigh or buttock cramping pain on walking, relieved by standing still within a few minutes. The W22 head clerk gets pain after 5 minutes and stops at 10 minutes.
- Rest pain in the toes and forefoot at night, relieved by hanging the leg over the bed.
- Cold feet, numbness, slow-healing wounds.

Signs

- Cool, pale skin; thin, shiny, hairless skin; thick, brittle nails; wasted calf (W22: "legs appear thin, skin dry").
- Weak or absent pulses below the block.
- Pallor when the leg is raised and dusky red colour (dependent rubor) when it hangs down.
- Delayed capillary refill; ulcers on the toes, heel or pressure points; gangrene.

Buerger's disease: extra features

- Young male smoker; starts in the feet and often the hands too.
- Foot (instep) claudication, cold sensitive fingers and toes, Raynaud's-like colour changes.
- Migrating red tender lines along superficial veins (superficial thrombophlebitis).
- Painful ulcers and gangrene of the fingertips and toes.

> **Acute limb ischaemia: the 6 P's (emergency)**
>
> **P**ain, **P**allor, **P**ulselessness, **P**araesthesia, **P**aralysis, **P**erishing cold. Stop and call the doctor at once.

Don't confuse: vascular and neurogenic claudication

Feature	Vascular (PAD)	Neurogenic (spinal canal stenosis)
Pain comes on	After a fixed walking distance	On walking or standing; distance varies
Relieved by	Standing still for a few minutes	Sitting or bending forward; takes longer
Uphill or cycling	Worse	Uphill or cycling (spine flexed) often easier
Pulses	Weak or absent	Normal

6 Investigations and special tests

"Three special tests to assess arterial insufficiency" was asked three times (W16, S18, W20). ABI was asked three more times.

Ankle–brachial index: how to do it

1. Patient lies flat and rests for 5–10 minutes.
2. Measure systolic pressure in both arms with a BP cuff and hand-held Doppler. Take the higher one.
3. Measure systolic pressure at the dorsalis pedis and posterior tibial arteries of each ankle, cuff just above the ankle. Take the higher of the two for each leg.
4. **ABI = ankle systolic pressure ÷ higher arm systolic pressure**, for each leg.
5. Interpret with the table in section 2.

Bedside special tests for arterial insufficiency

Test	How	Positive result
Pulse palpation	Feel femoral, popliteal, dorsalis pedis and posterior tibial pulses on both sides	Weak or absent. Grade 0 absent, 1+ weak, 2+ normal, 3+ full, 4+ bounding.
Buerger's test	Lying, raise the leg slowly; note the angle at which the foot turns pale (Buerger's angle). Then sit the patient up with the legs hanging down.	Pallor at a low angle (below about 20° means severe ischaemia); then slow, dusky red colour on hanging (dependent rubor)
Capillary refill time	Press the nail bed or pulp for a few seconds and release	Colour takes longer than about 2 seconds to return
Venous filling time	Raise the leg for about 1 minute, then hang it down; time how long the foot veins take to refill	Refilling takes longer than about 20 seconds
Treadmill walking test	Walk at a set speed and grade; record when pain starts and when the patient must stop	Pain-free (initial claudication) distance and maximum walking distance; used to measure progress
Allen's test (hand)	Press both radial and ulnar arteries, patient clenches, then release one	Hand stays pale when one artery is released: that artery is blocked (useful in Buerger's)

Other investigations

- **Doppler ultrasound (duplex):** shows the site and degree of the block.
- **Toe–brachial index** when the ABI is over 1.40.
- **CT or magnetic resonance angiography and conventional angiography** before angioplasty or bypass.
- **Blood tests:** blood sugar and HbA1c, lipids, kidney function.

7 Medical and surgical management

- **Stop smoking completely.** For Buerger's disease this is the only treatment proven to halt the disease; even a few cigarettes or bidis a day keep it active.
- Antiplatelet drug, statin, BP and diabetes control to cut the high risk of heart attack and stroke.
- **Cilostazol** can reduce claudication symptoms.
- **Supervised exercise therapy** is first-line treatment for claudication (section 12).
- **Revascularisation** when claudication still limits life after exercise therapy, or for limb-threatening ischaemia: angioplasty with or without a stent, or a bypass graft (for example femoro-popliteal bypass).
- Wound care, infection control, and amputation if the limb cannot be saved (see the amputation page).
- Acute limb ischaemia: urgent anticoagulation and removal of the clot (embolectomy) or bypass.

8 Complications and prevention

W22 asks for "complications which he is likely to face giving reason, along with their preventive measures".

Complication	Reason	Prevention
Arterial ulcer	Poor blood supply; minor injury cannot heal	Daily foot inspection, proper footwear, no barefoot walking, nail care
Infection and gangrene	Ischaemic tissue dies; diabetes worsens infection	Early treatment of any wound; blood-sugar control
Amputation	Non-healing gangrene or CLTI	Stop smoking, exercise, foot care, early referral
Heart attack and stroke	The same atherosclerosis affects the heart and brain	Risk-factor control, regular exercise
Deconditioning, weight gain, low mood	Pain makes him avoid walking	Graded walking programme, education

Complication	Reason	Prevention
Burns and pressure injury	Numb, ischaemic skin cannot sense heat or pressure	No hot-water bottles or heating pads on the feet; padded heels in bed
Neuropathic ulcer (with diabetes)	Loss of protective sensation	Monofilament check, offloading footwear

9 PT assessment

S21 asked to "enumerate evaluation parameters for peripheral arterial disorders". This list is the answer.

- **History:** claudication site, onset distance or time, time to stop, recovery time; rest pain; smoking (pack-years, bidi); diabetes, HTN, heart disease; occupation, walking demands, stairs at home.
- **Inspection:** colour, hair loss, nails, muscle wasting, ulcers and gangrene (site, size, base, edge), oedema.
- **Palpation:** skin temperature (compare sides), pulses with grades.
- **Special tests:** ABI, Buerger's test, capillary refill, venous filling time.
- **Sensation:** 10 g monofilament if diabetic.
- **ROM and strength:** ankle dorsiflexion and calf strength; joint stiffness from inactivity.
- **Walking capacity:** treadmill test or 6-minute walk test with pain-free and maximum distance; claudication pain scale.
- **Questionnaires:** Walking Impairment Questionnaire; quality of life (VascuQoL).

10 PT diagnosis by ICF

Draw the six-box ICF model (see the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (W22, 5 + 3 + 7 marks)

> "A fifty year old male head clerk, chain smoker, non-vegetarian, diabetic has come with chief complaint of pain on walking for five minutes and is unable to continue walking after 10 minutes. His legs appear thin & leg skin dry. Discuss the clinical reasoning for the impairments, state the complications which he is likely to face giving reason along with their preventive measures if any. Discuss the management in details."

Spot the ICF clues before you write

pain on walking for five minutes is intermittent claudication with a pain-free walking time of 5 minutes. chain smoker and diabetic are personal factors and risk factors. thin legs, dry skin are trophic changes of chronic ischaemia. head clerk means a sitting job, so walking to work and around the office matters.

ICF component	From this case	Clinical reasoning
Health condition	Peripheral arterial disease with intermittent claudication (Fontaine II), diabetes	Atherosclerosis worsened by smoking and diabetes
Body function and structure impairments	• Claudication pain after 5 minutes of walking • Reduced walking endurance (stops at 10 minutes) • Calf muscle wasting and weakness • Trophic skin changes: dry, thin skin • Possible reduced sensation (diabetes) • Reduced cardiorespiratory fitness	• Narrowed artery cannot meet exercise demand, so ischaemic pain • Pain-avoidance and chronic ischaemia waste the muscle • Poor blood supply starves the skin
Activity limitations	Walking more than about 10 minutes, climbing stairs, shopping, walking to the bus or office	Pain forces him to stop

ICF component	From this case	Clinical reasoning
Participation restrictions	Commuting to work, social outings, family errands, leisure walks	Limited walking distance restricts life outside home
Environmental factors	**Barriers:** easy access to cigarettes; non-vegetarian, possibly fatty diet at home. **Facilitators:** desk job, family support, access to a hospital programme.	Barriers become education goals
Personal factors	50-year-old man, chain smoker, diabetic	Readiness to quit smoking decides prognosis

One-line functional diagnosis

A 50-year-old diabetic head clerk and chain smoker with peripheral arterial disease, who has calf claudication after 5 minutes of walking, reduced walking endurance, calf wasting and trophic skin changes. These limit walking, stairs and outings, and restrict commuting and social life. Smoking and diet are barriers; a desk job and family support are facilitators.

Recall check. Fill the ICF boxes for the 25-year-old smoker with claudication and cold sweaty feet (S16, W17).

11 Problem list and goals

Goals for the W22 head clerk

Problem	Short-term (4 weeks)	Long-term (12 weeks and beyond)
Claudication	Understands the walking programme and claudication pain scale; walks within it 3 times a week	Pain-free walking time and maximum walking time clearly increased
Muscle weakness	Better calf endurance (heel raises)	Normal calf strength for his age
Skin and feet	Inspects his feet daily; no new wounds	No ulcers; good foot-care habits
Risk factors	Sets a quit date; referred for cessation support	Tobacco-free; HbA1c, BP and lipids controlled
Participation	Walks to the bus stop with planned rest stops	Commutes and does errands without stopping; walks for leisure

12 PT management

1. Supervised exercise therapy (the core treatment)

Memory hook: the walking prescription

30 / 3 / 12

- **30**–45 minutes of walking per session (build up to it)
- **3** sessions or more a week
- **12** weeks at least

Checked against the AHA exercise recommendations and the 2023 ESC-led consensus. The 2024 ACC/AHA guideline gives supervised exercise Class 1, Level A.

Interval walking programme

Part	Dose
Warm-up	**5–10 min** slow walking and calf stretches

Part	Dose
Walking bouts	Treadmill or level ground. Walk until claudication pain reaches **moderate to strong** (about 2–3 on the 4-point claudication scale), then stop.
Rest	Stand or sit until the pain has fully gone, then walk again
Total	Repeat walk–rest cycles to reach **30–45 min** of exercise time, **≥3 times a week**, for **≥12 weeks**
Progression	When he can walk about 8–10 minutes before moderate pain, raise the treadmill speed or slope
Cool-down	**5–10 min** slow walking and stretches

Claudication pain scale: 1 = pain starts, mild; 2 = moderate; 3 = intense; 4 = unbearable. Write "walk to moderate pain (grade 2–3), rest until it goes, repeat" in the exam.

- **If walking is not possible** (foot ulcer, arthritis): arm cranking or cycling also improve walking ability (2024 guideline, Class 2a).
- **If supervision is not available:** a structured home walking programme with a log book, at least 30 minutes, 3–5 times a week.
- **Add:** calf and thigh strengthening (heel raises, sit-to-stand), and stretching of the calf muscles.

2. Buerger's disease (7 marks in S17 and S22)

- **Absolute stop of all tobacco**, including bidi, chewing tobacco and passive smoke.
- Walking programme as above, within tolerance; foot (instep) muscle and calf exercises.
- Buerger–Allen exercises (classic, below) and position changes to aid circulation.
- Keep hands and feet warm with socks and gloves; avoid cold exposure, but never apply direct heat.
- Meticulous foot and hand care; protect ulcers; manage pain with positioning (leg slightly down at night).
- After amputation of digits or limb: stump care and prosthetic training.

Classic answer vs current evidence: Buerger–Allen exercises

Classic	Current	Write in the exam
1. Lying, raise the leg to about 45–90° until the skin blanches (about 1–2 minutes). 2. Sit with the leg hanging down until it turns red (about 2–5 minutes), with ankle movements. 3. Lie flat with the leg level for about 5 minutes. 4. Repeat the cycle several times, a few times a day.	There is little good evidence for Buerger–Allen exercises. **Supervised walking** to moderate claudication pain has the strongest evidence (Class 1, Level A) and should be the main treatment whenever the patient can walk.	"Supervised interval walking 30–45 min, ≥3 times a week, ≥12 weeks, is the main treatment; Buerger–Allen exercises may be added."

3. Rest pain, ulcers and limb-threatening ischaemia

- Sleep with the head of the bed slightly raised or the leg slightly lower; do not elevate an ischaemic leg.
- Protect heels with padding; turn regularly.
- Gentle active ankle and toe exercises within comfort; no walking on an open ulcer without offloading footwear.
- After angioplasty or bypass: early walking as the surgeon allows; protect groin wounds; then a walking programme.

By impairment (for "management in detail")

Impairment	Technique and dose	Why
Claudication, low walking tolerance	Interval walking to moderate pain (grade 2–3), rest, repeat; 30–45 min, ≥3 times a week, ≥12 weeks	Improves muscle oxygen use, collaterals and walking economy
Calf weakness and wasting	Heel raises, resisted plantarflexion, sit-to-stand: 2–3 sets of 10–15, 2–3 times a week	Stronger muscles do the same work with less blood
Reduced ankle range	Active ankle and toe ROM, calf stretches 30 s × 3	Keeps a normal gait pattern
Poor skin, ulcer risk	Daily foot inspection, moisturiser (not between toes), proper footwear	Prevents the small wounds that turn into ulcers
Smoking, diabetes	Brief advice at every session; referral to cessation services; exercise for sugar control	Removes the cause of progression

Recall check. Write the "30 / 3 / 12" walking prescription and the claudication pain scale from memory.

13 Precautions, contraindications and stop criteria

Take care

- **No heat** (hot packs, shortwave, hot water) on an ischaemic limb: blood flow cannot carry the heat away and the skin burns.
- **No compression bandaging** if the ABI is low; check ABI before any compression.
- Do not elevate the leg in rest pain.
- Most patients have heart disease too: check HR and BP, and watch for chest pain during walking.
- Diabetics: check blood sugar and feet before and after exercise.
- Do not walk on an open foot ulcer without offloading.

Stop and refer urgently if

- Any of the 6 P's appear suddenly (acute limb ischaemia)
- Chest pain, breathlessness or palpitations during exercise
- New rest pain, a new ulcer, or the foot turns black

14 Education, lifestyle and home programme

Foot care

- Look at your feet every day, including the soles and between the toes. Use a mirror or ask a family member.
- Wash in lukewarm water, test the temperature with your elbow, dry well between the toes.
- Cut nails straight across. Do not cut corns yourself.
- Always wear soft, well-fitting footwear and socks. Never walk barefoot, even at home or in temples.
- No hot-water bottles, heating pads or fire near the feet.
- Report any colour change, wound or pain at rest at once.

Lifestyle

- Stop all tobacco: this is the most important single step.
- Keep walking every day using the walk–rest method.
- Heart-healthy diet: less fried and fatty food, less salt; control weight.

- Take medicines regularly: antiplatelet, statin, BP and diabetes drugs.
- Avoid cold; keep hands and feet warm with socks and gloves.

15 Prognosis and discharge criteria

Prognosis: most people with claudication do not lose their leg, but they have a high risk of heart attack and stroke. Walking distance improves greatly with supervised exercise. Continued smoking is the strongest predictor of worsening disease and, in Buerger's disease, of amputation.

Discharge from the supervised programme: walks independently with the walk–rest method; can measure progress with a walking log; does daily foot care; has stopped or is actively stopping smoking; knows the warning signs.

16 Exam corner

Past questions (MUHS, 2016–2026)

S16 Q4(b), 5 + 5 + 5 marks; W17 Q4(b), 7 + 8 marks

25-year-old male chronic smoker with intermittent claudication and cold feet with sweating. Functional diagnosis and physiotherapy management.

W22 Q3(a), 5 + 3 + 7 marks

The worked example in section 10.

7-mark questions

Physiotherapy in Buerger's disease (S17, S22). Physiotherapy for claudication in arterial disease of the popliteal artery (S19). ABI and its significance (W17). Physiotherapy for peripheral arterial disorders (W24). Assessment and physiotherapy of PAD (S25).

3-mark questions

ABI (S17, W18). Three special tests for arterial insufficiency (W16, S18, W20). Intermittent claudication (S20). Evaluation parameters for peripheral arterial disorders (S21). Normal and abnormal grades of ABI (S23). Grades of ABI (S24). ABI (W25). Interpretation of ABI (S26).

Technique cards for this topic: exercise testing (p. 308), scales (p. 322), exercise physiology (p. 313), therapeutic exercise for circulation (p. 298), bedside vascular tests (p. 317).

Keywords to underline

intermittent claudication ABI ≤ 0.90 Buerger's test Fontaine supervised exercise therapy walk to moderate pain, rest, repeat 30–45 min, 3 times a week, 12 weeks collateral circulation stop smoking foot care no heat

Answer skeleton: claudication long answer

1. Name the condition from the case (PAD, or Buerger's disease in a young smoker) and its stage (Fontaine).
2. **[ICF or impairments, 5–7]** ICF diagram from the case; reasoning for claudication, wasting and skin changes.
3. **[Complications, 3]** if asked: ulcer, gangrene, amputation, heart attack and stroke, with prevention.
4. **[Management, 7–8]** Assessment tests (ABI, Buerger's); supervised walking with dose; strengthening; foot care; smoking cessation; precautions (no heat); progression and outcome measures.

Draw these

- The ABI number line and Fontaine ladder (section 2)
- A table of arterial vs venous ulcers (see the venous page)

Viva questions

1. How do you calculate the ABI, and what does 0.6 mean?
2. Why can the ABI be falsely high in a diabetic?
3. What is Buerger's angle?
4. How is vascular claudication different from neurogenic claudication?
5. Why must you not apply heat to an ischaemic leg?
6. What are the 6 P's?

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Three special tests to assess arterial insufficiency.**

Answer

1. **ABI:** ankle systolic ÷ higher arm systolic; ≤ 0.90 indicates PAD.
2. **Buerger's test:** pallor on raising the leg at a low angle, then dependent rubor on hanging it down.
3. **Capillary refill time:** longer than about 2 seconds. (Also accept pulse palpation or venous filling time over 20 seconds.)

3 m **Ankle–brachial index.**

Answer

Ratio of ankle to arm systolic pressure, measured with a Doppler after 5–10 minutes' rest. Normal 1.00–1.40; borderline 0.91–0.99; PAD ≤ 0.90 (severe below 0.40); over 1.40 means calcified arteries, so use the toe–brachial index.

7 m **Physiotherapy management in Buerger's disease.**

Answer

Define (inflammatory disease of small and medium vessels, young smokers). Aims. Then: absolute tobacco cessation; supervised interval walking with dose; Buerger–Allen exercises (classic); warmth without direct heat; foot and hand care; positioning for rest pain; ulcer protection; post-amputation care; education.

MCQ **A 55-year-old man's ABI is 1.6. What is the best next step?**

A. Reassure: normal arteries

B. Measure the toe–brachial index

C. Start compression bandaging

D. Diagnose severe PAD

Answer

B. An ABI above 1.40 means the ankle arteries are stiff and cannot be compressed, so the result is unreliable. The toe–brachial index (0.70 or less = PAD) is used instead.

MCQ **During supervised walking for claudication, the patient should:**

A. Stop at the first hint of pain

B. Walk until moderate-to-strong pain, rest until it goes, then repeat

C. Walk through unbearable pain

D. Avoid walking and use only Buerger–Allen exercises

Answer

B. Training to moderate claudication pain gives the best gains. Unbearable pain is unnecessary, and stopping at the first hint gives too little stimulus.

MCQ **Which is a contraindication in a patient with severe PAD?**

A. Calf strengthening

B. A hot pack on the calf

C. Daily foot inspection

D. Arm ergometry

Answer

B. Ischaemic tissue cannot get rid of heat, so it burns easily.

Last recall check. Write the ABI method and bands, three special tests, and the walking prescription.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

1. MUHS BPTh syllabus (2012), Cardio-Vascular & Respiratory Physiotherapy. muhs.ac.in
2. MUHS IV BPTh question papers, code 51413, Summer 2016 to Summer 2026: muhsonline.com (unofficial archive, 2016–2022) and the Dhole Patil College of Physiotherapy past-paper page (2019–2026)
3. Gornik HL, et al. 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VESS Guideline for the Management of Lower Extremity Peripheral Artery Disease. *J Am Coll Cardiol* 2024. Key points: acc.org; ABI thresholds as summarised in *American Family Physician* 2025.
4. Treat-Jacobson D, et al. Optimal Exercise Programs for Patients With Peripheral Artery Disease: AHA Scientific Statement. *Circulation* 2019.
5. Society for Vascular Surgery practice guidelines for asymptomatic PAD and intermittent claudication (2015).
6. ESC-led consensus document on exercise in PAD, *Eur Heart J* 2023.
7. Shionoya S. Diagnostic criteria of Buerger's disease. *Int J Cardiol* 1998. Olin JW. Thromboangiitis obliterans (Buerger's disease). *N Engl J Med* 2000.
8. Structure and classical content: Hillegass, *Essentials of Cardiopulmonary Physical Therapy*, 5th ed. (2022); Cash's Textbook of Chest, Heart and Vascular Disorders for Physiotherapists.

COPD: chronic bronchitis and emphysema

> COPD was in 13 of 20 papers (2016–2026). The long cases (2016–2020) describe a long-time smoker, breathless on daily activities, living upstairs with no lift; newer papers ask short questions on PFT, X-ray, ABG, hypoxic drive and leg-muscle training. The answer is the same each time: ICF from the story, then acute-stage and long-term management with pulmonary rehab doses.

Syllabus	Ch 6(b)(i–ii) Bronchitis, emphysema; pulmonary rehabilitation from Ch 8
Priority	**Tier 1.** In 13 of 20 papers (2016–2026). 5 long-answer cases and many short questions.
Asked as	LAQ W16 · S17 · W17 · S18 · S20 · 7m S19 · W24 · 3m S17 · W18 · S20 · W20 · W22 ×2 · W19 · W21 · W24 ×2 · W25
Usual split	5 + 5 + 5 (functional diagnosis or ICF, short-term, long-term), or 5 + 3 + 7
Checked against	GOLD 2026 report; ATS pulmonary rehabilitation guideline (2023); ATS home oxygen guideline (2020); BTS emergency oxygen guideline; BTS/ACPRC physiotherapy guideline (2009). Full list under Sources. Content checked 19 October 2026.

1 Definition

Chronic obstructive pulmonary disease (COPD) is a common, preventable lung condition with long-lasting breathlessness, cough and sputum, caused by damage to the airways (bronchitis, bronchiolitis) and the alveoli (emphysema). The airflow limitation is persistent and usually progressive.

The diagnosis needs spirometry: a **post-bronchodilator FEV1/FVC below 0.70**.

- **Chronic bronchitis** (clinical definition): cough with sputum on most days for at least 3 months a year, for 2 years in a row, with other causes excluded.
- **Emphysema** (structural definition): permanent enlargement of the air spaces beyond the terminal bronchioles, with destruction of their walls.

2 Classification and grading

GOLD spirometric grade (post-bronchodilator FEV1, when FEV1/FVC < 0.70)

Grade	Severity	FEV1, % predicted
GOLD 1	Mild	80 or more
GOLD 2	Moderate	50–79
GOLD 3	Severe	30–49
GOLD 4	Very severe	Below 30

GOLD A, B, E groups (symptoms and exacerbations)

- **Group A:** few symptoms (mMRC 0–1 or CAAT below 10), no exacerbation in the last year.
- **Group B:** more symptoms (mMRC 2 or more, or CAAT 10 or more), no exacerbation in the last year.
- **Group E:** exacerbations. GOLD 2026 puts a patient in group E after even **one moderate or severe exacerbation** in the past year. Older editions needed two moderate ones, or one needing hospital admission.

GOLD 2026 renamed the COPD Assessment Test as the Chronic Airways Assessment Test (**CAAT**). It still has 8 items scored 0–40.

Modified MRC dyspnoea scale

Grade	Breathless…
0	only with strenuous exercise
1	when hurrying on the level or walking up a slight hill
2	walks slower than people of the same age on the level, or has to stop for breath at own pace
3	stops for breath after about 100 m or a few minutes on the level
4	too breathless to leave the house, or when dressing

Don't confuse: chronic bronchitis and emphysema (the classic picture)

Feature	Chronic bronchitis ("blue bloater")	Emphysema ("pink puffer")
Main problem	Airway inflammation and mucus	Destroyed alveoli, loss of elastic recoil
Cough and sputum	Marked, productive	Little
Breathlessness	Moderate	Severe, early
Body build	Often overweight, cyanosed, ankle oedema (cor pulmonale)	Thin, pursed-lip breathing, barrel chest, uses accessory muscles
Blood gases	Low oxygen, high CO_2 early	Oxygen kept near normal until late

Most patients have a mix of both. The labels "pink puffer" and "blue bloater" are stigmatising and best used only to explain the classic textbook picture.

3 Causes and risk factors

- **Tobacco smoking:** cigarettes, bidi, hookah. Risk rises with pack-years (packs a day × years smoked).
- **Household air pollution:** cooking on wood, dung or crop waste in a poorly ventilated kitchen. A major cause in Indian women who have never smoked.
- **Occupational dust and fumes:** sweepers, farm workers, miners, textile and construction workers (the W17 sweeper, S17 labourer, S20 farmer).
- Outdoor air pollution.
- **Past tuberculosis**, repeated childhood chest infections, poor lung growth.
- **Alpha-1 antitrypsin deficiency:** a rare genetic cause of early emphysema.

Effects of smoking on pulmonary function (3 marks in S17 and W22)

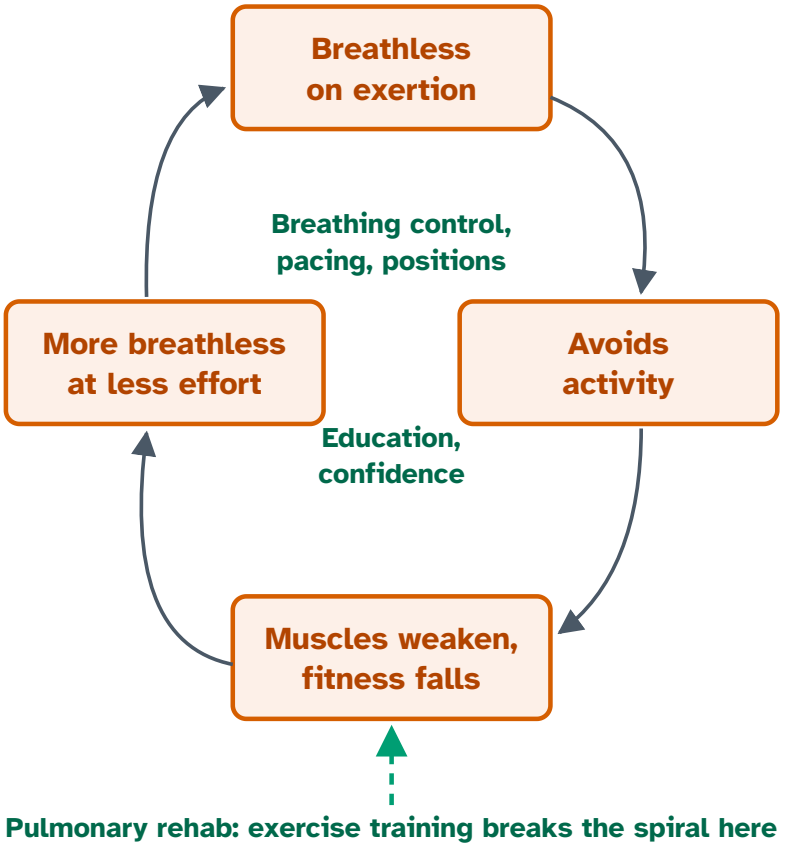
- Paralyses and destroys cilia, so mucus clearance fails.
- Enlarges mucus glands and goblet cells, so sputum increases.
- Inflames the airways, narrowing them and making them more reactive.
- Releases enzymes (proteases) that destroy alveolar walls: emphysema.
- FEV1 falls faster than with normal ageing; diffusing capacity falls.
- Carbon monoxide binds haemoglobin and lowers oxygen carriage; infections become more common.

4 Pathophysiology to impairments

1. **Inflammation** from smoke and dust narrows the small airways, enlarges mucus glands and damages cilia: cough, sputum, infections.
2. **Destruction of alveolar walls** (emphysema) removes the elastic recoil that holds small airways open. Airways collapse during breathing out.

3. Air gets trapped: **hyperinflation**. The diaphragm is pushed down and flattened, so it works at a mechanical disadvantage. Accessory muscles take over.
4. When breathing speeds up during exercise, there is less time to breathe out, so trapping worsens (**dynamic hyperinflation**). This is the main cause of breathlessness on exertion.
5. Mismatched ventilation and blood flow lowers oxygen; in advanced disease CO_2 rises.
6. Long-term low oxygen narrows the lung blood vessels, raising pressure: right heart failure (**cor pulmonale**).
7. Breathlessness leads to inactivity, which weakens the limb muscles, which makes breathlessness worse: the **breathlessness spiral**.

The breathlessness–inactivity spiral



Each turn of the loop makes the next one worse. Physiotherapy breaks the loop at the deconditioning step.

Recall check. Explain in three sentences why a COPD patient becomes more breathless when he walks faster.

5 Clinical features

- Breathlessness on exertion that slowly worsens over years; chronic cough, often worse in the morning; sputum.
- Wheeze, chest tightness, tiredness, weight loss in advanced disease.
- **Signs:** barrel chest (increased front-to-back diameter), use of accessory muscles, pursed-lip breathing, tripod (forward-lean) posture, reduced chest expansion, hyper-resonant percussion, quiet breath sounds with prolonged expiration and wheeze.
- Inward movement of the lower ribs on breathing in (Hoover's sign) from a flattened diaphragm.
- Cyanosis, ankle oedema and raised neck veins in cor pulmonale.

Acute exacerbation

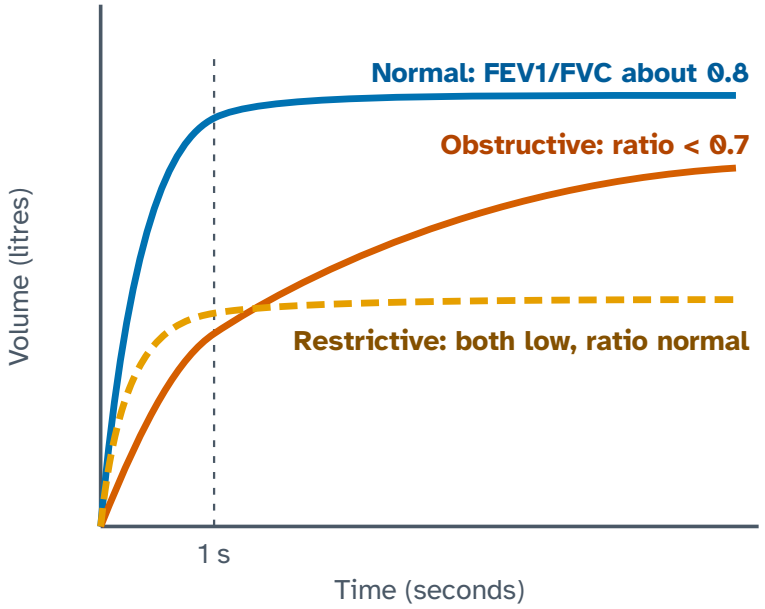
A sudden worsening of breathlessness, cough or sputum (more, or purulent) beyond day-to-day variation, usually triggered by infection or pollution. Look for breathlessness at rest, confusion or drowsiness (rising CO_2), and the use of accessory muscles. This is the picture in the W17 and S20 cases.

6 Investigations

Pulmonary function tests in obstructive disease (3 marks in W20; PFT in COPD, W19)

- **FEV1 reduced**; FVC normal or slightly reduced.
- **FEV1/FVC below 0.70** after bronchodilator.
- Little improvement after bronchodilator (asthma shows large reversibility).
- Peak expiratory flow reduced; flow–volume loop shows a "scooped" expiratory curve.
- Lung volumes: residual volume (RV), total lung capacity (TLC) and RV/TLC raised (air trapping).
- Diffusing capacity (DLCO) reduced in emphysema.

Spirometry: normal, obstructive, restrictive



The curve at 1 second is FEV1; the level where it flattens is FVC. Obstruction slows the rise; restriction lowers the plateau.

Chest X-ray in emphysema (3 marks in W18; X-ray in COPD, W24)

- Hyperinflated, very black (hyperlucent) lung fields with fewer vessel markings at the edges.
- Low, **flattened diaphragms**; more than 6 anterior or 10 posterior ribs visible above the diaphragm.
- Ribs more horizontal, intercostal spaces wider.
- Long, narrow ("tubular") heart shadow.
- Increased air space behind the sternum on the lateral view; bullae may be seen.

Chronic bronchitis often shows a near-normal film, or more prominent lung markings ("dirty chest") and thickened bronchial walls seen as parallel lines (tram lines). With pulmonary hypertension the pulmonary arteries at the hila look large; with cor pulmonale the heart shadow enlarges.

Other tests

Test	What it shows	Physio meaning
ABG	Low PaO_2; raised $PaCO_2$ in type 2 respiratory failure (W16: "mild hypoxaemia and hypercapnia")	Target SpO_2 88–92% if CO_2 is high; watch for drowsiness
Pulse oximetry at rest and on exercise	Desaturation with effort	Decides whether oxygen is needed during exercise
6MWT or shuttle walk test	Exercise capacity and desaturation	Sets walking intensity; outcome measure
ECG and echo	Right heart strain, heart disease	Cor pulmonale lowers exercise tolerance
Sputum culture, blood count	Infection, raised haemoglobin (polycythaemia)	Infection means exacerbation care, not training

7 Medical management

- **Stop smoking** and reduce dust and smoke exposure. This is the only measure that slows the loss of lung function.
- **Inhaled bronchodilators:** long-acting LAMA and LABA, with reliever salbutamol. Inhaled steroids added in group E when blood eosinophils are high.
- **Vaccines:** influenza, pneumococcal, COVID-19 and others as advised.
- **Pulmonary rehabilitation** for symptomatic patients and after a hospital admission.
- **Long-term oxygen therapy** (ATS 2020) if resting PaO_2 is 55 mmHg or less (SpO_2 88% or less), or 56–59 mmHg (SpO_2 89%) with oedema, polycythaemia or right heart strain. Used for **15 hours a day or more**.
- **Exacerbations:** nebulised bronchodilators, a short course of oral steroids, antibiotics if sputum is purulent, controlled oxygen, and **non-invasive ventilation (NIV)** for acidotic hypercapnic respiratory failure. Intubation and ventilation if NIV fails (S20 patient on SIMV).
- Surgery in selected patients: bullectomy, lung volume reduction, transplant.

Oxygen target

In COPD with a risk of CO_2 retention, aim for SpO_2 **88–92%**, not 94–98%. Too much oxygen can raise CO_2 and make the patient drowsy.

Classic answer vs current understanding: hypoxic drive (3 marks in W25)

Classic

In chronic CO_2 retention the brain stops responding to CO_2, so breathing is driven by low oxygen (the hypoxic drive). Give high oxygen and this drive is removed, breathing slows and CO_2 rises.

Current (BTS emergency oxygen guideline)

Reduced drive plays only a small part. Most of the CO_2 rise comes from **worse ventilation–perfusion matching**: oxygen opens vessels in poorly ventilated lung that were narrowed by low oxygen, so more blood goes where little air goes. The **Haldane effect** (oxygenated haemoglobin carries less CO_2) adds to it. Never withhold oxygen from a hypoxic patient; control it instead.

Write in the exam

Define hypoxic drive, then: "Uncontrolled oxygen can cause CO_2 narcosis. Give controlled oxygen (Venturi mask or low-flow nasal cannula) to SpO_2 88–92%, and check ABG."

8 Complications and prevention

Complication	What to look for	Physio role
Exacerbations, pneumonia	More sputum, colour change, fever, more breathlessness	Airway clearance, early action plan, vaccination advice
Respiratory failure	Drowsiness, confusion, flapping tremor, rising CO_2	Positioning, support NIV, report urgently
Cor pulmonale	Ankle oedema, raised neck veins	Lower training intensity, oxygen if desaturating
Pneumothorax (bullae rupture)	Sudden chest pain and breathlessness	Stop; report; avoid high-pressure techniques in bullous disease
Muscle weakness, weight loss	Low thigh strength, low BMI	Resistance training; refer for nutrition
Osteoporosis (steroids, inactivity)	Back pain, fractures	Weight-bearing exercise, careful manual techniques
Anxiety and depression	Panic with breathlessness, low mood	Breathing control for panic, relaxation, group rehab

9 PT assessment

- **Subjective:** breathlessness (mMRC), cough and sputum (amount, colour, morning load), wheeze, exacerbations in the last year, smoking pack-years, oxygen use, inhaler technique, sleep position, ADL that cause breathlessness, home (floor, lift, toilet), occupation and dust exposure, earning role.
- **Observation:** posture (forward lean), chest shape, breathing pattern and rate, accessory muscles, pursed-lip breathing, cyanosis, clubbing (suggests another cause), oedema.
- **Palpation:** chest expansion at three levels, trachea position.
- **Auscultation:** breath sounds, wheeze, crackles.
- **Vital signs and SpO_2** at rest and on exertion.
- **Exercise capacity:** 6MWT or incremental shuttle walk test with SpO_2 and Borg breathlessness.
- **Muscle strength:** quadriceps, grip; respiratory muscle strength (PImax) if available.

Outcome measures to name

Measure	What it tells you
mMRC dyspnoea scale	Breathlessness in daily life
CAAT (formerly CAT)	Symptom impact, 0–40
6MWT, incremental and endurance shuttle walk tests	Exercise capacity
Modified Borg (CR10) dyspnoea scale	Breathlessness during exercise
St George's Respiratory Questionnaire, CRQ	Disease-specific quality of life

10 PT diagnosis by ICF

Draw the six-box ICF model (see the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (W16, 5 + 5 + 5 marks; repeated in S18)

> "A 50 year old male patient, chronic smoker since last 30 years, stays on 2nd floor with no lift facility, is admitted with chief complaints of dyspnea on performing activities of daily living and productive cough since past two days. His pulmonary function test shows moderate obstructive, he has mild hypoxaemia and hypercapnia. Discuss functional diagnosis, short term and long term physiotherapy management."

Spot the ICF clues before you write

30 years of smoking is a personal factor and cause. 2nd floor with no lift is an environmental barrier. dyspnoea on ADL and productive cough since two days mean an acute exacerbation. moderate obstructive means GOLD 2. hypercapnia means an oxygen target of 88–92%.

ICF component	From this case	Clinical reasoning
Health condition	COPD, moderate airflow obstruction (GOLD 2), acute exacerbation with type 2 respiratory failure	Exacerbation sets the acute phase of the plan
Body function and structure impairments	• Breathlessness, altered breathing pattern • Excess sputum, ineffective clearance • Reduced airflow; hyperinflation • Impaired gas exchange (low O_2, high CO_2) • Reduced exercise tolerance; weak limb and breathing muscles • Anxiety	• Inflamed airways and mucus raise resistance • Loss of recoil traps air and flattens the diaphragm • Inactivity weakens muscles (the spiral)
Activity limitations	Walking, climbing to the 2nd floor, bathing, dressing, carrying loads	Each needs more ventilation than he can provide
Participation restrictions	Work, household roles, social and religious gatherings, leisure	Cannot leave home easily because of the stairs
Environmental factors	**Barriers:** 2nd floor without a lift, smoke and dust exposure. **Facilitators:** family support, medicines and oxygen, hospital rehab.	Plan rests on the stairs; energy conservation at home
Personal factors	50-year-old man, 30-year smoker, motivation, worry about income	Readiness to quit and motivation decide progress

One-line functional diagnosis

Impaired airway clearance and reduced ventilatory capacity leading to breathlessness and reduced exercise tolerance, which limit walking, stair climbing to his 2nd-floor home and ADL, and restrict work and social participation. The 2nd floor without a lift is a barrier; family support is a facilitator.

Recall check. Fill the ICF for the S17 labourer (only earning member, moderate obstruction with good reversibility). What does "good reversibility" suggest?

11 Problem list and goals

Goals for the W16 patient

Problem	Short-term (during admission, 1–2 weeks)	Long-term (after a 6–12 week rehab programme)
Breathlessness	Uses breathing control and relieving positions independently; SpO_2 88–92%	mMRC improved by at least one grade
Secretions	Clears sputum independently with ACBT	Daily airway-clearance routine; fewer exacerbations

Problem	Short-term (during admission, 1–2 weeks)	Long-term (after a 6–12 week rehab programme)
Exercise tolerance	Walks on the ward with rests; climbs one flight with pacing before discharge	6MWT distance improved; climbs to the 2nd floor with one rest
Weakness	Independent transfers and toileting	Stronger thigh and arm muscles; carries loads with energy-saving methods
Risk factors and self-management	Correct inhaler technique; smoking-cessation plan	Tobacco-free; written action plan for exacerbations

Using ABG, PFT and X-ray to set goals (7 marks in W24)

Test	Finding	Decision it drives
ABG	Acidosis with high CO_2 (acute respiratory failure)	Acute goals only: positioning, breathing control, airway clearance, support NIV; no exercise training yet
ABG	High CO_2 with normal pH (chronic, compensated)	Controlled oxygen to SpO_2 88–92%; may need NIV at night; monitor drowsiness during sessions
ABG and oximetry	Low PaO_2 at rest or falling SpO_2 on walking	Exercise with oxygen; interval training; long-term oxygen if criteria are met
PFT	FEV1 % predicted (GOLD 1–4) and reversibility	Severity and realistic goals: mild disease aims for full fitness and work; very severe disease aims for independence in self-care with pacing
PFT	Raised RV and RV/TLC (air trapping)	Pursed-lip breathing, forward-lean positions, slower breathing; avoid fast breathing during exercise
X-ray	Hyperinflation and flat diaphragm	Diaphragm works poorly: forward-lean positions, breathing control; do not force deep abdominal breathing
X-ray	Bullae	Avoid high-pressure techniques (IPPB, high PEP, vigorous bagging): pneumothorax risk
X-ray	Consolidation, collapse or pneumothorax	Consolidation or collapse: target airway clearance and positioning to that lobe. Undrained pneumothorax: no positive pressure; report

12 PT management

Acute exacerbation (in hospital)

- **Positions that ease breathlessness:** high side-lying; sitting leaning forward with forearms on a table or knees; standing leaning forward on a wall or railing. These lift the abdominal contents and let the flattened diaphragm work better.
- **Breathing control:** gentle, relaxed breathing using the lower chest, shoulders relaxed; pursed-lip breathing during effort.
- **Airway clearance** if sputum is a problem: ACBT with huffing; positive expiratory pressure devices (PEP, flutter) if helpful; postural drainage only if it clearly helps (see the classic-vs-current box).
- **Support NIV:** help the patient tolerate the mask, position upright, clear secretions during breaks.
- **Early mobilisation:** sit out, walk with oxygen if needed, keep SpO_2 88–92%.
- **On a ventilator** (S2Ø, SIMV with right lower lobe pneumonia): regular repositioning (affected lung up for drainage), suction, manual techniques where appropriate, careful hyperinflation (risk of air trapping and pneumothorax), early sitting and limb exercises, inspiratory muscle training during weaning.
- **Before discharge:** refer to pulmonary rehab to start within a few weeks.

Classic answer vs current guideline: postural drainage and percussion in COPD		
Classic Postural drainage with percussion and vibrations for every COPD patient with sputum.	**Current (BTS/ACPRC 2009)** Use active techniques first (ACBT, PEP). Add postural drainage only if it clearly helps and causes no harm. Head-down tilt can worsen breathlessness.	**Write in the exam** "ACBT and huffing as first choice; PEP devices if helpful; postural drainage only if effective and tolerated."

Pulmonary rehabilitation (long-term)

An individually prescribed programme of **exercise training, education and behaviour change**. Usually **6–12 weeks**, with at least **2 supervised sessions a week** plus home exercise. It improves breathlessness, exercise capacity and quality of life, and after a hospital admission it reduces readmissions (ATS 2023: strong recommendation).

Exercise prescription

Part	Endurance	Strength
Frequency	**3–5 days** a week	**2–3 days** a week
Intensity	Walking at about 80% of the speed reached on the 6MWT or shuttle test, or cycling at 60–80% of peak work; breathlessness **Borg 4–6** (moderate to strong)	About **60–70% of 1RM**, 8–12 repetitions
Time	**20–60 min**; intervals (for example 1 minute on, 1 minute off) if too breathless for continuous work	1–3 sets of the main muscle groups, legs first
Type	Walking, treadmill, cycle; arm ergometer for upper-limb endurance	Weights, bands, sit-to-stand, step-ups
Oxygen	Give oxygen during training if SpO_2 falls below about 88%	

- **Why strengthen the leg and arm muscles (3 marks in W24):** COPD weakens and wastes limb muscles (inactivity, low oxygen, inflammation, steroids, poor nutrition). Weak thighs tire early and produce lactic acid sooner, which drives breathing harder, so many patients stop exercise because of leg fatigue, not breathlessness. Strength training increases muscle strength and size, improves exercise tolerance and daily tasks (stairs, standing up, carrying), lowers breathlessness at the same workload, helps bone health and reduces falls. It also causes less breathlessness than endurance training, so very breathless patients can do it.
- **Inspiratory muscle training:** for weak breathing muscles, at least 30% of maximal inspiratory pressure, about 30 breaths twice a day.
- **Education:** disease, inhalers, airway clearance, energy conservation, action plan, smoking cessation, nutrition.
- **Maintenance:** keep exercising at home or in community groups after the programme ends.

Pursed-lip breathing (3 marks in S20 and W22)

Technique: breathe in through the nose, then breathe out slowly through loosely pursed lips (as if blowing on hot tea), taking about twice as long to breathe out as to breathe in. Do not force the air out.

Rationale and effects:

- Creates back-pressure in the airways, which splints the small airways open during breathing out.
- Less air trapping, so less hyperinflation.
- Slower, deeper breathing: lower breathing rate, better tidal volume and oxygen.
- Reduces breathlessness and anxiety, and helps during walking and stair climbing.

Energy conservation (7 marks in S19)

- **Plan:** spread heavy tasks across the week; do them at the time of day you feel best.

- **Pace:** slow, steady movements; rest before you get very breathless; breathe out on effort.
- **Prioritise:** drop or delegate tasks that are not essential.
- **Position:** sit to wash, dress and cook; keep often-used things between waist and shoulder height; avoid bending and reaching overhead.
- **Use aids:** trolley, long-handled sponge, shower stool, raised toilet seat.
- **Stairs:** climb slowly, one step per out-breath, rest on landings; plan trips so he climbs to the 2nd floor fewer times.

Why it matters: it lowers the oxygen cost of daily tasks, so the patient stays independent and less breathless.

By impairment

Impairment	Technique and dose	Why
Breathlessness	Relieving positions, breathing control, pursed-lip breathing during activity	Better diaphragm mechanics and less air trapping
Secretions	ACBT 1–2 times a day, more in exacerbations; PEP device if helpful	Huffs move mucus from small to large airways
Low exercise tolerance	Endurance training as above, Borg 4–6	Trained muscles need less oxygen and produce less lactate
Limb weakness	Resistance training 60–70% 1RM, 2–3 days a week	Reverses deconditioning
Weak breathing muscles	IMT ≥ 30% PImax, 30 breaths twice daily	Stronger inspiratory muscles reduce breathlessness
Reduced chest mobility, poor posture	Thoracic mobility exercises, posture correction	Better chest wall movement
Anxiety	Relaxation, breathing control for panic, group rehab	Breaks the fear–breathlessness cycle

Recall check. Write the pulmonary rehab endurance prescription and three effects of pursed-lip breathing.

13 Precautions, contraindications and stop criteria

Take care

- Keep SpO_2 at 88–92% in patients with high CO_2; do not give high-flow oxygen without a prescription.
- Avoid head-down positions in breathless patients, and after meals.
- Bullous emphysema: avoid high-pressure techniques (high PEP, aggressive manual hyperinflation); risk of pneumothorax.
- Steroid-related osteoporosis: no vigorous percussion over fragile ribs.
- Check heart disease, which is common in COPD.
- Do not start exercise training during an acute exacerbation; manage the exacerbation first.

Stop exercise if

- SpO_2 falls below about 85–88% despite oxygen
- Severe breathlessness (Borg above 7), chest pain, dizziness, confusion
- New irregular pulse, or a marked fall in BP

14 Education, lifestyle and home programme

- Stop all tobacco; avoid second-hand smoke, chulha smoke and dust. Use masks at work where possible.

- Learn correct inhaler and spacer technique; take medicines regularly.
- Daily walking at moderate breathlessness, plus home strengthening exercises.
- Daily airway clearance if you have sputum.
- Use energy-conservation and pacing methods for housework and stairs.
- Eat small, frequent meals; keep a healthy weight.
- Get vaccinated against influenza and pneumonia.
- **Action plan:** know the early signs of an exacerbation (more breathless, more or darker sputum) and when to start rescue medicines or see a doctor.

15 Prognosis and discharge criteria

Prognosis is worse with lower FEV1, frequent exacerbations, low BMI, low exercise capacity and continued smoking. Stopping smoking slows decline; pulmonary rehab improves symptoms and function; long-term oxygen improves survival when resting oxygen is severely low.

Discharge from hospital: stable on home medicines and oxygen if prescribed; independent with breathing control, airway clearance and inhalers; can walk and manage stairs with pacing; referred to pulmonary rehab.

16 Exam corner

Past questions (MUHS, 2016–2026)

W16 Q3(a) and S18 Q3(b), 5 + 5 + 5 marks

The worked example in section 10 (S18: age 45, same story).

S17 Q3(a), 5 + 5 + 5 marks

50-year-old labourer with COPD, acute exacerbation; smoker; only earning member; moderate obstruction with good reversibility. ICF, and physiotherapy in acute and chronic stages.

W17 Q3(b), 5 + 3 + 7 marks

50-year-old sweeper with COPD for 10 years; breathless at rest, thick sputum; on nasal-prong oxygen. Impairments, activity and participation; short- and long-term management.

S20 Q3(b), 5 + 3 + 7 marks

60-year-old farmer with a COPD exacerbation and respiratory failure, in ICU on SIMV; right lower lobe pneumonia. Impairments, activity and participation; short- and long-term management.

7-mark questions

Energy conservation techniques for COPD patients and their importance (S19). Relevance of ABG, PFT and X-ray in decision-making for goal setting in COPD (W24).

3-mark questions

Effects of smoking on pulmonary function (S17, W22). Radiological findings of emphysema (W18). Pursed-lip breathing effects (S20) and rationale in COPD (W22). PFT in obstructive lung disease (W20). PFT findings in COPD (W19). Rationale of pursed-lip breathing in COPD (W21). X-ray findings in COPD (W24). Importance of peripheral muscle strengthening in COPD (W24). Hypoxic drive in COPD (W25).

Technique cards for this topic: dyspnoea and energy conservation (p. 285), oxygen therapy (p. 290), ABG (p. 326), breathing exercises (p. 275), respiratory muscle training (p. 282), exercise testing (p. 308), pulmonary rehabilitation (p. 339), X-ray and PFT (p. 317).

Keywords to underline

FEV1/FVC < 0.70 GOLD grade hyperinflation dynamic hyperinflation breathlessness spiral SpO_2 88–92% pursed-lip breathing ACBT pulmonary rehabilitation energy conservation pacing action plan

Answer skeleton: COPD long answer

1. Restate: age, job, smoking, home, PFT grade, blood gases, acute or stable.
2. **[ICF, 5]** ICF diagram from the case; one-line diagnosis; reasoning (hyperinflation, sputum, deconditioning).
3. **[Short-term, 3–5]** Acute care: positioning, breathing control, airway clearance, controlled oxygen, NIV support, early mobilisation.
4. **[Long-term, 5–7]** Pulmonary rehab with doses (endurance, strength, IMT), education, energy conservation, stairs, smoking cessation, action plan, outcome measures.

Draw these

- The breathlessness spiral (section 4)
- Spirometry curves (section 6)
- Forward-lean positions as stick figures (sitting and standing)

Viva questions

1. Why is the oxygen target 88–92% in some COPD patients?
2. How does pursed-lip breathing help?
3. What is dynamic hyperinflation?
4. Why does leaning forward relieve breathlessness?
5. What are the components of pulmonary rehabilitation?
6. What are the X-ray features of emphysema?

Linked practical spots

- Chest X-ray of hyperinflation: flat diaphragms, horizontal ribs, narrow heart.
- PFT report with FEV1/FVC 0.55 and FEV1 60% predicted: moderate obstruction (GOLD 2).
- ABG with pH 7.32, $PaCO_2$ 58 mmHg, raised bicarbonate: acute-on-chronic respiratory acidosis.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Rationale of pursed-lip breathing in COPD.**

Answer

Back-pressure splints small airways open during expiration, reducing air trapping and hyperinflation; breathing slows and deepens, improving tidal volume and oxygenation; breathlessness and anxiety fall during activity.

3 m **Radiological findings of emphysema.**

Answer

Hyperlucent lungs with reduced peripheral markings; flattened low diaphragms (more than 6 anterior ribs above); horizontal ribs and wide spaces; narrow tubular heart; increased retrosternal air space; bullae.

7 m **Energy conservation techniques for COPD patients and their importance.**

Answer
Define; the 4 P's (plan, pace, prioritise, position); breathing with activity (exhale on effort); aids; stairs strategy; examples for bathing, dressing, cooking; importance (less oxygen cost, less breathlessness, more independence, fewer exacerbation triggers).

MCQ **A COPD patient with $PaCO_2$ 60 mmHg is on oxygen with SpO_2 98% and is becoming drowsy. The best action is:**

A. Increase oxygen

B. Report and reduce oxygen to target 88–92% as prescribed

C. Start a 6-minute walk test

D. Head-down postural drainage

Answer
B. Too much oxygen in a CO_2-retaining patient can raise CO_2 further and cause drowsiness.

MCQ **Which spirometry pattern confirms COPD?**

A. FEV1/FVC 0.85, FVC 55% predicted

B. Post-bronchodilator FEV1/FVC 0.60

C. FEV1/FVC 0.75 that rises to 0.85 after salbutamol

D. Normal FEV1, low DLCO only

Answer
B. A post-bronchodilator ratio below 0.70 confirms persistent airflow limitation. A is restrictive; C suggests asthma.

MCQ **The main aim of endurance training in pulmonary rehab is to:**

A. Improve FEV1

B. Reverse emphysema

C. Improve limb muscle function so daily tasks cause less breathlessness

D. Replace bronchodilators

Answer
C. Lung function does not change much, but trained muscles need less ventilation for the same work.

Last recall check. Write the GOLD grades, the oxygen target, and the pulmonary rehab prescription. Then check them against this page.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

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Lung surgery, thoracotomy, chest drain and chest trauma

> Thoracic surgery was in 12 of 20 papers (2016–2026): 7-mark questions on incisions, complications and chest drains, and long cases after lobectomy, segmentectomy and pneumonectomy. Learn each incision with the soft tissues cut, the drain, and a day-by-day post-op plan.

Syllabus	Ch 6(e) Pulmonary surgeries: traumatic and surgical conditions of the chest, lung, pleura and mediastinum
Priority	**Tier 1.** In 12 of 20 papers (2016–2026). 5 long-answer cases (3 since 2021) plus chest trauma within a burns case.
Asked as	LAQ W16 · S22 · W21 · S25 · S26 · 7m S17 · S18 · W20 · S22 · W22 · W19 · S23 · 3m W20 · W19 · S23 · S25
Usual split	5 + 5 + 5 (functional diagnosis, short-term, long-term) or 15 (diagnosis, goals, management in detail)
Checked against	BTS pleural disease guideline (2010, 2023 update); BTS/ACPRC physiotherapy guideline (2009); AARC incentive spirometry guideline (2011); Cochrane review of exercise before lung cancer surgery. Full list under Sources. Content checked 19 October 2026.

1 Definition

Thoracic surgery is surgery on the lungs, pleura, chest wall, oesophagus or mediastinum. Physiotherapy matters because every chest operation reduces lung volumes and cough strength and stiffens the shoulder and trunk on the operated side.

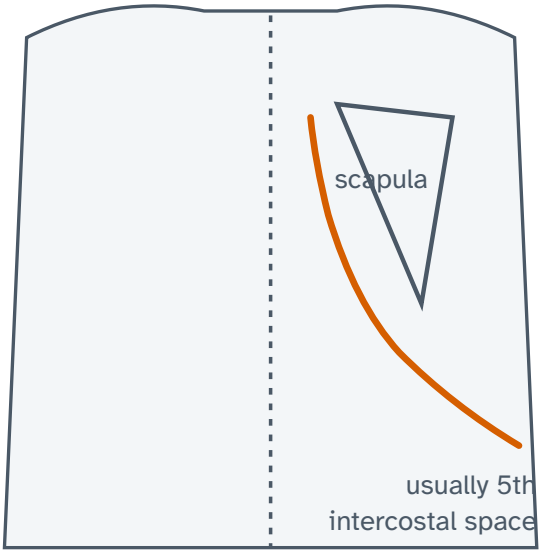
An **intercostal chest drain (ICD)** is a tube passed between the ribs into the pleural space to remove air, blood, fluid or pus, connected to an **underwater seal** that lets air and fluid out but not back in.

2 Classification: incisions and operations

Thoracic incisions (7 marks in S22; 3 marks in W19 and S23)

Incision	Where and what is cut	Used for
Posterolateral thoracotomy	Curved incision from below the nipple line round under the tip of the scapula and up between scapula and spine, usually through the 5th intercostal space. Cuts latissimus dorsi, serratus anterior (sometimes spared), trapezius and rhomboids (if extended up), and intercostal muscles. Most painful.	Lobectomy, pneumonectomy, oesophagectomy (W16 case)
Anterolateral thoracotomy	Under the breast (inframammary fold): skin, pectoralis major (and minor if extended), serratus anterior and intercostals	Emergency trauma, some lung and heart operations
Axillary (muscle-sparing) thoracotomy	In the armpit: skin; latissimus dorsi pulled back, serratus anterior split along its fibres, intercostals; muscles split or retracted rather than cut	Upper lobe surgery, pneumothorax
Median sternotomy	Midline skin and fat, pectoral fascia, then the sternum is split with a saw; no major muscle cut; closed with steel wires	Heart surgery, anterior mediastinum, both lungs
Clamshell (bilateral anterior)	Across the front of both sides of the chest and sternum	Lung transplant, major trauma
Thoracoabdominal	Chest and abdomen together, cutting the diaphragm	Oesophagus, aorta
VATS (video-assisted thoracoscopic surgery)	2–4 small ports between ribs through skin, small slips of serratus or latissimus, and intercostals; little muscle cut	Segmentectomy, lobectomy, biopsy, pleurodesis. Less pain, faster recovery.

Posterolateral thoracotomy: muscles cut (3 marks in W20)



Muscles cut:

1. Latissimus dorsi
2. Serratus anterior
3. Trapezius (lower part)
4. Rhomboids
5. Intercostal muscles

Right side, viewed from behind

The incision curves under the tip of the scapula. Because shoulder and trunk muscles are cut, shoulder exercises start from day 1.

Lung operations

- **Pneumonectomy:** whole lung removed.
- **Lobectomy:** one lobe removed (commonest for lung cancer); bilobectomy on the right.
- **Segmentectomy:** one or more segments removed (S22: stage 1 lung cancer).
- **Wedge resection:** a small wedge of lung.
- **Sleeve resection:** a lobe with a section of main bronchus, rejoined.
- **Decortication:** removal of a thick fibrous peel from the lung (chronic empyema).
- **Pleurectomy and pleurodesis:** to stop recurrent pneumothorax or effusion.
- **Bullectomy, lung volume reduction**, **thoracoplasty** (see the TB page), **lung transplant**.

3 Indications

- **Lung cancer** (commonest), carcinoid and other tumours; risk factors are smoking, occupational exposure (asbestos), air pollution and family history.
- **Localised bronchiectasis**, lung abscess, aspergilloma, complications of TB.
- **Hydatid cyst** (W16 case): a parasitic cyst from dog tapeworm, common in farming communities.
- **Empyema** needing decortication; recurrent pneumothorax.
- **Chest trauma:** rib fractures, flail chest, haemothorax, pneumothorax, lung contusion, penetrating wounds.

4 Pathophysiology to impairments after thoracic surgery

1. **Anaesthesia and lying flat** reduce functional residual capacity; small airways close at the lung bases.

2. **Pain** from the incision, rib spreading and drains makes breathing shallow and stops deep breaths and effective coughing.
3. **Lung tissue is removed**, so total lung capacity and exercise reserve fall; the remaining lung must expand to fill the space.
4. Sputum is increased (surgery, smoking history) but cannot be cleared: **sputum retention**, then **atelectasis** and **pneumonia**.
5. **Cut shoulder-girdle muscles** and guarding cause shoulder stiffness and a posture bent towards the operated side.
6. Bed rest and drains reduce mobility and fitness and raise DVT risk.

Recall check. List three reasons a patient cannot cough well on day 1 after thoracotomy.

5 Clinical features after surgery

- Incision and drain-site pain, worse on deep breathing and coughing.
- Shallow, fast breathing; reduced chest expansion and breath sounds on the operated side; crackles if secretions are retained.
- Weak, painful cough with sputum (W16: "cough with expectoration").
- Drains in place; oxygen; drips; possibly an epidural or nerve-block catheter.
- Stiff shoulder on the operated side, posture leaning towards the incision.
- Fatigue and breathlessness on exertion.

6 Investigations

Test	What it shows	Physio meaning
Pre-op spirometry and diffusing capacity (FEV1, DLCO)	Lung reserve; predicted post-op lung function	Low values mean high risk: pre-op training and close post-op care
Exercise test (stair climb, shuttle walk, CPET)	Fitness for surgery	Guides prehabilitation
Post-op chest X-ray	Lung re-expansion, drain position, collapse, effusion, surgical emphysema	Collapse means more expansion work and airway clearance
ABG and SpO_2	Oxygenation and ventilation	Oxygen during mobilisation if needed
Haemoglobin	Blood loss	Low Hb lowers exercise tolerance

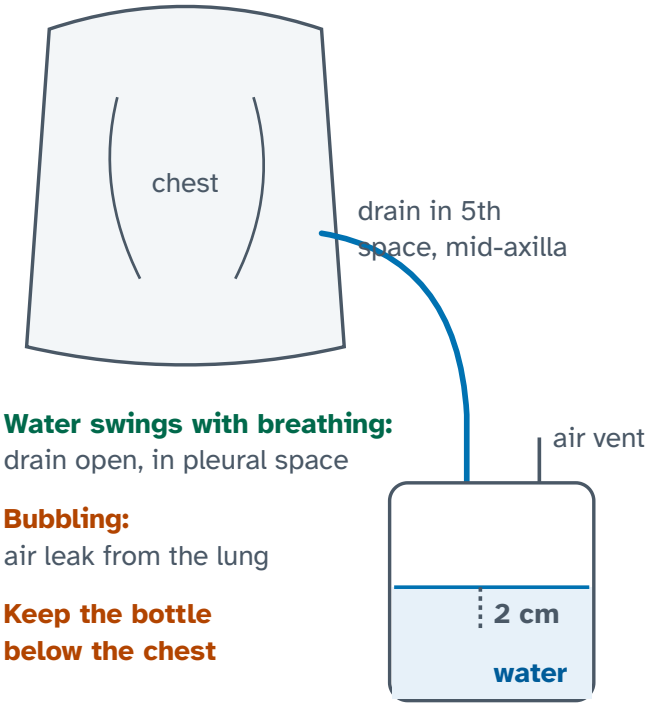
7 Medical and surgical management

Intercostal drain and underwater seal (7 marks in W20; part of W22)

- **Indications (3 marks in S25):** large or breathless pneumothorax; tension pneumothorax after needle decompression; any pneumothorax in a ventilated patient; haemothorax; empyema or infected effusion; large or malignant pleural effusion (and before pleurodesis); chylothorax; after thoracic or cardiac surgery.
- **Site:** usually the "safe triangle": bordered by the front edge of latissimus dorsi, the side edge of pectoralis major, and a line at the level of the nipple (about the 5th intercostal space), with its apex below the axilla.
- **Underwater seal:** the tube from the chest ends about 2 cm under water in the drainage bottle. Air and fluid can bubble out, but water blocks air from being sucked back. The bottle must stay **below the level of the chest** at all times.
- **Swinging** of the water in the tube with breathing shows the drain is open and in the pleural space. **Bubbling** shows an air leak from the lung. No swing may mean the tube is kinked or blocked, or the lung has fully expanded.

- Suction (low negative pressure) may be added to help re-expand the lung.

Intercostal drain with underwater seal



The water is the one-way valve. Lifting the bottle above the chest would siphon water back into the pleura.

Other medical care

- Pain control: epidural, paravertebral or intercostal nerve block, patient-controlled analgesia, regular oral analgesia.
- Oxygen, nebulisers, antibiotics as needed.
- Enhanced recovery programmes: early eating, early drain removal, walking on day 1.
- For cancer: chemotherapy or radiotherapy after surgery where needed.

8 Complications and prevention (7 marks in S17 and S18)

Complication	What to look for	Physio role
Sputum retention, atelectasis, pneumonia	Reduced breath sounds, crackles, fever, falling SpO_2	Deep breathing, ACBT, supported cough, early walking, positioning
Respiratory failure	Rising breathing rate, drowsiness	Positioning, airway clearance, support NIV, report
Persistent air leak, bronchopleural fistula	Ongoing bubbling; after pneumonectomy, coughing up a lot of thin fluid	Report; avoid high-pressure techniques; after pneumonectomy keep the operated side down if a fistula is suspected
Haemorrhage	Heavy blood in the drain, falling BP	Stop and report
Surgical emphysema	Crackling swelling under the skin of the chest and neck	Report; check the drain is working
Arrhythmias (atrial fibrillation)	Irregular pulse, palpitations	Check pulse before mobilising
Empyema, wound infection	Fever, cloudy drain fluid, red wound	Report; continue chest care
Nerve injury	Hoarse voice and weak cough (recurrent laryngeal); raised hemidiaphragm (phrenic)	Supported cough, huffing, breathing exercises

Complication	What to look for	Physio role
Shoulder stiffness, poor posture, scoliosis (long term)	Loss of shoulder elevation, trunk bent towards the operated side	Shoulder and trunk exercises from day 1, posture correction
Post-thoracotomy pain syndrome	Burning pain along the scar months later	Early mobilisation, desensitisation, TENS
DVT and PE	Calf swelling, sudden breathlessness	Early walking, ankle pumps

9 PT assessment

- **Pre-op:** lung function, exercise tolerance, smoking, sputum, shoulder range, occupation and hand dominance; teach the post-op plan.
- **Post-op chart:** operation, incision, drains (number, site, swing, bubbling, amount and type of drainage), analgesia, oxygen, X-ray.
- **Respiratory:** breathing pattern and rate, SpO_2, chest expansion, auscultation, cough and sputum.
- **Pain** at rest and on coughing (VAS).
- **Musculoskeletal:** shoulder range on the operated side, posture.
- **Function:** bed mobility, sitting, standing, walking distance with SpO_2.
- **Outcome measures:** 6MWT, shoulder ROM, pain scale, quality of life (for example EORTC questionnaire for cancer).

10 PT diagnosis by ICF

Draw the six-box ICF model (see the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (W16, 5 + 5 + 5 marks)

> "A 40 year old male labourer, diagnosed case of hydatid cyst of right lower lobe, underwent right lower lobe lobectomy two days back via a posterolateral thoracotomy. He is right hand dominant, is currently having cough with expectoration with an intercostal drain in situ. Discuss functional diagnosis, short term and long term physiotherapy management."

Spot the ICF clues before you write

right lower lobectomy via posterolateral thoracotomy means the right shoulder-girdle muscles are cut. right hand dominant labourer means his working arm is affected. cough with expectoration means airway clearance is needed. intercostal drain in situ adds drain precautions.

ICF component	From this case	Clinical reasoning
Health condition	Hydatid cyst of the right lower lobe; right lower lobectomy, day 2; ICD in place	Acute post-op phase
Body function and structure impairments	• Incisional and drain pain • Reduced lung volume, shallow breathing; risk of atelectasis of remaining lobes • Retained secretions, weak cough • Reduced right shoulder range and strength • Reduced exercise tolerance	• Pain and anaesthesia reduce deep breathing • Latissimus dorsi, serratus anterior and intercostals were cut • Lung tissue removed lowers reserve
Activity limitations	Bed mobility, walking with the drain, self-care with the right arm, lifting	Pain, drain and arm restriction

ICF component	From this case	Clinical reasoning
Participation restrictions	Cannot work as a labourer (heavy lifting, overhead work) for some weeks; family income	Heavy manual work needs full shoulder and lung capacity
Environmental factors	**Facilitators:** hospital care, family. **Barriers:** physically heavy job; dust exposure; daily wages.	Plan a graded return to work
Personal factors	40-year-old man, right-handed labourer	Young, good recovery potential

One-line functional diagnosis

Day 2 after right lower lobectomy through a posterolateral thoracotomy, with pain, reduced lung expansion, retained secretions and reduced right shoulder movement, limiting bed mobility, walking and use of his dominant arm, and restricting his work as a labourer.

Recall check. Fill the ICF for the S22 chemical engineer, day 5 after segmentectomy for stage 1 lung cancer.

11 Problem list and goals

Goals for the W16 labourer

Problem	Short-term (by discharge, about 1 week)	Long-term (6–8 weeks)
Secretions, lung expansion	Clear chest; effective supported cough; remaining lung expanded on X-ray	Normal breathing pattern; good chest mobility
Pain	Pain controlled enough for deep breathing and walking	Pain-free daily activity
Shoulder	Active right shoulder flexion and abduction to near full range	Full range and strength for overhead work
Mobility and fitness	Walks independently on the ward and climbs one flight	Improved 6MWT; back to labour work gradually

12 PT management

Before surgery

- Explain the operation, drains and the post-op routine; teach breathing exercises, ACBT, supported cough, bed mobility and shoulder exercises.
- **Stop smoking** as early as possible.
- **Prehabilitation:** aerobic and strength training for a few weeks before lung cancer surgery reduces post-op lung complications and hospital stay (Cochrane review).

After surgery (lobectomy or segmentectomy)

Day	Physiotherapy
Day 0–1	• Sit upright; high side-lying alternating sides (after lobectomy either side is allowed) • Deep breathing with a 3-second hold, thoracic expansion to the operated side, ACBT • Supported huff and cough: hold a pillow or the patient's arm over the incision and drain • Ankle pumps; sitting out of bed and marching on the spot or walking with the drain • Active-assisted shoulder movement on the operated side within pain

Day	Physiotherapy
Day 2–4	• Progress walking distance 3–4 times a day; stairs before discharge • Active shoulder flexion and abduction, trunk side-bending away from the incision and rotation • Posture correction in a mirror • Continue airway clearance until the chest is clear
After drain removal and discharge	• Home walking programme, increasing weekly • Shoulder stretches and strengthening; thoracic mobility • Pulmonary rehabilitation or exercise training for lung cancer survivors • Graded return to heavy work over about 6–8 weeks (no heavy lifting at first)

Pneumonectomy (7 marks in S23; long cases W21 and S26)

Surgical approach and soft tissues cut (S26): usually a **posterolateral thoracotomy** through the 4th or 5th intercostal space; a rib may be removed or divided for access. In order: skin and subcutaneous tissue → **latissimus dorsi** → **serratus anterior** (divided or retracted) → lower **trapezius** and **rhomboids** if the cut extends up → **intercostal muscles** (external, internal, innermost) → parietal pleura. The pulmonary artery, pulmonary veins and main bronchus are then divided and the bronchial stump is closed.

- **The empty space:** it slowly fills with fluid over weeks, and the mediastinum shifts towards it. A drain, if used, is usually clamped or balanced and removed early.
- **Positioning:** lie on the back, sit up, or turn a quarter towards the **operated** side. Avoid lying fully on the remaining lung side, because fluid in the space could spill through a leaking bronchial stump into the only lung. Avoid head-down tilt.
- **Breathing:** all gas exchange depends on one lung, so deep breathing, thoracic expansion of the remaining lung, supported huff and cough, and early walking matter even more.
- **Shoulder and posture:** active shoulder exercises on the operated side from day 1 (the cut muscles move the shoulder); patients lean and drop the shoulder towards the operated side, and a scoliosis can develop, so correct posture in a mirror and add trunk mobility.
- **Watch and report:** irregular pulse (atrial fibrillation is common), breathlessness with crackles in the remaining lung (fluid overload), fever, and coughing up thin, watery or blood-stained fluid, especially when lying on the good side. That last sign, or air still bubbling from a drain days later (as in W21), suggests a **bronchopleural fistula**: a leak at the bronchial stump.
- **Long term:** exercise capacity is lower but improves with a graded walking and exercise programme.

Physiotherapy with an intercostal drain

- Check before every session: swing, bubbling, type and amount of drainage, tubing not kinked.
- Keep the bottle upright and below chest level; carry it low when walking (or use a portable drain).
- Never clamp a drain that is bubbling.
- If the tube disconnects: put the chest end under sterile water or clamp briefly as per ward protocol, and call the nurse. If the drain falls out: cover the hole with a sealed dressing and call for help.
- Support the drain site during coughing; do shoulder exercises gently near the drain.

Classic answer vs current evidence: incentive spirometry		
Classic Every patient after thoracic or abdominal surgery uses an incentive spirometer hourly.	**Current (AARC 2011)** Not recommended alone or routinely. Deep breathing, airway clearance and early walking are the core. It can be an add-on to encourage deep breaths.	**Write in the exam** "Deep breathing exercises, ACBT and early mobilisation; incentive spirometer as an adjunct."

Chest trauma (W17 electrician with 3 rib fractures, on ventilator)

- Good pain control (nerve blocks, epidural) is the key to breathing and coughing.
- On the ventilator: positioning, suction, airway clearance, careful manual techniques away from fractures, limb exercises.
- After extubation: deep breathing, supported cough, early walking, shoulder exercises.
- No percussion or vibration over fractured ribs; flail chest is covered on the chest-wall page.

By impairment

Impairment	Technique and dose	Why
Reduced lung volume	Deep breaths with 3-second hold, 5–10 every hour; upright positions; early walking	Re-expands collapsed lung and fills the space
Retained secretions	ACBT with supported huff and cough; humidification	Clears mucus with less pain
Pain	Treat after analgesia; support the incision; TENS	Pain stops deep breathing and coughing
Shoulder stiffness	Active-assisted then active flexion and abduction, 10 reps 3–4 times a day	Cut muscles and guarding stiffen the shoulder
Poor posture	Mirror posture correction, trunk side-bending away from incision	Prevents long-term scoliosis
Low exercise tolerance	Graded walking; exercise training after discharge	Restores fitness after loss of lung tissue

Recall check. Write the five muscles cut in a posterolateral thoracotomy and four rules for physiotherapy with a chest drain.

13 Precautions, contraindications and stop criteria

Take care

- Drain precautions as above; never lift the bottle above the chest.
- After pneumonectomy: avoid full side-lying on the remaining lung, head-down tilt and vigorous manual techniques.
- Fresh bronchial stump: avoid high-pressure techniques such as aggressive manual hyperinflation or high-level PEP.
- No percussion over fractured ribs, incisions or drains.
- Epidural: check leg power and BP before standing.

Stop and report if

- Sudden breathlessness or chest pain (pneumothorax, PE)
- Heavy fresh blood in the drain or coughed up
- New irregular pulse or falling BP
- Rapidly spreading surgical emphysema

14 Education, lifestyle and home programme

- Keep doing deep breathing and huffing for a few weeks after going home.
- Walk every day, increasing the distance gradually.
- Do shoulder and trunk exercises until the range is full.
- No heavy lifting for the first weeks; return to heavy work gradually as advised.
- Stop smoking for good.
- Report fever, more breathlessness, wound redness or discharge.

15 Prognosis and discharge criteria

Prognosis: most patients recover well after lobectomy or segmentectomy, especially after VATS. Lung function settles at a lower level depending on how much lung was removed. Cancer prognosis depends on stage.

Discharge: drains out; chest clear; walking independently and managing stairs; shoulder range near full; knows home exercises and warning signs.

16 Exam corner

Past questions (MUHS, 2016–2026)

W16 Q4(b), 5 + 5 + 5 marks
The worked example in section 10.

S22 Q3(a), 15 marks
40-year-old male chemical engineer, segmentectomy for stage 1 lung carcinoma 5 days ago. Functional diagnosis, goals and physiotherapy management in detail.

W21 Q3(a), 15 marks
30-year-old man, left pneumonectomy for bronchiectasis 7 days ago; pain at the suture site; his ICD drains air and serous fluid. Functional diagnosis and management.

S25 Q3(b), 4 + 4 + 7 marks
35-year-old painter, right lower lobectomy 2 days ago; suture-site pain, cough and difficulty bringing up sputum; treated for TB 7 years ago. Pre-op PFT: FVC 60%, FEV1 82%, FEV1/FVC 94% predicted (a restrictive pattern). ICF; short- and long-term goals; physiotherapy.

S26 Q3(a), 3 + 5 + 7 marks
35-year-old housewife with left bronchiectasis and a destroyed lung; pneumonectomy 3 days ago; incision pain, restricted left shoulder movement, pain at the ICD site. (i) Surgical approach and soft tissues cut; (ii) ICF with short- and long-term goals; (iii) physiotherapy.

7-mark questions

Complications of pulmonary surgeries (S17). Complications and post-op management after right lower lobectomy (S18). Intercostal drainage and underwater seal (W20). Types of thoracic incisions and complications of thoracotomy (S22). Pneumothorax, and precautions and physiotherapy with an intercostal drain (W22). Physiotherapy after right middle lobectomy (W19). Physiotherapy after pneumonectomy (S23).

3-mark questions

Muscles incised in posterolateral thoracotomy (W20). Three incisions in pulmonary surgery (W19). Three thoracic incisions with the soft tissues dissected (S23). Indications of intercostal drainage (S25).

Technique cards for this topic: ACBT and autogenic drainage (p. 261), breathing exercises (p. 275), positioning (p. 236), suctioning (p. 231), chest X-ray (p. 317).

Keywords to underline

posterolateral thoracotomy latissimus dorsi, serratus anterior lobectomy underwater seal swing and bubbling bottle below chest supported cough early mobilisation shoulder exercises from day 1 prehabilitation

Answer skeleton: post-thoracotomy long answer

1. Restate: age, job, hand dominance, operation, incision, day, drains, sputum.
2. **[Functional diagnosis or ICF, 5]** ICF diagram from the case; reasons.
3. **[Short-term, 5]** Day-by-day: breathing, ACBT, supported cough, positioning, drain care, walking, shoulder ROM.
4. **[Long-term, 5]** Exercise training, shoulder and posture, return to work, smoking cessation.

Draw these

- The posterolateral thoracotomy incision with muscles (section 2)
- The underwater seal drain (section 7)

Viva questions

1. What does it mean if the water in the drain tube stops swinging?
2. Why must the bottle be kept below the chest?
3. Which side should a patient lie on after pneumonectomy?
4. Which muscles are cut in a posterolateral thoracotomy?
5. What is the safe triangle?
6. Why does VATS cause less pain than thoracotomy?

Linked practical spots

- Device: underwater seal drainage bottle, portable drain bag.
- X-ray: drain in the pleural space; surgical emphysema; post-pneumonectomy space.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Enumerate the muscles incised in posterolateral thoracotomy.**

Answer

Latissimus dorsi; serratus anterior (may be spared); trapezius (lower fibres); rhomboids (if extended up); intercostal muscles of the chosen space (usually the 5th).

7 m **Intercostal drainage and underwater seal.**

Answer
Define; indications; site (safe triangle); parts of the system with a diagram; principle of the underwater seal (tube about 2 cm under water, one-way); swing and bubbling meaning; suction; precautions (bottle below chest, never clamp a bubbling drain, no kinks); physiotherapy with a drain.

MCQ **A post-lobectomy patient's drain is bubbling continuously. During a walk, the best action is:**

A. Clamp the tube to stop the leak

B. Keep the bottle upright below chest level and do not clamp

C. Lift the bottle onto the bed

D. Avoid walking until the bubbling stops

Answer
B. Clamping a bubbling drain can cause a tension pneumothorax. Walking is encouraged with the bottle kept low.

MCQ **After a right pneumonectomy, which position should be avoided?**

A. Supine

B. Lying fully on the left side

C. Quarter-turn towards the right

D. Sitting upright

Answer
B. Lying fully on the remaining (left) lung risks fluid from the empty space spilling into it through a leaking stump.

MCQ **Which measure most reduces lung complications after lung cancer surgery?**

A. Routine incentive spirometry alone

B. Bed rest for 3 days

C. Pre-op exercise training, deep breathing and early mobilisation

D. Head-down postural drainage

Answer
C. Prehabilitation and early mobilisation reduce complications; incentive spirometry alone is not recommended.

Last recall check. Write the incision and muscles, the drain rules, and the day-by-day plan.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

1. MUHS BPTh syllabus (2012), Cardio-Vascular & Respiratory Physiotherapy. muhs.ac.in
2. MUHS IV BPTh question papers, code 51413, Summer 2016 to Summer 2026: muhsonline.com (unofficial archive, 2016–2022) and the Dhole Patil College of Physiotherapy past-paper page (2019–2026)
3. Havelock T, et al. Pleural procedures and thoracic ultrasound: BTS pleural disease guideline 2010. *Thorax* 2010. BTS clinical statement on pleural procedures (2023).
4. Restrepo RD, et al. AARC clinical practice guideline: Incentive spirometry. *Respir Care* 2011.
5. Cavalheri V, Granger C. Preoperative exercise training for patients with non-small cell lung cancer. Cochrane Review 2017.
6. Bott J, et al. BTS/ACPRC physiotherapy guideline for the adult, medical, spontaneously breathing patient. *Thorax* 2009.
7. Structure and classical content: Main & Denehy (2016); Frownfelter & Dean (2022); Cash's Textbook of Chest, Heart and Vascular Disorders for Physiotherapists.

Bronchiectasis, including after tuberculosis

> Bronchiectasis was in 11 of 20 papers (2016–2026), often after TB and often naming the lobe or side. Every answer needs the vicious cycle, a daily airway-clearance plan with the right drainage position for that lobe, and exercise.

Syllabus	Ch 6(c)(iv) Bronchiectasis; airway clearance from Ch 4
Priority	**Tier 1.** In 11 of 20 papers (2016–2026). 5 long-answer cases.
Asked as	LAQ S16 · S19 · S20 · S21 · W24 · 7m W16 · W17 · W18 · W22 · W19 · 3m W20 · W19
Usual split	5 + 5 + 5 (functional diagnosis, short-term, long-term), 5 + 10 (reasons for impairments, rehab programme) or 7 + 8 (ICF, management)
Checked against	ERS bronchiectasis guideline (2025); BTS bronchiectasis guideline (2019); BTS/ACPRC physiotherapy guideline (2009); Cystic Fibrosis Foundation postural drainage instructions. Full list under Sources. Content checked 19 October 2026.

1 Definition

Bronchiectasis is a chronic lung disease in which the bronchi are **permanently and abnormally widened**, with a daily wet cough, sputum and repeated chest infections.

The widened, scarred airways cannot clear mucus, so mucus pools and becomes infected. This is why airway clearance is the core of physiotherapy.

2 Classification and grading

By shape of the airway (Reid)

- **Cylindrical (tubular):** evenly widened bronchi that do not taper. Mildest.
- **Varicose:** irregular, beaded widening like varicose veins.
- **Cystic (saccular):** balloon-like sacs at the ends of bronchi, often with air–fluid levels. Most severe, most sputum.

By extent

- **Localised:** one lobe or segment (for example after TB, a foreign body or a single severe pneumonia). The S16 middle-lobe and S19 right-lower-lobe cases.
- **Diffuse:** both lungs (cystic fibrosis, immune deficiency, ciliary disorders).

Severity scores to name

The **Bronchiectasis Severity Index** and the **FACED** score combine FEV1, age, infections with *Pseudomonas*, number of lobes, breathlessness (mMRC) and admissions.

3 Causes and risk factors

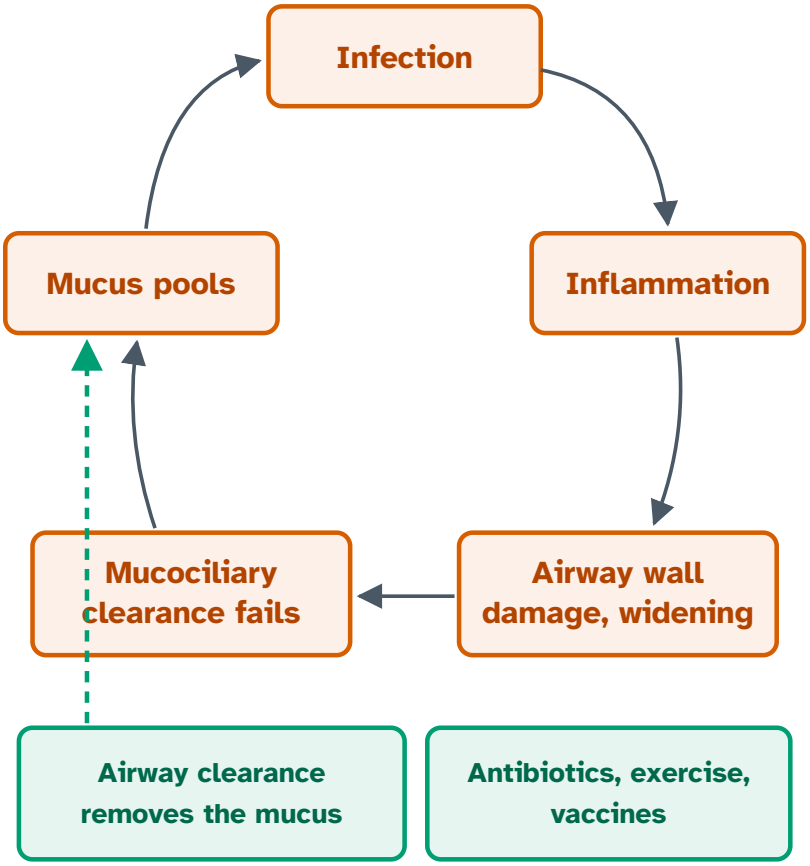
- **Post-infective: tuberculosis** (the commonest cause in India; four of the past cases are "post-Koch's" or "post-PTB"), severe pneumonia, measles or whooping cough in childhood.
- **Airway blockage:** foreign body, tumour, enlarged lymph nodes (for example the middle lobe).
- **Cystic fibrosis.**
- **Primary ciliary dyskinesia:** cilia do not beat. Kartagener's syndrome adds sinusitis and a reversed position of the organs.
- **Immune deficiency**, HIV.

- **Allergic bronchopulmonary aspergillosis (ABPA)** in asthma.
- Rheumatoid arthritis and other connective tissue diseases; inflammatory bowel disease; repeated aspiration.
- In many patients no cause is found (idiopathic).

4 Pathophysiology to impairments

S19 asked "discuss the reasons for her impairments" for 5 marks. The vicious cycle is the answer.

The vicious cycle (Cole)



Each turn of the cycle damages the airways more. Physio breaks it by removing the pooled mucus.

Text version (write this as the answer)

1. An infection (for example TB) damages the airway wall.
2. Inflammatory cells release enzymes that destroy elastic and muscle tissue, so the bronchi widen permanently.
3. Cilia are destroyed and the widened airways cannot clear mucus.
4. Mucus pools in the dilated bronchi and becomes colonised by bacteria.
5. This causes more infection and inflammation, so the cycle repeats.

Impairments and their reasons

- **Chronic productive cough and copious sputum:** pooled mucus in dilated airways.
- **Foul-smelling sputum (S19):** anaerobic bacteria growing in stagnant mucus.
- **Breathlessness and reduced exercise tolerance:** airflow obstruction from mucus and narrowed airways, plus deconditioning.
- **Haemoptysis:** enlarged, fragile bronchial arteries in the inflamed wall.
- **Stress urinary incontinence (S19: "laughing or talking ends in cough and stress incontinence"):** repeated coughing raises abdominal pressure and weakens the pelvic floor.

- **Fatigue, weight loss, low mood:** chronic infection and the social embarrassment of cough and sputum.
- **Reduced chest expansion and tracheal shift (S21):** fibrosis and collapse of a destroyed lobe pull the trachea towards that side.

Recall check. Draw the five-box vicious cycle from memory and mark where physiotherapy acts.

5 Clinical features (3 marks in W20)

- **Chronic cough** with large amounts of mucopurulent sputum, worse in the morning and when changing position.
- Sputum may separate into three layers in a pot: froth on top, mucus in the middle, pus at the bottom.
- **Recurrent chest infections**: more sputum, darker sputum, fever, tiredness.
- **Haemoptysis**, from blood-streaked sputum to large bleeds.
- Breathlessness, wheeze, chest pain, fatigue, weight loss; sinus symptoms.
- **Signs:** clubbing; coarse crackles over the affected area, which may change after coughing; wheeze; reduced expansion over a fibrosed lobe.
- Bad breath (halitosis) with foul sputum.

6 Investigations

Test	What it shows	Physio meaning
HRCT chest (gold standard)	Bronchus wider than its neighbouring artery ("signet ring" sign), airways that do not taper, cysts	Shows exactly which lobes and segments to drain
Chest X-ray	"Tram lines" and ring shadows, cysts with fluid levels; collapse or fibrosis pulling the trachea to that side (S21)	Choose positions for the affected lobe
Sputum culture	Bacteria such as *Pseudomonas* or *H. influenzae*; TB and fungi	Infection control; Pseudomonas means worse disease
Spirometry	Usually obstructive; sometimes mixed with restriction	Baseline and outcome
ABG and SpO_2	Low oxygen in advanced disease	Oxygen during exercise if needed
Tests for cause	Immunoglobulins, aspergillus tests, sweat test, TB tests	Treatable causes change management

7 Medical and surgical management

- **Airway clearance:** every patient with a productive cough should be taught it by a physiotherapist (ERS 2025: strong recommendation).
- **Pulmonary rehabilitation** for patients with reduced exercise capacity (ERS 2025: strong recommendation).
- **Antibiotics:** 14 days for exacerbations; long-term (for example macrolides) or inhaled antibiotics for frequent exacerbations; eradication of new *Pseudomonas*.
- **Mucoactive treatment:** nebulised hypertonic or normal saline before airway clearance when sputum is hard to clear.
- Bronchodilators if there is airflow obstruction or asthma.
- Treat the cause: TB treatment, immunoglobulin replacement, ABPA treatment.
- Vaccination against influenza and pneumococcus.
- **Surgery:** lobectomy for localised disease that does not respond to treatment, or for major haemoptysis (see the lung surgery page). Bronchial artery embolisation for bleeding.

8 Complications and prevention

Complication	What to look for	Physio role
Exacerbations, pneumonia	More, darker sputum; fever; breathlessness	Increase airway clearance frequency; teach the action plan
Haemoptysis	Fresh blood in sputum	Stop percussion, vibration and head-down tilt; gentle breathing control; report. Large bleed: emergency.
Respiratory failure, cor pulmonale	Low SpO_2, oedema	Oxygen during exercise; pace activity
Lung abscess, empyema, pneumothorax	Swinging fever, pleuritic pain, sudden breathlessness	Report; adapt positions
Stress urinary incontinence	Leakage when coughing or laughing	Pelvic floor exercises; "the knack" (tighten the pelvic floor before coughing)
Social isolation, low mood	Avoids people because of cough and sputum	Clear airways before going out; group rehab; reassurance

9 PT assessment

- **Sputum:** daily amount (in teaspoons or cups), colour, smell, thickness, blood; best time and position for clearing.
- Cough effectiveness; breathlessness (mMRC); exacerbations per year; current airway-clearance routine.
- Past TB and treatment; incontinence (ask directly, patients may not mention it); occupation and home.
- **Observation:** posture, clubbing, breathing pattern, chest shape (flattening over a fibrosed lobe).
- **Palpation:** chest expansion (reduced over the affected side), tracheal position.
- **Auscultation:** coarse crackles over the affected lobe; repeat after treatment to check effect.
- SpO_2 at rest and on exercise; 6MWT or shuttle test.
- **Outcome measures:** sputum weight or volume, Leicester Cough Questionnaire, Quality of Life–Bronchiectasis (QOL-B), St George's Respiratory Questionnaire, 6MWT, mMRC.

10 PT diagnosis by ICF

Draw the six-box ICF model (see the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (S19, 5 + 10 marks)

> "50 year old lady with history of pulmonary tuberculosis ten years back. She has completed full course of anti-Koch's medication. However it resulted in right lower lobe bronchiectasis. She had intermittent episodes of infection now and then for the last ten years. At present she complains of increased frequency of infective episodes with productive cough with foul smelling sputum. It is difficult for her to even laugh or talk for long as it ends up with cough and stress incontinence. She has dyspnoea on performing accustomed activities. She works as a clerk and wishes to continue with work but is worried if she would be able to. (i) Discuss the reasons for her impairments. (ii) Plan a rehabilitation program for her."

Spot the ICF clues before you write

right lower lobe sets the drainage positions. foul smelling sputum means infected, stagnant mucus. laugh or talk leading to cough and stress incontinence is a participation restriction and a pelvic-floor impairment. clerk who wishes to continue with work but is worried gives the personal factor and the long-term goal.

ICF component	From this case	Clinical reasoning
Health condition	Post-tuberculous bronchiectasis of the right lower lobe, with frequent infective exacerbations	TB scarred the lobe; the vicious cycle keeps it infected
Body function and structure impairments	• Copious, foul sputum and ineffective clearance • Paroxysmal cough • Breathlessness and reduced exercise tolerance • Pelvic floor weakness with stress incontinence • Reduced right lower chest expansion • Anxiety about work	• Dilated bronchi pool mucus that anaerobes colonise • Repeated cough raises abdominal pressure on the pelvic floor • Infection and inactivity reduce fitness
Activity limitations	Talking or laughing for long; walking and stair climbing at her usual pace; household tasks	Each triggers cough or breathlessness
Participation restrictions	Working as a clerk (talking with the public); social gatherings because of cough, sputum and leakage	Embarrassment and fatigue limit social and work roles
Environmental factors	**Facilitators:** desk job, completed TB treatment, access to care. **Barriers:** dust or crowded office, no private space to clear sputum at work.	Plan a clearance routine around office hours
Personal factors	50-year-old woman, motivated to work, worried about her future	Motivation helps adherence to a daily routine

One-line functional diagnosis

Post-TB right lower lobe bronchiectasis with excess infected sputum, ineffective airway clearance, breathlessness, reduced exercise tolerance and stress incontinence, limiting talking, walking and household tasks and restricting her clerical work and social life. Motivation and a desk job are facilitators.

Recall check. Fill the ICF for the S21 case: 34-year-old smoker, post-TB bronchiectasis, mMRC 3, copious sputum, trachea shifted to the left. Which side is diseased?

11 Problem list and goals

Goals for the S19 clerk

Problem	Short-term (2–4 weeks)	Long-term (3 months and beyond)
Sputum and cough	Independent with ACBT in the right lower lobe position twice a day; chest clearer on auscultation	Daily routine kept up; fewer exacerbations a year
Breathlessness, exercise tolerance	Breathing control during activity; walks 20 minutes at a steady pace	6MWT improved; normal daily activities without stopping
Stress incontinence	Correct pelvic floor contraction; uses "the knack" when coughing	Leakage stopped or much reduced
Participation	Clears her chest before leaving for work	Works full days confidently; social life resumed

12 PT management

1. Daily airway clearance (the core)

- Usually **once or twice a day** for about 10–30 minutes, more often during an exacerbation. Choose the technique the patient will actually do.

- If ordered, take a bronchodilator before, then nebulised saline (hypertonic or normal) to loosen sputum, then airway clearance, then inhaled antibiotics after.
- Combine a technique with a **gravity-assisted position** for the affected lobe when tolerated.

Active cycle of breathing technique (ACBT)

1. **Breathing control:** relaxed, gentle tidal breathing with the lower chest, shoulders relaxed.
2. **Thoracic expansion exercises:** 3–4 deep breaths in, with a 3-second hold, relaxed breath out. Air gets behind the mucus through collateral channels.
3. **Breathing control** again.
4. **Forced expiration technique:** 1–2 huffs from mid-lung volume (a long, medium breath out with the mouth open, as if steaming a mirror), then breathing control. A huff from high lung volume clears the upper airways; then cough when secretions reach them.
5. Repeat the cycle until the chest sounds clear or two cycles bring nothing up.

Other techniques

- **Autogenic drainage:** controlled breathing at three volumes: low lung volume to "unstick" peripheral mucus, mid volume to "collect" it, high volume to "evacuate" it. The patient breathes out through an open mouth and suppresses coughing until mucus reaches the large airways.
- **Positive expiratory pressure (PEP) and oscillating PEP** (flutter, Acapella): breathing out against resistance keeps airways open and moves mucus; the oscillating devices also vibrate the airway. Good for independent use at home and at work.
- **Postural drainage with percussion and vibration:** traditional; use when the patient finds it helpful and has no contraindication.

ACBT vs autogenic drainage (asked as 7 marks in W17)

Feature	ACBT	Autogenic drainage
Principle	Deep breaths get air behind mucus; huffs move it up	Expiratory airflow at different lung volumes moves mucus from small to large airways
Learning	Easy to learn; can be taught in one session	Needs training and practice; best for older children and adults
Positions	Can be combined with drainage positions	Usually sitting
Coughing	Huff, then cough	Cough suppressed until mucus is central, so less airway collapse
Best for	Most patients, post-op patients, exacerbations	Patients with unstable airways, cough-triggered bronchospasm or incontinence (useful for the S19 lady)
Equipment	None	None

2. Postural drainage positions (choose the affected lobe)

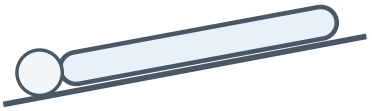
Principle: put the affected segment **uppermost** so gravity drains it towards the main bronchus.

Four drainage positions to draw

Upper lobes, apical
sit, lean back slightly



Right middle lobe
lie on LEFT side, ¼ turn back
foot of bed up about 15°



Lower lobes, posterior basal
lie prone, pillow under hips
foot of bed up about 20–30°



Lower lobe, lateral basal
lie on the OTHER side,
¼ turn forward, foot up



Head end is low on the left of each tilted panel. The words, not the stick figure, carry which side the patient lies on.

Positions for each lobe

Lobe and segment	Position	Bed
Upper lobes: apical	Sitting upright, leaning back slightly (about 30°)	Flat or sitting
Upper lobes: posterior	Sitting, leaning forward over a pillow (about 30°); or lying on the opposite side turned a quarter towards prone	Flat
Upper lobes: anterior	Lying flat on the back	Flat
Lingula (left)	Lying on the **right** side, turned a quarter backwards, pillow behind	Foot raised, about 15°
Middle lobe (right)	Lying on the **left** side, turned a quarter backwards, pillow behind	Foot raised, about 15°
Lower lobes: superior (apical)	Lying prone, two pillows under the hips	Flat
Lower lobes: anterior basal	Lying on the back, or on the opposite side turned slightly back	Foot raised, about 20–30°
Lower lobes: lateral basal	Lying on the **opposite** side, turned a quarter forward	Foot raised, about 20–30°
Lower lobes: posterior basal	Lying prone, pillow under the hips	Foot raised, about 20–30°

Tilt angles differ between books (raising the foot of the bed by about 35–45 cm). The direction of lying is what matters most.

Exam point: right lower lobe (W16, S19)

Drain the right lower lobe in three positions: prone with the foot raised (posterior basal), lying on the **left** side turned a quarter forward with the foot raised (lateral basal), and prone flat with pillows under the hips (superior segment). Do ACBT in each position.

Contraindications to head-down postural drainage (S18: any 6)

- Recent or active haemoptysis
- Raised intracranial pressure, recent head injury or neurosurgery
- Uncontrolled hypertension, unstable heart condition, arrhythmia, pulmonary oedema or heart failure
- Severe breathlessness or bronchospasm
- Gastro-oesophageal reflux, hiatus hernia, recent oesophageal surgery, just after meals
- Aortic aneurysm, recent eye surgery
- Gross obesity, abdominal distension, late pregnancy
- Untreated pneumothorax, large pleural effusion or empyema, rib fractures, painful recent surgery

Use modified (flat side-lying or sitting) positions instead.

3. Exercise and pulmonary rehabilitation

- Endurance and strength training as in COPD pulmonary rehab (6–12 weeks, at least twice a week supervised). Exercise also loosens sputum.
- Do airway clearance before exercise so the airways are clear.
- Inspiratory muscle training if the breathing muscles are weak.

4. Pelvic floor and other measures

- **Pelvic floor muscle training:** slow holds (up to 10 seconds × 10) and quick contractions, three times a day; tighten before coughing or huffing ("the knack").
- Breathing control and posture; thoracic mobility exercises.
- Drink enough water to keep sputum thin.

Ergonomic and energy-conservation advice for left lung bronchiectasis with a normal right lung (7 marks in W22)

- **Sleep and rest lying on the right side.** This puts the diseased left lung uppermost, so it drains, and the healthy right lung lowermost, where it gets more blood flow and better gas exchange.
- Clear the chest in the morning and before work or travel; plan a short huffing break during the day in private.
- Sit upright at work with the desk at elbow height; avoid long slumped sitting, which squashes the lower chest.
- Pace tasks, sit for tasks such as cooking and ironing, keep things within reach, use trolleys and avoid heavy lifting (as in COPD energy conservation).
- Avoid dust, smoke and crowded places during infections; wear a mask.

By impairment

Impairment	Technique and dose	Why
Excess sputum	ACBT or autogenic drainage in the lobe position, 10–30 minutes, 1–2 times a day; PEP device	Breaks the vicious cycle at mucus retention
Thick sputum	Nebulised saline before clearance (as prescribed); hydration	Thinner mucus moves more easily
Breathlessness, low exercise tolerance	Breathing control; endurance training 3–5 days a week at moderate breathlessness	Better fitness, less breathlessness
Stress incontinence	Pelvic floor training 3 times a day; tighten before coughing	Supports the bladder neck during raised abdominal pressure
Reduced chest expansion	Thoracic expansion exercises, mobility exercises	Improves ventilation of the affected lobe

Impairment	Technique and dose	Why
Anxiety, social withdrawal	Education, routine before outings, group rehab	Confidence and control over symptoms

Recall check. Write the ACBT cycle and the drainage positions for the right lower lobe and middle lobe.

13 Precautions, contraindications and stop criteria

Take care

- Haemoptysis: no percussion, vibration or head-down positions until bleeding settles.
- Head-down tilt: check the contraindication list above; avoid straight after meals.
- Bronchospasm: give the bronchodilator before clearance; avoid forced coughing bouts.
- Infection control: dispose of sputum safely; separate patients with *Pseudomonas* or other resistant organisms; clean devices daily.
- Past TB: confirm the patient is no longer infectious.

Stop and report if

- Fresh bleeding, more than streaks
- SpO_2 falls, severe breathlessness or wheeze during treatment
- Sudden chest pain (possible pneumothorax)

14 Education, lifestyle and home programme

- Do airway clearance every day, even on good days. Increase it when you have more sputum.
- Keep a sputum diary; know your action plan for exacerbations (more sputum, darker colour, fever).
- Stop smoking; avoid smoke and dust.
- Exercise every day; it helps you clear sputum and stay fit.
- Clean nebulisers and devices daily; wash hands.
- Do pelvic floor exercises daily.
- Get influenza and pneumonia vaccines.

15 Prognosis and discharge criteria

Prognosis varies widely. It is worse with *Pseudomonas* infection, frequent exacerbations, low FEV1, many lobes involved and low body weight. Daily airway clearance and exercise reduce symptoms and exacerbations.

Discharge from an episode of physio: independent and confident with the chosen airway-clearance technique and positions; daily exercise routine; knows the action plan; pelvic floor programme in place.

16 Exam corner

Past questions (MUHS, 2016–2026)

S16 Q3(b), 5 + 5 + 5 marks

35-year-old male labourer, middle lobe bronchiectasis, cough with sputum and breathlessness. Functional diagnosis; short- and long-term goals; physiotherapy management.

S19 Q3(a), 5 + 10 marks

The worked example in section 10.

S20 Q3(a), 5 + 5 + 5 marks

40-year-old male tailor, post-Koch's bronchiectasis of the right lung, copious mucopurulent sputum, breathless on exertion; attending pulmonary rehab for a week. Impairments, activity and participation; short- and long-term management.

S21 Q3(a), 7 + 8 marks

34-year-old male smoker, post-PTB bronchiectasis, mMRC 3, copious sputum; heterogeneous opacities throughout with the trachea shifted to the left. ICF and physiotherapy management.

W24 Q3(a), 15 marks

35-year-old man with post-TB bronchiectasis, recently married; foul-smelling sputum for 2 days; cannot lie on his right side because it brings up more sputum; embarrassed to meet people and feels unwell. Functional diagnosis and physiotherapy. Clue: lying on the right side drains the left lung, so the disease is mainly on the left.

7-mark questions

Physiotherapy for right lower lobe bronchiectasis (W16). ACBT vs autogenic drainage in bronchiectasis (W17). Physiotherapy management of bronchiectasis (W18). Ergonomic and energy-conservation advice, left lung bronchiectasis with a normal right lung (W22). Mucus removal techniques in chronic respiratory disease (W19).

3-mark question

Clinical features of bronchiectasis (W19, W20).

Technique cards for this topic: airway clearance (p. 258), ACBT and autogenic drainage (p. 261), postural drainage (p. 267), PEP devices (p. 271), pulmonary rehabilitation (p. 339).

Keywords to underline

permanent dilatation, vicious cycle, post-tuberculous, airway clearance, ACBT, forced expiration technique, autogenic drainage, oscillating PEP, gravity-assisted position, affected lobe uppermost, pelvic floor, action plan

Answer skeleton: bronchiectasis long answer

1. Restate: age, job, cause (TB), lobe, sputum, breathlessness grade, X-ray detail.
2. **[ICF or reasons, 5–7]** ICF diagram; vicious cycle diagram; reason for each impairment.
3. **[Short-term, 5]** Airway clearance plan for that lobe: technique, positions, dose; nebulisation; breathing control; pelvic floor if leakage.
4. **[Long-term, 5–8]** Pulmonary rehab doses; home routine; devices; energy conservation; action plan; outcome measures.

Draw these

- The vicious cycle (section 4)
- The drainage position for the lobe in the question (section 12)
- The ACBT cycle as a loop: breathing control → expansion → breathing control → huff

Viva questions

1. Why is TB a common cause of bronchiectasis in India?
2. What is the difference between a huff and a cough?
3. How do you drain the middle lobe?

4. Name six contraindications to head-down tilt.
5. Why do bronchiectasis patients develop stress incontinence?
6. Which side should a patient with left lung disease lie on, and why?

Linked practical spots

- Chest X-ray: tram lines and ring shadows; trachea pulled to the side of a fibrosed lung.
- Devices: flutter, Acapella, PEP mask, nebuliser.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Clinical features of bronchiectasis.**

Answer
Chronic cough with copious purulent sputum (worse in the morning and with position change); recurrent infections; haemoptysis; breathlessness and fatigue; clubbing; coarse crackles over the affected area; halitosis.

7 m **Physiotherapy management of right lower lobe bronchiectasis.**

Answer
Aims; assessment points; airway clearance (ACBT steps) in the right lower lobe positions (prone foot-up, left side-lying quarter forward foot-up, prone flat with pillows); nebulisation before; PEP devices; contraindications to tilt; exercise training; breathing control; education and action plan.

LAQ **S21: trachea shifted to the left with opacities throughout. What is the likely problem, and what does it change?**

Answer
The trachea is pulled towards a shrunken, fibrosed or collapsed lung, so the **left lung** is likely destroyed after TB. Expect reduced left chest expansion and ventilation. Drain the remaining affected areas, use breathing control, and expect lower exercise tolerance (mMRC 3). Oxygen may be needed on exercise.

MCQ **To drain the right middle lobe, the patient lies:**

A. On the right side, foot of bed raised

B. On the left side, turned a quarter back, foot of bed raised about 15°

C. Prone, flat

D. Sitting upright

Answer
B. The middle lobe is on the right and in front, so the patient lies on the left side, rolled slightly back, with the foot of the bed raised.

MCQ **A bronchiectasis patient coughs up fresh blood during a session. What do you do first?**

A. Increase percussion to clear it

B. Stop percussion and head-down tilt, sit the patient up, report

C. Continue with vibrations only

D. Start a walk test

Answer

B. Haemoptysis is a contraindication to percussion, vibration and head-down positions.

MCQ **In ACBT, the huff is done mainly to:**

A. Increase lung volume

B. Move secretions from smaller to larger airways with less airway collapse than a cough

C. Relax the patient

D. Measure peak flow

Answer

B. A medium-volume huff produces airflow that shears mucus from peripheral airways.

Last recall check. Write the vicious cycle, ACBT, and three right-lower-lobe positions. Then check them against this page.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

1. MUHS BPTh syllabus (2012), Cardio-Vascular & Respiratory Physiotherapy. muhs.ac.in
2. MUHS IV BPTh question papers, code 51413, Summer 2016 to Summer 2026: muhsonline.com (unofficial archive, 2016–2022) and the Dhole Patil College of Physiotherapy past-paper page (2019–2026)
3. European Respiratory Society guidelines for the management of adult bronchiectasis (2025): strong recommendations for airway clearance and pulmonary rehabilitation. European Lung Foundation summary.
4. Hill AT, et al. British Thoracic Society guideline for bronchiectasis in adults. *Thorax* 2019;74(Suppl 1):1–69.
5. Bott J, et al. BTS/ACPRC guidelines for the physiotherapy management of the adult, medical, spontaneously breathing patient. *Thorax* 2009.
6. Cystic Fibrosis Foundation. An Introduction to Postural Drainage and Percussion (consumer fact sheet), positions and table elevations.
7. Structure and classical content: Main & Denehy, *Cardiorespiratory Physiotherapy*, 5th ed. (2016); Frownfelter & Dean (2022); Pryor & Prasad, *Physiotherapy for Respiratory and Cardiac Problems* (2008).

Amputation and stump care

> Amputation was in 11 of 20 papers (2016–2026), usually a below-knee amputation in a diabetic or vascular patient who is the family's only earner: a driver, a milkman, a labourer. S23 asked about an above-knee amputee who lives alone. Answer with ICF, stump care, contracture prevention, gait training and a clear route back to work.

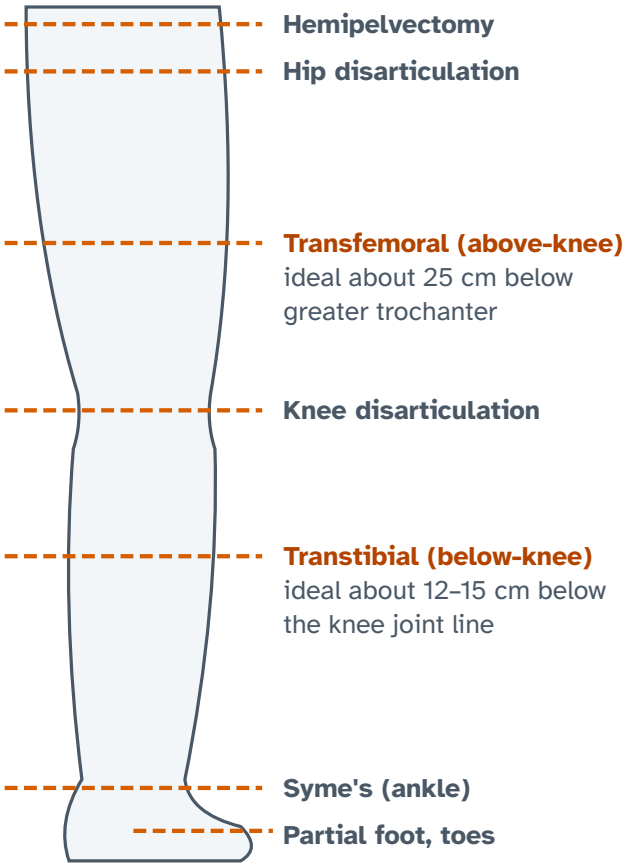
Syllabus	Ch 6(h) Diabetic and vascular ulcers / amputations (stump care only)
Priority	**Tier 1.** In 11 of 20 papers (2016–2026). 6 long-answer cases.
Asked as	LAQ S18 · S19 · S21 · W19 · S23 · S25 · 7m W16 · S17 · W20 · S26 · 3m W18
Usual split	7 + 8 (ICF, management), or 15 (ICF and management; short- and long-term goals and return to job)
Checked against	BACPAR guidance on pre- and post-operative physiotherapy for lower-limb amputees (2016); published data on residual limb length; Medicare functional classification (K-levels). Full list under Sources. Content checked 19 October 2026.

1 Definition

Amputation is the surgical removal of all or part of a limb. The part that remains is the **residual limb** (stump). In this syllabus the focus is lower-limb amputation for vascular disease and diabetes, and stump care.

2 Classification: levels

Lower-limb amputation levels



The higher the level, the more energy walking takes and the harder prosthetic use becomes. Keeping the knee is the single biggest advantage.

Ideal stump (3 marks in W18)

- **Adequate length:** transtibial about 12–15 cm below the knee joint line (longer in tall people); transfemoral about 25 cm from the greater trochanter, leaving room above the knee for a prosthetic knee joint.
- **Conical (cylindrical) shape**, firm, not bulbous or flabby.
- Well-healed, mobile scar that is not adherent and not over the weight-bearing end.
- Good soft-tissue padding over the bone ends; bone ends smooth and bevelled (in transtibial, the fibula a little shorter than the tibia).
- Muscles anchored to bone (myodesis) so they stay strong and the stump is stable.
- Good circulation and sensation; no neuroma or painful spots.
- **Full range of the joint above**, with no contracture.

Functional level (K-levels)

K-level	Ability
K0	Cannot walk or transfer safely with a prosthesis
K1	Transfers and walks on level surfaces at a fixed speed (household walker)
K2	Manages low barriers such as kerbs, stairs and uneven ground (limited community walker)
K3	Walks at varying speed in the community; can work or exercise
K4	High-impact activity beyond basic walking (children, athletes)

3 Causes and risk factors

- **Peripheral arterial disease and diabetes:** the commonest cause in adults (S18 diabetes with gangrene; S19 uncontrolled diabetes and hypertension; S21 PAD).
- **Trauma:** road and rail accidents, machinery, crush injuries; the commonest cause in young people.
- **Infection:** gas gangrene, severe diabetic foot infection.
- **Tumours** of bone or soft tissue.
- Burns, frostbite, congenital limb deficiency.

4 Pathophysiology to impairments

1. Loss of the limb removes the ankle and foot (and the knee in higher levels), so balance, push-off and shock absorption are lost.
2. The stump swells after surgery (oedema). It must shrink and shape before a socket can fit.
3. Sitting and lying with a pillow under the stump pull it into the strongest flexors' position: knee flexion after transtibial, hip flexion, abduction and external rotation after transfemoral. These **contractures** make prosthetic walking hard or impossible.
4. Cut nerves continue to send signals: **phantom limb sensation** (feeling the missing part) and **phantom limb pain**; a neuroma at a nerve end causes local pain.
5. Bed rest and underlying disease cause weakness and low fitness. Walking with a prosthesis uses much more energy, especially at higher levels and in patients with heart disease.
6. In vascular patients the **other leg** often has the same disease and is at risk.

Recall check. Which contractures follow transtibial and transfemoral amputation, and why?

5 Clinical features after amputation

- Stump pain and swelling; wound and drains in the first days.
- Phantom sensation (common) and phantom pain (burning, cramping, shooting in the missing limb).
- Reduced balance on one leg; risk of falls, especially at night when getting up to the toilet.
- Weakness of hip extensors and abductors and knee extensors; reduced fitness.
- Grief, anxiety about work and income, altered body image.

6 Investigations

Test	What it shows	Physio meaning
Pulses, ABI, Doppler of both legs	Blood supply (S21: right popliteal 1+, left foot pulses 2+)	Level choice; protect the remaining leg
Blood sugar, HbA1c, kidney function	Diabetes control	Uncontrolled sugar delays healing
Haemoglobin, albumin	Anaemia, nutrition	Affect wound healing and endurance
ECG, echo	Heart disease	Limits exercise intensity

7 Medical and surgical management

- The level is chosen to remove all dead or infected tissue while keeping as much length as will heal, ideally keeping the knee.
- Transtibial for vascular disease usually uses a **long posterior flap**; muscles are anchored to bone (myodesis) or to each other (myoplasty).

- Post-op: wound care, pain relief (including for phantom pain), control of diabetes and blood pressure, nutrition.
- Dressings: soft dressings with bandaging, a rigid removable dressing, or a shrinker sock once the wound has healed.
- Prosthetic fitting once the wound has healed and the stump volume has stabilised (often several weeks).

Prosthesis basics (transtibial; W19: suitable prosthesis with reasoning)

- **Socket:** patellar-tendon-bearing (PTB) or total-surface-bearing.
- **Suspension:** supracondylar cuff strap, sleeve, pin-lock liner or suction.
- **Pylon** (shank) and **foot:** SACH (solid ankle cushion heel) foot or, in India, the **Jaipur foot**, a flexible rubber foot that allows squatting, sitting cross-legged and walking barefoot on uneven ground.

Choosing for a diabetic patient with vascular disease (W19): his skin heals poorly and may be numb, so choose a socket that spreads pressure widely: a **total-surface-bearing socket with a gel or silicone liner** (less shear on fragile skin), or a PTB socket with a soft foam liner where cost matters. Use a secure, comfortable suspension (sleeve or pin-lock liner) and a light endoskeletal pylon. He is likely to be a limited community walker, so a stable **SACH or Jaipur foot** suits him; the Jaipur foot fits Indian floor-level living. Before the definitive limb, an early walking aid in physiotherapy sessions and an interim prosthesis help gait and shape the stump. Teach daily checks of the stump and the other foot.

8 Complications and prevention

Complication	What to look for	Physio role
Contractures	Loss of knee or hip extension	Positioning, prone lying, stump board, stretching, strengthening
Oedema, poor stump shape	Bulbous, swollen end	Correct bandaging or shrinker, exercise, elevation
Delayed healing, infection, wound breakdown	Discharge, redness, gaping wound	Report; protect the wound during exercise; sugar control
Phantom pain, neuroma	Pain in the missing limb; sharp tender spot	Desensitisation, mirror therapy, TENS, early prosthesis
Falls	Getting up at night forgetting the limb is gone	Balance training, safe transfers, night-time plan
Skin problems in the socket	Redness that lasts, blisters, sores	Daily skin checks, socket fit review
Problems in the other leg	Ulcer, gangrene, overuse pain	Foot care, footwear, avoid hopping long distances
Deconditioning, DVT, chest infection	Weakness, calf swelling, breathlessness	Early mobilisation, breathing exercises, exercise programme
Depression	Low mood, loss of role	Goal setting, peer amputees, counselling

9 PT assessment

- **History:** cause, level, date, side, other illnesses (diabetes, BP, heart), occupation and its demands (driving, walking door-to-door, lifting), home (floors, steps, toilet type), family support and earning role, previous walking level.
- **Stump:** length (from a bony landmark), shape, circumference at fixed points, wound, scar mobility, skin, sensation, temperature, pain, neuroma.
- **Range and strength:** hip and knee range on both sides (look for contracture), muscle power of both legs, trunk and arms.
- **Other leg:** pulses, skin, sensation, ulcers.
- **Balance and function:** sitting and standing balance, transfers, wheelchair skills, hopping with a walker.
- **Phantom sensation and pain.**

- **Outcome measures:** Amputee Mobility Predictor (AMPPRO), K-level, Timed Up and Go, 2- or 6-minute walk test, Houghton scale, SIGAM mobility grades.

10 PT diagnosis by ICF

Draw the six-box ICF model (see the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (S21, 7 + 8 marks)

> "48 year old driver operated for right BK amputation 7 days back, due to peripheral arterial disease. Popliteal artery on the same side is grade 1+. On the left side the dorsalis pedis and posterior tibial are 2+. Discuss ICF and PT management for him."

Spot the ICF clues before you write

driver with a right BKA cannot use a standard accelerator and brake: participation and an environmental adaptation. 7 days means the early post-op, pre-prosthetic phase. popliteal 1+ means reduced blood supply to the stump, so healing may be slow. left pulses 2+ means the remaining leg is perfused now but must be protected.

ICF component	From this case	Clinical reasoning
Health condition	Right transtibial amputation for PAD, day 7	Pre-prosthetic phase
Body structure	Loss of the right leg below the knee; healing wound; reduced popliteal pulse	Weak pulse means slower healing
Body function impairments	• Stump pain and oedema • Risk of knee flexion contracture • Phantom sensation or pain • Reduced balance on one leg • Weakness of right hip and knee extensors; reduced endurance	• Surgery and dependency cause swelling • Sitting with the knee bent shortens the hamstrings • Cut nerves still signal
Activity limitations	Standing, walking, stairs, toileting, bathing, driving	Lost one support limb; no prosthesis yet
Participation restrictions	Cannot work as a driver; family income; social life	Driving needs a right foot for pedals
Environmental factors	**Facilitators:** prosthetic services (Jaipur foot centres), automatic vehicles or left-foot accelerator conversion, family. **Barriers:** loss of income, possibly an Indian toilet and steps at home.	Plan vehicle adaptation and home changes
Personal factors	48-year-old man with PAD (likely smoker or diabetic), motivated to work	Risk-factor control protects the other leg

One-line functional diagnosis

Right transtibial amputation for PAD, day 7, with stump pain and oedema, risk of knee flexion contracture, poor single-leg balance and reduced strength, limiting walking, transfers and self-care, and restricting his work as a driver. Prosthetic services and vehicle adaptation are facilitators; loss of income is a barrier.

Recall check. Fill the ICF for the S19 milkman (right BKA, uncontrolled diabetes and BP, delivers milk door to door, only earner of five).

11 Problem list and goals

Goals for the S21 driver

Problem	Short-term (pre-prosthetic, 4–6 weeks)	Long-term (prosthetic, 3–6 months)
Stump oedema and shape	Wound healed; conical stump ready for casting	Stable stump volume; good socket fit
Contracture	Full knee extension maintained	Full hip and knee range
Strength and balance	Independent transfers; walks with a walker or crutches (hopping) on level ground	Walks with a prosthesis without aids in the community (K3)
Phantom pain	Pain controlled; desensitisation routine	Minimal phantom pain
Participation	Independent in self-care at home	Drives an adapted or automatic vehicle and returns to work

12 PT management

1. Before surgery (when planned)

- Assess, explain, meet another amputee if possible; teach transfers, wheelchair use and exercises; strengthen the arms and the other leg; check the other foot.

2. Early post-op (days 1–10)

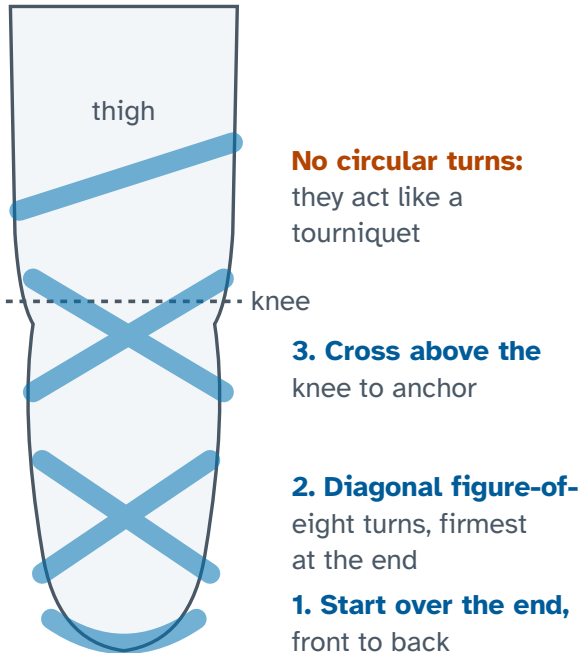
- Breathing exercises, ankle pumps of the other leg, early sitting out.
- **Positioning** (see the box below), prone lying as soon as tolerated.
- **Oedema control**: stump bandaging or shrinker as the team decides; elevation without bending the knee.
- **Exercises:** isometric then active quadriceps, hip extensors and abductors on the amputated side; full exercises for the other leg, trunk and arms.
- **Transfers and wheelchair** (with a stump board to keep the knee straight); standing balance in parallel bars; hopping with a walker over short distances only.

Contracture prevention: positions

- No pillow under the stump, under the knee, or between the thighs.
- Do not let the stump hang over the bed edge or rest on a crutch handle.
- Lie prone for about 20–30 minutes, two or three times a day, if possible.
- Transtibial: keep the knee straight; use a stump board in the wheelchair.
- Transfemoral: avoid long sitting; keep the hip straight and not rolled out or out to the side.

3. Stump bandaging (stump management, 7 marks in W20)

Figure-of-eight bandaging of a below-knee stump



Pressure should be greatest at the end and less towards the knee, so fluid moves upwards and the stump becomes conical.

- Use an elastic bandage about **10 cm** (4 inches) wide for a transtibial stump and about **15 cm** (6 inches) wide for a transfemoral stump.
- Diagonal, figure-of-eight turns; no wrinkles; firmest at the end, decreasing upwards.
- Transtibial: anchor above the knee. Transfemoral: include the groin and anchor round the waist (hip spica).
- Re-apply every 4–6 hours and whenever loose; wear day and night except for bathing and exercise.
- A **shrinker sock** is easier and more consistent once the wound has healed.

4. Stump pain, phantom pain and desensitisation (S25: severe pain at the amputation site)

- **Stump (residual-limb) pain** is felt in the stump itself. Look for a cause first: wound infection, poor blood supply, a tender neuroma, a bony end, or a tight or badly applied bandage. Report infection or ischaemia; give treatment after analgesia.
- **Phantom pain** is felt in the missing part (burning, cramping, shooting). It is common and usually lessens over months.
- Tapping, gentle massage and rubbing the stump with different textures (cotton, towel), several times a day.
- Mirror therapy: watch the intact leg moving in a mirror placed where the missing leg would be.
- TENS; wearing the shrinker or prosthesis; reassurance that phantom sensation is normal.

5. Prosthetic training

- Donning and doffing the prosthesis and socks; daily skin check of the stump.
- Weight shifting in parallel bars, standing on the prosthesis, stepping, then walking with fewer aids.
- Correct common gait deviations (short step, wide base, trunk lean).
- Stairs: up with the sound leg first, down with the prosthesis first ("good goes up, bad goes down").
- Uneven ground, ramps, kerbs, getting up from the floor; squatting and cross-legged sitting with a Jaipur foot.
- Build wearing time gradually (start with short periods, check the skin after each).

Above-knee (transfemoral) amputation (S23: 60-year-old, lives alone, one floor to climb)

- **Contracture risk:** hip flexion, abduction and external rotation. Prone lying 20–30 minutes two or three times a day; no pillow under or between the thighs; limit long sitting; keep the stump beside the other leg in lying.
- **Strengthen:** hip extensors and adductors (they hold the prosthesis under the body), abductors, trunk, arms and the other leg.
- **Stump care:** bandage with a hip spica up to the groin, or a shrinker; desensitisation.
- **Prosthesis:** ischial-containment or quadrilateral socket; suction or liner suspension; for an older, low-activity walker a **stance-control (weight-activated) knee** or a locked knee for safety, with a SACH or Jaipur foot.
- **Energy:** walking with an above-knee prosthesis takes far more energy than with a below-knee one, so build general fitness; in an older diabetic, independent wheelchair life comes first and prosthetic walking is a further goal if fitness, balance and the other leg allow.
- **Living alone, one floor up:** safe transfers, wheelchair skills, getting up from the floor after a fall; stairs with crutches and a rail (up with the sound leg, down with the crutches and the stump first), or sitting and shuffling on the stairs if unsafe; a home visit, rails, a phone within reach, help with shopping, and checks of the other foot.

6. Return to work

- **Driver (S21):** right-leg amputation: an automatic vehicle with a left-foot accelerator, or hand controls; check licensing rules.
- **Milkman going door to door (S19):** endurance walking, stairs, carrying loads with a prosthesis; possibly a two-wheeler with adaptations; plan rest breaks; good diabetes control.

By impairment

Impairment	Technique and dose	Why
Stump oedema	Figure-of-eight bandage or shrinker; isometric exercises	Shapes the stump for the socket
Contracture risk	Positioning; prone lying 20–30 min 2–3 times a day; stretching	Full extension is needed for prosthetic walking
Weakness	Hip extensor and abductor, quadriceps strengthening; arm strengthening for crutches	Controls the prosthesis and the pelvis
Poor balance	Sitting and standing balance, single-leg stance in bars	Reduces falls
Phantom pain	Desensitisation, mirror therapy, TENS	Retrains the brain's body map
Low endurance	Arm ergometer, graded walking	Prosthetic walking costs more energy

Recall check. Write the bandaging rules and five contracture-prevention instructions.

13 Precautions

Take care

- Check the wound before bandaging and exercise; stop if it opens or discharges.
- Vascular stumps heal slowly: no pressure on the wound end until healed.
- Diabetics: check sugar before exercise; inspect the other foot daily.
- Heart disease: monitor HR, BP and breathlessness, because amputee walking is energy-costly.
- Avoid long-distance hopping: it strains the other leg and risks falls.
- Never sleep with the prosthesis or with a tight bandage left unchecked overnight.

14 Education and stump care at home

- Inspect the stump daily with a mirror for redness, blisters or sores.
- Wash the stump daily with mild soap, dry well; wash socks and liner daily.
- Keep the bandage or shrinker on as taught; re-apply when loose.
- Do your positioning and exercises every day.
- Protect the other foot: proper footwear, daily checks, no barefoot walking.
- Control sugar and blood pressure; stop smoking.
- Keep a stable weight: stump volume changes with weight and affects the socket fit.
- At night, keep a walker or wheelchair next to the bed and a light on.

15 Prognosis and discharge criteria

Prognosis: most people with a transtibial amputation who could walk before can walk again with a prosthesis. Outcomes are poorer with transfemoral level, older age, heart and lung disease, and poor diabetes control. Vascular amputees have a high risk of problems in the other leg.

Discharge: safe transfers and wheelchair or walker use at home; stump healed and shaped; full knee and hip range; knows stump and foot care; referred for prosthetic fitting and follow-up.

16 Exam corner

Past questions (MUHS, 2016–2026)

S18 Q3(a), 15 marks

50-year-old man with a left transtibial amputation, referred 1 week after surgery; sole earner of a family of four; diabetes and gangrene. ICF and physiotherapy management.

S19 Q3(b), 15 marks

45-year-old milkman delivering door to door, right BKA days ago; uncontrolled diabetes and hypertension; only earner of five. Short- and long-term goals; how he can return to his previous job.

S21 Q3(b), 7 + 8 marks

The worked example in section 10.

W19 Q4(b), 5 + 7 + 3 marks

50-year-old man with diabetes and peripheral vascular disease, referred 24 hours after a right transtibial amputation for gangrene. Functional diagnosis; short-term programme; a suitable prosthesis with clinical reasoning.

S23 Q4(a), 15 marks

60-year-old man, unilateral above-knee amputation 1 month ago for uncontrolled diabetes; has one floor to climb and lives alone. ICF; short- and long-term physiotherapy. See "Above-knee amputation" in section 12.

S25 Q4(b), 4 + 4 + 7 marks

65-year-old diabetic smoker and manual labourer, below-knee amputation 8 days ago for a non-healing ulcer over the malleoli; severe pain at the amputation site; difficulty walking because of weakness. ICF; short- and long-term goals; physiotherapy.

7-mark questions

Physiotherapy after below-knee amputation (W16, S17, S26). Stump management of BKA (W20).

3-mark question

Characteristics of an ideal stump (W18).

Technique cards for this topic: exercise physiology (p. 313), bed rest (p. 241).

Keywords to underline

transtibial ideal stump figure-of-eight bandaging shrinker knee flexion contracture prone lying stump board phantom pain desensitisation pre-prosthetic Jaipur foot K-level care of the other foot

Answer skeleton: amputation long answer

1. Restate: age, side, level, cause, day, pulses of the other leg, job and family role.
2. **[ICF, 7]** ICF diagram; one-line diagnosis; reasons.
3. **[Management, 8]** Pre-prosthetic: positioning, bandaging, exercises, transfers, balance, phantom pain. Prosthetic: gait training, stairs, uneven ground. Return to job; other-foot care; outcome measures.

Draw these

- Amputation levels (section 2)
- Figure-of-eight stump bandaging (section 12)

Viva questions

1. What are the features of an ideal transtibial stump?
2. Why are circular turns avoided in stump bandaging?
3. Which contracture is common after transfemoral amputation?
4. What is the difference between phantom sensation and phantom pain?
5. What is special about the Jaipur foot?
6. On stairs, which leg goes first going up and going down?

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Characteristics of an ideal stump.**

Answer
Adequate length (transtibial about 12–15 cm below the knee joint line); conical shape; well-healed, mobile scar away from the weight-bearing end; good padding and bevelled bone ends; muscles anchored; good circulation and sensation; full range of the joint above; no neuroma.

7 m **Stump management of below-knee amputation.**

Answer
Aims; wound care; positioning against contracture; bandaging technique with diagram (or shrinker); oedema control; desensitisation and phantom pain; strengthening; skin hygiene and inspection; preparation for prosthesis.

MCQ **After a transtibial amputation, which advice is correct?**

A. Keep a pillow under the knee for comfort

B. Keep the knee straight; use a stump board in the wheelchair

C. Let the stump hang over the bed edge

D. Avoid prone lying

Answer
B. A pillow under the knee causes knee flexion contracture, which makes prosthetic walking very difficult.

MCQ **The correct pressure pattern in stump bandaging is:**

A. Firmest near the knee

B. Even circular turns

C. Firmest at the end, decreasing upwards, in figure-of-eight turns

D. As tight as possible everywhere

Answer
C. A pressure gradient pushes fluid upwards and shapes a conical stump.

MCQ **A diabetic transtibial amputee asks what matters most for his future. Your first priority is:**

A. Cosmetic cover for the prosthesis

B. Daily care and checks of the remaining foot

C. Running training

D. A heavier prosthesis

Answer
B. The other leg is at high risk of ulceration and amputation.

Last recall check. Write the ideal stump, bandaging rules, contracture prevention and the stair rule.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

1. MUHS BPTh syllabus (2012), Cardio-Vascular & Respiratory Physiotherapy. muhs.ac.in
2. MUHS IV BPTh question papers, code 51413, Summer 2016 to Summer 2026: muhsonline.com (unofficial archive, 2016–2022) and the Dhole Patil College of Physiotherapy past-paper page (2019–2026)
3. British Association of Chartered Physiotherapists in Limb Absence Rehabilitation (BACPAR). Pre and post amputation physiotherapy guidelines (2016). Recommendations
4. Residual limb length: cohort data reporting best function with tibial length 12–15 cm below the knee joint line; orthopaedic reference range 12.5–17.5 cm by body height.
5. Gailey RS, et al. The Amputee Mobility Predictor. *Arch Phys Med Rehabil* 2002. Medicare Functional Classification Levels (K-levels).
6. Structure and classical content: Cash's Textbook of General Medical and Surgical Conditions for Physiotherapists; O'Sullivan & Schmitz, *Physical Rehabilitation*.

Varicose veins, venous ulcer, DVT and thrombophlebitis

> Venous disease was in 11 of 20 papers (2016–2026): varicose veins in people who stand all day (a cook, an air hostess, a bus conductor), venous ulcers, and DVT prevention. Every answer rests on one idea: the calf muscle pump, and helping it with exercise, elevation and compression.

Syllabus	Ch 6(a)(vi–vii) Thrombosis, phlebitis and phlebothrombosis; varicose veins and ulcers; vascular ulcers from 6(h)
Priority	**Tier 1.** In 11 of 20 papers (2016–2026). 4 long-answer cases and many 7-mark questions.
Asked as	LAQ S17 · W18 · W21 · S26 · 7m W16 · S18 · W18 · S20 · W20 · S22 · W19 · W21 · S16 (2 tests)
Usual split	5 + 7 + 3 (ICF, management, foot care) or 5 + 5 + 5 (functional diagnosis, management, do's and don'ts)
Checked against	ESVS chronic venous disease guideline (2022); CEAP classification 2020 update; APTA clinical practice guideline on venous thromboembolism (2016, updated 2022). Full list under Sources. Content checked 19 October 2026.

1 Definition

- **Varicose veins:** dilated, twisted superficial veins of the legs, 3 mm or more in diameter when standing, caused by faulty valves.
- **Chronic venous insufficiency:** long-term high pressure in the leg veins (venous hypertension) causing swelling, skin changes and ulcers.
- **Venous ulcer:** a break in the skin of the lower leg, usually above the inner ankle, caused by venous hypertension, that has not healed within about 2 weeks.
- **Deep vein thrombosis (DVT):** a blood clot in a deep vein, usually of the calf or thigh. Part of it can break off and block the lung arteries: **pulmonary embolism (PE)**. DVT and PE together are called venous thromboembolism.
- **Thrombophlebitis** (superficial): inflammation of a superficial vein with a clot in it; a red, tender cord under the skin. **Phlebothrombosis** is the older term for a clot in a deep vein with little inflammation.

2 Classification and grading

CEAP clinical classes (2020)

Class	Signs
C0	No visible signs
C1	Thread veins (telangiectasia) or reticular veins under 3 mm
C2	Varicose veins, 3 mm or more
C3	Oedema
C4	Skin changes: (a) pigmentation or eczema; (b) lipodermatosclerosis (hard, tight skin) or atrophie blanche; (c) corona phlebectatica (fan of small veins at the ankle)
C5	Healed venous ulcer
C6	Active venous ulcer

CEAP also records the **E**tiology (primary, secondary such as after DVT, congenital), **A**natomy (superficial, deep, perforator veins) and **P**athophysiology (reflux, obstruction).

DVT by site

- **Distal** (calf veins only): lower risk of PE.
- **Proximal** (popliteal, femoral, iliac): higher risk of PE and of post-thrombotic syndrome.

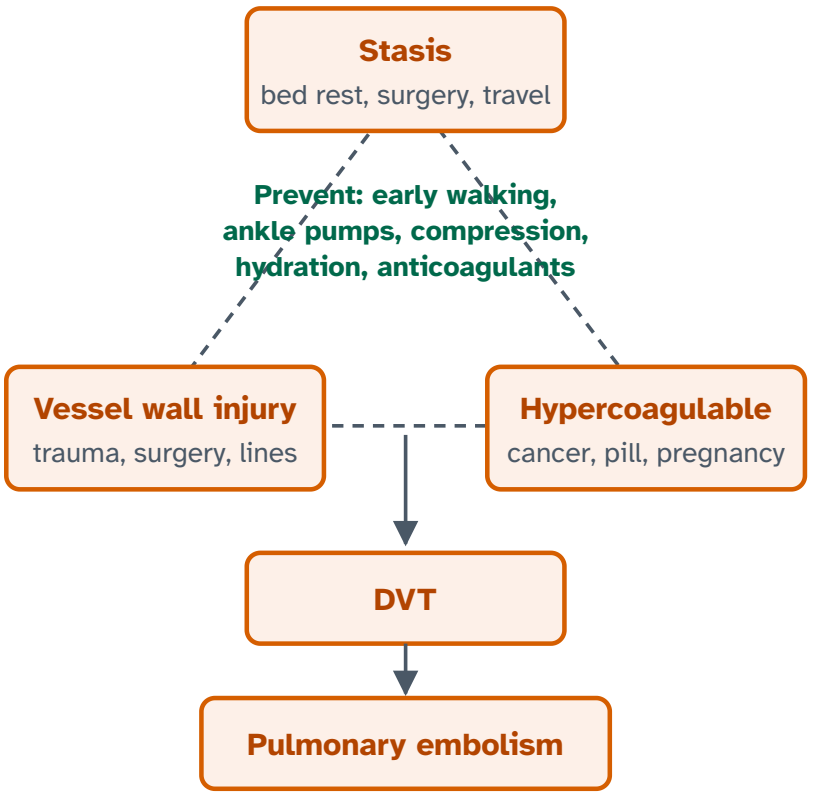
3 Causes and risk factors

Varicose veins and venous insufficiency

- Jobs with **prolonged standing**: cooks (S17), shop and factory workers, teachers (W18), traffic police.
- Female sex, pregnancy, family history, older age, obesity.
- Previous DVT damaging the deep valves (post-thrombotic syndrome).
- Reduced calf pump: inactivity, ankle stiffness, paralysis.

DVT: Virchow's triad

Virchow's triad, DVT and prevention



Any one corner of the triangle raises risk; most patients have two or three.

- **Stasis:** bed rest, surgery (especially hip, knee, abdominal and cancer surgery), long travel, paralysis, heart failure.
- **Vessel wall injury:** trauma, fractures, surgery, central lines, previous DVT.
- **Hypercoagulability:** cancer, pregnancy and the weeks after delivery, oral contraceptive pill and hormone therapy, dehydration, inherited clotting disorders, obesity.

4 Pathophysiology to impairments

The calf muscle pump

When the calf contracts during walking, it squeezes the deep veins and pushes blood up towards the heart. One-way valves stop it falling back. When the calf relaxes, the deep veins refill from the superficial veins through perforating veins.

1. If valves fail (in the saphenous veins, perforators or deep veins), blood falls back down when standing: **reflux**.
2. Pressure in the leg veins stays high even during walking: **venous hypertension**.
3. Superficial veins stretch and twist: **varicose veins**, with aching and heaviness that worsen as the day goes on.

4. Fluid, red cells and proteins leak into the tissues: **oedema**, brown **pigmentation** (haemosiderin), eczema.
5. Chronic inflammation hardens the skin and fat (**lipodermatosclerosis**); the skin breaks down after minor injury: **venous ulcer**.
6. A stiff ankle or weak calf (pain, ulcer, inactivity) makes the pump fail further, so healing slows.

Why the skin breaks down (S26: pathophysiology of varicose ulcer): two classic theories explain the step from venous hypertension to ulcer. In the **fibrin cuff theory**, high capillary pressure leaks fibrinogen, which forms cuffs of fibrin around the capillaries and blocks oxygen and nutrients from reaching the skin. In the **white-cell trapping theory**, white cells stick in the slowed capillaries and release enzymes and free radicals that damage the tissue. Either way the skin of the gaiter area becomes poorly nourished and inflamed, and a minor knock becomes an ulcer that will not heal while the venous pressure stays high.

? Why does walking help but standing still make it worse?

> Standing still lets venous pressure at the ankle build up to the height of the blood column from the heart. Walking makes the calf pump repeatedly, lowering ankle venous pressure. In venous insufficiency this fall is smaller, but exercise that strengthens the calf and keeps the ankle mobile still improves it.

Recall check. Explain, in four steps, how a faulty valve leads to a venous ulcer.

5 Clinical features

Varicose veins and venous insufficiency

- Dull aching, heaviness, tired legs, worse after long standing and in the evening; relieved by elevation (S17 case).
- Ankle swelling by evening; night cramps; itching.
- Visible dilated veins on standing; pigmentation, eczema, hard skin above the ankle.

Don't confuse: three kinds of leg ulcer

Ulcer	Site and look	Pain and position	Pulses, skin, ABPI
Venous	Gaiter area, above the **inner ankle**; shallow, irregular edge, moist, sloughy or red base, lots of exudate	Aching, moderate; **better with the leg raised**	Pulses present; pigmentation, oedema, lipodermatosclerosis; ABPI normal
Arterial	Toes, heel, outer ankle, pressure points; **punched out**, deep, pale dry base	Very painful, worse at night and with the leg raised; **better hanging down**	Weak or absent pulses; cold, shiny, hairless skin; ABPI low
Neuropathic (diabetic)	Sole, under the metatarsal heads or heel; punched out with a thick **callus rim**	Often **painless**	Pulses often present, foot warm and dry; loss of sensation; may be mixed with arterial disease

DVT

- One-sided calf or thigh swelling, pain, tenderness along the deep veins, warmth, redness, dilated superficial veins.
- Often silent, especially after surgery.
- **Pulmonary embolism:** sudden breathlessness, chest pain worse on breathing in, fast heart rate, low SpO_2, coughing blood, collapse.

Superficial thrombophlebitis

A red, hot, tender, hard cord along a superficial vein, often in a varicose vein or at a drip site.

6 Investigations and special tests

"Two special tests to assess venous insufficiency" (S16) is the classic short question.

Test	How	Result
Brodie–Trendelenburg test	Lying, raise the leg to empty the veins. Press over the saphenofemoral junction (or apply a tourniquet high on the thigh). Stand the patient up and watch.	Veins stay empty, then fill fast when pressure is released: **saphenofemoral valve incompetent**. Veins fill fast with pressure still on: **perforators below are incompetent**.
Perthes test	Tourniquet below the knee in standing; patient walks or does heel raises	Varicose veins empty: deep veins open. Veins swell and the leg hurts: **deep veins blocked** (do not strip superficial veins).
Tap (percussion) test	Tap the vein at the knee and feel at the groin	Impulse felt above: valves between are incompetent
Cough impulse	Feel the saphenofemoral junction while the patient coughs	A thrill means an incompetent junction
Duplex ultrasound (gold standard)	Ultrasound of veins with flow	Shows reflux and blocks; confirms DVT
ABPI	As in PAD	Must be done before compression to rule out arterial disease

DVT diagnosis

- **Wells score** to estimate likelihood; **D-dimer** blood test (a normal result helps rule DVT out when likelihood is low); **compression ultrasound** to confirm.
- Suspected PE: CT pulmonary angiogram.

Classic answer vs current practice: Homans' sign

Classic	Current	Write in the exam
Pain in the calf on passive dorsiflexion of the foot (Homans' sign) suggests DVT.	Homans' sign is neither sensitive nor specific and is no longer used to diagnose DVT. Use the Wells score, D-dimer and ultrasound.	"Homans' sign is unreliable; DVT is confirmed by Wells score, D-dimer and duplex ultrasound."

7 Medical and surgical management

Varicose veins

- Conservative: compression stockings, exercise, leg elevation, weight control.
- **Endovenous thermal ablation** (laser or radiofrequency) of the incompetent saphenous vein: first-line in current guidelines.
- Foam sclerotherapy; surgery (high ligation and stripping); removal of side branches (phlebectomy).

Venous ulcer

- **Compression** is the main treatment: about **40 mmHg** at the ankle with multilayer bandages or adjustable wraps, once ABPI shows adequate arterial supply.
- Wound cleaning, simple dressings, antibiotics only for spreading infection (W18 "infected" ulcer).
- Treat the underlying vein (ablation) to speed healing and prevent recurrence.

DVT

- **Anticoagulation**: heparin then oral anticoagulants (for example apixaban, rivaroxaban or warfarin), usually for at least 3 months.
- Clot-dissolving treatment or a vena cava filter in selected cases.

Superficial thrombophlebitis

- Anti-inflammatory drugs, warm compresses, compression, keep walking; anticoagulation if the clot is long or near the groin junction.

8 Complications and prevention

Complication	What to look for	Physio role
Venous ulcer	Skin breakdown above the ankle	Compression, calf-pump exercise, elevation, skin care
Bleeding from a varicose vein	Sudden bleeding after a minor knock	Lie flat, raise the leg, press firmly; seek help
Cellulitis, infected ulcer	Spreading redness, heat, fever	Report; antibiotics; continue gentle movement
Pulmonary embolism	Sudden breathlessness, chest pain	Emergency: stop, sit up, call for help
Post-thrombotic syndrome	Chronic swelling, pain, skin changes after a DVT	Exercise, elevation, compression for symptoms
Ankle stiffness, calf weakness	Reduced dorsiflexion, shuffling gait	ROM and calf strengthening to restore the pump

9 PT assessment

Asked as 7 marks in W19 (assessment of venous insufficiency) and as "evaluation" in the W21 air-hostess case.

- **History:** standing hours at work, pain pattern through the day, swelling, itching, previous DVT, pregnancies, surgery, diabetes (S17), smoking, home stairs.
- **Inspection standing:** varicose veins, oedema, pigmentation, eczema, lipodermatosclerosis, ulcer (site, size, depth, base, edge, exudate, smell).
- **Measurements:** calf and ankle circumference (compare sides), ulcer size (tracing or photograph).
- **Palpation:** temperature, pitting oedema, tenderness, pulses.
- **Special tests:** Trendelenburg, Perthes; ABPI before compression.
- **Ankle ROM** (dorsiflexion), calf strength (number of single-leg heel raises), gait.
- **Sensation** (monofilament) if diabetic.
- **Outcome measures:** Venous Clinical Severity Score, ulcer area, pain scale, quality of life (for example CIVIQ).

10 PT diagnosis by ICF

Draw the six-box ICF model (see the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (S17, 5 + 7 + 3 marks)

> "A 55 year old widow, works as a cook and stays on second floor with no lift facility. She is a known diabetic. She complains of dull aching pain in bilateral lower limb which increases with sustained standing and swelling which increases by the end of the day. She also presents with dilated veins in both her lower limbs. Discuss ICF, physiotherapy management and preventive foot care for her condition."

Spot the ICF clues before you write

cook means prolonged standing, the cause and the main barrier. widow suggests she may be the earner and lives alone. second floor with no lift is an environmental barrier. diabetic raises the risk of foot ulcers, so the add-on is "preventive foot care".

ICF component	From this case	Clinical reasoning
Health condition	Bilateral varicose veins with chronic venous insufficiency (CEAP C2–C3); type 2 diabetes	Valve failure with venous hypertension
Body function and structure impairments	• Dull aching leg pain, heaviness • Evening oedema of both legs • Dilated superficial veins • Reduced calf muscle pump endurance • Risk of skin changes and ulcers (worse with diabetes)	• Standing builds venous pressure; reflux stretches veins • High pressure pushes fluid into tissues by evening • Diabetes slows healing and dulls sensation
Activity limitations	Standing for long to cook, walking long distances, climbing to the 2nd floor	Pain and heaviness build up with time upright
Participation restrictions	Full working day as a cook; household and social roles	Her livelihood depends on standing
Environmental factors	**Barriers:** job with prolonged standing, hot kitchen, 2nd floor without lift, possibly no family help. **Facilitators:** regular work, access to care.	Plan work modifications and breaks
Personal factors	55-year-old widow, diabetic, possibly overweight	Diabetes changes foot-care needs

One-line functional diagnosis

A 55-year-old diabetic cook with bilateral varicose veins and venous insufficiency, causing aching, evening swelling and reduced calf pump endurance, which limit prolonged standing and stair climbing and threaten her ability to work. Prolonged standing at work and a 2nd-floor home are barriers.

Recall check. Fill the ICF for the W18 teacher with infected venous ulcers on the lower third of the leg.

11 Problem list and goals

Goals for the S17 cook

Problem	Short-term (2–4 weeks)	Long-term (3 months)
Pain and heaviness	Pain reduced with stockings and elevation	Works full shifts with little aching
Oedema	Evening ankle circumference reduced	Swelling controlled day to day
Calf pump	Does ankle pumps and heel raises through the day	Normal calf endurance (for example 20 single-leg heel raises)
Skin and feet	Daily foot inspection and moisturising	No ulcers; good foot-care habits

12 PT management

1. Varicose veins and venous insufficiency

- **Calf muscle pump exercises:** ankle pumps 10–20 times every hour while standing or sitting; heel raises 2–3 sets of 10–15; toe walking.
- **Walking** 30 minutes most days.
- **Ankle mobility:** dorsiflexion stretches if stiff.
- **Elevation:** lie with the legs above the level of the heart for about 30 minutes, 3–4 times a day; raise the foot of the bed slightly at night.

- **Compression stockings** (after checking ABPI): put on in the morning before getting up, take off at night.
- **Breathing:** deep diaphragmatic breaths during elevation help venous return.
- **Weight loss** and general fitness.
- **After vein ablation or surgery:** walk the same day, wear compression as advised, avoid long standing or sitting for the first weeks.

2. Venous ulcer (7 marks in W16 and W20)

- **Compression** (by nurse or doctor) about 40 mmHg at the ankle if ABPI allows.
- **Calf-pump exercise and walking** with the bandage on: they work together. Progressive resistance calf exercise helps healing.
- **Ankle ROM** to keep the pump working.
- **Elevation** at rest.
- **Electrotherapy** as an add-on for slow-healing ulcers (see the diabetes and wound-healing page): electrical stimulation, ultrasound, ultraviolet for infected wounds.
- **Skin care** around the ulcer: moisturise, avoid scratching and injury.

Compression and ABPI (guidelines differ in exact cut-offs)

ABPI	Compression
0.8 or more	Full compression (about 40 mmHg at the ankle) is safe
About 0.5–0.8	Reduced compression only, under specialist supervision (mixed arterial and venous disease)
Below about 0.5	No compression; refer to a vascular surgeon

3. DVT: prevention and management (7 marks in S18, W18, S22)

Prevention (anyone immobile or after surgery)

- Early mobilisation and walking.
- Ankle pumps 10 every hour while awake; deep breathing.
- Leg exercises in bed: static quadriceps, gluteal sets, knee bending.
- Hydration; avoid crossing legs and pillows behind the knees.
- Mechanical prophylaxis as prescribed: graduated compression stockings, intermittent pneumatic compression.
- Drug prophylaxis (heparin) as prescribed.

After a DVT is diagnosed

- Check the anticoagulant type and when it started. **Begin walking once anticoagulation has reached its therapeutic level** (APTA guideline, strong recommendation). Bed rest does not lower the risk of PE.
- Progress walking and leg exercises; elevate the leg at rest.
- Compression for symptom relief if swelling and pain persist.
- Watch for signs of PE and bleeding.

By impairment

Impairment	Technique and dose	Why
Pain and heaviness	Elevation 30 min × 3–4 a day; compression stockings; breaks from standing	Lowers venous pressure
Oedema	Elevation, ankle pumps, compression	Fluid returns to veins and lymphatics
Poor calf pump	Heel raises 2–3 × 10–15, walking 30 min a day	A stronger pump empties veins better
Ankle stiffness	Active dorsiflexion and plantarflexion, calf stretch 30 s × 3	A full ankle range is needed for the pump to work

Impairment	Technique and dose	Why
Ulcer	Compression plus exercise; electrotherapy as an add-on	Treats the cause and speeds healing

Recall check. Write the ABPI rule for compression and when a patient with a new DVT can start walking.

13 Precautions, contraindications and stop criteria

Take care

- **Always check ABPI before compression.** Compression on an ischaemic leg can cause gangrene.
- No compression in severe heart failure with fluid overload, severe neuropathy with fragile skin, or acute cellulitis until reviewed.
- No vigorous massage over a leg with a suspected or untreated DVT.
- No heat on an ulcerated or swollen leg.
- Anticoagulated patients bruise and bleed easily: gentle handling.

Stop and call for help if

- Sudden breathlessness, chest pain, fast heart rate or coughing blood (possible PE)
- Toes become pale, blue, cold or numb under a bandage (compression too tight)

14 Education, do's and don'ts, and foot care

"Patient education in varicose veins" (S20, 7 marks), "do's and don'ts" (W18) and "preventive foot care" (S17, 3 marks).

Do

- Walk every day; do ankle pumps and heel raises whenever you stand or sit for long.
- Raise your legs above heart level several times a day; sleep with the foot of the bed slightly raised.
- Wear compression stockings from morning till night; check the skin under them daily.
- Take short sitting or walking breaks at work; shift weight from foot to foot while standing; use a footrest.
- Keep a healthy weight; wear comfortable, low-heeled shoes.
- Moisturise the skin of the lower legs daily.

Don't

- Stand or sit still for long periods; cross your legs.
- Wear tight garters, socks with tight bands or tight clothing at the waist.
- Use hot water bags or sit close to a fire with bare legs.
- Scratch itchy skin or injure the legs.

Preventive foot care (especially with diabetes)

- Inspect feet and legs every day, including soles and between toes.
- Wash in lukewarm water, dry well, moisturise but not between the toes.
- Cut nails straight; never walk barefoot; wear well-fitting footwear with socks.
- Keep blood sugar controlled; report any wound, colour change or swelling early.

15 Prognosis and discharge criteria

Prognosis: varicose veins tend to progress slowly; compression and exercise control symptoms, and vein ablation treats the cause. Venous ulcers usually heal with good compression but often recur unless compression and exercise continue. After a DVT, early walking with anticoagulation is safe; many patients later develop some post-thrombotic symptoms.

Discharge: independent with the exercise and elevation routine; correct use of stockings; knows do's and don'ts and foot care; ulcer healed or improving with a plan.

16 Exam corner

Past questions (MUHS, 2016–2026)

S17 Q3(b), 5 + 7 + 3 marks

The worked example in section 10.

W18 Q3(b), 5 + 5 + 5 marks

50-year-old female schoolteacher with infected venous ulcers on the lower third of the leg, managed conservatively. Functional diagnosis; physiotherapy; do's and don'ts.

W21 Q4(a), 15 marks

40-year-old air hostess, 10 years in the job, with swelling of both feet and calf pain for 3 months; has noticed spider veins in the popliteal fossa. Evaluation; functional diagnosis; management.

S26 Q3(b), 15 marks (three parts)

40-year-old bus conductor with a non-healing, oozing ulcer above the right medial malleolus; pigmentation and swelling of the foot and ankle; dilated, tortuous veins; finds it hard to walk and work. (i) Pathophysiology of varicose ulcers; (ii) goals with ICF; (iii) physiotherapy.

7-mark questions

Physiotherapy of venous ulcers (W16) and varicose ulcers (W20). Prevention and physiotherapy of DVT (S18). Physiotherapy of DVT (W18). Patient education in varicose veins (S20). Complications of abdominal surgery and physiotherapy of DVT (S22). Two special tests for venous insufficiency (S16). Assessment of venous insufficiency (W19). Management of venous ulcers (W21).

Technique cards for this topic: bed rest (p. 241), exercise physiology (p. 313), compression and exercise for circulation (p. 298).

Keywords to underline

calf muscle pump, valve incompetence, venous hypertension, CEAP, Trendelenburg test, Perthes test, ABPI before compression, 40 mmHg, elevation, Virchow's triad, mobilise once anticoagulated

Answer skeleton: venous long answer

1. Restate: age, job with standing, sides, swelling and pain pattern, ulcer site and infection, diabetes.
2. **[ICF or functional diagnosis, 5]** ICF diagram; reasoning (valve failure, venous hypertension, calf pump).
3. **[Management, 5–7]** Assessment (special tests, ABPI); calf-pump exercise and walking with doses; elevation; compression; ulcer care; electrotherapy add-on; post-surgery care.
4. **[Add-on, 3–5]** Do's and don'ts, or preventive foot care.

Draw these

- Virchow's triad (section 3)
- The leg-ulcer comparison table (section 5)
- A leg showing the gaiter area above the inner ankle

Viva questions

1. How does the calf muscle pump work?
2. How do you do the Trendelenburg test, and what does a positive result mean?
3. Why must you check ABPI before compression?
4. Where does a venous ulcer usually occur, and why there?
5. When can a patient with a new DVT start walking?
6. Why is Homans' sign no longer used?

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Two special tests to assess venous insufficiency.**

Answer

1. **Brodie–Trendelenburg:** empty the veins, occlude the saphenofemoral junction, stand: fast filling on release = junction incompetent; fast filling with occlusion = perforators incompetent.
2. **Perthes:** tourniquet below the knee, walk: veins empty = deep veins patent; veins swell with pain = deep vein obstruction.

7 m **Physiotherapy management of venous ulcers.**

Answer

Assessment (ulcer, oedema, ABPI); compression after ABPI; calf-pump exercise and walking with doses; ankle ROM; elevation; electrotherapy add-on; skin care; education and do's and don'ts; prevention of recurrence.

MCQ **A patient with a leg ulcer has ABPI 0.45. What is correct?**

A. Apply full 40 mmHg compression

B. Avoid compression and refer to a vascular surgeon

C. Elevate the leg high all night

D. Apply a hot pack

Answer

B. Severe arterial disease: compression could cut off the blood supply, and elevation worsens arterial pain.

MCQ **A patient with a new calf DVT has reached therapeutic anticoagulation. The best advice is:**

A. Strict bed rest for 1 week

B. Begin walking and leg exercises

C. Calf massage to break up the clot

D. Avoid all leg movement

Answer

B. Early walking is safe once anticoagulated and does not increase PE risk. Massage is contraindicated.

MCQ **A shallow, moist ulcer above the inner ankle with brown pigmentation and good pulses is most likely:**

A. Arterial

B. Venous

C. Neuropathic

D. Pressure ulcer

Answer

B. The gaiter-area site, pigmentation and present pulses are typical of a venous ulcer.

Last recall check. Write the calf-pump explanation, two special tests, the ulcer comparison table and the DVT walking rule.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

1. MUHS BPTh syllabus (2012), Cardio-Vascular & Respiratory Physiotherapy. muhs.ac.in
2. MUHS IV BPTh question papers, code 51413, Summer 2016 to Summer 2026: muhsonline.com (unofficial archive, 2016–2022) and the Dhole Patil College of Physiotherapy past-paper page (2019–2026)
3. De Maeseneer MG, et al. European Society for Vascular Surgery (ESVS) 2022 Clinical Practice Guidelines on the Management of Chronic Venous Disease of the Lower Limbs. *Eur J Vasc Endovasc Surg* 2022.
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5. Hillegass E, et al. Role of physical therapists in the management of individuals at risk for or diagnosed with venous thromboembolism: evidence-based clinical practice guideline. *Phys Ther* 2016; updated 2022. APTA
6. Research summary on compression thresholds: ABPI and compression application review, *Wound Practice and Research*.
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Asthma in adults and children

> Asthma was in 10 of 20 papers (2016–2026): an acute attack in a farmer, a smoker who is the sole earner, a teacher with allergic attacks, and children kept from school and play. Learn the acute plan, the long-term plan built on breathing retraining and exercise, and how to counsel parents and teachers.

Syllabus	Ch 6(b)(iii) Bronchial asthma; 6(d)(v) Childhood asthma
Priority	**Tier 1.** In 10 of 20 papers (2016–2026). 5 long-answer cases, 2 of them children.
Asked as	LAQ W18 · W20 · W22 · W19 · S23 · 7m S17 · S18 · S22 · W21 · S25
Usual split	5 + 5 + 5 (functional diagnosis or ICF, short-term, long-term) or 5 + 7 + 3 (counselling, rehab, outcome measures)
Checked against	GINA 2026 strategy report; BTS/SIGN asthma guideline (acute severity); Cochrane reviews of breathing exercises (2020) and physical training in asthma. Full list under Sources. Content checked 19 October 2026.

1 Definition

Asthma is a chronic inflammatory disease of the airways with **variable** symptoms (wheeze, breathlessness, chest tightness, cough) and **variable airflow limitation** that changes over time and in intensity, often reversing with treatment or on its own.

An **acute exacerbation** (attack) is a worsening of symptoms and lung function beyond the person's usual, needing a change in treatment.

2 Classification and grading

Types

- **Allergic (extrinsic):** often starts in childhood; linked to eczema, allergic rhinitis and family history.
- **Non-allergic (intrinsic):** often adult-onset.
- **Occupational asthma:** caused by agents at work (flour, grain dust, chemicals). Relevant for the farmer cases.
- **Exercise-induced bronchoconstriction:** narrowing during or after exercise.
- **Asthma–COPD overlap** in long-term smokers (W20 case).

Severity of an acute attack (BTS/SIGN)

Level	Features
Moderate	Worse symptoms; peak flow more than 50–75% of best or predicted; no severe features
Acute severe	Any one: peak flow 33–50%; breathing rate 25 or more per minute (adult); heart rate 110 or more; cannot complete sentences in one breath
Life-threatening	Any one: peak flow below 33%; SpO_2 below 92%; silent chest; cyanosis; poor breathing effort; exhaustion; confusion; low BP; arrhythmia; a normal or rising CO_2

Asthma control

Asthma is well controlled when daytime symptoms and reliever use are rare, there is no night waking and no activity limitation. The **Asthma Control Test** (ACT, adults) and **Childhood ACT** score this.

3 Causes, risk factors and triggers

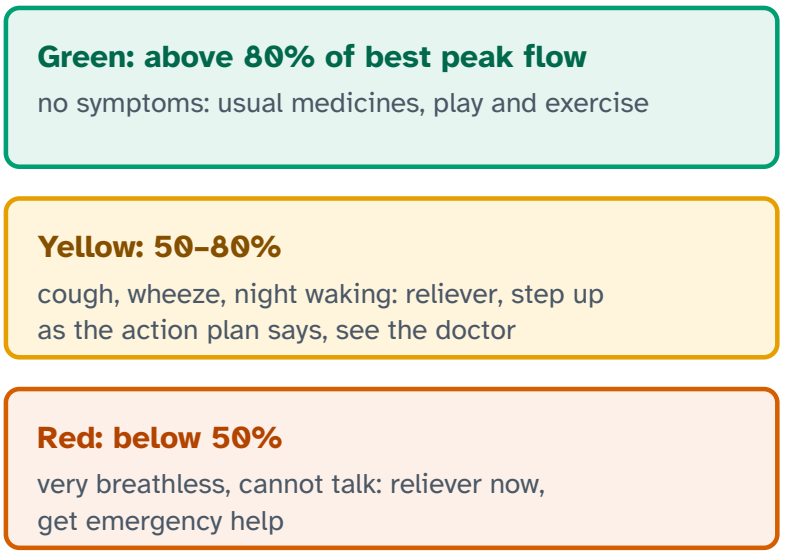
- **Risk factors:** family history, atopy (eczema, hay fever), obesity, smoking and passive smoke, prematurity and early viral infections.

- **Triggers:** respiratory infections; house dust mites, pollen, mould, animal fur, cockroaches; smoke, chulha smoke, mosquito coils, incense; cold air and weather change; exercise; strong smells; emotional stress; some medicines (aspirin, beta-blockers); occupational dusts.

4 Pathophysiology to impairments

1. A trigger activates inflammatory cells in the airway wall (mast cells, eosinophils).
2. Three things narrow the airways: **bronchospasm** (smooth muscle contraction), **mucosal swelling** and **thick mucus plugs**.
3. Airflow is obstructed mainly on breathing out, so air is trapped: hyperinflation, wheeze and breathlessness.
4. Uneven ventilation lowers oxygen; at first CO_2 falls because the patient breathes fast. A **normal or rising CO_2 in an attack is a danger sign** of tiring.
5. Repeated inflammation thickens the airway wall over years (remodelling), so some obstruction can become fixed.
6. Fear of attacks leads to avoiding exercise, so fitness falls and breathlessness rises (as in COPD). Some patients also develop a dysfunctional, fast upper-chest breathing pattern.

Asthma action plan: peak-flow zones



Traffic-light zones are a common way to write a personal action plan; the doctor sets the exact medicines for each zone.

Recall check. Name the three causes of airway narrowing in asthma and the danger sign on an ABG during an attack.

5 Clinical features

- Episodic wheeze, breathlessness, chest tightness and cough, often worse at night or early morning, with triggers.
- Symptoms come and go; the patient may be completely well between attacks (W22 girl).
- **During an attack:** fast breathing, use of accessory muscles, prolonged expiration with wheeze, difficulty speaking, anxiety; a "silent chest" means very little air is moving (life-threatening).
- Children: cough at night or with play, avoiding running games, missing school.

Don't confuse: asthma and COPD

Feature	Asthma	COPD
Onset	Often childhood or young adult	Usually over 40, after years of smoke
Symptoms	Variable, episodic, normal between attacks	Persistent and slowly progressive

Feature	Asthma	COPD
Reversibility	Large (FEV1 rises more than 12% and 200 mL)	Little; FEV1/FVC stays below 0.70
Allergy	Common	Uncommon

6 Investigations

Test	What it shows	Physio meaning
Spirometry with bronchodilator	Obstruction with a rise in FEV1 of more than 12% and more than 200 mL after a reliever (adults)	Confirms variable obstruction
Peak expiratory flow diary	Variability between morning and evening and day to day	Basis of the action plan; outcome measure
Exercise challenge	Fall in FEV1 after exercise	Confirms exercise-induced bronchoconstriction
FeNO, blood eosinophils, allergy tests	Type of airway inflammation, triggers	Trigger avoidance advice
ABG and SpO_2 in an attack	Low oxygen; low CO_2 early, normal or high CO_2 if tiring	Rising CO_2 means call the team at once
Chest X-ray	Usually normal or hyperinflated; rules out pneumothorax and pneumonia	Pneumothorax changes management

7 Medical management

- **Inhaled steroid-containing treatment is the base for everyone.** GINA no longer recommends a short-acting reliever such as salbutamol on its own at any step.
- Adults and adolescents: the preferred plan (GINA Track 1) uses **low-dose ICS–formoterol** as the reliever, and also as daily maintenance at higher steps.
- Add-on treatment for severe asthma: long-acting antimuscarinics, leukotriene antagonists, biologic injections.
- **Inhaler technique** and use of a **spacer** (with a mask for young children) are essential.
- **Acute attack:** repeated reliever by spacer or nebuliser, oral steroids, controlled oxygen (BTS target SpO_2 94–98%), ipratropium and magnesium in severe attacks; ICU and ventilation if life-threatening.
- Avoid triggers; stop smoking; treat rhinitis; written action plan.

The W22 girl takes only SOS salbutamol and has monthly attacks. Under current guidance this needs medical review for a steroid-containing plan. As a physio, encourage that review while you treat.

8 Complications and prevention

Complication	What to look for	Physio role
Life-threatening attack, respiratory failure	Silent chest, exhaustion, confusion, SpO_2 below 92%	Recognise, sit up, oxygen, call for help
Pneumothorax	Sudden chest pain and breathlessness	Stop treatment; report
Mucus plugging, collapse	Reduced breath sounds over a lobe	Airway clearance once bronchospasm settles
Deconditioning, obesity	Avoids activity, weight gain	Graded exercise, sport participation
Dysfunctional breathing	Sighing, upper-chest breathing, light-headedness	Breathing retraining
Steroid side effects	Weight gain, bone loss with frequent oral steroids	Weight-bearing exercise; good control reduces oral steroid courses
Missed school or work, anxiety	Absenteeism, fear of play	Education, counselling, confidence building

9 PT assessment

- **History:** triggers, frequency of attacks and admissions, night symptoms, reliever use, medicines and inhaler technique, smoking, occupation and dust, home environment, school or work absence, sport and play.
- **Observation:** breathing pattern and rate, accessory muscles, posture, ability to speak in sentences.
- **Auscultation:** wheeze, prolonged expiration, crackles.
- **Peak flow** and SpO_2; ABG in severe attacks.
- **Exercise capacity:** 6MWT or shuttle test (children: shuttle or step test), with SpO_2 and peak flow before and after.
- **Breathing pattern questionnaire** (Nijmegen) if dysfunctional breathing is suspected.
- **Outcome measures:** ACT or Childhood ACT, Asthma Quality of Life Questionnaire (PAQLQ for children), peak-flow diary, exercise test, school or workdays missed, attacks per year.

10 PT diagnosis by ICF

Draw the six-box ICF model (see the IHD page (p. 15)), then fill it from the case.

Worked example from a real paper (W22, 5 + 7 + 3 marks)

> "A seven year old girl diagnosed as having bronchial asthma since one year is referred for physiotherapy. This condition has affected her attendance at school as she gets breathless during acute exacerbation and her mother carries her bag to school every day. Her parents are worried about sending her to play with her friends and insist on playing indoor sedentary games. Her teachers also do not allow her to participate in outdoor play due to fear of breathlessness. The girl however wishes to play. She is prescribed SOS salbutamol inhaler and she is absolutely fine between acute episodes which come once a month. (i) Discuss the content of parent and teacher counseling. (ii) Plan rehabilitation program for this girl. (iii) Which outcome measures will show her improvement following rehabilitation?"

Spot the ICF clues before you write

mother carries her bag, parents insist on indoor sedentary games and teachers do not allow outdoor play are environmental **barriers** created by fear. The girl wishes to play is a personal facilitator. fine between episodes means good potential. SOS salbutamol only means she needs medical review.

ICF component	From this case	Clinical reasoning
Health condition	Bronchial asthma, partly controlled, monthly exacerbations	Variable airway obstruction
Body function impairments	Episodic breathlessness and wheeze; reduced exercise tolerance from inactivity; possible poor breathing pattern; anxiety about attacks	Bronchospasm in attacks; deconditioning between them
Activity limitations	Running and active play, carrying her school bag, climbing stairs during attacks	Breathlessness, and being stopped by adults
Participation restrictions	Missed school days; excluded from outdoor play and sports; reduced time with friends	Attacks plus over-protection
Environmental factors	**Barriers:** fearful parents and teachers, reliever-only medicine. **Facilitators:** caring family, school, health services.	Counselling turns barriers into facilitators
Personal factors	7-year-old girl who wants to play	Motivation for a play-based programme

One-line functional diagnosis

A 7-year-old girl with partly controlled asthma and monthly attacks, with episodic breathlessness and reduced fitness, limiting active play and carrying her bag, and restricting school attendance and outdoor play with friends. Fearful parents and teachers are the main barrier; her wish to play is a facilitator.

Recall check. Fill the ICF for the W20 farmer: 45 years, known asthmatic, admitted with wheeze and sputum, smoker, married, sole earner.

11 Problem list and goals

Goals for the W22 girl

Problem	Short-term (4–6 weeks)	Long-term (3–6 months)
Knowledge and fear	Parents and teacher understand asthma, triggers and the action plan	She plays outdoors with friends with adult support
Inhaler technique	Correct use of inhaler with spacer, checked by the physio	Uses the plan independently
Breathing and fitness	Uses relaxed nose breathing and breathing control; warms up before play	Better shuttle or 6-minute walk result; joins sports
School	Carries her own (lighter) bag	Fewer school days missed; fewer attacks

12 PT management

1. Acute exacerbation (7 marks in S22; W18 and W20 cases)

- **Positioning:** high sitting or forward-lean sitting with arms supported, to ease the work of breathing.
- **Breathing control and relaxation:** slow, gentle tidal breathing with the lower chest; relaxed shoulders; reassure and calm. Avoid deep, forced breathing during bronchospasm.
- Oxygen and nebulised bronchodilator as prescribed; time airway clearance **after** the bronchodilator.
- **Airway clearance** once bronchospasm eases, if mucus is present: ACBT with gentle huffs (see below).
- **Avoid percussion and vibration** during acute bronchospasm; they can worsen it.
- Encourage fluids; early mobilisation as the attack settles.
- Before discharge: inhaler technique, action plan, trigger avoidance, smoking cessation, follow-up.

Rationale of ACBT in asthma (7 marks in S17)

- **Breathing control** between other parts of the cycle prevents bronchospasm and fatigue; it is the "rest" in the cycle.
- **Thoracic expansion exercises** get air behind mucus plugs through collateral ventilation.
- **Huffs** move mucus without the airway narrowing and coughing fits that repeated forceful coughing can cause.
- The patient controls the pace, so it suits sensitive airways; stop and return to breathing control if wheeze increases.

2. Long-term management

- **Breathing retraining** (for example Papworth or Buteyko-based methods): nose breathing, slower rate, relaxed lower-chest pattern, reduced over-breathing. Improves asthma quality of life and symptoms (Cochrane 2020).
- **Exercise training:** aerobic training 2–3 times a week for at least 8–12 weeks improves fitness and quality of life without worsening asthma. Swimming is well tolerated (warm, humid air).
- **Exercise-induced bronchoconstriction:** warm up for 10–15 minutes (including short bursts), take the pre-exercise inhaler if prescribed, breathe through the nose, avoid cold dry air and pollution, cool down gradually.
- Inspiratory muscle training if the breathing muscles are weak.

- Posture and thoracic mobility; relaxation.
- Self-management: inhaler and spacer technique, peak-flow diary, action plan, trigger control.

Energy-conservation instructions (W19, 5 marks)

For the 25-year-old teacher recovering from an attack, energy conservation lowers breathlessness during daily tasks while fitness returns:

- **Pace:** slow, steady movement; breathe out during the effort part of a task ("blow as you go"); rest before becoming breathless; build activity up over 1–2 weeks after discharge.
- **Plan:** spread heavy jobs (washing, cleaning, shopping) through the week; do them at the best time of day; keep a reliever with you.
- **Prioritise:** drop or share tasks that are not essential while recovering.
- **Position:** sit for washing, ironing and cooking; keep things used often between waist and shoulder height; avoid prolonged bending and lifting overhead.
- **At work:** sit for part of each class; schedule classes to limit repeated stair climbs; use a whiteboard or dustless chalk; a microphone saves breath in large classes.
- **Avoid triggers while working:** damp dusting instead of sweeping, a mask for dusty jobs, no strong-smelling sprays.

3. Paediatric asthma rehabilitation (7 marks in S18; W22)

- **Assessment and goals** with the child and parents.
- **Education:** what asthma is, triggers, inhaler with spacer and mask, action plan, when to seek help.
- **Breathing exercises as games:** blowing bubbles, windmills, cotton-ball races (gentle long exhalation), "smell the flower, blow the candle".
- **Exercise through play:** warm-up games, interval play (short bursts with rest), skipping, cycling, swimming; 2–3 sessions a week, progressing time and intensity.
- Strengthening (body-weight games), posture and flexibility.
- **Family and school involvement** (counselling below).

Parent and teacher counselling (W22, 5 marks)

Message	Why
Asthma can be controlled; children with good control can play and do sports	Removes fear-based restriction
Exercise is good for her; a warm-up and a reliever before play prevent attacks	Exercise-induced symptoms are preventable
Know the early signs (cough, wheeze, breathlessness) and the action plan steps	Quick response prevents severe attacks
Keep her inhaler and spacer at school and in her bag; teachers know how to help her use it	Reliever always available
Reduce triggers: no smoking at home, dust control, avoid mosquito coils and incense	Fewer attacks
Let her carry a light bag; encourage independence	Over-protection restricts participation
Take her for medical review: reliever-only treatment is no longer recommended	A steroid-containing plan prevents monthly attacks
When to seek emergency help: cannot speak, lips blue, reliever not helping	Safety

Outcome measures (W22, 3 marks): Childhood Asthma Control Test; Paediatric Asthma Quality of Life Questionnaire; peak-flow diary; 6-minute walk or shuttle test; school days missed; number of attacks and reliever use; time spent in active play.

By impairment

Impairment	Technique and dose	Why
Breathlessness in an attack	Forward-lean sitting, breathing control, relaxation	Lowers work of breathing and anxiety
Mucus plugs	ACBT with gentle huffs after bronchodilator	Clears plugs without triggering bronchospasm
Dysfunctional breathing	Breathing retraining, daily practice 10–15 min	Better pattern, fewer symptoms
Low fitness	Aerobic training 2–3 times a week, 8–12 weeks	Higher threshold for breathlessness
Exercise-induced symptoms	10–15 min warm-up, pre-exercise inhaler as prescribed, nose breathing	Prevents bronchoconstriction
Poor technique, triggers	Inhaler and spacer training, trigger plan	Medicine reaches the airways; fewer attacks

Recall check. Write the acute plan in five lines, five counselling points for parents, and the four Ps of energy conservation.

13 Precautions, contraindications and stop criteria

Take care

- No percussion, vibration or forced breathing in acute bronchospasm.
- Keep the reliever within reach during every session.
- Avoid exercise in cold, dry air or high pollution; avoid exercising during an attack or a chest infection.
- Do not use head-down postural drainage in acute asthma.

Stop and get help if

- Unable to complete sentences, very fast breathing, SpO_2 below 92%
- Silent chest, blue lips, drowsiness or confusion
- Reliever does not help or is needed more than every 4 hours

14 Education, lifestyle and home programme

Why education matters (7 marks in W21): most severe attacks and asthma deaths are linked to poor control: preventers not taken, relievers over-used, poor inhaler technique, and no plan for worsening symptoms. Self-management education with a written action plan and regular review reduces hospital admissions, emergency visits, night symptoms and days off work or school. Physiotherapists teach inhaler technique, breathing control, exercise and trigger control, and check them at every visit.

- Take preventer (steroid-containing) medicine as prescribed, even when well.
- Use the inhaler with a spacer; rinse the mouth after steroid inhalers.
- Follow your written action plan; keep a peak-flow diary.
- Avoid triggers; no smoking; keep bedding dust-free (wash in hot water, sun-dry).
- Exercise regularly with a warm-up; practise breathing exercises daily.
- Get the influenza vaccine; keep a healthy weight.
- Farmers and workers: wear a mask in dust, and ask about occupational triggers.

15 Prognosis and discharge criteria

Prognosis: most people with asthma can be well controlled and lead normal lives, including sport. Many children improve in adolescence, but symptoms can return. Poor control, smoking and frequent oral steroid courses worsen the outlook.

Discharge after an attack: symptoms controlled, peak flow over about 75% of best, correct inhaler technique, written action plan, follow-up booked.

16 Exam corner

Past questions (MUHS, 2016–2026)

W18 Q3(a), 5 + 5 + 5 marks

30-year-old farmer with breathlessness, chest tightness, unproductive cough and audible wheeze: acute exacerbation of asthma. Functional diagnosis, short- and long-term management.

W20 Q3(a), 5 + 5 + 5 marks

45-year-old farmer, known asthmatic, in the chest ward with wheeze, breathlessness, cough and sputum for 3 days, on optimal medical therapy; chronic smoker; married; sole earner. ICF, short- and long-term management.

W22 Q4(a), 5 + 7 + 3 marks

The worked example in section 10.

W19 Q3(b), 5 + 5 + 5 marks

25-year-old teacher with allergic asthma and repeated exacerbations; admitted 2 days ago with a sudden attack, now stable. Functional diagnosis; short-term physiotherapy; energy-conservation instructions.

S23 Q3(a), 15 marks

12-year-old boy with allergic asthma who misses school in winter and the rainy season; 3 emergency visits in the past month. ICF with short- and long-term goals.

7-mark questions

Rationale of ACBT in bronchial asthma (S17). Paediatric pulmonary rehabilitation for children with asthma (S18). Physiotherapy in acute exacerbation of asthma (S22). Importance of patient education in controlling asthma (W21). Physiotherapy for a child with asthma for 2 years (S25).

Technique cards for this topic: breathing exercises (p. 275), nebulisers and inhalers (p. 294), dyspnoea and energy conservation (p. 285), scales (p. 322), pulmonary rehabilitation (p. 339), yoga (p. 302).

Keywords to underline

variable airflow obstruction reversibility triggers bronchospasm breathing control relaxation forward-lean sitting no percussion in acute bronchospasm breathing retraining warm-up action plan inhaler with spacer

Answer skeleton: asthma long answer

1. Restate: age, job, triggers, severity, smoking, social role.
2. **[ICF, 5]** ICF from the case with reasoning.
3. **[Short-term, 5]** Acute care: positioning, breathing control, relaxation, oxygen and nebuliser timing, gentle ACBT, precautions.

4. **[Long-term, 5–7]** Breathing retraining, exercise training with warm-up, inhaler technique, action plan, trigger control, smoking cessation, counselling, outcome measures.

Draw these

- The peak-flow action plan zones (section 4)
- Forward-lean sitting position (stick figure)

Viva questions

1. How do you tell asthma from COPD on spirometry?
2. Why is a normal CO_2 worrying during an asthma attack?
3. Why avoid percussion in acute asthma?
4. How do you prevent exercise-induced bronchoconstriction?
5. What do you tell a teacher about a child with asthma?
6. Name three outcome measures for childhood asthma.

Test yourself

Cover each answer and write yours on paper first, then check.

7 m **Physiotherapy management of acute exacerbation of asthma.**

Answer

Assessment of severity; positioning; breathing control and relaxation; oxygen and bronchodilator timing; ACBT with gentle huffs once settled; no percussion; hydration; mobilisation; education and action plan; danger signs.

7 m **Paediatric pulmonary rehabilitation for asthma.**

Answer

Aims; assessment; education with parents; inhaler with spacer; breathing games; play-based aerobic and interval exercise 2–3 times a week; warm-up for exercise-induced symptoms; strength and posture; school liaison; outcome measures.

MCQ **During an asthma attack, the patient's $PaCO_2$ rises from low to normal while she looks more tired. This means:**

A. She is improving

B. She is tiring and at risk of respiratory failure

C. The test is wrong

D. She is hyperventilating

Answer

B. Early in an attack CO_2 is low from fast breathing. A normal or rising CO_2 with fatigue is life-threatening.

MCQ **Best advice for a child who wheezes during football:**

A. Stop all sport

B. Warm up 10–15 minutes and use the prescribed pre-exercise inhaler

C. Run in cold air to toughen the lungs

D. Hold the breath while running

Answer

B. Exercise-induced bronchoconstriction is preventable; exercise is beneficial.

MCQ **Which finding confirms asthma rather than COPD?**

A. FEV1/FVC 0.62 that stays 0.62 after salbutamol

B. FEV1 rises by 18% and 400 mL after salbutamol

C. Low DLCO

D. Barrel chest

Answer

B. A rise of more than 12% and more than 200 mL shows significant reversibility.

Last recall check. Write the severity levels, the acute plan, ACBT rationale and counselling points.

Sources

Notes written in original words. Diagrams drawn for these notes. Numbers were checked against the sources below.

1. MUHS BPTh syllabus (2012), Cardio-Vascular & Respiratory Physiotherapy. muhs.ac.in
2. MUHS IV BPTh question papers, code 51413, Summer 2016 to Summer 2026: muhsonline.com (unofficial archive, 2016–2022) and the Dhole Patil College of Physiotherapy past-paper page (2019–2026)
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8. British Thoracic Society guideline for oxygen use in adults (2017).
9. Structure and classical content: Main & Denehy (2016); Frownfelter & Dean (2022).

Abdominal surgery: pre- and post-operative care

> Abdominal surgery was asked in 9 of 20 papers (2016–2026), as 3-mark incisions and 7-mark questions on pre-operative physiotherapy, exploratory laparotomy, cholecystectomy, gastrectomy and complications. The key idea: upper abdominal surgery stops deep breathing and coughing, so physiotherapy prevents lung complications, and one pre-operative session makes a big difference.

Syllabus	Ch 6(f) General abdominal and oncological surgeries: pre- and post-operative care; complications and management
Priority	**Tier 2.** In 9 of 20 papers (2016–2026), as short answers; in 6 of the 10 papers since 2021.
Asked as	7m S16 · W16 · W20 · S22 · W21 · S23 · S24 · W25 · 3m S16 · W16 · W21 · S26
Usual split	7-mark and 3-mark answers
Checked against	LIPPSMAck-POP randomised trial (BMJ 2018); AARC incentive spirometry guideline (2011); ERAS Society principles. Full list under Sources. Content checked 19 October 2026.

1 Definition

Abdominal surgery includes operations on the stomach, intestines, liver, gallbladder, pancreas, spleen, kidneys and abdominal wall, done open (laparotomy) or by keyhole surgery (laparoscopy). **Post-operative pulmonary complications (PPCs)** are lung problems after surgery, such as atelectasis and pneumonia, that delay recovery.

2 Classification: incisions (3 marks in S16 and W16)

Incision	Where	Used for
Midline (upper or lower)	Vertical, along the linea alba	Laparotomy, gastrectomy, bowel surgery, emergencies
Paramedian	Vertical, 2–3 cm to one side of the midline	Older approach for bowel and gastric surgery
Kocher's (right subcostal)	Oblique, below the right ribs	Open gallbladder and liver surgery
Rooftop (Chevron)	Below both rib margins	Liver, pancreas, stomach
McBurney's (gridiron) and Lanz	Right lower abdomen	Appendicectomy
Pfannenstiel	Low transverse, above the pubis	Caesarean section, gynaecological, bladder
Transverse	Across the abdomen	Children, some bowel surgery
Inguinal	Oblique, above the groin crease	Hernia repair
Laparoscopic ports	Several 5–12 mm cuts	Most gallbladder, appendix and many bowel operations; less pain

Exam point

Upper abdominal incisions (upper midline, subcostal, rooftop) cause more pain on breathing and more lung complications than lower or laparoscopic ones, because they are close to the diaphragm.

3 Risk factors for lung complications

- **Surgery:** upper abdominal or thoracic site, open rather than laparoscopic, long operation, emergency surgery, nasogastric tube.

- **Patient:** older age, smoking, COPD or asthma, obesity, poor fitness, poor nutrition, heart failure, sleep apnoea, poor general health.

4 Pathophysiology to impairments

1. General anaesthesia relaxes the diaphragm and reduces functional residual capacity; small airways at the lung bases close.
2. After surgery, wound pain stops deep breaths and coughing; the diaphragm works poorly for days after upper abdominal surgery.
3. Opioid painkillers and lying flat reduce breathing further; secretions are retained.
4. Result: **atelectasis**, low oxygen, and later **pneumonia**.
5. Bed rest adds DVT risk, weakness and loss of fitness; abdominal muscles cut in the incision weaken trunk support.

5 Clinical features after surgery

- Wound pain, shallow fast breathing, weak cough, reduced breath sounds at the bases, low SpO_2 if atelectasis.
- Drains, catheter, nasogastric tube, drips; nausea; bowel not working for a few days (ileus).
- Fatigue and weakness, especially after major cancer surgery.

6 Investigations

- Pre-op: lung function in high-risk patients, fitness tests, haemoglobin, albumin.
- Post-op: chest X-ray (basal atelectasis, raised diaphragm, effusion), ABG or SpO_2, blood counts, inflammatory markers.

7 Medical and surgical management

- Good pain control: epidural, wound catheters, regular analgesia.
- **Enhanced recovery after surgery (ERAS):** pre-op education, short fasting, minimal drains, early eating, early mobilisation on the day of surgery or day 1, early removal of tubes.
- DVT prophylaxis, antibiotics, nutrition.
- **Gastrectomy** (S16): removal of part or all of the stomach (often for cancer) through an upper midline or rooftop incision: high risk of lung complications; small frequent meals afterwards.

8 Complications and prevention (7 marks in S22)

Complication	Physio role
Atelectasis, sputum retention, pneumonia	Pre-op teaching; deep breathing, ACBT, supported cough, upright positions, early walking
DVT and pulmonary embolism	Early walking, ankle pumps, leg exercises, stockings as prescribed
Wound infection, dehiscence, incisional hernia	Support the wound when coughing; avoid heavy lifting for about 6 weeks; log-roll out of bed
Paralytic ileus, nausea	Early mobilisation helps bowel function
Urinary retention	Early mobilisation
Pain, weakness, deconditioning	Graded exercise, walking programme
Pressure sores	Turning and early sitting out

9 PT assessment

- **Pre-op:** type and site of surgery, risk factors (smoking, lung disease, obesity, fitness), breathing pattern, cough, baseline mobility.
- **Post-op:** chart (operation, incision, drains, analgesia), pain on coughing, breathing rate, SpO_2, chest expansion, auscultation, sputum, bed mobility, sitting and walking ability.
- **Outcome measures:** incidence of chest complications, SpO_2 on air, walking distance, time to independent mobility, length of stay.

10 PT diagnosis by ICF

Practice case built from past questions (not a single paper): a 58-year-old smoker, day 1 after open partial gastrectomy for cancer through an upper midline incision.

- **Impairments:** incisional pain, shallow breathing and reduced lung bases (atelectasis risk), weak cough with sputum, fatigue, weakness, DVT risk.
- **Activity limitations:** bed mobility, sitting up, walking, self-care.
- **Participation:** work and family roles suspended during recovery and any chemotherapy.
- **Environmental factors:** drains and tubes (barrier), ERAS ward routine and family support (facilitators).
- **Personal factors:** smoking, age, anxiety about cancer.

11 Problem list and goals

- **Short-term (days):** no chest complications; effective supported cough; sitting out on day 1; walking in the ward on day 1–2.
- **Long-term (weeks):** walking independently outdoors; back to daily activities; no hernia; return to work as advised.

12 PT management

Pre-operative physiotherapy (7 marks in W16 and W20)

- **Assessment** to find high-risk patients.
- **Education:** what will happen after surgery, why breathing and early walking matter, pain relief.
- **Teach and practise:** deep breathing with a 3–5 second hold, ACBT, supported cough (hands or pillow over the wound), log-roll bed mobility, ankle pumps, getting up from bed.
- **Prehabilitation** for high-risk patients: aerobic and strength training, and inspiratory muscle training, for a few weeks before surgery.
- Stop smoking as early as possible.

> **Exam point: the evidence for pre-op physio**
>
> In the LIPPSMAck-POP trial (441 patients having upper abdominal surgery), a single 30-minute pre-operative physiotherapy education and breathing-exercise session roughly halved post-operative pulmonary complications (adjusted hazard ratio 0.48). Quote this for "importance of pre-operative chest physiotherapy".

Rationale of pre-operative goals (3 marks in W21)

Goal	Rationale
Teach deep breathing, huffing and supported cough	Learned while the patient is pain-free and alert, so they are done correctly after surgery when pain and drowsiness make learning hard; reverses the fall in lung volume and prevents collapse and pneumonia
Teach log-roll, getting out of bed and early walking	Less pain and fear when moving, so the patient sits out and walks sooner; walking raises lung volumes and prevents DVT

Goal	Rationale
Assess and record a baseline	Finds high-risk patients (smoker, COPD, obesity, older age, upper abdominal incision) who need more treatment; gives a baseline to compare after surgery
Reduce anxiety with education	A patient who knows what to expect (drains, drips, pain relief) co-operates better and reports problems early
Prehabilitation in high-risk patients	Fitter, stronger patients (including stronger breathing muscles) tolerate surgery better and recover faster

Post-operative physiotherapy

When	Physiotherapy
Day 0–1	• Sit upright or high side-lying; avoid slumping • Deep breaths with a hold, 5–10 every hour while awake; ACBT if sputum • Supported huff and cough • Ankle pumps, quadriceps and gluteal sets • Log-roll and sit over the edge of the bed; sit out; walk with help
Day 2 to discharge	• Increase walking 3–4 times a day; stairs if needed at home • Continue breathing exercises until fully mobile • Gentle trunk mobility; posture correction
After discharge	• Daily walking, increasing gradually • Gentle abdominal (core) activation; progress after about 6 weeks • No heavy lifting for about 6 weeks

Principles of physiotherapy after abdominal surgery (7 marks in W25)

1. **Control pain first:** treat when analgesia is working; support the wound with a pillow or hands for breathing, coughing and moving.
2. **Restore lung volume:** upright positions, deep breathing with a hold, early walking.
3. **Clear secretions:** ACBT and supported huff and cough when sputum is present.
4. **Mobilise early:** sit out of bed on the day of surgery or the next day, then walk several times a day (the basis of enhanced recovery programmes).
5. **Prevent circulatory problems:** ankle pumps and walking against DVT; check for calf pain and swelling.
6. **Protect the wound:** log-roll to get up; no heavy lifting or sit-ups for about 6 weeks, or as the surgeon advises.
7. **Restore posture and trunk movement:** patients bend forward to protect the wound, so correct posture and add gentle trunk and core exercise, progressed after about 6 weeks.
8. **Watch for complications:** fever, low SpO_2, breathlessness, wound leakage, abdominal swelling; report them.
9. **Educate and plan discharge:** home walking programme, safe lifting, return to work.

Exploratory laparotomy (7 marks in W21 and S23)

A long vertical midline incision, often an emergency for peritonitis (perforation), obstruction, trauma or abdominal bleeding. These patients are often sicker than planned-surgery patients: sepsis, ICU stay, nasogastric tube, several drains, urinary catheter, sometimes a stoma, and a swollen abdomen (ileus) that pushes the diaphragm up. The lung complication risk is high.

- **If in ICU or ventilated:** positioning, airway clearance, suction and early mobilisation as on the ICU physiotherapy card (p. 223).
- **On the ward:** sit upright; deep breathing with a hold, 5–10 every hour while awake; ACBT and huffing with a pillow pressed over the wound; ankle pumps; log-roll to sit; sit out and walk from day 1 if stable, managing tubes and drains.

- **Wound protection:** the long midline wound has a real risk of opening (dehiscence) and of a later incisional hernia: support it when coughing, no straining or heavy lifting, abdominal binder if the surgeon uses one.
- **With a stoma:** support the stoma with a hand or pillow when coughing; gentle core work later, to lower the risk of a hernia around the stoma.
- **Later:** graded walking and endurance, posture, gradual core strengthening, return to work.

Day 2 after cholecystectomy: rationale and procedure (7 marks in S24)

Type	What it means for physiotherapy
Laparoscopic (usual)	Four small port wounds; low risk of lung complications; often home in 1–2 days. Gas used to inflate the abdomen irritates the diaphragm and causes **shoulder-tip pain** (referred along the phrenic nerve, C3–5); it settles in a few days, and walking helps
Open (right subcostal, Kocher's)	Upper abdominal incision: painful breathing, low lung volumes at the right base, high risk of lung complications; may have a drain

Day 2 procedure (open surgery, or a slow laparoscopic recovery)

Step	Procedure	Rationale
1. Assess	Pain score, SpO_2, temperature, breath sounds at the right base, sputum, drains, mobility	Finds collapse or infection early; sets the plan
2. Time it	Treat when analgesia is working	Pain stops deep breaths and coughs
3. Position	High sitting, or side-lying on the left with the right side up	Upright and side-lying positions increase lung volume at the bases
4. Breathing exercises	Deep breaths with a 3-second hold, thoracic expansion to the right lower chest; 5–10 breaths, repeated hourly while awake	Re-opens collapsed alveoli and restores lung volume
5. Secretion clearance	Huff and cough with a pillow pressed over the wound (ACBT if sputum)	Clears sputum with less pain and less strain on the wound
6. Mobilise	Log-roll to sit; walk 3–4 times a day, increasing the distance; stairs before discharge if needed	The best way to raise lung volume; prevents DVT, ileus and weakness
7. Circulation	Ankle pumps 10 every hour in bed	Prevents DVT
8. Posture and shoulder	Stand tall; gentle trunk movements; right shoulder movements	Patients bend towards the wound; this prevents stiffness and poor posture
9. Teach	Continue exercises at home; walk daily; avoid heavy lifting until the surgeon allows	Keeps recovery going after discharge

Classic answer vs current evidence: incentive spirometry

Classic	Current (AARC 2011)	Write in the exam
Incentive spirometry hourly for every upper abdominal surgery patient.	Not recommended alone or routinely. Deep breathing, coughing and early mobilisation are the core; the spirometer can be an add-on.	"Pre-op education and breathing exercises, then deep breathing, supported cough and early mobilisation; incentive spirometry only as an adjunct."

13 Precautions

- Always support the wound during coughing and moving.
- Check drains, catheter and lines before moving; keep bags below the wound.
- Epidural: check leg strength and BP before standing.
- Watch for postural hypotension, bleeding, sudden breathlessness (PE).

- No heavy lifting or strong abdominal exercises for about 6 weeks.
- No percussion near fresh abdominal wounds and drains.

14 Education and home programme

- Keep walking every day; increase gradually.
- Support the wound with a pillow when coughing or laughing.
- Get out of bed by rolling onto your side first.
- Avoid lifting heavy objects (such as water buckets or gas cylinders) for about 6 weeks.
- Stop smoking; eat well for wound healing.
- Report fever, wound discharge, calf pain or sudden breathlessness.

15 Prognosis

Most patients recover well with good pain control and early mobilisation. Lung complications lengthen hospital stay and can be largely prevented by pre-operative teaching and early activity.

16 Exam corner

7-mark questions

Physiotherapy after gastrectomy (S16). Importance of pre-operative chest physiotherapy in major abdominal surgery (W16, W20). Complications of abdominal surgery, and physiotherapy of DVT (S22). Physiotherapy after exploratory laparotomy (W21, S23). Rationale and procedure of physiotherapy on day 2 after cholecystectomy (S24). Principles of physiotherapy after abdominal surgery (W25).

3-mark questions

Three incisions in abdominal surgery (S16, W16, S26). Rationale of any three goals of pre-operative physiotherapy (W21).

Technique cards for this topic: breathing exercises (p. 275), ACBT and autogenic drainage (p. 261), bed rest (p. 241), ICU physiotherapy (p. 223).

Keywords to underline

upper abdominal incision, post-operative pulmonary complications, pre-operative education, supported cough, early mobilisation, ERAS, log-roll, no heavy lifting 6 weeks

Draw these

- An abdomen outline with the incisions marked (midline, Kocher's, McBurney's, Pfannenstiel)

Test yourself

3 m **Any three incisions for abdominal surgery.**

Answer
Midline (laparotomy); Kocher's right subcostal (gallbladder); McBurney's gridiron (appendix); Pfannenstiel (caesarean, pelvic). Also paramedian, rooftop, transverse.

7 m **Importance of pre-operative chest physiotherapy in major abdominal surgery.**

Answer

Why lungs suffer after upper abdominal surgery; risk factors; content of the pre-op session (education, deep breathing, ACBT, supported cough, bed mobility, early mobilisation plan, prehab, smoking cessation); evidence (one session halved complications in LIPPSMAck-POP); benefits (fewer complications, shorter stay, less anxiety).

MCQ **Which surgery carries the highest risk of lung complications?**

A. Laparoscopic appendicectomy

B. Open upper abdominal gastrectomy

C. Inguinal hernia repair

D. Pfannenstiel caesarean section

Answer

B. Upper abdominal open incisions most affect the diaphragm and coughing.

Sources

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3. Restrepo RD, et al. AARC clinical practice guideline: Incentive spirometry. *Respir Care* 2011.
4. ERAS Society guidelines for perioperative care (principles).
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Diabetes, diabetic foot, wound healing and diabetes insipidus

> Diabetes is part of many long cases (CABG, MI, PAD, amputation, varicose veins) and was asked on its own as 7-mark questions on glycaemic control and diabetic foot care. Newer papers ask about wound healing and pressure sores. Learn the exercise prescription, the precautions, foot care with its reasons, and wound care.

Syllabus	Ch 6(i) Metabolic syndrome: diabetes mellitus and insipidus; 6(h) diabetic ulcers
Priority	**Tier 2.** 7-mark questions on diabetes (S16, W18, S19) and on wounds (W19 pressure sore; S24 and S25 wound healing); a co-existing condition in 10 long cases.
Asked as	7m S16 · W18 · S19 · W19 · S24 · S25 · 3m S16 · S19 · S22
Usual split	7-mark answers; part of long answers on CABG, PAD, amputation and venous disease
Checked against	ADA Standards of Care (2026); ADA position statement on physical activity and diabetes (2016); IWGDF diabetic foot guidelines (2023); 2022 renaming of diabetes insipidus. Full list under Sources. Content checked 19 October 2026.

1 Definition

Diabetes mellitus is a group of metabolic disorders with a raised blood glucose, caused by lack of insulin, poor response to insulin, or both.

Diabetic foot is infection, ulceration or destruction of tissues of the foot in a person with diabetes, usually because of nerve damage (neuropathy) and/or peripheral arterial disease.

Diabetes insipidus (now called **arginine vasopressin (AVP) deficiency** or **AVP resistance**) is a different disease: the body cannot concentrate urine, so the person passes large volumes of dilute urine and is always thirsty. Blood sugar is normal.

2 Classification and grading

Types of diabetes mellitus

- **Type 1:** autoimmune destruction of insulin-producing cells; usually young; needs insulin; risk of ketoacidosis.
- **Type 2:** insulin resistance with failing insulin secretion; linked to obesity, inactivity and family history; the commonest type, and common in younger Indians.
- **Gestational diabetes** in pregnancy; other specific types (pancreatic disease, steroids).

Diagnosis (ADA)

Test	Diabetes	Prediabetes
Fasting plasma glucose	126 mg/dL or more	100–125 mg/dL
2-hour glucose after 75 g glucose drink	200 mg/dL or more	140–199 mg/dL
HbA1c	6.5% or more	5.7–6.4%
Random glucose with symptoms	200 mg/dL or more	–

Diabetic foot ulcer: Wagner grades

Grade	Description
0	At-risk foot, no open wound (callus, deformity)

Grade	Description
1	Superficial ulcer
2	Deep ulcer reaching tendon, capsule or bone, no abscess
3	Deep ulcer with abscess or osteomyelitis
4	Gangrene of part of the foot (toes or forefoot)
5	Gangrene of the whole foot

Foot risk categories (IWGDF)

- **0, very low:** no loss of protective sensation and no PAD. Screen yearly.
- **1, low:** loss of protective sensation or PAD. Screen every 6–12 months.
- **2, moderate:** two of: loss of sensation, PAD, foot deformity. Screen every 3–6 months.
- **3, high:** loss of sensation or PAD plus a past ulcer, past amputation or end-stage kidney disease. Screen every 1–3 months.

3 Causes and risk factors

- **Type 2:** obesity, especially abdominal; physical inactivity; family history; South Asian ethnicity (diabetes at lower BMI and younger age); unhealthy diet; age; gestational diabetes in the past.
- **Diabetic foot:** long duration, poor sugar control, neuropathy, PAD, foot deformity, smoking, kidney disease, walking barefoot, poor footwear, previous ulcer.

4 Pathophysiology to impairments

Why the diabetic foot ulcerates: three causes

- **Sensory neuropathy:** loss of protective feeling, so injuries, tight shoes and burns are not noticed.
- **Motor neuropathy:** weak small foot muscles cause claw toes and prominent metatarsal heads, so pressure under the forefoot rises and callus forms.
- **Autonomic neuropathy:** no sweating, so dry, cracked skin lets bacteria in.
- **Ischaemia** (PAD) slows healing; **high sugar** weakens immunity, so infection spreads fast.

Repeated pressure under a callus causes bleeding and breakdown beneath it: a neuropathic ulcer. Add infection and ischaemia, and the result can be gangrene and amputation.

Other complications that affect physiotherapy

- **Heart and vessels:** heart attack (often silent), stroke, PAD.
- **Eyes:** retinopathy (bleeding risk with straining and head-down positions).
- **Kidneys:** nephropathy.
- **Autonomic neuropathy:** postural hypotension, fixed heart rate, poor heat tolerance.
- **Musculoskeletal:** frozen shoulder, stiff hands, Charcot foot (collapse of foot joints).

Stages of wound healing (3 marks in S19)

Haemostasis (minutes to hours) → inflammation (about days 1–5) → proliferation (about day 4 to week 3) → remodelling (week 3 to 1–2 years). See the full table on the burns page (p. 34). Diabetes slows every stage, especially inflammation and proliferation.

Factors affecting wound healing (7 marks in S25)

- **Local:** infection; poor blood supply (arterial disease, pressure, tight dressings); oedema; dead tissue or foreign bodies; repeated shear, friction or pressure; a wound that is too dry or too wet; a large or deep wound; previous

radiotherapy.

- **General:** older age; poor nutrition (protein, vitamin C, zinc); diabetes with high sugar; smoking; anaemia and low oxygen; obesity; drugs (steroids, chemotherapy); kidney or liver failure; reduced immunity; stress and poor sleep.

In a diabetic foot the main ones are high sugar, ischaemia, unnoticed pressure from neuropathy, infection and smoking. Answer S25 with this list plus the modalities in section 12.

Recall check. Write the three neuropathy types and how each leads to a foot ulcer.

5 Clinical features

- **Diabetes:** thirst, frequent urination, tiredness, weight loss, blurred vision, slow-healing wounds, repeated infections; many type 2 patients have no symptoms.
- **Neuropathic foot:** numbness, tingling or burning, warm dry skin, callus, claw toes, good pulses (unless PAD), painless ulcer under the forefoot or heel.
- **Ischaemic foot:** cold, pale, absent pulses, painful ulcers at toes and heel.
- **Diabetes insipidus:** passing large amounts of pale urine (more than about 3 litres a day in adults), waking at night to pass urine, strong thirst, dehydration if water is not available.

Hypoglycaemia (blood sugar below 70 mg/dL)

Sweating, shaking, hunger, fast heart rate, confusion, slurred speech, unusual behaviour, seizures. Can happen during or many hours after exercise, especially with insulin or sulfonylureas.

6 Investigations and foot tests

Test	What it shows	Physio meaning
HbA1c	Average sugar over about 3 months	Poor control means slow healing, more care with exercise
Glucometer reading	Sugar now	Check before and after exercise if on insulin or sulfonylureas
10 g monofilament	Loss of protective sensation	Foot risk category; protective footwear
128 Hz tuning fork, ankle reflexes	Large-fibre neuropathy	Balance and fall risk
Pulses, ABI, toe pressures	PAD (ABI may be falsely high from calcified arteries)	Ischaemic foot needs vascular review
Lipids, kidney function, ECG, eye check	Complications	Modify exercise for heart, eye and kidney disease
Water deprivation test, sodium	Diabetes insipidus	Allow free access to water during sessions

7 Medical management

- **Lifestyle:** diet, physical activity, weight loss, stopping smoking.
- **Medicines:** metformin; sulfonylureas (cause hypoglycaemia); DPP-4 inhibitors; SGLT2 inhibitors and GLP-1 receptor agonists (also protect the heart and kidneys); insulin.
- Usual target HbA1c around 7% for many adults, set individually.
- Control BP and lipids; yearly eye, kidney and foot checks.
- **Diabetic foot ulcer:** off-loading (for example total contact cast), debridement, infection control, revascularisation if ischaemic, good sugar control.

- **Diabetes insipidus:** treat the cause; desmopressin for AVP deficiency.

8 Complications and prevention

Complication	What to look for	Physio role
Hypoglycaemia	Sweating, shaking, confusion during or after exercise	Check sugar; keep glucose tablets ready; rule of 15
Foot ulcer, infection, amputation	Callus, blisters, redness, warmth, smell	Foot checks every session, footwear advice, off-loading
Silent heart attack	Breathlessness or fatigue without chest pain	Monitor HR and BP; use RPE
Retinal bleeding	Sudden blurred vision	Avoid heavy straining, breath-holding, jumping and head-down positions in advanced retinopathy
Falls	Numb feet, poor balance, postural hypotension	Balance training, safe footwear, slow position changes
Frozen shoulder, stiff hands	Painful restricted shoulder	ROM exercises, mobilisation

9 PT assessment

- Type and duration of diabetes, medicines (insulin, sulfonylureas), recent HbA1c and readings, hypoglycaemia episodes, complications.
- Activity level, occupation, diet habits, smoking, weight, BMI and waist.
- **Feet:** skin, nails, callus, deformity, footwear, monofilament sensation, vibration, pulses, ulcer description (site, size, depth, Wagner grade).
- BP lying and standing; resting HR; fitness (6MWT); strength; balance (single-leg stance, Timed Up and Go).
- **Outcome measures:** HbA1c, weight and waist, 6MWT, ulcer area, balance tests, quality of life.

10 PT diagnosis by ICF

Diabetes appears inside other long cases. This practice case is built from those (not a single past paper).

> Practice case: a 55-year-old shopkeeper with type 2 diabetes for 12 years, HbA1c 9%, numb feet, callus under the first metatarsal head, sedentary, overweight. He walks barefoot at home and in the shop.

ICF component	From this case	Clinical reasoning
Health condition	Poorly controlled type 2 diabetes with peripheral neuropathy; high-risk foot	Long duration and high HbA1c
Body function impairments	Loss of protective sensation; callus and raised forefoot pressure; reduced fitness; excess weight; reduced balance	Neuropathy plus inactivity
Activity limitations	Safe walking on uneven ground; long standing in the shop	Numb feet and callus pain
Participation restrictions	At risk of losing work days to ulcers; limited social walking	Ulcer or amputation would stop work
Environmental factors	**Barriers:** barefoot habit, sweets and fried snacks at the shop. **Facilitators:** family, local clinic.	Education targets habits
Personal factors	55-year-old man, low awareness of foot risk	Teach "why", not just "what"

11 Problem list and goals

- **Short-term (4 weeks):** daily foot inspection habit; protective footwear always; walking 20 minutes on most days with safe sugar readings; knows hypoglycaemia signs and treatment.
- **Long-term (3–6 months):** 150 minutes a week of aerobic activity plus resistance training twice a week; HbA1c and weight reduced; no ulcers; better balance.

12 PT management

Exercise for glycaemic control (7 marks in S16 and W18)

Why exercise works: contracting muscle takes up glucose even without insulin; after training, insulin sensitivity improves for a day or more; exercise also reduces weight, BP, lipids and heart risk.

ADA prescription for adults with type 2 diabetes

Type	Dose
Aerobic	At least **150 min a week** of moderate-to-vigorous activity, spread over 3–7 days, **no more than 2 days in a row without exercise**
Resistance	**2–3 sessions a week** on non-consecutive days; 1–3 sets of 10–15 repetitions to near fatigue at first
Flexibility and balance	**2–3 days a week**; stretches held 10–30 s, 2–4 repetitions; balance work for older adults and those with neuropathy
Less sitting	A few minutes of light activity **every 30 minutes** of sitting

Memory hook: hypoglycaemia "rule of 15"

15 g, then 15 minutes

If sugar is below 70 mg/dL or symptoms appear: give 15 g of fast sugar (glucose tablets, 3 teaspoons of sugar in water, half a glass of juice), recheck after 15 minutes, repeat if still low, then a snack.

Diabetic foot care (7 marks in S19: "what is diabetic foot, how to take care of it, and why")

Care	How	Why
Inspect daily	Look at soles and between toes with a mirror or a family member	Numb feet do not feel injuries
Wash and dry	Lukewarm water (test with the elbow), dry between toes	Prevents burns and fungal infection
Moisturise	Cream on dry skin, not between the toes	Dry skin cracks and lets bacteria in
Nail care	Cut straight across; no home surgery on corns	Prevents ingrown nails and wounds
Footwear	Soft, roomy, closed shoes with socks; check inside before wearing; never barefoot (home, temple, beach)	Protects against unnoticed injury and pressure
Avoid heat	No hot-water bottles, heating pads or walking on hot surfaces	Burns are not felt
Control sugar, stop smoking	Medicines, diet, exercise	Better healing and circulation
Seek help early	Any blister, cut, redness or swelling	Small wounds become ulcers in days

- **With an active plantar ulcer:** off-load the foot (cast, boot, crutches); switch to non-weight-bearing exercise such as cycling with the other leg, arm ergometry or swimming once the wound allows.

- **Neuropathy without ulcer:** moderate weight-bearing walking is safe with proper footwear and daily checks.

Electrotherapy for wound healing (3 marks in S16 and S22; 7 marks in S24 and S25)

Typical textbook parameters (vary between sources; follow local protocol)

Modality	Typical parameters	Effect
High-voltage pulsed current (electrical stimulation)	Twin-peak monophasic pulses, about 100 pulses per second, sensory-level voltage (often 100–150 V), 45–60 minutes, daily or 5 days a week; negative pole on an infected wound, positive to promote healing	Attracts repair cells, improves blood flow, antibacterial effect
Therapeutic ultrasound	Pulsed 20%, about 0.5 W/cm², 3 MHz, 5–10 minutes on the skin around the wound (not into an open wound without a sterile coupling)	Stimulates inflammation to move on, collagen formation
Ultraviolet (UVC)	Short-wave UV (about 254 nm), short exposures by test dose, protecting surrounding skin and eyes	Kills bacteria in infected wounds
Low-level laser	Low dose, about 1–4 J/cm² to wound edges	Cell stimulation, healing

Exact parameters differ between electrotherapy books and trials. Write a range with the effect, and never use heat or electrotherapy over an ischaemic or numb area without checking circulation.

Pressure sores (pressure injuries) (7 marks in W19)

Damage to skin and the tissue under it from **pressure, or pressure with shear**, usually over a bony point: sacrum, heels, hips (greater trochanter), ischial tuberosities, back of the head, elbows. Patients at risk: bed- or chair-bound, numb skin (diabetes, spinal cord injury), ICU patients, poor nutrition, incontinence, older age.

Stages (NPIAP, 2016)

Stage	What you see
Stage 1	Intact skin with redness that does not fade (blanch) when pressed
Stage 2	Partial-thickness skin loss: shallow pink wound or blister; fat not visible
Stage 3	Full-thickness skin loss: fat visible; no bone, tendon or muscle seen
Stage 4	Full-thickness skin and tissue loss: muscle, tendon, ligament, cartilage or bone exposed or felt
Unstageable	Base covered by slough or eschar, so depth cannot be seen
Deep tissue pressure injury	Intact or broken skin with a persistent deep red, maroon or purple area

Risk assessment: Braden scale (sensory perception, moisture, activity, mobility, nutrition, friction and shear; total 6–23). A score of **18 or less** means the patient is at risk; the lower the score, the higher the risk.

Physiotherapy

- **Relieve pressure:** reposition on a schedule set for that patient; 30° tilted side-lying rather than lying straight on the hip; head of bed no higher than 30° unless breathing needs more; heels floated off the bed with a pillow under the calves.
- **Surfaces:** high-specification foam or alternating-pressure air mattress; pressure-relieving cushion in the chair; teach chair users to shift weight (push-ups or leaning) regularly.
- **Move:** bed mobility, sitting out and walking as early as possible; active and passive exercise to keep circulation and joint range.
- **Avoid shear:** lift, do not drag; use slide sheets.
- **Help the wound heal:** electrical stimulation has the best evidence of the physical modalities for stage 2 sores that are slow to heal and for stage 3–4 sores (parameters as in the table above); wound cleaning and dressing by the team; nutrition and continence care.
- **Teach:** daily skin checks, weight shifts, never sit or lie on a red area.

Classic answer vs current guideline: turning		
Classic Turn every patient every 2 hours.	**Current (international guideline, 2019)** Set the turning frequency for each patient from their risk, skin checks, comfort and the mattress they are on.	**Write in the exam** "Reposition at least every 2 hours, or as the patient's skin and support surface require."

Diabetes insipidus

- Allow free access to water and toilets during sessions; watch for dehydration (dizziness, dry mouth, fast pulse).
- Night-time urination raises the fall risk in older patients: night light, clear path.
- Seen in ICU after head injury or pituitary surgery: monitor fluid balance before mobilising.

Recall check. Write the ADA exercise prescription, five foot-care rules with their reasons, and the six pressure-sore stages.

13 Precautions

Exercise precautions in diabetes

- Check sugar before and after exercise if on insulin or sulfonylureas; carry fast sugar; avoid exercising at the peak action of insulin.
- Treat a low sugar first and do not start until it is back to normal.
- Avoid exercise when sugar is very high with ketones (mainly type 1).
- Advanced retinopathy: avoid vigorous exercise, heavy lifting, jumping, head-down positions and breath-holding.
- Neuropathy: proper footwear; check feet after each session; balance training.
- Autonomic neuropathy: change position slowly; use RPE instead of heart rate; avoid hot, humid days; drink water.
- Heart disease is often silent: report unusual breathlessness or fatigue.
- No heat or ice on numb skin; check circulation before any electrotherapy.

14 Education and lifestyle

- Be active every day; break up sitting every 30 minutes.
- Eat regular meals with fewer refined carbohydrates and sweets; more vegetables, pulses and fibre.
- Reach a healthy weight and waist (Indian targets: BMI below 23; waist below 90 cm in men, 80 cm in women).
- Take medicines regularly; know your hypoglycaemia plan.
- Daily foot care for life; yearly eye, kidney and foot checks.
- Stop smoking and tobacco.

15 Prognosis

Good control of sugar, BP and lipids, regular exercise and daily foot care greatly reduce complications. Most diabetic amputations start with a foot ulcer that could have been prevented.

16 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Physiotherapy for glycaemic control in diabetes (S16). Role of physiotherapy in diabetes (W18). What is diabetic foot, how to care for it, and why (S19). Physiotherapy for pressure sores (W19). Modalities for wound healing (S24). Factors affecting wound healing, with the modalities used (S25).

3-mark questions

Three electrotherapy modalities for an infected wound (S16). Stages of wound healing (S19). Three electrotherapy modalities with parameters for wound healing (S22).

Inside long cases

Diabetes with CABG (W18, S21), PAD (W22), amputation (S18, S19), varicose veins (S17), angina with EF 25% (S20), amputation (W19, S23, S25), MI (S26).

Technique cards for this topic: exercise physiology (p. 313), exercise testing (p. 308), electrotherapy and exercise (p. 298).

Keywords to underline

insulin sensitivity · 150 min a week · no more than 2 days without exercise · resistance 2–3 times a week · hypoglycaemia · rule of 15 · loss of protective sensation · monofilament · off-loading · daily foot inspection

Viva questions

1. How does exercise lower blood sugar?
2. What would you do if a patient becomes sweaty and confused during exercise?
3. Why do neuropathic ulcers form under the metatarsal heads?
4. Which exercises should a patient with proliferative retinopathy avoid?
5. How is diabetes insipidus different from diabetes mellitus?

Test yourself

Cover each answer and write yours on paper first, then check.

7 m **Physiotherapy management for glycaemic control in diabetes mellitus.**

Answer

Why exercise works; assessment and screening; aerobic and resistance prescription (ADA doses); flexibility and balance; breaking up sitting; precautions (hypoglycaemia, retinopathy, neuropathy, autonomic); foot care; education; monitoring outcomes (HbA1c, weight).

3 m **Three electrotherapeutic modalities with parameters for wound healing.**

Answer

1. High-voltage pulsed current: about 100 pps, sensory level, 45–60 min, negative pole for infected wound.
2. Pulsed ultrasound: 20%, about 0.5 W/cm², 3 MHz, 5–10 min around the wound.
3. UVC about 254 nm, short test-dosed exposure for bacterial kill. (Laser 1–4 J/cm² also accepted.)

MCQ **A patient on insulin feels shaky and sweaty mid-session; glucometer shows 60 mg/dL. Best action:**

A. Continue at lower intensity

B. Stop, give 15 g fast sugar, recheck in 15 minutes

C. Give insulin

D. Tell him to rest at home

Answer

B. The rule of 15 treats mild hypoglycaemia safely.

MCQ **Which footwear advice is correct for a diabetic with numb feet?**

A. Walk barefoot at home to strengthen the feet

B. Tight shoes for support

C. Soft, roomy closed shoes with socks, checked inside before wearing

D. Hot-water soaks every night

Answer

C. Numb feet cannot sense injury; protection and daily checks prevent ulcers.

Sources

Notes written in original words. Numbers were checked against the sources below.

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. American Diabetes Association. Standards of Care in Diabetes 2026. *Diabetes Care* 2026.
3. Colberg SR, et al. Physical Activity/Exercise and Diabetes: A Position Statement of the American Diabetes Association. *Diabetes Care* 2016;39:2065–79. Key points summary: PMC.
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5. Arima H, et al. Changing the name of diabetes insipidus: a position statement. 2022.
6. Electrical stimulation parameters: reviews of high-voltage pulsed current in chronic wounds (PMC); electrotherapy.org.
7. RSSDI clinical practice recommendations for type 2 diabetes (India), 2022.
8. National Pressure Injury Advisory Panel. Pressure injury stages (2016). npiap.com
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Obesity and metabolic syndrome

> Obesity appears inside many long cases (BMI 30–32 in CABG cases) and as 3-mark questions: grades of obesity (S21) and the definition of metabolic syndrome (W24). Know both WHO and Indian cut-offs, the metabolic syndrome criteria, and a safe exercise plan.

Syllabus	Ch 6(i)(ii) Obesity, within metabolic syndrome
Priority	**Tier 2.** Asked as 3-mark questions (S21 grades of obesity; W24 metabolic syndrome) and as a risk factor in several long cases.
Asked as	3m S21 (grades of obesity) · W24 (metabolic syndrome)
Usual split	3-mark add-on to cardiac long answers; 7 marks if asked alone
Checked against	WHO BMI classification; Asian Indian consensus (Misra 2009) and 2025 proposed staging; harmonised metabolic syndrome definition (2009); ACSM position stand on physical activity and weight (2009). Full list under Sources. Content checked 19 October 2026.

1 Definition

Obesity is abnormal or excessive body fat that harms health. In practice it is screened with the **body mass index**: weight (kg) ÷ height (m)^2. Waist circumference shows abdominal (central) fat, which carries the most risk.

Metabolic syndrome is a cluster of risk factors (central obesity, raised BP, raised sugar, high triglycerides, low HDL) that together raise the risk of heart disease and type 2 diabetes.

2 Classification and grading (3 marks in S21)

Category	WHO BMI (kg/m^2)	Asian Indian BMI (kg/m^2)
Underweight	Below 18.5	Below 18.5
Normal	18.5–24.9	18.5–22.9
Overweight	25–29.9	23–24.9
Obese	Class I 30–34.9; Class II 35–39.9; Class III 40 or more	25 or more

- **Abdominal obesity (Indian cut-offs):** waist **90 cm or more** in men, **80 cm or more** in women.
- A newer staged definition for Asian Indians was proposed in 2025 (stage 1 without and stage 2 with obesity-related disease or functional impairment). Use the cut-offs above in the exam and mention the proposal.

Metabolic syndrome: any 3 of 5 (harmonised definition)

1. Waist above the population cut-off (South Asians: 90 cm men, 80 cm women)
2. Triglycerides **150 mg/dL** or more (or on treatment)
3. HDL below **40 mg/dL** in men or **50 mg/dL** in women (or on treatment)
4. BP **130/85** or more (or on treatment)
5. Fasting glucose **100 mg/dL** or more (or on treatment)

3 Causes and risk factors

- Energy intake higher than energy use: calorie-dense food, sugary drinks, fried snacks, large portions; sitting jobs, screen time, little walking.
- Genetics and family habits; South Asians store more fat centrally at lower BMI.

- Sleep loss, stress, some medicines (steroids, some antidepressants), hypothyroidism, Cushing's syndrome, polycystic ovary syndrome.

4 Pathophysiology to impairments

- Excess abdominal fat releases fatty acids and inflammatory substances, causing insulin resistance, high BP and abnormal lipids: the route to diabetes and heart disease.
- A heavy chest wall and abdomen reduce lung volumes (especially lying flat) and raise the work of breathing; obstructive sleep apnoea and obesity hypoventilation can follow.
- Extra load on the knees, hips and back causes pain and osteoarthritis.
- Low fitness, heat intolerance, breathlessness and stigma reduce activity, so the cycle continues.

5 Clinical features

- High BMI and waist; breathlessness on exertion; snoring and daytime sleepiness; joint pain; fatigue.
- Skin changes: dark velvety skin folds (acanthosis nigricans, a sign of insulin resistance), stretch marks.
- Low self-esteem, depression.

6 Investigations

- BMI, waist circumference, waist-to-hip ratio; body fat (skinfolds, bioimpedance).
- Fasting glucose, HbA1c, lipid profile, liver tests (fatty liver), thyroid function.
- BP; sleep study if sleep apnoea is suspected.
- Fitness: 6MWT or step test; joint assessment.

7 Medical and surgical management

- **Lifestyle first:** diet with a calorie deficit, physical activity, behaviour change, sleep.
- A loss of **5–10% of body weight** already improves BP, sugar and lipids.
- Medicines (for example GLP-1 receptor agonists) in selected patients.
- Bariatric surgery for severe obesity with complications.

8 Complications and prevention

- Type 2 diabetes, hypertension, coronary heart disease, stroke, fatty liver.
- Sleep apnoea, obesity hypoventilation; post-operative chest complications (higher risk after CABG and abdominal surgery).
- Osteoarthritis, back pain, falls, DVT, sternal wound problems after CABG.
- Some cancers; depression.

Prevention: regular activity, healthy diet, weight and waist checks, early treatment of risk factors.

9 PT assessment

- Weight, height, BMI, waist; BP and resting HR (use a large cuff).
- Activity history, sitting time, diet habits, sleep, motivation and barriers.
- Joint pain and range; posture; balance.
- Fitness: 6MWT with Borg; strength (sit-to-stand test).
- Outcome measures: weight, waist, 6MWT, sit-to-stand, quality of life, step count.

10 PT diagnosis by ICF

Obesity is usually written into another case's ICF as a personal factor and body-function impairment. Example from S21 (businessman, CABG 20 days ago, type 2 diabetes, BMI 32):

- **Body function:** class I obesity (WHO; obese by Indian criteria), insulin resistance, low fitness, heavy chest wall reducing lung volumes after surgery.
- **Activity:** walking distance and stairs limited by breathlessness and weight.
- **Participation:** business travel and social roles.
- **Personal factors:** diet habits, sitting job, stress.
- **Environment:** family meals, availability of safe walking space.

11 Problem list and goals

- **Short-term (4–6 weeks):** walks 30 minutes on most days; sitting broken up every 30 minutes; keeps an activity and food diary.
- **Long-term (6 months):** loses 5–10% of body weight; waist reduced; 250 minutes or more of activity a week with resistance training twice a week; better 6MWT.

12 PT management

Exercise for weight management (ACSM 2009)

Weekly activity	Effect
Less than 150 min	Little weight loss
150–250 min	Prevents weight gain; modest weight loss
More than 250 min (about 225–420)	Larger, clinically important weight loss, especially with diet
About 200–300 min	Helps keep weight off after losing it

- **Type:** low-impact aerobic activity first: brisk walking, cycling, swimming or water exercise, which spare the knees.
- **Intensity:** moderate (talk test; RPE 12–13), progressing to some vigorous work if fit.
- **Build up:** start with 10-minute bouts several times a day; add 5 minutes a week.
- **Resistance training** 2–3 days a week: keeps muscle while losing fat; improves insulin sensitivity.
- **Lifestyle activity:** stairs instead of lifts, walking to shops, breaking up sitting.
- Combine with a dietitian's plan; behaviour change (goal setting, self-monitoring, family support).

13 Precautions

- Screen for heart disease, diabetes and sleep apnoea before vigorous exercise.
- Low-impact choices for painful knees and hips; good footwear.
- Heat intolerance: exercise in cooler hours, drink water.
- Use a large BP cuff; watch breathlessness; avoid lying flat for long.
- Chafing and skin-fold infections: breathable clothing.

14 Education and lifestyle

- Aim for slow, steady loss of about 0.5 kg a week.
- Fewer sugary drinks, sweets, fried snacks and refined flour; more vegetables, pulses, fruit and whole grains; smaller portions.
- Move more through the day; sleep 7–8 hours; manage stress.
- Track weight and waist monthly.

15 Prognosis

Even modest weight loss with regular activity lowers the risk of diabetes and heart disease. Fitness improves health even when weight changes little. Keeping weight off needs long-term activity.

16 Exam corner

S21 Q4(b), 3 marks

Give grades of obesity (inside a Phase II cardiac rehab case with BMI 32).

W24 Q1(a), 3 marks

Define metabolic syndrome.

Technique cards for this topic: exercise physiology (p. 313).

Keywords to underline

BMI Indian cut-offs 23 and 25 waist 90/80 cm metabolic syndrome 3 of 5 150–250 min a week low-impact resistance training 5–10% weight loss

Test yourself

3 m **Give grades of obesity.**

Answer
WHO: overweight 25–29.9; obese class I 30–34.9, class II 35–39.9, class III 40 or more. Asian Indian: normal 18.5–22.9, overweight 23–24.9, obese 25 or more; waist 90 cm men, 80 cm women.

MCQ **An Indian man has BMI 26 and waist 96 cm. By Indian criteria he is:**

A. Normal

B. Overweight only

C. Obese with abdominal obesity

D. Underweight

Answer
C. BMI 25 or more is obese for Asian Indians, and waist 90 cm or more is abdominal obesity in men.

MCQ **Which is NOT a metabolic syndrome criterion?**

A. Triglycerides 150 mg/dL or more

B. HDL below 40 mg/dL in men

C. LDL above 100 mg/dL

D. BP 130/85 or more

Answer
C. LDL is not one of the five criteria.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
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3. Misra A, et al. Consensus statement for diagnosis of obesity, abdominal obesity and the metabolic syndrome for Asian Indians. *JAPI* 2009. Revised definition of obesity in Asian Indians (2025 proposal).
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Tuberculosis, thoracoplasty and post-TB lung disease

> TB itself is rarely the question; its after-effects are. Four bronchiectasis cases were post-TB, one patient had a thoracoplasty, two had a destroyed lung (S22, S24) and a lobectomy patient had old TB (S25). Learn infection control, airway clearance and rehab for post-TB lung disease.

Syllabus	Ch 6(c)(i) Tuberculosis
Priority	**Tier 2.** 3 long-answer cases (W18 thoracoplasty; S22 and S24 destroyed lung) plus 4 post-TB bronchiectasis cases.
Asked as	LAQ W18 · S22 · S24
Usual split	5 + 5 + 5 (functional diagnosis, short-term, long-term) or 15 (diagnosis, goals, management in detail)
Checked against	India National TB Elimination Programme (NTEP) guidance; Clinical standards for post-tuberculosis lung disease (2021); NTEP pulmonary rehabilitation guidance (2024, draft). Full list under Sources. Content checked 19 October 2026.

1 Definition

Tuberculosis (TB) is an infectious disease caused by *Mycobacterium tuberculosis*, spread through the air when a person with lung TB coughs. It mainly affects the lungs (pulmonary TB) but can affect any organ.

Post-TB lung disease is lasting lung damage after TB is cured: fibrosis, cavities, bronchiectasis, airway narrowing, pleural thickening or a **destroyed lung** (one lung or lobe almost completely destroyed).

Thoracoplasty is an operation that removes parts of several ribs so the chest wall collapses inwards onto a diseased lung or an infected space. Once used to treat TB, it is now rare and used mainly for chronic empyema spaces.

2 Classification

- **Latent TB infection:** infected but no disease and not infectious.
- **Active pulmonary TB:** primary (children) or post-primary (reactivation in adults, usually in the upper lobes with cavities).
- **Extrapulmonary TB:** pleura (TB effusion), lymph nodes, spine (Pott's disease), meninges, abdomen.
- **Drug-sensitive or drug-resistant** (MDR-TB resists the two main drugs).
- **Post-TB sequelae:** bronchiectasis, fibrosis with volume loss, destroyed lung, aspergilloma in an old cavity, chronic obstruction.

3 Risk factors

- Close contact with an infectious patient; crowded, poorly ventilated housing.
- Undernutrition, diabetes, HIV, smoking, alcohol, silicosis, steroids or other immune suppression.
- Delayed diagnosis or incomplete treatment (risk of drug resistance and lung damage).

4 Pathophysiology to impairments

1. Inhaled bacilli cause granulomas; in active disease the centre breaks down (caseation) and is coughed out, leaving **cavities**.
2. Healing is by **fibrosis**, which shrinks the lobe, pulls the trachea and mediastinum to that side and distorts the airways (traction bronchiectasis).
3. Damaged airways cannot clear mucus, so the vicious cycle of bronchiectasis begins (see the bronchiectasis page).

4. Restriction (fibrosis), obstruction (airway damage) or both reduce exercise capacity; severe damage causes low oxygen and right heart strain.
5. After thoracoplasty: chest wall deformity, loss of ribs, paradoxical movement early on, and later scoliosis and shoulder problems on the operated side.

5 Clinical features

- **Active TB:** cough for more than 2 weeks, sputum, haemoptysis, evening fever, night sweats, weight loss, loss of appetite, fatigue.
- **Post-TB lung disease:** chronic cough with sputum, breathlessness, recurrent infections, haemoptysis, reduced chest expansion and flattening over the affected side, tracheal shift towards it.
- **Destroyed lung (S22):** breathlessness, chest pain, fever and cough with infective exacerbations.

6 Investigations

- **Sputum tests:** rapid molecular test (for example CBNAAT/Xpert, also detects rifampicin resistance), smear microscopy, culture.
- **Chest X-ray:** upper-lobe infiltrates, cavities, fibrosis with volume loss, calcification; tracheal shift; destroyed lung appears opaque and shrunken.
- CT for post-TB damage; spirometry (obstructive, restrictive or mixed); SpO_2, ABG, 6MWT.
- HIV and diabetes testing.

7 Medical and surgical management

- **Drug-sensitive TB (NTEP, India):** about 6 months of daily fixed-dose combination tablets: an intensive phase with four drugs (isoniazid, rifampicin, pyrazinamide, ethambutol) for 2 months, then a continuation phase (isoniazid, rifampicin, ethambutol) for 4 months.
- Drug-resistant TB needs longer, specialised regimens.
- Treatment support, nutrition support (Ni-kshay Poshan Yojana), contact tracing.
- **Surgery** for complications: lobectomy or pneumonectomy for destroyed lung or aspergilloma; decortication for empyema; thoracoplasty for chronic spaces.

8 Complications and prevention

- Haemoptysis (can be massive), pneumothorax, empyema, bronchiectasis, aspergilloma, respiratory failure, cor pulmonale.
- Drug side effects: liver injury, peripheral neuropathy (isoniazid), visual problems (ethambutol), joint pain.
- Post-thoracoplasty: scoliosis, shoulder stiffness, chest infection, paradoxical chest movement.

9 PT assessment

- Infectious status (on treatment how long? sputum results?), treatment adherence, nutrition and weight.
- Cough, sputum amount and colour, haemoptysis, breathlessness (mMRC), fever.
- Chest shape, expansion, tracheal position, auscultation; posture and spine; shoulder range (after thoracoplasty).
- SpO_2 at rest and on exercise; 6MWT; muscle strength.
- Work and home: crowding, ventilation, smoke exposure, earning role.
- **Outcome measures:** 6MWT, mMRC, St George's Respiratory Questionnaire, weight.

10 PT diagnosis by ICF

Worked example from a real paper (S22, 15 marks)

> "A 45 year old farmer, diagnosed case of destroyed lung secondary to tuberculosis. He complains of shortness of breath, chest pain, fever and cough since 4 days. Write functional diagnosis, goals and physiotherapy management in detail."

ICF component	From this case	Clinical reasoning
Health condition	Post-TB destroyed lung with an infective exacerbation	Damaged lung harbours infection
Impairments	Breathlessness; productive cough with retained sputum; pleuritic chest pain; fever; reduced lung volume; low exercise tolerance; possible low SpO_2	One lung barely works; infection adds sputum and pain
Activity limitations	Walking, lifting, field work, self-care during the exacerbation	Ventilatory reserve is low
Participation	Farming and family income; social life	Farm work is heavy and dusty
Environmental factors	**Barriers:** dust and smoke in farming, distance to hospital. **Facilitators:** family, NTEP services.	Plan masks and pacing at work
Personal factors	45-year-old man; possibly smoker and undernourished	Nutrition and smoking affect recovery

One-line functional diagnosis

Post-TB destroyed lung with an infective exacerbation, causing breathlessness, sputum retention, chest pain and low exercise tolerance, limiting walking and farm work and restricting his earning role.

How the S24 case differs (25-year-old student, right destroyed lung)

- **Clues:** crackles over the right upper chest (secretions), pain on deep breathing and coughing on the right (pleural involvement), HR 105 and RR 28 (fast and shallow breathing at rest), FEV1 60% and FVC 56% predicted (both reduced by a similar amount: a mainly restrictive loss of lung volume).
- **Participation:** a student who cannot run or play sport, so his goals are studies, sport and fitness rather than farm work.
- **Plan:** pain relief and supported breathing; ACBT with the right upper lobe drained in sitting; thoracic expansion of the left lung and the healthy parts of the right; then pulmonary rehabilitation with interval training and strengthening, aiming at a graded return to sport; infection control and completion of TB follow-up.

11 Problem list and goals

- **Short-term (1–2 weeks):** clears sputum independently; breathlessness controlled with breathing control and positioning; walks in the ward with stable SpO_2.
- **Long-term (2–3 months):** completes pulmonary rehab; better 6MWT and mMRC; returns to lighter farm work with pacing and a mask; fewer exacerbations.

12 PT management

During an exacerbation or active disease

- Airway clearance (ACBT, autogenic drainage, PEP devices) in positions that drain the affected area; avoid head-down tilt with haemoptysis.

- Breathing control, relaxation, positioning to ease breathlessness. For gas exchange, lie with the **better lung down** when one lung is destroyed.
- Pain relief and supported coughing for pleuritic pain.
- Early mobilisation as fever settles; hydration and nutrition advice.

Pulmonary rehabilitation for post-TB lung disease

- Endurance and strength training as in COPD (6–12 weeks, 2–3 supervised sessions a week), with oxygen if SpO_2 falls on exercise.
- Daily airway clearance routine; inspiratory muscle training if weak.
- Thoracic mobility and posture exercises; energy conservation for farm and household work.
- Education: complete TB treatment, recognise relapse, nutrition, smoking cessation, dust protection.

After thoracoplasty (W18 nurse, day 2)

- Deep breathing, ACBT and supported cough; early sitting and walking.
- Early on, a firm pad may be used over the chest wall to limit paradoxical movement; support the area when coughing.
- **Shoulder exercises** on the operated side from day 1 (muscles cut over the scapula).
- **Posture correction** in front of a mirror and trunk exercises to prevent scoliosis and a dropped shoulder.
- Later: exercise training, thoracic mobility, return to work.

13 Precautions and infection control

Infection control

- Until the doctor confirms the patient is no longer infectious (often after about 2 weeks of effective treatment for drug-sensitive TB): staff wear N95 respirators; the patient wears a surgical mask; treat in a well-ventilated room or outdoors.
- Teach cough etiquette and safe sputum disposal (covered container, disinfectant).
- Do not use shared devices without cleaning; avoid sputum induction in open areas.

Stop and report

- Fresh haemoptysis: stop percussion, vibration and head-down positions; sit up; report. Massive bleeding: lie on the bleeding side and call for help.
- Sudden breathlessness or chest pain (pneumothorax).

14 Education

- Take every dose of TB medicine for the full course, even when you feel better.
- Cover your mouth when coughing; keep windows open; dispose of sputum safely.
- Eat enough protein and calories; stop smoking and alcohol.
- Continue daily breathing exercises, airway clearance and walking after cure.
- Report return of fever, weight loss or blood in sputum.

15 Prognosis

Drug-sensitive TB is curable with full treatment. Many survivors are left with some lung damage, which causes breathlessness and infections for life; pulmonary rehabilitation improves exercise capacity and quality of life.

16 Exam corner

W18 Q4(b), 5 + 5 + 5 marks

52-year-old nurse, thoracoplasty 2 days ago for tuberculosis. Functional diagnosis; short- and long-term management.

S22 Q4(b), 15 marks

The worked example in section 10.

S24 Q4(a), 15 marks

25-year-old student with a right destroyed lung after TB; crackles over the right upper chest; pain on deep breathing and coughing on the right; cannot run or play sport; HR 105/min, RR 28/min; FEV1 60% and FVC 56% predicted. Functional diagnosis and management.

Related

Post-TB bronchiectasis cases S19, S20, S21 and W24 (see the bronchiectasis page). Pneumonectomy for a destroyed lung (S26) and lobectomy after old TB (S25): see the lung surgery page.

Technique cards for this topic: airway clearance (p. 258), ACBT and autogenic drainage (p. 261), exercise testing (p. 308), pulmonary rehabilitation (p. 339).

Keywords to underline

infection control N95 NTEP post-TB lung disease destroyed lung fibrosis, tracheal shift airway clearance pulmonary rehabilitation thoracoplasty: posture and shoulder

Test yourself

LAQ **W18: thoracoplasty day 2. List the short-term goals.**

Answer
Clear chest with effective supported cough; pain controlled; sitting out and walking in the ward; maintain shoulder range on the operated side; good posture habits; no paradoxical movement problems.

MCQ **A post-TB patient with a destroyed right lung is breathless. Best side-lying position for oxygenation:**

A. Right side down
B. Left side down
C. Head-down prone
D. Flat supine

Answer
B. Lying with the good (left) lung down sends more blood and ventilation to the working lung.

MCQ **A patient with smear-positive TB, started on treatment 3 days ago, needs physio. Which is correct?**

A. No precautions needed
B. Staff wear N95 respirators; treat in a ventilated room; patient wears a mask
C. Delay all physio for 6 months
D. Use shared nebulisers

Answer
B. The patient is still infectious early in treatment.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
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Interstitial and occupational lung disease

> Restrictive lung disease was asked in 4 of 20 papers (2016–2026), all between 2017 and 2021: a cotton-mill worker with ILD, a farmer with mMRC 2, and short questions on spirometry and goals. The key physio point: these patients desaturate on exertion, so rehab must be dosed and monitored with oxygen.

Syllabus	Ch 6(b)(v–vi) Occupational lung diseases, interstitial lung diseases
Priority	**Tier 2.** In 4 of 20 papers (2016–2026). 1 long-answer case.
Asked as	LAQ S18 · 7m S21 · 3m S17 · S19
Usual split	15 (ICF, goals, management) or 7-mark management
Checked against	ATS pulmonary rehabilitation guideline (2023: strong recommendation for ILD); ATS/ERS/JRS/ALAT IPF guideline (2022); ATS home oxygen guideline (2020). Full list under Sources. Content checked 19 October 2026.

1 Definition

Interstitial lung disease (ILD) is a group of disorders that inflame and scar (fibrose) the lung tissue between the air sacs (the interstitium). The lungs become stiff and small, and oxygen passes poorly into the blood.

Occupational lung disease is lung disease caused by breathing in dusts, fibres, fumes or organic material at work.

2 Classification

Group	Examples
Idiopathic	Idiopathic pulmonary fibrosis (IPF), the commonest fibrosing ILD in older adults
Connective tissue disease	Rheumatoid arthritis, systemic sclerosis, myositis
Hypersensitivity pneumonitis	Farmer's lung (mouldy hay or grain), bird fancier's lung
Pneumoconioses (mineral dusts)	**Silicosis** (stone cutting, mining, sandblasting, glass and ceramics), **asbestosis**, coal workers' pneumoconiosis
Organic dust airway disease	**Byssinosis** (cotton, flax, jute: chest tightness on the first working day of the week, "Monday chest tightness")
Others	Sarcoidosis, drug- or radiation-induced fibrosis, post-COVID fibrosis

3 Causes and risk factors

- Years of exposure to dusts at work without masks or ventilation: cotton mills (S18), stone quarries, mines, construction, agriculture (S21 farmer).
- Smoking, older age and male sex for IPF; family history; autoimmune disease; some drugs (amiodarone, methotrexate); radiotherapy.

4 Pathophysiology to impairments

1. Repeated injury causes inflammation and scarring of the interstitium.
2. Stiff lungs need more effort to inflate: lung volumes and compliance fall (**restriction**), so breathing becomes rapid and shallow.
3. Thickened alveolar walls slow oxygen transfer (low DLCO). During exercise, blood moves through the lungs faster, so there is even less time to pick up oxygen: **marked exertional desaturation**.
4. Low oxygen narrows lung vessels: pulmonary hypertension and right heart failure.

5. Breathlessness causes inactivity and muscle weakness; a dry cough is often troublesome.

5 Clinical features

- Progressive breathlessness on exertion; dry cough; fatigue; weight loss.
- Fine, "Velcro-like" crackles at the lung bases at the end of inspiration; clubbing (especially IPF); cyanosis late.
- Rapid, shallow breathing; reduced chest expansion; SpO_2 that falls sharply on walking.
- Signs of the cause: joint disease, skin changes, work history.

6 Investigations

Three spirometric findings in restrictive disease (3 marks in S19)

1. **FVC reduced** (and FEV1 reduced in proportion).
2. **FEV1/FVC normal or high** (0.70 or more, often above 0.80).
3. **Total lung capacity reduced** on lung volume testing, with **low DLCO** in ILD.

See the spirometry curve on the COPD page (p. 76): restriction lowers the plateau, obstruction slows the rise.

- **HRCT:** reticular lines, honeycombing, ground-glass areas, traction bronchiectasis (S21: "heterogeneous opacities").
- **Chest X-ray:** small lungs, reticular or nodular shadows; silicosis shows upper-zone nodules and "eggshell" calcified nodes.
- **6MWT with SpO_2:** distance and the lowest SpO_2 (key for rehab and oxygen prescription).
- ABG, echo (pulmonary hypertension), autoimmune blood tests.

7 Medical management

- **Remove exposure:** leave or change the dusty job; masks and dust control.
- **Antifibrotic drugs** (nintedanib, pirfenidone) slow decline in IPF and other progressive fibrosis.
- Steroids or immunosuppressants for inflammatory types (hypersensitivity pneumonitis, connective tissue disease).
- **Oxygen:** long-term if resting oxygen is low; ambulatory oxygen for marked exertional desaturation.
- **Pulmonary rehabilitation:** strong recommendation for ILD (ATS 2023).
- Vaccinations, smoking cessation, treatment of reflux and cough; lung transplant in selected patients; palliative care in advanced disease.

8 Complications

- Respiratory failure; pulmonary hypertension and cor pulmonale.
- Acute exacerbations (sudden worsening, often fatal in IPF); infections; pneumothorax.
- Lung cancer (higher with IPF, asbestosis and silicosis); TB (higher with silicosis).
- Deconditioning, anxiety, depression.

9 PT assessment

- Work history (type and years of exposure, masks), smoking, symptoms (mMRC), cough.
- Breathing pattern, chest expansion, crackles, clubbing.
- **SpO_2 at rest and during a 6MWT**; heart rate; Borg breathlessness.
- Muscle strength; function in ADL and work.
- **Outcome measures:** 6MWT distance and lowest SpO_2, mMRC, St George's Respiratory Questionnaire (IPF version), King's Brief ILD questionnaire.

10 PT diagnosis by ICF

Worked example from a real paper (S18, 15 marks)

> "A 40 year old female, working in cotton mill since past 15 years was admitted for complaint of grade II dyspnoea with diagnosis of ILD. Discuss ICF. Write physiotherapy goals and management for the same."

ICF component	From this case	Clinical reasoning
Health condition	Occupational interstitial lung disease after 15 years of cotton dust exposure	Long dust exposure caused fibrosis
Impairments	Breathlessness (grade II), reduced lung volumes, exertional desaturation, dry cough, reduced exercise tolerance, weakness	Stiff lungs and slow diffusion
Activity limitations	Walking fast, stairs, carrying loads, housework	Oxygen falls with effort
Participation	Continuing mill work; household and family roles	The workplace is the cause
Environmental factors	**Barriers:** continued cotton dust exposure, financial need to work. **Facilitators:** hospital care, possible workplace transfer or masks.	Removing exposure is the most important step
Personal factors	40-year-old woman, earning member	Work change needs counselling

> **One-line functional diagnosis**
>
> Occupational ILD with restrictive impairment, exertional breathlessness (grade II) and desaturation, limiting walking, stairs and housework, and restricting her mill work; continued dust exposure is the main barrier.

11 Problem list and goals (3 marks in S17)

- **Short-term:** safe exercise with SpO_2 kept at about 88% or more (with oxygen if needed); breathing control and pacing during ADL; cough control strategies.
- **Long-term:** better exercise tolerance (6MWT) and quality of life; independent home exercise; energy conservation; avoidance of further dust exposure; return to suitable work.

12 PT management

Pulmonary rehabilitation for ILD (7 marks in S21)

- **Aerobic training:** walking or cycling 3–5 days a week, 20–30 minutes, at moderate breathlessness (Borg 3–4); use **interval training** (short bouts with rests) if SpO_2 drops quickly.
- **Oxygen during exercise:** titrate to keep SpO_2 at about 88–90% or more; many ILD patients need it.
- **Resistance training** of legs and arms 2–3 days a week.
- **Breathing control** and recovery positions (forward lean); pacing; energy conservation.
- **Cough control:** sip water, swallow, nose breathing, suppression techniques for a dry cough.
- Thoracic mobility and posture exercises; inspiratory muscle training may help.
- Education: disease, oxygen safety (no smoking or flames near oxygen), when to seek help, workplace protection.

> **Why ILD rehab is different from COPD rehab**
>
> Airway clearance is rarely needed (dry cough). The main limit is **oxygen transfer**, so monitor SpO_2 continuously and use oxygen and interval training. Progress more gently, especially with pulmonary hypertension.

13 Precautions and stop criteria

- Stop or reduce if SpO_2 falls below about 85% despite oxygen, or with chest pain, dizziness or severe breathlessness.
- Pulmonary hypertension: avoid sudden heavy exertion, breath-holding and heavy lifting; watch for syncope.
- Steroid use: bone fragility.
- Oxygen safety around flames and smoking.

14 Education and prevention

- Avoid the dust that caused the disease; ask about job transfer; wear a fitted respirator mask if exposure continues.
- Stop smoking.
- Keep active every day within safe SpO_2 limits.
- Get vaccinated; report fever or sudden worsening breathlessness early.
- Silicosis: screening for TB.

15 Prognosis

Varies by type: IPF progresses over a few years even with treatment; some hypersensitivity and inflammatory types improve if exposure stops and treatment starts early. Pulmonary rehabilitation improves exercise capacity, breathlessness and quality of life.

16 Exam corner

S18 Q4(b), 15 marks

The worked example in section 10.

S21 Q2(d), 7 marks

Physiotherapy for a 43-year-old male farmer with ILD, heterogeneous opacities in the right lung, mMRC grade 2, dry cough and generalised weakness.

3-mark questions

Goals of management in restrictive lung disease (S17). Three spirometric findings in restrictive lung disease (S19).

Technique cards for this topic: oxygen therapy (p. 290), exercise testing (p. 308), ABG (p. 326), dyspnoea and energy conservation (p. 285), pulmonary rehabilitation (p. 339).

Keywords to underline

restrictive pattern FEV1/FVC normal or high low TLC and DLCO exertional desaturation oxygen during exercise interval training remove exposure byssinosis silicosis

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Test yourself

3 m **Three spirometric findings in restrictive lung disease.**

> **Answer**
> Reduced FVC; normal or raised FEV1/FVC; reduced TLC (with low DLCO in ILD).

MCQ **An ILD patient's SpO_2 falls from 95% to 84% after 2 minutes of walking. Best plan:**

A. Stop all exercise

B. Interval training with oxygen to keep SpO_2 about 88% or more

C. Walk faster to build tolerance

D. Postural drainage instead

Answer

B. Exercise remains beneficial when oxygen is used and bouts are shortened.

MCQ **Chest tightness worse on the first day of the working week in a textile worker suggests:**

A. Silicosis

B. Byssinosis

C. Asbestosis

D. Farmer's lung

Answer

B. Byssinosis from cotton dust classically causes "Monday chest tightness".

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Rochester CL, et al. Pulmonary Rehabilitation for Adults with COPD (and ILD recommendations). ATS Clinical Practice Guideline, *Am J Respir Crit Care Med* 2023. ATS press release.
3. Raghu G, et al. Idiopathic Pulmonary Fibrosis (an Update) and Progressive Pulmonary Fibrosis in Adults. ATS/ERS/JRS/ALAT guideline, 2022.
4. Jacobs SS, et al. Home Oxygen Therapy for Adults with Chronic Lung Disease. ATS guideline, 2020.
5. Structure and classical content: Main & Denehy (2016); Hillegass (2022).

Pleural effusion, pneumothorax, empyema and pleurisy

> Pleural disease was in 4 of 20 papers (2016–2026): X-ray of pleural effusion, pneumothorax with chest drain precautions, and a long case of empyema with a chest drain (S25). Learn the X-ray signs, the drain rules and why positive pressure is dangerous with an undrained pneumothorax.

Syllabus	Ch 6(c)(v–x) Pneumothorax, hydropneumothorax, pleuritis, pleural effusion, empyema and other pleural disorders
Priority	**Tier 2.** In 4 of 20 papers (2016–2026): 1 long-answer case (S25 empyema) and short answers.
Asked as	LAQ S25 · 7m W20 · W22 · 3m S16 · S25
Usual split	15 (ICF, goals, physiotherapy); 7-mark and 3-mark answers
Checked against	BTS pleural disease guideline (2023); Light's criteria; BTS/ACPRC physiotherapy guideline (2009). Full list under Sources. Content checked 19 October 2026.

1 Definitions

- **Pleural effusion:** abnormal fluid in the pleural space.
- **Pneumothorax:** air in the pleural space, so the lung collapses partly or fully.
- **Hydropneumothorax:** both air and fluid in the pleural space.
- **Empyema:** pus in the pleural space.
- **Pleurisy (pleuritis):** inflammation of the pleura, causing sharp pain on breathing.
- **Haemothorax:** blood in the pleural space (usually trauma).

2 Classification

Pleural effusion: transudate or exudate (Light's criteria)

The fluid is an **exudate** if any one of these is present:

1. Pleural fluid protein ÷ serum protein more than **0.5**
2. Pleural fluid LDH ÷ serum LDH more than **0.6**
3. Pleural fluid LDH more than **two-thirds** of the upper normal limit for serum LDH

Type	Mechanism	Common causes
Transudate	Pressure imbalance; pleura normal	Heart failure, cirrhosis, kidney disease, low albumin
Exudate	Inflamed or diseased pleura leaks protein	Pneumonia (parapneumonic), TB (common in India), cancer, pulmonary embolism

Pneumothorax types

- **Primary spontaneous:** no known lung disease; typically tall, thin young men; smoking raises risk.
- **Secondary spontaneous:** underlying lung disease (COPD with bullae, TB, cystic fibrosis, ILD); more dangerous.
- **Traumatic** (rib fractures, stab wounds) and **iatrogenic** (after procedures, central lines, ventilation).
- **Tension pneumothorax:** a one-way valve lets air in but not out; pressure pushes the mediastinum across and stops venous return. A medical emergency.

3 Pathophysiology to impairments

- Fluid or air in the pleural space stops the lung expanding: **reduced lung volume, breathlessness, low oxygen**.

- Large collections push the mediastinum to the **opposite** side; collapse (atelectasis) pulls it **towards** the affected side.
- Inflamed pleura hurts when it moves, so breathing becomes shallow on that side (pleurisy).
- Untreated empyema or haemothorax leaves a thick peel (fibrothorax) that traps the lung, reduces chest wall movement and can cause scoliosis.

4 Clinical features

Sign	Pleural effusion	Pneumothorax
Expansion on the affected side	Reduced	Reduced
Percussion	**Stony dull**	**Hyper-resonant**
Breath sounds	Reduced or absent (bronchial breathing above the fluid)	Reduced or absent
Trachea (large)	Pushed away	Pushed away (tension)
Onset	Gradual breathlessness	Sudden one-sided chest pain and breathlessness

- **Pleurisy:** sharp pain on breathing in and coughing; pleural rub (creaking sound).
- **Empyema:** swinging fever, sweats, chest pain, not improving on antibiotics.
- **Tension pneumothorax:** severe distress, low BP, fast heart rate, tracheal shift away, distended neck veins.

5 Investigations and X-ray findings

Radiological findings of pleural effusion (3 marks in S16)

- **Blunting of the costophrenic angle** (needs about 200 mL or more on a PA film; less on a lateral film).
- A homogeneous white (opaque) area at the lung base with a **curved upper border rising towards the axilla** (meniscus sign).
- Loss of the diaphragm outline on that side.
- Large effusion: mediastinum and trachea pushed to the opposite side; whole hemithorax white ("white-out").
- Lying films: diffuse haziness rather than a level.

Pneumothorax on X-ray

- A thin visible pleural line with **no lung markings beyond it**.
- Tension: mediastinal shift away, flattened diaphragm.
- **Hydropneumothorax:** a straight horizontal **air–fluid level** (air above, fluid below).
- Ultrasound to find and mark fluid; CT for complex cases.
- Pleural tap (aspiration) for protein, LDH, glucose, pH, cells, culture, TB tests and cancer cells.

6 Medical and surgical management

- **Effusion:** treat the cause; therapeutic aspiration or an intercostal drain for large or symptomatic effusions; pleurodesis or an indwelling catheter for recurrent malignant effusion.
- **Pneumothorax (BTS 2023):** management depends mainly on **symptoms and stability**, not just size: observation for minimal symptoms; needle aspiration, ambulatory device or chest drain for symptomatic patients; surgery (VATS pleurodesis) for recurrence or persistent air leak.
- **Tension pneumothorax:** immediate needle decompression, then a chest drain.

- **Empyema:** antibiotics, chest drain, sometimes drugs to break down the fluid; surgery (VATS or decortication) if it does not drain.
- **Pleurisy:** treat the cause; pain relief.

Classic answer vs current guideline: pneumothorax size

Classic	Current (BTS 2023)	Write in the exam
A rim of more than 2 cm on X-ray means a chest drain.	Decisions are led by symptoms and stability. Many patients with few symptoms can be observed or treated as outpatients, even with a larger rim.	"Management depends on symptoms: observation, aspiration or intercostal drain."

7 Complications

- Trapped lung, fibrothorax, chest wall stiffness, scoliosis.
- Recurrent pneumothorax; tension pneumothorax; persistent air leak.
- Re-expansion pulmonary oedema after rapid drainage of a large effusion.
- Infection of the drain site; sepsis from empyema.

8 PT assessment

- Breathlessness, pain on breathing, cough; drain details (site, swing, bubbling, drainage).
- Breathing pattern, chest expansion (compare sides), percussion note, auscultation, tracheal position.
- SpO_2, X-ray, posture (leaning towards the painful side), shoulder range.
- Mobility and exercise tolerance.

9 PT diagnosis by ICF (practice)

Worked example from a real paper (S25, 3 + 4 + 8 marks)

The question

"A 45 year old male driver by occupation, tobacco chewer is a diagnosed case of empyema, has intercostal drainage tube since 15 days. He complains of pain at the site of ICD, cough with expectoration and difficulty in doing his ADLs. Write ICF, short term goals and long term goals and physiotherapy treatment for the same."

(i) ICF, 3 marks

Part	From this case
Health condition	Empyema with an intercostal drain for 15 days
Impairments	Pain at the drain site; reduced expansion of the affected side (risk of a trapped lung and pleural thickening); sputum with a weak, painful cough; reduced exercise tolerance after 15 days of illness; stiff shoulder and trunk on the drain side; leaning towards the affected side
Activity limitations	Bathing, dressing and toileting; walking any distance and stairs; lifting
Participation restrictions	Cannot work as a driver, so income is lost; family and social roles
Environmental factors	Facilitators: hospital team, family. Barriers: drain tubing limits movement; easy access to tobacco
Personal factors	45-year-old man; tobacco chewer; working age

(ii) Goals, 4 marks. Short-term (about 2 weeks): pain at the drain site 3/10 or less during breathing and coughing; clears sputum on his own with ACBT; better expansion of the affected side; independent in self-care and walking on the ward with the drain; full shoulder movement on the drain side. Long-term (6–12 weeks): full chest expansion and upright posture without pleural thickening; fitness to sit and drive for long hours; tobacco-free; independent home programme.

(iii) Physiotherapy, 8 marks.

- **Before each session:** check the drain (swing, bubbling, amount and type of fluid), temperature, SpO_2; treat after analgesia.
- **Positioning:** side-lying on the good side, so the affected side is uppermost and expands; high sitting; change position regularly.
- **Lung expansion:** thoracic expansion exercises with a 3-second hold, hands on the affected lower ribs for feedback, 5–10 breaths every hour while awake.
- **Sputum:** ACBT with huffing; supported cough with a hand or pillow over the drain site.
- **Mobility:** walking with the bottle carried below chest level, increasing distance daily; stairs before discharge.
- **Shoulder and trunk:** active shoulder flexion and abduction on the drain side; trunk side-bending away from the drain; posture correction in a mirror.
- **After the drain is removed:** thoracic mobility, strengthening and aerobic training; graded return to driving once he can turn his trunk and move his shoulder without pain.
- **Education:** stop tobacco (refer to a cessation service); good nutrition; report fever, chest pain or more breathlessness (the empyema can re-collect); home exercises. If decortication is planned, prepare him with pre-operative physiotherapy.

Practice case (not from a paper): a 30-year-old tailor with a right TB pleural effusion after drainage. Impairments: reduced right basal expansion, pleuritic pain, breathlessness, reduced fitness, posture leaning to the right. Activity: walking fast, lifting. Participation: work at the sewing machine for long hours. Environment: crowded workshop (barrier), TB treatment programme (facilitator).

10 Goals

- **Short-term:** pain controlled; full use of the lung on the affected side during deep breaths; safe mobilisation with a drain.
- **Long-term:** full chest expansion and posture; prevention of pleural thickening and scoliosis; normal exercise tolerance; prevention of recurrence (pneumothorax: stop smoking).

11 Physiotherapy for pleural effusion and empyema

- **Positioning:** lie on the **unaffected** side, so the affected lung is uppermost and can expand, and the healthy lung below has better ventilation and blood flow.
- **Localised thoracic expansion exercises** to the affected side (hands on the lower ribs for feedback), with a 3-second hold.
- Supported coughing; ACBT if secretions are present.
- Early mobilisation; walking with the drain.
- Trunk side-bending away from the affected side and posture correction; shoulder exercises.
- After resolution: exercise training to restore fitness.

12 Physiotherapy for pneumothorax and with an intercostal drain (7 marks in W22)

- Check the drain before and after each session: swing, bubbling, drainage, no kinks; bottle upright and below chest level; never clamp a bubbling drain (full detail on the lung surgery page (p. 89)).

- Breathing control; deep breathing exercises and thoracic expansion to help the lung re-expand once drained.
- Supported cough at the drain site; adequate pain relief.
- Early mobilisation with the drain; shoulder ROM on the drain side.
- Posture correction.

Precautions and contraindications

- **Undrained pneumothorax:** do not use positive-pressure techniques (IPPB, NIV, PEP, manual hyperinflation) until a drain is in. They can turn it into a tension pneumothorax.
- Avoid vigorous percussion over the drain and fresh wounds.
- Avoid heavy straining and breath-holding until healed.
- No flying until the doctor confirms resolution; no scuba diving unless definitive surgery has been done.

Stop and call for help if

- Sudden worsening breathlessness, falling BP, tracheal shift (tension)
- The drain falls out or disconnects
- Sudden heavy blood in the drain

13 Physiotherapy for pleurisy

- Pain relief first; lying on the painful side may splint it and ease pain.
- Supported breathing: hand pressure over the painful area during deep breaths and coughing.
- Gentle thoracic expansion as pain allows, to prevent basal collapse.
- Early mobilisation.

14 Education

- Stop smoking (strongly linked to pneumothorax recurrence).
- Complete TB or other treatment.
- Return to work and sport gradually; know the warning signs of recurrence (sudden chest pain and breathlessness).

15 Prognosis

Most effusions and pneumothoraces resolve with treatment of the cause and drainage. Pneumothorax can recur, especially in smokers and those with lung disease. Early physiotherapy helps prevent pleural thickening and chest stiffness.

16 Exam corner

S25 Q4(a), 3 + 4 + 8 marks

45-year-old driver who chews tobacco, empyema with an intercostal drain for 15 days; pain at the drain site, cough with sputum, difficulty with daily activities. ICF; short-term goals; long-term goals and physiotherapy. See the worked example in section 9.

W22 Q2(f), 7 marks

Pneumothorax; precautions, contraindications and physiotherapy for a patient with an intercostal drain.

W20 Q2(a), 7 marks

Intercostal drainage and underwater seal (see the lung surgery page).

S16 Q1(f), 3 marks

Radiological findings of pleural effusion.

S25 Q1(a), 3 marks

Indications of intercostal drainage (see the lung surgery page).

Technique cards for this topic: positioning (p. 236), breathing exercises (p. 275), chest X-ray (p. 317).

Keywords to underline

costophrenic angle blunting · meniscus sign · stony dull · hyper-resonant · tension pneumothorax · air–fluid level · Light's criteria · lie on the unaffected side · no positive pressure before a drain

Test yourself

3 m **Radiological findings of pleural effusion.**

Answer

Blunted costophrenic angle; homogeneous basal opacity with a meniscus rising laterally; loss of the diaphragm outline; large effusions push the mediastinum away; supine films show diffuse haze.

MCQ **A patient with an undrained pneumothorax becomes breathless. Which technique is contraindicated?**

A. Breathing control

B. IPPB or manual hyperinflation

C. Sitting upright

D. Relaxation

Answer

B. Positive pressure can enlarge the pneumothorax or cause tension.

MCQ **X-ray shows a straight horizontal line with air above and fluid below in the pleural space. This is:**

A. Simple effusion

B. Hydropneumothorax

C. Consolidation

D. Collapse

Answer

B. An air–fluid level needs both air and fluid.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Roberts ME, et al. British Thoracic Society Guideline for pleural disease. *Thorax* 2023. PMC
3. Light RW. Pleural effusions: the diagnostic separation of transudates and exudates. *Ann Intern Med* 1972.
4. Bott J, et al. BTS/ACPRC physiotherapy guideline. *Thorax* 2009.

Hypertension

> Hypertension was asked on its own as a 7-mark question (S18) and as a long case (S23: a teacher admitted with chest pain and hypertension), and it appears in many other long cases as a risk factor. Know the classification, the exercise dose, the lifestyle changes with their effect, and when not to exercise.

Syllabus	Ch 6(a)(i) Hypertension
Priority	**Tier 2.** 1 standalone 7-mark question (S18) and 1 long case (S23); a co-existing condition in many cardiac and vascular cases.
Asked as	LAQ S23 · 7m S18
Usual split	7-mark answer; 15 as a case with chest pain; lifestyle part of cardiac long answers
Checked against	2025 AHA/ACC hypertension guideline; ACC/AHA 2017 non-drug treatment table; ESC 2024 and ISH 2020 guidelines; BJSM network meta-analysis of exercise and blood pressure (2023). Full list under Sources. Content checked 19 October 2026.

1 Definition

Hypertension is persistently raised arterial blood pressure, measured correctly on two or more occasions. Most guidelines use **140/90 mmHg or more** (office reading) as hypertension; the ACC/AHA uses 130/80 or more.

2 Classification

ACC/AHA categories (2017, kept in 2025)

Category	Systolic	Diastolic
Normal	Below 120	and below 80
Elevated	120–129	and below 80
Stage 1 hypertension	130–139	or 80–89
Stage 2 hypertension	140 or more	or 90 or more
Hypertensive crisis	Above 180	and/or above 120 (emergency if organ damage)

- **ESC 2024:** non-elevated below 120/70; **elevated BP** 120–139/70–89; hypertension 140/90 or more.
- **ISH 2020 and Indian guidelines:** hypertension 140/90 or more; grade 1 140–159/90–99; grade 2 160/100 or more.
- **Primary (essential):** about 90% or more, no single cause. **Secondary:** kidney disease, hormonal causes, sleep apnoea, drugs.

3 Causes and risk factors

- Age, family history, obesity (especially abdominal), high salt intake (pickles, papad, packaged snacks), low potassium intake, physical inactivity, alcohol, stress, smoking, diabetes, sleep apnoea.
- Secondary causes: chronic kidney disease, renal artery stenosis, hyperaldosteronism, thyroid disease, coarctation of the aorta, pregnancy, steroids and NSAIDs.

4 Pathophysiology to impairments

- BP = cardiac output × peripheral resistance. Most hypertension comes from stiff, narrowed small arteries (raised resistance), with salt and water retention and nervous system overactivity.

- Long-term high pressure damages **target organs**: heart (left ventricular hypertrophy, heart failure, coronary disease), brain (stroke, dementia), kidneys, eyes (retinopathy) and arteries (PAD, aneurysm).
- Physio-relevant impairments: reduced exercise tolerance, risk during straining, drug side effects.

? Why does regular exercise lower BP?

> Training improves the function of the artery lining, relaxes small arteries, lowers sympathetic nervous activity and helps weight loss, so resting resistance falls. A single session also lowers BP for several hours afterwards (post-exercise hypotension).

5 Clinical features

Usually **no symptoms** (the "silent killer"). Sometimes headache, dizziness or nosebleeds. Very high BP can cause severe headache, blurred vision, chest pain, breathlessness or confusion (an emergency).

6 Investigations

- Correct BP measurement: seated, back supported, arm at heart level, correct cuff size, rested 5 minutes, two readings; home or 24-hour readings to confirm.
- Blood and urine tests for kidney function, sugar, lipids; ECG for left ventricular hypertrophy; echo; eye examination.

7 Medical management

- **Lifestyle change** for everyone (section 14).
- **Drugs:** ACE inhibitors or ARBs, calcium channel blockers, thiazide-type diuretics; beta-blockers for specific indications. Often two drugs in one tablet.
- **Targets:** below 130/80 for most adults (ACC/AHA 2025); ESC 2024 aims for a systolic of 120–129 if tolerated.

Drugs and exercise

Drug	Effect on exercise	What to do
Beta-blockers	Lower heart rate response, fatigue	Use RPE, not HR targets
Diuretics	Dehydration, low potassium, cramps	Fluids; avoid heat
Alpha-blockers, vasodilators, calcium channel blockers	Drop in BP after exercise	Longer cool-down; rise slowly
ACE inhibitors, ARBs	Few exercise effects; occasional dizziness	Usual monitoring

8 Complications

Stroke, heart attack, heart failure, kidney failure, retinopathy, aortic aneurysm and dissection, PAD, vascular dementia. Good BP control lowers all of these.

9 PT assessment

- BP both arms at rest (correct cuff), HR, medicines and timing, target-organ disease (heart, kidney, eyes), other risk factors.
- Activity level, diet and salt intake, alcohol, smoking, stress, sleep.
- BMI, waist; fitness (6MWT or step test) with BP before and after.
- **Outcome measures:** resting BP (home readings), weight and waist, 6MWT, activity minutes a week.

10 PT diagnosis by ICF

Real paper (S23, 15 marks): schoolteacher, 50, chest pain and hypertension, treated conservatively

Functional diagnosis: a 50-year-old schoolteacher with hypertension and chest pain of probable cardiac origin (angina), managed with medicines, with reduced exercise tolerance and fear of exertion, limiting walking, stairs and standing to teach, and restricting her work and family roles.

- **Before treating:** confirm the cause of the chest pain with the doctor (ECG, troponin); resting BP below 180/110; no chest pain at rest.
- **Plan:** treat it like stable angina with hypertension: education on BP and angina, GTN use and warning signs; graded walking at RPE 11–13 (moderate), building to 30 minutes on most days, kept below the heart rate that brings on angina; light resistance training with breathing out on effort; relaxation and stress management; a low-salt diet rich in fruit and vegetables, and weight control; regular medicines and home BP checks. The angina sections of the IHD page give the doses and stop criteria.
- **Participation:** pacing at school (sitting for part of the class, planning stair use), return to full duties as symptoms settle.

Practice case (not from a paper): a 52-year-old bank clerk with stage 2 hypertension, BMI 28, sedentary, high salt intake and work stress. Impairments: raised BP, low fitness, excess weight. Activity: limited brisk walking and stair climbing. Participation: at risk of losing work through stroke or heart disease. Environment: desk job and fried canteen food (barriers); family and a nearby park (facilitators). Personal: stress, low awareness.

11 Goals

- **Short-term (4 weeks):** correct home BP measurement; walking 30 minutes on most days; salt reduced.
- **Long-term (3–6 months):** BP at target; 5% or more weight loss; regular aerobic, resistance and isometric training; stress management habits.

12 PT management (7 marks in S18)

Exercise prescription for hypertension

Type	Dose
Aerobic	Brisk walking, cycling, swimming; moderate intensity (talk test, RPE 12–13); about 30 minutes on most days. ACC/AHA 2025: **75–150 min a week** of aerobic and/or resistance exercise.
Dynamic resistance	2–3 days a week; major muscle groups; moderate loads (about 50–80% of 1RM), about 3 sets of 10; breathe out on effort
Isometric	Handgrip **4 × 2 minutes** at **30–40%** of maximum grip, 1–3 minutes' rest between, 3 days a week; or wall squats in the same pattern. Gave the largest BP fall in a 2023 analysis of 270 trials (about 8/4 mmHg).
Flexibility and relaxation	Stretching; yoga with slow breathing; relaxation training for stress

- Warm-up and a **longer cool-down** (10 minutes) to avoid a sudden BP drop.
- Progress duration first, then intensity.
- Combine with weight loss and diet change for the best effect.

13 Precautions and stop criteria

Do not start exercise if

- Resting BP is above about 180/110: recheck after rest and refer before exercising.
- There are symptoms of a hypertensive emergency (severe headache, chest pain, blurred vision, confusion).

Stop exercise if

- Systolic rises above about 250 or diastolic above about 115 mmHg during exercise
- Systolic BP falls as the workload rises
- Chest pain, severe breathlessness, dizziness or headache

- Avoid breath-holding (Valsalva), very heavy lifting and long head-down positions.
- Isometric holds: keep breathing throughout; avoid in uncontrolled hypertension.

14 Lifestyle modification

Change	Target (ACC/AHA 2025)	Why
Less salt	Sodium below **2,300 mg a day**, ideally 1,500 mg (avoid pickles, papad, packaged snacks, added salt)	Less fluid retention and arterial stiffness
Weight loss	**5% or more** of body weight if overweight	BP falls with weight loss
Healthy diet	More vegetables, fruit, pulses, low-fat dairy (DASH-style)	Potassium and fibre lower BP
Physical activity	75–150 min a week of aerobic and/or resistance exercise	Lowers resting BP and heart risk
Alcohol	Avoid; if drinking, no more than 2 a day (men) or 1 (women)	Alcohol raises BP
Stop smoking; manage stress; sleep well	Tobacco-free; relaxation, yoga; treat sleep apnoea	Reduces overall cardiovascular risk

15 Prognosis

Treated, well-controlled hypertension greatly reduces stroke and heart disease. Lifestyle change alone can control stage 1 hypertension in many people and reduces the number of drugs needed in others.

16 Exam corner

S23 Q3(b), 15 marks

50-year-old schoolteacher admitted with chest pain and hypertension, treated conservatively. Functional diagnosis and physiotherapy management. Use this page with the angina sections of the IHD page.

S18 Q2(a), 7 marks

Physiotherapy management of hypertension.

Inside long cases

"Known hypertensive" in CABG and MI cases (S17, S18, W18, W19, W20, S24, S26), amputation (S19) and others.

Technique cards for this topic: exercise physiology (p. 313), exercise testing (p. 308), cardiac rehabilitation (p. 334), drugs (p. 352), yoga (p. 302).

Keywords to underline

140/90 stage 1 and 2 target organ damage aerobic most days isometric handgrip avoid Valsalva longer cool-down salt below 2,300 mg do not exercise above 180/110

Test yourself

7 m **Physiotherapy management of hypertension.**

Answer
Define and classify; assessment; aims; exercise prescription (aerobic, resistance, isometric with doses); warm-up and cool-down; precautions and stop criteria; drug effects; lifestyle advice table; outcome measures.

MCQ **Resting BP before a session is 196/112. You should:**

A. Start with light exercise

B. Recheck after rest; do not exercise and refer if it stays above about 180/110

C. Do isometric handgrip

D. Do a maximal exercise test

Answer
B. Uncontrolled high BP is a contraindication to starting exercise.

MCQ **A patient on an alpha-blocker feels dizzy after stopping exercise suddenly. Best prevention:**

A. Longer, gradual cool-down

B. Exercise harder

C. Hold the breath when lifting

D. Lie head-down after exercise

Answer
A. Vasodilating drugs plus post-exercise vasodilation can drop BP; a gradual cool-down prevents it.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. 2025 AHA/ACC Guideline for the Prevention, Detection, Evaluation and Management of High Blood Pressure in Adults; summary at HCPLive.
3. Whelton PK, et al. 2017 ACC/AHA high blood pressure guideline (non-pharmacological interventions table).
4. McEvoy JW, et al. 2024 ESC Guidelines for the management of elevated blood pressure and hypertension. Unger T, et al. 2020 ISH Global Hypertension Practice Guidelines.
5. Edwards JJ, et al. Exercise training and resting blood pressure: a large-scale pairwise and network meta-analysis of randomised controlled trials. *Br J Sports Med* 2023.

Valve disease, heart failure, cardiomyopathy, pacemaker and ICD

> These topics were rarely asked alone, but a 2022 long answer asked for pre- and post-implantation physiotherapy for a teacher getting a pacemaker and ICD, W21 asked for cardiac rehab after valve replacement (7 marks), and NYHA grades were asked as 3 marks. Learn the device precautions, the valve-surgery precautions and how to exercise a weak heart safely.

Syllabus	Ch 6(a)(iii) Valvular heart disease; 6(a)(v) Acquired heart disease
Priority	**Tier 2.** 1 long-answer case (S22), a 7-mark question (W21, rehab after valve replacement) and a 3-mark question (S21, NYHA); heart failure appears inside cardiac cases (S20, EF 25%; S25, EF 40%).
Asked as	LAQ S22 · 7m W21 (valve replacement) · 3m S21 (NYHA)
Usual split	15 (pre- and post-implantation assessment and management)
Checked against	ESC heart failure guideline (2021, focused update 2023); Universal definition of heart failure (2021); ACSM guidance on exercise with implantable devices; device clinic patient guidance. Full list under Sources. Content checked 19 October 2026.

1 Definitions

- **Valvular heart disease:** a heart valve that does not open fully (**stenosis**) or does not close fully (**regurgitation**).
- **Rheumatic heart disease (RHD):** permanent valve damage after rheumatic fever, which follows untreated streptococcal throat infection. Still common in India; the mitral valve is most often affected.
- **Heart failure:** a condition in which the heart cannot pump enough blood for the body's needs, or can do so only at high filling pressures, causing breathlessness, fatigue and fluid retention.
- **Cardiomyopathy:** disease of the heart muscle itself (dilated, hypertrophic or restrictive).
- **Pacemaker:** a device that sends electrical impulses to keep the heart rate from going too low. **ICD** (implantable cardioverter-defibrillator): a device that detects dangerous fast rhythms and treats them with pacing or a shock.

2 Classification and grading

Heart failure by ejection fraction (universal definition 2021)

- **Reduced EF:** 40% or less.
- **Mildly reduced EF:** 41–49%.
- **Preserved EF:** 50% or more (stiff ventricle that fills poorly).

NYHA functional classes (3 marks in S21)

Class	Symptoms
I	No limitation; ordinary activity causes no undue breathlessness, fatigue or palpitations
II	Slight limitation; comfortable at rest, ordinary activity causes symptoms
III	Marked limitation; comfortable at rest, less than ordinary activity causes symptoms
IV	Symptoms at rest; any activity increases discomfort

Valve lesions and their murmurs

Lesion	Murmur	Exercise point
Mitral stenosis (commonest in RHD)	Low rumbling mid-diastolic murmur at the apex	Fast heart rates shorten filling time: breathlessness, pulmonary oedema; atrial fibrillation common
Mitral regurgitation	Pansystolic murmur at the apex, spreading to the axilla	Fatigue, breathlessness; AF
Aortic stenosis	Ejection systolic murmur at the right second intercostal space, spreading to the neck	**Symptomatic severe aortic stenosis is a contraindication to exercise**: risk of fainting and sudden death
Aortic regurgitation	Early diastolic murmur at the left sternal edge	Usually tolerates moderate exercise until the ventricle fails

Memory rule for murmurs

Open valves stenose, closed valves leak

A murmur happens when a valve that should be **open** is narrow (stenosis), or a valve that should be **shut** leaks (regurgitation). In systole the aortic valve is open and the mitral valve shut, so systolic murmurs are aortic stenosis or mitral regurgitation; in diastole it is the reverse.

3 Causes and risk factors

- **Valve disease:** rheumatic fever (poverty, crowding, untreated sore throats), age-related calcification (aortic stenosis), infective endocarditis, congenital (bicuspid aortic valve), heart attack damaging the mitral support.
- **Heart failure:** coronary disease and MI, hypertension, valve disease, cardiomyopathy, arrhythmias, diabetes, alcohol, some chemotherapy.
- **Cardiomyopathy:** genetic, viral myocarditis, alcohol, pregnancy (peripartum), unknown.
- **Pacemaker:** slow heart rhythms (heart block, sick sinus syndrome). **ICD:** survivors of cardiac arrest, or people at high risk (for example EF 35% or less with heart failure symptoms despite treatment).

Rheumatic fever (Jones criteria): major features are carditis, migrating polyarthritis, chorea, a ring-shaped rash (erythema marginatum) and nodules under the skin; minor features are joint pain, fever, raised inflammatory markers and a prolonged PR interval.

4 Pathophysiology to impairments

1. The weak or overloaded heart cannot raise its output with exercise, so muscles get less oxygen: **fatigue and low exercise tolerance**.
2. Back-pressure into the lungs causes **breathlessness**, worse lying flat (orthopnoea) and at night; back-pressure into the body causes **leg oedema** and an enlarged liver.
3. The body retains salt and water to compensate, which worsens congestion.
4. Skeletal muscles weaken and change (fewer mitochondria), so even daily tasks become hard: the main target of exercise training.
5. Arrhythmias (AF, ventricular tachycardia) cause palpitations, stroke risk and sudden death.

5 Clinical features

- Breathlessness on exertion, orthopnoea, waking breathless at night; fatigue; ankle swelling; weight gain from fluid.
- Fast pulse, low BP in advanced disease, raised neck veins, basal crackles, third heart sound.
- Valve disease: murmurs, palpitations (AF), fainting on exertion (aortic stenosis), coughing blood (mitral stenosis).

6 Investigations

- **Echocardiogram:** EF, valve area and gradients, chamber size (the key test).
- **ECG:** rhythm (AF), conduction blocks, ventricular hypertrophy; Holter monitoring.
- **Chest X-ray:** large heart, pulmonary congestion, effusions; mitral stenosis shows a large left atrium.
- **BNP or NT-proBNP:** raised in heart failure.
- **Exercise test or CPET:** peak oxygen uptake, safe exercise range (not in symptomatic severe aortic stenosis).
- **INR** for patients on warfarin (mechanical valves, AF).

7 Medical and surgical management

- **Heart failure with reduced EF:** four main drug groups (ACE inhibitor/ARB/ARNI, beta-blocker, mineralocorticoid antagonist, SGLT2 inhibitor), diuretics for congestion; devices (ICD, cardiac resynchronisation therapy); exercise training.
- **Rheumatic heart disease:** regular benzathine penicillin injections to prevent recurrence; balloon mitral valvotomy; valve repair or replacement.
- **Valve replacement:** mechanical (lifelong warfarin) or tissue valve; open surgery via sternotomy or catheter valve (TAVI for aortic stenosis).
- **Pacemaker or ICD:** implanted under the skin below the collarbone (usually left), with leads passed through a vein into the heart.

8 Complications

- Acute pulmonary oedema, arrhythmias, stroke (AF), sudden death, kidney impairment.
- Valve: infective endocarditis, valve thrombosis, bleeding on warfarin.
- Device: lead displacement (early, from arm movement), pocket haematoma or infection, pneumothorax at insertion, inappropriate ICD shocks, anxiety after shocks.

9 PT assessment

- Diagnosis, EF, NYHA class, medicines, device type and settings (pacing mode, rate-response, ICD detection rate), recent shocks.
- Symptoms at rest and on exertion; daily weight trend; oedema; orthopnoea.
- HR and rhythm, BP, SpO_2, RR at rest and with activity; auscultation for crackles.
- Wound and shoulder range on the device side.
- Exercise capacity (6MWT with Borg), muscle strength, function.
- **Outcome measures:** 6MWT, NYHA class, Minnesota Living with Heart Failure or Kansas City Cardiomyopathy Questionnaire, HADS.

10 PT diagnosis by ICF

Worked example from a real paper (S22, 15 marks)

> "A 34 year old female school teacher is proposed for pacemaker and intracardiac defibrillator following cardiomyopathy. What would be pre-operative and post-implantation assessment and physiotherapy management?"

ICF component	From this case	Clinical reasoning
Health condition	Cardiomyopathy with risk of dangerous arrhythmia; pacemaker and ICD implantation	Weak heart muscle and rhythm risk

ICF component	From this case	Clinical reasoning
Impairments	Reduced exercise tolerance, fatigue, breathlessness; after implant: wound pain and restricted shoulder on the device side; anxiety about shocks	Low cardiac output; arm restriction protects the leads
Activity limitations	Lifting, reaching overhead (writing on the blackboard), stairs, brisk walking	Device-side arm restriction for 4–6 weeks; low fitness
Participation	Teaching full days, household roles, travel, driving (rules after ICD)	Stamina and arm use needed at work
Environmental factors	**Facilitators:** device clinic, family, school support. **Barriers:** classroom equipment near the device (rarely a problem), stairs at school.	Plan a graded return to work
Personal factors	Young woman; fear of shocks; motivated to keep working	Education reduces fear and supports activity

11 Goals

- **Short-term (first 4–6 weeks):** wound healed; no lead displacement; shoulder kept mobile below shoulder height; walking daily; knows device precautions and shock plan.
- **Long-term (3 months):** full shoulder range; better 6MWT and NYHA class; regular exercise below the ICD detection rate; back to teaching.

12 PT management

Pre-implantation

- Assessment as in section 9; baseline shoulder range and 6MWT if stable.
- Education: what the device does, arm precautions after implantation, what to do if the ICD shocks.
- Breathing exercises and early mobilisation plan; reduce anxiety.

Post-implantation

- **First 4–6 weeks:** no lifting of the device-side arm above shoulder height and no heavy lifting with that arm, so the leads can fix in place. Gentle active movement of the shoulder below shoulder height, elbow, wrist and hand from day 1 to prevent a stiff shoulder.
- Early walking from the day after implantation; breathing exercises.
- After 4–6 weeks: restore full shoulder range; start structured exercise training.

Exercise training in heart failure and cardiomyopathy

- Exercise training is recommended for all stable heart failure patients who can exercise (ESC 2021, Class I): it improves exercise capacity and quality of life and reduces hospital admissions.
- **Aerobic:** walking or cycling, 3–5 days a week, building to 20–60 minutes; start low (RPE 11–13), progress slowly; **interval training** suits low-fitness patients.
- **Resistance:** light loads, 2–3 days a week, 10–15 repetitions, breathe out on effort.
- **Inspiratory muscle training** for weak breathing muscles.
- **With an ICD:** keep training heart rate **at least 10 beats below the ICD's programmed detection rate** (ask the device clinic for the setting).
- **With a pacemaker:** if the heart rate cannot rise (chronotropic incompetence) or the device is fixed-rate, use RPE and BP to guide intensity.
- **Valve disease:** moderate exercise in mild to moderate lesions; no exercise training in symptomatic severe aortic stenosis until it is treated; after valve surgery follow the CABG sternotomy plan.

Cardiac rehab after valve replacement (7 marks in W21)

Valve surgery is usually through a median sternotomy, so Phase I follows the CABG plan (sternal precautions, breathing exercises, early walking). Add these valve-specific points:

Phase	Physiotherapy
Phase 1 (in hospital, about 5–7 days)	• Before surgery: education, breathing exercises, supported cough, baseline walking • Day 0–1: deep breathing with a hold, supported huff and cough, positioning, ankle pumps, sitting out of bed • Day 2 to discharge: walking 3–4 times a day from the room to the corridor, stairs before discharge; RPE below 13, HR no more than rest + 30 • Monitor rhythm: atrial fibrillation is common, and heart block can follow aortic valve surgery (pacing wires, sometimes a permanent pacemaker) • Sternal precautions; posture and gentle shoulder movement
Phase 2 (outpatient, about 8–12 weeks)	• Exercise test or 6MWT first; many patients were inactive for months before surgery, so start low • Aerobic training 3–5 days a week, building to 30–60 minutes at RPE 11–14; use RPE and the talk test when atrial fibrillation or beta-blockers make heart rate unreliable • Leg resistance work early; arm resistance once the sternum has healed (about 6–8 weeks), progressing slowly • Education: anticoagulation, infection prevention, symptoms to report, return to work and driving

- **Mechanical valve:** lifelong warfarin with regular INR checks. Bruising and bleeding are easy, so handle gently, prevent falls, avoid contact sports and vigorous percussion, and report any bleeding.
- **Infective endocarditis:** keep teeth and skin healthy; report fever, night sweats or new breathlessness early.
- **Mitral valve disease** often leaves pulmonary hypertension and atrial fibrillation: progress more slowly.
- **Transcatheter valve (TAVI):** no sternotomy, so no sternal precautions; older, frailer patients; groin-puncture care; walk within a day and follow the same Phase 2 plan.

If the ICD shocks

- Stop, sit or lie down.
- One shock and feeling well: inform the device clinic.
- More than one shock, or feeling unwell (chest pain, breathlessness, fainting): call emergency services.

13 Precautions and contraindications

Devices

- **Avoid shortwave and microwave diathermy** in patients with pacemakers or ICDs.
- Avoid TENS, interferential and other electrical stimulation near the device or across the chest; use only after device-team advice.
- Do not apply therapeutic ultrasound directly over the device.
- Keep strong magnets away from the device.

Do not exercise if

- Symptoms of decompensated heart failure: breathless at rest, new orthopnoea, rapid weight gain (for example 2 kg in 3 days), increasing oedema
- Symptomatic severe aortic stenosis, uncontrolled arrhythmia, chest pain at rest
- Recent ICD shocks until reviewed

- Warfarin: gentle handling; avoid vigorous manual techniques.
- Hypertrophic cardiomyopathy: avoid dehydration and very heavy straining; exercise plans are individual.

14 Education

- Weigh yourself every morning; report a sudden gain.
- Follow salt and fluid advice; take medicines regularly.
- Keep active every day within your plan.
- Rheumatic heart disease: never miss penicillin injections; keep teeth healthy.
- Device: carry the device card; keep mobile phones away from the device pocket; tell staff before any procedure; follow driving rules after ICD implantation.

15 Prognosis

Modern drugs and devices have greatly improved survival in heart failure. Exercise training improves symptoms and quality of life. Valve disease has a good outlook when treated before the heart muscle is damaged.

16 Exam corner

S22 Q4(a), 15 marks
The worked example in section 10.

S21 Q1(f), 3 marks
Grades of NYHA scale.

W21 Q2(d), 7 marks
Phase 1 and 2 of cardiac rehabilitation after valve replacement (see section 12).

Related
S20 Q4(a): stable angina with EF 25% (see the IHD page).

Technique cards for this topic: exercise testing (p. 308), scales (p. 322), exercise physiology (p. 313), cardiac rehabilitation (p. 334), cardiac drugs (p. 352).

Keywords to underline

NYHA I–IV EF 40% or less exercise training Class I interval training 4–6 weeks arm restriction 10 beats below ICD detection rate no diathermy daily weights severe aortic stenosis

Test yourself

3 m **Grades of NYHA scale.**

Answer
I no limitation; II slight limitation, ordinary activity causes symptoms; III marked limitation, less than ordinary activity causes symptoms; IV symptoms at rest.

MCQ **Two days after a left-sided pacemaker, which is correct?**

A. Full overhead pulley exercises for the left shoulder

B. Gentle left shoulder movement below shoulder height; no overhead lifting for 4–6 weeks

C. Complete immobilisation of the left arm for 3 months

D. Shortwave diathermy for shoulder pain

Answer

B. Early gentle movement prevents stiffness while protecting the leads; diathermy is contraindicated.

MCQ **An ICD is set to detect ventricular tachycardia at 170 bpm. The training heart rate should stay below about:**

A. 180

B. 160

C. 175

D. 200

Answer

B. At least 10 beats below the detection rate avoids inappropriate shocks.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. McDonagh TA, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure; 2023 focused update. *Eur Heart J*.
3. Bozkurt B, et al. Universal definition and classification of heart failure. *J Card Fail* 2021.
4. Abraham LN, et al. Exercise-based cardiac rehabilitation for adults after heart valve surgery. Cochrane Review (CD010876), 2021.
5. ACSM's Guidelines for Exercise Testing and Prescription: exercise with pacemakers and ICDs.
6. Liverpool Heart and Chest Hospital patient guidance on living with an ICD (arm movement in the first 4–6 weeks). lhch.nhs.uk
7. Structure and classical content: Hillegass (2022); Cash's Textbook of Chest, Heart and Vascular Disorders for Physiotherapists.

Congenital heart disease: tetralogy of Fallot, ASD, VSD, PDA

> Tetralogy of Fallot was asked three times as 3 marks: clinical features (W16, S26) and radiological features (S21). Learn the four defects, the boot-shaped heart, the tet spell and the basics of physiotherapy after paediatric heart surgery.

Syllabus	Ch 6(a)(iv) Congenital heart disease
Priority	**Tier 2.** 3 short questions (W16, S21, S26).
Asked as	3m W16 S21 S26
Usual split	3-mark answers; 7 marks if paediatric cardiac surgery is asked
Checked against	Standard paediatric cardiology texts; paediatric cardiac surgery physiotherapy practice. Full list under Sources. Content checked 19 October 2026.

1 Definition

Congenital heart disease (CHD) is a structural abnormality of the heart or great vessels present at birth. It is the commonest birth defect.

2 Classification

Group	Examples	Main effect
Acyanotic, left-to-right shunt	Atrial septal defect (ASD), ventricular septal defect (VSD), patent ductus arteriosus (PDA)	Too much blood to the lungs: breathlessness, chest infections, poor growth; later pulmonary hypertension
Acyanotic, obstructive	Coarctation of the aorta, aortic stenosis, pulmonary stenosis	Pressure overload of a ventricle
Cyanotic (right-to-left)	Tetralogy of Fallot, transposition of the great arteries, tricuspid atresia, truncus arteriosus, total anomalous pulmonary venous return	Deoxygenated blood reaches the body: cyanosis

Memory hook: the big cyanotic lesions

The 5 T's

Tetralogy of Fallot, **T**ransposition of the great arteries, **T**ricuspid atresia, **T**runcus arteriosus, **T**otal anomalous pulmonary venous return. These are the main five, not every cyanotic lesion.

3 Causes and risk factors

- Most have no single cause.
- Genetic syndromes (Down syndrome, often with an atrioventricular septal defect), family history.
- Maternal rubella (PDA), diabetes, alcohol, some medicines, older maternal age.
- Prematurity (PDA).

4 Tetralogy of Fallot: pathophysiology

The four parts (all from one developmental fault: the outflow septum is displaced forwards):

1. **Ventricular septal defect** (large).
2. **Overriding aorta** sitting over the VSD.
3. **Right ventricular outflow tract obstruction** (pulmonary stenosis).
4. **Right ventricular hypertrophy** (a result of the obstruction).

The narrowed outflow pushes deoxygenated blood from the right ventricle through the VSD into the aorta: **cyanosis**. The more severe the obstruction, the bluer the child.

? Why does squatting help a child with a tet spell?

> Squatting (or knee–chest position) kinks the leg arteries and raises resistance in the body's circulation. This makes it harder for blood to cross from right to left through the VSD, so more blood goes to the lungs and oxygen rises.

5 Clinical features

Tetralogy of Fallot (3 marks in W16)

- Cyanosis, often appearing in the first months of life and increasing with crying or exertion.
- **Hypercyanotic ("tet") spells:** sudden deep cyanosis, fast breathing, irritability, sometimes fainting or fits, often after waking, feeding or crying.
- **Squatting** after exertion in older children.
- Clubbing of fingers and toes; poor growth; exercise intolerance.
- Harsh ejection systolic murmur at the left upper sternal edge (from the pulmonary stenosis); single second heart sound.

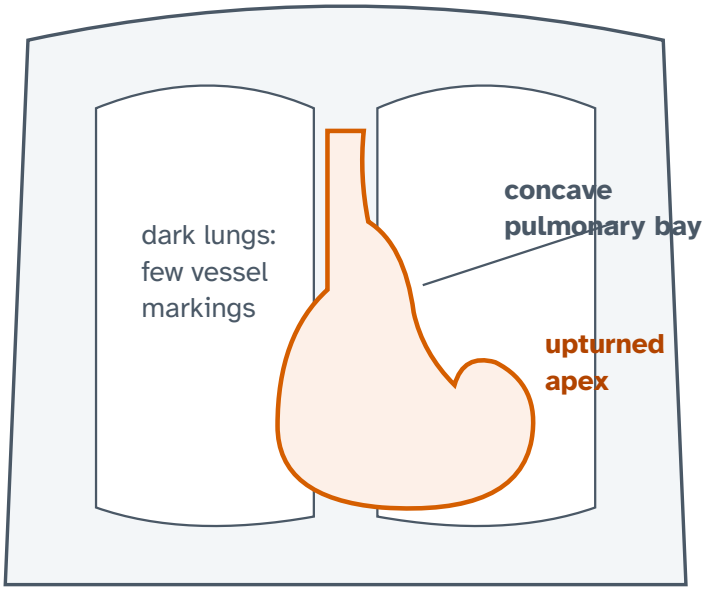
Other lesions

- **VSD:** loud pansystolic murmur at the left lower sternal edge; large defects cause breathlessness, sweating on feeding and poor weight gain.
- **ASD:** often no symptoms until adulthood; fixed split second heart sound.
- **PDA:** continuous "machinery" murmur below the left clavicle; bounding pulses.
- **Coarctation:** weak, delayed femoral pulses; high BP in the arms.
- **Eisenmenger syndrome:** a long-standing left-to-right shunt causes pulmonary hypertension that reverses the shunt, so cyanosis appears late.

6 Investigations

Radiological features of tetralogy of Fallot (3 marks in S21)

Boot-shaped heart (coeur en sabot)



Sketch for learning only; real X-rays vary. The heart is usually normal in size.

- **Boot-shaped heart** (coeur en sabot): upturned, rounded apex from right ventricular hypertrophy.
- **Concave pulmonary bay** (small pulmonary artery segment) on the left heart border.
- **Oligaemic lung fields:** reduced lung vessel markings, dark lungs.
- Heart size usually **normal**.
- A **right-sided aortic arch** in some patients.

Other tests

- **Echocardiogram:** the main diagnostic test for all lesions.
- ECG (right ventricular hypertrophy in TOF); pulse oximetry (newborn screening detects many critical lesions); haemoglobin (raised in chronic cyanosis); cardiac catheterisation or CT/MRI before surgery.

7 Medical and surgical management

- **TOF:** complete repair in infancy (VSD patch closure and relief of the outflow obstruction); a temporary shunt (Blalock–Taussig) in some small infants.
- **Tet spell first aid:** calm the child, knee–chest position (or hold the baby with knees to chest), oxygen, and call for medical help; doctors may give morphine, fluids and other drugs.
- **VSD, ASD, PDA:** many small defects close on their own; device closure by catheter or surgical closure for larger ones; drugs to close a PDA in preterm babies.
- **Transposition:** arterial switch operation in the newborn period.
- **Single-ventricle hearts:** staged surgery (Glenn, Fontan).
- Heart failure medicines, nutrition support, endocarditis prevention, lifelong follow-up.

8 Complications

- Heart failure, failure to thrive, repeated chest infections (left-to-right shunts).

- Polycythaemia, stroke, brain abscess (cyanotic lesions); infective endocarditis.
- Pulmonary hypertension and Eisenmenger syndrome.
- After repair: arrhythmias, leaking pulmonary valve (TOF), need for further surgery.
- Developmental delay; reduced exercise capacity.

9 PT assessment (child after heart surgery)

- Diagnosis, operation, day, lines and drains, pacing wires, ventilation, oxygen and the child's **target SpO_2 range** (it may be lower than normal in cyanotic or palliated hearts).
- Breathing rate and effort, recession, chest expansion, auscultation, secretions, cough.
- Behaviour, pain (age-appropriate scale), developmental level, posture, shoulder movement.
- Parents' understanding and concerns.

10 PT diagnosis by ICF (practice)

Practice case (not from a paper): a 3-year-old girl, day 2 after complete repair of tetralogy of Fallot. Impairments: sternotomy pain, reduced lung volumes, retained secretions, reduced mobility and fear. Activity: playing, walking, being lifted. Participation: play with family, nursery. Environment: anxious parents (barrier when they restrict movement), paediatric ICU team and play staff (facilitators). Personal: age and temperament.

11 Goals

- **Short-term:** clear chest; normal breathing pattern for age; sitting, standing and walking within days; parents confident with handling.
- **Long-term:** normal developmental milestones; active play; age-appropriate exercise and sport as advised by the cardiologist.

12 PT management after paediatric heart surgery

- **Positioning:** head of bed raised; side-lying; regular position changes. Avoid head-down positions.
- **Airway clearance only when secretions are a problem:** gentle manual techniques, assisted huffing, suction if needed; play-based deep breathing (blowing bubbles, windmills, party blowers).
- **Early mobilisation:** sitting out, standing, walking and play as soon as stable.
- **Sternal care:** lift the child with a scoop under the bottom and back rather than under the armpits for the first weeks; avoid pulling on the arms.
- Shoulder and trunk movement through play; posture.
- Developmental and play activities for infants; parent education and involvement.
- Later: encourage normal active play and fitness; individual advice on competitive sport.

13 Precautions

- Know the child's target SpO_2; do not chase normal values in a cyanotic heart.
- Pulmonary hypertension: avoid crying, straining and drops in oxygen; handle calmly; pre-oxygenate before suction.
- Pacing wires and drains: secure before moving.
- Anticoagulated children: gentle handling.
- Cyanotic children: avoid dehydration (thick blood) and fatigue.

14 Education for parents

- How to lift and handle the child after sternotomy; breathing games.

- Recognise a tet spell and the knee–chest position; when to seek help.
- Good dental hygiene (endocarditis prevention), vaccinations, follow-up appointments.
- Encourage normal play; avoid over-protection.

15 Prognosis

Most children with repaired tetralogy of Fallot, ASD, VSD or PDA grow up to lead active lives, with lifelong cardiology follow-up. Complex single-ventricle hearts have a more limited exercise capacity.

16 Exam corner

W16 Q1(f), 3 marks
Clinical features of tetralogy of Fallot.

S21 Q1(e), 3 marks
Radiological features seen in tetralogy of Fallot.

S26 Q1(f), 3 marks
Any three clinical features of tetralogy of Fallot.

Technique cards for this topic: breathing exercises (p. 275).

Keywords to underline

VSD, overriding aorta, pulmonary stenosis, RVH cyanosis tet spell squatting boot-shaped heart oligaemic lungs acyanotic vs cyanotic Eisenmenger

Test yourself

3 m **Clinical features of tetralogy of Fallot.**

Answer
Cyanosis increasing with crying or exertion; hypercyanotic (tet) spells; squatting; clubbing; poor growth and exercise intolerance; harsh ejection systolic murmur at the left upper sternal edge with a single second sound.

3 m **Radiological features of tetralogy of Fallot.**

Answer
Boot-shaped heart with an upturned apex; concave pulmonary bay; oligaemic lung fields; normal heart size; sometimes a right-sided aortic arch.

MCQ **A child with TOF suddenly becomes very blue and breathless after crying. First action:**

A. Lay flat with legs straight
B. Knee–chest position, calm the child, oxygen, call for help
C. Chest percussion
D. Encourage running to improve fitness

Answer
B. The knee–chest position raises systemic resistance and reduces the right-to-left shunt.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Standard paediatric cardiology texts (for example Park's *Pediatric Cardiology for Practitioners*) for lesion features and X-ray signs.
3. Structure and classical content: Main & Denehy (2016), paediatric chapters; Pryor & Prasad (2008), paediatric cardiac surgery.

Pneumonia, lung abscess and atelectasis

> Pneumonia came once as a long case (W24: atypical pneumonia of both lungs on a ventilator) and inside the S20 COPD ICU case; lung abscess and atelectasis were not asked on their own in 2016–2026. The key exam point: physiotherapy is not routine in uncomplicated pneumonia; it is for sputum retention, collapse, ventilated patients and mobility.

Syllabus	Ch 6(c)(ii) Pneumonia, (iii) Lung abscess, (vii) Atelectasis
Priority	**Tier 3.** 1 long-answer case (W24, on a ventilator) and inside the S20 COPD ICU case.
Asked as	LAQ W24 · Inside S20 LAQ
Usual split	15 (functional diagnosis and physiotherapy of a ventilated patient); short notes for 3 or 7 marks
Checked against	BTS/ACPRC physiotherapy guideline (2009); Cochrane review of chest physiotherapy for pneumonia (2022); CURB-65 (NICE quality standard). Full list under Sources. Content checked 19 October 2026.

1 Pneumonia

Definition and types

Infection of the lung tissue in which the alveoli fill with inflammatory fluid and cells (consolidation).

- **By setting:** community-acquired; hospital-acquired (starting 48 hours or more after admission); ventilator-associated; aspiration pneumonia.
- **By pattern:** lobar (one lobe), bronchopneumonia (patchy, both lungs), interstitial (viral, atypical).
- **By organism:** bacterial (pneumococcus commonest), viral, atypical, fungal, TB.

Key features

- Fever, cough, sputum (rusty in pneumococcal), pleuritic chest pain, breathlessness, fast breathing; confusion in the elderly.
- Signs over the consolidation: dull percussion, bronchial breathing, crackles, increased vocal resonance.
- **X-ray:** white consolidation with **air bronchograms** (dark airways inside white lung) and **no loss of volume**.

Severity score: CURB-65 (one point each)

C U R B 65

- **C**onfusion (new)
- **U**rea above 7 mmol/L (about 42 mg/dL as reported in Indian labs)
- **R**espiratory rate 30 or more
- **B**lood pressure: systolic below 90 or diastolic 60 or less
- Age **65** or more

0–1 low risk (often treated at home); 2 moderate (hospital); 3–5 severe (consider ICU).

Medical point that matters for physio

Antibiotics, oxygen (target SpO_2 94–98%, or 88–92% if at risk of CO_2 retention), fluids. Early consolidation is solid exudate, not loose sputum, so percussion and drainage do not clear it.

Classic answer vs current guideline: chest physiotherapy in pneumonia

Classic	Current (BTS/ACPRC 2009; Cochrane 2022)	Write in the exam
Postural drainage, percussion and vibrations for every pneumonia patient.	Airway clearance is **not routine** in uncomplicated community-acquired pneumonia; evidence for benefit is of very low certainty. Patients should **sit out of bed for at least 20 minutes within the first 24 hours** and mobilise progressively. Treat sputum retention, weakness and underlying lung disease when present.	"Early mobilisation and positioning for all; airway clearance (ACBT) only when sputum is retained or in pre-existing lung disease."

ICF summary

Impairments: fever, breathlessness, low oxygen, pleuritic pain, sputum (later), weakness. Activity: walking, self-care. Participation: work and family roles during illness. Environment: hospital or home care.

PT management by problem

- **Low oxygen:** positioning; in one-sided pneumonia, lie with the **good lung down** to improve oxygenation.
- **Deconditioning:** sit out within 24 hours (20 minutes or more), then progressive walking.
- **Sputum retention (resolving stage):** ACBT with huffing, 1–2 times a day; affected lung uppermost for drainage when tolerated; humidification.
- **Pleuritic pain:** supported breathing and coughing; pain relief before treatment.
- **On a ventilator (S20, W24):** see the worked plan below.

Ventilated pneumonia (W24: 35-year-old, both lungs, CMV, FiO_2 0.6, PEEP 10 for 3 days)

Read the settings first. FiO_2 0.6 and PEEP 10 cmH_2O are high: his lungs are stiff and poorly oxygenated, close to ARDS. Losing PEEP even briefly (disconnecting the circuit) can collapse alveoli and drop his SpO_2.

Functional diagnosis by ICF

Part	From this case
Health condition	Severe bilateral atypical pneumonia with acute respiratory failure, sedated on controlled ventilation for 3 days
Impairments	Poor oxygenation needing high FiO_2 and PEEP; stiff lungs with low volumes; risk of collapse, retained secretions and ventilator-associated pneumonia; muscle wasting and ICU-acquired weakness; joint stiffness; pressure-injury and DVT risk; reduced consciousness from sedation
Activity limitations	Fully dependent for bed mobility, sitting, transfers and self-care
Participation restrictions	Off work; family roles
Contextual factors	Young and previously fit (good outlook); ICU team and family (facilitators); tubes, lines and sedation (barriers)

- **Safety check every session:** SpO_2, FiO_2 and PEEP trend, HR, BP, sedation level, and the doctor's plan. At FiO_2 0.6 and PEEP 10 he is at the upper limit for active mobilisation, so start with in-bed treatment.
- **Positioning:** head of bed up 30–45° (prevents aspiration and ventilator pneumonia); regular side-lying turns to change ventilation and drain secretions; prone positioning for severe hypoxaemia is decided by the doctors, and physiotherapists help to turn the patient safely.
- **Secretions:** atypical pneumonia often produces little sputum, so suction only when needed, using a **closed (in-line) suction** system to keep PEEP. Avoid bagging off the ventilator at this PEEP; if hyperinflation is needed, the

ventilator can deliver it (ventilator hyperinflation).

- **Limbs:** passive then active-assisted movement of all joints, ankle splints or positioning against foot drop, pressure care with regular turns.
- **Progress** as oxygen needs fall and sedation lightens: active exercise, sitting over the edge of the bed, in-bed cycling, then standing and walking with the ventilator.
- **After extubation:** deep breathing, ACBT, incentive spirometry as an add-on, early walking, strengthening and inspiratory muscle training if weak.

Ventilator modes and weaning, suction and positioning are covered in more detail on the mechanical ventilation (p. 216), suctioning (p. 231) and positioning (p. 236) cards.

Precautions

- Watch for sepsis (low BP, confusion), worsening hypoxia and fatigue.
- Infection control: hand hygiene, masks for viral pneumonia.

Exam points

- Consolidation vs collapse: consolidation keeps lung volume; collapse loses it.
- Physio is not routine in uncomplicated pneumonia.

2 Lung abscess

Definition and types

A localised area of lung necrosis forming a cavity filled with pus.

Causes and key features

- **Aspiration** of mouth contents (alcohol, seizures, reduced consciousness, poor teeth, swallowing problems). Commonest sites: posterior segment of the right upper lobe and superior segment of the right lower lobe (the lowest parts when lying on the back).
- Other causes: severe pneumonia (Staphylococcus, Klebsiella), blocked bronchus (tumour, foreign body).
- Swinging fever, large amounts of **foul-smelling sputum**, haemoptysis, weight loss, clubbing.
- **X-ray:** a thick-walled cavity with an **air–fluid level**.

Medical point that matters for physio

Several weeks of antibiotics; drainage or surgery if it does not resolve. Once the abscess drains into a bronchus, sputum is copious.

ICF summary

Impairments: copious foul sputum, fever, fatigue, weight loss. Activity and participation: reduced work and social life (embarrassment from sputum smell).

PT management by problem

- **Copious sputum:** gravity-assisted drainage of the affected segment with ACBT, 2–3 times a day, once the abscess communicates with a bronchus (the patient is coughing it up).
- Encourage strong huffs and coughs during drainage so pus is cleared, not spilled into other lobes.
- Breathing exercises, mobilisation, nutrition and mouth hygiene advice.

Precautions

- Haemoptysis: stop drainage and percussion.
- Weak or drowsy patients may not clear a sudden large release of pus: have suction ready.

Exam points

- Cavity with an air–fluid level; foul sputum; aspiration risk factors; drain the affected segment.

3 Atelectasis (collapse)

Definition and types

Incomplete expansion or collapse of part or all of a lung.

- **Obstructive (resorption):** mucus plug, tumour or foreign body; air beyond is absorbed.
- **Compressive:** effusion, pneumothorax, tumour pressing on the lung.
- **Passive (micro-atelectasis):** shallow breathing after surgery or with pain.
- **Adhesive:** surfactant loss (neonatal RDS, ARDS). **Cicatrisation:** scarring (TB, fibrosis).

Key features

- May be silent when small; larger collapse: breathlessness, low SpO_2, reduced breath sounds.
- **X-ray:** increased opacity with **loss of volume**: trachea and mediastinum pulled **towards** the collapse, raised hemidiaphragm, displaced fissures, crowded ribs.

Medical point that matters for physio

Post-operative and mucus-plug collapse respond well to physiotherapy; bronchoscopy for a plug that will not clear.

ICF summary

Impairments: reduced lung volume, low oxygen, retained secretions. Activity: mobility limited by breathlessness and the underlying cause.

PT management by problem

- **Reduced lung volume:** deep breathing with a 3-second inspiratory hold (sustained maximal inspiration), thoracic expansion to the affected area, 5–10 breaths every hour while awake.
- **Positioning:** affected area uppermost to ventilate and drain it.
- **Mucus plug:** humidification, ACBT with huffs, suction if needed.
- **Early mobilisation**: the best "deep breathing" exercise.
- In ventilated or very weak patients: positive pressure (CPAP, manual hyperinflation, IPPB) as prescribed; incentive spirometry only as an add-on.

Precautions

- Positive pressure is contraindicated with an undrained pneumothorax.
- Compressive collapse needs the fluid or air drained first.

Exam points

- Collapse pulls structures towards; effusion and pneumothorax push away; consolidation keeps volume.

4 Exam corner

W24 Q4(a), 15 marks

35-year-old man with infective atypical pneumonia of both lungs, on CMV with FiO_2 0.6 and PEEP 10 for 3 days, referred for physiotherapy. Functional diagnosis and physiotherapy management. Worked plan in section 1.

S20 Q3(b), 5 + 3 + 7 marks

60-year-old farmer with a COPD exacerbation and respiratory failure, in ICU on SIMV, with right lower lobe pneumonia (see the COPD page).

Technique cards for this topic: ICU physiotherapy (p. 223), mechanical ventilation (p. 216), suctioning (p. 231), positioning (p. 236).

Don't confuse: X-ray patterns

Condition	Opacity and volume	Trachea and mediastinum
Consolidation	White, air bronchograms; volume normal	Central
Collapse	White; volume lost	Pulled towards
Large effusion	White with a meniscus	Pushed away
Tension pneumothorax	Black, no lung markings	Pushed away

Test yourself

3 m **CURB-65.**

Answer
Confusion; urea above 7 mmol/L; respiratory rate 30 or more; systolic BP below 90 or diastolic 60 or less; age 65 or more. One point each: 0–1 low, 2 moderate, 3–5 severe.

MCQ **Day 1 of uncomplicated community-acquired pneumonia, no sputum. Best physiotherapy:**

A. Percussion and postural drainage

B. Sit out of bed for at least 20 minutes and progress mobility

C. Bed rest for 3 days

D. IPPB

Answer
B. Early mobilisation is recommended; routine airway clearance is not.

MCQ **An X-ray shows a white right lower zone, raised right hemidiaphragm and trachea shifted right. This is most likely:**

A. Right pleural effusion

B. Right lower lobe collapse

C. Right tension pneumothorax

D. Normal

Answer
B. Volume loss pulls the trachea towards the collapse.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Bott J, et al. BTS/ACPRC physiotherapy guideline. *Thorax* 2009. Concise version.
3. Cochrane Review: Chest physiotherapy for pneumonia in adults (CD006338), 2022 update.
4. NICE quality standard QS110: CURB-65. nice.org.uk
5. British Thoracic Society guideline for oxygen use (2017).

Neonatal and paediatric respiratory conditions

Neonatal chest physiotherapy was asked as 7 marks (W18: precautions and contraindications for postural drainage in a neonate) and signs of respiratory distress in children as 3 marks (W24). The theme of this page: babies are fragile, so treat only when secretions are a real problem, avoid head-down positions, and handle minimally.

Syllabus	Ch 6(d) Neonatal and paediatric respiratory infection: ARDS, meconium aspiration, pneumonitis, pneumonia, childhood asthma, cystic fibrosis and chronic lung disease
Priority	**Tier 3.** 2 short questions (W18, W24) plus the paediatric asthma questions (see the asthma page).
Asked as	7m W18 · S18 (asthma) · 3m W24
Usual split	Short notes; 7-mark answers
Checked against	Cochrane reviews of chest physiotherapy in bronchiolitis (2023) and of positioning in infants with reflux; ILCOR neonatal resuscitation recommendations on meconium (2015, 2020). Full list under Sources. Content checked 19 October 2026.

1 Neonatal chest physiotherapy: precautions and contraindications

Asked as 7 marks in W18. Learn this section first.

Precautions

- **Treat only when needed:** retained secretions, collapse or a blocked tube; not as routine.
- **Minimal handling:** cluster care with nursing; short sessions; let the baby rest between.
- **Monitor continuously:** SpO_2, heart rate, breathing, colour and temperature; keep the baby warm.
- **Timing:** before feeds or well after them, to avoid vomiting and aspiration.
- **Positioning:** avoid head-down tilt; use modified (flat or head-up) positions. In infants with reflux, a 30° head-up position caused fewer reflux episodes than head-down (Cochrane).
- **Gentle techniques:** very preterm babies have fragile ribs (osteopenia) and fragile brain blood vessels; vigorous percussion has been linked to injury, so use gentle, modified techniques only when clearly indicated.
- **Suction:** correct catheter size and pressure, pre-oxygenate if needed, limit duration; watch for bradycardia and desaturation.

Contraindications (do not do chest physiotherapy, or only after senior review)

- Unstable baby: hypotension, uncontrolled arrhythmia, severe hypoxia with handling
- Recent or extending intraventricular haemorrhage, very preterm baby in the first days of life
- Persistent pulmonary hypertension of the newborn (handling triggers desaturation)
- Undrained pneumothorax or air leak
- Rib fractures, osteopenia of prematurity, low platelets or bleeding disorder (for percussion and vibration)
- Raised intracranial pressure; immediately after feeds; some post-operative states (for example after abdominal or cardiac surgery) without surgical clearance

2 Neonatal respiratory distress syndrome (RDS)

Definition and cause

Breathing failure in premature babies because the lungs lack **surfactant**, so alveoli collapse at the end of each breath. Commoner with more premature birth and maternal diabetes.

Key features

- Within hours of birth: fast breathing, **grunting**, nasal flaring, rib and sternal recession, cyanosis.
- **X-ray:** small lungs with a "ground-glass" appearance and air bronchograms.

Medical point

Antenatal steroids to the mother, CPAP, surfactant given into the airway, ventilation if needed.

PT management

- Routine chest physiotherapy is not needed; the lungs are stiff, not full of secretions.
- Positioning for comfort and oxygenation (prone positioning is used only under continuous monitoring in neonatal units), developmental positioning, minimal handling.
- Airway clearance only if secretions build up later (for example on a ventilator).

Exam points

- Surfactant deficiency; grunting; ground-glass X-ray; CPAP and surfactant; minimal handling.

3 Meconium aspiration syndrome

Definition and cause

Breathing problems in a term or post-term baby who inhaled meconium (first stool) passed into the amniotic fluid, often after fetal distress.

Key features

- Meconium-stained fluid and skin; breathing distress from birth.
- Meconium blocks airways (with air trapping and pneumothorax risk), inflames the lungs (chemical pneumonitis) and inactivates surfactant; it can cause persistent pulmonary hypertension.
- **X-ray:** patchy shadows and over-inflation; possible pneumothorax.

Medical point

Supportive: oxygen, CPAP or ventilation, surfactant, nitric oxide for pulmonary hypertension. Current resuscitation guidance (ILCOR) no longer recommends **routine** laryngoscopy and tracheal suction of non-vigorous babies born through meconium; start effective breathing support promptly.

PT management

- Airway clearance (gentle positioning, modified drainage, suction) only when secretions are clearly retained.
- Minimal handling, especially with pulmonary hypertension.

Exam points

- Term or post-term; air trapping, pneumothorax, chemical pneumonitis, pulmonary hypertension.

4 Pneumonitis and pneumonia in children

Definition and causes

- **Pneumonitis:** lung inflammation from aspiration of milk or stomach contents (reflux, poor swallowing), chemicals (kerosene ingestion in children is a classic Indian example), or infection.
- **Pneumonia:** bacterial or viral infection; a major cause of death in children under 5 in India.
- **Bronchiolitis** (viral, under 2 years) often looks similar: wheeze, crackles, feeding difficulty.

Key features

Fast breathing for age, chest in-drawing, fever, cough, poor feeding; low SpO_2 in severe cases.

Signs of respiratory distress in children (3 marks in W24)

Fast breathing for age (WHO), counted for a full minute while the child is calm

Age	Fast breathing
Under 2 months	**60 or more** breaths a minute
2–11 months	**50 or more**
1–5 years	**40 or more**

- **Work of breathing:** nasal flaring; chest in-drawing (subcostal, intercostal, suprasternal); tracheal tug; use of neck muscles; head bobbing in infants; see-saw (paradoxical) movement of chest and abdomen.
- **Noises:** grunting (the baby breathes out against a closed glottis to keep alveoli open), stridor, wheeze.
- **Effects:** cyanosis and low SpO_2; fast heart rate; cannot feed or talk; sweating.
- **Danger signs (tiring):** drowsiness or poor response, pauses in breathing (apnoea), slowing breathing in an exhausted child, silent chest. Call for urgent help.

Memory hook

60, 50, 40

Fast-breathing cut-offs fall by 10 as the child grows: under 2 months, under 1 year, under 5 years.

PT management

- **Bronchiolitis (Cochrane 2023):** conventional chest physiotherapy (vibration, percussion) does not improve severity; forced expiratory techniques cause vomiting and breathing instability. Slow passive expiratory techniques may give a small benefit in moderate cases.
- **Pneumonia:** no routine airway clearance; use positioning, mobilisation and play; airway clearance if secretions are retained (for example a collapsed lobe).
- **Aspiration:** head-up positioning, feeding advice with speech and language therapy.

Exam points

- Chest physiotherapy is not routine for bronchiolitis or uncomplicated pneumonia in children.

5 Chronic lung disease of prematurity (bronchopulmonary dysplasia)

Definition and cause

Long-term lung damage in premature babies who needed oxygen and ventilation; usually defined by an ongoing oxygen need at 36 weeks after conception.

Key features

Fast breathing, oxygen dependence, recurrent wheeze and infections, poor growth.

PT management

- Airway clearance during infections; positioning; developmental care (tummy time when awake and monitored, head control, motor milestones).
- Parent education on home oxygen, signs of infection and feeding.
- Later: active play and exercise as tolerated.

6 Paediatric ARDS

Acute, severe lung inflammation in a child after pneumonia, sepsis, drowning or trauma, with low oxygen and bilateral shadows on X-ray, needing ventilation. Physiotherapy: positioning (prone positioning as directed by the ICU team), careful secretion clearance, minimal disconnection from the ventilator, early mobilisation and rehabilitation after recovery.

7 Childhood asthma and cystic fibrosis

See the asthma page (p. 132) (paediatric rehabilitation, parent and teacher counselling) and the cystic fibrosis page (p. 208).

8 Exam corner

W18 Q2(a), 7 marks

Precautions and contraindications for postural drainage in a neonate.

S18 Q2(d), 7 marks

Paediatric pulmonary rehabilitation for children with bronchial asthma (asthma page).

W24 Q1(e), 3 marks

Signs and symptoms of respiratory distress in children (section 4).

Technique cards for this topic: postural drainage (p. 267), airway clearance (p. 258), oxygen therapy (p. 290).

Answer skeleton: precautions and contraindications in a neonate

1. Indications: retained secretions, collapse; not routine.
2. Precautions: minimal handling, monitoring, warmth, timing with feeds, head-up or flat positions, gentle techniques, safe suction.
3. Contraindications: instability, intraventricular haemorrhage, pulmonary hypertension, undrained pneumothorax, fragile ribs or bleeding, raised intracranial pressure.
4. Modified techniques: positioning, gentle vibration, assisted breathing, suction.

Test yourself

MCQ **A preterm baby on CPAP has reflux. Best drainage position:**

A. Head-down tilt

B. Flat or head-up (about 30°)

C. Upside down

D. No position change ever

Answer

B. Head-down positions increase reflux and are avoided in neonates.

MCQ **For a 6-month-old with moderate bronchiolitis, which statement is true?**

A. Percussion and vibration shorten illness

B. Forced expiratory techniques are safe and effective

C. Conventional chest physiotherapy does not improve severity

D. Head-down drainage is first-line

Answer

C. Cochrane (2023) found no benefit from conventional techniques, and harm from forced expiratory techniques.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Roqué-Figuls M, et al. Chest physiotherapy for acute bronchiolitis in paediatric patients between 0 and 24 months old. Cochrane Review 2023. PMC
3. Cochrane Review: Positional therapy for infants with gastro-oesophageal reflux (CD010297): head-up versus head-down positioning.
4. International Liaison Committee on Resuscitation (ILCOR) neonatal recommendations on meconium-stained amniotic fluid (2015, 2020).
5. World Health Organization. Integrated Management of Childhood Illness (IMCI): fast-breathing cut-offs by age; revised WHO classification of childhood pneumonia (2014).
6. Structure and classical content: Main & Denehy (2016), neonatal and paediatric chapters; Pryor & Prasad (2008).

Cystic fibrosis

> Cystic fibrosis was not asked in any of the 20 papers from 2016 to 2026, but it is in the syllabus twice (adults and children) and is the classic disease for airway clearance. Learn the cause, the sweat test, and a lifelong daily physiotherapy routine.

Syllabus	Ch 6(b)(iv) Cystic fibrosis; 6(d)(vi) Cystic fibrosis and chronic lung disease
Priority	**Tier 3.** Not asked in 2016–2026.
Asked as	Not asked 2016–2026
Usual split	Short notes; possible 3- or 7-mark questions
Checked against	Cystic Fibrosis Trust physiotherapy standards (5th edition, July 2026); Cystic Fibrosis Foundation diagnosis consensus (2017). Full list under Sources. Content checked 19 October 2026.

1 Definition and cause

Cystic fibrosis (CF) is an inherited disease (autosomal recessive: both parents carry one faulty gene) caused by mutations in the **CFTR** gene. The faulty CFTR protein cannot move chloride and water properly across cell surfaces, so secretions become thick and sticky in the lungs, pancreas, liver, intestines and reproductive tract, and sweat becomes very salty.

2 Key features and diagnosis

- **Lungs:** thick mucus, chronic infection (*Staphylococcus aureus*, later *Pseudomonas*), progressive bronchiectasis, haemoptysis, pneumothorax, clubbing.
- **Gut and pancreas:** meconium ileus at birth, fatty bulky stools, malabsorption, poor weight gain; CF-related diabetes later.
- **Other:** nasal polyps and sinusitis, liver disease, male infertility, salt loss in heat, low bone density.
- **Diagnosis:** newborn screening where available; **sweat chloride 60 mmol/L or more** supports the diagnosis (30–59 is intermediate and needs further testing); genetic testing.
- In India CF is under-diagnosed and often found late, in children with repeated chest infections and poor growth.

3 Medical point that matters for physio

- **CFTR modulator drugs** (for example elexacaftor–tezacaftor–ivacaftor) correct the faulty protein in many patients and have greatly improved lung function and survival where available.
- Mucoactive treatments: dornase alfa and hypertonic saline, inhaled before airway clearance.
- Antibiotics (oral, inhaled, intravenous), pancreatic enzymes, high-calorie diet, salt supplements.
- **Usual order:** bronchodilator → hypertonic saline or dornase alfa → airway clearance → inhaled antibiotic.
- Lung transplant in end-stage disease.

4 ICF summary

Impairments: thick secretions, recurrent infections, reduced lung function and exercise tolerance, poor growth, posture and back pain. Activity: sport and play, daily treatment time burden. Participation: school, work, social life (time-consuming treatments, cross-infection rules). Environment: family support and CF centre (facilitators); cost and access to drugs (barriers). Personal: adherence in adolescents.

5 PT management by problem

Daily airway clearance (lifelong)

- **Infants:** modified postural drainage **without head-down tilt**, gentle percussion or vibration, assisted autogenic drainage, PEP mask, and active play (bouncing on a ball, rolling).
- **Children:** PEP mask, oscillating PEP (flutter, Acapella), blowing games, ACBT once they can huff.
- **Teenagers and adults:** ACBT, autogenic drainage, PEP or oscillating PEP, chest-wall oscillation vest; choose what the person will actually do.
- Usually once or twice daily, more during infections; review technique regularly.

Exercise

- Regular aerobic and strength exercise and sport for all ages: improves fitness, helps clear mucus, supports bone health and quality of life.
- Exercise does not replace airway clearance for most people.

Other

- Posture and thoracic mobility; management of back pain.
- Pelvic floor training (urinary leakage with coughing is common in women with CF).
- Inhaled therapy technique; NIV in advanced disease; rehabilitation before and after transplant.

6 Precautions

- **Cross-infection:** people with CF must be kept apart from each other (no shared rooms, gyms or group sessions); clean equipment between patients; hand hygiene.
- **Haemoptysis:** stop percussion and positive-pressure devices; report.
- **Pneumothorax:** avoid positive-pressure techniques until drained.
- Exercise in heat: extra fluids and salt.
- CF-related diabetes: check sugar around exercise.
- Avoid head-down positions in infants and with reflux.

7 Exam points

- Autosomal recessive; CFTR; thick secretions; sweat chloride 60 mmol/L or more.
- Lifelong daily airway clearance adapted to age; exercise; strict cross-infection control.

Technique cards for this topic: airway clearance (p. 258), ACBT and autogenic drainage (p. 261), PEP devices (p. 271), nebulisers and inhalers (p. 294).

Test yourself

3 m **Airway clearance techniques for a 6-month-old with CF.**

Answer
Modified postural drainage without head-down tilt; gentle percussion or vibration; assisted autogenic drainage; PEP mask; active play. Treat before feeds.

MCQ **Two teenagers with CF want to train together in the hospital gym. Best advice:**

A. Train together for motivation

B. Train separately because of cross-infection risk

C. Share the same flutter device

D. Avoid exercise

Answer

B. People with CF can pass resistant bacteria to each other, so they must be kept apart.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Cystic Fibrosis Trust. Standards of care and good clinical practice for the physiotherapy management of cystic fibrosis, 5th edition (July 2026). cysticfibrosis.org.uk
3. Farrell PM, et al. Diagnosis of cystic fibrosis: consensus guidelines from the Cystic Fibrosis Foundation. *J Pediatr* 2017. CFF clinical care guide.
4. Structure and classical content: Main & Denehy (2016); Pryor & Prasad (2008).

Flail chest, scoliosis and kyphosis

> These chest-wall problems were not asked on their own in 2016–2026, but rib fractures appeared in a ventilated burns case (W17). The common thread: a stiff or unstable chest wall reduces lung volumes, so physiotherapy protects breathing and mobility.

Syllabus	Ch 6(j) Musculoskeletal dysfunction: flail chest, scoliosis, kyphosis
Priority	**Tier 3.** Not asked on its own in 2016–2026; rib fractures in the W17 case.
Asked as	Inside W17 LAQ
Usual split	Short notes; possible 3- or 7-mark questions
Checked against	Radiological definition of flail chest; SOSORT guidelines (2016); BTS/ACPRC physiotherapy guideline (2009). Full list under Sources. Content checked 19 October 2026.

1 Flail chest

Definition

A segment of chest wall that has lost its bony connection to the rest of the rib cage, usually defined as **3 or more neighbouring ribs, each broken in 2 or more places**. The loose segment moves in the opposite direction to the rest of the chest.

Memory hook

3 ribs, 2 breaks, 1 floating segment

Causes and key features

- Major blunt trauma: road accidents, falls from height (W17 electrician), crush injuries; in older people, lesser falls.
- **Paradoxical movement:** the flail segment is sucked **in** during inspiration and pushed **out** during expiration.
- Severe pain, shallow breathing, poor cough, low oxygen.
- The underlying **lung contusion** is the main cause of low oxygen; haemothorax or pneumothorax may coexist.
- **X-ray/CT:** multiple rib fractures, contusion (patchy opacity), air or fluid in the pleura.

Medical point that matters for physio

- **Pain control** is the key treatment: epidural, paravertebral or other nerve blocks, regular analgesia.
- Oxygen; non-invasive ventilation in selected patients; intubation and ventilation for respiratory failure (positive pressure splints the segment from inside).
- Surgical rib fixation in selected patients; chest drain for pneumothorax or haemothorax.

ICF summary

Impairments: pain, reduced lung volumes, ineffective cough, low oxygen, reduced shoulder movement. Activity: bed mobility, walking, self-care. Participation: work (often manual) for weeks.

PT management by problem

- **Pain:** treat after analgesia takes effect; supported positions; splint the chest with a hand or pillow during cough.
- **Reduced lung volumes:** sit upright; deep breathing with a 3-second hold, 5–10 breaths every hour while awake; thoracic expansion avoiding pressure on the fractures.
- **Secretions:** ACBT with gentle huffs and supported cough; suction if needed.

- **On a ventilator:** positioning, suction, careful handling of the chest, early mobilisation when stable.
- **Mobility:** early sitting out and walking; shoulder range on the injured side.

Precautions

- No percussion, vibration or shaking over fractured ribs.
- Undrained pneumothorax: no positive-pressure techniques.
- Watch for worsening hypoxia (contusion often worsens over the first day or two), haemoptysis and spinal or head injuries.

Exam points

- Definition (3 ribs, 2 places), paradoxical movement, lung contusion, pain control first, no percussion over fractures.

2 Scoliosis

Definition and types

A sideways curvature of the spine with rotation of the vertebrae, defined as a **Cobb angle of 10° or more** on an X-ray.

- **Idiopathic:** infantile, juvenile, adolescent (commonest; mostly girls).
- **Congenital:** vertebral malformations.
- **Neuromuscular:** cerebral palsy, muscular dystrophy, spinal muscular atrophy, spinal cord injury.
- **Degenerative** (adult) and secondary (for example after thoracoplasty or empyema).

Key features

- Uneven shoulders and waist, prominent shoulder blade; **rib hump on forward bending** (Adam's test); scoliometer measures the trunk rotation.
- **Breathing:** large thoracic curves, especially early-onset and neuromuscular curves, stiffen the chest wall and reduce lung growth and volumes (restrictive pattern), sometimes leading to breathing failure, especially at night.

Medical point that matters for physio

Management depends on curve size, growth remaining and progression (SOSORT 2016): observation, physiotherapeutic scoliosis-specific exercises, bracing for moderate curves in growing children, and surgery (spinal fusion) for large or progressive curves.

PT management by problem

- **Curve and posture:** scoliosis-specific exercises (for example Schroth or SEAS methods): self-correction, stabilisation and breathing into the concave side.
- **Restricted breathing:** thoracic expansion exercises directed to the concave side, rib mobility, inspiratory muscle training; aerobic fitness.
- **Neuromuscular scoliosis:** seating and positioning, airway clearance and assisted cough, night ventilation as prescribed.
- **After fusion surgery:** breathing exercises, early mobilisation, log-roll and spinal precautions as advised.

Precautions

- Follow brace-wearing schedules; avoid forceful spinal mobilisation in progressive or post-surgical curves.

Exam points

- Cobb angle 10° or more; rib hump; restrictive pattern in large thoracic curves; breathing into the concave side.

3 Kyphosis

Definition and types

Excessive forward curvature of the thoracic spine (normal thoracic kyphosis is about 20–40°).

- **Postural:** flexible, corrects on extension.
- **Scheuermann's:** rigid kyphosis in adolescence with wedging of 5° or more in 3 neighbouring vertebrae.
- **Osteoporotic:** from vertebral wedge fractures in older adults.
- **Ankylosing spondylitis:** rigid kyphosis with fused costovertebral joints, so chest expansion is very reduced.
- **Tuberculosis of the spine** (Pott's disease): sharp angular kyphosis (gibbus); congenital kyphosis.

Key features

Rounded back, forward head, back pain; reduced chest expansion and vital capacity in severe or rigid kyphosis; breathlessness on exertion; falls risk in osteoporosis.

PT management by problem

- **Posture:** spinal extension and back-extensor strengthening, chest stretching, scapular retraction, posture awareness.
- **Breathing:** thoracic expansion (lateral costal) and diaphragmatic breathing; in ankylosing spondylitis, daily chest expansion and spinal mobility exercises.
- **Osteoporosis:** weight-bearing and balance exercise, falls prevention; avoid loaded spinal flexion.
- Bracing for growing patients with Scheuermann's as advised.

Precautions

- No forceful flexion or heavy loading in osteoporotic spines; careful manual techniques on a rigid, fragile thorax.

Exam points

- Types of kyphosis; reduced chest expansion; extension and breathing exercises.

4 Exam corner

W17 Q3(a), 5 + 5 + 5 marks

30-year-old electrician with electrical burns and a fall, 3 fractured ribs on the right, unconscious and ventilated (see the burns page). Impairments, activity and participation; short-term management.

Technique cards for this topic: breathing exercises (p. 275), ICU physiotherapy (p. 223).

Test yourself

3 m **Define flail chest and explain paradoxical movement.**

Answer

Three or more neighbouring ribs each fractured in two or more places, creating a free segment. On inspiration the segment moves in as pressure in the chest falls; on expiration it moves out. This reduces effective ventilation and is painful.

MCQ **In a patient with multiple rib fractures, which technique should be avoided over the fractures?**

A. Deep breathing with supported cough

B. Percussion and vibration

C. Upright positioning

D. Early mobilisation

Answer

B. Percussion and vibration over fractures cause pain and can displace fragments.

MCQ **Breathing exercises in thoracic scoliosis are directed mainly to:**

A. The convex side

B. The concave side

C. The abdomen only

D. Neither side

Answer

B. The concave side is compressed, so expansion exercises target it.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
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3. Negrini S, et al. 2016 SOSORT guidelines: orthopaedic and rehabilitation treatment of idiopathic scoliosis during growth. *Scoliosis Spinal Disord* 2018.
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Part B

Technique cards

Skills, devices, tests and scales that the short answers ask about.

ICU and ventilation

Mechanical ventilation: indications, modes, CPAP and PEEP, complications, weaning 216

Physiotherapy in the ICU: assessment, safety, equipment and ventilated cases 223

Suctioning and manual hyperinflation 231

Therapeutic positioning, V/Q and the zones of the lung 236

Effects of bed rest and early mobilisation 241

ICU conditions: head injury, tetanus, spinal cord injury and neuromuscular disease 245

ICU conditions: pulmonary oedema, shock, organ failure, poisoning, aspiration and drowning 252

Airway clearance

Airway clearance techniques: overview and choice 258

ACBT, forced expiration, huffing and autogenic drainage 261

Postural drainage, percussion and vibration 267

PEP, flutter, Acapella and other airway-clearance devices 271

Breathing and breathlessness

Breathing exercises, lung expansion and breathing retraining 275

Neurophysiological facilitation and PNF of respiration 279

Respiratory muscle training 282

Dyspnoea, relieving positions, energy conservation and pacing 285

Oxygen, humidification and aerosols

Oxygen therapy: devices, targets and hazards 290

Humidification, nebulisers and inhalers 294

Electrotherapy, exercise, yoga and handling skills

Electrotherapy and therapeutic exercise in heart, lung and vascular conditions 298

Applied yoga in cardio-respiratory conditions 302

Handling skills and exercise equipment: stockings, bandaging, taping, splints, monitors 305

Exercise testing and exercise physiology

Exercise testing: contraindications, stop criteria, Bruce, cycle, walk and step tests 308

Exercise physiology and prescription: METs, energy systems, intensity, training 313

Investigations and scales

Investigations: chest X-ray, PFT, ECG and bedside tests 317

Scales: Borg, mMRC, NYHA, CCS and quality-of-life questionnaires 322

ABG interpretation and why patients desaturate 326

Abnormal breathing patterns and chest signs 330

Mechanical ventilation: indications, modes, CPAP and PEEP, complications, weaning

> Ventilated patients now appear in long answers (GBS on SIMV in S23, ARDS on SIMV in S24, pneumonia on CMV in W24). You do not set the ventilator, but you must read its settings, know what each mode means for your treatment, and know when the patient is ready to come off it.

Syllabus	Ch 9(c) Mechanical ventilator modes; 9(b) Principles of ICU monitoring
Priority	**High yield.** In 7 of 20 papers (2016–2026); an ICU question appears in every paper since 2023.
Asked as	LAQ S23 · 7m S17 · W19 · W20 · 3m S22 · W21 ×2 · S24
Usual split	3-mark lists (indications, complications, CPAP vs PEEP); 7-mark modes or weaning; part of ventilated long cases
Checked against	International weaning consensus (2007); ACCP/ATS liberation guideline (2017); ARDS Network low tidal volume trial (2000); ICU mobilisation safety consensus (2014). Full list under Sources. Content checked 19 October 2026.

1 What it is

Mechanical ventilation means a machine moves gas into the lungs with positive pressure when the patient cannot breathe well enough alone. It buys time while the cause is treated; it does not cure anything.

- **Invasive:** through an endotracheal tube (ETT) or a tracheostomy.
- **Non-invasive (NIV):** through a tight face or nasal mask: CPAP or bilevel pressure (BiPAP).

Words on the ventilator screen

Term	Meaning	Usual adult value
FiO_2	Fraction of oxygen in the gas (0.21 = room air, 1.0 = 100%)	The lowest that keeps SpO_2 in the target range
Tidal volume (VT)	Volume of each breath	**6–8 mL/kg** of predicted body weight; 6 mL/kg in ARDS
Rate (f)	Machine breaths per minute	About **12–20**
PEEP	Pressure kept in the lungs at the end of breathing out	About **5 cmH_2O**; higher in ARDS
Peak and plateau pressure	Highest airway pressure in a breath; pressure in the alveoli during a pause	Plateau kept at **30 cmH_2O or less**
I:E ratio	Time breathing in : time breathing out	About 1:2; longer expiration in COPD and asthma
Trigger	The patient's own effort that starts a breath	Set by flow or pressure sensitivity
Pressure support	Extra pressure added to each breath the patient starts	5–8 cmH_2O near the end of weaning

Minute ventilation = tidal volume × rate (about 5–8 L/min at rest).

2 Indications (3 marks in S24)

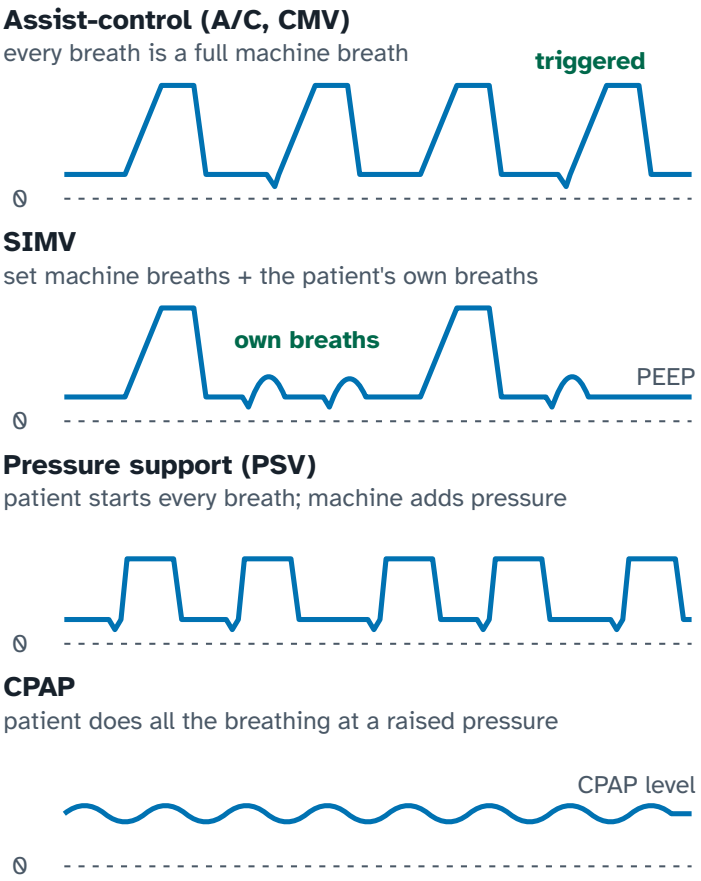
Group	Examples
Type 1 respiratory failure (low oxygen)	PaO_2 below about 60 mmHg despite high oxygen: pneumonia, ARDS, pulmonary oedema
Type 2 respiratory failure (high CO_2 with acidosis)	COPD exacerbation when NIV fails, severe asthma, drug overdose

Group	Examples
Failure of the breathing pump	Guillain–Barré syndrome, myasthenic crisis, high spinal cord injury, flail chest, exhausted breathing muscles
Airway protection	Coma (GCS 8 or less), head injury, stroke, poisoning, burns of the airway
Heart and circulation	Cardiac arrest, cardiogenic shock, severe pulmonary oedema
Planned	During and after major surgery; when heavy sedation or paralysis is needed; to control CO_2 in raised intracranial pressure

Warning signs that lead to ventilation: respiratory rate above about 35, exhaustion, falling SpO_2 despite oxygen, rising CO_2 with falling pH, drowsiness, and in neuromuscular disease a falling vital capacity.

3 Modes (7 marks in S17 and W20)

Airway pressure in four common modes



Blue line = airway pressure. A small dip below the baseline is the patient trying to breathe in (a trigger). The baseline is above zero in all four because of PEEP.

Mode	How it works	Used for
Controlled mandatory ventilation (CMV)	Set volume (or pressure) at a set rate; in strict CMV the patient cannot add breaths	Deeply sedated or paralysed patients (W24 pneumonia on CMV)
Assist-control (A/C)	Set breath and a minimum rate; if the patient triggers, each extra breath is a full machine breath. Most modern "CMV" settings work this way	Resting tired muscles early in illness. Fast breathing can cause over-breathing and air trapping

Mode	How it works	Used for
Synchronised intermittent mandatory ventilation (SIMV)	A set number of machine breaths, timed with the patient's effort; between them the patient breathes alone, usually with pressure support	Partial support (S20, S23, S24 cases). Once used for weaning by lowering the rate, but weaning is slower than with pressure support or daily breathing trials
Pressure control (PCV)	Set pressure and inspiratory time; the volume varies with lung stiffness	Limits pressure in stiff lungs (ARDS)
Pressure support (PSV)	The patient starts every breath and controls rate and depth; the machine adds a set pressure	Comfortable partial support; weaning by lowering the support to 5–8 cmH_2O
CPAP	One continuous positive pressure; no machine breaths; the patient does all the breathing	Weaning on the ventilator; mask CPAP for sleep apnoea, cardiogenic pulmonary oedema, post-operative collapse
Bilevel NIV (BiPAP)	Higher pressure on breathing in (IPAP), lower on breathing out (EPAP), by mask	COPD exacerbation with acidosis, after extubation in high-risk patients, neuromuscular disease at night

Rescue modes in very severe ARDS, such as high-frequency oscillation, are run by the doctors; patients on them are not mobilised out of bed.

? Why does the mode matter to a physiotherapist?

> It tells you how much work the patient's own muscles do. On CMV or A/C at a high rate the muscles are resting and wasting, so you rely on positioning, passive movement and secretion clearance. On SIMV, PSV or CPAP the patient is breathing, so active exercise, breathing exercises and inspiratory muscle training become possible, and you watch for fatigue.

4 CPAP and PEEP (3 marks in W21)

Point	CPAP	PEEP
What it is	A **mode**: positive pressure through the whole breath in a patient who breathes on their own	A **setting**: positive pressure kept at the end of breathing out, added to any mode with machine breaths
Breaths	All spontaneous; no machine breaths	Machine or supported breaths on top of it
How given	Mask, nasal interface or ventilator	Ventilator (or a PEEP valve on a bag)
Typical use	Sleep apnoea, cardiogenic pulmonary oedema, post-op collapse, weaning	Almost every ventilated patient (about 5 cmH_2O); higher in ARDS

Memory hook

CPAP = PEEP for a patient who breathes alone

Effects of both: raise the functional residual capacity (FRC), keep small airways and alveoli open, improve oxygenation and lung compliance, and reduce collapse. CPAP also lowers the work of breathing and, in heart failure, the load on the left ventricle.

Side effects of high levels: less blood returns to the heart, so BP can fall; over-stretching and pneumothorax (barotrauma); raised intracranial pressure; mask discomfort, air swallowing and pressure sores with CPAP.

Do not confuse with auto-PEEP (intrinsic PEEP): trapped air when the patient cannot breathe out fully before the next breath, common in COPD and asthma; it lowers BP and makes triggering harder.

5 Complications (3 marks in S22 and W21)

System	Complications	Physio role
Lungs	Ventilator-induced lung injury (over-stretch, high pressure, repeated opening and closing of alveoli); pneumothorax; ventilator-associated pneumonia (VAP); collapse; oxygen toxicity; auto-PEEP; fighting the ventilator	Head-up position, turning, airway clearance; report sudden high pressures or falling SpO_2
Airway	Tube blocked by secretions, displaced or pulled out; tube in the right main bronchus; damage to the larynx and trachea, later narrowing (stenosis); sinusitis	Humidification, suction; secure the tube during every turn and transfer
Heart and circulation	Lower venous return and cardiac output (hypotension), fluid retention	Check BP before and during treatment
Muscles and joints	Weak, wasted diaphragm; ICU-acquired weakness; contractures; pressure injuries; DVT	Early mobilisation, ROM, positioning, inspiratory muscle training
Other	Stress ulcers and gut bleeding; delirium; poor sleep; anxiety and post-traumatic stress after ICU	Orientation, communication, day–night routine, follow-up rehab

Long-term ventilation (W21): tracheal stenosis or softening, repeated VAP, diaphragm wasting and ventilator dependence, ICU-acquired weakness, contractures, pressure injuries, and psychological effects. Patients who need the ventilator for more than about 1–2 weeks often get a tracheostomy, which is more comfortable, needs less sedation and makes weaning easier.

6 Weaning (7 marks in W19)

Weaning is the process of freeing the patient from the ventilator. It starts with a daily check of readiness, then a **spontaneous breathing trial (SBT)**, then extubation.

Ready to try? (daily screen)

- The cause of respiratory failure is improving.
- **Oxygenation:** SpO_2 90% or more on FiO_2 **0.4–0.5 or less** and PEEP **5–8 cmH_2O or less** (PaO_2/FiO_2 ratio above about 150–200).
- **Circulation:** stable heart rate and BP, no or low-dose drugs to support BP, no active ischaemia.
- **Brain:** awake or easily roused, follows commands; can cough and protect the airway.
- **Breathing:** breathes on their own; pH 7.25 or more; few secretions; no fever, normal electrolytes.

Bedside weaning measures

Measure	Favours weaning
Rapid shallow breathing index (RSBI = rate ÷ tidal volume in litres)	Below **105** breaths/min/L
Maximal inspiratory pressure (MIP)	Stronger than about −20 to −30 cmH_2O
Vital capacity	More than about 10 mL/kg
Minute ventilation	Below about 10 L/min

? Why does a high RSBI predict failure?

> A tiring patient breathes fast and shallow: rate goes up, tidal volume goes down, so the ratio rises. Example: 35 breaths of 0.25 L gives 140 (likely to fail); 20 breaths of 0.5 L gives 40 (likely to succeed).

The breathing trial

- **30–120 minutes** on low pressure support (5–8 cmH_2O) or a T-piece. The 2017 ACCP/ATS guideline suggests using low pressure support for the trial.

- **Stop the trial if:** breathing rate above 35 for more than 5 minutes; SpO_2 below 90%; heart rate above 140 or a change of more than 20%; systolic BP above 180 or below 90 mmHg; agitation, sweating, use of accessory muscles or see-saw breathing.
- Passed: the team considers extubation (with a cough and secretion check). High-risk patients may go onto NIV straight after extubation.

How weaning went (international consensus, 2007)

Group	Meaning
Simple	Extubated after the first trial
Difficult	Needed up to 3 trials or up to 7 days
Prolonged	More than 3 trials or more than 7 days

The physiotherapist's part in weaning

- Keep the airways clear: positioning, manual techniques, hyperinflation and suction.
- Sit the patient upright for trials: the diaphragm works better and lung volume is larger.
- **Inspiratory muscle training** for patients who fail weaning: it strengthens the inspiratory muscles and may shorten weaning (systematic review, 2015).
- Early mobilisation builds the strength and endurance needed to breathe alone.
- After extubation: deep breathing, huffing and supported cough, early walking.

7 Reading the settings in a case

Before treatment, note the mode, FiO_2, PEEP, set and actual tidal volume, rate, peak pressure, SpO_2 and the latest ABG, and whether these are getting better or worse. Rising FiO_2 or PEEP means the lungs are getting worse.

The MUHS cases

Case	Settings	What it tells you
S23: GBS, 45-year-old teacher	SIMV, PEEP 6, FiO_2 0.4	Moderate support for weak breathing muscles; lungs not badly affected; safe for active treatment as strength returns
S24: ARDS, 30-year-old woman	SIMV, PEEP 5, FiO_2 0.4	Mild to moderate support; oxygenation improving or mild disease; in-bed and out-of-bed activity possible if other signs are stable
W24: pneumonia, 35-year-old man	CMV, PEEP 10, FiO_2 0.6, day 3	High support, deeply sedated; at the limit for mobilising; avoid disconnection (closed suction)

Alarms during treatment

- **High-pressure alarm:** coughing, secretions, biting the tube, kinked tubing, bronchospasm, the patient fighting the ventilator, or a pneumothorax.
- **Low-pressure or low-volume alarm:** disconnection or a leak around the cuff.
- If the cause is not obvious, or SpO_2 falls, stop and call the nurse or doctor. Never silence an alarm you do not understand.

8 Exam corner

Past questions (MUHS, 2016–2026)

S23 Q4(b), 15 marks

45-year-old schoolteacher with Guillain–Barré syndrome on a ventilator: SIMV, PEEP 6, FiO_2 0.4. Discuss his impairments and ventilator management. Worked on the ICU physiotherapy card (p. 226).

7-mark questions

Modes of ventilation and the role of physiotherapy in ICU (S17). Criteria for weaning a patient off the ventilator (W19). Modes of mechanical ventilation (W20).

3-mark questions

Respiratory complications of mechanical ventilation (S22). Complications of long-term mechanical ventilation (W21). Difference between CPAP and PEEP (W21). Three indications for mechanical ventilation (S24).

Answer skeletons

1. **[Modes, 7]** Definition of mechanical ventilation → the four traces sketched → table of CMV/A-C, SIMV, pressure control, pressure support, CPAP, BiPAP with one use each → one line on why the mode matters to physiotherapy.
2. **[Weaning, 7]** Definition → readiness criteria (cause, oxygenation, circulation, brain, breathing) → RSBI and MIP with values → breathing trial and its stop criteria → simple, difficult, prolonged → physiotherapy role.
3. **[CPAP vs PEEP, 3]** Three rows: mode vs setting, spontaneous vs machine breaths, typical use.

Test yourself

3 m **Give three differences between CPAP and PEEP.**

Answer
CPAP is a mode, PEEP is a setting. CPAP applies pressure throughout the breath in a patient who breathes alone, with no machine breaths; PEEP is pressure kept at the end of expiration during machine or supported breaths. CPAP is often given by mask (sleep apnoea, pulmonary oedema); PEEP is given through the ventilator to almost every ventilated patient.

3 m **A patient on a breathing trial breathes 32 times a minute with a tidal volume of 0.4 L. Calculate the RSBI and interpret it.**

Answer
32 ÷ 0.4 = 80 breaths/min/L. Below 105, so it favours successful weaning (other criteria must also be met).

MCQ **In which mode does the patient breathe spontaneously between a set number of machine breaths?**

A. CMV
B. SIMV
C. CPAP
D. Pressure control

Answer
B. SIMV gives a set number of synchronised machine breaths; between them the patient breathes alone, usually with pressure support.

MCQ **During a spontaneous breathing trial, which finding means the trial should be stopped?**

A. SpO_2 94%

B. Breathing rate 38 for 6 minutes

C. Heart rate rising from 90 to 100

D. Systolic BP 130 mmHg

Answer

B. A rate above 35 for more than 5 minutes means the patient is tiring.

Sources

Notes written in original words. The mode diagram is a simplified drawing made for these notes.

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
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Physiotherapy in the ICU: assessment, safety, equipment and ventilated cases

> Four long answers since 2020 put the patient on a ventilator (COPD on SIMV, GBS on SIMV, ARDS on SIMV, pneumonia on CMV), and ICU short questions now come in every paper. The answer always has the same backbone: read the settings and monitors, check safety, then treat the chest, the limbs and function.

Syllabus	Ch 9(a) Basic evaluation; 9(b) Principles of ICU monitoring; 9(e) Therapeutic intervention (ARDS, GBS, neuromuscular disease and others)
Priority	**High yield.** In 9 of 20 papers (2016–2026), including 4 ventilated long cases.
Asked as	LAQ S20 · S23 · S24 · W24 · 7m S17 · W17 · S21 · W24 · S25 · S26
Usual split	15 (impairments or functional diagnosis, complications, physiotherapy); 7-mark ICU physiotherapy, goals, equipment
Checked against	ICU mobilisation safety consensus (Hodgson 2014); Berlin definition of ARDS (2012); early mobilisation trials (2009, 2022); prone positioning trial (2013). Full list under Sources. Content checked 19 October 2026.

1 ICU assessment

Write this first in any ICU answer. It shows the examiner you check before you treat.

From the chart and the monitors

- **Diagnosis and course:** reason for admission, surgery, days on the ventilator, trend (better or worse since yesterday).
- **Ventilator:** mode, FiO_2, PEEP, rate, tidal volume, pressures (see the ventilation card (p. 220)).
- **Vital signs:** heart rate and rhythm, BP and mean arterial pressure (MAP), SpO_2, respiratory rate, temperature; intracranial pressure if monitored.
- **ABG and chest X-ray:** oxygenation, CO_2 and pH; collapse, consolidation, effusion, pneumothorax, tube position.
- **Drugs:** sedation, pain relief, drugs supporting BP, paralysing agents.
- **Blood tests:** haemoglobin, platelets and clotting (bleeding risk), lactate, electrolytes, sugar.
- **Lines and drains:** what is attached and where (femoral lines limit hip bending).

At the bedside

- **Conscious level:** Glasgow Coma Scale; sedation score (Richmond Agitation–Sedation Scale, RASS, –5 to +4); delirium screen (CAM-ICU).
- **Chest:** breathing pattern, synchrony with the ventilator, chest movement, auscultation, secretions (amount, colour, thickness), cough on suction.
- **Limbs:** range of motion, tone, early contractures, skin and pressure areas, oedema.
- **Strength and function (when awake):** MRC sum score of 12 muscle groups (0–60; **below 48 = ICU-acquired weakness**); bed mobility, sitting balance; ICU Mobility Scale (0 = lying in bed, 10 = walking independently).
- **Pain:** self-report if possible, or a behavioural pain scale.

2 Principles of ICU monitoring

Monitor	Usual target	Why the physiotherapist watches it
ECG and heart rate	60–100, regular	Suction can slow the heart (vagal reflex); new arrhythmia means stop

Monitor	Usual target	Why the physiotherapist watches it
Arterial line: BP and MAP	MAP usually 65 mmHg or more, as the team sets	Turning, sitting up and bagging can drop BP
Pulse oximetry (SpO_2)	Usually 92–96%; 88–92% if at risk of CO_2 retention	Falls with suction, disconnection and exertion
Respiratory rate and ventilator graphics	12–20; patient in time with the ventilator	Fast rate or fighting the ventilator signals distress
Capnography (end-tidal CO_2)	About 35–45 mmHg	Confirms the tube is in the airway; shows ventilation changes
Temperature	36–38 °C	Fever raises oxygen demand and heart rate
Intracranial pressure (head injury)	Normal 5–15 mmHg; treated above about 22; cerebral perfusion pressure (MAP – ICP) 60–70	Suction, coughing, head-down tilt and neck flexion raise it
Urine output and fluid balance	At least 0.5 mL/kg/hour	Low output or heavy fluid loss suggests poor circulation
ABG	pH 7.35–7.45, $PaCO_2$ 35–45, PaO_2 80–100 mmHg (on room air)	Sets treatment aims; repeated after major changes

3 Safety before mobilising a ventilated patient

An international consensus (Hodgson, 2014) uses traffic lights: **green** = low risk, go ahead; **yellow** = possible risk, discuss with the team; **red** = high risk, do not mobilise actively without senior agreement.

Area	Green for in-bed and out-of-bed exercise	Red (out of bed)
Respiratory	Endotracheal tube or tracheostomy in place (not a barrier); FiO_2 **0.6 or less**; SpO_2 **90% or more**; respiratory rate **30 or less**; PEEP **10 cmH_2O or less**	**SpO_2 below 90%; high-frequency oscillation; prone positioning**
Cardiovascular	MAP above the lower target with no or low-dose support; ventricular rate below 120 in a stable tachyarrhythmia; stable rhythm with a pacemaker	**IV drugs for a hypertensive emergency; low MAP causing symptoms; bradycardia needing drugs or emergency pacing; ventricular rate above 150; active cardiac ischaemia**
Neurological	Drowsy, calm or restless (RASS –1 to +1); no delirium, or delirium but follows simple commands	**Very agitated or combative (RASS above +2); deeply sedated (RASS below –2); raised ICP being actively treated; uncleared spine; uncontrolled seizures**
Other	ICU-acquired weakness; dialysis lines; femoral venous and arterial lines; drains, catheters, nasogastric tube	**Uncontrolled active bleeding; unstable pelvic, spinal or long-bone fracture; large open chest or abdominal wound (surgeon decides); femoral sheaths**

If a patient is near the limit in several areas at once (for example FiO_2 0.6, PEEP 10 and SpO_2 90%, like the W24 case), decide with an experienced ICU doctor before mobilising.

Stop the session if

- SpO_2 falls below about 88–90% and does not recover with rest
- Heart rate rises or falls sharply, a new arrhythmia appears, or MAP falls below target
- Respiratory rate stays above about 35, the patient fights the ventilator, or looks distressed
- The tube or a line is displaced, or the patient falls

4 ICU equipment (7 marks in W24)

Equipment	Purpose	What it means for physiotherapy
Mechanical ventilator	Breathes for or with the patient	Read settings; avoid disconnection when PEEP or FiO_2 is high
Endotracheal tube or tracheostomy	Artificial airway with a cuff	Secure it when turning; suction through it; speaking valve later with a tracheostomy
Bedside monitor	ECG, SpO_2, BP, respiratory rate, temperature, end-tidal CO_2	Check before, during and after treatment
Arterial line	Beat-by-beat BP; blood gas samples	Avoid kinking at the wrist; reading is wrong if the arm is moved or the transducer level changes
Central venous catheter	Drugs, fluids, central venous pressure	Neck or groin site; avoid pulling during turns
Intercostal drain	Drains air, blood or fluid from the pleura	Bottle below the chest; check swing and bubbling (details (p. 89))
Nasogastric tube	Feeding, drainage	Pause feeds before head-down positions; keep head up 30–45°
Urinary catheter	Measures urine output	Bag below the bladder when moving
Infusion pumps	Sedation, BP-support drugs, insulin, heparin	High doses of BP-support drugs limit mobilisation
Suction unit, oxygen and humidifier	Airway clearance, oxygen delivery, warm humid gas	Check suction pressure before use
Self-inflating bag (Ambu) and resuscitation trolley with defibrillator	Manual ventilation; emergencies	Must be at every ventilated bedside
Intracranial pressure monitor or external ventricular drain	Measures or drains CSF pressure	Clamp the drain as instructed before moving; avoid head-down positions
Intra-aortic balloon pump, ECMO, dialysis machine	Support for heart, lungs or kidneys	Femoral cannulas limit hip flexion; in-bed exercise only unless the team agrees
Pressure-relieving mattress, compression devices, tilt table, bed cycle	Prevent pressure injuries and DVT; early rehabilitation	Tools for the mobilisation plan

5 Physiotherapy for the ventilated patient (S17, S21, S26; goals W17)

Short-term goals for a patient on a ventilator (W17)

- Keep the airways clear and the lungs expanded; improve oxygenation.
- Prevent ventilator-associated pneumonia, collapse and aspiration.
- Keep joint range; prevent contractures, pressure injuries and DVT.
- Limit muscle wasting; start active exercise as soon as safe.
- Support weaning; keep the patient oriented and able to communicate.

Chest physiotherapy on SIMV (7 marks in S26)

1. **Check:** settings, vital signs, ABG, X-ray, sedation and safety criteria; time the session with pain relief and away from feeds.
2. **Positioning:** head of bed 30–45°; side-lying with the affected area uppermost for drainage; turns every 2–4 hours; upright sitting as soon as possible.

3. **Manual techniques** on expiration: chest shaking, vibrations, percussion where secretions are retained (avoid with fractures, low platelets, unstable patients).
4. **Hyperinflation** by bag (manual) or by the ventilator to re-open collapsed areas and move secretions, then **suction** (see the suction card (p. 231)).
5. **Breathing exercises** when awake on SIMV: deep breaths with a hold on the patient's own breaths; inspiratory muscle training during weaning.
6. **Limbs:** passive ROM of all joints when sedated; active-assisted and active exercise when awake; splints for ankles and hands where needed.
7. **Mobilisation ladder:** rolling → sitting up in bed → sitting over the edge → standing → marching on the spot → transfer to a chair → walking with the ventilator, as safety criteria allow.
8. **Reassess:** breath sounds, SpO_2, secretions, ventilator pressures; record.

Precautions on the ventilator

- High PEEP or FiO_2: avoid disconnecting the circuit; use closed suction and ventilator hyperinflation.
- Undrained pneumothorax: no hyperinflation or other positive-pressure techniques.
- Unstable BP or arrhythmia: limit to positioning and gentle treatment.
- Raised ICP: head up 30°, neutral neck, short treatments, no head-down tilt.
- Low platelets or clotting problems: no percussion; gentle suction.
- Recent lung surgery: no deep suction or high pressures near the bronchial stump.

6 Worked cases from the papers

GBS on SIMV (S23 Q4(b), 15 marks)

The question

"A 45 year old school teacher is admitted as GBS and is on mechanical ventilator SIMV mode with PEEP 6 and FiO_2 of 0.4. Discuss his impairments and ventilator management."

Read the case: Guillain–Barré syndrome is an immune attack on peripheral nerves causing rapidly rising, symmetrical flaccid weakness. He needs a ventilator because his breathing muscles are weak, not because his lungs are diseased, so his FiO_2 and PEEP are modest.

Impairment	Why
Weak breathing muscles, low vital capacity	Phrenic and intercostal nerve involvement; a vital capacity below 20 mL/kg is one sign of the need for ventilation (the "20/30/40" rule: VC below 20 mL/kg, inspiratory pressure weaker than −30, expiratory pressure below 40 cmH_2O)
Weak cough, retained secretions, risk of collapse and pneumonia	Weak expiratory muscles; lying still
Swallowing weakness, aspiration risk	Cranial nerve involvement
Unstable BP and heart rate	Autonomic nerve involvement; suction can cause sudden bradycardia
Flaccid paralysis, loss of reflexes, sensory loss, nerve pain	Demyelination of motor and sensory nerves
Risk of contractures (especially ankle plantarflexion), pressure injuries, DVT	Immobility

Impairment	Why
Anxiety and difficulty communicating	Awake but unable to move or speak with a tube

Activity and participation: fully dependent for bed mobility, self-care and communication; off work as a teacher.

Ventilator management, physiotherapy part:

- **Lungs:** 2–4-hourly turns, head up 30–45°, manual or ventilator hyperinflation with vibrations, suction when secretions are present (watch the heart rate; pre-oxygenate).
- **Track strength:** vital capacity and inspiratory pressure trends tell when weaning can start; a tracheostomy is common because recovery takes weeks.
- **Weaning:** as strength returns, gradually more spontaneous breathing (lower SIMV rate, pressure support, breathing trials); inspiratory muscle training in short, non-fatiguing sets; upright sitting for trials.
- **Limbs:** gentle passive ROM within pain, ankles at 90° with splints or pillows, hands in a functional position; then active-assisted and active exercise, avoiding fatigue (recovering muscles weaken further if overworked).
- **Other:** pressure care, DVT prevention, tilt table for low BP on sitting up, a communication board, reassurance (most patients recover over months).

ARDS on SIMV (S24 Q3(b), 15 marks; acute phase, S25 Q2(c), 7 marks)

ARDS (Berlin definition, 2012): within 1 week of a known insult (pneumonia, sepsis, trauma, aspiration); white shadows in both lungs on X-ray; not fully explained by heart failure or fluid overload; PaO_2/FiO_2 on a PEEP of 5 or more: **mild 201–300, moderate 101–200, severe 100 or less** (mmHg). A 2024 global definition also includes patients on high-flow oxygen and allows SpO_2/FiO_2 when blood gases are not available.

- **Impairments (S24):** poor oxygenation; stiff, low-volume lungs; possible secretions; inactive breathing muscles; reduced consciousness from sedation; limb stiffness and weakness; anxiety.
- **Likely complications:** lung injury from the ventilator, pneumothorax, ventilator-associated pneumonia, ICU-acquired weakness, contractures, pressure injuries (especially if nursed prone), DVT and pulmonary embolism, delirium, and later reduced exercise capacity, anxiety and memory problems (post-intensive care syndrome).
- **Medical point:** lung-protective ventilation (tidal volume 6 mL/kg predicted body weight, plateau pressure 30 or less), PEEP, careful fluids, and **prone positioning** for 16 hours or more a day when PaO_2/FiO_2 is below 150 (this lowered deaths in the PROSEVA trial).

Acute-phase physiotherapy (S25):

- **Minimal, clustered handling** while oxygenation is poor; agree the plan with the ICU team.
- **No disconnection:** closed suction only when secretions are present; ventilator hyperinflation rather than bagging; avoid high pressures (the lungs tear easily).
- **Positioning:** head up 30–45°; regular turns; help the team with safe prone turning (protect eyes, face, shoulders, knees and lines).
- **Limbs:** passive ROM and positioning against contractures and pressure injury.
- **As she improves (S24 settings, FiO_2 0.4 and PEEP 5, are within the green zone):** active exercise, sitting over the edge of the bed, standing and walking with the ventilator; inspiratory muscle training during weaning; after extubation, breathing exercises and a graded exercise programme, with follow-up rehabilitation.

Other ventilated cases

- W24: atypical pneumonia of both lungs on CMV, FiO_2 0.6, PEEP 10: worked plan on the pneumonia page (p. 198).
- S20: COPD exacerbation with right lower lobe pneumonia on SIMV: COPD page (p. 73).
- W17: electrician with burns and 3 rib fractures on a ventilator: burns page (p. 31) and chest-wall page (p. 211).

The other ICU conditions in the syllabus are on two pages: head injury, tetanus, spinal cord injury and neuromuscular disease (p. 245), and pulmonary oedema, shock, multiple organ failure, poisoning, aspiration and near

drowning (p. 252).

7 Early mobilisation: what the evidence says

Classic answer vs current evidence: how early, how much

Classic	Current	Write in the exam
Keep ventilated patients in bed; passive movements only until they come off the ventilator.	Mobilising ventilated patients is safe within the safety criteria. Starting physiotherapy and occupational therapy during daily sedation breaks led to more patients walking and independent at discharge (59% vs 35%) and less delirium (Schweickert, 2009). A large 2022 trial (TEAM) found that pushing **more** early active mobilisation than usual care did not improve survival or days at home at 6 months and caused more adverse events, so the dose must suit the patient.	"Early mobilisation as soon as safety criteria are met, progressed step by step, with monitoring before, during and after each session."

Mobilisation is one part of the ICU **ABCDEF bundle**: Assess and manage pain; Both spontaneous awakening and breathing trials; Choice of sedation; Delirium care; Early mobility and exercise; Family involvement.

8 Exam corner

Past questions (MUHS, 2016–2026)

S20 Q3(b), 5 + 3 + 7 marks

60-year-old farmer with a COPD exacerbation and respiratory failure, in ICU on SIMV, with right lower lobe pneumonia. Impairments, activity and participation; short- and long-term management.

S23 Q4(b), 15 marks

45-year-old teacher with GBS on SIMV, PEEP 6, FiO_2 0.4. Impairments and ventilator management (section 6).

S24 Q3(b), 15 marks

30-year-old housewife with ARDS, ventilated for 2 days: SIMV, PEEP 5, FiO_2 0.4. Impairments; likely complications; physiotherapy (section 6).

W24 Q4(a), 15 marks

35-year-old man with atypical pneumonia of both lungs on CMV, FiO_2 0.6, PEEP 10 for 3 days. Functional diagnosis and physiotherapy.

7-mark questions

Modes of ventilation and the role of physiotherapy in ICU (S17). Short-term goals for a patient on a ventilator (W17). Physiotherapy on a ventilator (S21). Equipment used in ICU (W24). Acute-phase physiotherapy in mechanically ventilated ARDS (S25). Chest physiotherapy for a patient on SIMV (S26).

Answer skeleton: any ventilated long case

1. **[Read the case, 1–2]** Diagnosis, why ventilated, what the settings mean (low, moderate or high support).
2. **[Impairments or ICF, 4–5]** Chest (gas exchange, secretions, lung volume), muscles and joints, circulation and skin, consciousness and communication; activity; participation; context.

3. **[Complications, if asked, 3]** Lung injury, VAP, ICU-acquired weakness, contractures, pressure injuries, DVT, delirium.
4. **[Physiotherapy, 7–8]** Assessment and safety check → positioning → airway clearance (manual techniques, hyperinflation, suction) → limbs → mobilisation ladder → weaning support → after extubation → precautions and stop criteria.

Test yourself

3 m **List five safety criteria that make a ventilated patient "green" for out-of-bed mobilisation.**

Answer
FiO_2 0.6 or less; SpO_2 90% or more; respiratory rate 30 or less; PEEP 10 cmH_2O or less; MAP above the lower target with no or low-dose support; calm, drowsy or restless (RASS –1 to +1); no active bleeding or unstable fracture.

3 m **Give the Berlin severity grades of ARDS.**

Answer
PaO_2/FiO_2 with PEEP 5 or more: mild 201–300, moderate 101–200, severe 100 or less (mmHg).

MCQ **A patient with GBS on SIMV becomes suddenly bradycardic during suction. The most likely reason is:**

A. Pneumothorax
B. Autonomic instability with a vagal reflex
C. Low PEEP
D. Fever

Answer
B. GBS affects autonomic nerves; tracheal stimulation triggers a vagal slowing of the heart. Pre-oxygenate, keep passes short, and stop if the heart rate falls.

MCQ **An MRC sum score below which value defines ICU-acquired weakness?**

A. 60
B. 48
C. 36
D. 24

Answer
B. Below 48 out of 60.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Hodgson CL, et al. Expert consensus and recommendations on safety criteria for active mobilization of mechanically ventilated critically ill adults. *Crit Care* 2014;18:658. Traffic-light tables read from the original figures.
3. ARDS Definition Task Force. Acute respiratory distress syndrome: the Berlin Definition. *JAMA* 2012;307:2526–33. Matthay MA, et al. A new global definition of acute respiratory distress syndrome. *Am J Respir Crit Care Med* 2024.
4. Guérin C, et al. Prone positioning in severe acute respiratory distress syndrome (PROSEVA). *N Engl J Med* 2013;368:2159–68.
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6. TEAM Study Investigators and ANZICS Clinical Trials Group. Early active mobilization during mechanical ventilation in the ICU. *N Engl J Med* 2022;387:1747–58.
7. De Jonghe B, et al. Paresis acquired in the intensive care unit: a prospective multicenter study. *JAMA* 2002 (MRC sum score below 48).

8. Lawn ND, et al. Anticipating mechanical ventilation in Guillain–Barré syndrome. *Arch Neurol* 2001 (the 20/30/40 rule).

9. Structure and classical content: Hough, *Physiotherapy in Respiratory and Cardiac Care*; Main & Denehy (2016); Frownfelter & Dean (2022).

Suctioning and manual hyperinflation (bagging)

> "Hazards of suctioning" is one of the most repeated 3-mark questions (W16, S17, S18, W19, S26), and suction with bagging was asked as 7 marks twice (S22, S24). Learn the indications, the numbers (15 seconds, –200 mmHg, half the tube width) and a clean list of hazards.

Syllabus	Ch 9(d) Suctioning and humidification; practical: airway clearance techniques, suctioning
Priority	**High yield.** In 7 of 20 papers (2016–2026).
Asked as	7m S22 · S24 · 3m W16 · S17 · S18 · W19 · S26
Usual split	3-mark hazards or complications; 7-mark procedure, indications and precautions with bagging
Checked against	AARC clinical practice guideline on artificial airway suctioning (2022); systematic review of manual hyperinflation (2012). Full list under Sources. Content checked 19 October 2026.

1 What it is

Suctioning removes secretions from the airway with a catheter attached to negative pressure, when the patient cannot clear them by coughing.

Type	Route	Notes
Endotracheal or tracheostomy, open	Disconnect the ventilator and pass a sterile catheter	Loses PEEP and oxygen during the pass
Closed (in-line)	Catheter inside a sealed sleeve, attached to the circuit	Keeps PEEP and FiO_2; less infection spread; preferred with high PEEP or FiO_2 and infectious patients
Nasopharyngeal or nasotracheal	Through the nose, in a patient without an artificial airway	Use a lubricated catheter or a nasopharyngeal airway; uncomfortable
Oropharyngeal	Mouth, with a Yankauer sucker	Clears mouth and throat secretions above the cuff

Shallow suction passes the catheter only to the length of the tube; **deep** suction goes until resistance is felt, then pulls back about 1 cm before suction. Use shallow suction routinely and deep suction only when shallow suction does not work (AARC 2022).

2 Indications

Suction only when there are signs of secretions, not on a fixed schedule:

- Secretions seen or heard in the tube; coarse crackles over the trachea and main bronchi.
- A saw-tooth pattern on the ventilator's flow graph; rising peak pressure (volume modes) or falling tidal volume (pressure modes).
- Falling SpO_2 or worsening blood gases that are likely due to secretions.
- A weak or absent cough after other airway clearance has moved secretions up; suspected aspiration.
- To take a sputum sample.

3 Contraindications and precautions

Suction has no absolute contraindication when an artificial airway is blocked. Take extra care in:

- **Raised intracranial pressure:** extra sedation as prescribed, short passes, few passes.
- **High PEEP or FiO_2, severe hypoxaemia:** closed suction, pre-oxygenate.
- **Bleeding risk:** low platelets or clotting problems, fresh haemoptysis: gentle, shallow suction.

- **Unstable heart rhythm or BP;** spinal cord injury or GBS (vagal bradycardia).
- **Bronchospasm** (suction can trigger it).
- **Recent lung resection:** no deep suction near the bronchial stump.
- **Nasal route not used** with a base-of-skull fracture or CSF leak, facial fractures, nasal bleeding or blocked nasal passages, and used with care in stridor (laryngospasm).

4 Procedure (endotracheal, open)

The three numbers (AARC 2022)

15 s / −200 mmHg / half the tube

- Each pass no longer than **15 seconds**.
- Suction pressure no stronger than **−200 mmHg** in adults (−120 mmHg in children and newborns).
- Catheter fills less than **half** of the tube's inner width in adults and children (less than 70% in newborns). For an 8.0 mm tube, a 12 French catheter.

1. **Assess and explain:** need for suction, vital signs, SpO_2; tell the patient (even if sedated) what you will do.
2. **Prepare:** hand hygiene, apron, eye protection, sterile glove on the dominant hand; check the suction pressure; choose the catheter size.
3. **Pre-oxygenate** if the patient drops their SpO_2 with suction (raise FiO_2 briefly, often for 30–60 seconds, using the ventilator's oxygen button).
4. **Insert** the catheter **without suction** to the planned depth (shallow: tube length).
5. **Withdraw with suction** applied, rolling the catheter between the fingers, in under 15 seconds.
6. **Reconnect** the ventilator at once; let SpO_2 and heart rate recover before another pass; usually no more than 2–3 passes.
7. **Oral suction** last, with a separate catheter or Yankauer (never go from the mouth into the tube).
8. **Reassess and record:** breath sounds, SpO_2, amount, colour and thickness of secretions, any bleeding, how the patient tolerated it.

- **No routine saline** down the tube: it does not thin secretions well and may raise infection risk. Thick secretions need better humidification and hydration.
- With a **closed system**, advance the catheter through the sleeve, suction while withdrawing, then flush the catheter.

5 Hazards and complications (3 marks in W16, S17, S18, W19, S26)

Memory hook: the big three first

Low oxygen, slow heart, high head pressure

Then trauma, spasm, collapse, infection.

Hazard	Why it happens	Prevention
Hypoxaemia	Oxygen is sucked out with the air; disconnection loses PEEP	Pre-oxygenate, short passes, closed suction with high PEEP
Bradycardia and other arrhythmias	Vagal stimulation of the trachea; low oxygen	Watch the monitor; stop if the heart rate falls
Raised intracranial pressure; BP swings	Coughing and distress	Sedation as prescribed; limit passes

Hazard	Why it happens	Prevention
Mucosal trauma and bleeding	Catheter hitting the airway wall; strong suction	Correct pressure; no suction on the way in; shallow suction
Atelectasis	Negative pressure sucks air out of alveoli	Small catheter, short passes, hyperinflation afterwards if needed
Bronchospasm	Airway irritation in reactive airways	Gentle technique; bronchodilator as prescribed
Infection	Bacteria carried into the lower airway	Sterile technique, closed system
Pain, anxiety, coughing fits	An unpleasant procedure	Explain; suction only when needed
Laryngospasm, nasal bleeding	Nasal route	Lubricate; nasopharyngeal airway
Cardiac arrest (rare)	Severe hypoxaemia and vagal reflex together	Stop at the first warning sign

6 Manual hyperinflation ("bagging") (7 marks in S22 and S24)

Manual hyperinflation (MHI) means disconnecting the ventilator and giving larger, slower breaths with a rebreathing bag, followed by a quick release. It imitates a cough.

Aims

- Re-open collapsed alveoli (air passes behind secretions through collateral channels).
- Move secretions towards the large airways: the quick release creates a fast expiratory flow.
- Improve lung compliance and oxygenation for a short time.

Technique

1. Two-person technique is common: one bags, one applies chest shaking or vibration.
2. A 2-litre bag with oxygen at about 15 L/min, a pressure manometer and a PEEP valve if the patient needs PEEP.
3. Patient in side-lying with the affected area uppermost (or supine).
4. **Slow, deep inspiration** (larger than the ventilator breath), an **inspiratory hold of about 2–3 seconds**, then a **quick release** of the bag so the patient breathes out fast; add vibrations or shaking during that expiration.
5. Keep the peak airway pressure at about **40 cmH_2O or less** on the manometer.
6. Repeat a few breaths in sets, suction when secretions reach the large airways, then reconnect and reassess.

Contraindications and precautions

- **Undrained pneumothorax** (absolute); bullae; recent lung surgery with a fresh bronchial stump.
- Unstable BP or arrhythmia (large breaths reduce venous return).
- High PEEP (above about 10) or high FiO_2: disconnection collapses the lung.
- Raised intracranial pressure not under control; severe bronchospasm; frank haemoptysis; pulmonary oedema.

Classic answer vs current evidence: bag or ventilator?

Classic	Current	Write in the exam
Bag every ventilated patient with secretions, by hand.	A systematic review (Paulus, 2012) found short-term improvements in oxygenation, compliance and secretion clearance, but no proof of better long-term outcomes. **Ventilator hyperinflation** (larger breaths set on the ventilator) works about as well and avoids disconnection, so it is preferred with high PEEP or FiO_2.	"Manual hyperinflation with a manometer, pressure kept at about 40 cmH_2O or less, with vibrations and suction; ventilator hyperinflation when PEEP or FiO_2 is high."

7 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Suctioning and bagging: indication, procedure and precautions (S22). Suctioning and manual hyperinflation: procedure, indication and precaution (S24).

3-mark questions

Hazards of suctioning (W16, S18, S26). Any six complications of suctioning (S17). Complications with suctioning (W19).

Answer skeleton: suctioning and bagging, 7 marks

1. **[1]** Definitions: suction, open and closed, shallow and deep; manual hyperinflation.
2. **[1.5]** Indications for each.
3. **[3]** Procedure: suction steps with the three numbers; MHI technique (slow breath, hold, quick release, 40 cmH_2O).
4. **[1.5]** Precautions and contraindications; hazards.

Test yourself

3 m **List six hazards of suctioning.**

Answer
Hypoxaemia; bradycardia and arrhythmias; raised intracranial pressure; mucosal trauma and bleeding; atelectasis; bronchospasm; infection; pain and anxiety (any six).

MCQ **For an adult with an 8.0 mm endotracheal tube, the most suitable suction setting is:**

A. 18 Fr catheter, −300 mmHg, 30 s

B. 12 Fr catheter, −150 mmHg, under 15 s

C. 14 Fr catheter, −250 mmHg, 20 s

D. 8 Fr catheter, −50 mmHg, 60 s

Answer
B. The catheter fills no more than half the tube, the pressure is within −200 mmHg, and the pass is under 15 seconds.

MCQ **Manual hyperinflation is absolutely contraindicated in:**

A. Right lower lobe collapse

B. Undrained pneumothorax

C. Thick secretions

D. Pneumonia

Answer

B. Positive pressure can turn it into a tension pneumothorax.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Blakeman TC, et al. AARC Clinical Practice Guidelines: Artificial Airway Suctioning. *Respir Care* 2022;67(2):258–71. aarc.org
3. Paulus F, et al. Benefits and risks of manual hyperinflation in intubated and mechanically ventilated intensive care unit patients: a systematic review. *Crit Care* 2012;16:R145.
4. Structure and classical content: Hough, *Physiotherapy in Respiratory and Cardiac Care*; Pryor & Prasad (2008); Main & Denehy (2016).

Therapeutic positioning, ventilation–perfusion and the zones of the lung

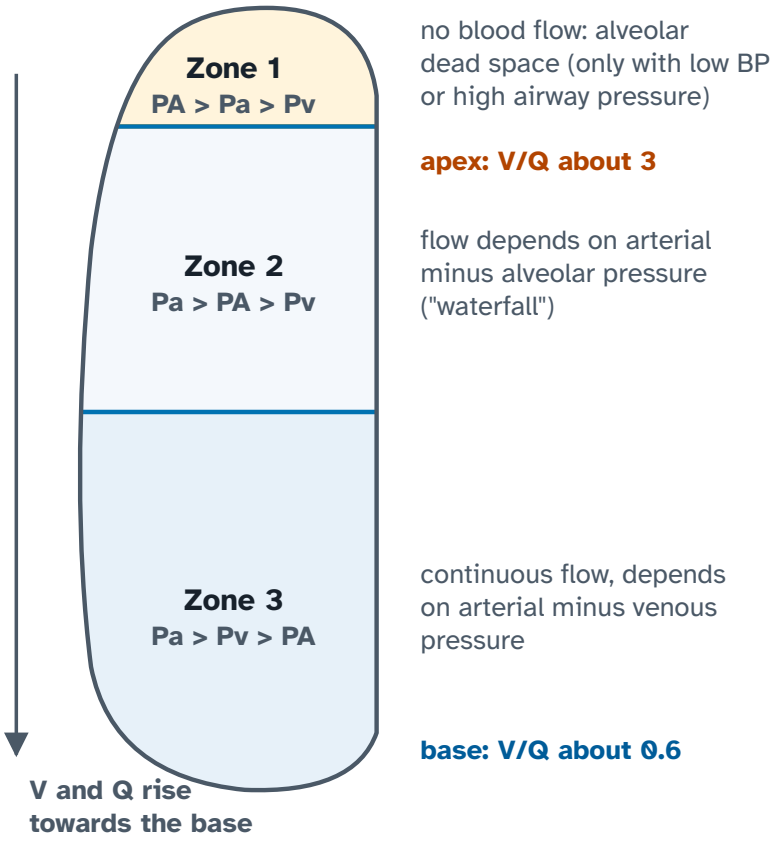
> Positioning is the cheapest treatment in the ICU and the papers keep asking about it: positioning in ICU (W22, W25), sitting upright (W21), zones of the lung (S25), causes of altered V/Q (S23) and positions that ease breathlessness (S22). One idea runs through all of them: gravity decides where air and blood go.

Syllabus	Ch 4(c) Therapeutic positioning to improve ventilation and perfusion matching; 4(d) positioning to alleviate dyspnoea
Priority	**High yield.** In 6 of 20 papers (2016–2026), 3 of them as 7 marks.
Asked as	7m W21 · W22 · W25 · 3m S22 · S23 · S25
Usual split	7-mark positioning in ICU or sitting upright; 3-mark zones, V/Q causes, positions for dyspnoea
Checked against	West's zones and regional V/Q (standard respiratory physiology); prone positioning trial (2013); positioning evidence in adults and infants. Full list under Sources. Content checked 19 October 2026.

1 Ventilation and perfusion in the upright lung

- **Ventilation (V)** is the air reaching the alveoli (about 4 L/min at rest); **perfusion (Q)** is the blood flowing past them (about 5 L/min). The overall V/Q ratio is therefore about **0.8**.
- In upright lungs both ventilation and perfusion are greater at the **base** than the apex, but perfusion rises much more steeply, because gravity pulls blood down.
- So V/Q is **high at the apex (about 3)** and **low at the base (about 0.6)**.
- Basal alveoli get more ventilation because they start smaller (the pleural pressure there is less negative), so they are on the steep, easily filled part of the pressure–volume curve.

West's zones of the upright lung (3 marks in S25)



PA = alveolar pressure, Pa = pulmonary arterial pressure, Pv = pulmonary venous pressure. In a healthy upright person there is normally no zone 1; it appears when blood pressure falls or airway pressure rises (for example high PEEP).

? If the question means the airway zones instead

> Some books describe the **conducting zone** (trachea to terminal bronchioles, airway generations 0–16: no gas exchange; anatomical dead space about 150 mL in an adult) and the **respiratory zone** (respiratory bronchioles, alveolar ducts and alveolar sacs, generations 17–23: gas exchange). Write both if the wording is unclear.

2 Causes of altered V/Q (3 marks in S23)

Type	Meaning	Causes
Low V/Q (shunt when no ventilation)	Blood passes alveoli that get too little air	Secretions and mucus plugs; bronchospasm; collapse (atelectasis); consolidation (pneumonia); pulmonary oedema; ARDS; lying flat, obesity, pain-limited breathing after surgery
High V/Q (dead space when no perfusion)	Air reaches alveoli that get too little blood	Pulmonary embolism; low cardiac output or blood loss; emphysema (capillaries destroyed); high PEEP or over-stretched alveoli; pulmonary vascular disease

Both lower PaO_2. Low V/Q improves with oxygen and with positions that send air to the affected area; a true shunt (no ventilation at all) barely responds to oxygen until the alveoli are re-opened.

3 Effects of body positions

Position	Physiological effect	Used for
Upright sitting or standing	Largest lung volume (FRC); diaphragm moves best; least airway closure at the bases	Almost everyone: post-op, ICU, weaning trials, breathlessness

Position	Physiological effect	Used for
Supine	Abdominal contents push the diaphragm up; FRC falls; the back of the lungs collapses; aspiration risk rises	Avoid long periods; exception: tetraplegia, where vital capacity is often better lying than sitting (use an abdominal binder in sitting)
Side-lying, breathing spontaneously (adult)	The lower (dependent) lung gets more ventilation and more perfusion	**Good lung down** to improve oxygenation in one-sided lung disease; affected lung up to drain it
Side-lying, on a controlled ventilator	Air goes more to the upper lung, blood to the lower lung, so matching can worsen	Watch SpO_2 after every turn
Side-lying, infant	Ventilation goes preferentially to the **upper** lung (soft chest wall)	In babies with one-sided disease, put the good lung **up**
Prone	More even ventilation; re-opens the large back (dorsal) areas of lung; better V/Q	Severe ARDS (PaO_2/FiO_2 below 150), 16 hours or more a day: lowered deaths in the PROSEVA trial
Head-down tilt	Gravity drains the lower lobes; but increases work of breathing, reflux and intracranial pressure	Postural drainage only when no contraindication
Head-up 30–45°	Less reflux and aspiration; better FRC than flat	All ventilated patients (prevents ventilator-associated pneumonia); raised ICP
Forward-lean sitting or standing	Abdomen pushes a flattened diaphragm up into a better shape; arms fixed so accessory muscles help	Breathlessness in COPD

Exceptions to "good lung down": a lung abscess or bleeding in the affected lung (pus or blood could spill into the good lung below) and infants.

4 Sitting upright as a therapeutic position (7 marks in W21)

System	Physiological application
Respiratory	Diaphragm descends and abdominal contents fall away: larger FRC and tidal volume; fewer collapsed basal alveoli; better V/Q and oxygenation; lower work of breathing; a deeper breath before a cough, so a stronger cough; less reflux and aspiration
Cardiovascular	Keeps the blood pressure reflexes working (prevents the fall in BP on standing that follows bed rest); with legs down it lowers venous return, which eases breathlessness in pulmonary oedema and heart failure (orthopnoea)
Neurological and other	Wakefulness, orientation, communication and safer swallowing; head-up lowers intracranial pressure; first step towards standing and walking; relieves pressure on the back

Used in: post-operative patients, ICU and ventilated patients (including breathing trials), COPD and asthma (with forward lean), pulmonary oedema, neurological patients, older people on bed rest. Watch for sliding into a slumped position, which squashes the abdomen and loads the sacrum.

5 Positioning in the ICU (7 marks in W22 and W25)

Physiological effects and benefits (W25)

- Improves V/Q matching and oxygenation; increases lung volume; re-opens collapsed areas.
- Drains secretions with gravity; lowers the work of breathing.
- Head-up positions reduce aspiration and ventilator-associated pneumonia.
- Regular turns prevent pressure injuries, contractures and nerve compression.
- Upright positions keep the blood pressure reflexes and muscles working; they prepare for mobilisation and raise alertness.
- Comfort, communication and dignity.

Examples (W22)

Situation	Position	Reason
Right lower lobe pneumonia, breathing spontaneously	Left side-lying (good lung down)	Better oxygenation; the right lung is uppermost and drains
Left lower lobe collapse	Right side-lying, slightly turned, foot of bed raised if allowed	Air and gravity drainage to the collapsed lobe
Severe ARDS	Prone, 16 hours or more a day	Re-opens the back of the lungs; better oxygenation and survival
Head injury with raised ICP	Head up 30°, neck straight, no tight ties, no head-down tilt	Venous drainage from the brain lowers ICP
Every ventilated patient	Head of bed 30–45°	Less aspiration and pneumonia
Pulmonary oedema	High sitting with legs down	Less venous return; easier breathing
Abdominal distension or obesity	Sitting or reverse Trendelenburg (whole bed tilted head-up)	Lets the diaphragm descend
Tetraplegia	Lying or semi-reclined when breathless; binder when sitting	Abdominal contents support the diaphragm
Pressure injury risk	30° tilted side-lying; heels floated; turns on a schedule	Off-loads sacrum, hips and heels
Burns, hemiplegia, unconscious patients	Anti-deformity and tone-reducing positions	Prevents contractures

Before and after every turn

Check SpO_2, heart rate and BP; secure the tube, lines and drains; give the patient time to settle (oxygenation can dip for several minutes after a turn).

6 Positions to ease breathlessness (3 marks in S22)

- **High side-lying** with pillows, top arm supported.
- **Forward-lean sitting** with forearms on the knees or on pillows on a table.
- **Relaxed sitting** back in a chair with forearms supported.
- **Forward-lean standing** with forearms on a wall, rail or trolley; **leaning back** against a wall with feet slightly apart.

Rationale: in forward lean, the abdominal contents push a flat diaphragm up into a dome, so it contracts more effectively; fixing the arms lets pectoral and other accessory muscles help breathing; relaxed, supported positions lower muscle work and anxiety. Details on the dyspnoea card (p. 285).

7 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Physiological application of sitting upright as a therapeutic position (W21). Positioning of the patient in ICU with examples (W22). Physiological effects and benefits of body positioning in ICU (W25).

3-mark questions

Rationale of therapeutic positioning to alleviate dyspnoea (S22). Three causes of altered ventilation–perfusion ratio (S23). Physiological zones of ventilation (S25).

Answer skeleton: positioning in ICU, 7 marks

1. **[1]** Definition: therapeutic positioning uses gravity to improve breathing, circulation and comfort, and to prevent complications.
2. **[2]** Physiology: V/Q gradient and zones (draw the zones), good lung down, prone, head-up.
3. **[3]** Examples table: pneumonia, collapse, ARDS, raised ICP, VAP prevention, pressure care.
4. **[1]** Precautions: monitor before and after turns; secure tubes and lines.

Test yourself

3 m **Write the pressure relationships of West's three zones.**

Answer
Zone 1: alveolar > arterial > venous (no flow). Zone 2: arterial > alveolar > venous (flow depends on arterial minus alveolar). Zone 3: arterial > venous > alveolar (continuous flow).

MCQ **An adult with right lower lobe pneumonia, breathing on his own, has SpO_2 89%. Which position will most likely raise his SpO_2?**

A. Right side-lying

B. Left side-lying

C. Flat supine

D. Head-down tilt

Answer
B. Good lung down: the healthy left lung below gets more ventilation and more blood.

MCQ **The V/Q ratio at the apex of the upright lung is about:**

A. 0.6

B. 0.8

C. 1.0

D. 3

Answer
D. Little blood reaches the apex, so ventilation exceeds perfusion.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. West JB, Luks AM. *West's Respiratory Physiology: The Essentials* (zones, regional V/Q).
3. Guérin C, et al. Prone positioning in severe acute respiratory distress syndrome (PROSEVA). *N Engl J Med* 2013;368:2159–68.
4. Heaf DP, et al. Postural effects on gas exchange in infants. *N Engl J Med* 1983;308:1505–8.
5. Structure and classical content: Frownfelter & Dean (2022), chapters on positioning and mobilisation; Hough, *Physiotherapy in Respiratory and Cardiac Care*; Main & Denehy (2016).

Effects of bed rest and immobilisation, and early mobilisation

> Bed rest was asked five times: as 3 marks (one effect on three systems, S19 and W21) and as 7 marks (complications of bed immobility in ICU, W17; complications on different systems, S24; effects on the cardiovascular and pulmonary systems, S26). Learn it system by system, with the mechanism and the prevention for each.

Syllabus	Ch 3(d) Complications of bed rest and immobilisation, and prevention
Priority	**High yield.** In 5 of 20 papers (2016–2026), 3 of them as 7 marks.
Asked as	7m W17 · S24 · S26 · 3m S19 · W21
Usual split	3-mark list (one effect per system); 7-mark system-wise effects with prevention
Checked against	Dallas bed rest study (1968); muscle wasting in critical illness (2013); ICU-acquired weakness definition (2002); early mobilisation trials. Full list under Sources. Content checked 19 October 2026.

1 Why bed rest harms

The body adapts to the load it carries. Lying flat removes two loads at once: **gravity along the body** (which keeps blood-pressure reflexes and fluid balance trained) and **muscle work against gravity** (which keeps muscle, bone and fitness). Within days every system starts to lose what it does not use. Older and critically ill patients lose it fastest.

Two classic numbers

3 weeks: fitness down about a quarter

In the Dallas bed rest study (1968), 3 weeks of bed rest cut young men's maximal oxygen uptake by about 27%. In critically ill patients the thigh muscle (rectus femoris) shrank by about 12.5% in the first week (Puthucheary, 2013).

2 Cardiovascular and pulmonary effects (7 marks in S26)

Effect	Mechanism	Prevention
Higher resting heart rate, smaller stroke volume	Blood volume falls and the heart is less filled; the heart adapts to less work	Sit out and stand early; active exercise
Orthostatic (postural) hypotension	Lying flat shifts fluid to the chest; the body passes more urine and plasma volume falls; blood-pressure reflexes are not used	Sit up in stages; dangle legs; tilt table; ankle pumps before standing; compression stockings as prescribed
Lower aerobic fitness (VO_2max)	Smaller stroke volume and cardiac output; muscles lose capillaries and mitochondria	Bed cycling, walking, graded endurance exercise
Deep vein thrombosis and pulmonary embolism	Stasis (no calf pump), plus a sticky blood state after illness or surgery	Ankle pumps 10 every hour, early walking, compression devices, anticoagulants as prescribed
Lower lung volumes, collapse at the bases	Abdominal contents push the diaphragm up when lying flat; shallow breathing; small airways close	Upright positions, deep breathing with a hold, turning, walking
Retained secretions and pneumonia	Secretions pool in dependent areas; weaker cough; slower mucus clearance; aspiration when flat	Turn 2–4-hourly, head up 30–45°, airway clearance, huffing and coughing

Effect	Mechanism	Prevention
Weaker breathing muscles	Disuse; on a ventilator the diaphragm itself wastes	Spontaneous breathing as soon as possible; inspiratory muscle training

3 Effects on other systems (S24; W17 in ICU)

System	Effects	Prevention
Muscles	Wasting and weakness, mostly in antigravity muscles (quadriceps, glutes, calf, back); lower endurance. A classic estimate is a strength loss of about 1–1.5% a day of strict bed rest; critical illness is faster	Active and resisted exercise, functional activity, electrical muscle stimulation in sedated patients
Joints and soft tissue	Contractures (ankle plantarflexion, hip and knee flexion, shoulder), stiffness	Daily ROM, positioning, splints
Bone	Bone loss (calcium leaves bone within days), raised calcium in urine	Weight-bearing (standing, tilt table) as early as possible
Skin	Pressure injuries over the sacrum, heels, hips and back of the head, within hours in high-risk patients	Turning schedule, pressure-relieving mattress, skin checks, nutrition
Metabolic and endocrine	Insulin resistance, protein loss (negative nitrogen balance), lower metabolic rate	Exercise; nutrition support
Kidney and bladder	Urinary stasis, infection and stones (with high urine calcium); retention	Upright positions, fluids, early toileting
Gut	Constipation, poor appetite, reflux	Mobility, fluids, fibre, sitting up for meals
Nervous system and mind	Poor balance and coordination; delirium and confusion; low mood, anxiety; poor sleep; sensory deprivation	Day–night routine, orientation, family visits, sitting out, early walking

ICU-acquired weakness (W17: complications of bed immobility in ICU)

Generalised, symmetrical limb weakness that develops during critical illness with no other cause. Diagnosed at the bedside with the **MRC sum score** (6 muscle groups each side, 0–5 each, total 60): **below 48 = ICU-acquired weakness**. Risk factors: sepsis, multi-organ failure, long ventilation, immobility, high blood sugar, steroids and paralysing drugs. It slows weaning and can affect walking and work for months, so prevention by early mobilisation matters.

4 Prevention: early mobilisation

1. **Assess safety** first (see the ICU safety criteria (p. 224)).
2. **Positioning:** turns every 2–4 hours, head up, upright sitting in bed.
3. **Range of motion:** passive when sedated, then active-assisted and active.
4. **In-bed exercise:** bed cycling, bridging, resisted limb exercise.
5. **Sitting over the edge of the bed**, then standing (tilt table if BP falls or the patient is weak).
6. **Transfer** to a chair; **marching on the spot**; **walking**, with the ventilator if needed.
7. **Breathing:** deep breathing, airway clearance, inspiratory muscle training.
8. **Monitor** before, during and after: HR, BP, SpO_2, respiratory rate, symptoms.

Classic answer vs current evidence: rest after illness

Classic	Current	Write in the exam
Strict bed rest for days or weeks after a heart attack, surgery or serious illness.	Bed rest is a treatment with side effects, used only when needed and for as short a time as possible. Early mobilisation is safe within criteria and shortens delirium and recovery; the amount is matched to the patient.	"Bed rest affects every system within days; prevention is early, graded mobilisation with positioning, breathing exercises and ROM."

5 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Complications of bed immobility in an ICU patient (W17). Complications of bed rest on different physiological systems (S24). Adverse effects of bed rest on the cardiovascular and pulmonary systems (S26).

3-mark questions

One effect of bed rest on any three systems (S19). Effects of long-term bed rest on any three systems (W21).

Memory hook for a 3-mark answer: one effect each

Heart, lungs, legs

Heart: postural hypotension. Lungs: basal collapse and pneumonia. Legs: muscle wasting and DVT. Add skin (pressure sores) or bone (osteoporosis) if asked for more.

Answer skeleton: 7 marks

1. **[1]** One-line principle: unloading and lying flat cause every system to adapt downwards within days.
2. **[4]** Table: system, effect, mechanism (cardiovascular and pulmonary in detail for S26).
3. **[2]** Prevention: positioning, breathing exercises, ROM, early graded mobilisation, DVT and pressure care.

Test yourself

3 m **Why does a patient feel faint when first sat up after 10 days in bed?**

Answer

Plasma volume has fallen (fluid shifts to the chest when lying flat and is passed as urine), the heart is less filled, and blood-pressure reflexes are out of practice. On sitting, blood pools in the legs and BP falls: postural hypotension. Sit up in stages, use ankle pumps and a tilt table.

MCQ **The MRC sum score that defines ICU-acquired weakness is:**

A. Below 60

B. Below 48

C. Below 30

D. Below 12

Answer

B. Below 48 out of 60.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
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5. Dittmer DK, Teasell R. Complications of immobilization and bed rest. *Can Fam Physician* 1993;39:1428–37.
6. Structure and classical content: Frownfelter & Dean (2022); Hough, *Physiotherapy in Respiratory and Cardiac Care*; Kisner & Colby, *Therapeutic Exercise*.

ICU conditions: head injury, tetanus, spinal cord injury and neuromuscular disease

> These ICU conditions are in the syllabus (Ch 9e) but were not asked on their own in the 20 papers from 2016 to 2026. Ventilated patients now appear in every paper, so a head injury or spinal cord injury case could come next. Each condition has seven parts: what it is, what you see in the ICU, the medical point that matters, ICF, physiotherapy by problem, precautions and exam points.

Syllabus	Ch 9(e) Therapeutic intervention in tetanus, head injury, neuromuscular disease, Guillain–Barré syndrome, spinal cord injury and other acute respiratory disorders
Priority	**Syllabus.** Not asked on its own in 2016–2026; ICU cases are rising.
Asked as	Not asked 2016–2026
Usual split	Possible 7-mark notes or a 15-mark ventilated case
Checked against	Brain Trauma Foundation guidelines (4th edition, 2016); ICU mobilisation safety consensus (2014); study of chest physiotherapy and ICP (2013); tetanus magnesium trial (2006); Cochrane review of respiratory muscle training in tetraplegia (2013); NIV trial in motor neurone disease (2006). Full list under Sources. Content checked 19 October 2026.

1 Traumatic head injury

What it is

Brain injury from an external force (road accidents, falls, assault). **Primary injury** happens at impact (contusion, diffuse axonal injury, extradural or subdural haematoma). **Secondary injury** follows over hours and days from low oxygen, low BP, raised intracranial pressure (ICP), seizures, fever and high or low blood sugar. The whole ICU plan, including physiotherapy, is built to prevent secondary injury.

Severity by Glasgow Coma Scale (GCS, 3–15)

Severity	GCS
Mild	13–15
Moderate	9–12
Severe	3–8 (usually intubated to protect the airway: "GCS 8, intubate")

What you see in the ICU

- Reduced consciousness; ventilated and sedated; ICP monitor or external ventricular drain; sometimes part of the skull removed (decompressive craniectomy).
- Abnormal posturing: **decorticate** (arms flexed, legs extended) or **decerebrate** (arms and legs extended); changing tone; seizures.
- Episodes of fast heart rate, high BP, sweating, fever and posturing (paroxysmal sympathetic hyperactivity).
- Chest problems: aspiration at the time of injury, ventilator-associated pneumonia, collapse; agitation and confusion as the patient wakes.

Medical point that matters for physio

- ICP is treated when it rises above about **22 mmHg**; cerebral perfusion pressure (MAP – ICP) is kept at about **60–70 mmHg** (Brain Trauma Foundation, 2016).
- Avoid low oxygen and low BP; keep CO_2 normal (high CO_2 widens brain vessels and raises ICP); head up 30°; sedation; drugs such as mannitol or hypertonic saline; surgery for haematomas.

ICF summary

Impairments: reduced consciousness, retained secretions, abnormal tone and posture, weakness, contracture and pressure-sore risk, later cognitive and behaviour changes. Activity: fully dependent. Participation: work, family and study roles, often for months. Context: young adults; family support and access to long-term rehab matter.

Physiotherapy by problem

- **Chest:** head up 30° with the neck straight; side-lying turns as ICP allows; chest vibrations and compressions on expiration for secretions (in a study of sedated head-injury patients they did not change ICP or cerebral perfusion pressure); suction only when needed, after sedation and pre-oxygenation, with short passes. Suction raised ICP (about 20 to 26 mmHg) but it settled within about 10 minutes, and perfusion pressure was kept.
- **Tone and contractures:** positioning out of the abnormal posture (for example side-lying with flexed hips and knees for extensor tone), ankle splints, gentle passive range of motion; serial casting later for fixed plantarflexion.
- **Mobilisation:** once ICP is stable and not being actively treated: sitting up in bed, sitting over the edge, tilt table, standing; watch BP and ICP.
- **As the patient wakes:** calm, consistent handling for agitation; orientation; family involvement; then neurological rehabilitation (see Neurophysiotherapy).

Precautions

- ICP above target or being actively treated: no active mobilisation; minimal, clustered handling.
- No head-down tilt; avoid neck flexion and rotation, tight tube ties and high-pressure bagging (all raise ICP).
- Craniectomy: no pressure on the side without bone; helmet when mobilising.
- External ventricular drain: clamp or level it as the team instructs before moving the patient.
- Spine not yet cleared: log-roll and spinal precautions; check for other fractures.

Exam points

- GCS grades; ICP 22 and perfusion pressure 60–70; head up 30°; suction raises ICP briefly, so sedate and keep it short; prevent contractures from day one.

2 Tetanus

What it is

An illness caused by the toxin of *Clostridium tetani*, whose spores enter through wounds (puncture wounds, burns, injections, unclean delivery or cord care in newborns). The toxin travels up the nerves and blocks the **inhibitory** nerve cells in the spinal cord and brainstem, so muscles contract without control and the autonomic nervous system overacts. Symptoms usually start 3–21 days after the injury. It is prevented by vaccination; India was declared free of maternal and neonatal tetanus in 2015, but adult cases still occur after injuries.

What you see in the ICU

- **Lockjaw (trismus)**, a fixed grin (risus sardonicus), stiff neck, back and abdomen; arching of the back (opisthotonus).
- **Painful spasms** set off by touch, noise, light or moving the patient; the patient stays fully conscious, which is very frightening.
- Spasm of the larynx and a rigid chest wall cause respiratory failure; difficulty swallowing and aspiration.
- Severe cases: unstable BP, fast or irregular heart rate, sweating, fever (autonomic instability).

Medical point that matters for physio

- Wound cleaning, human tetanus immunoglobulin, metronidazole; sedation (benzodiazepines); magnesium sulphate reduces spasms and autonomic swings (trial, 2006); paralysing drugs with ventilation and an early tracheostomy in severe cases.

- A dark, quiet room. The illness lasts several weeks. The patient needs vaccination after recovery, because having tetanus does not give immunity.

ICF summary

Impairments: rigidity and spasms, respiratory failure, retained secretions, aspiration, pain, autonomic instability, later weakness and deconditioning. Activity: fully dependent. Participation: off work for weeks to months. Context: often a manual worker with an untreated wound; fear and anxiety.

Physiotherapy by problem

- **Plan with the team:** treat when sedation is adequate (often just after a dose); keep sessions short; no sudden touch or noise.
- **Chest:** positioning and regular turns, gentle chest vibrations and shaking, suction with sedation; humidification for a tracheostomy.
- **Limbs:** positioning and pressure care; slow, gentle passive movement only when spasms are controlled.
- **Recovery phase:** as sedation and spasms settle: active exercise, jaw-opening exercises, sitting, standing, walking; build strength and endurance after weeks in bed; reassurance.

Precautions

- Any stimulus can trigger spasms and laryngospasm: warn the patient, use firm, slow handling, and stop if spasms start.
- Watch BP and heart rate (autonomic swings); avoid fatigue.
- Strong spasms can fracture vertebrae: report new back pain.

Exam points

- Toxin blocks inhibition; trismus, risus sardonicus, opisthotonus; spasms triggered by stimuli; dark, quiet room; coordinate physiotherapy with sedation.

3 Acute spinal cord injury: the respiratory picture

What it is

Damage to the spinal cord, usually from trauma (falls from trees and roofs, road accidents, diving). The higher the level, the more breathing muscles are lost.

Level	Breathing muscles lost	Effect
C1–C2	Diaphragm and all others	Needs a ventilator (or phrenic pacing) for life
C3–C5	Diaphragm partly (phrenic nerve C3, 4, 5), intercostals, abdominals	Often ventilated at first; some wean as the diaphragm recovers
C6–C8	Intercostals and abdominals; diaphragm works	Low vital capacity, very weak cough, paradoxical upper-chest movement
T1–T12	Some intercostals and abdominals (abdominals T6–T12)	Weak cough; the higher the level, the weaker

Memory hook

C3, 4, 5 keeps the diaphragm alive

What you see in the ICU

- **Spinal shock:** flaccid paralysis and no reflexes below the level for days to weeks.
- **Neurogenic shock** (injuries at T6 and above): low BP with a slow heart rate.

- More bronchial secretions (unopposed vagal activity), a weak cough, collapse and pneumonia (the commonest early complications); gut distension that pushes the diaphragm up.
- Vital capacity falls sharply after a cervical injury and often improves over the following weeks to months as spinal shock settles.

Medical point that matters for physio

- The spine is stabilised (collar, traction or surgery); until then strict spinal precautions and log-rolling.
- BP support (a mean arterial pressure of about 85–90 mmHg is often targeted for the first week); ventilation and early tracheostomy in high injuries; DVT prophylaxis.

ICF summary

Impairments: paralysis and sensory loss, low vital capacity, weak cough, secretion retention, low BP, risk of pressure sores, contractures and DVT. Activity: dependent for all mobility and self-care. Participation: work, family and social life for life; major adjustment. Context: home access, carers, funding for equipment.

Physiotherapy by problem

- **Weak cough:** manually **assisted cough** (an upward and inward push on the abdomen or lower ribs timed with the cough), cough assist (mechanical insufflation–exsufflation), suction.
- **Low lung volume:** diaphragmatic breathing; lying flat or semi-reclined when breathless (the abdominal contents support the diaphragm, so vital capacity is often better lying than sitting); an abdominal binder when sitting up; respiratory muscle training (it improves vital capacity and respiratory muscle strength in tetraplegia, Cochrane 2013); manual or ventilator hyperinflation if ventilated.
- **Secretions:** positioning and turns (log-roll), chest vibrations and shaking, assisted cough.
- **Limbs and skin:** passive range of motion within spinal limits, positioning and splints, 2–3-hourly turns, pressure care.
- **Mobilisation** once the spine is stable: sitting up in stages, tilt table for low BP, then wheelchair and rehabilitation.

Precautions

- Spine not yet stable: no active mobilisation; log-roll only.
- Suction or turning can cause severe bradycardia (atropine at hand).
- **Autonomic dysreflexia** (injuries at T6 and above, after spinal shock): sudden very high BP, pounding headache, flushing and sweating above the level, slow pulse, usually triggered by a full bladder or bowel. Sit the patient up, loosen tight clothing, check the catheter and bowel, call for help.
- Postural hypotension when sitting up; poor temperature control; no assisted cough over a fresh abdominal wound or ileus.

Exam points

- Level decides breathing; C3, 4, 5; assisted cough; better vital capacity lying than sitting; abdominal binder; autonomic dysreflexia at T6 and above.

4 Neuromuscular disease in the ICU

What it is

Weakness of the breathing and swallowing muscles from disease of the nerves, the neuromuscular junction or the muscles. Guillain–Barré syndrome is worked on the ICU physiotherapy card (p. 226); the others in this section are:

- **Myasthenic crisis:** myasthenia gravis (antibodies block acetylcholine receptors) worsening until breathing fails, often triggered by infection, surgery or some drugs.
- **Motor neurone disease (ALS):** progressive loss of motor neurones; breathing failure is the usual cause of death.

- **Muscular dystrophies** such as Duchenne: progressive muscle weakness, scoliosis, chest infections and night-time hypoventilation in the teens and twenties.
- **Critical illness polyneuropathy and myopathy:** weakness that develops during critical illness itself (ICU-acquired weakness; see the bed rest card (p. 242)).
- **Neurotoxic snakebite** (krait, cobra): see the poisoning section (p. 254).

Warning signs of breathing failure

- Falling vital capacity (below about 20 mL/kg is a warning sign), a weak cough, choking on food or saliva, a nasal voice (bulbar weakness).
- Rising breathing rate, accessory muscle use, the abdomen moving in on inspiration (diaphragm weakness), breathlessness lying flat.
- Morning headaches, daytime sleepiness and poor sleep (hypoventilation at night).
- A rising $PaCO_2$ is a **late** sign; do not wait for it.
- **Peak cough flow:** below about **270 L/min** the cough may fail during a chest infection, so teach cough augmentation; below about **160 L/min** the cough is ineffective.

Medical point that matters for physio

- **Myasthenic crisis:** ventilation, IV immunoglobulin or plasma exchange, steroids. Too much anticholinesterase drug can cause a cholinergic crisis (weakness with excess secretions).
- **Motor neurone disease:** non-invasive ventilation improves survival and quality of life in patients without severe bulbar weakness (trial, 2006); decisions about invasive ventilation are planned in advance with the patient.
- **Muscular dystrophy:** night-time NIV, cough assist, scoliosis surgery, heart monitoring.

ICF summary

Impairments: weak breathing and cough muscles, retained secretions, swallowing weakness and aspiration risk, limb weakness, fatigue, contracture risk. Activity: mobility and self-care limited. Participation: depends on the disease course; work and family roles. Context: equipment (NIV, cough assist) at home, carers, advance care planning.

Physiotherapy by problem

- **Weak cough:** manually assisted cough, cough assist, **breath stacking** (several breaths in without breathing out, with a bag and one-way valve, to reach a larger volume before coughing), suction when needed.
- **Low lung volume and night hypoventilation:** support NIV (mask fit, timing); positioning; breath stacking for lung volume.
- **Secretions and aspiration:** positioning, airway clearance; swallowing assessment by the speech and language therapist.
- **Limbs:** range of motion and positioning against contractures; gentle, non-fatiguing strengthening in slowly progressive disease; energy conservation.
- **Myasthenia:** short sessions with rests, timed when the drug is working (often about an hour after a pyridostigmine dose); stop at the first sign of fatigue.

Precautions

- Avoid fatigue: overworked weak muscles get weaker.
- Watch for the warning signs above and report early; aspiration risk with weak swallowing.
- In progressive disease, respect the patient's choices about ventilation and comfort care.

Exam points

- Warning signs of failure; peak cough flow 270 and 160 L/min; cough augmentation and breath stacking; NIV in motor neurone disease; avoid fatigue.

5 Exam corner

None of these conditions was asked on its own in the 20 papers from 2016 to 2026. Guillain–Barré syndrome on a ventilator was asked in S23 (15 marks; see the ICU physiotherapy card (p. 226)), and the paper says any syllabus topic can be asked.

Practice questions (not from a paper)

- 7 marks: physiotherapy for a patient with severe head injury on a ventilator, with precautions.
- 7 marks: respiratory management of a patient with C5 tetraplegia.
- 3 marks: three signs of respiratory failure in neuromuscular disease.
- 15 marks: a 24-year-old after a fall from a tree, C5 complete injury, day 3, weak cough, SpO_2 falling. ICF, goals, physiotherapy.

Test yourself

3 m **Why does vital capacity often improve when a C6 tetraplegic patient lies flat?**

Answer
With paralysed abdominal muscles, sitting lets the abdominal contents drop and the diaphragm flatten. Lying flat lets the abdominal contents push the diaphragm up into a domed, longer position, so it contracts more effectively. An abdominal binder copies this in sitting.

3 m **Give the signs of autonomic dysreflexia and the first action.**

Answer
Sudden very high BP, pounding headache, flushing and sweating above the level of injury, slow pulse, in an injury at T6 or above. Sit the patient up, loosen tight clothing, check the bladder catheter and bowel for the trigger, and call for help.

MCQ **In severe head injury, which part of chest physiotherapy raises ICP the most?**

A. Chest vibrations
B. Positioning head up 30°
C. Airway suction
D. Passive ankle movements

Answer
C. Suction raised ICP briefly in a study of sedated patients; vibrations did not. Sedate and pre-oxygenate, and keep suction short.

MCQ **A patient with tetanus has spasms whenever touched. Best approach:**

A. Vigorous percussion every 2 hours
B. Short, calm sessions timed with sedation in a quiet, dim room
C. No physiotherapy at all
D. Early resisted exercise

Answer
B. Coordinate with sedation, minimise stimuli, and keep the chest clear and joints mobile.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
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10. Structure and classical content: Hough, *Physiotherapy in Respiratory and Cardiac Care*; Main & Denehy (2016); Pryor & Prasad (2008).

ICU conditions: pulmonary oedema, shock, multiple organ failure, poisoning, aspiration and near drowning

> These ICU conditions are in the syllabus (Ch 9e) but were not asked on their own in the 20 papers from 2016 to 2026. The physiotherapy message is similar in all of them: recognise when the patient is too unstable for active treatment, protect the lungs, joints and skin while they are, and start rehabilitation as soon as they are not.

Syllabus	Ch 9(e) Therapeutic intervention in pulmonary oedema, multiple organ failure, smoke inhalation, poisoning, aspiration and near drowning, shock, and other acute respiratory disorders
Priority	**Syllabus.** Not asked on its own in 2016–2026; ICU cases are rising.
Asked as	Not asked 2016–2026
Usual split	Possible 7-mark notes or a 15-mark ICU case
Checked against	Sepsis-3 definitions (2016); Surviving Sepsis Campaign (2021); ICU mobilisation safety consensus (2014); organophosphate poisoning reviews and treatment protocol (2004, 2008); drowning definition (WHO, 2002); BTS emergency oxygen guideline (2017); Cochrane review of CPAP and NIV in cardiogenic pulmonary oedema (2013). Full list under Sources. Content checked 19 October 2026.

1 Pulmonary oedema

What it is

Fluid in the lung tissue and alveoli, so gas exchange fails.

- **Cardiogenic:** a failing left ventricle (heart attack, heart failure, valve disease, arrhythmia) or fluid overload raises the pressure in the lung capillaries and pushes fluid out.
- **Non-cardiogenic:** leaky capillaries with normal pressure: ARDS, near drowning, smoke inhalation, high altitude, after severe head injury (neurogenic).

What you see

- Severe breathlessness, worse lying flat (orthopnoea), waking breathless at night; anxiety, sweating.
- **Frothy, pink sputum**; fine crackles at both bases, sometimes wheeze; low SpO_2.
- **X-ray:** haziness spreading from both hila ("bat's wing"), Kerley B lines at the edges, a large heart and pleural effusions in cardiogenic oedema.

Medical point that matters for physio

Sit upright, oxygen to target, diuretics, nitrates, treatment of the cause. **CPAP or bilevel NIV** by mask relieves breathlessness and acidosis quickly and reduces the need for intubation in cardiogenic pulmonary oedema (Cochrane, 2013).

Physiotherapy by problem

- **Breathlessness:** high sitting with the legs down (less venous return), breathing control, reassurance; help apply and tolerate the CPAP mask.
- **Sputum:** the frothy fluid is oedema, not secretions, so airway clearance does not help and only tires the patient. Treat secretions only if there is also a chest infection.
- **After it settles:** early mobilisation, then heart failure rehabilitation (see the heart failure page (p. 188)).

Precautions

- Never lie the patient flat or head-down; watch BP (diuretics and nitrates lower it) and fluid balance; do not exercise until breathing and oxygen are stable at rest.

Exam points

- Cardiogenic vs non-cardiogenic; pink frothy sputum; bat's wing and Kerley B lines; sit up with legs down; CPAP; no airway clearance for oedema fluid.

2 Shock

What it is

Circulatory failure: the tissues do not get enough oxygen. Signs: fast pulse, low BP (systolic below about 90 or MAP below 65 mmHg), cold clammy skin (warm in early septic shock), urine below 0.5 mL/kg/hour, confusion, raised blood lactate (above 2 mmol/L).

Type	Cause	Examples
Hypovolaemic	Too little blood or fluid	Bleeding, burns, severe diarrhoea and vomiting
Cardiogenic	The heart cannot pump	Large MI, severe arrhythmia, end-stage heart failure
Distributive	Vessels widen and leak	Septic shock, anaphylaxis, neurogenic shock after spinal cord injury (low BP with a slow pulse)
Obstructive	Blood flow is blocked	Tension pneumothorax, cardiac tamponade, massive pulmonary embolism

Sepsis (Sepsis-3, 2016): life-threatening organ dysfunction caused by a dysregulated response to infection. **Septic shock:** sepsis needing drugs to keep MAP at 65 mmHg or more, with lactate above 2 mmol/L despite adequate fluids.

Medical point that matters for physio

Fluids, drugs that raise BP (for example noradrenaline), antibiotics within an hour in sepsis, treatment of the cause; MAP usually kept at 65 mmHg or more (Surviving Sepsis Campaign, 2021).

Physiotherapy by problem

- **While unstable:** no active mobilisation; positioning, pressure care and gentle passive movement only if BP allows; chest physiotherapy only for a clear need (retained secretions, collapse), kept short.
- **As support is weaned:** follow the safety criteria (MAP above target on no or low support is green; high support or rising doses is a reason to wait) and start the mobilisation ladder (see the ICU safety table (p. 224)).
- **Recovery:** survivors of sepsis are often very weak and tired for months; progressive strength and endurance training and follow-up.

Precautions

- Turning and sitting up can drop BP sharply: monitor MAP during and after; stop if it falls below target, the pulse races or the patient becomes confused or clammy.

Exam points

- Four types with examples; MAP 65; lactate above 2; sepsis and septic shock definitions; no active mobilisation while unstable.

3 Multiple organ failure (MODS)

What it is

Multiple organ dysfunction syndrome: progressive failure of two or more organ systems in an acutely ill patient, so that normal balance cannot be kept without support. Common causes: sepsis, major trauma, burns, pancreatitis, major surgery, shock of any cause.

Severity is tracked with the **SOFA score**, which scores six systems: breathing (PaO_2/FiO_2), clotting (platelets), liver (bilirubin), circulation (MAP and BP-support drugs), brain (GCS) and kidneys (creatinine or urine output).

What you see

A ventilated, sedated patient on BP-support drugs, often on dialysis, with jaundice, bleeding tendency, swelling, confusion or coma. Long stays lead to ICU-acquired weakness.

Physiotherapy by problem

Failing organ	What it means for physiotherapy
Lungs (ARDS)	See the ARDS plan (p. 226): no disconnection, closed suction, positioning, prone help
Circulation	Mobilise only when MAP is stable on low support
Clotting (low platelets) and liver	No percussion; gentle suction; watch for bleeding and bruising
Kidneys (dialysis)	Femoral dialysis lines do not stop exercise (green in the safety table), but secure them; watch fluid shifts and BP during dialysis
Brain (delirium, coma)	Orientation, family, day–night routine; passive movement and positioning if unresponsive
Muscles (ICU-acquired weakness)	Range of motion, positioning and splints early; active exercise and the mobilisation ladder as soon as safe

After the ICU: many survivors have **post-intensive care syndrome**: weakness, fatigue, breathlessness, memory and concentration problems, anxiety, depression and nightmares. Plan follow-up rehabilitation.

Exam points

- Definition; causes; SOFA systems; organ-by-organ precautions; prevent ICU-acquired weakness; post-ICU follow-up.

4 Poisoning

In India the commonest ICU poisonings are **pesticides** (organophosphates, often deliberate self-poisoning), **snakebite**, and drug overdose. Every self-poisoning patient needs a mental health assessment before discharge.

Organophosphate poisoning

Organophosphates block acetylcholinesterase, so acetylcholine builds up at every cholinergic junction.

Effects	Features
Muscarinic (glands, gut, heart, eyes)	Pinpoint pupils, sweating, salivation, watering eyes, vomiting, diarrhoea, urination, slow heart; **bronchorrhoea** (floods of bronchial secretions) and bronchospasm
Nicotinic (muscles)	Twitching (fasciculations), then weakness and paralysis, including breathing muscles
Brain	Confusion, seizures, coma, depressed breathing
Intermediate syndrome	About 24–96 hours after poisoning, after the first crisis: weakness of **neck flexors**, proximal limb muscles, cranial nerves and breathing muscles, which can need (re-)ventilation

- **Medical point:** decontamination, oxygen, intubation if needed, and **atropine** given until the chest is clear and dry, the heart rate is above 80, the systolic BP above 80, the pupils no longer pinpoint and the armpits dry; oximes in some centres; ventilation for weakness.
- **Physiotherapy:** early on, frequent suction and positioning for the bronchorrhoea and aspiration (pneumonia is common), head up; ventilator care. Each day, **test neck flexion** (can the patient lift the head off the pillow?): new neck weakness is an early warning of the intermediate syndrome. Then mobilise and rebuild strength.

Other poisonings

Poison	Main problem	Physio point
Neurotoxic snakebite (krait, cobra)	Descending paralysis: drooping eyelids first, then swallowing and breathing muscles; treated with antivenom and ventilation	Manage like neuromuscular weakness: airway clearance, positioning, range of motion; recovery usually within days
Paraquat (herbicide)	Progressive lung fibrosis; high oxygen speeds the lung damage	Give oxygen only when saturation is very low, as prescribed (BTS)
Kerosene and other hydrocarbons (children)	Aspiration pneumonitis	See the paediatric page (p. 204); positioning; airway clearance only for retained secretions
Sedative or opioid overdose	Slow breathing, aspiration while unconscious, pressure damage to muscles and nerves from lying still for hours	Chest care for aspiration; check for limb swelling, weakness and nerve palsies (for example foot drop) after prolonged lying

Exam points

- Muscarinic, nicotinic and brain effects; bronchorrhoea needs suction; intermediate syndrome at 24–96 hours with neck flexor weakness; mental health assessment.

5 Aspiration and near drowning

Aspiration

- **Aspiration pneumonitis:** chemical injury from inhaled stomach contents (Mendelson's syndrome), causing sudden breathlessness and low oxygen within hours.
- **Aspiration pneumonia:** infection from inhaled mouth bacteria, developing over days.
- **Who:** reduced consciousness (head injury, overdose, alcohol, seizures), stroke and other swallowing problems, vomiting, tube feeding while flat, poor oral hygiene.
- **Where it goes:** usually the **right lung** (the right main bronchus is wider and more vertical): the lower lobe basal segments when upright, the posterior segment of the upper lobe and the superior segment of the lower lobe when lying supine.
- **Physiotherapy:** head up 30–45° and side-lying for at-risk patients; suction and airway clearance for the affected segments; mobilise; refer for a swallowing assessment; good oral care.

Drowning and near drowning

- **Drowning** (WHO, 2002) is "the process of experiencing respiratory impairment from submersion or immersion in liquid"; the outcome may be death, survival with damage, or survival without damage. "Near drowning" is the older term for survivors.
- **Problems:** severe hypoxaemia; aspirated water washes out surfactant, causing collapse and pulmonary oedema that can progress to ARDS; aspiration pneumonia; brain damage from lack of oxygen; hypothermia; neck injury in diving accidents.
- **Physiotherapy:** as for ARDS and aspiration in the ventilated stage (positioning, careful airway clearance, no high pressures); spinal precautions until the neck is cleared; then neurological rehabilitation if the brain was injured.

6 Smoke inhalation

- Heat injures the upper airway (swelling can close it within hours, so early intubation is common); smoke particles and chemicals injure the lower airways (bronchospasm, sloughing, casts, pneumonia, ARDS).
- **Carbon monoxide:** binds haemoglobin; the pulse oximeter reads **falsely normal**, so a blood carboxyhaemoglobin level is needed; give high-flow oxygen through a reservoir mask. **Cyanide** from burning plastics also blocks oxygen use.
- **Physiotherapy:** upright positioning, humidified oxygen, deep breathing, ACBT, suction and early mobilisation; details and precautions on the burns page (p. 40) (inhalation burns, 7 marks in W22).

7 Other acute respiratory disorders

Condition	Where it is covered
ARDS and ventilated pneumonia	ICU physiotherapy card (p. 226); pneumonia page (p. 198)
Acute severe asthma	Asthma page (p. 132)
COPD exacerbation on NIV or a ventilator	COPD page (p. 73)
Pneumothorax and chest drains	Pleural page (p. 174); lung surgery page (p. 89)
Pulmonary embolism	Sudden breathlessness, pleuritic pain, fast pulse, low SpO_2, often after DVT. Treated with anticoagulants (clot-dissolving drugs if shocked). Mobilise once anticoagulation has started and the patient is stable, as the team advises
Flail chest and chest trauma	Chest-wall page (p. 211)

8 Exam corner

None of these conditions was asked on its own in the 20 papers from 2016 to 2026; inhalation burns (W22, 7 marks) and ventilated pneumonia and ARDS cases (W24, S24) come closest. The paper says any syllabus topic can be asked.

Practice questions (not from a paper)

- 7 marks: physiotherapy in organophosphate poisoning on a ventilator.
- 7 marks: types of shock, and physiotherapy precautions in a patient in shock.
- 3 marks: three X-ray features of cardiogenic pulmonary oedema.
- 3 marks: define multiple organ dysfunction syndrome and name the six SOFA systems.

Test yourself

3 m **Why is airway clearance not used for the pink frothy sputum of pulmonary oedema?**

Answer
It is oedema fluid leaking into the alveoli from high capillary pressure, not retained secretions. It clears with treatment of the cause (diuretics, nitrates, CPAP). Airway clearance would only tire the patient.

3 m **Name the four types of shock with one example each.**

Answer
Hypovolaemic (bleeding); cardiogenic (large MI); distributive (septic shock; anaphylaxis; neurogenic after spinal cord injury); obstructive (tension pneumothorax).

MCQ **On day 3 after organophosphate poisoning, a patient cannot lift his head off the pillow. This suggests:**

A. Recovery

B. Intermediate syndrome

C. Atropine excess

D. Normal fatigue

Answer

B. Neck flexor weakness 24–96 hours after poisoning is the early sign; breathing muscles may follow, so report it at once.

MCQ **In carbon monoxide poisoning, the pulse oximeter reading is usually:**

A. Falsely low

B. Falsely normal

C. Accurate

D. Zero

Answer

B. The oximeter cannot tell carboxyhaemoglobin from oxyhaemoglobin.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Singer M, et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA* 2016;315:801–10.
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10. Structure and classical content: Hough, *Physiotherapy in Respiratory and Cardiac Care*; Main & Denehy (2016).

Airway clearance techniques: overview and choice

> Four papers asked a broad 7-mark question: "various airway clearance techniques with application, advantages and disadvantages" (S19, W24) and "mucus removal techniques in chronic respiratory disease" (W19), plus indications of bronchial hygiene therapy (W25). This card is the one-page answer; the detail of each technique is on its own card.

Syllabus	Ch 4(a) Bronchial hygiene therapy; 4(b) adjunct therapy
Priority	**High yield.** Asked as a 7-mark overview in 3 papers and as 3 marks in 1 (2016–2026).
Asked as	7m S19 · W19 · W24 · 3m W25
Usual split	7-mark overview table; 3-mark indications
Checked against	BTS/ACPRC physiotherapy guideline (2009); ERS bronchiectasis guideline (2025); Cystic Fibrosis Trust physiotherapy standards (2026). Full list under Sources. Content checked 19 October 2026.

1 Why airway clearance is needed

Healthy airways clear mucus by the **mucociliary escalator** (cilia sweep a thin mucus layer upwards) and by **coughing**. Airway clearance techniques (ACTs) are needed when:

- there is too much mucus, or it is too thick (bronchiectasis, cystic fibrosis, COPD with chronic bronchitis, chest infection);
- the cilia work poorly (smoking, cystic fibrosis, ciliary dyskinesia, dry gas through a tracheal tube);
- the cough is weak or painful (neuromuscular disease, after surgery, sedation).

Aims: remove mucus, reduce infections and the "vicious cycle" of infection and inflammation, improve ventilation and gas exchange, reduce cough and breathlessness, and improve quality of life.

2 Classification

- **Breathing techniques:** active cycle of breathing technique (ACBT), forced expiration technique and huffing, autogenic drainage, directed or assisted cough.
- **Gravity and manual techniques:** postural drainage, percussion, vibration, shaking.
- **Positive expiratory pressure devices:** PEP mask, oscillating PEP (Flutter, Acapella), bottle PEP.
- **Mechanical devices:** chest wall oscillation vest, intrapulmonary percussive ventilation, cough assist (insufflation–exsufflation), IPPB.
- **Exercise** (increases ventilation and loosens mucus).
- **Adjuncts:** hydration and humidification, nebulised saline (normal or hypertonic), mucolytics, bronchodilator before treatment.
- **Suction** when the patient cannot cough secretions out.

3 Application, advantages and disadvantages (S19, W24)

Technique	Best applied in	Advantages	Disadvantages
ACBT (p. 261)	Most patients with sputum; post-op; exacerbations; asthma with mucus	Easy to learn; no equipment; independent; adaptable to any position	Needs co-operation; not for young children or confused patients

Technique	Best applied in	Advantages	Disadvantages
Autogenic drainage (p. 263)	Bronchiectasis, cystic fibrosis (older children and adults); unstable, wheezy airways	Gentle: little airway collapse, bronchospasm or desaturation; no equipment	Hard to learn; slow (30–45 minutes); needs concentration
Postural drainage with percussion and vibration (p. 267)	Localised disease in a known lobe; infants; dependent or ventilated patients	Targets a lobe; can be done by a carer	Time-consuming; head-down tilt has many contraindications (reflux, breathlessness, raised ICP); usually needs a helper
PEP mask (p. 271)	Cystic fibrosis, bronchiectasis; post-op lung expansion	Independent, quick, portable; can combine with a nebuliser	Needs a device and training; not with untreated pneumothorax
Oscillating PEP (Flutter, Acapella) (p. 271)	Bronchiectasis, cystic fibrosis, COPD with sputum	Adds vibration; small; easy; motivating	Cost; Flutter only works upright; must be cleaned well
Chest wall oscillation vest	Cystic fibrosis, neuromuscular disease	Hands-free; several areas at once	Very expensive; not portable
Cough assist and assisted cough	Weak cough: neuromuscular disease, spinal cord injury	Produces an effective cough when muscles cannot	Equipment and training; care with bullae and pneumothorax
Exercise	All mobile patients, as part of the routine	Also improves fitness; enjoyable	Does not replace airway clearance for most people with daily sputum
Suction (p. 231)	Intubated, unconscious or very weak patients	Removes central secretions directly	Invasive; hazards (hypoxaemia, bradycardia, trauma)

4 Choosing a technique

- **No single technique is best for everyone** (BTS/ACPRC 2009); short-term studies show they all help clear sputum. Choose by the patient's age, condition, ability, preference and what they will actually keep doing.
- **Children:** infants need modified postural drainage, gentle percussion and active play; older children PEP, bubble PEP and ACBT.
- **Unstable airways or wheeze with forced breaths:** autogenic drainage, PEP.
- **Weak cough:** assisted cough, cough assist, suction.
- **Order with medicines:** bronchodilator → mucoactive drug (for example hypertonic saline) → airway clearance → inhaled antibiotic or steroid.
- Review the technique regularly; adjust frequency during exacerbations.

5 Mucus removal in chronic respiratory disease (7 marks in W19)

1. **Assess:** daily sputum amount and colour, how easily it comes up, X-ray or CT (which lobes), breathlessness, previous techniques.
2. **Prepare the mucus:** enough fluids; humidified or nebulised saline if sputum is thick; bronchodilator first in reactive airways.
3. **Daily routine:** one main technique (ACBT, autogenic drainage or oscillating PEP), in a gravity-assisted position for the worst lobe if tolerated; 10–30 minutes, once or twice a day.
4. **Exacerbation plan:** increase to 2–3 times a day; early medical review for antibiotics.
5. **Exercise:** regular aerobic exercise as part of pulmonary rehabilitation.
6. **Pelvic floor:** ask about urine leakage with coughing (common in women) and teach pelvic floor contraction before huffs and coughs.
7. **Review:** outcome measures such as sputum weight, cough or bronchiectasis quality-of-life scores, exacerbation count.

6 Indications of bronchial hygiene therapy (3 marks in W25)

- Large amounts of sputum (classically more than about 25–30 mL a day) or difficulty clearing it.
- Mucus plugging with lobar or segmental collapse.
- Weak or painful cough: neuromuscular disease, after surgery, sedated or ventilated patients.
- Chronic suppurative lung disease (bronchiectasis, cystic fibrosis) as a daily routine.

7 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Various airway clearance techniques: application, advantages and disadvantages of each (S19). Mucus removal techniques in chronic respiratory disease (W19). Various airway clearance techniques (W24).

3-mark question

Three indications of bronchial hygiene therapy (W25).

Answer skeleton: "various airway clearance techniques", 7 marks

1. **[1]** Why: mucociliary clearance and cough fail; aims of ACTs.
2. **[1]** Classification list.
3. **[4]** The comparison table for 5–6 techniques.
4. **[1]** Choosing: no single best technique; patient factors; order with medicines.

Test yourself

3 m **Classify airway clearance techniques with one example of each.**

Answer

Breathing techniques (ACBT); gravity and manual (postural drainage with percussion); PEP devices (Acapella); mechanical devices (cough assist); exercise; adjuncts (nebulised hypertonic saline); suction.

MCQ **Which technique is most suitable for a 3-month-old with cystic fibrosis?**

A. Autogenic drainage done by the infant

B. Modified postural drainage with gentle percussion and active play

C. Flutter

D. Incentive spirometry

Answer

B. Infants cannot do active techniques; drainage without head-down tilt, gentle percussion and play are used.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
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5. Structure and classical content: Pryor & Prasad (2008); Main & Denehy (2016); Frownfelter & Dean (2022).

ACBT, forced expiration, huffing and autogenic drainage

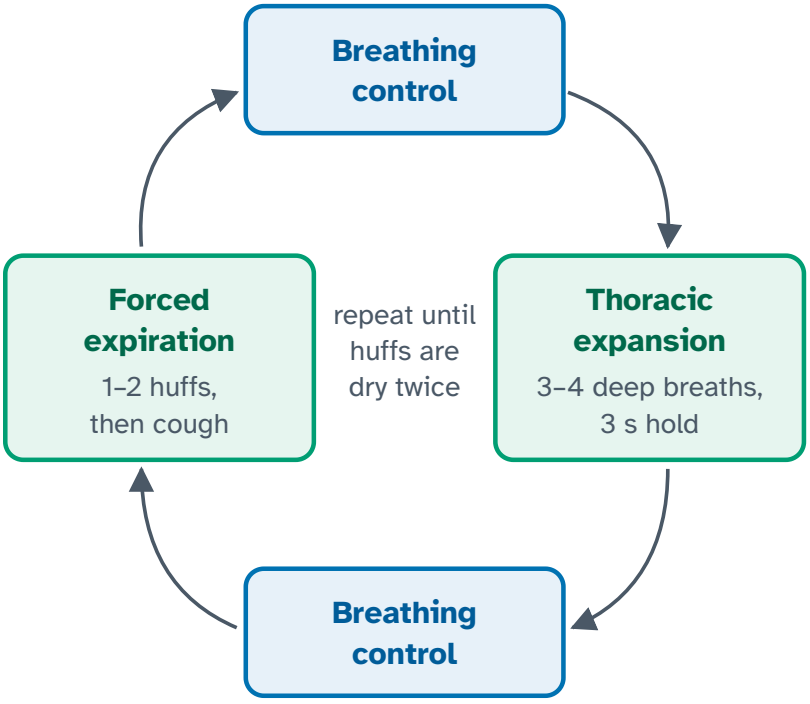
> The active cycle of breathing technique is the most-asked technique in this paper: 7 marks in W16, S23 and S26, 3 marks in S19 and W20, its rationale in asthma (S17), plus FET (W17), huffing vs coughing (W16, S20) and autogenic drainage (S20, W19, and against ACBT in W17). Learn the cycle as a drawing and the physiology behind each part.

Syllabus	Ch 4(a) Bronchial hygiene therapy: forced expiratory technique, ACBT, autogenic drainage
Priority	**High yield.** In 10 of 20 papers (2016–2026), the most-asked technique.
Asked as	7m W16 · S17 · W17 · S20 · S23 · S26 · 3m W16 · W17 · S19 · W19 · S20 · W20 · W25
Usual split	7-mark description with principle, indications and contraindications; 3-mark components, huff vs cough, AD phases
Checked against	BTS/ACPRC physiotherapy guideline (2009); ACPRC airway clearance leaflets; classic descriptions of ACBT (Pryor) and autogenic drainage (Chevaillier). Full list under Sources. Content checked 19 October 2026.

1 ACBT: what it is

The **active cycle of breathing technique (ACBT)** is a flexible cycle of three breathing exercises that the patient does independently to loosen and clear sputum: **breathing control**, **thoracic expansion exercises** and the **forced expiration technique**.

The ACBT cycle



Breathing control always sits between the active parts. The order and the number of each part are adapted to the patient (for example, more breathing control in asthma).

2 Components and physiological principle (S23)

Component	How to do it	Physiological principle
Breathing control (BC)	Gentle, relaxed tidal breathing with the lower chest; shoulders and upper chest relaxed; as long as the patient needs between the active parts	Rest; prevents bronchospasm, fatigue and desaturation that deep or forced breaths can cause
Thoracic expansion exercises (TEE)	3–4 deep breaths in, with a 3-second hold (or a sniff) at the top, then a relaxed breath out; the therapist's or patient's hand on the chest wall over the area for feedback; percussion or shaking can be added during them	A large lung volume lets air travel through collateral channels (pores of Kohn, canals of Lambert, channels of Martin) and get behind mucus; the hold evens out filling between lung units; expanding one area helps open its neighbours (interdependence)
Forced expiration technique (FET)	1–2 huffs (forced breaths out through an open mouth and glottis, like steaming a mirror) from mid to low lung volume, followed by breathing control; a huff or cough from high volume clears secretions that have reached the large airways	During a forced expiration the airways narrow downstream of the **equal pressure point**, and fast air there shears mucus towards the mouth. Starting from a lower lung volume moves the equal pressure point further out, so a longer huff clears more peripheral airways

? What is the equal pressure point?

> In a forced breath out, pressure inside the airways falls from the alveoli to the mouth. At some point it equals the pressure outside the airway (the pleural pressure): that is the equal pressure point. Beyond it the airway is squeezed narrower, the air speeds up, and mucus is carried along. At high lung volume this point is in the large central airways; as lung volume falls it moves towards smaller, more peripheral airways. That is why a long, medium-volume huff reaches further than a cough from a full breath.

3 Indications and contraindications

Indications

- Excess or retained bronchial secretions: bronchiectasis, cystic fibrosis, COPD with sputum, chest infection, asthma with mucus plugging.
- Collapse (atelectasis) and prevention of chest complications after surgery.
- To make coughing more effective and less tiring; patients who become breathless or wheezy with coughing.
- Independent daily airway clearance at home (no equipment).

Contraindications and precautions

- Few true contraindications. Not possible in patients who cannot follow instructions (unconscious, confused, very young children).
- **Bronchospasm:** lengthen breathing control; stop if wheeze increases.
- **Haemoptysis:** avoid vigorous huffing and added percussion; gentle breathing control.
- **Untreated pneumothorax, unstable heart, severe pain after surgery:** modify (supported huffing) or avoid forced parts.
- Fatigue and desaturation: shorter cycles, oxygen as prescribed.

Indications of bronchial hygiene therapy in general (3 marks in W25)

- Large amounts of sputum (classically more than about 25–30 mL a day) or difficulty clearing it.
- Mucus plugging with lobar or segmental collapse.
- Weak or painful cough: neuromuscular disease, after surgery, ventilated patients.
- Chronic suppurative lung disease (bronchiectasis, cystic fibrosis) as a daily routine.

4 Procedure and dose

1. Explain the purpose; take a bronchodilator first if prescribed.
2. Position: comfortable sitting, or a gravity-assisted position for the affected lobe if tolerated (see the postural drainage card (p. 267)).
3. Start with breathing control until relaxed.
4. 3–4 thoracic expansion breaths with a 3-second hold.
5. Breathing control.
6. 1–2 huffs from mid lung volume, then breathing control; when secretions are in the large airways, a high-volume huff or one cough.
7. Repeat the cycle. Stop when the huff sounds dry and non-productive in two consecutive cycles, or when the patient tires.

Dose: usually 10–30 minutes, once or twice a day in chronic sputum production, more often during an infection. After surgery, a few cycles every hour or two while awake.

Rationale in asthma (S17): breathing control prevents bronchospasm and fatigue; thoracic expansion gets air behind plugs; huffing moves mucus without the coughing fits that narrow sensitive airways; the patient controls the pace and stops if wheeze increases.

5 Huffing and coughing (3 marks in W16 and S20)

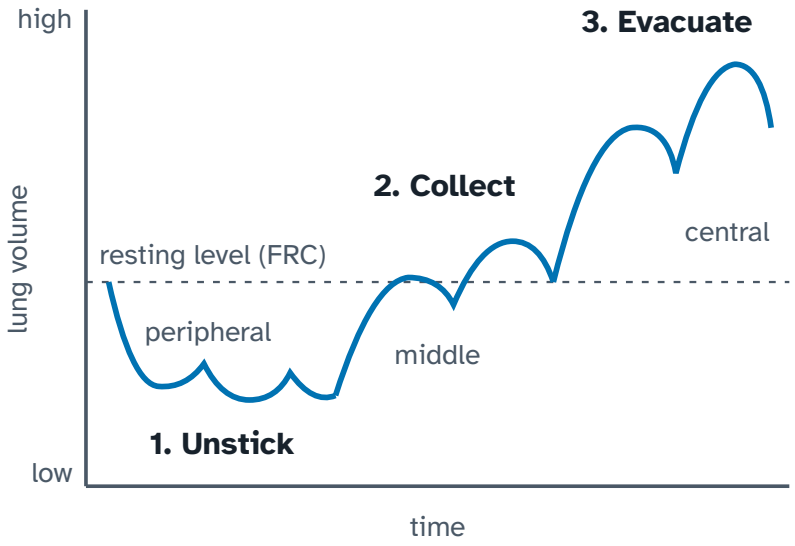
Feature	Huff	Cough
Glottis	Open throughout	Closes, pressure builds, then opens suddenly
Pressure in the chest	Moderate	High
Lung volume	Usually mid to low: clears smaller, peripheral airways	High: clears large, central airways
Control	Voluntary, graded by the patient	Reflex or voluntary
Effort and side effects	Less tiring; less airway collapse, bronchospasm, pain and urine leakage	Tiring when repeated; can cause airway collapse in COPD, bronchospasm, wound pain and stress incontinence
Best for	Moving mucus up from the periphery; after surgery (with wound support); unstable airways	Clearing mucus once it reaches the trachea and main bronchi

Teaching a huff: medium breath in, then breathe out fast but not violently through an open "O"-shaped mouth, as if steaming up a mirror; the sound should be a breathy rush, not a cough. A mouthpiece or tube helps keep the glottis open. Support a wound with a pillow or hands.

6 Autogenic drainage (7 marks in S20; phases 3 marks in W19)

Autogenic drainage (AD) (Chevaillier, Belgium) uses controlled breathing at different lung volumes to create the highest airflow the airways can take without collapsing, so mucus moves from the small airways to the large ones, where it is cleared with a gentle huff.

The three phases of autogenic drainage



Blue line = lung volume. Each phase shifts the breathing to a higher volume, and the mucus moves from peripheral to middle to central airways, then a gentle huff clears it.

Phase	Breathing	Effect
1. Unstick	Small breaths at **low lung volume** (in the expiratory reserve range)	Loosens mucus in the peripheral airways
2. Collect	Breaths at **mid lung volume** (around normal tidal breathing)	Gathers mucus in the middle airways
3. Evacuate	Deeper breaths at **high lung volume**; then a gentle huff or cough	Moves mucus to the central airways and out

Technique points

- Sitting upright (or lying); slow breath in through the nose; hold for 2–3 seconds; breathe out through an open mouth and open throat, actively but without forcing, like a sigh, as fast as possible without wheezing or airway squeezing.
- Feel or hear the mucus crackle to judge which phase to move to.
- **Suppress the urge to cough** until mucus reaches the large airways.
- Sessions usually take 30–45 minutes; the patient learns by feedback over several sessions.

Indications, advantages and disadvantages

- **Indications:** bronchiectasis, cystic fibrosis (from about 8–12 years), COPD with sputum; patients whose airways collapse or wheeze with forced techniques; patients with cough-induced incontinence.
- **Advantages:** independent, no equipment, gentle (less airway collapse, bronchospasm and desaturation), can be done anywhere quietly.
- **Disadvantages:** hard to learn; needs concentration and good body awareness; time-consuming; not for young children or confused patients (a therapist-assisted form exists for infants).

ACBT and autogenic drainage are compared in a table on the bronchiectasis page (p. 102) (W17, 7 marks).

7 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Active cycle of breathing technique (W16). Rationale of ACBT in bronchial asthma (S17). Compare ACBT and autogenic drainage in bronchiectasis (W17). Indications and phases of autogenic drainage (S20). ACBT: physiological principle, indication and contraindication (S23). Describe ACBT (S26).

3-mark questions

Huffing and coughing techniques (W16). Forced expiratory techniques (W17). ACBT (S19, W20). Difference between huffing and coughing (S20). Phases of autogenic drainage (W19). Three indications of bronchial hygiene therapy (W25).

Answer skeleton: ACBT, 7 marks (S23 wording)

1. **[1]** Definition and the cycle drawing.
2. **[2.5]** Each component: how to do it and its principle (collateral ventilation, equal pressure point, rest).
3. **[1.5]** Indications.
4. **[1]** Contraindications and precautions.
5. **[1]** Dose and end point; advantages (independent, no equipment, adaptable).

Test yourself

3 m **Name the three components of ACBT with one purpose of each.**

Answer
Breathing control: rest, prevents bronchospasm and fatigue. Thoracic expansion exercises: get air behind mucus through collateral ventilation. Forced expiration technique (huffs): moves mucus from smaller to larger airways.

3 m **Name the three phases of autogenic drainage and the lung volume of each.**

Answer
Unstick (low lung volume), collect (mid, tidal volume), evacuate (high lung volume), followed by a gentle huff.

MCQ **A huff from low lung volume clears mainly:**

A. The trachea

B. Peripheral (smaller) airways

C. The nasal passages

D. The pleural space

Answer
B. At low lung volume the equal pressure point lies further out, in the smaller airways.

MCQ **ACBT should be stopped or changed when:**

A. The huff sounds wet

B. Wheeze increases

C. The patient is sitting

D. Breathing control lasts more than 20 seconds

Answer
B. Increase breathing control and stop forced breaths if wheeze increases.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Bott J, et al. Guidelines for the physiotherapy management of the adult, medical, spontaneously breathing patient (BTS/ACPRC). *Thorax* 2009;64 Suppl 1:i1–51.
3. Association of Chartered Physiotherapists in Respiratory Care (ACPRC). The active cycle of breathing techniques; Autogenic drainage (patient leaflets). acprc.org.uk
4. Pryor JA, Webber BA, et al. Evaluation of the forced expiration technique as an adjunct to postural drainage. *BMJ* 1979.
5. Structure and classical content: Pryor & Prasad, *Physiotherapy for Respiratory and Cardiac Problems* (2008); Main & Denehy (2016); Frownfelter & Dean (2022).

Postural drainage, percussion, vibration and shaking

> Postural drainage was asked as contraindications (S18, any six) and for a neonate (W18), and it belongs in every answer on airway clearance or bronchiectasis. Learn the positions by lobe, the head-down contraindications, and when percussion is not allowed.

Syllabus	Ch 4(a) Bronchial hygiene therapy: postural drainage; practical: postural drainage and percussions
Priority	**Core technique.** In 2 of 20 papers on its own (2016–2026), and part of every airway-clearance answer.
Asked as	7m W18 (neonate) · 3m S18
Usual split	3-mark contraindications; part of 7-mark airway-clearance and bronchiectasis answers
Checked against	BTS/ACPRC physiotherapy guideline (2009); Cystic Fibrosis Foundation drainage positions; classic positions (Pryor & Prasad). Full list under Sources. Content checked 19 October 2026.

1 What it is

Postural drainage (gravity-assisted positioning) places the patient so that the bronchus of the affected segment points downwards towards the trachea, letting gravity help secretions drain to the large airways, where they are huffed or coughed out. It is almost always combined with a breathing technique (ACBT) and sometimes with percussion, vibration or shaking.

- **Classic:** 11–12 positions, many with the foot of the bed raised (head-down tilt).
- **Modified postural drainage:** the same side-lying and prone positions **without head-down tilt**, used when tilting is contraindicated or not tolerated (infants, reflux, breathlessness).

Dose: usually about 10 minutes in each productive position, with the whole session kept to about 20–40 minutes; once or twice a day for chronic sputum, more during infection. Not straight after meals.

2 Positions for each lobe

Lobe and segment	Position	Bed
Upper lobes: apical	Sitting upright, leaning back slightly (about 30°)	Flat or sitting
Upper lobes: posterior	Sitting, leaning forward over a pillow (about 30°); or lying on the opposite side turned a quarter towards prone	Flat
Upper lobes: anterior	Lying flat on the back	Flat
Lingula (left)	Lying on the **right** side, turned a quarter backwards, pillow behind	Foot raised, about 15°
Middle lobe (right)	Lying on the **left** side, turned a quarter backwards, pillow behind	Foot raised, about 15°
Lower lobes: superior (apical)	Lying prone, two pillows under the hips	Flat
Lower lobes: anterior basal	Lying on the back, or on the opposite side turned slightly back	Foot raised, about 20–30°
Lower lobes: lateral basal	Lying on the **opposite** side, turned a quarter forward	Foot raised, about 20–30°
Lower lobes: posterior basal	Lying prone, pillow under the hips	Foot raised, about 20–30°

Tilt angles differ between books (raising the foot of the bed by about 35–45 cm). The direction of lying is what matters most. Stick-figure drawings of four key positions are on the bronchiectasis page (p. 102).

Memory hook

Bad side up, bronchus down

Lie on the healthy side for a lateral segment; turn towards prone for posterior segments and towards supine for anterior segments; raise the foot for lower lobes and the middle lobe or lingula.

3 Indications

- Large amounts of sputum: bronchiectasis, cystic fibrosis, lung abscess (affected segment uppermost).
- Retained secretions or collapse of a known lobe or segment (on X-ray or by auscultation).
- Patients too weak or too young to do active techniques alone (infants, ventilated or unconscious patients) using modified positions.

4 Contraindications (3 marks in S18)

To head-down tilt

- Recent or active haemoptysis
- Raised intracranial pressure, recent head injury or neurosurgery
- Uncontrolled hypertension, unstable heart condition, arrhythmia, pulmonary oedema or heart failure
- Severe breathlessness or bronchospasm
- Gastro-oesophageal reflux, hiatus hernia, recent oesophageal surgery, just after meals or feeds
- Aortic aneurysm, recent eye surgery
- Gross obesity, abdominal distension, late pregnancy
- Untreated pneumothorax, large pleural effusion or empyema, rib fractures, painful recent surgery

Use modified (flat side-lying or sitting) positions instead.

To percussion, vibration and shaking

- Rib fractures, flail chest, osteoporosis, bone metastases, recent spinal surgery
- Haemoptysis; low platelets or clotting problems
- Unstable BP or heart rhythm; bronchospasm (percussion can worsen it)
- Burns, fresh skin grafts, open wounds or a recently inserted pacemaker over the area; surgical emphysema
- Pain the patient cannot tolerate

Newborns (W18, 7 marks): head-down tilt is avoided (reflux, intraventricular haemorrhage risk in preterm babies); handle minimally; see the neonatal chest physiotherapy page (p. 203) for the full list.

5 Percussion, vibration and shaking

Technique	How	Effect
Percussion (clapping)	Rhythmic clapping with cupped hands and relaxed wrists over the affected segment, through a thin towel or clothing; for about 30 seconds to a few minutes, alternating with breathing control or thoracic expansion	The air cushion transmits an energy wave that loosens mucus from the airway walls

Technique	How	Effect
Vibration	Fine, rapid shaking from the therapist's tensed arms with gentle compression, during the breath out only	Increases expiratory flow and moves mucus towards larger airways
Shaking	Coarser, rhythmic bouncing compression of the chest wall during the breath out, usually after a deep breath in	Stronger version of vibration; helps move thick mucus

- Use them only when the patient has secretions and finds them helpful; they are added to, not instead of, a breathing technique.
- Infants: gentle percussion with fingertips or a soft cup; no head-down tilt.
- Ventilated patients: vibrations or shaking timed with expiration, often after manual or ventilator hyperinflation.

Classic answer vs current guideline: percussion and head-down tilt

Classic	Current (BTS/ACPRC 2009)	Write in the exam
Postural drainage with head-down tilt and percussion for every patient with sputum.	Gravity-assisted positions help when secretions are in a known area, but active, independent techniques (ACBT, autogenic drainage, PEP) are preferred; head-down tilt is used only when it helps and is safe; percussion is not routine. Choose the method the patient will actually do.	"Gravity-assisted positions for the affected lobe, combined with ACBT; modified positions without head-down tilt where contraindicated; percussion and vibration as adjuncts if helpful."

6 Exam corner

Past questions (MUHS, 2016–2026)

S18 Q1(a), 3 marks

Any six contraindications for postural drainage.

W18 Q2(a), 7 marks

Precautions and contraindications for postural drainage in a neonate (see the paediatric page).

Related

Drainage positions are part of the bronchiectasis long answers (S16 middle lobe, S19 and W16 right lower lobe, W24 left lung).

Test yourself

3 m **Give six contraindications to head-down postural drainage.**

Answer

Haemoptysis; raised intracranial pressure; uncontrolled hypertension or unstable heart disease; pulmonary oedema; severe breathlessness; gastro-oesophageal reflux or just after meals; aortic aneurysm; late pregnancy or abdominal distension; untreated pneumothorax (any six).

MCQ **To drain the right middle lobe, the patient lies:**

A. On the right side, turned a quarter forward, flat

B. On the left side, turned a quarter backwards, foot raised

C. Prone, foot raised

D. Sitting upright

Answer

B. Left side down, turned back onto a pillow, foot of the bed raised about 15°.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Bott J, et al. Guidelines for the physiotherapy management of the adult, medical, spontaneously breathing patient (BTS/ACPRC). *Thorax* 2009;64 Suppl 1:i1–51.
3. Cystic Fibrosis Foundation. Postural drainage and percussion positions (patient education).
4. Structure and classical content: Pryor & Prasad (2008); Frownfelter & Dean (2022); Hough, *Physiotherapy in Respiratory and Cardiac Care*.

PEP, flutter, Acapella and other airway-clearance devices

> PEP devices were asked in seven papers: the PEP mask (W17), Acapella and Flutter (S18), two PEP devices (W18), principle with types (S21, S22), assistive devices (W21) and three uses of a PEP device (S26). Learn one principle (breathing out against resistance), three devices, and the contraindications.

Syllabus	Ch 4(b) Adjunct therapy: flutter and PEP therapy; practical: mechanical assistive devices (PEP, flutter, vest)
Priority	**High yield.** In 7 of 20 papers (2016–2026).
Asked as	7m S21 · W21 · 3m W17 · S18 · W18 · S22 · S26
Usual split	3-mark principle and two devices, or three uses; 7-mark principle, classification and types, or assistive devices
Checked against	BTS/ACPRC physiotherapy guideline (2009); NHS and Cystic Fibrosis Trust PEP patient guides; AARC guideline on positive airway pressure adjuncts. Full list under Sources. Content checked 19 October 2026.

1 PEP: the principle

Positive expiratory pressure (PEP) therapy means breathing out against a resistance, so pressure inside the airways stays above atmospheric pressure during expiration. The usual target with a PEP mask is about **10–20 cmH_2O** during the breath out.

- **Splints the airways open:** back-pressure stops small, floppy airways collapsing early in expiration (the same idea as pursed-lip breathing).
- **Gets air behind mucus:** lung volume rises a little and air moves through collateral channels into blocked units; on breathing out, that air pushes mucus towards the larger airways.
- **Oscillating PEP** adds vibration: the expiratory pressure flutters on and off many times a second, shaking mucus off the airway walls and making it less sticky.

The loosened mucus is then cleared with huffs and a cough.

2 Types of PEP device (S21: classification and types)

Type	Device and how it works	Points
Low-pressure PEP	**PEP mask** (or mouthpiece): one-way valve, a fixed resistor of chosen size on the expiratory side, and a manometer to check pressure	Pressure set by the physiotherapist; works in sitting or drainage positions; can be combined with a nebuliser
Oscillating PEP	**Flutter:** a pipe-shaped device with a steel ball in a cone; breathing out lifts the ball, which drops back repeatedly, causing oscillation	Must be held at the right angle, so used sitting; small and cheap
Oscillating PEP	**Acapella:** a counterweighted lever and magnet open and close the airflow; a dial adjusts resistance and frequency	Works in any position (lying, drainage positions); suits patients with low expiratory flow
Oscillating PEP	Other designs: RC-Cornet (curved tube with a flexible inner hose), Aerobika (oscillating valve)	Position-independent
Low-cost PEP	**Bottle (bubble) PEP:** breathing out through a tube into a bottle of water. The depth of the tube under water in cm gives the pressure in cmH_2O (10 cm deep ≈ 10 cmH_2O); bubbling adds some oscillation	Very cheap, child-friendly; keep the water clean and changed daily
High-pressure PEP	Forced expiration through a PEP mask at much higher pressures	Specialist use in some cystic fibrosis patients with unstable airways

3 Technique (PEP mask)

1. Sit upright with elbows resting on a table (or in a drainage position).
2. Hold the mask firmly over the nose and mouth (or seal the lips round the mouthpiece).
3. Breathe in slightly deeper than normal; breathe out actively but not forcefully through the device, keeping the manometer at about 10–20 cmH_2O.
4. Do about 8–10 such breaths.
5. Remove the mask; do 2–3 huffs, then a cough to clear loosened mucus; rest with breathing control.
6. Repeat the cycle until the chest feels clear, usually for about 15–20 minutes, once or twice a day (more during infections).

Flutter or Acapella: the same cycle; with the Flutter, tilt the device until the vibration is felt most in the chest. Clean devices after each use and do not share them (cross-infection, especially in cystic fibrosis).

4 Uses of a PEP device (3 marks in S26)

- **Airway clearance** in bronchiectasis, cystic fibrosis and COPD with sputum: independent, portable, quick.
- **Unstable, collapsible airways** (COPD, tracheomalacia): back-pressure keeps airways open during expiration.
- **Lung expansion after surgery**: helps prevent and reverse collapse.
- **Better aerosol delivery**: a nebuliser can be connected so the drug is breathed in while PEP keeps airways open.
- Children: play-based breathing practice (bubble PEP).

5 Contraindications and precautions

- **Untreated pneumothorax** (the main contraindication).
- Active haemoptysis.
- Raised intracranial pressure; haemodynamic instability.
- Recent facial, oral, skull or oesophageal surgery or injury; acute sinusitis; nosebleeds; middle-ear problems or a perforated eardrum.
- Patients who cannot tolerate the extra work of breathing (very breathless, exhausted).
- Large bullae: use lower pressures with care.

6 Other assistive devices for airway clearance (7 marks in W21)

Device	How it works	Used for
Incentive spirometer	Visual feedback (a rising piston or ball) encourages slow, deep breaths with a hold	Lung expansion after surgery, as an add-on to deep breathing and walking
IPPB (Bird ventilator)	Positive pressure through a mouthpiece boosts each breath the patient starts	Weak or tired patients who cannot take deep breaths; now used less. Not with an undrained pneumothorax
High-frequency chest wall oscillation (vest)	An inflatable vest rapidly squeezes and releases the chest wall	Cystic fibrosis, neuromuscular disease; hands-free but expensive
Intrapulmonary percussive ventilation	Rapid small bursts of air through a mouthpiece, with nebulised saline	Thick secretions, neuromuscular disease
Mechanical insufflation–exsufflation (cough assist)	A positive-pressure breath in, then a quick switch to negative pressure, often up to about ±40 cmH_2O in adults, to imitate a cough	Weak cough in neuromuscular disease and spinal cord injury
Manually assisted cough	The therapist pushes inwards and upwards on the abdomen or lower ribs as the patient coughs	Spinal cord injury, weak abdominal muscles (not over a fresh abdominal wound)

Device	How it works	Used for
Suction	Negative pressure through a catheter	Patients who cannot cough secretions out (see the suction card (p. 231))

7 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

PEP devices: principle, classification and types (S21). Various assistive devices used for airway clearance (W21).

3-mark questions

PEP mask and its use (W17). Acapella and Flutter device and their use (S18). Two PEP devices (W18). Principle of a PEP device with two devices (S22). Any three uses of a PEP device (S26).

Answer skeleton: PEP devices, 7 marks

1. **[2]** Principle: expiratory resistance, 10–20 cmH_2O, airway splinting, collateral ventilation, oscillation.
2. **[2.5]** Types: PEP mask; oscillating (Flutter, Acapella, others); bottle PEP; high-pressure PEP; one line each.
3. **[1.5]** Technique cycle: breaths, huffs, cough, duration.
4. **[1]** Indications and contraindications.

Test yourself

3 m **Write the principle of PEP and name two devices.**

Answer
Breathing out against a resistance keeps airway pressure positive (about 10–20 cmH_2O), splinting small airways open and letting air get behind mucus through collateral channels; oscillating devices also vibrate the airways. Devices: PEP mask, Flutter, Acapella (any two).

MCQ **A patient needs oscillating PEP while lying in a drainage position. The best device is:**

A. Flutter

B. Acapella

C. Incentive spirometer

D. Peak flow meter

Answer
B. The Acapella works in any position; the Flutter depends on gravity acting on its steel ball.

MCQ **The main contraindication to PEP therapy is:**

A. Bronchiectasis

B. Undrained pneumothorax

C. Cystic fibrosis

D. Post-operative collapse

Answer
B. Positive pressure could enlarge the pneumothorax.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Bott J, et al. Guidelines for the physiotherapy management of the adult, medical, spontaneously breathing patient (BTS/ACPRC). *Thorax* 2009;64 Suppl 1:i1–51.
3. Cystic Fibrosis Trust. How to use a PEP mask (patient leaflet); NHS trust PEP patient guides (target pressure about 10–20 cmH_2O).
4. AARC Clinical Practice Guideline: Use of positive airway pressure adjuncts to bronchial hygiene therapy. *Respir Care* 1993;38:516–21 (contraindications).
5. Cystic Fibrosis Trust. Standards of care and good clinical practice for the physiotherapy management of cystic fibrosis, 5th edition (2026).
6. Structure and classical content: Pryor & Prasad (2008); Frownfelter & Dean (2022).

Breathing exercises, lung expansion and breathing retraining

> Breathing exercises turn up as diaphragmatic breathing (W18), diaphragm recruitment and strengthening (S21), thoracic mobility exercises (W17), breathing retraining (S23) and pursed-lip breathing (S20, W21, W22). Each answer needs the technique in steps, the position, the dose and the reason it works.

Syllabus	Ch 4(f) Lung expansion therapy; practical: breathing control strategies (pursed-lip breathing, sustained maximal inspiration, deep breathing), relaxation, positioning
Priority	**High yield.** In 7 of 20 papers (2016–2026), counting pursed-lip breathing.
Asked as	7m S21 · S23 · 3m W17 · W18 · S20 · W21 · W22
Usual split	3-mark technique; 7-mark diaphragm recruitment or breathing retraining
Checked against	BTS/ACPRC physiotherapy guideline (2009); AARC incentive spirometry guideline (2011); Cochrane reviews of breathing exercises in COPD (2012) and asthma (2020); breathing retraining trial in asthma (2018). Full list under Sources. Content checked 19 October 2026.

1 Types and aims

Exercise	Main aim
Breathing control	Relaxed, efficient tidal breathing; calm breathlessness and panic
Diaphragmatic breathing	More diaphragm and less accessory-muscle work; better basal ventilation
Pursed-lip breathing	Slower breathing and less air trapping in COPD
Thoracic expansion (localised) exercises	Re-expand a specific area; loosen secretions
Sustained maximal inspiration and incentive spirometry	Lung expansion after surgery or immobility
Thoracic mobility exercises	Free a stiff chest wall and spine so the lungs can expand
Breathing retraining	Correct dysfunctional breathing (over-breathing, upper-chest pattern) in asthma and anxiety

2 Diaphragmatic breathing and diaphragm recruitment (W18; 7 marks in S21)

Technique

1. **Position:** half-lying with the head and knees supported (the diaphragm is dome-shaped and at a good length here); later sitting, standing and walking.
2. Relax the shoulders and upper chest. The therapist's (or patient's) hand rests on the upper abdomen just below the front of the ribs.
3. Breathe in slowly through the nose, letting the hand rise as the abdomen moves out; the upper chest stays quiet.
4. Breathe out gently and relaxed; do not force the air out.
5. About 5–10 breaths at a time, several times a day, then rest with normal breathing (watch for dizziness from over-breathing).

Recruiting and strengthening the diaphragm (S21)

- **Facilitation:** a quick stretch inwards and upwards on the upper abdomen at the end of expiration, then light resistance as the patient breathes in; a short sniff in also recruits the diaphragm strongly.
- **Positioning:** half-lying or supine first (easiest), progressing to sitting and standing as it gets stronger; in tetraplegia an abdominal binder helps in sitting.
- **Resistance:** manual resistance on the upper abdomen; the classic method of small weights on the abdomen in supine; best evidence for inspiratory muscle training (p. 282) with a threshold device at 30% or more of maximal inspiratory pressure.
- **Function:** diaphragmatic breathing during walking, stairs and speech; combine with breathing control.
- **Indications:** post-operative patients, neuromuscular disease and spinal cord injury (the diaphragm may be the main breathing muscle left), weaning from the ventilator, dysfunctional upper-chest breathing.

Classic answer vs current evidence: diaphragmatic breathing in COPD

Classic	Current	Write in the exam
Teach every COPD patient diaphragmatic breathing.	In severe COPD with a flat, hyperinflated diaphragm, deliberate abdominal breathing can increase the work of breathing and breathlessness (Gosselink, 1995). Pursed-lip breathing, forward-lean positions and breathing control usually help more.	"Diaphragmatic breathing for basal expansion and relaxed breathing; in severe hyperinflation, prefer pursed-lip breathing and positions of ease, and stop if breathlessness increases."

3 Pursed-lip breathing (3 marks in S20, W21, W22)

Breathe in through the nose; breathe out slowly through loosely pursed lips (as if blowing on hot tea), taking about twice as long to breathe out as to breathe in, without forcing.

- **Effects and rationale:** back-pressure splints small airways open during expiration, so less air is trapped; breathing becomes slower and deeper; tidal volume and oxygen saturation improve; breathlessness and anxiety fall; it helps during walking and stairs.
- Most patients with COPD find it themselves. More detail on the COPD page (p. 80).

4 Thoracic expansion (localised) exercises

- **Hand placement** over the area to be expanded gives feedback: lateral costal (lower ribs at the side), posterior basal (lower ribs at the back), apical (below the clavicle), unilateral or bilateral.
- **Technique:** gentle pressure at the end of expiration; ask the patient to "breathe into my hand"; hold for 3 seconds at the top; relaxed breath out. 3–4 breaths, then breathing control.
- **Uses:** after surgery, pleural effusion and empyema, collapse, chest wall stiffness, scoliosis (concave side).

5 Lung expansion therapy: SMI and incentive spirometry

- **Sustained maximal inspiration (SMI):** a slow, deep breath in to the top of the lungs, held for 3–5 seconds, then a relaxed breath out. The hold lets air reach collapsed areas through collateral channels.
- **Incentive spirometer:** a device that shows the breath as a rising piston or ball, encouraging a slow, deep SMI. **Volume-oriented** devices (piston shows volume) encourage slow deep breaths better than **flow-oriented** ones (balls rise with flow, which tempts patients to breathe fast).
- **Dose:** commonly 5–10 breaths every hour while awake after surgery, with early walking.
- **Not suitable** when the patient cannot be taught or supervised, will not co-operate, or cannot breathe deeply enough to use it (vital capacity below about 10 mL/kg or inspiratory capacity below a third of predicted); then use

IPPB or CPAP as the team decides.

- **Other lung expansion methods:** deep breathing exercises and early mobilisation (first choice), PEP, IPPB, CPAP by mask.

Classic answer vs current guideline: incentive spirometry

Classic	Current (AARC 2011)	Write in the exam
Incentive spirometry hourly for every post-operative patient.	Not recommended alone or routinely; deep breathing, coughing and early mobilisation are the core, and the spirometer is an add-on.	"Deep breathing with a 3–5 second hold (SMI), supported cough and early walking; incentive spirometer as an adjunct."

6 Exercises to increase thoracic wall mobility (3 marks in W17)

1. **Trunk rotation with breathing:** sitting, hands behind the head or arms crossed; rotate the trunk and breathe out, return to the middle and breathe in. 10 each way.
2. **Side-bending with arm overhead:** sitting, lift one arm overhead and bend to the other side while breathing in, to stretch the lengthened side. Useful after thoracotomy, pleural disease and in scoliosis.
3. **Bilateral arm elevation with breathing:** raise both arms forward and up while breathing in; lower them while breathing out.
4. **Pectoral stretch:** in a doorway or corner, arms at shoulder height, lean forward and breathe in deeply; hold 20–30 seconds.
5. **Thoracic extension:** lying over a rolled towel placed across the upper back, arms overhead, deep breaths; or sitting, hands behind the head, extend over the chair back.

Uses: ankylosing spondylitis, kyphosis, scoliosis, after thoracic surgery and burns of the chest, long-standing COPD posture.

7 Breathing retraining (7 marks in S23)

Breathing retraining teaches a slower, calmer, more efficient breathing pattern to people whose breathing has become dysfunctional: fast, upper-chest, mouth breathing with frequent sighs or over-breathing (hyperventilation). It is common in asthma and with anxiety, and causes breathlessness, chest tightness, dizziness and tingling out of proportion to lung disease. The Nijmegen questionnaire helps screen for it.

Components

- **Education:** how over-breathing lowers CO_2 and causes the symptoms; reassurance.
- **Nose breathing** at rest and during light activity (warms and humidifies air, slows breathing).
- **Lower-chest (diaphragmatic) breathing** with relaxed shoulders.
- **Slower rate and smaller, gentler breaths;** a pause after breathing out.
- **Relaxation** and control of sighs and yawns; speech with breathing (phrasing).
- **Practice:** about 10–15 minutes once or twice a day, then in daily activities (Papworth method; Buteyko-based methods add reduced-breathing exercises).

Evidence: in asthma, breathing exercises improve quality of life and symptoms (Cochrane, 2020), and a physiotherapy breathing retraining programme, including a DVD version, improved asthma quality of life in a large UK trial (2018). They do not replace inhaled medicines.

8 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Diaphragm recruitment and strengthening (S21). Breathing retraining (S23).

3-mark questions

Three exercises to increase thoracic wall mobility (W17). Diaphragmatic breathing exercises (W18). Effects of pursed-lip breathing (S20). Rationale of pursed-lip breathing in COPD (W21, W22).

Test yourself

3 m **Describe diaphragmatic breathing in three steps.**

Answer

Half-lying with head and knees supported, hand on the upper abdomen; breathe in slowly through the nose letting the abdomen rise with the upper chest relaxed; breathe out gently without forcing; 5–10 breaths, progressing to sitting and walking.

MCQ **The main reason for the 3–5 second hold in sustained maximal inspiration is:**

A. To raise CO_2

B. To let air reach collapsed alveoli through collateral channels

C. To slow the heart

D. To strengthen the abdominal muscles

Answer

B. The hold evens out filling and lets air move through collateral channels into collapsed units.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Restrepo RD, et al. AARC Clinical Practice Guideline: Incentive spirometry: 2011. *Respir Care* 2011;56:1600–4.
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7. Bott J, et al. BTS/ACPRC physiotherapy guideline. *Thorax* 2009.
8. Structure and classical content: Frownfelter & Dean (2022); Pryor & Prasad (2008); Main & Denehy (2016).

Neurophysiological facilitation and PNF of respiration

> Facilitation of breathing was asked three times as 7 marks, all recently: neurophysiological facilitation of respiration (W22), three PNF techniques for facilitation of respiration (W24) and PNF techniques to improve lung expansion (S25). NPF works on patients who cannot follow commands; PNF works on patients who can.

Syllabus	Ch 4(g) Neurophysiologic facilitation of respiration; practical: PNF for breathing facilitation and inhibition
Priority	**Rising.** In 3 of 20 papers (2016–2026), all since 2022.
Asked as	7m W22 · W24 · S25
Usual split	7-mark description of techniques with application and response
Checked against	Bethune's description of neurophysiological facilitation (1975) as summarised in later studies; recent clinical studies in ventilated and stroke patients; PNF principles (Adler, Beckers and Buck). Full list under Sources. Content checked 19 October 2026.

1 Neurophysiological facilitation (NPF): what it is

Neurophysiological facilitation of respiration uses external touch and pressure (proprioceptive and tactile stimuli) to produce **reflex** changes in breathing: deeper breaths, more movement in a chosen area, and changes in rate. It was described by Bethune (1975) for unconscious adults.

- **Who:** patients who cannot follow instructions: unconscious or sedated (including ventilated patients with some own breathing), after head injury or stroke, confused patients, young children.
- **Aims:** increase depth of breathing and lung expansion, improve basal and segmental ventilation, help secretions move, improve oxygenation.
- **Dose:** each technique held through several breaths (often 30 seconds to a minute), repeated as long as a good response is seen; watch breathing, SpO_2 and heart rate.

2 NPF techniques (7 marks in W22)

Technique	How to apply	Response seen
Perioral pressure	Firm, maintained pressure with a finger on the upper lip, between the nose and the lip	More movement of the upper abdomen, deeper breaths, sighing; mouth closure and swallowing (linked to the sucking reflex)
Intercostal stretch	Pressure on the upper border of a rib, pushing it down to widen the intercostal space above; hold through several breaths	More expansion of the stretched area (intercostal stretch reflex)
Vertebral pressure	Firm pressure over the upper thoracic (about T2–T5) or lower thoracic (about T7–T10) spinous processes	Upper level: more upper-abdominal movement and deeper breathing. Lower level: more upper-chest (apical) movement
Co-contraction of the abdomen	Pressure on the lower ribs and the pelvis on one side, drawing them towards each other	More upper-abdominal movement and posterior basal expansion; firmer abdominal wall
Anterior stretch basal lift	Patient supine; the therapist's hands under the lower posterior ribs lift them gently upwards and hold	Posterior basal expansion and upper-abdominal movement
Maintained (moderate) manual pressure	Firm, steady hand pressure over an area of the chest wall	More movement under the hand during breathing in

Precautions: no pressure techniques over fractured ribs or an unstable spine, osteoporosis, painful wounds or burns; stop if the patient becomes agitated, the heart rate or BP becomes unstable, or SpO_2 falls. In raised intracranial pressure keep handling short and gentle.

Evidence: small studies in ventilated, stroke and paediatric patients show short-lasting improvements in breathing pattern, chest expansion and some gas-exchange measures. The effects last minutes, so NPF is used together with positioning and other techniques.

3 PNF for respiration (7 marks in W24 and S25)

Proprioceptive neuromuscular facilitation (PNF) applies the same principles used for limbs (manual contacts, stretch, resistance, verbal commands, irradiation from strong patterns) to the breathing muscles, in patients who can co-operate.

Technique	How to apply	Effect
Repeated quick stretch with resistance	Hands on the lower lateral ribs (or upper abdomen for the diaphragm); a quick stretch inwards and downwards at the end of expiration, then light resistance through inspiration; repeat the stretch during the breath in ("breathe in, push my hands out")	Stretch reflex recruits the intercostals or diaphragm; more expansion of that area
Upper-limb D2 flexion pattern with inspiration	Bilateral D2 flexion (arms move up and out, as if reaching overhead) while breathing in; return in D2 extension while breathing out; can be resisted	Irradiation from strong arm and scapular muscles; lifts and expands the upper chest
Trunk lift and chop	"Lift" pattern (diagonal upward reach with trunk extension and rotation) with inspiration; "chop" pattern (diagonal downward with trunk flexion) with expiration	Trunk extension opens the chest for inspiration; flexion helps expiration and coughing
Scapular posterior elevation with inspiration	Side-lying; resisted scapular posterior elevation as the patient breathes in	Upper-chest and lateral expansion
Rhythmic initiation	Passive, then active-assisted, then active and resisted breathing movement in a rhythm	Teaches a coordinated breathing pattern in patients who cannot start the movement
Approximation and timing for emphasis	Compressing the ribs at end-expiration before the breath in; holding a strong pattern while emphasising the weaker area	Facilitates expiration and directs expansion to the weak area

For "three PNF techniques" (W24): write repeated quick stretch with resistance, bilateral D2 flexion with inspiration, and the trunk lift and chop, each with the method and the effect.

Indications: weak or uncoordinated breathing after stroke, spinal cord injury and neuromuscular disease; post-operative and post-ICU patients with low lung expansion; collapse at a lung base. **Precautions:** rib fractures, fresh wounds and drains under the hands, pain, unstable cardiovascular status.

4 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

What is neurophysiological facilitation of respiration? (W22). Any three PNF techniques for facilitation of respiration (W24). PNF techniques to improve lung expansion (S25).

Answer skeleton: NPF, 7 marks

1. **[1]** Definition and who it is for (cannot follow commands).

2. **[4]** The six techniques: application and response (a table).
3. **[1]** Aims and dose.
4. **[1]** Precautions and contraindications.

Test yourself

3 m **Name six NPF techniques.**

Answer
Perioral pressure; intercostal stretch; vertebral pressure (upper and lower thoracic); co-contraction of the abdomen; anterior stretch basal lift; maintained manual pressure.

MCQ **Perioral pressure is applied:**

A. Over the sternum

B. On the upper lip between the nose and lip

C. Over T7–T10

D. Under the posterior ribs

Answer
B. Firm, maintained pressure on the upper lip.

MCQ **Which PNF limb pattern is paired with inspiration to expand the upper chest?**

A. D1 extension

B. D2 flexion

C. D2 extension

D. Chop

Answer
B. Bilateral D2 flexion lifts the arms up and out with the breath in.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Bethune D. Neurophysiological facilitation of respiration in the unconscious adult patient. *Physiother Can* 1975;27:241–5.
3. Immediate effects of perioral pressure and intercostal stretch in mechanically ventilated patients: a randomised crossover pilot study (University of Benin Teaching Hospital). International Journal of Medicine and Medical Research.
4. The effects of neurophysiological facilitation techniques on respiration in stroke patients (randomised trial, registered as NCT05601934). Impact of neurophysiological respiratory facilitation techniques on arterial blood gas parameters in paediatric patients with bronchial pneumonia. *Bull Fac Phys Ther* 2026.
5. Adler SS, Beckers D, Buck M. *PNF in Practice: An Illustrated Guide* (patterns, techniques and their use for breathing).
6. Structure and classical content: Frownfelter & Dean (2022), chapter on facilitating ventilation patterns.

Respiratory muscle training

> Asked as "two types of inspiratory muscle training" (S18, 3 marks), "therapeutic exercise programme to strengthen respiratory muscles" (W22, 7 marks) and within diaphragm strengthening (S21). Know the types, the dose (30% or more of maximal inspiratory pressure) and who benefits.

Syllabus	Ch 4(j) Therapeutic exercise program to strengthen respiratory muscles; practical: ventilatory muscle training
Priority	**Moderate.** In 3 of 20 papers (2016–2026).
Asked as	7m S21 · W22 · 3m S18
Usual split	3-mark types; 7-mark programme
Checked against	ATS/ERS statement on respiratory muscle testing (2002); meta-analysis of inspiratory muscle training in COPD (2011); IMT in COPD trial (2018); IMT and weaning review (2015). Full list under Sources. Content checked 19 October 2026.

1 Why train the breathing muscles

Breathing muscles can be **weak** (low strength) or **overloaded** (stiff lungs, airway obstruction, hyperinflation), and both cause breathlessness, poor exercise tolerance, a weak cough and failure to wean from a ventilator. Like limb muscles, they get stronger with progressive overload.

- **Measure:** maximal inspiratory pressure (MIP or PImax) and maximal expiratory pressure (MEP) with a mouth pressure meter. A MIP stronger than about −80 cmH_2O usually rules out important inspiratory weakness; in COPD, a MIP weaker than about 60 cmH_2O marks the patients who gain most from training.
- **Who benefits:** COPD with inspiratory muscle weakness, heart failure, before high-risk heart, lung or abdominal surgery, weaning failure in ICU, neuromuscular disease and spinal cord injury (early stages), dysfunctional breathing, athletes.

2 Types of training (3 marks in S18)

Type	How it works	Points
Threshold (pressure-threshold) loading	A spring-loaded valve opens only when the patient breathes in hard enough to reach a set pressure	Load does not depend on how fast the patient breathes; easy to set as a % of MIP; the most used type (handheld threshold trainers)
Inspiratory resistive (flow-resistive) training	Breathing in through a small opening; smaller holes give more resistance	The load depends on flow, so the patient must keep a set breathing pattern (or use a target device)
Normocapnic hyperpnoea (endurance training)	Breathing at a high minute ventilation for 10–30 minutes, with a rebreathing bag to keep CO_2 normal	Trains endurance; needs equipment and supervision
Electronic tapered-flow resistive loading	A device adjusts the resistance through each breath	Allows heavier loads comfortably; costly
Expiratory muscle training	Threshold device on the breath out	Cough, voice and swallowing in neuromuscular disease, Parkinson's disease, multiple sclerosis

Older textbooks also describe **abdominal weights** (small weights on the upper abdomen in supine) and diaphragmatic resisted breathing by hand.

3 Dose

Part	Inspiratory muscle training (threshold)
Intensity	Start at about **30% of MIP**; progress towards 50–60% or more as tolerated; re-measure MIP every 1–2 weeks
Volume	About **30 breaths** (or about 15 minutes), **twice a day**
Frequency and duration	5–7 days a week for at least 6–8 weeks, then a maintenance dose
Technique	Sitting upright, nose clip if needed, a quick strong breath in through the device, then a slow relaxed breath out; rest between sets

4 Precautions and contraindications

- Recent or untreated pneumothorax; recent lung or chest-wall surgery not yet healed.
- Acute exacerbation of COPD or asthma; unstable heart failure; recent heart attack.
- Perforated eardrum; raised intracranial pressure; active haemoptysis.
- Watch for fatigue (especially in neuromuscular disease, where overwork can harm weak muscles), dizziness and breathlessness; stop if they appear.

Classic answer vs current evidence: IMT in COPD

Classic	Current	Write in the exam
Give inspiratory muscle training to every COPD patient.	On its own, IMT improves inspiratory strength, endurance, breathlessness and walking distance, most in patients with weak inspiratory muscles (meta-analysis, 2011). Added to a full pulmonary rehabilitation programme, it improved muscle strength but not exercise capacity in a 2018 trial. It is an add-on for weak patients, not a replacement for whole-body exercise. In ICU, IMT helps patients who fail weaning (review, 2015).	"IMT at 30% or more of MIP, about 30 breaths twice a day for at least 8 weeks, for patients with inspiratory muscle weakness, alongside general exercise training."

5 Exercise programme to strengthen respiratory muscles (7 marks in W22)

1. **Assess:** MIP and MEP, breathing pattern, chest expansion, peak cough flow, breathlessness (Borg, mMRC), posture.
2. **Diaphragmatic breathing** with facilitation and manual resistance, progressing from half-lying to sitting and standing.
3. **Inspiratory muscle training** with a threshold device at the dose above.
4. **Expiratory muscle training** and huffing practice where cough is weak; abdominal muscle strengthening (bracing, curl-ups as allowed).
5. **Accessory and postural muscles:** scapular and upper-limb exercise (arm ergometry, resisted arm patterns), thoracic mobility.
6. **Whole-body aerobic training** (walking, cycling): the ventilatory load of exercise also trains the breathing muscles.
7. **Endurance:** normocapnic hyperpnoea where available; singing or wind-instrument games for children.
8. **Progress and review** every 1–2 weeks; re-measure MIP; adjust load.

6 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Diaphragm recruitment and strengthening (S21; see the breathing exercises card (p. 275)). Therapeutic exercise programme to strengthen respiratory muscles (W22).

3-mark question

Write two types of inspiratory muscle training (S18).

Test yourself

3 m **Differentiate threshold loading from resistive training.**

Answer

Threshold: a spring valve opens at a set pressure, so the load is fixed whatever the flow. Resistive: breathing through a small hole, so the load rises with faster flow and the breathing pattern must be controlled.

MCQ **A common starting intensity for inspiratory muscle training is:**

A. 5% of MIP

B. 30% of MIP

C. 90% of MIP

D. 150% of MIP

Answer

B. About 30% of MIP, progressed as tolerated.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. American Thoracic Society/European Respiratory Society. ATS/ERS statement on respiratory muscle testing. *Am J Respir Crit Care Med* 2002;166:518–624.
3. Gosselink R, et al. Impact of inspiratory muscle training in patients with COPD: what is the evidence? *Eur Respir J* 2011;37:416–25.
4. Charususin N, et al. Randomised controlled trial of adjunctive inspiratory muscle training for patients with COPD. *Thorax* 2018;73:942–50.
5. Elkins M, Dentice R. Inspiratory muscle training facilitates weaning from mechanical ventilation among patients in the intensive care unit: a systematic review. *J Physiother* 2015.
6. Hulzebos EHJ, et al. Preoperative intensive inspiratory muscle training to prevent postoperative pulmonary complications in high-risk patients undergoing CABG surgery. *JAMA* 2006.
7. Structure and classical content: Frownfelter & Dean (2022); Pryor & Prasad (2008).

Dyspnoea, relieving positions, energy conservation and pacing

> Breathlessness questions come in many forms: definition, mechanism and treatment (W25), positions that ease it (S22), energy conservation (W17, S19, S22, W22, S25) and pacing of activities (S20, W21). One idea links them: lower the demand, improve the efficiency, and calm the fear.

Syllabus	Ch 4(d) Therapeutic positioning to alleviate dyspnoea; 4(k) ergonomic advice, energy conservation advice, home exercise program
Priority	**High yield.** In 8 of 20 papers (2016–2026).
Asked as	7m S19 · S22 · W22 · W25 · 3m W17 · S20 · S22 · W21 · S25
Usual split	7-mark dyspnoea (definition, mechanism, treatment) or energy conservation; 3-mark positions, pacing, energy conservation
Checked against	ATS statement on dyspnoea (2012); BTS/ACPRC physiotherapy guideline (2009); handheld fan trial (2010). Full list under Sources. Content checked 19 October 2026.

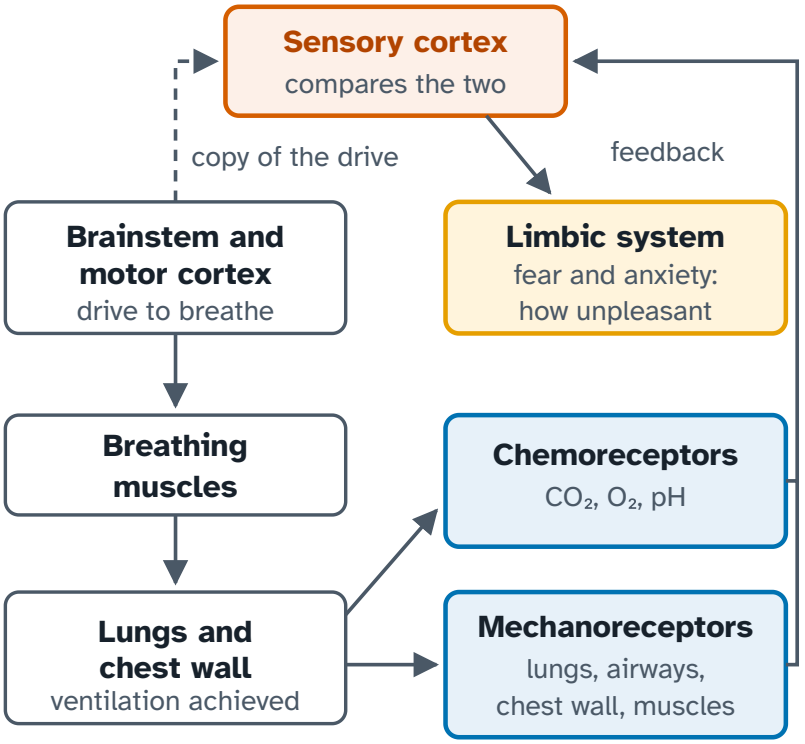
1 Definition

Dyspnoea (breathlessness) is "a subjective experience of breathing discomfort that consists of qualitatively distinct sensations that vary in intensity" (American Thoracic Society, 2012). It is what the patient feels, so it is measured by the patient's own rating.

- **Different sensations:** air hunger (cannot get enough air), work or effort of breathing, chest tightness (bronchospasm).
- **Two parts:** how strong it is (sensory) and how distressing it is (emotional).
- **Measure:** modified Borg (0–10) during activity; mMRC grade (0–4) for daily life; see the scales card (p. 322).
- Related terms: **orthopnoea** (breathless lying flat: heart failure), **paroxysmal nocturnal dyspnoea** (waking breathless at night), **platypnoea** (breathless sitting or standing, relieved by lying).

2 How breathlessness is perceived (W25)

Drive versus feedback: where breathlessness comes from



Breathlessness is the brain noticing that the effort sent out is not matched by the breathing that results (neuromechanical dissociation), coloured by fear.

- **Motor drive and sense of effort:** a copy of the motor command to the breathing muscles (corollary discharge) reaches the sensory cortex. Weak or overloaded muscles need a bigger command for the same breath, which is felt as effort.
- **Chemoreceptors:** rising CO_2 and acidity (central, in the medulla) and low oxygen (peripheral, carotid and aortic bodies) increase the drive and produce air hunger.
- **Mechanoreceptors:** stretch and irritant receptors in the airways and lungs, J receptors (stiff, wet lungs), and muscle spindles and tendon organs in the chest wall report how well the lungs and chest are moving.
- **Mismatch:** when the drive is high but the lungs cannot respond (hyperinflation, stiff lungs, weak muscles), the cortex senses a mismatch: this is the main source of breathlessness.
- **Emotion:** the limbic system adds fear and anxiety; anxiety speeds breathing, which worsens hyperinflation and breathlessness (the breathlessness–anxiety cycle).

Common clinical causes: higher demand (exercise, fever, acidosis, low oxygen), higher load (airway obstruction, stiff lungs or chest wall, hyperinflation), weaker muscles (neuromuscular disease, deconditioning), anxiety.

3 Treatment of breathlessness (W25)

Approach	Examples
Treat the cause	Bronchodilators, treatment of heart failure, infection, anaemia; oxygen if hypoxaemic
Lower the work of breathing	Positions of ease; pursed-lip breathing; breathing control; airway clearance; NIV in selected patients
Lower the demand	Energy conservation, pacing, weight loss, aids for daily tasks

Approach	Examples
Build capacity	Exercise training (pulmonary rehabilitation) desensitises to breathlessness and lowers ventilation at each workload; inspiratory muscle training if muscles are weak
Calm the brain	Education, relaxation, a "breathlessness plan"; a handheld fan blown on the face (reduces breathlessness through facial cool-air receptors); anxiety management
Medical palliation	Low-dose opioids in advanced disease, as prescribed

4 Positions to ease breathlessness (3 marks in S22)

1. **High side-lying:** head and shoulders well supported on pillows, top arm on a pillow.
2. **Forward-lean sitting:** forearms on the thighs, or on pillows on a table.
3. **Relaxed sitting:** leaning back in a supportive chair with forearms on armrests.
4. **Forward-lean standing:** forearms on a wall, window sill, rail or walking frame.
5. **Relaxed standing:** back and hips against a wall, feet a little apart, shoulders relaxed.

Rationale: leaning forward lets the abdominal contents push a flattened diaphragm up into a better dome shape, so each contraction moves more air; fixing the arms lets the pectoral and other accessory muscles pull on the ribs; supported, relaxed positions cut muscle work and anxiety; upright positions increase lung volume.

5 Energy conservation (S19, S22, W22; 3 marks W17, S25)

Memory hook

The 4 Ps: Plan, Pace, Prioritise, Position

Principle	Examples
Plan	Spread heavy jobs (washing clothes, cleaning, shopping) through the week; do them at the best time of day (often late morning, after airway clearance and medicines); gather everything before starting
Pace	Slow, steady movement; rest before becoming very breathless; breathe out on the effort part ("blow as you go"): lifting, standing up, reaching
Prioritise	Drop or share tasks that are not essential; ask family to help with heavy work
Position and ergonomics	Sit to bathe, dress, iron and cook; keep things used often between waist and shoulder height; avoid bending, lifting and working with the arms above the head; use a trolley or bag on wheels; long-handled sponge, shower stool, raised toilet seat
Breathing with activity	Pursed-lip breathing while walking; stairs one step per breath out, resting on landings
Environment	Avoid smoke, dust and extreme heat or cold; loose clothes; fan nearby

Ergonomic advice for one-sided lung disease (W22, left lung bronchiectasis with a normal right lung): rest and sleep lying on the **right** side: the healthy right lung is below, where it gets more blood and air, and the diseased left lung is on top, where it drains. Sit upright at work with the desk at elbow height; plan huffing breaks; avoid dust and smoke. Full answer on the bronchiectasis page (p. 102).

6 Activity pacing (3 marks in S20 and W21)

Pacing means doing activity in planned, manageable amounts with rests, so the patient stays below the level of breathlessness that forces a stop, and finishes the task without "boom and bust".

- **Break tasks into steps:** make the bed in stages, sitting between them.
- **Alternate heavy and light tasks** through the day.

- **Rest before exhaustion:** "stop, rest, recover, restart" when breathlessness reaches a moderate level (Borg about 3–4).
- **Stairs:** one step on each breath out, rest every few steps or at the landing.
- **Walking:** interval walking (walk a set distance, rest, repeat) and slower speed with pursed-lip breathing.
- **Bathing and dressing:** sit to wash and dress; rest between bathing and dressing; dressing gown instead of towelling off.

7 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Energy conservation techniques for COPD and their importance (S19; see the COPD page). Methods of energy conservation (S22). Ergonomic and energy-conservation advice in left lung bronchiectasis (W22). Define dyspnoea, explain the mechanism of its perception and its treatment (W25).

3-mark questions

Three energy conservation techniques for a breathless patient (W17). Pacing of activities in a breathless patient (S20). Rationale of therapeutic positioning to alleviate dyspnoea (S22). Activity pacing with examples (W21). Energy conservation techniques (S25).

Answer skeleton: dyspnoea, 7 marks (W25)

1. **[1]** ATS definition; sensations; measurement.
2. **[3]** Mechanism: motor drive and corollary discharge, chemoreceptors, mechanoreceptors, mismatch, emotion; draw the diagram.
3. **[3]** Treatment table: cause, work of breathing, demand, capacity, brain.

Test yourself

3 m **Give the rationale of forward-lean sitting in a breathless COPD patient.**

Answer
The abdominal contents push the flattened diaphragm up into a domed, longer position, so it contracts more effectively; fixing the arms lets accessory muscles help lift the ribs; the supported posture reduces muscle work and anxiety.

3 m **What is activity pacing? Give two examples.**

Answer
Doing activities in planned amounts with rests so breathlessness stays manageable. Examples: climbing stairs one step per breath out with rests at the landing; making the bed in stages with sitting breaks.

MCQ **The main source of breathlessness when a hyperinflated COPD patient walks fast is:**

A. Low haemoglobin

B. A mismatch between high drive and limited lung expansion

C. Raised blood sugar

D. Low heart rate

Answer

B. Neuromechanical dissociation: the drive rises but the hyperinflated chest cannot expand further.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Parshall MB, et al. An official American Thoracic Society statement: update on the mechanisms, assessment, and management of dyspnea. *Am J Respir Crit Care Med* 2012;185:435–52.
3. Galbraith S, et al. Does the use of a handheld fan improve chronic dyspnea? A randomized, controlled, crossover trial. *J Pain Symptom Manage* 2010;39:831–8.
4. Bott J, et al. BTS/ACPRC physiotherapy guideline. *Thorax* 2009.
5. Structure and classical content: Frownfelter & Dean (2022); Main & Denehy (2016); occupational therapy energy conservation guidance.

Oxygen therapy: devices, targets and hazards

> Oxygen was asked as low-flow oxygen therapy (S20, 7 marks), methods of delivery (S23), hazards (S25) and hypoxic drive (W25). Learn the device ladder with flows and approximate FiO_2, the two target ranges (94–98% and 88–92%), and the hazards.

Syllabus	Practical: medication delivery including oxygen; Ch 9(b) ICU monitoring
Priority	**Moderate.** In 4 of 20 papers (2016–2026), 3 of them since 2023.
Asked as	7m S20 · 3m S23 · S25 · W25
Usual split	7-mark low-flow oxygen therapy; 3-mark methods of delivery or hazards
Checked against	BTS guideline for oxygen use in healthcare and emergency settings (2017, with later updates); ATS home oxygen guideline (2020). Full list under Sources. Content checked 19 October 2026.

1 What oxygen is for

Oxygen is a **drug** given to correct or prevent **hypoxaemia** (low oxygen in the blood). It does not treat breathlessness itself, and it should be prescribed with a target saturation.

Target SpO_2 (BTS)

Patient	Target
Most acutely ill adults	**94–98%**
At risk of high CO_2 (COPD, severe obesity, chest wall deformity, neuromuscular disease, cystic fibrosis with high CO_2)	**88–92%**, adjusted with blood gases
Critically ill, peri-arrest, carbon monoxide poisoning	High concentration first (reservoir mask 15 L/min), then titrate

- **Indications:** low SpO_2 in acute illness (pneumonia, asthma, heart failure, pulmonary embolism, after surgery); critical illness; carbon monoxide poisoning; desaturation on exercise; long-term oxygen in chronic hypoxaemia (criteria on the COPD page (p. 77)).
- **Not routinely** for breathless patients with normal saturation, or in heart attack when SpO_2 is 90% or more.

2 Methods of delivery (3 marks in S23)

Devices are grouped by whether they supply all the air the patient breathes in. **Low-flow (variable-performance)** devices give less than the patient's inspiratory flow, so room air mixes in and the FiO_2 changes with the breathing pattern. **High-flow (fixed-performance)** devices supply all of it, so the FiO_2 stays fixed.

Device	Flow	Approximate FiO_2	Points
Nasal cannula (prongs)	1–4 L/min (up to 6)	About 24–40% (roughly 4% more per litre)	Comfortable; can eat and talk; FiO_2 falls with fast or mouth breathing
Simple face mask	5–10 L/min	About 35–60%	Never below 5 L/min (CO_2 is rebreathed); not for patients at risk of high CO_2
Reservoir (non-rebreather) mask	15 L/min, bag kept inflated	About 60–90%	Emergencies, critical illness, carbon monoxide
Venturi mask (fixed performance)	The flow printed on each colour-coded valve	Fixed: 24, 28, 31, 35, 40 or 60%	Best for patients at risk of high CO_2 (start 24% or 28%); raise the flow by half if breathing faster than 30

Device	Flow	Approximate FiO_2	Points
High-flow nasal cannula	Up to about 60 L/min, heated and humidified	21–100%, set	Comfortable; gives a little PEEP and washes out CO_2 from the upper airway; type 1 respiratory failure, after extubation
Tracheostomy mask or T-piece	Humidified flow	Set by the system	Patients with a tracheostomy or tube breathing on their own
Head box, incubator	Set flow	Measured by an analyser	Newborns
NIV, CPAP, ventilator	Oxygen blended into the circuit	21–100%, set	With respiratory support

FiO_2 figures for low-flow devices are approximate and vary between sources, because they depend on the patient's breathing. Sources of oxygen: piped wall supply, cylinders, oxygen concentrators at home.

3 Low-flow oxygen therapy (7 marks in S20)

- **Definition:** oxygen given at a flow lower than the patient's own inspiratory flow, so the gas breathed is a mix of oxygen and room air and the FiO_2 varies.
- **Devices:** nasal cannula, simple face mask, reservoir mask, tracheostomy mask (oxygen catheter in older texts).
- **Uses:** mild to moderate hypoxaemia in stable patients; home and ambulatory oxygen; during exercise in patients who desaturate.
- **Advantages:** simple, cheap, comfortable (cannula), patient can eat, talk and walk.
- **Disadvantages:** FiO_2 is not known exactly (higher with slow, shallow breathing, lower with fast breathing), so it is less suitable when the CO_2 level must be controlled precisely; nasal dryness at higher flows.
- **Humidification:** not needed for low-flow oxygen; consider it for high flows used for more than about 24 hours, for a tracheostomy, or if the nose and mouth feel dry.
- **Monitoring:** SpO_2 against the prescribed target, breathing rate, drowsiness (rising CO_2), ABG in those at risk.

4 Hazards of oxygen therapy (3 marks in S25)

Hazard	Details
CO_2 retention (hypercapnic respiratory failure)	In COPD and other at-risk patients, uncontrolled high oxygen raises CO_2 (worse V/Q matching and the Haldane effect; reduced drive plays a small part): drowsiness, confusion, acidosis. See hypoxic drive (p. 77)
Absorption atelectasis	High FiO_2 washes nitrogen out of alveoli; the oxygen is absorbed and poorly ventilated alveoli collapse
Oxygen toxicity	Lung injury from prolonged high concentrations (FiO_2 above about 0.6 for long periods)
Retinopathy of prematurity	High oxygen damages developing retinal vessels in preterm babies
Fire	Oxygen feeds fire: no smoking, no flames or chulha nearby, no oil or grease on fittings; burns of the face in smokers using oxygen
Masking deterioration	A good SpO_2 on oxygen can hide worsening breathing; watch the rate and the patient
Local problems	Dry nose and mouth, nosebleeds, pressure sores behind the ears and on the nose; masks cause claustrophobia and aspiration risk if vomiting
Equipment	Trips over tubing; empty cylinders during walks; heavy cylinders

5 The physiotherapist's role

- Check the prescription and target before treatment; never stop or change oxygen without it.

- **Exercise:** measure SpO_2 during walking; give oxygen (as prescribed) to keep SpO_2 above about 88–90% during training; use portable cylinders or concentrators for ambulatory oxygen.
- Keep oxygen on during airway clearance and positioning when needed; watch SpO_2 after turns and suction.
- Teach long-term oxygen users: at least 15 hours a day, safety (no smoking), cleaning the cannula, and pacing.
- **Pulse oximetry limits:** wrong readings with cold fingers, poor circulation, movement, nail polish and carbon monoxide (falsely normal); oximeters can over-read in darker skin, so act on the whole clinical picture.

6 Exam corner

Past questions (MUHS, 2016–2026)

S20 Q2(c), 7 marks

Low-flow oxygen therapy.

3-mark questions

Methods of delivery of oxygen therapy (S23). Hazards of oxygen therapy (S25). Hypoxic drive in COPD (W25; see the COPD page).

Test yourself

3 m **List three low-flow and two high-flow oxygen devices with flows.**

Answer

Low-flow: nasal cannula 1–4 L/min; simple face mask 5–10 L/min; reservoir mask 15 L/min. High-flow (fixed performance): Venturi mask (flow printed on the valve, FiO_2 24–60%); high-flow nasal cannula (up to about 60 L/min).

MCQ **A COPD patient at risk of CO_2 retention needs oxygen. The best first choice is:**

A. Reservoir mask 15 L/min

B. Simple face mask 3 L/min

C. 24% or 28% Venturi mask, target SpO_2 88–92%

D. No oxygen

Answer

C. A fixed, low FiO_2 with the 88–92% target; a simple mask below 5 L/min risks rebreathing CO_2.

MCQ **Which hazard of oxygen is caused by nitrogen washout?**

A. Absorption atelectasis

B. Retinopathy of prematurity

C. Nosebleeds

D. Fire

Answer

A. Without nitrogen to hold them open, poorly ventilated alveoli collapse as the oxygen is absorbed.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. O'Driscoll BR, et al. BTS guideline for oxygen use in adults in healthcare and emergency settings. *Thorax* 2017;72 Suppl 1:i1–90. brit-thoracic.org.uk

3. Jacobs SS, et al. Home oxygen therapy for adults with chronic lung disease: an official ATS clinical practice guideline. *Am J Respir Crit Care Med* 2020.
4. Abdo WF, Heunks LMA. Oxygen-induced hypercapnia in COPD: myths and facts. *Crit Care* 2012;16:323.
5. Byrne RA, et al. 2023 ESC Guidelines for the management of acute coronary syndromes (oxygen only if SpO_2 below 90%).
6. Structure and classical content: Hough, *Physiotherapy in Respiratory and Cardiac Care*; Main & Denehy (2016).

Humidification, nebulisers and inhalers

> These were asked in seven papers: humidification and its uses (S17), humidification in intubated patients (W17), define humidification and nebulisation (S21), nebulisers in three conditions (S19), jet and ultrasonic nebulisers (S20), nebulisation indications, contraindications and types (S24), and metered-dose inhalers (W25).

Syllabus	Ch 4(e) Nebulisation and humidification; 9(d) Suctioning and humidification
Priority	**High yield.** In 7 of 20 papers (2016–2026).
Asked as	7m S20 · S24 · 3m S17 · W17 · S19 · S21 · W25
Usual split	3-mark definitions, uses, inhalers; 7-mark nebuliser types with indications and contraindications
Checked against	AARC humidification guideline (2012); BTS oxygen guideline (2017, air-driven nebulisers in hypercapnia); Cochrane review of spacers versus nebulisers (2013). Full list under Sources. Content checked 19 October 2026.

1 Definitions (3 marks in S21)

- **Humidification:** adding water vapour (a gas) to inspired air or oxygen, to warm and moisten it.
- **Nebulisation:** turning a liquid (usually a drug solution) into an aerosol of fine droplets that can be breathed into the airways.
- **Aerosol:** tiny liquid or solid particles suspended in a gas.

2 Humidification (3 marks in S17 and W17)

Normal airway conditioning

The nose and upper airway warm and humidify inspired air so that, a few centimetres below the carina, it is at body temperature (37 °C) and fully saturated (100% relative humidity, about 44 mg of water per litre). This point is called the **isothermic saturation boundary**.

Why intubated patients need it (W17)

An endotracheal tube or tracheostomy bypasses the nose and throat, so dry, cold gas reaches the trachea. This causes thick, sticky secretions, slows the cilia, damages the airway lining, blocks the tube, causes collapse and loses body heat and water. The AARC (2012) strongly recommends humidification for every invasively ventilated patient.

Type	How it works	Points
Heat and moisture exchanger (HME, "artificial nose")	Passive: traps heat and moisture from each breath out and returns it on the next breath in	Should deliver at least **30 mg H_2O/L**; cheap, simple. Not with thick, copious or bloody secretions, a large leak, hypothermia, during nebulisation, or with the low tidal volumes of lung-protective ventilation (adds dead space)
Heated humidifier	Active: gas passes over a heated water chamber; heated-wire circuits limit condensation	Target **33–44 mg H_2O/L** at **34–41 °C** at the Y-piece, 100% relative humidity (AARC 2012); used for long ventilation, thick secretions, high-flow nasal cannula, NIV comfort
Bubble humidifier	Oxygen bubbles through water at room temperature	For oxygen; adds little real humidity
Saline nebulisation	An aerosol of saline droplets	Loosens sputum; tracheostomy care

Uses (S17): intubated and tracheostomy patients; high-flow oxygen used for long periods; thick secretions; NIV comfort; dry upper airway.

Hazards: overheating and airway burns (heated systems); water collecting in the tubing (can be washed into the patient, spreads infection, triggers false breaths); infection from contaminated water; extra resistance and dead space with an HME, which can block with secretions.

3 Nebulisation (7 marks in S20 and S24)

Particle size decides where the drug lands

Droplet size	Where it deposits
Larger than 10 µm	Mouth, nose and throat (swallowed)
5–10 µm	Large central airways
1–5 µm (the respirable range)	Small airways and alveoli: the target for most drugs
Smaller than 1 µm	Mostly breathed out again

Types of nebuliser

Type	Principle	Advantages	Disadvantages
Jet (pneumatic)	Compressed air or oxygen (about 6–8 L/min) rushes through a narrow opening; the low pressure (Bernoulli or Venturi effect) draws the liquid up a fine tube and shatters it into droplets; a **baffle** knocks out the large ones	Cheap, widely available, works with most drugs and suspensions	Noisy; slow (10–15 minutes); some drug left in the cup; needs a compressor or gas supply
Ultrasonic	A **piezoelectric crystal** vibrates at high frequency; waves on the liquid surface throw off droplets	Quiet; fast; large output	Can warm and damage some drugs; poor for suspensions (for example budesonide); costly
Vibrating mesh	A vibrating plate pushes the liquid through a mesh with thousands of tiny holes	Very efficient; little drug wasted; quiet, small, battery-powered; used in ventilator circuits	Expensive; mesh needs careful cleaning

Indications (S24) and three conditions with purpose (S19)

- **Acute severe asthma or COPD exacerbation:** bronchodilators (salbutamol, ipratropium) to relieve bronchospasm.
- **Cystic fibrosis and bronchiectasis:** hypertonic saline or dornase alfa to loosen sputum before airway clearance; inhaled antibiotics to control infection.
- **Ventilated patients:** bronchodilators given in-line.
- **Children with croup:** nebulised adrenaline or steroid as prescribed.
- **Sputum induction** for TB and other tests (hypertonic saline).
- Anyone who cannot use an inhaler: very young, very old, confused or very breathless patients.

Contraindications and precautions

- Known allergy to the drug.
- **Hypertonic saline** can cause bronchospasm: give a bronchodilator first and check peak flow or spirometry with the first dose.
- **Patients at risk of high CO_2** (COPD): drive the nebuliser with compressed air, not oxygen, and give oxygen separately by nasal cannula (BTS).
- Side effects: fast heart rate and tremor (salbutamol), low potassium with frequent high doses.
- Infection control: nebulising can spread infection (aerosol-generating); clean, dry and do not share equipment.

Technique

1. Sit upright; fill the cup (usually 2–5 mL).
2. Use a mouthpiece where possible (less drug lost to the face and eyes); a mask for children and very breathless patients.
3. Breathe normally through the mouth, with an occasional slow deep breath.
4. Continue until the nebuliser starts to splutter.
5. Rinse the mouth after steroids; wash and dry the parts.

4 Inhalers (3 marks in W25)

Metered-dose inhaler (MDI)

A pressurised canister holds the drug with a propellant; a **metering valve** releases a fixed dose with each press (for example 100 µg of salbutamol per puff).

1. Remove the cap and shake the inhaler.
2. Breathe out gently, away from the inhaler.
3. Seal the lips around the mouthpiece.
4. Start a slow, deep breath in (about 3–5 seconds) and press the canister once at the start.
5. Hold the breath for up to 10 seconds, then breathe out slowly.
6. Wait about 30–60 seconds before a second puff; rinse the mouth after a steroid inhaler.

- **Advantages:** small, quick, cheap, many drugs available.
- **Disadvantages:** needs hand–breath coordination; much of the dose hits the throat; cold spray can stop the breath.
- **Spacer (holding chamber):** the puff goes into the chamber and is breathed in over a few breaths, so coordination matters less and less drug lands in the throat. With a face mask for young children. In acute asthma an MDI with a spacer works as well as a nebuliser (Cochrane, 2013).

Dry powder inhaler (DPI)

Drug powder released by the patient's own fast, deep breath in (for example Rotahaler with capsules, Accuhaler, Turbuhaler). No propellant and no coordination needed, but the patient must breathe in quickly and strongly, so it does not suit young children or a severe attack. Do not breathe out into it (moisture spoils the powder).

5 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Principle of working of jet and ultrasonic nebulisers (S20). Indications and contraindications of nebulisation and types of nebulisation systems (S24).

3-mark questions

Humidification and its uses (S17). Need for humidification in an intubated patient (W17). Three conditions in which nebulisers are used, with purpose (S19). Define humidification and nebulisation (S21). Metered-dose inhalers (W25).

Test yourself

3 m **Why must gas be humidified for an intubated patient?**

Answer
The tube bypasses the nose and upper airway that normally warm and humidify air. Dry, cold gas thickens secretions, slows the cilia, damages the airway lining, blocks the tube and causes collapse. Humidification restores near-normal conditions (heated humidifier 33–44 mg/L, or an HME giving at least 30 mg/L).

MCQ **The droplet size best for reaching small airways is:**

A. Over 10 µm

B. 5–10 µm

C. 1–5 µm

D. Under 0.1 µm

Answer
C. 1–5 µm is the respirable range.

MCQ **The part of a jet nebuliser that removes large droplets is the:**

A. Piezoelectric crystal

B. Baffle

C. Mesh

D. Metering valve

Answer
B. Large droplets hit the baffle and fall back into the cup.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Restrepo RD, Walsh BK. Humidification during invasive and noninvasive mechanical ventilation: 2012. AARC clinical practice guideline. *Respir Care* 2012;57:782–8.
3. O'Driscoll BR, et al. BTS guideline for oxygen use in adults in healthcare and emergency settings. *Thorax* 2017 (air-driven nebulisers in patients at risk of hypercapnia).
4. Cates CJ, et al. Holding chambers (spacers) versus nebulisers for beta-agonist treatment of acute asthma. Cochrane Review 2013.
5. Global Initiative for Asthma (GINA) 2026: inhaler technique guidance.
6. Structure and classical content: Hough, *Physiotherapy in Respiratory and Cardiac Care*; Frownfelter & Dean (2022).

Electrotherapy and therapeutic exercise in cardiovascular and respiratory conditions

> Wound-healing modalities were asked four times (S16, S22, S24, S25), and exercise for circulation runs through the PAD, venous and mastectomy answers. This card collects the electrotherapy and exercise that the syllabus lists for heart, lung and vascular patients, with the precautions that matter most in them (pacemakers, ischaemic limbs, anticoagulants).

Syllabus	Ch 4(h) Electrotherapeutic modalities for pain, swelling and wound healing; 4(i) Therapeutic exercise to alleviate pain, achieve mobility, correct posture and improve peripheral circulation
Priority	**Moderate.** Wound-healing modalities in 4 papers; exercise for circulation inside PAD, venous and mastectomy answers.
Asked as	7m S24 · S25 · 3m S16 · S22
Usual split	3-mark modalities with parameters; 7-mark modalities or exercise programmes
Checked against	Systematic review of TENS after thoracic surgery (2012); reviews of neuromuscular electrical stimulation in COPD (2014, 2016); compression thresholds by ABPI used in venous ulcer care; International Society of Lymphology consensus (2020). Full list under Sources. Content checked 19 October 2026.

1 Pain

- **TENS after chest surgery:** electrodes on either side of and parallel to a thoracotomy, sternotomy or abdominal wound (not on it); conventional (high-frequency, about 80–100 Hz, short pulses, strong comfortable tingling) during breathing exercises and coughing. A systematic review of 11 trials found TENS gave better pain relief after thoracic surgery than standard pain relief alone; it is an add-on to medicines, not a replacement.
- **Supported cough and positioning:** a pillow over the wound, side-lying with the wound uppermost, relaxed upright sitting.
- **Cold** for acute musculoskeletal chest-wall pain and after minor trauma; **superficial heat** for muscle spasm in the neck, shoulders and back of breathless patients, with care (see precautions).
- **Relaxation and breathing control** lower pain-related muscle guarding and anxiety.

2 Swelling

Method	Use	Care
Elevation and muscle-pump exercise	Post-operative, venous and lymphatic swelling	Not elevation in severe arterial disease (worsens rest pain)
Compression bandages, stockings and garments	Venous disease and ulcers, lymphoedema, after DVT, burns scars	Check ABPI first: full compression usually needs an ABPI of about 0.8 or more; reduced compression with specialist advice between about 0.5 and 0.8; none below about 0.5
Intermittent pneumatic compression	DVT prevention in bed-bound patients; add-on in lymphoedema; some arterial disease (specialist devices)	Not over a known untreated DVT, infection or uncontrolled heart failure
Manual lymph drainage and decongestive therapy	Lymphoedema after mastectomy	Details on the mastectomy page (p. 52)
Cold	Acute traumatic swelling	Not on numb or ischaemic skin

3 Wound healing (S16, S22, S24, S25)

Electrical stimulation (high-voltage pulsed current), therapeutic ultrasound, ultraviolet and low-level laser, with typical parameters and effects, are tabulated on the diabetes and wound page (p. 153), together with the factors that affect healing and pressure-sore care. Electrical stimulation has the best evidence of these for slow-healing pressure ulcers.

4 Electrical stimulation of weak muscles

- **Neuromuscular electrical stimulation (NMES)** of the quadriceps produces visible contractions without breathlessness, so it suits very breathless or bed-bound patients: severe COPD, heart failure, ICU and after major surgery.
- **Evidence is mixed:** a 2016 meta-analysis of nine trials found better quadriceps strength and exercise capacity in moderate to severe COPD, while a 2014 review found no clear benefit; a 2023 review after heart and abdominal surgery found a small gain in leg strength. Use it as a bridge to active exercise.
- **Practical:** electrodes over the quadriceps; a frequency that gives a smooth, strong contraction; on–off cycles; about 20–30 minutes, several times a week; progress intensity as tolerated.

5 Precautions for electrotherapy in these patients

Check before every modality

- **Pacemakers and ICDs:** no shortwave or microwave diathermy; TENS and other stimulation only away from the chest and device, with the device team's advice.
- **Ischaemic limbs (PAD, diabetes):** no heat (the skin cannot lose heat and burns); check sensation and circulation first.
- **Anticoagulants and bleeding:** gentle handling; avoid vigorous percussion and deep massage.
- **DVT:** no massage or heat over the affected limb until treated.
- **Cancer:** no heat, ultrasound or stimulation over a known tumour.
- Never over the front of the neck (carotid sinus), the pregnant abdomen, broken or numb skin, or in a confused patient who cannot report discomfort.

6 Therapeutic exercise programmes

Aim	Examples
Relieve pain	Gentle active range of motion, relaxation, breathing control, posture correction, graded activity
Mobility	Shoulder range after thoracotomy, mastectomy and CABG (within sternal and drain precautions); thoracic mobility exercises (see the breathing exercises card (p. 277)); scar stretching after burns
Correct posture	COPD: rounded shoulders and forward head; after thoracotomy: leaning to the operated side; after mastectomy: raised, protracted shoulder; kyphosis: thoracic extension, scapular retraction, pectoral stretches, mirror feedback
Arterial circulation	Supervised walking to moderate claudication pain, rest, repeat (30–45 minutes, 3 times a week, for at least 12 weeks; see the PAD page (p. 61)); Buerger–Allen exercises (below)
Venous circulation	Ankle pumps and heel raises (calf pump), walking, leg elevation, ankle range to free a stiff ankle; with compression
Lymphatic circulation	Gentle rhythmic exercise of the swollen limb with compression on, deep breathing, then slowly progressive resistance training

Buerger–Allen exercises (classic, for arterial disease)

1. Lie with the legs raised (about 45° or more on a support) until the feet go pale, usually 1–3 minutes.

2. Sit with the legs hanging down until the feet turn pink, about 2–5 minutes, moving the ankles and toes.
3. Lie flat with the legs level, covered, for about 5 minutes.
4. Repeat the cycle several times, a few times a day.

These postural exercises are traditional and still asked about, but the evidence for them is weak; supervised walking has strong evidence and is the first-line exercise for claudication. Avoid them in severe rest pain, ulceration or gangrene.

7 Exam corner

Past questions (MUHS, 2016–2026)

3-mark questions

Three electrotherapy modalities to treat an infected wound (S16). Three modalities with parameters for wound healing (S22).

7-mark questions

Modalities used for wound healing (S24). Factors affecting wound healing and modalities (S25). See the diabetes page (p. 149).

Related

Physiotherapy in Buerger's disease (S17, S22) and claudication (S19) on the PAD page (p. 61); DVT and venous ulcers on the venous page (p. 121).

Test yourself

3 m **How would you use TENS after a posterolateral thoracotomy?**

Answer

Electrodes on either side of and parallel to the incision, not on the wound or drain site; conventional high-frequency TENS at a strong, comfortable tingle during breathing exercises and coughing; alongside analgesia. Check for a pacemaker and intact sensation first.

MCQ **Before applying compression bandaging for a leg ulcer you must check:**

A. Blood sugar

B. ABPI

C. Peak flow

D. Body temperature

Answer

B. Arterial disease must be ruled out; compression on an ischaemic leg can cause ulcers or worse.

MCQ **Heat is contraindicated over:**

A. A tight trapezius in COPD

B. An ischaemic foot in PAD

C. Normal skin

D. A healed scar with normal sensation

Answer

B. Ischaemic skin cannot dissipate heat and burns easily.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Sbruzzi G, et al. Transcutaneous electrical nerve stimulation after thoracic surgery: systematic review and meta-analysis of 11 randomized trials. *Rev Bras Cir Cardiovasc* 2012.
3. Meta-analyses of neuromuscular electrical stimulation in COPD: *Respirology* 2014 (no clear benefit) and *Int J Chron Obstruct Pulmon Dis* 2016 (improved quadriceps strength); thoracic and abdominal surgery review (2023).
4. International Society of Lymphology. The diagnosis and treatment of peripheral lymphedema: 2020 consensus document.
5. Venous ulcer guidance on ABPI thresholds for compression (for example SIGN and wound-care consensus documents).
6. Structure and classical content: Watson T, *Electrotherapy: Evidence-Based Practice*; Frownfelter & Dean (2022); Kisner & Colby, *Therapeutic Exercise*.

Applied yoga in cardio-respiratory conditions

> Yoga is in the syllabus (Ch 4l) and is widely used by Indian patients, but it was not asked on its own in 2016–2026. Know what it can add (better quality of life, breathing control, relaxation), what it cannot replace (exercise training and medicines), and which practices are unsafe for heart and lung patients.

Syllabus	Ch 4(l) Applied yoga in cardio-respiratory conditions; practical: relaxation and yoga skills
Priority	**Syllabus.** Not asked on its own in 2016–2026; possible short note or viva.
Asked as	Not asked 2016–2026
Usual split	Possible 3- or 7-mark note
Checked against	Yoga-CaRe trial (2020); Cochrane review of yoga for asthma (2016); meta-analysis of yoga in COPD (2019); Ministry of AYUSH Common Yoga Protocol. Full list under Sources. Content checked 19 October 2026.

1 What yoga includes

- **Asanas** (postures): stretching, strength and balance.
- **Pranayama** (breathing practices): slow deep breathing, alternate-nostril breathing (anulom-vilom), humming breath (bhramari), ujjayi; forceful practices such as kapalbhati and bhastrika; breath retention (kumbhaka).
- **Relaxation and meditation:** shavasana, yoga nidra, guided relaxation.
- India's Ministry of AYUSH publishes a Common Yoga Protocol of gentle loosening practices, postures, pranayama and meditation.

How it may help: slower, deeper breathing and better breathing control; less anxiety and sympathetic drive; flexibility and posture; a culturally familiar way to stay active.

2 What the evidence shows

Condition	Evidence
Coronary heart disease after MI	Yoga-CaRe (India, 3,959 patients, 2020): yoga-based cardiac rehab improved self-rated health and return to pre-infarct activities, but did not significantly reduce major cardiac events. A good option where usual cardiac rehab is not available
Asthma	Cochrane (2016): moderate-quality evidence that yoga probably gives small improvements in quality of life and symptoms; no clear change in lung function
COPD	Meta-analysis (2019, 11 trials, 586 patients): about 25 m more on the 6MWT and slightly better FEV1, mainly with breathing-focused yoga rather than postures
Hypertension	Trials show modest falls in BP, best with regular breathing and relaxation practice; an add-on to medicines and lifestyle change
Heart failure	Small trials suggest better quality of life and exercise tolerance with gentle, supervised yoga

Claim vs evidence

Common claim	Evidence	Write in the exam
Yoga cures asthma, reverses heart disease and replaces medicines.	Yoga gives small to moderate gains in symptoms, quality of life and fitness. It does not replace medicines, aerobic and strength training or risk-factor control.	"Yoga as an adjunct to standard rehabilitation: breathing-focused pranayama, gentle asanas and relaxation, avoiding breath-holding, straining and head-down postures."

3 A sample supervised session (stable COPD or after cardiac rehab Phase II)

1. **Loosening (5–10 minutes):** neck, shoulder, wrist, ankle and gentle trunk movements with relaxed breathing.
2. **Asanas (15–20 minutes), held comfortably while breathing normally:** tadasana (standing tall), vrikshasana (tree pose, near a wall), ardha kati chakrasana (side bend), supported trikonasana, vajrasana or sitting forward bends if comfortable, bhujangasana (if allowed by surgery and back), with rests.
3. **Pranayama (10 minutes):** slow deep breathing with a longer breath out, anulom-vilom without breath retention, bhramari.
4. **Relaxation (10 minutes):** shavasana or yoga nidra.
5. 3–5 times a week, as part of, not instead of, aerobic and strength training.

4 Precautions and contraindications

Avoid or adapt

- **Breath-holding (kumbhaka) and straining:** they raise BP sharply (Valsalva): avoid in heart disease, hypertension, aneurysm, after eye or brain surgery.
- **Forceful pranayama (kapalbhati, bhastrika):** avoid in uncontrolled hypertension, heart disease, recent abdominal or chest surgery, hernia, pregnancy, epilepsy, and during an asthma or COPD flare (can trigger bronchospasm or hyperventilation).
- **Inversions and head-down postures** (sirsasana, sarvangasana, halasana): avoid in hypertension, heart failure, glaucoma, reflux, neck problems and in breathless patients.
- **After sternotomy:** no weight-bearing on the arms, prone lying on the chest or strong back bends for about 6–8 weeks; follow sternal precautions.
- **Osteoporosis** (long-term steroids): avoid loaded spinal flexion and twisting.
- **Unstable states:** unstable angina, uncontrolled arrhythmia, acute exacerbations, fever: no session.
- Keep taking all medicines; teach the warning signs (chest pain, dizziness, marked breathlessness) and stop if they appear; learn from a trained teacher in a supervised group.

5 Exam corner

Not asked on its own in the 20 papers (2016–2026). It fits the "long-term management" part of COPD, asthma and cardiac answers in one or two lines, with its precautions.

Test yourself

3 m **Give three precautions when using yoga in a patient after myocardial infarction.**

Answer

No breath-holding or straining (Valsalva); no forceful pranayama such as kapalbhati; no inversions; after CABG keep sternal precautions; keep intensity light to moderate and stop for chest pain or dizziness; continue medicines.

MCQ **The Yoga-CaRe trial after MI showed:**

A. Fewer deaths and heart attacks

B. Better self-rated health and return to activities, without a significant fall in major events

C. Harm from yoga

D. No effect on anything

Answer

B.

Sources

1. MUHS BPTh syllabus (2012), CVR Ch 4(l).
2. Prabhakaran D, et al. Effect of yoga-based cardiac rehabilitation on health outcomes following acute myocardial infarction (Yoga-CaRe). *J Am Coll Cardiol* 2020.
3. Yang ZY, et al. Yoga for asthma. Cochrane Review 2016.
4. Cramer H, et al. Yoga for chronic obstructive pulmonary disease: a systematic review and meta-analysis. *Clin Rehabil* 2019.
5. Ministry of AYUSH, Government of India. Common Yoga Protocol.

Handling skills and exercise equipment

> The practical syllabus lists handling techniques (stockings, bandaging, dressings, taping, splints) and equipment skills that are easy to be asked to demonstrate in the short case and viva, but rarely written about. This card gives the steps and the safety checks for each.

Syllabus	Practical 3 (handling: stockings, bandaging, dressing, taping, splints and orthotics); 5 (exercise and monitoring equipment); 7 (monitoring compliance and contextual factors)
Priority	**Practical.** Short-case demonstrations and viva questions; not asked in the theory papers.
Asked as	Practical exam
Usual split	Short-case skill (10 marks)
Checked against	Standard compression therapy and wound-care practice; ACSM guidance on exercise equipment and monitoring. Full list under Sources. Content checked 19 October 2026.

1 Compression stockings

- **Two kinds:** anti-embolism stockings for bed-bound patients (DVT prevention) and graduated compression stockings for walking patients with venous disease or lymphoedema (pressure highest at the ankle, falling up the leg). Pressure classes differ between national standards: follow the prescription (light, medium or strong).
- **Check first:** ABPI (no compression below about 0.5; specialist advice between about 0.5 and 0.8), skin (infection, fragile skin, ulcers), sensation, and uncontrolled heart failure.
- **Measure in the morning**, when swelling is least: ankle at its narrowest, calf at its widest, and leg length (heel to just below the knee crease for below-knee stockings).

Putting on (donning)

1. In the morning, before getting out of bed (or after 15–20 minutes of leg elevation).
2. Turn the stocking inside out down to the heel, leaving the foot part inside.
3. Slide the foot in and centre the heel in the heel pocket.
4. Ease the stocking up the leg a little at a time, without pulling at the top band; rubber gloves help the grip; a frame-type donning aid helps weak hands, arthritis or obesity.
5. Smooth out all wrinkles; the top band sits about two fingers below the knee crease (for below-knee stockings) and is never rolled or folded down (that acts like a tourniquet).

Taking off and care

- Peel down from the top, turning it inside out, then off over the heel; at night unless prescribed day and night.
- Check the skin daily (red marks, blisters, colour, pain, numbness); stop and report problems.
- Wash as instructed; replace every few months, when the elastic slackens.

2 Bandaging

- **Venous ulcer (compression bandaging):** after ABPI; ankle at 90°; padding over bony points; bandage from the base of the toes to just below the knee, including the heel, in a spiral or figure-of-eight with about 50% overlap, firmest at the ankle; check toe colour and warmth after applying.
- **Lymphoedema:** multilayer short-stretch bandaging (see the mastectomy page (p. 52)).
- **Amputation stump:** figure-of-eight, firmest at the end, no circular turns (see the amputation page (p. 115)).
- **Always:** loosen and report if there is pain, numbness, tingling, a pale or blue toe or finger, or swelling above the bandage.

3 Working around dressings

- Physiotherapists do not usually change complex dressings, but must work safely around them: avoid shear and friction on dressings and grafts, support the wound during coughing, and keep dressings dry.
- Time exercise with dressing changes in burns, when the wound can be seen and analgesia is working (see the burns page (p. 40)).
- **Report:** blood or fluid soaking through, a loose or wet dressing, redness spreading from the wound, heat, pus, a bad smell, fever, or a wound opening.
- **Common types:** gauze, clear film, hydrocolloid, foam (heavy exudate), alginate (very wet wounds), silver (infected wounds).

4 Taping

- **Kinesiology tape** is sometimes used after breast cancer surgery for mild lymphoedema and for posture (scapular retraction), and on the chest wall for musculoskeletal pain. Evidence is limited; it does not replace compression for lymphoedema.
- **Before taping:** clean, dry, intact skin; test a small piece for allergy; not over wounds, drains, grafts, radiotherapy-damaged or very fragile skin, or areas with poor sensation.
- **Remove** if itching, redness or blistering appears; peel slowly along the hair, supporting the skin.

5 Splints, binders and orthoses

Device	Used for	Check
Burns splints (resting hand, neck conformer, axillary, knee and ankle)	Hold anti-deformity positions	Pressure areas and fit daily; swelling changes the fit
Ankle–foot splint or orthosis	Prevent foot drop and heel sores in ICU, GBS, head and spinal injuries	Heel off the mattress; skin checks every shift
Abdominal binder	Tetraplegia (better breathing and BP when sitting); support after abdominal surgery if the surgeon advises	Over the lower ribs and abdomen, not over the chest; remove for skin checks
Sternal support vest	Some patients after sternotomy (large breasts, obesity, coughing)	Comfort; does not replace sternal precautions
Cervical collar, thoracolumbar brace	Spinal stability after injury or surgery	Worn as the surgeon orders; log-roll; skin under the collar
Pressure garments	Burns scars, lymphoedema	Worn about 23 hours a day; replaced as they loosen

6 Exercise and monitoring equipment

Equipment	How to use it safely
Treadmill	Show the emergency stop and safety clip; start slow and flat; light hand support only; increase speed or gradient in small steps; slow down before stopping
Cycle and arm ergometers	Seat height so the knee is slightly bent at the bottom of the pedal stroke; start with little resistance; arm ergometer at shoulder height or below (after sternotomy, only when allowed)
Heart-rate monitor	Chest-strap monitors are more accurate than wrist devices during exercise; check against the pulse; heart rate is unreliable with beta-blockers and atrial fibrillation
Pulse oximeter	Warm finger, no nail polish, wait for a steady trace; during walking use an ear probe or recheck at rest if the signal is poor
Sphygmomanometer	Right cuff size, arm supported at heart level, patient seated and rested; recheck high readings

Equipment	How to use it safely
Free weights and resistance bands	Choose the load from 1RM or the repetitions possible; breathe out on effort; progress when 2 extra repetitions are easy in 2 sessions
Pressure biofeedback unit	An inflatable pad with a gauge, placed under the lower back (supine, commonly inflated to about 40 mmHg) or the abdomen (prone); the patient keeps or lowers the pressure while drawing in the lower abdomen. Used for deep abdominal (core) training and posture, for example after abdominal surgery once healed and in back pain with breathing pattern problems
Balance boards, Swiss (therapy) balls	Balance and trunk training in deconditioned and older patients, after ICU stays and in pulmonary rehab; near a rail or wall with supervision; avoid in severe breathlessness, unstable BP or high fall risk

7 Monitoring adherence and contextual factors

- **How to monitor:** exercise and symptom diaries, step counts on a phone or pedometer, attendance, checking inhaler and breathing technique at each visit, and asking open questions ("How many days did you walk last week?").
- **Common barriers:** cost and travel, work hours, fear of breathlessness or chest pain, low mood, poor understanding, family beliefs ("rest is best"), other illnesses (arthritis, diabetes, memory problems).
- **Helps:** goals the patient chose; a short written or picture programme in the patient's language; linking exercise to daily routines; family involvement; phone or WhatsApp reminders and video reviews; praise for small gains; reviewing and adjusting the plan.

Sources

1. MUHS BPTh syllabus (2012), CVR practical skills list.
2. Compression therapy and venous ulcer guidance (for example SIGN guideline on chronic venous leg ulcers; wound-care consensus documents on ABPI thresholds).
3. International Society of Lymphology. The diagnosis and treatment of peripheral lymphedema: 2020 consensus document.
4. American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription* (equipment, monitoring).
5. Structure and classical content: Kisner & Colby, *Therapeutic Exercise*; Hillegass (2022).

Exercise testing: contraindications, stop criteria, Bruce, cycle, walk and step tests

> Exercise testing is in 9 of 20 papers. The single most repeated question is "absolute and relative contraindications to exercise testing" (W18, W20, W19, S23, S26). Add the Bruce protocol (S16), cycle ergometer (S22), 6MWT versus shuttle walk (W22) and the Harvard step test from the syllabus.

Syllabus	Ch 2(b) Stress testing: six-minute walk test and Harvard step test (skill and interpretation); shuttle walk test and modified Bruce protocol (interpretation only)
Priority	**High yield.** In 9 of 20 papers (2016–2026).
Asked as	7m W18 · S19 · W20 · S26 · 3m S16 · S22 · W22 · W19 · S23
Usual split	7-mark contraindications (absolute and relative) or tests with advantages and disadvantages; 3-mark Bruce, cycle ergometer, 6MWT vs shuttle
Checked against	AHA exercise standards for testing and training (2013) as used by ACSM; ATS six-minute walk test statement (2002); ERS/ATS field walking test standard (2014). Full list under Sources. Content checked 19 October 2026.

1 Why test exercise capacity

- **Diagnosis:** exercise-induced ischaemia, arrhythmia, desaturation or bronchospasm.
- **Prognosis and risk stratification:** before cardiac rehab (see the IHD page (p. 15)), before surgery.
- **Functional capacity:** how much the patient can do (METs, distance).
- **Exercise prescription:** safe heart-rate range, walking speed, need for oxygen.
- **Outcome measure:** response to rehabilitation or treatment.

Types: laboratory tests (treadmill and cycle ergometer, with ECG; cardiopulmonary exercise test with gas analysis) and field tests (six-minute walk, shuttle walks, step tests).

2 Absolute and relative contraindications (7 marks in W18, W20, S26; 3 marks in W19, S23)

Absolute contraindications (do not test)

- Acute myocardial infarction within 2 days
- Ongoing unstable angina
- Uncontrolled cardiac arrhythmia with symptoms or poor circulation
- Active endocarditis
- Symptomatic severe aortic stenosis
- Decompensated (uncontrolled) heart failure
- Acute pulmonary embolism, pulmonary infarction or deep vein thrombosis
- Acute myocarditis or pericarditis
- Acute aortic dissection
- Physical disability that makes safe, adequate testing impossible

Relative contraindications (test only if benefit outweighs risk)

- Known left main coronary artery stenosis
- Moderate to severe aortic stenosis with uncertain symptoms
- Fast heart rhythm with uncontrolled ventricular rate
- Complete or advanced heart block
- Hypertrophic obstructive cardiomyopathy with a severe resting gradient
- Recent stroke or transient ischaemic attack
- Mental impairment limiting co-operation
- Resting BP above **200/110 mmHg**
- Uncorrected conditions: significant anaemia, electrolyte imbalance, overactive thyroid

Memory hook for the absolute list

"Acute heart, acute clot, tight valve"

Acute heart: MI within 2 days, unstable angina, arrhythmia, heart failure, myocarditis, pericarditis, endocarditis. Acute clot or tear: PE, DVT, aortic dissection. Tight valve: symptomatic severe aortic stenosis. Plus: unable to exercise safely.

3 When to stop a test

Absolute (stop)	Relative (consider stopping)
ST elevation over 1 mm in leads without old Q waves (except aVR, aVL, V1)	Marked ST depression (over 2 mm, horizontal or downsloping)
Systolic BP falls more than 10 mmHg despite rising workload, with other signs of ischaemia	Systolic BP falls more than 10 mmHg without other signs of ischaemia
Moderate to severe angina	Increasing chest pain
Dizziness, near fainting, unsteadiness	Fatigue, breathlessness, wheeze, leg cramps or claudication
Signs of poor perfusion: pallor, cyanosis	Hypertensive response: systolic over 250 or diastolic over 115 mmHg
Sustained ventricular tachycardia or other serious arrhythmia	Other arrhythmias (frequent or multifocal ectopics, supraventricular tachycardia, slow rhythms); new bundle branch block
Technical failure of ECG or BP monitoring; the patient asks to stop	–

4 Treadmill and cycle tests

Bruce protocol (3 marks in S16)

A symptom-limited treadmill test in which **speed and gradient both increase every 3 minutes**, with ECG, BP and symptoms monitored. The result is the exercise time, the stage reached (METs), and any ischaemic changes, arrhythmia or BP problems.

Stage (3 min each)	Speed	Gradient	Approximate METs
1	1.7 mph (2.7 km/h)	10%	about 5
2	2.5 mph (4.0 km/h)	12%	about 7
3	3.4 mph (5.5 km/h)	14%	about 10

Stage (3 min each)	Speed	Gradient	Approximate METs
4	4.2 mph (6.8 km/h)	16%	about 13
5	5.0 mph (8.0 km/h)	18%	about 15
6 and 7	5.5 and 6.0 mph	20% and 22%	athletes

Modified Bruce (for older, deconditioned or cardiac patients, early after MI): two gentler starting stages at 1.7 mph with 0% and then 5% gradient, before stage 1 of the standard protocol. It avoids the big jump in workload at the start.

Cycle ergometer testing (3 marks in S22)

Advantages	Disadvantages
Workload is known exactly in watts and does not depend on body weight	Thigh muscles tire before the heart and lungs reach their limit, so peak oxygen uptake is often about 10–20% lower than on a treadmill
Upper body is still, so ECG and BP are easier to record	Unfamiliar to people who do not cycle; needs a steady pedalling rate
Safer for poor balance, obesity, gait or orthopaedic problems; cheaper, smaller and quieter	Less like walking, so less useful for walking-based prescription

5 Field walking tests (3 marks in W22)

Six-minute walk test (6MWT)

- Walk back and forth along a flat, straight course of at least **30 m**, as far as possible in 6 minutes, at the patient's own pace; rests are allowed; standard encouragement every minute.
- Record distance, SpO_2 and heart rate (before, during, after), Borg breathlessness and fatigue before and after, and any stops. Do a practice test where possible.
- **Do not test (ATS 2002):** unstable angina or MI in the previous month. **Caution:** resting heart rate above 120, BP above 180/100 mmHg.
- **Stop for:** chest pain, intolerable breathlessness, leg cramps, staggering, sweating, pale or ashen appearance.
- **Meaning of a change:** in chronic respiratory disease, a change of about **25–33 m** (often quoted as 30 m) is the smallest that matters to patients.

Shuttle walk tests

- **Incremental shuttle walk test (ISWT):** walk around two cones on a **10 m** course in time with audio beeps; speed increases every minute through **12 levels**; the test ends when the patient cannot reach the cone in time or stops with symptoms. A maximal, symptom-limited test; distance relates closely to peak oxygen uptake.
- **Endurance shuttle walk test (ESWT):** constant speed set from the ISWT result (a high percentage of peak); measures endurance time; sensitive to change after rehab.

6MWT versus shuttle walk test

Feature	6MWT	ISWT
Pacing	Self-paced	Externally paced by beeps
Intensity	Submaximal; reflects daily walking	Incremental to maximum
Course	30 m or more	10 m
Result	Distance in 6 minutes	Distance or level reached
Encouragement	Standard phrases only	None needed; the beeps set the pace
Best use	Functional capacity, desaturation, outcome	Peak capacity, setting training intensity

6 Harvard step test

- **Method (original):** step up and down a **20-inch (about 50 cm)** step at **30 steps a minute** for 5 minutes or until exhausted (keeping time with a metronome). Lower steps are used in modified versions for women and patients.
- **Recovery pulse:** count the pulse for 30 seconds at 1–1.5, 2–2.5 and 3–3.5 minutes after stopping.
- **Fitness index (long form)** = (duration in seconds × 100) ÷ (2 × sum of the three pulse counts).
- **Short form** = (duration in seconds × 100) ÷ (5.5 × the 1–1.5 minute pulse count).
- **Interpretation (long form):** below 55 poor; 55–64 low average; 65–79 high average; 80–89 good; 90 or more excellent.
- **Use:** a quick fitness test for healthy people; not suitable for most cardiac or respiratory patients (high, uncontrolled intensity).

> **Worked example**
>
> Completes 300 seconds; pulse counts 70, 60, 50. Index = (300 × 100) ÷ (2 × 180) = 30000 ÷ 360 = **83**: good.

7 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Absolute and relative contraindications of the exercise tolerance test (W18). Various exercise tests, with advantages and disadvantages of each (S19). Absolute and relative contraindications for exercise testing (W20, S26).

3-mark questions

Bruce protocol (S16). Three advantages and disadvantages of cycle ergometer stress testing (S22). Compare the six-minute walk test and shuttle walk test (W22). Absolute contraindications for exercise testing (W19). Three relative and absolute contraindications (S23).

Test yourself

3 m **List five absolute contraindications to exercise testing.**

> **Answer**
>
> MI within 2 days; unstable angina; uncontrolled symptomatic arrhythmia; symptomatic severe aortic stenosis; decompensated heart failure; acute PE or DVT; acute myocarditis or pericarditis; active endocarditis; acute aortic dissection (any five).

MCQ **Stage 1 of the Bruce protocol is:**

A. 1.7 mph at 0%

B. 1.7 mph at 10%

C. 2.5 mph at 12%

D. 3.4 mph at 14%

> **Answer**
>
> **B.** 1.7 mph at 10%, for 3 minutes. 1.7 mph at 0% is the first stage of the modified Bruce.

MCQ **Which is a relative, not absolute, contraindication?**

A. Acute pulmonary embolism

B. Resting BP 210/115 mmHg

C. Unstable angina

D. Acute aortic dissection

Answer

B. Resting BP above 200/110 is relative; the others are absolute.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Fletcher GF, et al. Exercise standards for testing and training: a scientific statement from the American Heart Association. *Circulation* 2013;128:873–934 (contraindications and termination criteria, as used in ACSM's Guidelines).
3. ATS statement: guidelines for the six-minute walk test. *Am J Respir Crit Care Med* 2002;166:111–7.
4. Holland AE, Spruit MA, et al. An official ERS/ATS technical standard: field walking tests in chronic respiratory disease. *Eur Respir J* 2014;44:1428–46; Singh SJ, et al. Development of a shuttle walking test of disability in patients with chronic airways obstruction. *Thorax* 1992;47:1019–24.
5. Brouha L. The step test: a simple method of measuring physical fitness for muscular work in young men. *Res Q* 1943 (Harvard step test).
6. Structure and classical content: ACSM's Guidelines for Exercise Testing and Prescription; Hillegass (2022).

Exercise physiology and prescription: METs, energy systems, intensity, training

> Exercise physiology appears as 3-mark definitions (MET in W22, S24 and S25; aerobic vs anaerobic in W22; Karvonen in W25) and 7-mark explanations (energy systems in S24; methods of measuring exercise intensity in S23; ageing and fitness in S16). These numbers also run through every cardiac and pulmonary rehab answer.

Syllabus	Ch 3 Exercise physiology: bioenergetics, MET, acute and chronic adaptations, aerobic and anaerobic training, principles of exercise prescription
Priority	**Moderate.** In 6 of 20 papers (2016–2026).
Asked as	7m S16 · S23 · S24 · 3m W22 ×2 · S24 · S25 · W25
Usual split	3-mark MET, Karvonen, aerobic vs anaerobic; 7-mark energy systems, intensity methods, ageing
Checked against	2024 Adult Compendium of Physical Activities (MET values); ACSM intensity classification; WHO physical activity guidelines (2020). Full list under Sources. Content checked 19 October 2026.

1 The MET (3 marks in W22, S24, S25)

One **metabolic equivalent (MET)** is the energy used at rest, sitting quietly: about **3.5 mL of oxygen per kg of body weight per minute** (roughly 1 kcal per kg per hour). An activity of 4 METs uses four times the resting energy.

MET values of daily activities (2024 Adult Compendium)

Activity	METs
Eating, sitting	1.5
Bathing, sitting	1.5
Showering and towelling off, standing	2.0
Grooming (washing hands, shaving, brushing teeth)	2.0
Using the toilet	2.3
Dressing and undressing	2.8
Walking slowly, about 2 mph (3.2 km/h)	2.8
Walking at a moderate pace, about 3 mph (5 km/h)	3.8
Climbing stairs slowly	4.5
Brisk walking, about 3.5–3.9 mph	4.8
Climbing stairs, general	6.8

Uses: grading activity intensity (light below 3 METs, moderate 3–5.9, vigorous 6 or more); Phase I cardiac rehab goes up to about 3 METs; a good functional capacity is 7 METs or more; sexual activity is reasonable when 3–5 METs are tolerated.

2 Energy systems (7 marks in S24)

Muscles run on ATP, but they store only enough for a few seconds. Three systems remake it, overlapping all the time; which one leads depends on how hard and how long the effort is.

System	Fuel and process	Leads for	Examples
ATP–PC (phosphagen)	Creatine phosphate quickly remakes ATP; no oxygen needed	Maximal efforts up to about 10 seconds	100 m sprint, a jump, a heavy single lift, a fast bowler's run-up and delivery
Anaerobic glycolysis (lactic acid system)	Glucose and glycogen broken down without oxygen to lactate; fast but little ATP	Hard efforts of about 10 seconds to 2 minutes	400 m run, 100 m swim, climbing several flights of stairs fast
Aerobic (oxidative)	Carbohydrate and fat (some protein) fully broken down with oxygen in mitochondria; slow but large ATP yield	Efforts longer than about 2 minutes, and all rest	Walking, jogging, cycling, marathon, most daily activities

Fuel mix: fat dominates at rest and low intensity; carbohydrate takes over as intensity rises. The respiratory quotient (CO_2 produced ÷ O_2 used) is about 1.0 for carbohydrate and 0.7 for fat.

3 Aerobic and anaerobic exercise (3 marks in W22)

Feature	Aerobic	Anaerobic
Oxygen	Energy made with oxygen	Energy made mainly without oxygen
Intensity and duration	Low to moderate, long (minutes to hours)	High, short (seconds to about 2 minutes)
Fuel and by-products	Fat and carbohydrate; CO_2 and water	Creatine phosphate and glucose; lactate
Muscle fibres	Type I (slow-twitch)	Type II (fast-twitch)
Training effect	Endurance: more capillaries and mitochondria, higher VO_2max	Power, speed, strength, tolerance to lactate
Examples	Walking, cycling, swimming	Sprinting, weightlifting, jumping

4 Methods of measuring exercise intensity (7 marks in S23)

ACSM intensity categories

Intensity	% heart-rate reserve	% maximum heart rate	RPE (6–20)	METs
Light	30–39%	57–63%	9–11	2.0–2.9
Moderate	40–59%	64–76%	12–13	3.0–5.9
Vigorous	60–89%	77–95%	14–17	6.0 or more

1. **Heart-rate reserve (Karvonen) method (W25):** target HR = resting HR + intensity × (maximum HR – resting HR). Maximum HR is estimated as 220 – age (or 208 – 0.7 × age, Tanaka). Example: age 60, resting HR 70: maximum 160, reserve 90; 50% = 70 + 45 = **115**.
2. **Percentage of maximum heart rate:** simple but less precise.
3. **Oxygen uptake:** % of VO_2max or VO_2 reserve, from a cardiopulmonary exercise test (the gold standard).
4. **METs:** choose activities by their MET value; on a treadmill or cycle, set the workload in METs or watts.
5. **Rating of perceived exertion:** Borg 6–20 (moderate 12–13) or CR10 breathlessness (moderate to strong, 3–5 in pulmonary rehab).
6. **Talk test:** moderate if the patient can talk but not sing; vigorous if only a few words.
7. **Field-test speed:** walking at about 80% of the speed reached on the 6MWT or shuttle test.
8. **Thresholds:** ventilatory or lactate threshold; ischaemic threshold (train at least 10 beats below the heart rate where angina or ST change appears).
9. **Strength training:** % of one-repetition maximum (1RM) or the number of repetitions possible.

Use RPE and the talk test when heart rate is unreliable: beta-blockers, atrial fibrillation, pacemakers, heart transplant.

5 Principles of training and adaptations

- **FITT-VP:** Frequency, Intensity, Time, Type, Volume, Progression.
- **Principles:** overload (work above usual levels), specificity (you improve what you train), progression, individuality, reversibility (gains fade within weeks of stopping), warm-up and cool-down.

System	Acute response (during exercise)	Chronic adaptation (after weeks of training)
Heart	HR, stroke volume and cardiac output rise; systolic BP rises, diastolic stays similar	Lower resting and submaximal HR; larger stroke volume; higher maximal cardiac output; lower resting BP
Lungs	Breathing rate and depth rise; minute ventilation rises	Less ventilation needed for the same work; less breathlessness
Muscles	Blood flow to working muscles rises; more oxygen extracted	More capillaries, mitochondria and oxidative enzymes; higher VO_2max; strength and size with resistance training
Metabolism	Glucose taken up by muscle	Better insulin sensitivity, lipids (higher HDL), weight control

6 Cardio-respiratory changes with ageing and a fitness programme (7 marks in S16)

System	Changes with age
Heart and vessels	Lower maximum heart rate; stiffer arteries, so higher systolic BP; slower heart-rate response; VO_2max falls (faster in inactive people)
Lungs and chest wall	Lungs lose elastic recoil; chest wall stiffens; residual volume rises and vital capacity falls; small airways close earlier; weaker breathing muscles; slightly lower PaO_2
Muscle and bone	Loss of muscle mass and strength (sarcopenia), especially fast fibres; bone loss; stiffer joints; poorer balance

Fitness programme for older adults (WHO 2020): 150–300 minutes a week of moderate aerobic activity (or 75–150 minutes vigorous); muscle-strengthening on 2 or more days; varied activity that emphasises balance and strength on 3 or more days a week to prevent falls; start low and progress slowly; longer warm-up and cool-down; check medicines and heart conditions first. Regular exercise slows the loss of fitness, strength and independence with age.

7 Exam corner

Past questions (MUHS, 2016–2026)

7-mark questions

Cardio-respiratory changes with ageing and a fitness programme (S16). Methods of measuring exercise intensity (S23). Types of energy systems with examples (S24).

3-mark questions

Define one MET and state its value (W22). Three differences between aerobic and anaerobic exercise (W22). Define MET and give the METs of two daily activities (S24). Metabolic equivalent (S25). Karvonen's formula in cardiac rehabilitation (W25; worked on the IHD page).

Test yourself

3 m **Define a MET and give the METs of two daily activities.**

Answer
One MET is the resting energy use, about 3.5 mL O_2/kg/min. Eating sitting 1.5 METs; showering 2.0; dressing 2.8; walking at a moderate pace 3.8 (any two).

3 m **A 50-year-old with a resting HR of 80 trains at 60% intensity. Find the target HR by Karvonen.**

Answer
Maximum HR = 220 − 50 = 170. Reserve = 170 − 80 = 90. Target = 80 + 0.6 × 90 = 80 + 54 = **134** beats per minute.

MCQ **A fast bowler's single delivery mainly uses:**

A. Aerobic system

B. ATP–PC system

C. Fat oxidation

D. Lactic acid system only

Answer
B. A maximal effort of a few seconds runs on stored ATP and creatine phosphate.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Herrmann SD, et al. 2024 Adult Compendium of Physical Activities: a third update of the energy costs of human activities. *J Sport Health Sci* 2024. pacompendium.com
3. American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription* (intensity classification, FITT-VP).
4. Tanaka H, Monahan KD, Seals DR. Age-predicted maximal heart rate revisited. *J Am Coll Cardiol* 2001;37:153–6.
5. World Health Organization. WHO guidelines on physical activity and sedentary behaviour (2020).
6. Structure and classical content: McArdle, Katch & Katch, *Exercise Physiology*; Hillegass (2022).

Investigations: chest X-ray, PFT, ECG and bedside tests

> Investigations are asked as 3-mark questions (X-ray of pleural effusion, emphysema, COPD and tetralogy of Fallot; PFT in obstructive and restrictive disease; ECG criteria for MI; ABI) and they are the spots in the practical exam (chest X-ray, ABG, PFT, ECG). Learn a fixed order for reading each one and the physiotherapy meaning of each finding.

Syllabus	Ch 2(a) Investigation and clinical implication: X-ray, PFT, ABG, ECG, ABI, claudication time, pulses, auscultation, postural hypotension
Priority	**High yield.** 3-mark questions in most papers, and four of the five practical spots.
Asked as	7m W24 (ABG, PFT, X-ray in COPD) · 3m S16 · W18 · S19 · W19 · W20 · S21 · S24 · W24
Usual split	3-mark findings; practical spots
Checked against	ERS/ATS interpretive strategies for lung function tests (2022); consensus definition of orthostatic hypotension (2011); Fourth Universal Definition of MI (2018); standard chest radiograph and ECG interpretation. Full list under Sources. Content checked 19 October 2026.

1 Chest X-ray

A fixed order for reading

1. **Details and quality:** name, date, PA or AP (AP films, taken in bed, make the heart look bigger); rotation (the spinous processes sit midway between the clavicle ends); inspiration (about 6 anterior or 10 posterior ribs above the right diaphragm); exposure.
2. **A, Airway:** trachea central or shifted.
3. **B, Breathing (lungs and pleura):** compare left and right zone by zone; lung markings reaching the edge; pleural lines; costophrenic angles sharp.
4. **C, Cardiac:** heart size (cardiothoracic ratio below 0.5 on a PA film), borders, mediastinum.
5. **D, Diaphragm:** the right is usually slightly higher; flattened in hyperinflation; free air under it.
6. **E, Everything else:** bones (fractures, sternal wires), soft tissues (surgical emphysema), lines and tubes.

Silhouette sign: when a white area touches a border and blurs it, it lies next to that border. Loss of the right heart border means right middle lobe disease; loss of a hemidiaphragm means lower lobe disease.

Finding	What you see	Physio meaning
Consolidation	White area with air bronchograms; volume kept	Early pneumonia: positioning and mobility; airway clearance only when sputum is retained
Collapse (atelectasis)	White, smaller lobe; fissure, trachea and mediastinum pulled towards it; raised hemidiaphragm	Target that lobe: positioning, deep breaths with a hold, airway clearance
Pleural effusion	Dense white base with a concave upper edge (meniscus); blunted costophrenic angle; large ones push the mediastinum away	Lie on the good side, thoracic expansion; drainage first if large
Pneumothorax	Black area with no lung markings beyond a thin pleural line; tension: mediastinum pushed away	No positive-pressure techniques until drained; tension is an emergency
Hyperinflation (COPD)	Dark lungs, flat diaphragms, more than 6 anterior ribs visible, narrow heart	Forward-lean positions, pursed-lip breathing; bullae mean no high pressures

Finding	What you see	Physio meaning
Pulmonary oedema	Haziness spreading from both hila ("bat's wing"), Kerley B lines, large heart, effusions	Sit up; no airway clearance for oedema fluid
Interstitial lung disease	Fine net-like (reticular) shadows at the bases, small lungs	Expect exercise desaturation: oxygen during training
Bronchiectasis	Thick-walled airways seen as tram lines and ring shadows	Daily airway clearance for the affected lobes
Lung abscess	Round cavity with an air–fluid level	Drainage with the abscess uppermost; never the good lung down if it may spill
Lines and tubes	Endotracheal tube tip a few cm (about 5) above the carina; nasogastric tube below the diaphragm; chest drain in the pleural space	A tube in the right main bronchus collapses the left lung; check before treating

Condition-specific films: emphysema and COPD on the COPD page (p. 76); pleural effusion and pneumothorax on the pleural page (p. 174); tetralogy of Fallot (boot-shaped heart) on the congenital page (p. 192).

2 Pulmonary function tests

Measure	Meaning
FVC	The total volume blown out as hard and fast as possible after a full breath in
FEV1	The volume blown out in the first second
FEV1/FVC	Normally about 0.75–0.80 in adults; low in obstruction
PEF	Peak expiratory flow: the fastest flow; used for home asthma diaries
TLC, RV, FRC	Lung volumes (body box or gas dilution): high in hyperinflation, low in restriction
DLCO	Diffusing capacity: low in emphysema, ILD and pulmonary vascular disease; normal in asthma

Patterns

Pattern	FEV1	FVC	FEV1/FVC	TLC	Examples
Obstructive	Low	Normal or low	Low (below 0.70, or below the lower limit of normal)	Normal or high	COPD, asthma, bronchiectasis
Restrictive	Low	Low	Normal or high	Low (needed to confirm)	ILD, pleural thickening, kyphoscoliosis, obesity, neuromuscular weakness, after lung resection
Mixed	Low	Low	Low	Low	Post-TB lung disease, cystic fibrosis

- **Bronchodilator response:** traditionally an FEV1 or FVC rise of at least 12% and 200 mL after salbutamol (ATS/ERS 2005); the 2022 ERS/ATS standard uses a rise of more than 10% of the predicted value. Large responses suggest asthma.
- **Flow–volume loop:** a "scooped" (concave) expiratory curve in obstruction; a small, narrow loop in restriction; flattening in upper airway obstruction.
- **Peak flow variability** through the day supports asthma; a personal best guides the action plan.
- **Read the S25 case:** FVC 60%, FEV1 82%, FEV1/FVC 94% predicted: both volumes reduced with a high ratio, so restrictive (old TB scarring).

Severity grading in COPD (GOLD 1–4 by FEV1 % predicted) is on the COPD page (p. 73); the spirometry curves are drawn on the COPD page (p. 76).

3 ECG

Basics

- Paper speed 25 mm a second: one small square = **0.04 s**, one large square = **0.2 s**; 10 mm = 1 mV.
- **P wave** = atria depolarise; **QRS** = ventricles depolarise; **T wave** = ventricles repolarise.
- **Normal intervals:** PR 0.12–0.20 s (3–5 small squares); QRS below 0.12 s (under 3 small squares); QTc below about 0.44 s in men and 0.46 s in women.
- **Heart rate:** 300 ÷ number of large squares between two R waves (regular rhythm), or the number of QRS complexes in a 10-second strip × 6 (irregular rhythm).
- **Order:** rate, rhythm (regular?), P before every QRS?, PR, QRS width, ST and T changes.

Rhythm strips to recognise (simplified)

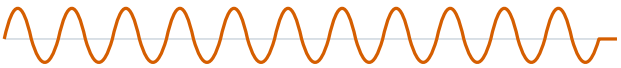
Normal sinus rhythm (75 a minute)
P before every QRS; regular



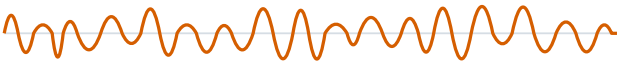
Atrial fibrillation
no P waves; irregularly irregular; wavy baseline



Ventricular tachycardia
fast, wide, regular complexes: emergency



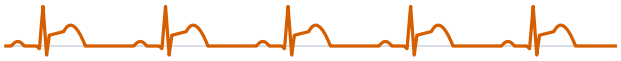
Ventricular fibrillation
chaotic, no QRS: start CPR, get the defibrillator



Complete (third-degree) heart block
P waves and slow wide QRS beat independently



ST elevation (acute MI)
ST segment raised and merging into the T wave



Drawings, not real tracings. In the practical spot, say the rate, whether it is regular, whether P waves are present, the QRS width, and the name.

Rhythm or finding	Key features	Physiotherapy action
Sinus tachycardia	Normal beats above 100	Look for a cause (pain, fever, anaemia, low fluid, anxiety); resting rate above 120: postpone exercise
Sinus bradycardia	Normal beats below 60	Common with beta-blockers and in fit people; stop if dizzy or the BP falls
Atrial fibrillation	No P waves; irregularly irregular	Exercise if the rate is controlled; use RPE, not heart rate; anticoagulants mean bleeding risk

Rhythm or finding	Key features	Physiotherapy action
Ectopic beats	Early extra beats; ventricular ones are wide	Occasional ones are harmless; frequent, multiform or in runs during exercise: stop
Ventricular tachycardia	Fast, wide, regular complexes	Emergency: check pulse, call for help; if no pulse, start CPR
Ventricular fibrillation	Chaotic line, no QRS	Cardiac arrest: CPR and defibrillation (see the BLS card (p. 356))
Heart block	First degree: PR above 0.20 s. Second degree: some P waves not followed by QRS (Mobitz I: PR lengthens then a beat drops; Mobitz II: beats drop suddenly). Third degree: P and QRS unrelated	First degree and Mobitz I: usually safe. Mobitz II and third degree: no exercise until treated (pacemaker)
ST elevation or depression	Elevation in neighbouring leads: acute MI; depression: ischaemia	New changes: stop and report (criteria on the IHD page (p. 10))
Paced rhythm	Sharp pacing spikes before the P wave or QRS	Device precautions; heart rate may not rise normally

4 Bedside cardiovascular tests

Test	How	Interpretation
Pulses	Feel both sides: radial, brachial, carotid (one side at a time), femoral, popliteal, dorsalis pedis, posterior tibial	Graded 0 absent, 1+ weak, 2+ normal, 3+ bounding (some scales go to 4+). A weak or absent foot pulse suggests arterial disease (S21 case: popliteal 1+)
Postural (orthostatic) hypotension	BP after 5 minutes lying, then at 1 and 3 minutes of standing	A sustained fall of 20 mmHg or more systolic or 10 mmHg or more diastolic within 3 minutes (consensus, 2011). Common after bed rest, with diuretics and antihypertensives, in diabetes and spinal cord injury
Ankle–brachial index (ABI)	Doppler pressures at both ankles and both arms; highest ankle ÷ highest arm, for each leg	0.91–1.40 normal, 0.90 or less PAD, below 0.40 severe; above 1.40 calcified arteries (full table on the PAD page (p. 61))
Claudication distance or time	Treadmill at a fixed speed and gradient (often 3.2 km/h at 12%), or a graded protocol, or a 6-minute walk	Record the pain-free (initial) claudication distance and the maximal walking distance; used to set and review the walking programme
Capillary refill	Press a nail bed for 5 seconds, release	Colour should return within about 2–3 seconds; slower suggests poor circulation or shock
Auscultation	Breath sounds and added sounds	See the chest signs card (p. 331)
ABG	Arterial sample	See the ABG card (p. 326)

5 Exam corner

Past questions (MUHS, 2016–2026)

3-mark questions

Radiological findings of pleural effusion (S16). Radiological findings of emphysema (W18). Three spirometric findings in restrictive lung disease (S19). PFT findings in COPD (W19). PFT in obstructive lung disease (W20). Radiological features of tetralogy of Fallot (S21). ECG criteria for acute and evolved MI (S24). X-ray findings in COPD (W24). ABI (S17, W18, S23, S24, W25, S26).

7-mark question

Relevance of ABG, PFT and X-ray in decision-making for goal setting in COPD (W24; COPD page (p. 78)).

Practical exam

Spots: chest or cardiac X-ray, ABG, PFT, ECG and devices (5 × 2 marks); see the practical exam card (p. 365).

Test yourself

3 m **Give three X-ray signs of a left lower lobe collapse.**

Answer

A triangular white shadow behind the heart (the "sail sign"); loss of the medial part of the left hemidiaphragm outline (silhouette sign); raised left hemidiaphragm and the mediastinum shifted slightly to the left; smaller left lung.

3 m **FEV1 45% predicted, FVC 80% predicted, FEV1/FVC 0.45, with only a small change after salbutamol. Interpret.**

Answer

Obstructive pattern (low ratio), severe by FEV1 (GOLD 3, 30–49%), with little reversibility: typical of COPD.

MCQ **An ECG shows a regular rhythm with 4 large squares between R waves. The heart rate is:**

A. 50

B. 75

C. 100

D. 150

Answer

B. 300 ÷ 4 = 75 beats a minute.

MCQ **Postural hypotension is a fall within 3 minutes of standing of at least:**

A. 10 mmHg systolic

B. 20 mmHg systolic or 10 mmHg diastolic

C. 40 mmHg systolic

D. 5 mmHg diastolic

Answer

B. 20 mmHg systolic or 10 mmHg diastolic.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Stanojevic S, et al. ERS/ATS technical standard on interpretive strategies for routine lung function tests. *Eur Respir J* 2022;60:2101499. Pellegrino R, et al. Interpretative strategies for lung function tests. *Eur Respir J* 2005.
3. Freeman R, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Clin Auton Res* 2011;21:69–72.
4. Thygesen K, et al. Fourth Universal Definition of Myocardial Infarction. *Circulation* 2018.
5. Gerhard-Herman MD, et al. 2016 AHA/ACC guideline on the management of patients with lower extremity peripheral artery disease (ABI interpretation).
6. Structure and classical content: Corne & Kumaran, *Chest X-Ray Made Easy*; Hampton, *The ECG Made Easy*; Hough, *Physiotherapy in Respiratory and Cardiac Care*.

Scales: Borg, mMRC, NYHA, CCS and quality-of-life questionnaires

> Scales are quick 3-mark questions: Borg CR10 (S16, S18), modified Borg (S26), NYHA (S21) and quality-of-life questionnaires for COPD, heart failure and PAD (W24). Write the whole scale with its anchors, and say what it is used for.

Syllabus	Ch 10 Introduction to functional scales: generic and disease-specific; patient's perception compared with therapist's evaluation
Priority	**Moderate.** In 5 of 20 papers (2016–2026), all 3-mark questions.
Asked as	3m S16 · S18 · S21 · W24 · S26
Usual split	3-mark: write the scale, or name the questionnaires
Checked against	Borg scales (Borg 1998); ATS six-minute walk statement (2002) for the modified Borg anchors; GOLD (mMRC); Canadian Cardiovascular Society angina grading. Full list under Sources. Content checked 19 October 2026.

1 Borg rating of perceived exertion (6–20)

Rates the overall effort of exercise. The numbers roughly match heart rate ÷ 10 in healthy young adults (13 ≈ 130 beats per minute).

Rating	Words
6	No exertion at all
7	Extremely light
9	Very light
11	Light
13	Somewhat hard
15	Hard (heavy)
17	Very hard
19	Extremely hard
20	Maximal exertion

The even numbers in between have no words. **Use:** Phase I cardiac rehab below 13; Phase II training 11–16 (start 11–13); moderate intensity 12–13.

2 Modified Borg and CR10 breathlessness scales (3 marks in S16, S18, S26)

A **category–ratio scale from 0 to 10** for breathlessness (or fatigue, pain) during activity. The **modified Borg dyspnoea scale** below is the version used in the six-minute walk test.

Rating	Breathlessness
0	Nothing at all
0.5	Very, very slight (just noticeable)
1	Very slight
2	Slight
3	Moderate

Rating	Breathlessness
4	Somewhat severe
5	Severe
6	–
7	Very severe
8	–
9	Very, very severe (almost maximal)
10	Maximal

- **Borg's own CR10** uses the same 0–10 frame with slightly different words ("extremely weak" at 0.5, "weak" at 2, "strong" at 5, "very strong" at 7, "extremely strong" at 10), and allows ratings above 10 for an absolute maximum. Either version scores the S16 and S18 question.
- **Why "category–ratio":** the numbers have word anchors (categories) and are spaced so that a 4 feels about twice a 2 (ratio), so it can be used to compare intensities.
- **Uses:** breathlessness before and after the 6MWT; pulmonary rehab training at 3–5 (moderate to severe); stop criteria (for example above 7); pacing advice.

3 mMRC dyspnoea scale

Grade	Description
0	Breathless only with strenuous exercise
1	Short of breath when hurrying on the level or walking up a slight hill
2	Walks slower than people of the same age on the level because of breathlessness, or has to stop for breath when walking at own pace on the level
3	Stops for breath after walking about 100 metres or after a few minutes on the level
4	Too breathless to leave the house, or breathless when dressing or undressing

Use: grades breathlessness in daily life (not during a test); GOLD uses mMRC 2 or more as "more symptoms". MUHS cases quote it (S21 bronchiectasis mMRC 3; S21 ILD mMRC 2).

4 NYHA classes (3 marks in S21) and CCS angina grades

New York Heart Association functional classes (heart failure)

Class	Description
I	No limitation; ordinary activity causes no undue breathlessness, fatigue or palpitations
II	Slight limitation; comfortable at rest, ordinary activity causes symptoms
III	Marked limitation; comfortable at rest, less than ordinary activity causes symptoms
IV	Unable to do any activity without discomfort; symptoms may be present at rest

Canadian Cardiovascular Society (CCS) angina grades

Grade	Description
I	Ordinary activity (walking, climbing stairs) does not cause angina; angina only with strenuous, rapid or prolonged exertion
II	Slight limitation: angina walking or climbing stairs rapidly, uphill, after meals, in cold or wind, under stress; or walking more than two blocks on the level or more than one flight of stairs at a normal pace
III	Marked limitation: angina walking one or two blocks on the level or one flight of stairs at a normal pace

Grade	Description
IV	Unable to do any physical activity without discomfort; angina may occur at rest

5 Quality-of-life questionnaires (3 marks in W24)

Condition	Questionnaire	Key facts
COPD	St George's Respiratory Questionnaire (SGRQ)	Symptoms, activity and impacts; 0–100, higher = worse; a change of 4 points matters
COPD	Chronic Respiratory Questionnaire (CRQ)	Dyspnoea, fatigue, emotion, mastery; a change of 0.5 per item matters
COPD	COPD Assessment Test (CAT)	8 items scored 0–5 (total 0–40); 10 or more = high symptom impact (used by GOLD)
Heart failure	Minnesota Living with Heart Failure Questionnaire (MLHFQ)	21 items, 0–105; higher = worse
Heart failure	Kansas City Cardiomyopathy Questionnaire (KCCQ)	23 items, 0–100; higher = better
Peripheral arterial disease	Walking Impairment Questionnaire (WIQ)	Walking distance, speed and stair climbing; 0–100, higher = better
Peripheral arterial disease	Vascular Quality of Life Questionnaire (VascuQoL)	25 items covering pain, symptoms, activities, social and emotional life
Any condition (generic)	SF-36, EQ-5D, WHOQOL-BREF	Compare across diseases and with healthy people

Generic or disease-specific (syllabus 10a): generic tools compare different conditions but are less sensitive to change in one disease; disease-specific tools detect small changes after rehab. Use one of each when possible, and compare the patient's own rating with your assessment (syllabus 10b).

6 Exam corner

Past questions (MUHS, 2016–2026)

3-mark questions

Borg's CR10 scale (S16, S18). Grades of the NYHA scale (S21). One quality-of-life questionnaire each for COPD, heart failure and peripheral arterial disease (W24). Write the modified Borg scale (S26).

Memory hook for the modified Borg

3 moderate, 5 severe, 7 very severe, 10 maximal

Test yourself

3 m **Write the NYHA classes.**

Answer

I: no limitation. II: slight limitation, ordinary activity causes symptoms. III: marked limitation, less than ordinary activity causes symptoms. IV: symptoms with any activity or at rest.

MCQ **A patient stops for breath after walking about 100 m on the level. mMRC grade:**

A. 1

B. 2

C. 3

D. 4

Answer

C. Grade 3.

MCQ **Which questionnaire is specific to heart failure?**

A. SGRQ

B. MLHFQ

C. WIQ

D. CAT

Answer

B. Minnesota Living with Heart Failure Questionnaire (KCCQ is the other).

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Borg G. *Borg's Perceived Exertion and Pain Scales*. Human Kinetics, 1998.
3. ATS statement: guidelines for the six-minute walk test. *Am J Respir Crit Care Med* 2002 (modified Borg dyspnoea scale wording).
4. Global Initiative for Chronic Obstructive Lung Disease (GOLD) 2026 report (mMRC, CAT).
5. Campeau L. Grading of angina pectoris (Canadian Cardiovascular Society). *Circulation* 1976;54:522–3.
6. The Criteria Committee of the New York Heart Association. *Nomenclature and Criteria for Diagnosis of Diseases of the Heart and Great Vessels*, 9th ed., 1994.
7. Structure and classical content: Hillegass (2022); Main & Denehy (2016).

ABG interpretation and why patients desaturate

> ABG appears as causes of respiratory acidosis (S24), reasons for desaturation with exercise (S21), the relevance of ABG to goal setting in COPD (W24), and in every ICU and COPD case that quotes "hypoxaemia and hypercapnia". Learn the normal values, a four-step method and the causes.

Syllabus	Ch 2(a) Investigation and clinical implication: X-ray, PFT, ABG, ECG
Priority	**Moderate.** In 3 of 20 papers (2016–2026), and inside COPD and ICU cases.
Asked as	7m W24 (COPD goals) · 3m S21 · S24
Usual split	3-mark causes; part of COPD and ICU long answers
Checked against	Standard acid–base physiology and compensation rules; BTS emergency oxygen guideline (2017); Berlin ARDS definition (PaO_2/FiO_2). Full list under Sources. Content checked 19 October 2026.

1 Normal values

Measure	Normal (adult, room air)	Meaning
pH	**7.35–7.45**	Below 7.35 acidaemia; above 7.45 alkalaemia
$PaCO_2$	**35–45 mmHg** (4.7–6.0 kPa)	The respiratory part: set by ventilation
HCO_3^- (bicarbonate)	**22–26 mmol/L**	The metabolic part: set by the kidneys and metabolism
PaO_2	**80–100 mmHg** (10.7–13.3 kPa); a little lower in older people	Oxygenation; below 60 mmHg (8 kPa) = respiratory failure
SaO_2	95–98%	Haemoglobin saturation
Base excess	−2 to +2 mmol/L	Negative in metabolic acidosis, positive in metabolic alkalosis

2 Four steps to interpret

1. **pH:** acidaemia or alkalaemia?
2. **Which caused it?** If $PaCO_2$ moves in the **opposite** direction to pH, the cause is respiratory; if HCO_3^- moves in the **same** direction as pH, it is metabolic.
3. **Compensation:** the other system moves the same way as the cause to bring pH back. Partial compensation leaves pH abnormal; full compensation brings pH into the normal range (but never over-corrects).
4. **Oxygenation:** PaO_2 against the FiO_2 the patient is breathing (PaO_2/FiO_2 ratio).

Memory hook

ROME

Respiratory **O**pposite (pH and $PaCO_2$ move opposite ways); **M**etabolic **E**qual (pH and HCO_3^- move the same way).

Expected compensation (rules of thumb)

Disorder	Expected change
Acute respiratory acidosis	HCO_3^- rises about 1 mmol/L for every 10 mmHg rise in $PaCO_2$
Chronic respiratory acidosis (days)	HCO_3^- rises about 3.5–4 mmol/L for every 10 mmHg rise in $PaCO_2$
Acute respiratory alkalosis	HCO_3^- falls about 2 mmol/L for every 10 mmHg fall in $PaCO_2$

Disorder	Expected change
Metabolic acidosis	Expected $PaCO_2$ = 1.5 × HCO_3^- + 8 (± 2) (Winter's formula)
Metabolic alkalosis	$PaCO_2$ rises about 0.7 mmHg for every 1 mmol/L rise in HCO_3^-

Worked examples

Values	Interpretation
pH 7.28, $PaCO_2$ 62, HCO_3^- 28	Acidaemia; $PaCO_2$ high (opposite to pH), so respiratory acidosis. Expected HCO_3^- about 26 if acute, about 32 if chronic; 28 lies between, so acute-on-chronic: a COPD exacerbation
pH 7.48, $PaCO_2$ 30, HCO_3^- 22	Alkalaemia; $PaCO_2$ low, so acute respiratory alkalosis: anxiety and over-breathing, early asthma attack, pulmonary embolism, pain
pH 7.25, $PaCO_2$ 28, HCO_3^- 12	Acidaemia; HCO_3^- low (same direction), so metabolic acidosis. Expected $PaCO_2$ = 1.5 × 12 + 8 = 26 ± 2, so the respiratory compensation is appropriate: for example diabetic ketoacidosis (Kussmaul breathing)

3 Causes of each disorder (3 marks in S24)

Disorder	Causes
Respiratory acidosis (hypoventilation, CO_2 retained)	**Brain:** sedatives, opioids, head injury, stroke. **Nerves and muscles:** Guillain–Barré syndrome, myasthenia gravis, high spinal cord injury, muscular dystrophy, exhausted breathing muscles. **Chest wall:** flail chest, severe kyphoscoliosis, obesity hypoventilation. **Lungs and airways:** COPD exacerbation, severe asthma (late), pneumonia, pulmonary oedema, airway obstruction. **Ventilator** set too low
Respiratory alkalosis (over-breathing)	Anxiety, pain, fever, hypoxaemia (pneumonia, pulmonary embolism, early asthma), high altitude, pregnancy, ventilator set too high
Metabolic acidosis	Diabetic ketoacidosis, lactic acidosis (shock, sepsis, severe exercise), kidney failure, severe diarrhoea, some poisonings
Metabolic alkalosis	Vomiting or nasogastric suction, diuretics, low potassium, excess antacids or bicarbonate

4 Oxygenation and respiratory failure

Type	Blood gases	Problem and examples
Type 1 (hypoxaemic)	PaO_2 below 60 mmHg; $PaCO_2$ normal or low	Gas exchange (V/Q mismatch, shunt): pneumonia, pulmonary oedema, ARDS, pulmonary embolism, ILD
Type 2 (hypercapnic)	PaO_2 below 60 mmHg and $PaCO_2$ above about 45–50 mmHg	Ventilation (the pump fails or the load is too high): COPD, neuromuscular disease, chest wall disease, sedation

- **PaO_2/FiO_2 ratio:** normal about 400–500 (for example 95 ÷ 0.21 ≈ 450). In ARDS: 201–300 mild, 101–200 moderate, 100 or less severe.
- **Acute or chronic type 2:** a low pH means acute (or acute-on-chronic) and needs urgent action, often NIV; a normal pH with high HCO_3^- means chronic, compensated CO_2 retention.

5 Why patients desaturate on exercise (3 marks in S21)

- **Worse V/Q matching:** more blood goes through poorly ventilated lung as cardiac output rises.
- **Diffusion limitation:** faster blood flow shortens the time red cells spend in the capillary; with a thickened or reduced membrane (ILD, emphysema with low DLCO) oxygen cannot load fully.
- **Hypoventilation:** dynamic hyperinflation and tired breathing muscles in COPD limit ventilation.
- **Low mixed venous oxygen:** working muscles extract more oxygen, so blood returns to the lungs emptier; worse when cardiac output is low.

- **Pulmonary hypertension and shunt:** vessels cannot open up; a right-to-left shunt through a patent foramen ovale may open.
- **Measurement error:** cold fingers, movement and poor circulation give falsely low readings; use an ear or forehead probe.

For physiotherapy: measure SpO_2 during the 6MWT; give oxygen during training if it falls below about 88%; use interval training; ILD patients desaturate most.

6 What the ABG means for physiotherapy

- **Acute type 2 failure (low pH):** treat the chest gently, support NIV, no exercise training; report drowsiness.
- **Chronic CO_2 retention:** controlled oxygen to SpO_2 88–92%; nocturnal NIV may be in use.
- **Low PaO_2 at rest or on exertion:** oxygen during treatment and exercise; positions that improve V/Q.
- **Metabolic acidosis with Kussmaul breathing:** this is compensation, not a breathing problem to treat; inform the team.
- Recheck gases after big changes in oxygen, ventilator settings or the patient's state. More on COPD goal setting on the COPD page (p. 78) (W24).

7 Exam corner

Past questions (MUHS, 2016–2026)

3-mark questions

Reasons for desaturation with exercise (S21). Causes of respiratory acidosis (S24).

Related

Relevance of ABG, PFT and X-ray in goal setting in COPD (W24, 7 marks; COPD page). COPD case with "mild hypoxaemia and hypercapnia" (W16).

Test yourself

3 m **Interpret: pH 7.30, $PaCO_2$ 58 mmHg, HCO_3^- 26 mmol/L, PaO_2 55 mmHg on room air.**

Answer
Acidaemia with high $PaCO_2$ (opposite direction): respiratory acidosis. Expected HCO_3^- if acute ≈ 24 + 1.8 ≈ 26, so acute and uncompensated. PaO_2 55: hypoxaemia. Acute type 2 respiratory failure.

MCQ **Which causes respiratory acidosis?**

A. Anxiety

B. Guillain–Barré syndrome with weak breathing muscles

C. Vomiting

D. Diabetic ketoacidosis

Answer
B. Weak breathing muscles cause hypoventilation and CO_2 retention.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. O'Driscoll BR, et al. BTS guideline for oxygen use in adults in healthcare and emergency settings. *Thorax* 2017.
3. ARDS Definition Task Force. The Berlin Definition. *JAMA* 2012.

4. West JB, Luks AM. *West's Respiratory Physiology: The Essentials* (gas exchange, exercise desaturation).

5. Structure and classical content: Hough, *Physiotherapy in Respiratory and Cardiac Care*; Main & Denehy (2016); standard acid–base texts for compensation rules.

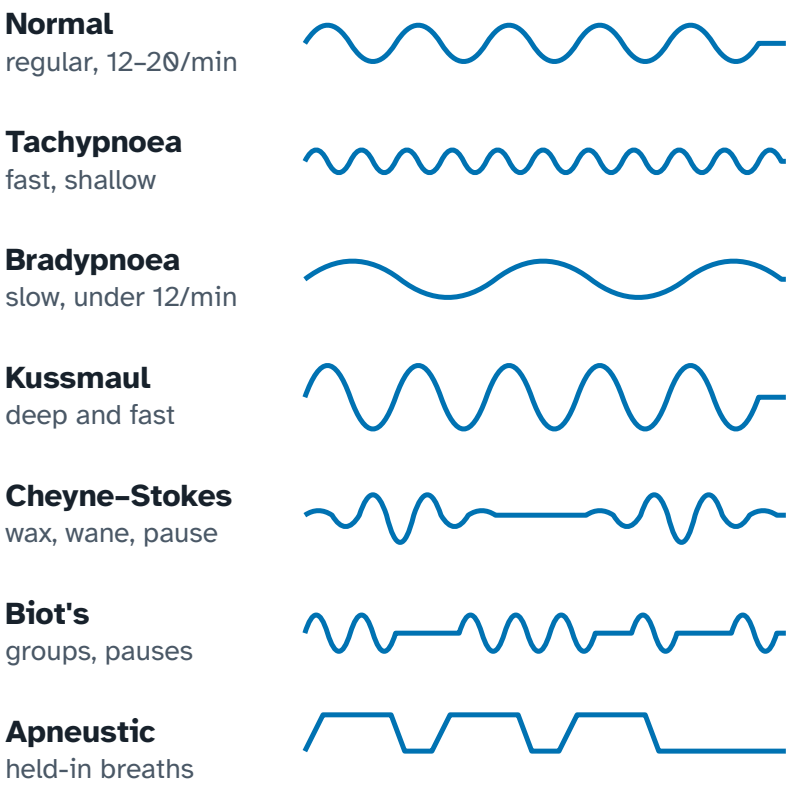
Abnormal breathing patterns and chest signs

Breathing patterns were asked as Cheyne–Stokes respiration (S16) and three abnormal breathing patterns (S22). A small sketch of each pattern earns marks quickly. The chest signs below are the observations every case answer starts with.

Syllabus	Ch 2(a) Investigation and clinical implication: auscultation; chest assessment
Priority	**Quick marks.** In 2 of 20 papers (2016–2026).
Asked as	3m S16 · S22
Usual split	3-mark patterns with a sketch and causes
Checked against	Standard clinical examination texts and respiratory physiotherapy texts. Full list under Sources. Content checked 19 October 2026.

1 Breathing patterns (3 marks in S16 and S22)

Sketches of breathing patterns



Up = breathing in. Draw the shape and write one cause; that is a full 3-mark answer for each pattern.

Pattern	Description	Causes
Tachypnoea	Rate above about 20 a minute in adults	Fever, pneumonia, pulmonary oedema, pulmonary embolism, anxiety, pain, ILD
Bradypnoea	Rate below about 12 a minute	Opioids and sedatives, raised intracranial pressure, hypothermia, severe hypothyroidism
Hyperpnoea and Kussmaul breathing	Deep breaths; Kussmaul is deep, fast and laboured, regular	Metabolic acidosis: diabetic ketoacidosis, kidney failure (the body blows off CO_2 to compensate)

Pattern	Description	Causes
Cheyne–Stokes respiration (S16)	Cycles of breaths that gradually deepen, then gradually become shallow, followed by a pause (apnoea) of seconds; then repeats	Heart failure (slow circulation delays the feedback from chemoreceptors), stroke and other brain injury, high altitude, opioids; can be normal in sleeping infants and the elderly
Biot's (ataxic) breathing	Groups of quick, equal-depth breaths with irregular pauses; ataxic breathing is completely irregular	Damage to the medulla: meningitis, raised intracranial pressure, brainstem stroke, opioids. A sign of serious brain injury
Apneustic breathing	Deep gasping breath in, held, then a short breath out	Damage to the pons
Central neurogenic hyperventilation	Sustained deep, rapid breathing	Midbrain and upper pons lesions
Paradoxical breathing	Part of the chest or abdomen moves the wrong way	Flail chest (segment moves in on inspiration); diaphragm paralysis or fatigue (abdomen moves in on inspiration); airway obstruction
Prolonged expiration, pursed lips	Breath out much longer than breath in	COPD, asthma
Sighing and erratic upper-chest breathing	Irregular, with frequent sighs and yawns	Dysfunctional breathing and hyperventilation (see breathing retraining)

Related positional terms: **orthopnoea** (breathless lying flat), **paroxysmal nocturnal dyspnoea** (waking breathless), **platypnoea** (breathless upright, better lying).

2 Chest signs in assessment

Sign	What it suggests
Barrel chest (front-to-back diameter increased)	Hyperinflation: emphysema, chronic asthma
Pectus excavatum or carinatum; kyphoscoliosis	Chest wall deformity; restrictive pattern when severe
Accessory muscle use, intercostal indrawing, tracheal tug	Increased work of breathing
Reduced expansion on one side	The disease is on that side: effusion, collapse, pneumothorax, consolidation, post-surgery
Trachea shifted	Pulled towards collapse or fibrosis; pushed away by a large effusion or tension pneumothorax
Clubbing	Bronchiectasis, cystic fibrosis, lung abscess, empyema, lung cancer, ILD, cyanotic heart disease (not COPD alone)
Central cyanosis (tongue, lips)	Low arterial oxygen; peripheral cyanosis alone (fingers) suggests poor circulation or cold

Percussion and breath sounds

Finding	Percussion	Breath sounds
Normal lung	Resonant	Vesicular
Consolidation	Dull	Bronchial breathing; crackles; increased vocal resonance
Collapse	Dull	Reduced or absent
Pleural effusion	Stony dull	Reduced or absent; reduced vocal resonance

Finding	Percussion	Breath sounds
Pneumothorax	Hyper-resonant	Reduced or absent
Emphysema	Hyper-resonant	Quiet, prolonged expiration

Added sounds

Sound	Meaning
Wheeze	Musical; narrowed airways: asthma, COPD; a fixed single wheeze suggests a local obstruction
Coarse crackles	Secretions in larger airways: bronchiectasis, chest infection; often change after a cough (a sign for airway clearance)
Fine crackles	Late inspiratory, like Velcro: pulmonary fibrosis; fluid: pulmonary oedema
Pleural rub	Creaking with breathing: inflamed pleura (pleurisy)
Stridor	Harsh inspiratory noise: upper airway narrowing (emergency)

3 Exam corner

Past questions (MUHS, 2016–2026)

3-mark questions

Cheyne–Stokes respiration (S16). Three abnormal breathing patterns (S22).

Test yourself

3 m **Describe Cheyne–Stokes respiration with two causes.**

> **Answer**
> Cycles of gradually deepening then gradually shallower breaths, followed by a pause, repeating (draw it). Causes: heart failure, stroke or other brain injury, high altitude, opioids.

MCQ **Deep, rapid, regular breathing in a patient with diabetic ketoacidosis is called:**

A. Biot's breathing

B. Kussmaul breathing

C. Apneustic breathing

D. Cheyne–Stokes respiration

> **Answer**
> **B.** Kussmaul breathing compensates for metabolic acidosis.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Structure and classical content: Macleod's Clinical Examination; Hough, *Physiotherapy in Respiratory and Cardiac Care*; Frownfelter & Dean (2022), chapter on assessment.

Part C

Other syllabus chapters

Rehabilitation programmes, the revision chapter and basic life support.

Cardiac and pulmonary rehabilitation

Cardiac rehabilitation: definition, indications, contraindications, phases, outcome measures 334

Pulmonary rehabilitation: definition, indications, contraindications, components, outcome measures 339

Basics: anatomy, physiology and drugs

Respiratory anatomy, physiology and mechanics of the thorax 343

Cardiac anatomy and physiology for physiotherapists 348

Cardiac and respiratory drugs: what physiotherapists need to know 352

Basic life support

Basic life support (CPR) for physiotherapists 356

Cardiac rehabilitation: definition, indications, contraindications, phases and outcome measures

> Cardiac rehab is part of the most common long answer in this paper (16 IHD and CABG cases in 2016–2026) and of several 7-mark questions (Phase I, Phase II after PTCA, rehab after valve replacement, risk stratification). This page is the programme itself for any heart condition; the day-by-day doses after MI and CABG are on the IHD page.

Syllabus	Ch 7 Cardiac rehabilitation (AHA/ACSM guidelines): definition, indications, contraindications, phases I–IV, outcome measures
Priority	**High yield.** Inside the most-asked long answer, plus 7-mark questions in W16, W17, S21, W21 and W24.
Asked as	16 IHD long cases · 7m W16 · W17 · S21 · W21 · W24 · 3m S20 · W25
Usual split	7-mark phases or programme; 15-mark IHD and CABG cases
Checked against	AHA/AACVPR core components (2024 update); ACSM guidelines (indications and contraindications); WHO definition; Cochrane review update (2026). Full list under Sources. Content checked 19 October 2026.

1 Definition and aims

Cardiac rehabilitation is a coordinated, multi-part programme that helps people with heart disease recover their physical, psychological and social function, and that slows the underlying disease through exercise, education and risk-factor control (AHA/AACVPR). The WHO describes it as all the activities needed so that patients, by their own efforts, regain as normal a place as possible in the community.

- **Aims:** safe return to daily activities, work and leisure; better exercise capacity; control of risk factors; fewer future heart attacks and admissions; better quality of life and mood; self-management for life.

2 Indications (who should be referred)

- Myocardial infarction or acute coronary syndrome.
- After coronary artery bypass grafting (CABG) or angioplasty with stent (PCI).
- Stable angina.
- Stable heart failure with reduced ejection fraction (and increasingly other heart failure).
- After heart valve repair or replacement (including transcatheter valves).
- Heart or heart–lung transplant.
- Others who benefit: pacemaker or ICD, cardiomyopathy, congenital heart disease in adults, after cardiac surgery of any kind, peripheral arterial disease (supervised walking programme), people at high risk of coronary disease (diabetes, metabolic syndrome).

Older age, a low ejection fraction in stable heart failure, an ICD and diabetes are **not** reasons to exclude; they are reasons to stratify risk and supervise more closely.

3 Contraindications to exercise in cardiac rehab (ACSM)

Do not start (or stop) exercise training if

- Unstable angina
- Resting systolic BP above **180** or diastolic above **110 mmHg** (decide case by case)
- BP falls more than **20 mmHg** on standing, with symptoms
- Significant aortic stenosis
- Uncontrolled atrial or ventricular arrhythmias; uncontrolled sinus tachycardia above **120**
- Uncompensated (decompensated) heart failure
- Third-degree heart block without a pacemaker
- Active pericarditis or myocarditis
- Recent embolism (pulmonary or systemic); acute thrombophlebitis; aortic dissection
- Acute illness or fever
- Uncontrolled diabetes; other uncorrected metabolic problems (for example abnormal potassium)
- Severe orthopaedic or psychological problems that prevent exercise

Stop criteria during sessions are on the IHD page (p. 22) and the exercise testing card (p. 309).

4 Core components (AHA/AACVPR, 2024 update)

Component	What it includes
Patient assessment	History, examination, ECG, exercise test or 6MWT, risk stratification, goals
Nutritional counselling	Heart-healthy diet, salt below 5 g a day, less fried food and sugar
Weight management and body composition	Waist and BMI targets (Indian cut-offs), diet plus activity
Cardiovascular risk-factor management	BP, lipids, diabetes, tobacco cessation, medicines and adherence
Psychosocial management	Screening for depression and anxiety, stress management, family support, return to work
Aerobic exercise training	FITT-VP prescription, monitored and progressed
Strength training	Major muscle groups, 2–3 days a week
Physical activity counselling	At least 150 minutes a week of moderate activity for life; less sitting
Programme quality improvement	Measuring outcomes and improving the service

5 Phases I–IV

Phase	When and where	Aims and content	Intensity
I: inpatient	From the first stable day to discharge (days)	Prevent bed-rest effects; breathing exercises (after surgery); graded self-care and walking; stairs before discharge; education, warning signs, GTN use; home plan; referral to Phase II	Up to about 3 METs; RPE below 13; HR within rest + 20 (MI) or rest + 30 (surgery)
II: early outpatient (supervised)	Starts within a few weeks of discharge; about 8–12 weeks; hospital, centre, home or hybrid	Exercise test, risk stratification, monitored aerobic and strength training, risk-factor control, psychological support, return to work, driving and sexual activity	40–80% heart-rate reserve or RPE 11–16, starting low; 3–5 days a week; resistance 2–3 days
III: community	After Phase II, for several months	Less supervision; keep and build fitness; self-monitoring with pulse and RPE	As in Phase II, progressed

Phase	When and where	Aims and content	Intensity
IV: lifelong maintenance	For life	Independent exercise and healthy lifestyle; periodic review	At least 150 min a week moderate activity plus 2 days of strength work

Detailed Phase I progression, Phase II FITT-VP, Karvonen examples and sternal precautions: IHD page, section 12 (p. 18). Rehab after PCI for sport (W24) and after valve replacement (W21): IHD page (p. 18) and valve page (p. 188).

6 Exercise across heart conditions

Condition	Key adjustment
After MI or CABG	Phase I limits; sternal precautions for 6–8 weeks after CABG; legs first in resistance training
After PCI	No sternal precautions; protect the radial or femoral puncture site for about a week; can progress faster
Heart failure	Start low (RPE 11–13), interval training, longer warm-up and cool-down; watch weight gain and breathlessness (fluid); inspiratory muscle training if weak
Valve surgery	Sternal precautions; atrial fibrillation is common; anticoagulation (bleeding, falls)
Pacemaker or ICD	Device-side arm below shoulder height for 4–6 weeks; keep heart rate at least 10 beats below the ICD detection rate
Heart transplant	The heart has no nerve supply, so heart rate rises slowly and late: use RPE, long warm-up and cool-down
Atrial fibrillation, beta-blockers	Heart rate is unreliable: use RPE and the talk test
Peripheral arterial disease	Supervised walking to moderate claudication pain, rest, repeat (see the PAD page (p. 61))

7 Risk stratification

Before Phase II, patients are grouped as **low, moderate or high risk** for events during exercise (AACVPR), using ejection fraction, exercise-test results (functional capacity in METs, ischaemia, arrhythmias), symptoms, complications in hospital and depression. Risk sets the amount of ECG monitoring and supervision and how fast to progress. The full table (asked as 7 marks in S21) is on the IHD page (p. 15).

Memory hook

EF 50 / 40, METs 7 / 5

Low risk: EF 50% or more and capacity 7 METs or more. High risk: EF below 40%, or symptoms below 5 METs, or complex arrhythmias, or a complicated course.

8 Outcome measures (syllabus 7d)

Domain	Measures
Exercise capacity	Exercise test (METs, peak oxygen uptake), 6-minute walk test, incremental shuttle walk test, sit-to-stand test
Risk factors	BP, lipid profile, HbA1c, weight, BMI, waist, smoking status
Symptoms	Angina frequency (CCS grade), breathlessness (NYHA, Borg), fatigue
Quality of life	MacNew heart disease questionnaire, Seattle Angina Questionnaire, MLHFQ or KCCQ in heart failure, SF-36, EQ-5D
Mood	Hospital Anxiety and Depression Scale (HADS), PHQ-9

Domain	Measures
Physical activity and behaviour	Step count, activity questionnaires, medicine adherence, return to work
Events	Readmissions, repeat heart attacks, revascularisation

9 Evidence and ways of delivering it

- **Benefit:** compared with no programme, exercise-based cardiac rehab reduces heart attacks by about 28% and hospital admissions by about 35%, probably reduces deaths, and improves quality of life (Cochrane review update, 2026: 107 trials, 26,886 people).
- **Settings:** home-based and digitally supported programmes are about as effective as centre-based ones, and suit patients who live far away or work.
- **Yoga-based rehab:** the Indian Yoga-CaRe trial improved self-rated health and return to activities, but did not significantly reduce major events (see the yoga card (p. 302)).
- **The gap:** few eligible patients join, especially in India and among women. Physiotherapists can refer at discharge, start Phase I early and offer home programmes.

10 Exam corner

Past questions (MUHS, 2016–2026)

Long answers

16 IHD, MI and CABG cases ask for cardiac rehab phases with doses; S26 Q4(a) asked the phases with their aims, inpatient care, convalescence to 6 weeks and rehab after 3 months (see the IHD exam corner (p. 25)).

7-mark questions

Phase I cardiac rehabilitation after MI (W16, W17). Risk assessment and stratification for CAD (S21). Phase 1 and 2 after valve replacement (W21). Phase 2 after PTCA to the LAD for a patient who wants to play sport (W24).

3-mark questions

Phase I of cardiac rehabilitation (S20). Karvonen's formula in cardiac rehabilitation (W25).

Answer skeleton: "cardiac rehabilitation", 7 marks

1. **[1]** Definition and aims.
2. **[1]** Indications.
3. **[1]** Contraindications (any six).
4. **[3]** Phases I–IV table with timing, content and intensity.
5. **[1]** Outcome measures and benefits.

Test yourself

3 m **List six indications for cardiac rehabilitation.**

> **Answer**
> MI or acute coronary syndrome; after CABG; after PCI; stable angina; stable heart failure with reduced EF; after valve surgery; heart transplant; pacemaker or ICD (any six).

3 m **Name three outcome measures used in cardiac rehabilitation.**

Answer

6-minute walk test or exercise test (METs); MacNew or Seattle Angina Questionnaire; HADS; risk factors such as BP, lipids and HbA1c (any three).

MCQ **Which is a contraindication to starting exercise training?**

A. Age 75

B. Stable heart failure with EF 35%

C. Unstable angina

D. An ICD

Answer

C. The others need risk stratification and closer supervision, not exclusion.

MCQ **After heart transplant, exercise intensity is best set by:**

A. Heart rate

B. RPE

C. Blood sugar

D. Respiratory rate alone

Answer

B. The transplanted heart has no nerve supply, so heart rate responds slowly.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Brown TM, et al. Core components of cardiac rehabilitation programs: 2024 update. A scientific statement from the AHA and AACVPR. *Circulation* 2024;150:e328–47.
3. American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription* (indications and contraindications for inpatient and outpatient cardiac rehabilitation).
4. World Health Organization. Rehabilitation after cardiovascular diseases, with special emphasis on developing countries. Technical Report Series 831, 1993.
5. Exercise-based cardiac rehabilitation for coronary heart disease. Cochrane Review CD001800, update 2026. Cochrane news summary.
6. Prabhakaran D, et al. Effect of yoga-based cardiac rehabilitation on health outcomes (Yoga-CaRe). *J Am Coll Cardiol* 2020.
7. Structure and classical content: Hillegass (2022); AACVPR *Guidelines for Cardiac Rehabilitation Programs*.

Pulmonary rehabilitation: definition, indications, contraindications, components and outcome measures

> Pulmonary rehab is the long-term half of every COPD, bronchiectasis, ILD, post-TB and asthma long answer, and it was asked directly for children with asthma (S18) and a post-TB patient "coming for pulmonary rehabilitation" (S20). This page is the programme itself; disease-specific doses are on the condition pages.

Syllabus	Ch 8 Pulmonary rehabilitation (AACVPR/ATS guidelines): definition, indications, contraindications, components of management, outcome measures
Priority	**High yield.** Inside the long-term management of every chronic lung disease case (COPD, bronchiectasis, ILD, TB, asthma).
Asked as	S20 (post-TB, in rehab) · 7m S18 (children, asthma) · S19 · S21 · W24
Usual split	Long-term part (5–8 marks) of chronic lung disease cases; 7-mark programme questions
Checked against	ATS/ERS statement (2013); ATS pulmonary rehabilitation guideline (2023); BTS guideline (2013); Cochrane reviews of pulmonary rehabilitation (2015), after exacerbations (2016) and telerehabilitation (2021). Full list under Sources. Content checked 19 October 2026.

1 Definition and aims

Pulmonary rehabilitation (ATS/ERS, 2013) is "a comprehensive intervention based on a thorough patient assessment followed by patient-tailored therapies that include, but are not limited to, exercise training, education, and behavior change, designed to improve the physical and psychological condition of people with chronic respiratory disease and to promote the long-term adherence to health-enhancing behaviors."

- **Aims:** less breathlessness and fatigue; better exercise capacity and daily function; better quality of life and mood; fewer exacerbations and admissions; lasting self-management.
- **Why it works when the lungs cannot be fixed:** it treats what can change: weak, deconditioned muscles, poor breathing patterns, fear of breathlessness, inactivity and poor self-management (the breathlessness spiral on the COPD page (p. 74)).

2 Indications (who should be referred)

- **COPD:** anyone limited by breathlessness (usually mMRC 2 or more), and after a hospital admission for an exacerbation, starting soon after discharge (ATS 2023: strong recommendation).
- **Interstitial lung disease** (ATS 2023: strong recommendation).
- **Bronchiectasis, cystic fibrosis, post-TB lung disease**.
- **Asthma** with persistent breathlessness or poor fitness despite good treatment.
- **Pulmonary hypertension** (ATS 2023: conditional recommendation, supervised).
- **Lung cancer**; before and after lung resection, lung volume reduction or transplant.
- Chest wall and neuromuscular disease (selected), after ICU stays and after COVID-19.

Age, continued smoking and low oxygen are **not** reasons to exclude: help the smoker quit during the programme, and give oxygen during exercise if needed.

3 Contraindications

- Unstable heart disease: unstable angina, recent heart attack, uncontrolled arrhythmia, severe aortic stenosis, decompensated heart failure.
- Locomotor or neurological problems that make exercise impossible or unsafe.

- Severe cognitive or psychiatric illness that prevents taking part.
- Acute exacerbation or infection: treat first, then start or resume.
- Severe pulmonary hypertension with fainting on exertion, or uncontrolled hypertension: only in specialist, supervised settings.

4 Assessment before the programme

- History, comorbidities (heart disease, diabetes, osteoporosis, anxiety and depression), medicines and inhaler technique, smoking.
- Spirometry (severity); breathlessness (mMRC); symptoms and quality of life (CAT, CRQ or SGRQ); mood (HADS).
- **Exercise test with SpO_2:** 6MWT or incremental shuttle walk test; sets the training speed and shows whether oxygen is needed (SpO_2 falling below about 88%).
- Muscle strength (quadriceps, grip, sit-to-stand), balance and falls risk.
- The patient's own goals (for example "climb to my 2nd-floor flat without stopping").

5 Components of management

Exercise training (the core)

Part	Endurance	Strength
Frequency	**3–5 days** a week	**2–3 days** a week
Intensity	Walking at about 80% of the 6MWT or shuttle-test speed, or cycling at 60–80% of peak work; breathlessness **Borg 4–6**	About **60–70% of 1RM**, 8–12 repetitions
Time	20–60 min; intervals (for example 1 minute on, 1 minute off) if too breathless for continuous work	1–3 sets, legs first, then arms and trunk
Type	Walking, treadmill, cycling; arm ergometry for upper-limb endurance	Weights, bands, sit-to-stand, step-ups; water bottles and sand bags at home

- **Add-ons:** inspiratory muscle training for weak breathing muscles; neuromuscular electrical stimulation of the thighs for very weak or bed-bound patients; flexibility, posture and balance work; oxygen during training if SpO_2 falls below about 88%.

Education and self-management

- Understanding the disease; inhaler and spacer technique; written action plan for exacerbations.
- Breathing strategies (pursed-lip breathing, breathing control, positions of ease) and airway clearance where there is sputum.
- Energy conservation and pacing (see the dyspnoea card (p. 285)).
- Smoking cessation; vaccinations; nutrition (weight gain if underweight, weight loss if obese); travel and oxygen use.
- Managing anxiety and panic; advance care planning in severe disease.

Psychosocial and nutritional support

Screening for and treating anxiety and depression; group sessions for peer support; dietary advice; involving family.

Programme structure

Usually **6–12 weeks**, with at least **2 supervised sessions a week** plus home exercise; centre-based, home-based or by telerehabilitation (ATS 2023: offer the choice). Then a maintenance plan, because gains fade within months if exercise stops.

6 Outcome measures (syllabus 8e)

Domain	Measure	Smallest change that matters
Exercise capacity	6-minute walk test	About 25–33 m (often quoted as 30 m)
Exercise capacity	Incremental shuttle walk test; endurance shuttle walk test; cycle test	ISWT about 47.5 m
Breathlessness	mMRC; modified Borg after exercise; CRQ dyspnoea domain	mMRC one grade; CRQ 0.5 per item
Health status	CAT; SGRQ; CRQ	CAT 2 points; SGRQ 4 points
Mood	Hospital Anxiety and Depression Scale	–
Function and strength	Sit-to-stand test, quadriceps strength, grip, balance	–
Activity and events	Step count; exacerbations, admissions	–

7 Evidence and delivery

- In COPD, pulmonary rehab improves breathlessness, fatigue, quality of life and exercise capacity by more than the smallest important changes (Cochrane, 2015).
- Started soon after an admission for an exacerbation, it reduces readmissions and improves quality of life (Cochrane, 2016).
- Telerehabilitation gives similar results to centre-based rehab (Cochrane, 2021), useful where travel is hard.
- **In India:** few centres offer it; a home programme using stairs, a chair, water-bottle weights and walking, with breathing exercises and pranayama (see the yoga card (p. 302)), reviewed by phone or video, brings the main parts to most patients.

Classic answer vs current evidence: breathing exercises alone

Classic	Current	Write in the exam
Pulmonary rehab means breathing exercises and postural drainage.	Exercise training of the legs and arms is the core; education, breathing strategies and airway clearance are added as needed.	"Pulmonary rehabilitation: assessment, endurance and strength training with doses, education and self-management, psychosocial support, outcome measures."

8 Exam corner

Past questions (MUHS, 2016–2026)

S20 Q3(a), 5 + 5 + 5 marks

40-year-old tailor with post-Koch's bronchiectasis of the right lung, copious sputum, breathless on exertion, coming for pulmonary rehabilitation for a week. Impairments, activity and participation; short- and long-term management (see the bronchiectasis page (p. 98)).

7-mark questions

Paediatric pulmonary rehabilitation for children with asthma (S18; asthma page (p. 132)). Energy conservation in COPD and its importance (S19). Physiotherapy for a farmer with ILD (S21; ILD page (p. 169)). Relevance of ABG, PFT and X-ray in goal setting in COPD (W24).

Long answers

Every COPD, bronchiectasis, ILD and post-TB case asks for long-term management: write pulmonary rehab with doses.

Answer skeleton: "pulmonary rehabilitation", 7 marks

1. **[1]** ATS/ERS definition and aims.
2. **[1]** Indications and contraindications.
3. **[1]** Assessment.
4. **[3]** Components: exercise table with doses, education, psychosocial and nutrition; programme length.
5. **[1]** Outcome measures.

Test yourself

3 m **What are the components of pulmonary rehabilitation?**

Answer
Patient assessment; exercise training (endurance, strength, upper limb, with IMT or NMES where needed); education and self-management (inhalers, action plan, breathing strategies, airway clearance, energy conservation, smoking cessation, nutrition); psychosocial support; outcome measurement and maintenance.

MCQ **A COPD patient still smokes. For pulmonary rehab he should be:**

A. Excluded until he quits
B. Accepted, with help to quit during the programme
C. Given breathing exercises only
D. Referred for surgery

Answer
B. Smoking is not an exclusion; the programme is a good time to support quitting.

MCQ **The smallest important change in 6-minute walk distance in chronic lung disease is about:**

A. 5 m
B. 30 m
C. 100 m
D. 300 m

Answer
B. About 25–33 m.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Spruit MA, et al. An official American Thoracic Society/European Respiratory Society statement: key concepts and advances in pulmonary rehabilitation. *Am J Respir Crit Care Med* 2013;188:e13–64.
3. Rochester CL, et al. Pulmonary rehabilitation for adults with chronic respiratory disease: an official ATS clinical practice guideline. *Am J Respir Crit Care Med* 2023. ATS summary
4. Bolton CE, et al. British Thoracic Society guideline on pulmonary rehabilitation in adults. *Thorax* 2013;68 Suppl 2.
5. McCarthy B, et al. Pulmonary rehabilitation for chronic obstructive pulmonary disease. Cochrane Review 2015. Puhan MA, et al. Pulmonary rehabilitation following exacerbations of COPD. Cochrane Review 2016. Cox NS, et al. Telerehabilitation for chronic respiratory disease. Cochrane Review 2021.
6. Holland AE, et al. ERS/ATS technical standard: field walking tests (2014); Singh SJ, et al. Minimum clinically important improvement for the incremental shuttle walking test. *Thorax* 2008.
7. Structure and classical content: AACVPR *Guidelines for Pulmonary Rehabilitation Programs*; Main & Denehy (2016).

Respiratory anatomy, physiology and mechanics of the thorax

> Chapter 1 is revision of earlier years, applied to physiotherapy. It is rarely asked on its own, but every technique answer rests on it: why a breath hold reopens alveoli, why huffing clears small airways, why a flat diaphragm fails. Learn the numbers and the drawings that turn up again and again.

Syllabus	Ch 1(a) Pulmonary anatomy and physiology; 1(d) Biomechanics of thorax (revision)
Priority	**Foundation.** Not asked on its own in 2016–2026; underpins technique and condition answers.
Asked as	Not asked 2016–2026
Usual split	Revision; possible 3-mark definitions or diagrams
Checked against	Standard physiology texts (Guyton and Hall; West) and anatomy texts. Full list under Sources. Content checked 19 October 2026.

1 Airways, lobes and segments

- **Upper airway** (nose, pharynx, larynx) warms, humidifies and filters air. The **trachea** divides at the **carina**, at the level of the sternal angle (about T4–T5).
- The **right main bronchus** is wider, shorter and more vertical than the left, so inhaled objects and aspirated fluid usually go right.
- About 23 generations of branching: generations 0–16 are the **conducting zone** (no gas exchange; anatomical dead space about 150 mL); 17–23 are the **respiratory zone** (respiratory bronchioles, alveolar ducts and sacs), with hundreds of millions of alveoli.
- **Alveolar cells:** type I (thin, for gas exchange) and type II (make **surfactant**, which lowers surface tension and stops small alveoli collapsing); alveolar macrophages clear particles.
- **Mucociliary escalator:** cilia sweep a thin mucus layer upwards to the throat; slowed by smoking, dry gas and infection.
- **Lobes:** right lung has 3 (upper, middle, lower; oblique and horizontal fissures), left has 2 (upper with the lingula, and lower; oblique fissure).

Bronchopulmonary segments (the basis of postural drainage)

Lobe	Right lung (10)	Left lung (8–10)
Upper	Apical, posterior, anterior	Apicoposterior, anterior
Middle or lingula	Lateral, medial (middle lobe)	Superior and inferior lingular
Lower	Superior (apical), medial basal, anterior basal, lateral basal, posterior basal	Superior, anteromedial basal, lateral basal, posterior basal

Surface markings (for hand placement and auscultation)

- **Oblique fissure:** from about the T3 spinous process at the back, round the side, to the 6th costal cartilage at the front. So the lower lobes are mostly at the back, the upper lobes mostly at the front.
- **Horizontal fissure (right):** along the 4th costal cartilage; the middle lobe lies between it and the oblique fissure, at the front.
- **Lower lung border** in quiet breathing: 6th rib in the mid-clavicular line, 8th in the mid-axillary line, 10th at the back. The pleura reaches about two ribs lower (8th, 10th, 12th).

Nerves: vagus (bronchoconstriction, mucus secretion); sympathetic β2 (bronchodilation, the target of salbutamol); phrenic nerve C3–C5 to the diaphragm; intercostal nerves to the intercostals. **Blood supply:** the low-pressure pulmonary circulation (pulmonary artery pressure about 25/10 mmHg) for gas exchange, and bronchial arteries for the airway walls.

2 Mechanics of breathing and biomechanics of the thorax

How the chest gets bigger on breathing in

Movement	Ribs	Diameter increased
Pump handle	Upper ribs (about 1–6): the front ends rise and the sternum moves forwards and up	Front-to-back (anteroposterior)
Bucket handle	Lower ribs (about 7–10): the middle of each rib swings up and out	Side-to-side (transverse)
Caliper	Floating ribs (11–12): swing outwards	Side-to-side, lower chest
Diaphragm descent	Dome moves down	Vertical (the biggest share of quiet breathing)

- **The diaphragm** is the main muscle of quiet breathing (often quoted as about three-quarters of the tidal volume). Where it lies against the inside of the lower ribs (the **zone of apposition**), its contraction, with the abdominal contents as a fulcrum, also lifts and widens the lower ribs.
- **In hyperinflation** (COPD) the diaphragm is flat and short, the zone of apposition shrinks, and contraction can pull the lower ribs **in** (Hoover's sign). This is why forward-lean positions, which push the diaphragm back into a dome, ease breathlessness.
- **Posture:** slumped sitting and lying flat reduce lung volume; kyphosis, scoliosis and a stiff chest wall (ankylosing spondylitis) restrict it.

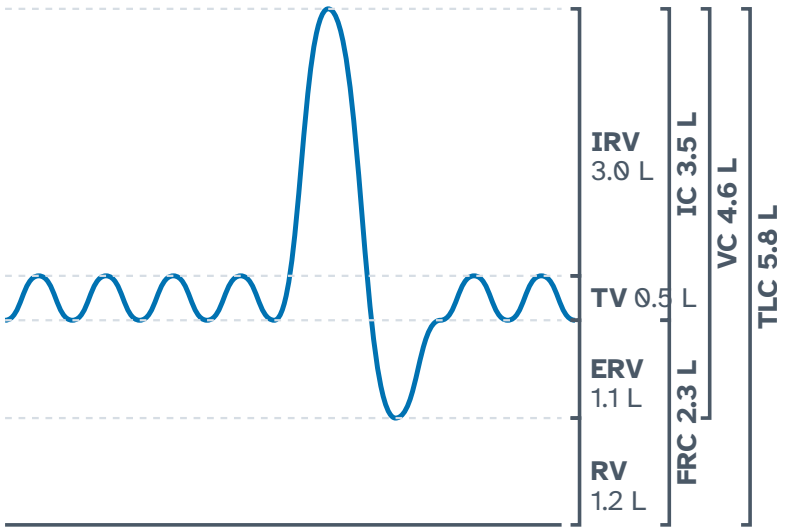
Muscles of breathing

Phase	Muscles
Quiet inspiration	Diaphragm; external intercostals and parasternal internal intercostals; scalenes
Forced inspiration (accessory)	Sternocleidomastoid, scalenes, upper trapezius, pectoralis major and minor (with arms fixed), serratus anterior, back extensors
Quiet expiration	None: passive elastic recoil of the lungs and chest wall
Forced expiration and cough	Abdominals (rectus, obliques, transversus), internal intercostals, latissimus dorsi

- **Pressures:** pleural pressure is about −5 cmH_2O at rest and about −7.5 cmH_2O at the end of a quiet breath in; alveolar pressure is about −1 cmH_2O during inspiration (air flows in) and +1 cmH_2O during expiration.
- **Compliance** (stretchiness: volume change per pressure change): about 200 mL/cmH_2O for the lungs alone and about 110 mL/cmH_2O for lungs and chest wall together. Low in fibrosis, oedema, obesity and chest wall deformity; high lung compliance in emphysema (loss of elastic recoil).
- **Airway resistance** sits mainly in the medium-sized bronchi; it rises with bronchospasm, mucus and airway collapse.
- **Work of breathing** takes only about 3–5% of the body's energy at rest, but much more in lung disease, so breathing muscles compete with leg muscles for oxygen during exercise.

3 Lung volumes and capacities

Spirogram: volumes and capacities (typical young adult)



Values for a young adult man (they vary with size, age and sex). Residual volume, FRC and TLC cannot be measured by simple spirometry, because the residual air never leaves the lungs.

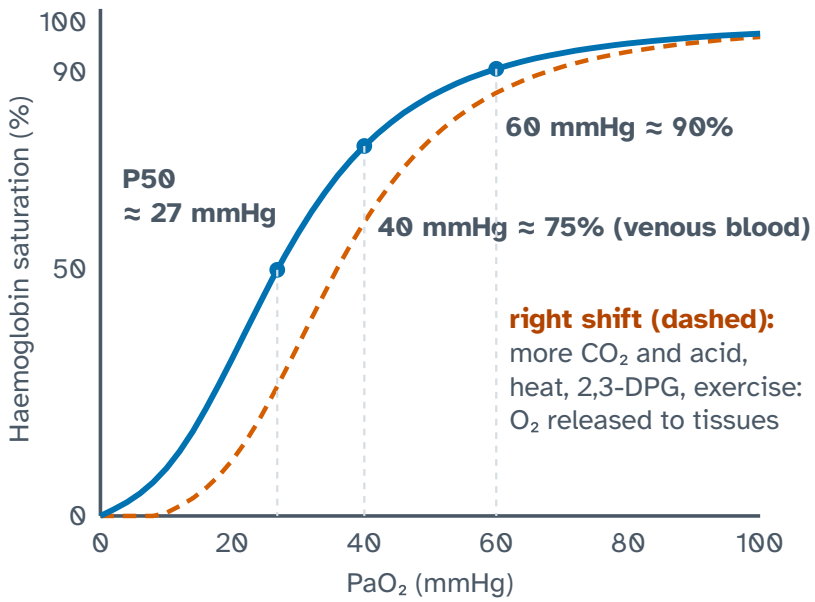
Term	Definition	About
Tidal volume (TV)	Breathed in or out in a quiet breath	0.5 L
Inspiratory reserve volume (IRV)	Extra air that can be breathed in after a quiet breath in	3.0 L
Expiratory reserve volume (ERV)	Extra air that can be breathed out after a quiet breath out	1.1 L
Residual volume (RV)	Air left after a maximal breath out	1.2 L
Inspiratory capacity (IC)	TV + IRV	3.5 L
Functional residual capacity (FRC)	ERV + RV: air left at the end of a quiet breath out	2.3 L
Vital capacity (VC)	IRV + TV + ERV	4.6 L
Total lung capacity (TLC)	All four volumes	5.8 L

FRC matters most to physiotherapists: it falls after surgery, with pain, obesity and lying flat, and small airways at the bases then close (collapse). Upright positions, deep breaths and walking restore it. Minute ventilation = tidal volume × rate (about 6 L/min); alveolar ventilation subtracts the dead space.

4 Gas exchange and transport

- Oxygen and CO_2 cross the very thin alveolar–capillary membrane by **diffusion**, which slows if the membrane thickens (fibrosis, oedema) or its area falls (emphysema, lung resection).
- **Oxygen** is carried about 98.5% bound to haemoglobin and 1.5% dissolved; 1 g of haemoglobin carries about 1.34 mL of oxygen, so anaemia lowers oxygen delivery even with a normal SpO_2.
- **CO_2** is carried as bicarbonate (about 70%), bound to proteins as carbamino compounds (about 23%) and dissolved (about 7%).
- **Bohr effect:** CO_2 and acid make haemoglobin release oxygen in the tissues. **Haldane effect:** oxygen binding makes haemoglobin release CO_2 in the lungs (one reason high oxygen raises CO_2 in COPD).

Oxyhaemoglobin dissociation curve



The flat top means SpO_2 stays above 90% until PaO_2 falls to about 60 mmHg; below that, saturation falls fast. This is why 60 mmHg (8 kPa) defines respiratory failure.

Ventilation–perfusion matching and West's zones are on the positioning card (p. 236).

5 Control of breathing

Part	Role
Medulla (respiratory centres)	Generates the basic rhythm; inspiratory and expiratory neurones
Pons	Fine-tunes the rhythm (pneumotaxic centre limits inspiration)
Central chemoreceptors (medulla)	Respond to CO_2 through acidity of the cerebrospinal fluid: the main drive in health
Peripheral chemoreceptors (carotid and aortic bodies)	Respond mainly to low oxygen (strongly below about 60 mmHg), also to CO_2 and acid
Lung and airway receptors	Stretch receptors (Hering–Breuer reflex), irritant receptors (cough, bronchospasm), J receptors (stiff, wet lungs: breathlessness)
Cortex and limbic system	Voluntary control (breath-holding, speech, breathing exercises) and emotion (anxiety speeds breathing)

6 Physiology behind physiotherapy techniques

Physiology	Technique it explains
Collateral ventilation (pores of Kohn, canals of Lambert, channels of Martin)	Deep breaths with a hold, SMI, PEP: air gets behind mucus and into collapsed units
Equal pressure point in forced expiration	Huffing from mid or low lung volume clears peripheral airways (ACBT card (p. 261))
Gravity on ventilation and perfusion	Positioning for oxygenation and postural drainage
Zone of apposition and diaphragm length	Forward-lean positions in COPD; abdominal binder in tetraplegia
FRC and airway closure	Upright positions, early walking after surgery
Surfactant and alveolar stability	Regular deep breaths and sighs prevent collapse in shallow breathers

Physiology	Technique it explains
Mucociliary clearance	Humidification, hydration, stopping smoking
Chemoreceptors and the Haldane effect	Controlled oxygen in COPD (88–92%)
Cortical control of breathing	Breathing control, pursed-lip breathing, breathing retraining

7 Exam corner

Not asked on its own in the 20 papers (2016–2026); the related questions are on the zones of the lung (S25) and causes of altered V/Q (S23) on the positioning card (p. 236), and spirometry in obstructive and restrictive disease (W20, S19) on the investigations card (p. 318).

Test yourself

3 m **Differentiate the pump-handle and bucket-handle movements of the ribs.**

Answer
Pump handle: upper ribs, front ends rise and the sternum moves forwards and up, increasing the front-to-back diameter. Bucket handle: lower ribs, the middle of each rib swings up and out, increasing the side-to-side diameter.

3 m **Draw a spirogram and label the volumes and capacities.**

Answer
See the figure in section 3: TV 0.5, IRV 3.0, ERV 1.1, RV 1.2 L; IC 3.5, FRC 2.3, VC 4.6, TLC 5.8 L.

MCQ **At a PaO_2 of 60 mmHg, haemoglobin saturation is about:**

A. 50%

B. 75%

C. 90%

D. 100%

Answer
C. About 90%; below this the curve falls steeply.

Sources

1. MUHS BPTh syllabus (2012), CVR Ch 1.
2. Hall JE, Hall ME. *Guyton and Hall Textbook of Medical Physiology* (volumes, pressures, compliance, gas transport).
3. West JB, Luks AM. *West's Respiratory Physiology: The Essentials*.
4. Severinghaus JW. Simple, accurate equations for human blood O_2 dissociation computations. *J Appl Physiol* 1979 (used to draw the curve).
5. Standard anatomy texts (Gray's Anatomy; Chaurasia) for segments and surface markings; Frownfelter & Dean (2022) for chest wall mechanics.

Cardiac anatomy and physiology for physiotherapists

> Chapter 1 is revision, but every cardiac answer leans on it: which artery feeds which wall (ECG leads in MI cases), why a fast heart rate starves the heart muscle, why stroke volume rises with training, why patients faint when stood up after bed rest. Learn these links rather than isolated facts.

Syllabus	Ch 1(b) Cardiac anatomy and physiology
Priority	**Foundation.** Not asked on its own in 2016–2026; underpins the cardiac pages.
Asked as	Not asked 2016–2026
Usual split	Revision; possible 3-mark definitions or diagrams
Checked against	Standard physiology and anatomy texts (Guyton and Hall). Full list under Sources. Content checked 19 October 2026.

1 Anatomy: chambers, valves and coronary arteries

- The heart lies in the middle mediastinum; the apex beat is normally in the left 5th intercostal space in the mid-clavicular line.
- **Right side** (right atrium → tricuspid valve → right ventricle → pulmonary valve → lungs) pumps at low pressure; **left side** (left atrium → mitral valve → left ventricle → aortic valve → body) pumps at high pressure, so the left ventricular wall is the thickest.
- **Pericardium:** a two-layered sac; fluid in it (effusion, tamponade) restricts filling.

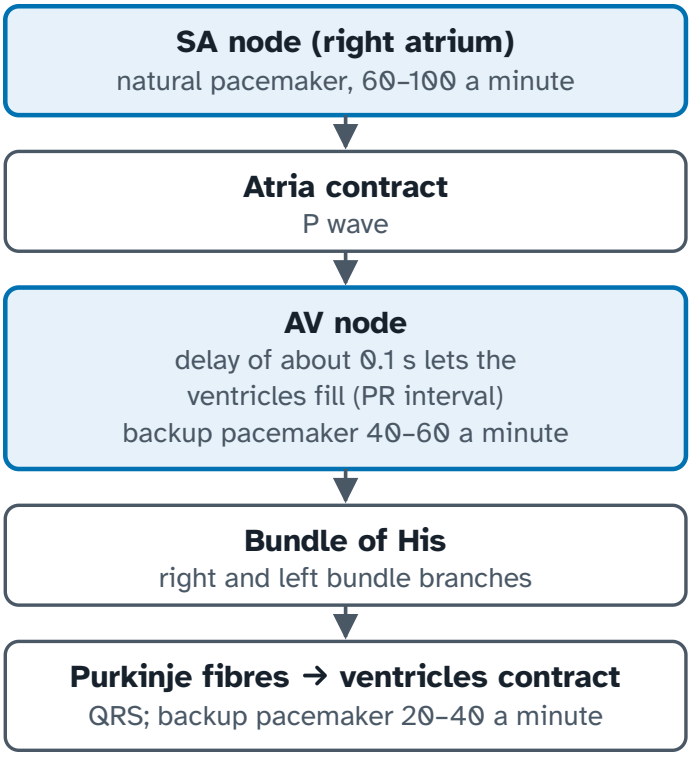
Coronary arteries and what they supply

Artery	Supplies	Blocked: ECG leads
Left main → left anterior descending (LAD)	Front wall of the left ventricle, front two-thirds of the septum, apex	V1–V4 (anteroseptal)
Left main → circumflex	Side and back of the left ventricle, left atrium	I, aVL, V5–V6 (lateral)
Right coronary artery (RCA)	Right atrium and ventricle, inferior wall of the left ventricle, back of the septum; the SA node in most people and the AV node in most (right-dominant) hearts	II, III, aVF (inferior); watch for slow heart rhythms and heart block

Veins drain into the coronary sinus. **Nerves:** sympathetic (β1) speeds the heart and strengthens contraction; the vagus (parasympathetic) slows it.

2 Conduction system

How each heartbeat spreads



The T wave is ventricular recovery (repolarisation). When the higher pacemakers fail, the lower ones take over at slower rates: this is why complete heart block gives a slow pulse.

3 Cardiac cycle and heart sounds

- **Systole** (contraction): isovolumetric contraction, then ejection. **Diastole** (relaxation): isovolumetric relaxation, then filling (rapid filling, then atrial contraction tops it up).
- Typical pressures: left ventricle 120/0–10 mmHg, aorta 120/80; right ventricle 25/0–5, pulmonary artery 25/10.
- **S1** ("lub"): mitral and tricuspid valves close (start of systole). **S2** ("dub"): aortic and pulmonary valves close (start of diastole); it splits on breathing in.
- **S3:** rapid filling into a failing, dilated ventricle (heart failure; can be normal in the young). **S4:** atrial contraction into a stiff ventricle (hypertension, ischaemia).
- **The left ventricle's own blood supply flows mainly in diastole**, because contraction squeezes the coronary vessels. At rest diastole is about two-thirds of each beat; a fast heart rate shortens it, so less blood reaches the heart muscle while it needs more. This is why tachycardia brings on angina and why beta-blockers help.

4 Cardiac output and its regulation

Term	Meaning	Typical value
Cardiac output (CO)	Heart rate × stroke volume	About 5 L/min at rest; about 20–25 L/min in maximal exercise in fit young adults
Stroke volume (SV)	End-diastolic volume – end-systolic volume	About 70 mL at rest
Ejection fraction (EF)	SV ÷ end-diastolic volume	About 55–70%; 40% or below is reduced (heart failure grades on the valve and heart failure page (p. 185))

Term	Meaning	Typical value
Preload	Stretch of the ventricle before it contracts (filling)	Raised by venous return: muscle pump, breathing, lying down
Afterload	Resistance the ventricle pumps against	Raised by high BP and aortic stenosis
Contractility	Strength of contraction at a given stretch	Raised by sympathetic drive; lowered by heart muscle damage

- **Frank–Starling law:** the more the ventricle is filled (stretched), the more strongly it contracts, up to a limit. A failing heart sits on a flatter curve.
- **Blood pressure** = cardiac output × total peripheral resistance. Mean arterial pressure ≈ diastolic + one-third of (systolic − diastolic); for 120/80, about 93 mmHg.
- **Baroreceptor reflex** (carotid sinus and aortic arch): a fall in BP on standing quickly raises heart rate and narrows vessels. After bed rest, with some drugs and with autonomic damage it fails, giving postural hypotension.
- **Myocardial oxygen demand** rises with heart rate, BP, wall tension and contractility. Heart rate × systolic BP (the rate–pressure product) is a bedside guide; angina appears at a fairly fixed product for each patient.

Responses to exercise and training are on the exercise physiology card (p. 315).

5 Physiology behind physiotherapy decisions

Physiology	What it means in practice
Coronary flow in diastole; rate–pressure product	Keep exercise below the heart rate that brings on angina; long warm-up; avoid sudden bursts
Valsalva raises chest pressure then BP swings	Breathe out on effort; no breath-holding in resistance training
Coronary territories	Inferior MI (RCA): watch for slow heart rate and heart block during mobilisation
Frank–Starling and the failing heart	Lower intensity, interval training, watch weight gain and breathlessness (fluid)
Baroreceptor reflex	Sit up in stages after bed rest; check postural BP
Muscle pump and venous return	Ankle pumps against DVT and swelling; cool down slowly so blood does not pool in the legs
Autonomic control of heart rate	Heart rate is unreliable with beta-blockers, pacemakers and after transplant: use RPE

6 Exam corner

Not asked on its own in the 20 papers (2016–2026). The ECG localisation table is used in MI cases (W20, W22, S25: changes in II, III and aVF mean an inferior MI).

Test yourself

3 m **Why does tachycardia worsen myocardial ischaemia?**

> **Answer**
> The heart muscle is perfused mainly in diastole; a fast rate shortens diastole, so supply falls while demand (rate × pressure) rises.

MCQ **ST elevation in II, III and aVF usually means a block in the:**

A. LAD

B. Circumflex only

C. Right coronary artery

D. Left main

Answer

C. Inferior wall, usually the right coronary artery.

MCQ **Cardiac output at rest is about:**

A. 1 L/min

B. 5 L/min

C. 15 L/min

D. 25 L/min

Answer

B. About 70 beats × 70 mL ≈ 5 L/min.

Sources

1. MUHS BPTh syllabus (2012), CVR Ch 1.
2. Hall JE, Hall ME. *Guyton and Hall Textbook of Medical Physiology* (cardiac cycle, output, regulation).
3. Standard anatomy texts (Gray's Anatomy; Chaurasia) for coronary anatomy.
4. Structure and classical content: Hillegass (2022), chapters on cardiovascular anatomy and physiology.

Cardiac and respiratory drugs: what physiotherapists need to know

> Patients in every long case are on several medicines, and each one changes what you see and do: beta-blockers flatten the heart-rate response, diuretics cause postural dizziness, anticoagulants make percussion risky, a bronchodilator before airway clearance makes it work better. This page is organised by what the drug means for physiotherapy, not by pharmacology.

Syllabus	Ch 1(c) Cardiac and respiratory pharmacology
Priority	**Foundation.** Not asked on its own in 2016–2026; needed in every case answer and viva.
Asked as	Not asked 2016–2026
Usual split	Revision; viva questions
Checked against	Standard pharmacology references (BNF, Goodman and Gilman) and ACSM guidance on medications and exercise. Full list under Sources. Content checked 19 October 2026.

1 Heart and blood-pressure drugs

Group (examples)	What it does	Physiotherapy meaning
Beta-blockers (metoprolol, bisoprolol, atenolol, carvedilol)	Slow the heart and lower BP; reduce angina and arrhythmias	Lower resting and exercise heart rate: Karvonen targets overestimate, so use RPE and the talk test; fatigue; non-selective ones can trigger bronchospasm in asthma
ACE inhibitors and ARBs (ramipril, enalapril, losartan, telmisartan); sacubitril–valsartan	Lower BP; protect the heart and kidneys in heart failure and after MI	Postural and post-exercise hypotension: longer cool-down; dry cough with ACE inhibitors
Diuretics (furosemide, torsemide, spironolactone, hydrochlorothiazide)	Remove salt and water (heart failure, oedema, BP)	Dehydration, dizziness on standing, low potassium (cramps, arrhythmias); urgency: plan sessions after the morning dose has worked
Nitrates (glyceryl trinitrate spray or tablet under the tongue, isosorbide)	Widen veins and coronary arteries; relieve angina	Patient should carry GTN to sessions; take it sitting (it drops BP); headache; use before a known trigger if prescribed
Calcium-channel blockers (amlodipine; diltiazem, verapamil)	Widen arteries; diltiazem and verapamil also slow the heart	Ankle swelling (amlodipine), hypotension; slower heart-rate response (diltiazem, verapamil)
Antiplatelets (aspirin, clopidogrel, ticagrelor)	Stop platelets clumping; protect stents and grafts	Bruising and bleeding; never advise stopping them
Anticoagulants (heparin, enoxaparin, warfarin, apixaban, rivaroxaban, dabigatran)	Prevent or treat clots (DVT, PE, atrial fibrillation, mechanical valves)	Bleeding risk: gentle handling, care with percussion, suction and falls; check INR with warfarin (often 2–3; higher for mechanical valves)
Clot-dissolving drugs (streptokinase, tenecteplase, alteplase)	Dissolve the clot in STEMI, stroke, massive PE	High bleeding risk for the first day: no vigorous chest physiotherapy or injections
Statins (atorvastatin, rosuvastatin)	Lower LDL cholesterol; stabilise plaque	Muscle aches or weakness: report new unexplained muscle pain
Digoxin	Slows the rate in atrial fibrillation; mild boost to contraction	Toxicity (nausea, confusion, visual changes, slow or irregular pulse), especially with low potassium
Anti-arrhythmics (amiodarone)	Control atrial and ventricular arrhythmias	Slow heart rate; long-term lung fibrosis and thyroid problems (new breathlessness or cough)

Group (examples)	What it does	Physiotherapy meaning
Inotropes and vasopressors (dobutamine, noradrenaline, adrenaline)	Support BP and cardiac output in ICU	High or rising doses: no active mobilisation (see the ICU safety table (p. 224))

2 Respiratory drugs

Group (examples)	What it does	Physiotherapy meaning
Short-acting β2-agonists (salbutamol, levosalbutamol)	Quick relief of bronchospasm (reliever)	Give 15–30 minutes before airway clearance or exercise if prescribed; keep the reliever at hand; tremor and fast heart rate
Long-acting β2-agonists (salmeterol, formoterol)	12–24-hour bronchodilation (with an inhaled steroid in asthma)	Not for sudden relief (except formoterol combinations in asthma, as prescribed)
Antimuscarinics (ipratropium short-acting; tiotropium, glycopyrronium long-acting)	Bronchodilation, less mucus secretion (mainly COPD)	Dry mouth; check inhaler technique
Inhaled corticosteroids (budesonide, fluticasone, beclometasone)	Reduce airway inflammation (asthma preventer; COPD with frequent exacerbations)	Rinse the mouth after use (thrush, hoarse voice); use a spacer
Oral steroids (prednisolone)	Treat exacerbations; some chronic diseases (ILD, sarcoidosis)	Long courses: osteoporosis (care with percussion and falls), muscle weakness, high sugar, thin skin, poor wound healing
Methylxanthines (theophylline, doxofylline)	Weak bronchodilator	Narrow safety margin: palpitations, nausea, tremor, seizures if levels are high
Leukotriene antagonists (montelukast)	Add-on in asthma and allergic rhinitis	Occasional mood or sleep changes
Mucoactive drugs (N-acetylcysteine, carbocisteine; hypertonic saline; dornase alfa in cystic fibrosis)	Thin or loosen sputum	Time airway clearance after them; hypertonic saline can cause bronchospasm, so give a bronchodilator first
Antibiotics (oral, IV, inhaled)	Treat infection	Inhaled antibiotics go **after** airway clearance, so they reach clear airways
Oxygen	A drug: corrects hypoxaemia	Prescribed with a target; see the oxygen card (p. 290)

Usual order in cystic fibrosis and bronchiectasis: bronchodilator → mucoactive drug → airway clearance → inhaled antibiotic or steroid.

3 Other drugs common in these patients

Group (examples)	Physiotherapy meaning
Insulin and sulfonylureas (glimepiride, gliclazide)	Risk of low blood sugar during and hours after exercise: check sugar, carry glucose, avoid exercising at peak insulin action without food (see the diabetes page (p. 149))
Metformin, gliptins, SGLT2 inhibitors (dapagliflozin, empagliflozin)	Low hypoglycaemia risk on their own; SGLT2 inhibitors also help heart failure but can cause dehydration
Anti-TB drugs (isoniazid, rifampicin, pyrazinamide, ethambutol)	Isoniazid: tingling and numbness of the feet (pyridoxine prevents it); rifampicin: orange urine and tears, many drug interactions; ethambutol: blurred vision; liver toxicity (jaundice, vomiting): report

Group (examples)	Physiotherapy meaning
Opioids and other painkillers (morphine, tramadol, paracetamol); epidurals and nerve blocks	Treat after the dose works (often 30–60 minutes after oral doses); opioids cause drowsiness, slow breathing, nausea and constipation; epidurals can cause leg weakness and low BP when sitting up
Sedatives and paralysing drugs (ICU)	No voluntary movement or cough under paralysis; daily sedation breaks are the time for active work
Anticholinesterases (pyridostigmine in myasthenia)	Treat when the dose is working (often about an hour after it); too much causes secretions and weakness

4 Timing physiotherapy around medicines

- **Bronchodilator** 15–30 minutes before airway clearance or exercise in reactive airways.
- **Pain relief** before post-operative chest physiotherapy and mobilisation.
- **Insulin and meals:** exercise 1–2 hours after a meal; check sugar before and after.
- **GTN** before a known angina trigger, if prescribed.
- **Diuretics:** after the morning dose has worked, so that toilet trips do not interrupt the session.
- **Sedation in tetanus or ICU:** coordinate with the nurse so the patient is comfortable but able to take part.

5 Exam corner

Not asked on its own in the 20 papers (2016–2026). In long cases, a line on the drug's effect on your plan (for example "on beta-blockers, so intensity is set by RPE 11–13") shows clinical reasoning.

Test yourself

3 m **How do beta-blockers affect an exercise prescription?**

Answer
They lower resting and peak heart rate, so heart-rate targets from formulas are too high; use RPE (11–13 or 12–16) and the talk test, or a heart rate from an exercise test done on the medicine.

MCQ **A patient with cystic fibrosis uses salbutamol, hypertonic saline and an inhaled antibiotic. The best order is:**

A. Antibiotic, airway clearance, saline, salbutamol

B. Salbutamol, saline, airway clearance, antibiotic

C. Airway clearance, salbutamol, antibiotic, saline

D. Saline, antibiotic, salbutamol, airway clearance

Answer
B. Open the airways, loosen the mucus, clear it, then deliver the antibiotic to clean airways.

MCQ **Which drug most raises the bleeding risk of chest percussion?**

A. Salbutamol

B. Warfarin

C. Amlodipine

D. Montelukast

Answer

B. Warfarin; check the INR and handle gently.

Sources

1. MUHS BPTh syllabus (2012), CVR Ch 1(c).
2. British National Formulary (BNF), current edition; Brunton LL, et al. *Goodman and Gilman's The Pharmacological Basis of Therapeutics*.
3. American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription*, appendix on common medications and their effects on exercise.
4. Cystic Fibrosis Trust physiotherapy standards (2026) for the order of inhaled treatments.

Basic life support (CPR) for physiotherapists

> Basic life support is syllabus Chapter 11 and a practical skill; it was not asked in the theory papers from 2016 to 2026, but a physiotherapist exercising heart and lung patients must be able to start CPR within seconds. Learn the steps, the numbers and the 2025 changes, and practise on a manikin.

Syllabus	Ch 11 Basic life support (CPCR); practical: skill of administering basic life support
Priority	**Skill.** Not asked in the 2016–2026 theory papers; a practical skill and a viva favourite.
Asked as	Not asked 2016–2026
Usual split	Possible 3- or 7-mark note; practical demonstration
Checked against	American Heart Association CPR and ECC guidelines (2025 update, as summarised in published secondary sources); Indian Society of Anaesthesiologists compression-only life support guideline (2017); ILCOR. Full list under Sources. Content checked 19 October 2026.

1 Recognising cardiac arrest and the chain of survival

Cardiac arrest = the heart stops pumping effectively: the person is **unresponsive** and has **no normal breathing** (no breathing, or only occasional gasps). Gasping is not normal breathing; start CPR. Brain damage begins within minutes, so every minute without CPR lowers the chance of survival.

Chain of survival (adult, outside hospital): call for help early → high-quality CPR → early defibrillation → advanced care → care after the arrest → recovery. In hospital, the first link is recognising deterioration early.

2 Adult BLS: the steps

1. **Safety:** make sure the scene is safe for you and the patient.
2. **Check response:** tap the shoulders and shout "Are you all right?". Do not shake the neck.
3. **Shout for help** and get the emergency response going: in hospital, call the emergency (code) team; outside, call **108** (ambulance) or 112, put the phone on speaker; send someone for an **AED**.
4. **Check breathing** (and, if trained, the carotid pulse at the same time) for **no more than 10 seconds**.
5. No normal breathing (no pulse): **start chest compressions** at once, then **30 compressions : 2 breaths**.
6. When the AED arrives, switch it on and follow its voice prompts; restart compressions immediately after each shock.
7. Continue until the person starts breathing normally, the team takes over, or you are exhausted. Swap compressors every 2 minutes.

Memory hook

Check, Call, Compress, Shock

3 High-quality CPR: the numbers

Element	Adult
Hand position	Heel of one hand on the centre of the chest (lower half of the sternum), other hand on top, arms straight, shoulders over the hands
Rate	**100–120** a minute
Depth	At least **5 cm** but not more than **6 cm**

Element	Adult
Recoil	Let the chest come back fully after each compression; do not lean on it
Pauses	Keep interruptions under **10 seconds**
Ratio	**30:2** without an advanced airway (one or two rescuers)
Breaths	Head tilt–chin lift (jaw thrust if neck injury is suspected); each breath over about 1 second, just enough to see the chest rise; avoid over-ventilation
With an advanced airway	Continuous compressions with 1 breath every 6 seconds (10 a minute)
Surface	Start at once, even on a bed (2025 update); a backboard helps but do not delay compressions to move the patient

Compression-only CPR: untrained bystanders, or anyone unwilling or unable to give breaths, should give continuous hard and fast compressions in the centre of the chest until help arrives. This is the basis of the Indian compression-only life support (COLS) guideline for lay rescuers.

4 Using an AED

1. Switch it on and follow the voice prompts.
2. Bare and dry the chest. Pads: one below the right collarbone, the other on the left side of the chest below the armpit (follow the pictures on the pads); not over a pacemaker bump.
3. Stand clear while it analyses; nobody touches the patient.
4. If a shock is advised: say "Stand clear", check, press the shock button.
5. Restart compressions **immediately** for 2 minutes; do not wait to check the pulse. The AED reanalyses every 2 minutes.

Defibrillation works for ventricular fibrillation and pulseless ventricular tachycardia; each minute of delay lowers survival, which is why AEDs are placed in public places and wards.

5 Children and infants (outline)

Element	Child (1 year to puberty)	Infant (under 1 year)
Depth	About one-third of the chest depth, about 5 cm	About one-third of the chest depth, about 4 cm
Technique	One or two hands	Two thumbs with the hands encircling the chest, or the heel of one hand (2025 update; the two-finger method is no longer advised)
Ratio	30:2 with one rescuer, 15:2 with two	30:2 with one rescuer, 15:2 with two
Rate	100–120 a minute	100–120 a minute

Children usually arrest because of a breathing problem, so breaths matter more than in adults.

6 Choking (airway obstruction by a foreign body)

- **Mild** (can cough, speak, breathe): encourage coughing; stay with them.
- **Severe** (cannot speak, breathe or cough; hands at the throat) in an adult or child: **5 back blows** between the shoulder blades with the person leaning forward, then **5 abdominal thrusts** (inward and upward, above the navel); keep alternating until the object comes out or the person becomes unresponsive (2025 AHA update adds back blows to abdominal thrusts).
- **Pregnant or very obese:** chest thrusts instead of abdominal thrusts.
- **Infant:** 5 back blows with the baby face-down along your forearm, then 5 chest thrusts; no abdominal thrusts.

- **Becomes unresponsive:** lower to the floor, call for help, start CPR; look in the mouth before giving breaths and remove a visible object (no blind finger sweeps).

7 Recovery position

For a person who is unresponsive but **breathing normally** (and has no suspected spinal injury): roll them onto their side, upper leg bent at hip and knee in front for stability, hand under the cheek, head tilted back slightly so the airway stays open and vomit can drain. Keep checking breathing; if it stops, roll back and start CPR.

8 Cardiac arrest in physiotherapy settings

- **Be ready:** know where the AED and emergency trolley are, the emergency number, and who in the gym is trained; keep a phone at hand during cardiac and pulmonary rehab.
- **Warning signs before an arrest:** chest pain, sudden breathlessness, dizziness, pallor and sweating, a new irregular pulse, falling BP: stop exercise, sit or lie the patient down, check vital signs and call for help early.
- **On a monitor (ICU, cardiac rehab):** ventricular fibrillation on the screen with an unresponsive patient means arrest: call, start CPR, defibrillate. A loose ECG lead also looks chaotic, so check the patient.
- **After heart surgery:** follow the unit's protocol (some units shock or re-open the chest first), but never withhold compressions out of fear of the sternal wound.
- **Do-not-resuscitate orders:** check the patient's documented wishes in advance.
- **Training:** BLS certificates usually last 2 years; practise on a manikin with feedback.

9 Exam corner

Not asked in the theory papers from 2016 to 2026. In the practical exam, BLS can be asked as a skill demonstration and in the viva.

Test yourself

3 m **List the features of high-quality CPR in an adult.**

Answer
Rate 100–120 a minute; depth 5–6 cm; full chest recoil; interruptions under 10 seconds; 30:2 compressions to breaths without an advanced airway; avoid excessive ventilation; change compressor every 2 minutes.

MCQ **A patient collapses in the gym, is unresponsive and gasping occasionally. The next step after calling for help is:**

A. Wait and watch the breathing for a minute

B. Start chest compressions

C. Give water

D. Put him in the recovery position

Answer
B. Gasping is not normal breathing; treat as cardiac arrest.

MCQ **A conscious adult is choking and cannot speak or cough. Current AHA advice is:**

A. Abdominal thrusts only

B. Alternate 5 back blows and 5 abdominal thrusts

C. Blind finger sweep

D. Give water

Answer

B. The 2025 update adds back blows, alternating with abdominal thrusts.

Sources

1. MUHS BPTh syllabus (2012), Ch 11 and practical skills list.
2. American Heart Association. Guidelines for cardiopulmonary resuscitation and emergency cardiovascular care, 2025 update (Part 7, adult basic life support, *Circulation*, October 2025), as summarised in published secondary sources; the primary text was not available to check.
3. Ahmed SM, et al. Compression-only life support (COLS) for cardiopulmonary resuscitation by layperson outside the hospital. *Indian J Anaesth* 2017;61:867–73 (Indian Resuscitation Council, Indian Society of Anaesthesiologists).
4. International Liaison Committee on Resuscitation (ILCOR) consensus on science and treatment recommendations.

Part D

Exam skills

Answer writing with ICF and goals, and the practical exam.

Exam skills

Answer writing: the paper, ICF, goals and documentation 361

Practical exam pack: long case, short case, spots and journal 365

Answer writing: the paper, ICF, goals and documentation

> Every one of the 20 papers (2016–2026) had the same shape, and almost every long answer asks for ICF, a functional diagnosis or impairments, goals and management. Marks follow structure as much as knowledge. This page turns the notes into answers that score.

Syllabus	Ch 5 Application of ICF model: short- and long-term goals; patient-specific goals with time frames and clinical reasoning; documentation
Priority	**Essential.** ICF and goals are asked in nearly every long answer; structure decides marks in all sections.
Asked as	Nearly every long answer (ICF, goals)
Usual split	Applies to every question
Checked against	WHO ICF (2001); MUHS question papers 2016–2026; SOAP documentation conventions. Full list under Sources. Content checked 19 October 2026.

1 The paper and a time plan

All 20 papers from 2016 to 2026 used this pattern (no MCQs, although the syllabus prints a different scheme): 80 marks in 3 hours, one answer book.

Question	Choice	Marks	Time to spend
Read and choose	Tick your choices; plan the long answers	–	5 min
Q1 short answers	Any 5 of 6	5 × 3 = 15	About 6 min each (30 min)
Q2 short answers	Any 5 of 6	5 × 7 = 35	About 14 min each (70 min)
Q3 and Q4 long answers	Any 1 of 2 each (patient cases)	2 × 15 = 30	About 30 min each (60 min)
Revise	Underline keywords, add missing doses and precautions	–	15 min

- About **2 minutes a mark**. Never spend 20 minutes on a 3-mark answer.
- If you write slowly, do the two long answers first, so 30 marks are safe; never leave them for the last 20 minutes.
- Choose the long-answer option whose **condition** you know best, not the shorter-looking question.

2 Short answers: 3 and 7 marks

3 marks (about half a page)

1. One-line definition.
2. Three to five points, a small table, or a labelled sketch (numbers, values, names).
3. One line of physiotherapy relevance if it fits.

Example: "Ankle–brachial index" (3 marks)

Definition: ratio of ankle to arm systolic pressure, measured with a Doppler. Formula: highest ankle ÷ highest brachial pressure, for each leg. Interpretation: 0.91–1.40 normal, 0.90 or less PAD, below 0.40 severe, above 1.40 non-compressible arteries. Use: diagnosis of PAD and a check before compression bandaging.

7 marks (1.5–2 pages)

1. Definition or introduction (2 lines).
2. Principle, classification or "why" (a flowchart is ideal).

3. The main body as 6–8 headed points or a table: technique with dose, or management by phase.
4. A labelled diagram where one exists (cycle, positions, curve, timeline).
5. Precautions and contraindications.
6. One concluding line (evidence or outcome measure).

Each technique card and condition page has an answer skeleton with marks for the questions actually asked.

3 Long answers: the 15-mark case

1. **Read the split** (for example 5 + 5 + 5 or 3 + 4 + 8) and give each part space in proportion.
2. **Underline the case clues:** age, occupation, earning role, home (stairs, lives alone), day after surgery, drains, settings, test results, what the patient wants.
3. **Functional diagnosis** in one sentence, then the **ICF table** with a "clinical reasoning" column (section 4).
4. **Problem list** in order of priority.
5. **Goals:** short-term and long-term, SMART, with time frames (section 5).
6. **Management:** by phase or by impairment, with technique, dose and reason; precautions and stop criteria; education and home programme.
7. **Outcome measures** and discharge or progression criteria in one or two lines.

Write doses with units (repetitions, sets, times a day, minutes, RPE, % HRR); a plan without doses loses marks.

4 Applying the ICF model (Ch 5)

The WHO **International Classification of Functioning, Disability and Health (ICF)** describes a person's health as the interaction of the health condition with body functions and structures, activities and participation, and contextual factors. The model diagram is drawn on the IHD page, section 10 (p. 15); draw it in every long answer.

Component	Meaning	Typical clues in the question
Health condition	The diagnosis and its stage	"CABG 3 days back", "post-Koch's bronchiectasis", "40% burns"
Body functions and structures (impairments)	Problems in body parts or functions	Pain VAS 6, breathlessness, sputum, reduced expansion, weakness, stiff shoulder, oedema, low SpO_2
Activity limitations	Difficulty doing tasks	Cannot sit or sleep, cannot climb stairs, cannot walk to the field, difficulty with self-care
Participation restrictions	Difficulty in life roles	Cannot work as a driver, teacher or cook; only earning member; misses school; embarrassed to meet people
Environmental factors	Facilitators and barriers around the person	2nd floor with no lift, lives alone, family support, cost, dusty job, distance to hospital
Personal factors	The person's own features	Age, smoker, motivation, fear, "wants to resume job early", "ignored advice"

> **Functional diagnosis template**
>
> "[Age]-year-old [occupation] with [health condition, day or stage], presenting with [3–4 key impairments], leading to [activity limitations], restricting [participation], with [main facilitator] and [main barrier]."

Common mistakes: listing symptoms as participation; leaving out contextual factors; writing a disease description instead of function; not linking each impairment to a treatment.

5 Goals with time frames (Ch 5a, 5b)

Goals should be **SMART**: Specific, Measurable, Achievable, Relevant to the patient, Time-bound. Write them as what the patient will do.

Type	Time frame	Example
Short-term	Days to about 2 weeks (in hospital, or early rehab)	"Clears sputum independently with ACBT within 3 days"; "walks 100 m on the ward with RPE below 13 by discharge"
Long-term	About 6–12 weeks or more (end of the programme)	"Returns to driving to work (1 hour) by 6 weeks"; "climbs to the 2nd floor with one rest by 8 weeks"

Patient-specific goals from real MUHS cases

Case	Long-term goal
Bank manager after angioplasty who drives 1 hour to work (S20)	Drives to work and completes a full working day without symptoms by about 4 weeks, after clearance
Tiffin cook after mastectomy (S19)	Cooks for ten clients twice a day, lifting cooking vessels with the operated arm, by about 8–12 weeks, with lymphoedema precautions
Farmer with COPD (S17)	Walks to the field and does light farm work with pacing, without an exacerbation, by 12 weeks
Milkman after below-knee amputation (S19)	Delivers door to door with a prosthesis, managing stairs, within the rehabilitation programme
Schoolchild with asthma (S23, W22)	Attends school every day and joins outdoor play with a warm-up and reliever plan

Clinical reasoning: link each goal to an impairment and each treatment to a goal ("because…").

6 Documentation (Ch 5c): SOAP notes

Part	What to write	Example: day 3 after CABG
S: Subjective	What the patient reports	"Wound pain 4/10 on coughing; slept 4 hours; wants to walk to the toilet."
O: Objective	Measured findings and treatment given	HR 92, BP 130/84, SpO_2 95% on air; reduced breath sounds at the left base; walked 40 m with RPE 12; ACBT 3 cycles
A: Assessment	Your interpretation and progress towards goals	Left basal collapse improving; tolerating Phase I progression
P: Plan	Next steps	Continue ACBT 2-hourly; walk 3–4 times a day, progress to 80 m; stairs trial tomorrow

Also record consent, precautions, adverse events and outcome measures with dates. The practical-exam journal (5 marks) follows the same headings plus ICF, prognosis and intervention.

7 Presentation that earns marks

- Headings, numbered points and tables; one idea per line.
- Underline the keywords listed in each exam corner (for example ICF, FITT, RPE, sternal precautions).
- Labelled diagrams: ICF model, ACBT cycle, drainage positions, CR phases timeline, rule of nines, spirogram.
- Always include precautions and stop criteria in management answers.
- Keep pathology short unless it is asked; spend the space on physiotherapy.
- Leave a few lines after each answer for additions during revision.

Checklist before you hand in

- Every part of every split answered?
- Doses with units? Time frames on goals?
- Precautions written? Diagram labelled?
- Question numbers written correctly in the margin?

Sources

1. MUHS BPTh syllabus (2012), CVR Ch 5; MUHS IV BPTh question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. International Classification of Functioning, Disability and Health (ICF), 2001.
3. Weed LL. Medical records that guide and teach (problem-oriented record and SOAP notes). *N Engl J Med* 1968.

Practical exam pack: long case, short case, spots and journal

> The CVR practical is 80 marks: a long case (45), a short case with two skills (20), five spots (10) and the journal (5), marked on knowledge, hands-on skill and attitude. This pack follows that scheme exactly and links each skill to the card that explains it.

Syllabus	Practical syllabus: positioning and breathing control, airway clearance and devices, handling techniques, PNF for breathing, exercise equipment and prescription, ICU equipment, physical and electrical agents, basic life support
Priority	**Essential.** 80 practical marks plus 20 internal assessment.
Asked as	Long case 45 · Short case 20 · Spots 10 · Journal 5
Usual split	Long case 10 + 10 + 10 + 15; short case 2 × 10; spots 5 × 2; journal 5
Checked against	MUHS BPTh syllabus (2012), CVR practical examination scheme; the cards on this site for each skill. Full list under Sources. Content checked 19 October 2026.

1 The marking scheme

Part	What is marked	Marks
Long case	(a) Subjective and physical examination 10; (b) evaluation and physiotherapy diagnosis (ICF) 10; (c) plan of care and goal setting 10; (d) demonstration of one important test and one treatment on the patient 15. Marked in cognitive, psychomotor and affective domains	45
Short case	Demonstration of two physiotherapy skills for effective patient management	2 × 10 = 20
Spots	Chest or cardiac X-ray, ABG, PFT, ECG, adjuncts and devices	5 × 2 = 10
Journal	Documentation: assessment, evaluation, diagnosis, prognosis and intervention of cases, with ICF	5
Total	Plus internal assessment 20 (terminal and preliminary exams)	80

In the college terminal and preliminary practicals, spots carry 15 marks (5 × 3) and the journal carries no marks.

2 Long case: assessment proforma

Subjective

- **Demographics:** name, age, sex, occupation (what the job demands), hand dominance, address (stairs, distance).
- **Chief complaints** in the patient's words, with duration.
- **History of present illness:** breathlessness (onset, triggers, mMRC or NYHA grade, orthopnoea, waking at night), cough and sputum (amount, colour, smell, blood), chest pain (character, radiation, relief), palpitations, fainting, swelling of the feet, calf pain on walking (claudication distance), fever, weight loss.
- **Past history:** TB, diabetes, hypertension, heart disease, surgery, admissions, ventilation.
- **Personal history:** smoking in pack-years (cigarettes a day ÷ 20 × years smoked), bidis and chewing tobacco, alcohol, diet, sleep, bladder and bowel.
- **Drug history:** inhalers and technique, cardiac drugs (beta-blockers change your exercise plan), oxygen at home.
- **Family and social history:** who lives at home, support, finances, home layout, work.
- **The patient's goals** ("I want to go back to driving").

Objective

- **Vital signs:** HR (rate and rhythm), BP, respiratory rate, SpO_2 (on air or oxygen), temperature, pain score.
- **General observation:** build and BMI, posture, cyanosis, clubbing, pallor, oedema, lines, drains, oxygen device, scars.
- **Chest:** shape, breathing pattern and rate, accessory muscle use, chest movement; palpation (trachea, expansion: tape measurements at the axilla, nipple line and xiphisternum, compared side to side); percussion; auscultation (breath sounds, added sounds); cough strength; sputum.
- **Cardiovascular:** pulses (including foot pulses), capillary refill, oedema (pitting grade), postural BP, ABI if vascular.
- **Musculoskeletal and function:** shoulder and thoracic range, posture, muscle strength (MRC grade), bed mobility, transfers, gait, balance.
- **Exercise tolerance:** 6MWT or shuttle test with SpO_2 and Borg, or a sit-to-stand test.
- **Scales:** mMRC, modified Borg, NYHA, CCS, pain scale, quality of life (CAT, SGRQ, MLHFQ).
- **Investigations:** X-ray, ABG, PFT, ECG, echo (EF), blood tests.

Then

1. ICF table and one-line functional diagnosis (see the answer-writing card (p. 362)).
2. Problem list; short- and long-term goals with time frames.
3. Plan of care with doses and precautions.
4. Demonstrate **one test** (for example chest expansion, 6MWT set-up, ABI, auscultation, postural BP) and **one treatment** (for example ACBT, drainage position, incentive spirometry, Phase I walking with monitoring).

3 Presenting the long case (3–5 minutes)

> **Template**
>
> "Mr X, 55, a bank manager from Pune, is on day 3 after CABG for triple vessel disease. His main complaints are wound pain and breathlessness on walking. He is a smoker of 20 pack-years with hypertension. On examination: HR 92 regular, BP 130/84, SpO_2 95% on air, reduced breath sounds at the left base, weak cough, walks 40 m with RPE 12. My functional diagnosis is … . My short-term goals are … ; long-term goals … . My plan is … . I will now demonstrate a 6-minute walk set-up and ACBT."

Likely viva questions: why this goal; what you would do if SpO_2 falls; stop criteria; how you would progress; what the X-ray or ECG shows; why this dose; what the drugs mean for exercise; outcome measures; home programme.

4 Short case: demonstrating skills (2 × 10 marks)

Steps for any skill demonstration

1. Introduce yourself, check the patient's identity, explain and take consent.
2. Hand hygiene; privacy and draping; mask in suspected TB.
3. Check vital signs and contraindications for this skill.
4. Position the patient (and yourself) correctly.
5. Explain each step in simple words; demonstrate; let the patient do it; correct.
6. Watch the response (breathing, SpO_2, pain, colour) and stop if needed.
7. State the dose (repetitions, sets, frequency) and the home programme.
8. Reassess, thank the patient, and document.

Skills likely to be asked

Skill	Card

Skill	Card
ACBT, huffing, supported cough, autogenic drainage	ACBT (p. 261)
Postural drainage positions, percussion, vibration, shaking	Postural drainage (p. 267)
PEP mask, Flutter, Acapella; MDI with spacer; nebuliser set-up	PEP devices (p. 271); nebulisers and inhalers (p. 294)
Diaphragmatic and pursed-lip breathing, thoracic expansion, SMI with incentive spirometer, thoracic mobility	Breathing exercises (p. 275)
NPF and PNF for breathing	NPF and PNF (p. 279)
Positions of ease, energy conservation, pacing, relaxation	Dyspnoea (p. 285); yoga and relaxation (p. 302)
Inspiratory muscle training set-up	Respiratory muscle training (p. 282)
6MWT, shuttle walk, Harvard step test; exercise prescription with heart-rate monitor and oximeter	Exercise testing (p. 308); exercise physiology (p. 313)
ABI, pulses, postural BP, claudication distance; Buerger–Allen exercises	Investigations (p. 320); therapeutic exercise (p. 299)
Stump bandaging; compression bandaging and fitting stockings; burns splints and positioning; taping; exercise and monitoring equipment	Handling skills (p. 305); amputation (p. 110); burns (p. 31)
Sternal precautions, log-roll, early mobilisation after surgery	IHD and CABG (p. 7); bed rest (p. 241)
Suctioning, manual hyperinflation, ICU positioning (on a manikin)	Suctioning (p. 231); positioning (p. 236)
TENS for wound pain; wound-healing modalities	Electrotherapy (p. 298)
Basic life support on a manikin	BLS (p. 356)

5 Spots (5 × 2 marks, a few minutes each)

Spot	What to write
Chest X-ray	PA or AP; the main finding (for example "left lower lobe collapse": sail sign, raised left hemidiaphragm); two features; physiotherapy meaning. Order: investigations card (p. 317)
ABG	pH → respiratory or metabolic → compensation → oxygenation; name it (for example "acute type 2 respiratory failure"). ABG card (p. 326)
PFT	FEV1, FVC, ratio → obstructive, restrictive or mixed; severity; reversibility; a likely condition. Investigations card (p. 318)
ECG	Rate, rhythm, P waves, QRS width, ST changes → name; safe to exercise or not. Investigations card (p. 318)
Device	Name; parts; principle; one indication; one contraindication or precaution (table below)

Devices that appear as spots

Device	Principle or use	One precaution
Incentive spirometer (volume or flow type)	Visual feedback for slow deep breaths with a hold (SMI); post-operative lung expansion	Add-on to deep breathing and walking; not for patients who cannot take a deep breath
PEP mask; Flutter; Acapella	Expiratory resistance 10–20 cmH_2O (plus oscillation) for airway clearance	Not with an undrained pneumothorax
Metered-dose inhaler, spacer, dry powder inhaler	Deliver bronchodilators and steroids to the airways	Rinse the mouth after steroids; DPI needs a fast, strong breath in
Jet nebuliser	Gas through a narrow opening makes droplets; baffle removes large ones	Air-driven in patients at risk of high CO_2

Device	Principle or use	One precaution
Nasal cannula, simple mask, Venturi mask, reservoir mask	Oxygen at about 24–40%, 35–60%, fixed 24–60%, and 60–90%	Simple mask never below 5 L/min; target SpO_2 prescribed
Heat and moisture exchanger	Passive humidification for intubated patients	Not with thick or bloody secretions or during nebulisation
Suction catheter, Yankauer sucker	Remove secretions from the airway	Under 15 s, no stronger than –200 mmHg in adults
Self-inflating bag (Ambu) and manual hyperinflation circuit	Manual ventilation; bagging with a slow breath, hold and quick release	Not with an undrained pneumothorax; pressure about 40 cmH_2O or less
Endotracheal tube, tracheostomy tube (cuffed)	Artificial airway; the cuff seals and limits aspiration	Secure the tube during every turn
Intercostal drain with underwater seal	Drains air or fluid; the water acts as a one-way valve	Keep the bottle below the chest; never clamp a bubbling drain
Peak flow meter; spirometer	Peak expiratory flow; FEV1 and FVC	Best of three blows
Pulse oximeter	Measures SpO_2 and pulse by light absorption	Wrong with cold fingers, movement, nail polish and carbon monoxide
Hand-held Doppler and sphygmomanometer	ABI measurement	ABI above 1.40 is unreliable (calcified arteries)
Threshold inspiratory muscle trainer	Spring valve opens at a set pressure (start about 30% of MIP)	Not after recent pneumothorax or with a perforated eardrum
Compression stockings, bandages, pressure garments, thermoplastic splints	Venous disease, lymphoedema, DVT prevention; burns scar and contracture control	Check ABPI before compression; check splints for pressure areas

6 The journal (5 marks)

Each case written up under the same headings: assessment (subjective, objective), evaluation (investigations, scales), ICF and functional diagnosis, prognosis, goals, intervention with doses and progression, outcome measures, and a SOAP follow-up note. Keep it neat and complete; examiners read the ICF and the doses first. Template on the answer-writing card (p. 363).

7 The affective domain: what examiners watch

- Greeting, introduction, consent, respectful language, privacy and draping.
- Hand hygiene; infection control (mask for TB, cleaning shared devices).
- Safety: checking vital signs, oxygen, lines and drains before moving the patient; stopping when the patient is unwell.
- Explaining in the patient's language (Marathi or Hindi where needed) and checking understanding.
- Calm, honest answers in the viva: if you do not know, say so and say how you would find out.
- Thanking the patient and leaving them comfortable.

Last two weeks

1. Present three practice long cases aloud (a cardiac, a respiratory and a vascular or surgical case), timed.
2. Practise each short-case skill on a friend with the 8-step sequence.
3. Do spot drills: one X-ray, ABG, PFT, ECG and device in 10 minutes, then check.
4. Finish the journal.

Sources

1. MUHS BPTh syllabus (2012), CVR physiotherapy: practical examination scheme and practical skills list.
2. The technique cards and condition pages on this site, with their sources.

Abbreviations

1RM: One-repetition maximum

6MWT: Six-minute walk test

AACVPR: American Association of Cardiovascular and Pulmonary Rehabilitation

AARC: American Association for Respiratory Care

ABG: Arterial blood gas

ABI: Ankle–brachial index

ABPI: Ankle–brachial pressure index

ACBT: Active cycle of breathing technique

ACC: American College of Cardiology

ACCP: American College of Chest Physicians

ACE: Angiotensin-converting enzyme

ACPRC: Association of Chartered Physiotherapists in Respiratory Care

ACSM: American College of Sports Medicine

ADA: American Diabetes Association

ADL: Activities of daily living

AED: Automated external defibrillator

AF: Atrial fibrillation

AHA: American Heart Association

AKA: Above-knee amputation

ALAT: Latin American Thoracic Association

APTA: American Physical Therapy Association

ARB: Angiotensin receptor blocker

ARDS: Acute respiratory distress syndrome

ARNI: Angiotensin receptor–neprilysin inhibitor

ATP: Adenosine triphosphate

ATS: American Thoracic Society

AV: Atrioventricular

AYUSH: Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy (Indian ministry)

BACPAR: British Association of Chartered Physiotherapists in Amputee Rehabilitation

BK: Below-knee

BKA: Below-knee amputation

BLS: Basic life support

BMI: Body mass index

BNP: Brain natriuretic peptide

BP: Blood pressure

BRCA: Breast cancer genes 1 and 2

BTS: British Thoracic Society

CABG: Coronary artery bypass grafting

CAD: Coronary artery disease

CAT: COPD Assessment Test

CBNAAT: Cartridge-based nucleic acid amplification test (for TB)

CCS: Canadian Cardiovascular Society

CEAP: Clinical, aetiological, anatomical, pathophysiological classification (venous disease)

CHD: Coronary heart disease

CIVIQ: Chronic Venous Insufficiency Quality of Life Questionnaire

CK: Creatine kinase

CK-MB: Creatine kinase, heart muscle (MB) form

CMV: Controlled mandatory ventilation

COPD: Chronic obstructive pulmonary disease

CPAP: Continuous positive airway pressure

CPET: Cardiopulmonary exercise test

CPR: Cardiopulmonary resuscitation

CR: Cardiac rehabilitation

CR10: Borg category–ratio 0–10 scale

CRQ: Chronic Respiratory Questionnaire

CSF: Cerebrospinal fluid

CT: Computed tomography

CVC: Central venous catheter

CXR: Chest X-ray

DLCO: Diffusing capacity of the lung for carbon monoxide

DM: Diabetes mellitus

DNA: Deoxyribonucleic acid

DPI: Dry-powder inhaler

DPP-4: Dipeptidyl peptidase 4

DVT: Deep vein thrombosis

ECC: Emergency cardiovascular care

ECG: Electrocardiogram

ECMO: Extracorporeal membrane oxygenation

EF: Ejection fraction

EMG: Electromyography

EORTC: European Organisation for Research and Treatment of Cancer

EQ-5D: EuroQol five-dimension questionnaire

ERS: European Respiratory Society

ESC: European Society of Cardiology

ESVS: European Society for Vascular Surgery

ESWT: Endurance shuttle walk test

ETT: Endotracheal tube

EU: European Union

FET: Forced expiration technique

FEV1: Forced expiratory volume in 1 second

FIM: Functional Independence Measure

FiO_2: Fraction of inspired oxygen (0.21 = room air)

FITT: Frequency, intensity, time and type

FRC: Functional residual capacity

FVC: Forced vital capacity

GBS: Guillain–Barré syndrome

GCS: Glasgow Coma Scale

GINA: Global Initiative for Asthma

GLP-1: Glucagon-like peptide 1

GOLD: Global Initiative for Chronic Obstructive Lung Disease

GTN: Glyceryl trinitrate (nitroglycerin)

HADS: Hospital Anxiety and Depression Scale

HbA1c: Glycated haemoglobin (3-month blood sugar)

HDL: High-density lipoprotein (good cholesterol)

HFCWO: High-frequency chest wall oscillation

HFNC: High-flow nasal cannula

HIV: Human immunodeficiency virus

HME: Heat and moisture exchanger

HR: Heart rate

HRCT: High-resolution CT scan

HRR: Heart-rate reserve

HTN: Hypertension

ICF: International Classification of Functioning, Disability and Health

ICMR: Indian Council of Medical Research

ICP: Intracranial pressure

ICS: Inhaled corticosteroid

ICU: Intensive care unit

ICU-AW: ICU-acquired weakness

IHD: Ischaemic heart disease

ILCOR: International Liaison Committee on Resuscitation

ILD: Interstitial lung disease

IMA: Internal mammary artery

IMT: Inspiratory muscle training

INR: International normalised ratio (warfarin control)

IP: Interphalangeal

IPPB: Intermittent positive pressure breathing

ISBI: International Society for Burn Injuries

ISH: International Society of Hypertension

ISWT: Incremental shuttle walk test

IWGDF: International Working Group on the Diabetic Foot

JRS: Japanese Respiratory Society

KCCQ: Kansas City Cardiomyopathy Questionnaire

LABA: Long-acting β2-agonist

LAD: Left anterior descending artery

LAMA: Long-acting muscarinic antagonist

LAQ: Long-answer question

LDH: Lactate dehydrogenase

LDL: Low-density lipoprotein

LIMA: Left internal mammary artery

LTG: Long-term goals

MAP: Mean arterial pressure

MCP: Metacarpophalangeal

MDI: Metered-dose inhaler

MDR-TB: Multidrug-resistant tuberculosis

MET: Metabolic equivalent

METs: Metabolic equivalents (1 MET = resting energy use)

MHI: Manual hyperinflation (bagging)

MI: Myocardial infarction

MIP: Maximal inspiratory pressure

MLHFQ: Minnesota Living with Heart Failure Questionnaire

mMRC: Modified Medical Research Council dyspnoea scale

MONA: Morphine, oxygen, nitrates, aspirin

MRC: Medical Research Council

MRI: Magnetic resonance imaging

MRM: Modified radical mastectomy

NHS: National Health Service (UK)

NICE: National Institute for Health and Care Excellence (UK)

NIV: Non-invasive ventilation

NMES: Neuromuscular electrical stimulation

NPF: Neurophysiological facilitation of respiration

NRS: Numeric rating scale

NSAID: Non-steroidal anti-inflammatory drug

NYHA: New York Heart Association

$PaCO_2$: Arterial carbon dioxide pressure

PAD: Peripheral arterial disease

PaO_2: Arterial oxygen pressure

PAQLQ: Paediatric Asthma Quality of Life Questionnaire

PCI: Percutaneous coronary intervention (angioplasty)

PE: Pulmonary embolism

PEEP: Positive end-expiratory pressure

PEF: Peak expiratory flow

PEP: Positive expiratory pressure

PFT: Pulmonary function test

PHQ-9: Patient Health Questionnaire (9 items)

PIP: Proximal interphalangeal

PNF: Proprioceptive neuromuscular facilitation

PSV: Pressure support ventilation

PT: Physiotherapy

QRS: QRS complex (ventricular depolarisation on the ECG)

RASS: Richmond Agitation–Sedation Scale

RDS: Respiratory distress syndrome

RIMA: Right internal mammary artery

RMT: Respiratory muscle training

ROM: Range of motion

RPE: Rating of perceived exertion (Borg scale)

RR: Respiratory rate

RSBI: Rapid shallow breathing index (breathing rate ÷ tidal volume in litres)

RVH: Right ventricular hypertrophy

SA: Sinoatrial

SBT: Spontaneous breathing trial

SEAS: Scientific Exercise Approach to Scoliosis

SF-36: Short Form 36 health survey

SGLT2: Sodium–glucose co-transporter 2

SGRQ: St George's Respiratory Questionnaire

SIGAM: Special Interest Group in Amputee Medicine (mobility grades)

SIGN: Scottish Intercollegiate Guidelines Network

SIMV: Synchronised intermittent mandatory ventilation

SMART: Specific, measurable, achievable, relevant, time-bound

SMI: Sustained maximal inspiration

SOAP: Subjective, objective, assessment, plan

SOFA: Sequential Organ Failure Assessment score

SOS: When needed (si opus sit)

SOSORT: International Society on Scoliosis Orthopaedic and Rehabilitation Treatment

SpO_2: Oxygen saturation by pulse oximeter

STEMI: ST-elevation myocardial infarction

STG: Short-term goals

SVS: Society for Vascular Surgery

TB: Tuberculosis

TBSA: Total body surface area

TEE: Thoracic expansion exercises

TENS: Transcutaneous electrical nerve stimulation

TLC: Total lung capacity

UK: United Kingdom

UN: United Nations

VAP: Ventilator-associated pneumonia

VAS: Visual analogue scale

VATS: Video-assisted thoracoscopic surgery

VC: Vital capacity

VHI: Ventilator hyperinflation

VO_2max: Maximal oxygen uptake

WHO: World Health Organization

WHOQOL-BREF: WHO Quality of Life questionnaire, short version

WIQ: Walking Impairment Questionnaire